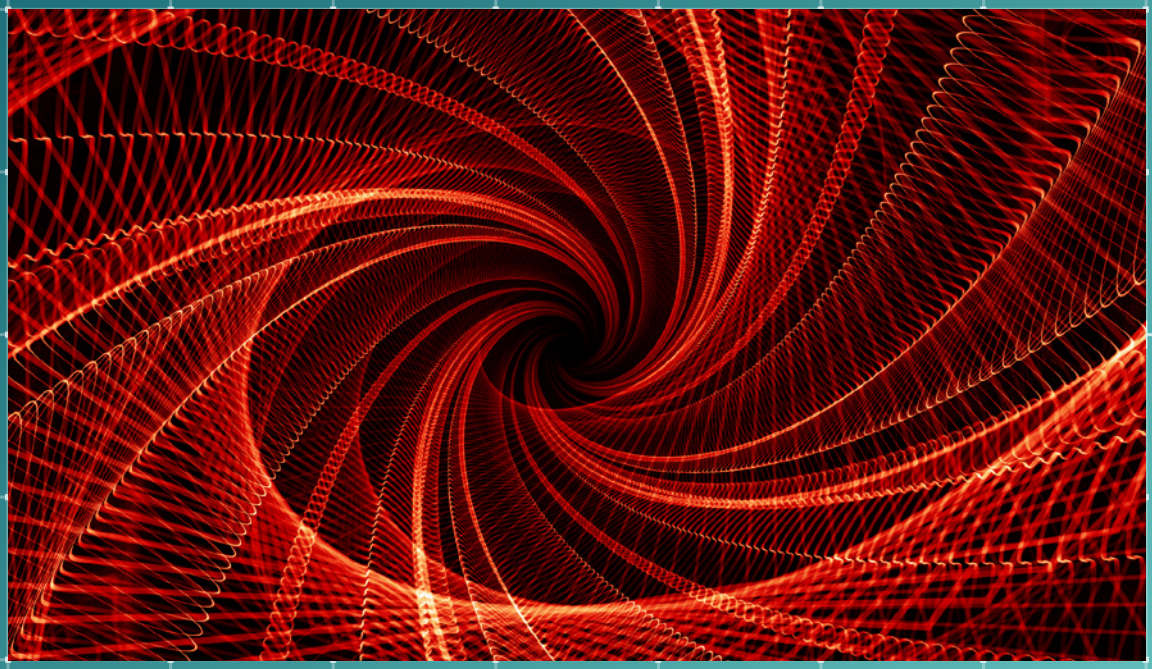
American Water Works Association

Manual of Water Supply Practices

M36

Fifth Edition



Water Audits and Loss Control Programs

This page intentionally blank.

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Manual of Water Supply Practices

M36

Water Audits and Loss Control Programs

Fifth Edition

Errata incorporated Feb. 2025

American Water Works Association

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Manual of Water Supply Practices—M36, fifth Edition

Water Audits and Loss Control Programs

Copyright © 2024 American Water Works Association

All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including scanning, recording, or any information or retrieval system. Reproduction and commercial use of this material is prohibited, except with written permission from the publisher.

Disclaimer
The authors, contributors, editors, and publisher do not assume responsibility for the validity of the content or any consequences of their use. In no event will AWWA be liable for direct, indirect, special, incidental, or consequential damages arising out of the use of information presented in this book. In particular, AWWA will not be responsible for any costs, including, but not limited to, those incurred as a result of lost revenue. In no event shall AWWA's liability exceed the amount paid for the purchase of this book.

Senior Editorial Manager – Books Products: Suzanne Snyder
Senior Manuals Specialist: Molly Beach
Manager – Publishing Operations: Gillian Wink
Copyright and Permissions Manager: Peggy Tyler
Technical Editor: Anne Lanctôt
Technical Illustrations: Michael Labruyere
Cover Image: agsandrew/Shutterstock.com
Production: Innodata

Library of Congress Cataloging-in-Publication Data
Names: American Water Works Association.
Title: M36 - water audits and loss control programs / by American Water Works Association.
Other titles: Water audits and loss control programs
Description: Fifth edition. | Denver, CO : American Water Works Association, [2024] | Revised edition of: Water audits and loss control programs / by George A. Kunkel, Jr. Denver : American Water Works Association, 2016. | Includes bibliographical references and index. |
Identifiers: LCCN 2024001564 (print) | LCCN 2024001565 (ebook) | ISBN 9781647171544 (paperback) | ISBN 9781613006832 (adobe pdf)
Subjects: LCSH: Water consumption--Measurement. | Water--Waste--Measurement. | Water leakage--Prevention. | Loss control.
Classification: LCC TD499 .W38 2024 (print) | LCC TD499 (ebook) | DDC 628.1/440287--dc23/eng/20240206
LC record available at https://lccn.loc.gov/2024001564
LC ebook record available at https://lccn.loc.gov/2024001565

Printed in the United States of America

ISBN 978-1-64717-154-4 ISBN, electronic 978-1-61300-683-2

http://doi.org/10.12999/AWWA.M36ed5

All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including scanning, recording, or any information or retrieval system. Reproduction and commercial use of this material is prohibited, except with written permission from the publisher. Please send any requests or questions to permissions@awwa.org.

American Water Works Association

American Water Works Association
6666 West Quincy Avenue
Denver, CO 80235-3098
303.794.7711
awwa.org

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Contents

List of Figures, v

List of Tables, xi

Preface, xiii

Acknowledgments, xv

Chapter 1 Introduction: Auditing Water Supply Operations and Controlling Losses 1

The Water Audit and Water Balance, 2
The Importance of Water Audits and Nonrevenue Water Management, 3
Getting Started, 5
North American Governmental Agency Nonrevenue Water Policy, 5
References and Resources, 7

Chapter 2 Conducting the Water Audit 9

The Water Audit, 11
The Water Balance Calculation, 11
Compiling the Top-Down Water Audit Data, 14
Considerations for Wholesale Water Systems, 83
Summary, 86
References, 86

Chapter 3 The Occurrence and Impacts of Apparent Losses. 89

How Apparent Losses Occur, 90
Monitoring Meter Performance, 99
Problems Created by Apparent Losses, 101
Summary, 102
References, 102

Chapter 4 Controlling Apparent Losses: Optimized Revenue Capture and Customer Data Integrity 105

Quantifying Apparent Losses in the Water Audit, 106
Customer Billing System, 109
Automatic Meter Reading and Advanced Metering Infrastructure Systems, 109
SDHE, 113
CMI, 125
UC, 141
Developing the Apparent Loss Control Strategy, 148
Developing a Revenue Protection Plan to Control Apparent Losses, 153
Summary, 156
References, 156

Chapter 5 Understanding Real Losses: The Occurrence and Impacts of Leakage 159

The Impact of Leakage, 160
The How and Why of Leakage, 162
The Effect of Time on Leakage Losses, 163

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Characterizing Leakage Events, 166
A Further Word on Customer Service Line Piping Leakage, 169
Water Pressure and Leakage, 170
How to Quantify Leakage, 173
Estimating Losses from Discovered Leaks, 180
How to Locate Leaks, 182
Starting A Leakage Management Program, 188
The Economics of Leakage Management, 200
Leak Detection Interventions, 202
CAM, 208
The Benefit–Cost Ratio Approach, 212
References, 213

Chapter 6 Approaches for Controlling Real Losses: Leakage and Pressure Management. 217

Active Leakage Control, 219
Distribution System Knowledge and Field Operations Savvy for Detecting Leaks, 235
Flow Measurement and Analysis to Quantity and Manage Leakage, 235
Summary, 264
References, 270

Chapter 7 Planning and Sustaining the Nonrevenue Water Management Program 273

Identifying Desired Outcomes and Benefits of an NRW Management Program, 274
Establishing A Cross-Functional NRW Management Group or Task Force, 275
Identifying Resource Needs, 277
Launching the NRW Management Program, 278
Reporting NRW Management Progress Within the Water Utility, 280
Communicating with Community Stakeholders—Public Relations, 282
Sustaining the NRW Management Program, 283
Reference, 284

Chapter 8 Considerations for Small Systems. 285

Obtaining Technical and Financial Resources for Water Loss Control, 285
Implementing Water Loss Control Programs in Small Systems, 286
Developing a Nonrevenue Management Program in a Small System, 291
Managing Unbilled Authorized Consumption, 292
Addressing Apparent Losses, 293
Addressing Real Losses, 295
Analyzing and Processing Water Audit Results, 297
Accountability in Small Water Utilities, 299
References, 299

Appendix A Validating Production Flowmeter Data and the Annual Water Supplied Volume. 301

Appendix B Assessing Water Resources Management 339

Glossary, 345

Index, 353

List of AWWA Manuals, 359

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Figures

1-1 North American States and Provinces with AWWA-based Water Loss Control Policies and Programs, 6

2-1 Water balance, 12
2-2 Water balance for County Water Company—2020 calendar year using the AWWA Free Water Audit Software (Version 6.0), 12
2-3a Identifying system boundaries for a water audit conducted on a wholesale transmission water supply system, 14
2-3b Identifying system boundaries for a treated water distribution system, 15
2-3c Identifying system boundaries for a discrete pressure zone or DMA, 15
2-4 AWWA Free Water Audit Software—Start Page completed for County Water Company, 18
2-5 AWWA Free Water Audit Software—Worksheet completed for County Water Company, 19
2-6 AWWA Free Water Audit Software—Interactive Data Grading for Volume from Own Sources (VOS) completed for County Water Company, 20
2-7 AWWA Free Water Audit Software—Dashboard for County Water Company, 20
2-8 AWWA Free Water Audit Software—Water Loss Control Planning Guide Worksheet completed for County Water Company, 23
2-9 Detailed meter lag correction, 37
2-10 Customer meter flow recorder, 53
2-11 Determining the average length of customer service line, Lp, for customer meter located at the curb stop, where line ownership changes, 63
2-12 Determining the average length of customer service line, Lp, for customer meter located inside customer premise, 64
2-13 Determining the average length of customer service line, Lp, for unmetered customer properties, 64
2-14 Map of fire hydrants in a section of a water utility distribution system (used to assist the calculation of average operating pressure), 67
2-15 AWWA Free Water Audit Software—Dashboard for County Water Company, 77

3-1 Metered consumption data archival path, 91
3-2 Accumulation of moisture inside a water meter register obscuring the dial and preventing visual meter readings, 93
3-3 A water meter installed upside down is an example of a poorly installed meter; water meters should be installed upright and usually in the horizontal position, 98
3-4 A bank of water meters showing numerous meters improperly installed at an angle instead of upright and horizontal, 99
3-5 The impact of uncontrolled apparent losses in a water utility over time, 101

Copyright © 2024 by American Water Works Association. All Rights Reserved

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

4-1 This meter was listed as a 3-in meter in the utility billing system software, but a quick field audit revealed it to be a 4-in turbine meter (identified by the eight flange bolts). There is also a small brass tag with "4-in" riveted to the body next to the register. The total monthly base charges for a 3-in meter at this utility are $598.49, but the total monthly base charges for a 4-in meter are $1,042.33. Thus, the utility was losing $5,326.08 a year in unbilled base charge revenue from this one meter., 106

4-2 Primary activities for reducing apparent water loss, 107

4-3 Apparent water loss in gallons from 32 validated billing system audits over 25 years, 108

4-4 A "cheater" or "jumper" pipe (top) that was illegally used to replace the top meter in the meter setter: in the lower part of this meter pit is a typical ⅝-in residential meter, 108

4-5 Sources of SDHE, 113

4-6 Billing process chart for a utility with manual meter reading, 115

4-7 Billing process chart for a utility with AMR meter reading, 116

4-8 Billing process chart for a utility with AMI meter reading, 117

4-9 Meter register showing active six digits on the meter, along with erroneous AMR touch-probe meter reading of only five digits, 119

4-10 Data-mining analysis chart for the City of Wylie, Tex., showing the number of active ¾-in meter single-family residential accounts in water consumption ranges, 122

4-11 Data-mining analysis chart for the City of Wylie, Tex., showing annual billed water consumption for ¾-in residential meter customer accounts in various annual water consumption billing ranges, 123

4-12 Data-mining analysis chart for the 3-in meters in Wylie, Tex., 124

4-13 Meter test bench for testing small water meters, 129

4-14 The pressure-reducing valve installed immediately upstream of this compound meter disturbed the flow profile entering the meter despite the presence of the plate strainer, causing the meter to overregister usage (102.4% overall weighted average accuracy), 129

4-15 The 90° elbow just upstream of this compound meter disturbed the flow profile entering the measuring chamber, and the absence of a strainer allowed debris to clog the bypass meter measuring chamber. This meter tested at 74.7% accuracy at the maximum flow rate through the main meter and 68.1% accurate at the intermediate flow rate through the main meter. Low-flow rate testing registered zero flow (0.0% accuracy) for three test regimens: minimum flow rate through the bypass meter, maximum flow rate through the bypass meter (below changeover), and during the changeover. It registered 16.3% accuracy above the changeover level. On a composite basis, the meter performed very poorly., 130

4-16 Truck equipped with large meter testing apparatus for in situ meter testing at LWC, 131

4-17 Hose connected to large meter test port in underground meter pit and ready for testing, 132

4-18 Interior of large meter test truck showing calibrated test meter, 132

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

4-19 Water being discharged to waste during large meter testing operation. Utilities should be certain to follow applicable state and local requirements for the safe discharge of drinking water to surfaces., 132
4-20 Graph produced from Wylie, Tex, big box retail store consumption meter data logging showing minimum/average/maximum flow rates, 135
4-21 Graph produced from Wylie, Tex, big box retail store customer consumption meter data logging showing percentage of flow in given flow ranges, 136
4-22 Public relations flyer cautioning communities about the impacts of unauthorized water consumption, 142
4-23 The open bypass line valve on this 4-in compound meter setting was costing the utility approximately $127,100 USD annually in unbilled water and sewer revenue., 143
4-24 The bypass meter isolation valves on this 3-in dual-body compound meter have been closed, which renders the meter incapable of registering any usage below 8.5 gpm., 144
4-25 The bypass meter isolation valves on this 6-in fire service meter serving a condominium have been closed, which renders the meter incapable of registering any usage below 45 gpm., 144
4-26 A finishing nail has been driven through the meter register between the face and the flow indicator, impeding the ability of the impeller to rotate on the spindle of this single-jet meter., 145
4-27 Unmetered 2-in tap upstream of a 3-in compound meter. There also is a single check valve backflow prevention device upstream of the meter. Backflow prevention devices should never be installed upstream of a water meter., 146
4-28 Illegal bypass at a residential meter service discovered during a water meter and mobile AMR system project, 147
4-29 Four-pillar approach to the control of apparent losses, 149
4-30 Cost curve for a mechanical water meter replacement program, 150
4-31 Economic balance for an apparent loss reduction by mechanical meter replacement, 150
4-32 Steps to execute an apparent loss control strategy, 152

5-1 Tank leaks and overflows are a minor component of real losses, 160
5-2 Large water main break, 164
5-3 Small leak on customer service piping, 165
5-4 Time periods in the life of a leak, 165
5-5 Example of various leakage types and impacts of time on the volume of loss, 166
5-6 Example leak noise signature suggesting increasing leak flow rate over time, 167
5-7 Components of leakage and appropriate intervention tools, 168
5-8 Distribution system flow metering and DMA design configuration options, 174
5-9 Preliminary design sketch for installation of a PRV chamber metered for daily monitoring with emergency bypass, 177
5-10 Leak correlation is used as a standard pinpointing tool of the leak detection squad in many water utilities, 184
5-11 Leak noise monitors help to automate the leak survey process and display leak candidates daily., 185

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

5-12 Hydrophone retrofitted to fire hydrant base, 186
5-13 Four-pillar approach to the control of real (leakage) losses, 190
5-14 WRF Project 4372-b, *Leak Repair Data Collection Guide* data selection options, 194
5-15 WRF Project 4372-b, *Leak Repair Data Collection Guide* water main failures data worksheet, 195
5-16 LCA model: leakage losses stemming from reported failures on primary water main piping in CWC (WRF 2014a), 196
5-17 LCA model: Leakage losses stemming from reported failures on mains fittings (fire hydrants, valves, meter, etc.) in CWC (WRF 2014a), 196
5-18 LCA model: Leakage losses stemming from reported failures on customer service connections in CWC (WRF 2014a), 197
5-19 LCA model: background leakage volume calculation for CWC (WRF 2014a), 199
5-20 LCA model summary table showing the steps used to calculate the component totals, 200
5-21 LCA model: component analysis summary CWC (WRF 2014a), 200
5-22 RR of leakage and changing leakage levels with an active leak detection program, 204
5-23 Economic unreported real losses for regular survey, 206
5-24 Economic intervention frequency for leak detection by survey for CWC, 207
5-25 Excavated break detected by correlating acoustic monitoring, 210
5-26 Impact of CAM on RR and leakage reduction, 211

6-1 Four-pillar approach to the control of real (leakage) losses, 218
6-2 Leak sound velocities in pipelines of different materials, 220
6-3 The Sahara technology is a tethered inline leak detection system that is effective in pinpointing leaks on large-diameter piping., 227
6-4 Tethered inline leak detection insertion apparatus, 227
6-5 Use of tethered inline leak detection technology in a 48-in water main, 228
6-6 Free-flowing leak detection sensor inside the retrieval tool, 228
6-7 Pressure and flow variations in a DMA without specific pressure management controls, 241
6-8 Pressure management changes water mains repair frequency in the Durban CBD in South Africa, 243
6-9 Pressure zones and DMA in the Philadelphia (United States) Water Department water service area, 246
6-10 Pressure zones in the water service area of Ann Arbor, Michigan, United States, 247
6-11 Pressure control devices, such as PRVs, provide consistent outlet pressure., 248
6-12 Fixed outlet flow and pressure in zone, 248
6-13 System schematic using fixed outlet control mode, 249
6-14 Flow-based dynamic modulation mode via a PRV combats high head loss in the distribution system and ensures a smooth pressure profile at the weaker points in the system., 250
6-15 Using flow-based dynamic modulation mode results in PRV outlet pressure variation closely matching the water demand flow., 251

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

6-16 Typical pressure management and DMA supply installation, 254
6-17 DMA 5 in Philadelphia baseline hydrograph of flow and pressure prior to DMA and pressure management, 258
6-18 DMA 5 in Philadelphia after configuration into a DMA with basic pressure management and completion of leak detection survey, 259
6-19 DMA 5 in Philadelphia after refinement of pressure management strategy, 259
6-20 DMA 5 in Philadelphia in an optimized state after final leak detection survey, 260
6-21 Repair clamps are commonly used to repair circumferential ruptures on distribution piping since they are quick to install and are highly durable., 263

8-1 Annual volumes of water supplied, authorized consumption, and nonrevenue water, 298
8-2 Monthly volumes of water supplied, authorized consumption, and nonrevenue water, 298
A-1 Magnetic flowmeter awaiting installation on a 60-in.-diameter raw water pipeline in Philadelphia, Pa., 303
A-2 Insertion magnetic flowmeter in use on a 30-in. pipeline in Birmingham, Ala., 304
A-3 Cross-sectional view of an orifice plate flowmeter installation, 305
A-4 Bank of differential pressure transmitters (DP cells) at the effluent pumping facility of the Crescent Hill Water Treatment Plant, Louisville, Ky., 306
A-5 Chart recorder and bank of chart recorders at the Queen Lane Pumping Station in Philadelphia, Pa., 306
A-6 Flowmeter installation with compression check valve located close to propeller flowmeter., 308
A-7 Venturi flowmeter installation with butterfly valve connected to downstream flange of the flowmeter, and used to throttle flow., 309
A-8 Schematic of the treated water clearwell and water supply piping configuration for County Water Company's treatment works from the Mountain Creek Source, 312
A-9 Insertion Pitot rods in-place to measure flows for comparison of water flow in large flowmeters., 316
A-10 Velocity profile in a smooth pipeline with unobstructed, uniform flow at ultra-low velocities, 317
A-11 Velocity profile in pipeline with moderate internal corrosion, typical of measurable turbulent flow, 318
A-12 Velocity profile upstream of a butterfly valve in a pipeline, 318
A-13 Velocity profile upstream of a partially closed gate valve in a pipeline, 319
A-14 Velocity profile skewed by close proximity to a bend in the pipeline at ultra-low velocities., 319
A-15 Schematic of the water supply configuration of the Queen Lane Water Treatment Plant in Philadelphia, Pa., 323
A-17 Ames, Iowa, SCADA system water supply tracking report, 332
B-1 Standard AWWA water balance, 340
B-2 Modified AWWA water balance showing raw water withdrawal, utilization, and losses, 341

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Tables

2-1 Water balance terms and definitions, 13

2-2 Metering locations in drinking water supply systems, 16

2-3 Acronym key for terms in the AWWA Free Water Audit Software, 21

2-4 Source water measuring devices for CWC, 25

2-5 Total water supply in million gallons for CWC (uncorrected), 25

2-6 Changes in reservoir/tank storage for CWC, 27

2-7 Volume from Own Sources in million gallons for CWC (adjusted for meter error), 30

2-8 Number of customer accounts and metered consumption by meter size for County Water Company: Jan. 1–Dec. 31, 2020, 34

2-9 Total metered water consumption by category for County Water Company (uncorrected), 34

2-10 Sum of individual estimates of Unbilled Unmetered Authorized Consumption, 41

2-11 Distorted customer consumption data due to customer billing adjustments triggered by using negative consumption values (example data for a ⅝-in residential meter account), 48

2-12 Utilizing separate fields for registered and billed consumption in the customer billing system (example data for a ⅝-in residential water meter account; see 2-11), 50

2-13 Example of weighting factors for flow rates related to volume percentages for ⅝-in water meters in County Water Company, 56

2-14 Meter testing data from a random sample of 50 meters for County Water Company, 57

2-15 Sample calculation of residential water-meter error, 57

2-16 Calculation of residential water meter error, 57

2-17 Example volume percentages for large meters for County Water Company, 58

2-18 Meter test data for large meters for County Water Company, 58

2-19 Calculation of large water meter error, 58

2-20 Determining average operating pressure in a water utility distribution system with varying topography, 65

2-21 Determining average system pressure in a section of a water utility distribution system (see Figure 2-14 for reference), 66

2-22 Volume-weighted average calculation of CRUC for County Water Company using billed volumes by billing category and billing rate, 71

2-23 Volume-weighted average calculation of CRUC for County Water Authority starting with total volumetric revenue, 71

2-24 Component values of the UARL calculation, 75

2-25 Standard unit values used for the UARL calculation, 75

2-26 System Attributes and Performance Indicators of the AWWA water audit method (from Trachtman et al. 2019 and Kunkel et al. 2020), 79

Copyright © 2024 by American Water Works Association. All Rights Reserved

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

2-27 Range of Values of the AWWA Performance Indicators in the AWWA Water Audit Reference Dataset (WARD) that are embedded in the "Gauge" or "Speedometer" Charts in the Dashboard of the Audit Software (US dollars), 81

4-1 Single-family residential (¾-in meter) water consumption for FY 2019 in Wylie, Tex, 121
4-2 Annual water consumption in gallons by range of water consumption in Wylie, Tex, 121
4-3 Annual water consumption for accounts with 3-in meters in Wylie, Tex, 123
4-4 Customer meter population demographics and metered consumption for CWC as of Dec. 31, 2020, 127
4-5a Compound meter test data for large meters for CWC, 133
4-5b Turbine meter test data for large meters for CWC, 133
4-6 Data logging results for 4-in horizontal turbine meters for an Illinois water utility, 138

5-1 Summary of AOP in psi in the WARD, 171
5-2 Leakage flow rates for metallic piping systems, 180
5-3 Leakage losses for circular holes under different pressures, 181
5-4 Leakage losses for joints and cracks, 182
5-5 Steps for initiating a sustainable leakage management program, 189
5-6 Leakage unit rates from the WARD, 189

6-1 Multiple benefits of pressure management in water distribution systems, 240

7-1 NRW management program activity planning matrix, 278
7-2 Monthly NRW Report for CWC, 281

8-1 Example data sources for water volume inputs to the AWWA water audit, 287
8-2 SOPs to track and reduce nonrevenue water, 291

A-1 Recommendations for desired upstream, unobstructed straight pipe length for common water utility flowmeters, 310
A-2 Example tabulations of a drop test conducted at County Water Company's water treatment plant at Mountain Creek, 313
A-3 Production flowmeter system procedure for inline meter accuracy verification, 321
A-4 Mass balance report for the Queen Lane Water Treatment Plant in Philadelphia, Pa., showing consistent production flow metering data reflecting soperations and good meter accuracy., 324
A-5 Mass balance report for the Queen Lane Water Treatment Plant in Philadelphia, Pa., showing unusual data variation suggesting unsoperations or compromised meter accuracy for the period July 25–31, 2012, 325
A-6 Water supply balancing report for County Water Company, 329
A-7 Water storage tank report for County Water Company showing corrections for Monument Road storage tank overflow event on September 3, 2020, 330
A-8 Example of a data gap and adjustments for pumped water flow to a typical pressure zone operated by a large water utility., 335

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Preface

Managing Non revenue Water—An Imperative for Reliable Water Supplies

North American water utilities have been successful in providing safe, reliable water supplies as a foundation for growth and prosperity for over two centuries. However, many utilities operate with uncertain accountability controls and high losses of both treated drinking water and revenue. These losses are collectively termed *nonrevenue water* (NRW).

The component of NRW that occurs as physical *real losses* (leakage) wastes valuable water and energy resources as more water is extracted from sources than is needed by the customer population. *Apparent* losses are the nonphysical losses from customer metering inaccuracies, reading and/or billing errors, and unauthorized consumption, which result in uncaptured revenue and customer equity issues. Water suppliers have a responsibility to manage water resources wisely and efficiently. The accountable and efficient management of water supplies by utilities is the central focus of the American Water Works Association (AWWA) water audit methodology and NRW management programs detailed in this manual.

The world's rapidly changing climate is causing weather extremes, spurring drought, floods, wildfires, tornadoes, sea level rise, and other impacts that can impair the provision of safe water supplies. In 2021, for the first time in history, the US Bureau of Reclamation, the federal agency that oversees operations of the Colorado River, declared a water shortage at Lake Mead, the massive lake formed by Hoover Dam which spans the Nevada–Arizona border. Mandatory water restrictions were enacted in 2022 for Arizona, California, Nevada, and Mexico. Water utilities in these locations must be prepared to adapt to the reduced water availability that has occurred largely due to the changing climate. With an increasing potential for water scarcity in many regions, water utilities have an urgent need to manage precious water supplies with greater efficiency than in the past. For many older water systems in North America, aging infrastructure is also exerting a toll as failures and high leakage rates increase and compromise system efficiency while often interrupting water service and causing damage to streets and property.

Fortunately, while water utility challenges continue to increase, so has innovation in methodology and technology that has enabled water supply operations to improve in a cost-effective manner. For more than 30 years, the M36 manual has provided the tools and guidance water utilities need to audit their supply and contain NRW to economically achievable levels. This fifth edition includes many updates and adds innovations developed since the fourth edition was published in 2016.

The first edition of this manual was published in 1990 and detailed the water audit method advocated by the California Department of Water Resources and adopted by the California-Nevada Section of AWWA. The second edition was published in 1999 and provided relatively minor updates to the first edition. The third edition (2009) included a major advancement in water audit methodology, giving water utilities greater guidance in improving accountability and economically controlling water and revenue losses. The fourth edition (2016) significantly enhanced all of the chapters and outlined new details on production

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

flowmeter testing and leakage and pressure management. It also provided descriptions of several free software tools and examples of regulatory approaches. This fifth edition features detailed instructions on Version 6.0 of the AWWA Free Water Audit Software released in 2020 and updated guidance on the use of key performance indicators (KPIs), instructions for water audits for wholesale water suppliers, and updated leakage and pressure management methods and technology.

As in the third and fourth editions, the fifth edition explains the AWWA water audit methodology in a user-friendly manner and provides an overview of some of the best loss control techniques used to implement a sustainable NRW management program. Chapter 1 provides a brief introduction, while Chapter 2 gives detailed instruction on the water audit process and explains the use of the Version 6.0 AWWA Free Water Audit Software. Chapters 3 and 4 describe the impacts and the methods to control apparent losses, especially ways to recoup missing revenues by controlling these losses. Chapters 5 and 6 discuss the impacts of real losses, which are largely due to leakage, and the methods to control these losses, including proactive leak detection, flow measurement in district metered areas, and advanced pressure management. Chapter 7 provides the planning and organizational steps a water utility can take to implement and sustain the NRW management program, while Chapter 8 offers valuable insights for small water systems in managing their losses.

A glossary of terms and definitions is also provided. Two appendixes are included. Appendix A provides detailed guidance on production flowmeter management, including flowmeter accuracy testing and flowmeter data integrity. Appendix B discusses NRW impacts on water resources. The AWWA Free Water Audit Software can be downloaded directly from the Water Loss Control resource page at the AWWA website and used to obtain a preliminary quantity of losses and their costs. This step can be followed up by field measurements and investigations to gradually enhance and validate the water audit, procedures that are clearly described throughout this manual. Examples are included within the manual for the fictitious County Water Company, illustrating the means to compile the water audit and initiate control of apparent and real losses.

AWWA provides water utilities with effective tools and methods to promote accountability and efficiency in their supply operations. Now more than ever, water utility managers are being called upon to carefully evaluate inefficiencies in their water system and to take corrective actions to control excessive NRW. The methods contained in this manual will help them achieve these goals efficiently.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Acknowledgments

The AWWA Technical & Educational Council, the Distribution & Plant Operations Division, and the Water Loss Control Committee gratefully acknowledge the contributions made by those volunteers who drafted, edited, and provided the significant and critical commentary essential to updating M36. The M36 Steering Committee members, chapter authors, and peer reviewers dedicated many hours in the preparation for this fifth edition to ensure the overall technical quality, consistency, and accuracy of the manual.

Chairs

George A. Kunkel Jr., Chair, Kunkel Water Efficiency Consulting, Hershey, Pa.
Tory Wagoner, Co-Chair, Cavanaugh, Winston-Salem, N.C.

Steering Committee Members

George A. Kunkel Jr., Chair, Kunkel Water Efficiency Consulting, Hershey, Pa.
Tory Wagoner, Co-Chair, Cavanaugh, Winston-Salem, N.C.
Andrew Chastain-Howley, Black & Veatch, Camden, Maine
David Hale, AWWA, Denver, Colo.
Jim Siriano, Water Research Foundation, Denver, Colo.
Gary B. Trachtman, ARCADIS U.S., Inc., Birmingham, Ala.

Authors, Peer Reviewers, and Contributors to the 5th Edition

George A. Kunkel Jr., Chair, Kunkel Water Efficiency Consulting, Hershey, Pa.
Tory Wagoner, Co-Chair, Cavanaugh, Winston-Salem, N.C.
William P. Bulloch, Utility Revenue Management, Houston, Tex.
John Eisnor, Halifax Water, Halifax, N.S. Canada
Jonathan Gunn, Johnson Controls Inc., Canandaigua, N.Y.
Craig Hannah, Johnson Controls Inc., Lubbock, Tex.
Colleen Heath, CDM Smith, Boston, Mass.
Ashlee Hollifield, Cavanaugh & Associates, P.A., Tampa, Fla.
Aaron Horbovetz, M.E. Simpson Company, Inc., Valparaiso, Ind.
David M. Hughes, Modernizing Distribution, Parker, Colo.
Margaret E. Hunter, American Water, Camden, N.J.
Will J. Jernigan, Cavanaugh, Asheville, N.C.
Thomas Kennedy, Kennedy Water Consulting, LLC, Escondido, Calif.
James Kessen, Illinois Department of Natural Resources, Chicago, Ill.
Thomas Kiger, City of Tarpon Springs, Tarpon Springs, Fla.
Marcin P. Kloc, Echologics, Toronto, Ont., Canada
Allan Lambert (Ret), Water Loss Research and Analysis, Ltd., Huddersfield, Yorkshire, United Kingdom
Keith Larson, Bowen Collins & Associates, Salt Lake City, Utah
Bruce A. Macler (Ret), United States Environmental Protection Agency, Orinda, Calif.
Sofia Marcus, Los Angeles Department of Water and Power, Los Angeles, Calif.
Chris Mastic, MARS Company, Ocala, Fla.
Edward R. Osann, Natural Resources Defense Council, Washington, D.C.
Michele L. Piotrowski, Engineering Enterprises, Inc., Sugar Grove, Ill.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Stephen K. Rupar, WSP USA, Shelton, Conn.
Michael D. Simpson, M.E. Simpson Company, Inc., Valparaiso, Ind.
Brian M. Skeens, Jacobs, Atlanta, Ga.
Travis Smith, Kamstrup, Cumming, Ga.
Reinhard Sturm, E Source, Boulder, Colo.
Russell G. Titus (Ret.), New Jersey American Water, Hillsborough, N.J.
Gary B. Trachtman, ARCADIS U.S., Inc., Birmingham, Ala.
John H. Van Arsdel, M.E. Simpson Company, Inc., Valparaiso, Ind.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 1

Introduction: Auditing Water Supply Operations and Controlling Losses

Community water supply systems around the world have been instrumental in improving the human condition by providing essential water to promote public health and safety and serving as a basis for economic development. For hundreds of years, societies have constructed infrastructure to withdraw water from available sources, to treat it to an acceptable standard, and to distribute it to communities, typically through buried piping distribution systems. Yet for all their success in quenching human needs, many water utilities operate with considerable inefficiencies in terms of water and revenue losses, resulting in nonrevenue water (NRW). NRW is that portion of a utility's water production that does not produce any revenue. As the world grapples with the dilemma of a growing population and a finite amount of water, these inefficiencies need to be brought under a reasonable level of control. This manual offers water utilities a set of tools and approaches to instill accountability and control losses, including

- step-by-step procedures to conduct a water audit to assess the efficiency of the water distribution system and water accounting practices;
- definitions and implications of apparent (nonphysical) losses and real (physical) losses;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- specific techniques to identify, measure, and verify flow from all water sources, consumption, and losses;
- examples of data inputs and sample calculations for each step of the water audit;
- references to freely available software tools to compile the water audit and plan leakage management activities;
- a road map to control apparent losses in metering and billing operations and to recover missed revenues;
- steps to implement a leakage and pressure management program to control real losses and preserve source water resources;
- planning steps to assemble the proper resources, information, and equipment to launch and sustain the NRW management program;
- approaches for short-term and long-term goal setting for the loss control program;
- considerations for small water systems; and
- discussions and listings of validated water audit data and descriptions of successful water loss control programs being applied by North American water utilities.

Water utilities typically experience several types of losses concurrently. Most operators recognize distribution system leakage, categorized under the heading *real losses*, as a primary type of loss. However, water suppliers also suffer revenue losses from data-handling errors, customer metering inaccuracies, and unauthorized consumption. These losses are collectively labeled *apparent losses* and have a negative impact on utility revenue and consumption data accuracy. While it is essential that system operators employ means to control such losses, the initial step is to assemble a water audit to quantify the volumes of losses existing in a water utility and the water resources and financial impacts caused by these losses.

THE WATER AUDIT AND WATER BALANCE

Good management of any resource requires that the supplier maintain accurate records of transactions and deliveries of the commodity provided to its customers. The *audit* is a common function in the world of finance and accounting, and it typically denotes activities that systematically review an organization's financial records and accounts to confirm their accuracy. Similarly, the *water audit* involves a review of records and data generated as water flows from its source and treatment through the water distribution system and into customer properties. The water audit usually exists in the form of a worksheet or spreadsheet that details the variety of supply, consumption, and loss volumes that exist in a community water system. The *water balance* summarizes these components and provides accountability, as all water entering a distribution system is accounted for.

In 2000, the International Water Association (IWA) published the manual *Performance Indicators for Water Supply Services* (Alegre et al. 2000). This publication included a description of a standardized water audit method developed between 1997 and 2000 by the IWA Water Loss Task Force (now the Water Loss Specialist Group), which at the time was a five-country group that included participation by the American Water Works Association (AWWA). Until then, a multitude of different water auditing practices had existed around the world, and the primary focus of the task force was to draw on the best practices of the various approaches and craft them into a single, standard best-management practice methodology. The method needed to be applicable worldwide, across the spectrum of differing system characteristics and units of measure. Many of the features of the resulting best practice methodology were drawn from the first edition of Manual M36, *Water Audits and Leak Detection*, published in 1990 (with a minimally revised second edition published in 1999). Shortly after the IWA Task Force published its report with the improved

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

methodology, the AWWA Water Loss Control Committee (WLCC) voiced support for it in its committee report "Applying Worldwide Best Management Practices in Water Loss Control" published in the August 2003 edition of *Journal AWWA* (AWWA Water Loss Control Committee 2003). In support of this approach, the WLCC comprehensively revised and expanded Manual M36 in creating the third edition (2009). This edition detailed the best practice water audit method and provided significant guidance on innovative loss control technologies and approaches for planning the loss control program. The fourth edition (2016) included additional refinements to the water audit process including the AWWA Free Water Audit Software, data grading, and data validation, and provided summaries of data gathered across North America.

This fifth edition of Manual M36 continues to promote the standardized water audit method, which is explained in detail in Chapter 2. An updated array of key performance indicators (KPIs) is also detailed in this chapter. KPIs are a set of metrics drawn from the water audit that provide insight into the most important aspects of system performance pertaining to water loss. The WLCC conducted research on KPIs and established a new position on their use in their committee report "Key Performance Indicators for Non-revenue water: AWWA's 2020 Position," published in the January 2020 edition of *Journal AWWA* (Kunkel et al. 2020). Effective new KPIs are included in the M36 water audit methodology, while percentage indicators are excluded from it. Because of this change, no percentage indicators are included in the fifth edition of M36 or the Version 6.0 AWWA Free Water Audit Software. Reasoning that percentage indicators are imprecise, misleading, and not actionable in managing NRW, AWWA recommends against their use in NRW assessments.

In addition to reliably quantifying water consumption and losses in the water audit, water utilities also have a variety of means to economically control apparent and real losses. Many new techniques are explained in Chapters 4, 5, and 6. The final chapters of this manual provide guidance on planning and sustaining the loss control program and considerations for small water systems.

This fifth edition has been streamlined without sacrificing content while including many relevant industry updates. For example, the fifth edition incorporates the latest AWWA Free Water Audit Software (v6.0, 2020), and its use is carefully explained. Initially released in 2006 and regularly updated, the software has been recommended by the WLCC to compile water audits annually as a standard business practice. Version 6.0 includes significant improvements over previous versions, notably a refined input data grading process, output dashboard, and updated KPIs. Additionally, this fifth edition of M36 includes information about validation of water audits and updated guidance on the validation process from the Water Research Foundation (Project 5057) (Sturm et al. 2021).

Results from validated water audits of hundreds of water utilities across North America are highlighted throughout the manual, including the Water Audit Reference Dataset (WARD) (Kunkel et al. 2021), the largest repository of validated water audit data in North America with data from 1,124 water audits. Recent reports from AWWA and the Water Research Foundation on performance indicators and means for developing and implementing a successful NRW management plan are referenced and integrated into their relevant chapters.

THE IMPORTANCE OF WATER AUDITS AND NONREVENUE WATER MANAGEMENT

Strong NRW management produces benefits in four primary ways:

1. through water resources management, by limiting unnecessary or wasteful source water withdrawals;
2. financially, by optimizing revenue recovery and promoting equity among ratepayers;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

3. operationally, by minimizing distribution system disruptions, optimizing supply efficiency, and generating reliable performance data; and
4. through system integrity, by reducing the potential for contamination in the water distribution system.

Water suppliers have obligations in all these areas. Properly executed water audits and NRW management help water utilities meet their obligations in these areas to the benefit of their customers and their own bottom line. The following are specific benefits of water auditing and loss control.

- *Reducing apparent losses improves utility finances* by recovering lost revenue from customers who have been undercharged or have taken water in an unauthorized manner.
- *Reducing real losses* saves bulk water purchase costs and operating costs for electric power, maintenance, and treatment. In some cases, water supply infrastructure expansion might be deferred by leakage reduction, and better use of existing resources can reduce drought vulnerability or allow economic development to be supported without exploiting new water resources. Lower leakage means less strain on treatment and pumping equipment, extending the interval between scheduled maintenance.
- *Improved data integrity* ensures the accuracy and integrity of water system input volumes and customer consumption. Knowing true water consumption patterns promotes better water resources management, confirms water conservation benefits, and aids long-term planning.
- *Preservation of available water resources* stretches existing supplies to meet increasing needs, thus avoiding or reducing the exploitation of new water sources. Environmental impacts of current withdrawals are moderated, given that avoidable leakage is reduced.
- *Increased knowledge of the distribution system* allows utility staff to become more familiar with the location of meters, valves and mains; pressure levels and variations; and utility operations. This knowledge allows them to respond quickly to emergencies, such as water main breaks, while providing a basis for optimization of supply operations.
- *Increased knowledge of the customer metering and billing systems* results in improved revenue collection. Water audits require utility staff to review the workings of the customer billing system. For many water utilities, inadvertent procedural or programming gaps exist in billing operations, resulting in errors that underbill customers for their water usage.
- *Improved public relations* are the result of efficient service. Consumers expect an uninterrupted water supply and high-quality service at a reasonable price. Efficient delivery of high-quality water, along with affordable, equitable water rates, creates a strong reputation for the water utility in the minds of its customers.
- *Reduced liability* is an important outcome of strict auditing procedures. By protecting public property and health and providing detailed information about the distribution system, NRW management activities help protect the utility from expensive lawsuits.
- *Reduced disruption to customers* is valuable in the market. Repairing more leaks on a proactive basis helps prevent them from developing into large leaks or main breaks that disrupt service, cause damage, and elicit negative customer sentiment.
- *Improved asset management* minimizes the likelihood of system leakage. By effectively managing leakage and optimizing pressure in the water distribution

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

system, water main and service connection leakage can be reduced, and pipeline asset life can be extended.

- *Benchmarking progress* allows utilities to track ongoing system improvements. With recent federal and state policies bringing unprecedented attention and resources to the repair and replacement of water infrastructure, credible and consistent water auditing can document baseline conditions and chart the progress and results from these new investments.
- *Favorable reviews from the financial community* lend credibility to utilities in the marketplace. Effective operations and accountability help the utility to secure financing at reasonable rates to sustain sound system upkeep well into the future.

Water and revenue losses are wasteful to the water utility, its customers, the environment, and society at large, while strict accountability and NRW management offer many benefits. This manual will serve as a valuable guide for North American water utilities to improve their level of accountability and implement NRW management practices.

GETTING STARTED

Just as a proactive water utility carefully tracks its finances, effective utilities should also track the water supply that they manage, and AWWA recommends that water utilities compile the standard water audit on an annual basis as proper business practice. Many North American water utilities are mandated by regulations to perform an annual water audit, but for other utilities, getting started can be the largest hurdle. Outstanding resources—including this manual, the AWWA Free Water Audit Software, and information on AWWA's website (www.awwa.org)—exist to assist all utilities. Using these resources, utility personnel can readily begin to quantify their losses and impacts. The most important step is to just get started, and this manual gives utilities everything they need to do this.

NORTH AMERICAN GOVERNMENTAL AGENCY NONREVENUE WATER POLICY

In 2001, AWWA commissioned an extensive survey of state and regional water resource and environmental agencies in the United States to uncover the extent and usefulness of their water accountability statutes and regulations. The project, titled *Survey of State Agency Water Loss Reporting Practices (*also known as the "States Survey Project") was successful in garnering valuable information from 46 jurisdictions, including 43 state agencies and three regional agencies (Beecher 2002). The results of the survey found that widely varying language existed throughout many regulations and statutes of these agencies. Many organizations defined water losses as some form of unaccounted-for water and employed a single, simplistic volumetric percentage performance indicator. The final report of the States Survey Project was astute in its recommendation that "a better system of accounting is necessary if accountability is to be instilled in water utilities" (Beecher 2002). With the creation of the AWWA Free Water Audit Software in 2006 and third edition of M36 in 2009, the "better system of accounting" became a reality when governmental agencies began requiring use of this system.

In 2021, AWWA again conducted a survey of North American governmental agencies to learn of their policies, practices, and regulations around NRW with a specific focus of identifying agencies employing the AWWA water audit methodology. The findings of the survey are documented in the report *Governmental Policies for Drinking Water Utility Water Loss Control* (AWWA 2022). Key findings of the survey show that 12 US states and the Province of Quebec have embraced the AWWA water audit methodology (Figure 1-1). Four states (California, Georgia, Hawaii, and Indiana) and the Province of Quebec employ

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

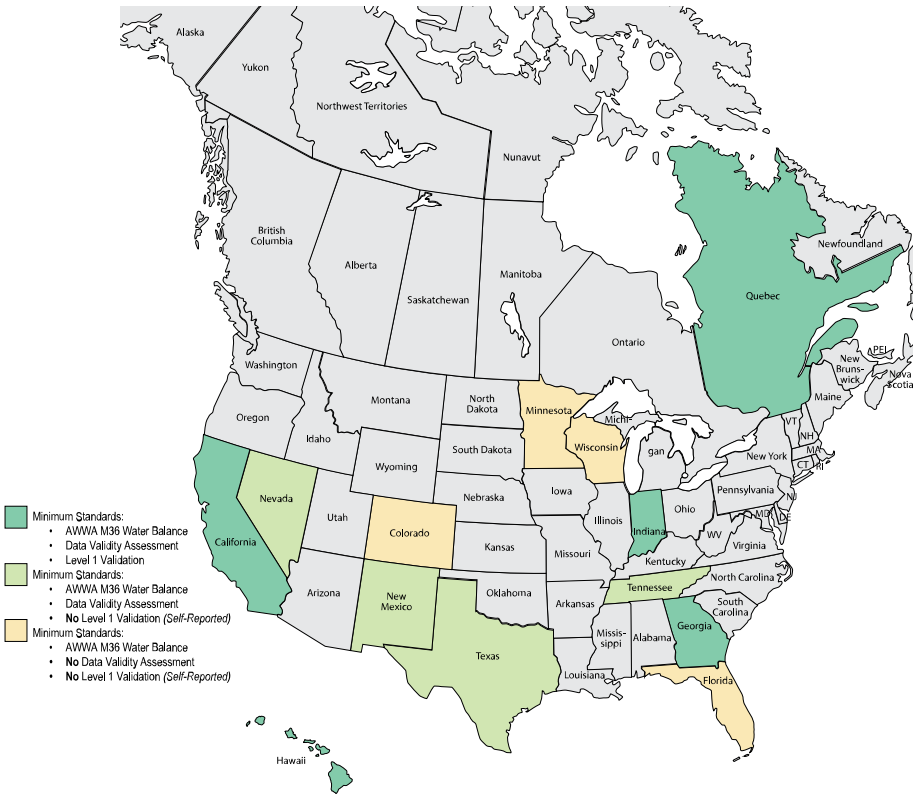


Figure 1-1 North American States and Provinces with AWWA-based Water Loss Control Policies and Programs

comprehensive programs and require AWWA Water Audit reporting using the AWWA Free Water Audit Software and data validation conducted independently of the auditor. Eight other states (Colorado, Florida, Minnesota, Nevada, New Mexico, Tennessee, Texas, and Wisconsin) collect audits using the AWWA water audit structure but either without a process to assess data quality or without data validation. These findings reflect a steady embrace of the AWWA water audit methodology by governmental regulatory agencies.

As more government agencies consider adopting NRW management policies around the AWWA water audit methodology, the rulemaking process can encourage utility stewardship and sustainability of water resources while the details of means and methods for achieving these goals are left to the utility. Government agencies approach the rulemaking process in ways appropriate to their mission, the focus of the new rule, and stakeholder concerns. Essential elements of this rulemaking should include a statement of the background and basis of the rulemaking, a list of stakeholders to be involved in the process, the best practice methods to be implemented, performance metrics to be tracked, targets and timelines for utilities to achieve, terms of utility compliance with the requirements of the rule, consequences of noncompliance, and any financial or technical support mechanisms available to water utilities.

A sound rulemaking process requires time and deliberation among many stakeholders, but a deliberate approach with the elements described previously can assist in implementing a workable program that results in measurable reduction in water losses over time.

Water accountability and loss control are garnering increasing prominence in water resources management, particularly as available water resources are decreasing in many parts of the world. New water sources will continue to be difficult and costly to develop.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Climate change impacts, water quality regulations, and customer expectations will affect the value of water. Growing populations and economies will need adequate water supplies on a continuous basis. These and other drivers will create an increased focus on water accountability, efficiency, and conservation. By employing the methods included in this manual, water utilities will have the tools to meet these growing challenges.

REFERENCES AND RESOURCES

Alegre, H.; Hirner, W.; Baptista, J.; & Parena R. 2000. *Performance Indicators for Water Supply Services*. Manual of Best Practice series. London: IWA Publishing.

AWWA (American Water Works Association).

AWWA (American Water Works Association). 2022. Governmental Policies for Drinking Water Utility Water Loss Control. Denver, CO: AWWA.

AWWA Water Loss Control Committee. 2003. "Water Loss Control Committee Report: Applying Worldwide BMPs in Water Loss Control." Journal AWWA, 95(8):65-79.

Beecher, J. 2002. *Survey of State Agency Water Loss Reporting Practices*. Final Report to the Technical and Educational Council of the American Water Works Association. Indianapolis, Ind.: Beecher Policy Research.

Kunkel, G.; Trachtman, G.; Wyatt, A.; & Jernigan, W. 2020. "Water Loss Control Committee Report: Key Performance Indicators for Non-revenue water—AWWA's 2020 Position." *Journal AWWA*. https://awwa.onlinelibrary.wiley.com/doi/full/10.1002/awwa.1428

Kunkel, G.; Jernigan, W.; & Trachtman, G. 2021. AWWA WLCC Water Audit Reference Dataset (WARD). AWWA. Denver, Colo. https://www.awwa.org/Resources-Tools/Resource-Topics/Water-Loss-Control/Free-Water-Audit-Software

Sturm, R.. 2021. *Level 1 Water Audit Validation Guidance Manual*. Project 5057. Denver, Colo.: Water Research Foundation.

Trachtman, G.; Cooper, J.; Sriboonlue, S.; Wyatt, A.; Davis, S.; & Kunkel, G. 2019. *Guidance on Implementing an Effective Water Loss Control Plan*. Project 4695. Denver, Colo.: Water Research Foundation.

WRF (Water Research Foundation). 2014. *Real Loss Component Analysis: A Tool for Economic Water Loss Control*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 2

Conducting the Water Audit

This chapter details the best practice American Water Works Association (AWWA) water audit method for quantifying customer consumption and volumes of real (leakage) losses and apparent (customer) losses. This method allows the operator to quantify the components of water supplied downstream of the entry point(s) to the distribution system. The first version of this method was published in the International Water Association (IWA) publication *Performance Indicators for Water Supply Services* (Alegre et al. 2000), which AWWA participated in. The AWWA water audit method follows the original IWA approach with several modifications.

AWWA recommends that drinking water utilities employ this method to conduct a water audit on an annual basis as a standard business practice. The auditing process can be conducted with one or both of these approaches:

1. *Top-down approach*: the initial desktop process of gathering information from existing records, procedures, data, and other information systems.
2. *Bottom-up approach*: validating the top-down results with actual field measurements such as field testing of supply meters. Similarly, physical inspections of customer points of delivery can uncover apparent losses from defective or vandalized customer meters or unauthorized consumption (UC). Process flowcharting of customer billing systems can be used to identify systematic billing errors.

The top-down approach is the recommended starting point for water utilities compiling their initial water audit, and it is described in this chapter. Descriptions of bottom-up approaches are given in Chapter 4 for apparent losses and Chapter 5 for real (leakage) losses. See the "Water Auditing: Top-Down vs. Bottom-Up Approach" sidebar for an explanation of the distinctions between the two methods.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Water Auditing: Top-Down vs. Bottom-Up Approach

Water systems that have matured in the water audit process employ both the top-down and bottom-up approaches to water system auditing to achieve long-term success in data validity and water loss management.

The top-down auditing approach represents a relatively quick assembly of available records and data regarding system parameters, water production, and customer consumption volumes and losses. Estimates can be applied where data are not readily available. Bottom-up auditing entails a relatively slow and deliberate extraction of detailed supporting data from both the office and the field, which improves the top-down audit. Where the top-down approach typically marks the beginning of water accountability for most water systems, the bottom-up approach constitutes a continuum of improvement, taking place over successive years of top-down auditing. Bottom-up auditing complements top-down auditing and ultimately strengthens the overall audit reliability. Where the top-down approach may often require the use of some educated assumptions, bottom-up auditing tests and validates those assumptions, increasing the accuracy and confidence of the audit inputs and outputs. For example, many water systems may not have accuracy test data for their customer meters and will use an assumed value for this input in their first top-down audit based on knowledge of the existing meter population's age and condition. The bottom-up activity for this input involves conducting accuracy tests on a statistically significant number of customer water meters to validate the assumed top-down input value.

The bottom-up approach, discussed in greater detail in Chapters 4 and 5, can involve an array of activities, including

- flow verification tests for production, import, and export meters;
- field or bench accuracy testing of customer meters;
- detailed analysis of customer consumption history for profiling of meters, customer classes, zero consumption and inactive accounts, consumption by rate code, registered vs. billed consumption, meter rightsizing analysis, and verification of consumption summary report totals;
- detailed analysis of customer account inventory for verification of revenue-dependent field values such as rate code and meter size;
- field investigation of inactive accounts;
- cross-reference of Geographic Information System (GIS) and Customer Information System (CIS) databases for indicators of UC, such as sewer-only customers that have no well and lie within the water service area;
- temporary fire hydrant metering for verification of estimated flow rates used in determining authorized, unmetered consumption volumes;
- utilization of district metered areas for zonal water balance and night flow analysis to quantify leakage rates;
- pressure logging to enable refinement of average system operating pressure calculations; and
- leak detection surveys, step testing, and real loss component analysis.

It can take several years for most water systems to obtain a mature level of validity in their water audit process. It is necessary that water systems be critical and candid when assessing the data grading questions for each audit input, as this will largely inform which next-step bottom-up activities will best serve the water system.

If the water auditing process is instituted as a standard, annual business practice—as it should be—a two-part goal should exist to both compile the top-down audit and incrementally utilize bottom-up audit methods to improve the completeness and quality of the data, which will lead to better decisions in managing nonrevenue water.

The water audit addresses the questions "How much water is being lost?" and "How much are these losses costing the water utility?" With relatively modest effort, the top-down method can provide a good preliminary assessment of water loss standing and insight to the quality of available water supply data. The top-down audit also helps to

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

identify components that require further validation. Ultimately, the auditor can better validate and improve the accuracy of the water audit when it is augmented by leakage component analysis, bottom-up field measurements, or both.

THE WATER AUDIT

The water auditing process is an effective tool available for utilities to quantify customer consumption and losses that occur in the distribution system and the billing data management processes of the water utility. The auditing process is a revealing undertaking that provides great insight to the auditor on the types and amounts of loss occurring in the utility. Also, launching a water audit often begins a culture change within the utility necessary to shift employee focus onto water-efficient practices. The top-down water audit is assembled in two steps: (1) quantifying, via measurement or estimation, individual water customer consumption, and water loss components and (2) undertaking the water balance calculation.

This chapter explains the AWWA water audit approach, which includes example data from the fictitious water utility, County Water Company (CWC). Step-by-step instructions are given to compile the water audit using the AWWA Free Water Audit Software (Audit Software) version 6 (Jernigan et al. 2020), including the required information, accessing that information, entering it into the software worksheets, and calculating performance indicators. The audit software was created and is maintained by the AWWA Water Loss Control Committee (WLCC) and is available for free download from AWWA's Water Loss Control Resource Community website (AWWA 2021), which also includes considerable additional information on water loss control. Select worksheets of the audit software are also shown as figures in this chapter. The reader is urged to download the audit software from the AWWA website and to view it in tandem with reading the discussions in this publication. This will ensure that the reader can adequately view the audit software and test its workings while reading the content in this chapter.

THE WATER BALANCE CALCULATION

A preliminary assessment of water loss can be obtained by gathering available records and placing data into the water audit worksheet. The summary data from the water audit is shown in the water balance, which compares the distribution system input volume with the sum of customer consumption and losses (estimated or known). The sum of all components in each column of the water balance is equal and therefore in balance, as shown in Figure 2-1. The water balance for CWC in the audit software is given in Figure 2-2. Most water utilities have readily available data on production, water imported from or exported to other utilities, and customer consumption. Utilities often have less data available to quantify leakage, meter error, and unauthorized consumption (UC). The water balance provides a guide for estimating how much water is lost because of customer metering inaccuracies (CMI), systematic data-handling errors (SDHE), and UC (collectively, apparent losses), as well as leakage (real losses) or in some cases tank overflow.

The two most powerful features of the best practice water audit method are its rational terms and definitions (Table 2-1) and standard set of performance indicators (listed in Table 2-25 later in this chapter). On the broadest level, water system input volume goes to two places: authorized consumption or losses. The water audit method advances the concept that all water should be quantified via measurement or estimate as either authorized consumption or losses. Hence, no water is unaccounted for.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

<table>
<tr><td rowspan="9">Volume From Own Sources (corrected for known errors)</td><td rowspan="10">System Input Volume</td><td rowspan="10">Water Exported (corrected for known errors)</td><td colspan="4">Billed Water Exported</td><td>Revenue Water</td></tr>
<tr><td rowspan="10">Water Supplied</td><td rowspan="3">Authorized Consumption</td><td rowspan="2">Billed Authorized Consumption</td><td>Billed Metered Consumption</td><td rowspan="2">Revenue Water</td></tr>
<tr><td>Billed Unmetered Consumption</td></tr>
<tr><td rowspan="2">Unbilled Authorized Consumption</td><td>Unbilled Metered Consumption</td><td rowspan="8">Non-Revenue Water</td></tr>
<tr><td>Unbilled Unmetered Consumption</td></tr>
<tr><td rowspan="6">Water Losses</td><td rowspan="3">Apparent Losses</td><td>Systematic Data-Handling Errors</td></tr>
<tr><td>Customer Metering Inaccuracies</td></tr>
<tr><td>Unauthorized Consumption</td></tr>
<tr><td rowspan="3">Real Losses</td><td>Leakage on Transmission and Distribution Mains</td></tr>
<tr><td>Leakage and Overflows at Utility's Storage Tanks</td></tr>
<tr><td>Leakage on Service Connections up to the Point of Customer Metering</td></tr>
<tr><td>Water Imported (corrected for known errors)</td></tr>
</table>

NOTE: All components of the water balance are expressed as volumes of water (not to scale) for the period of reference, typically one year.

Figure 2-1 Water balance

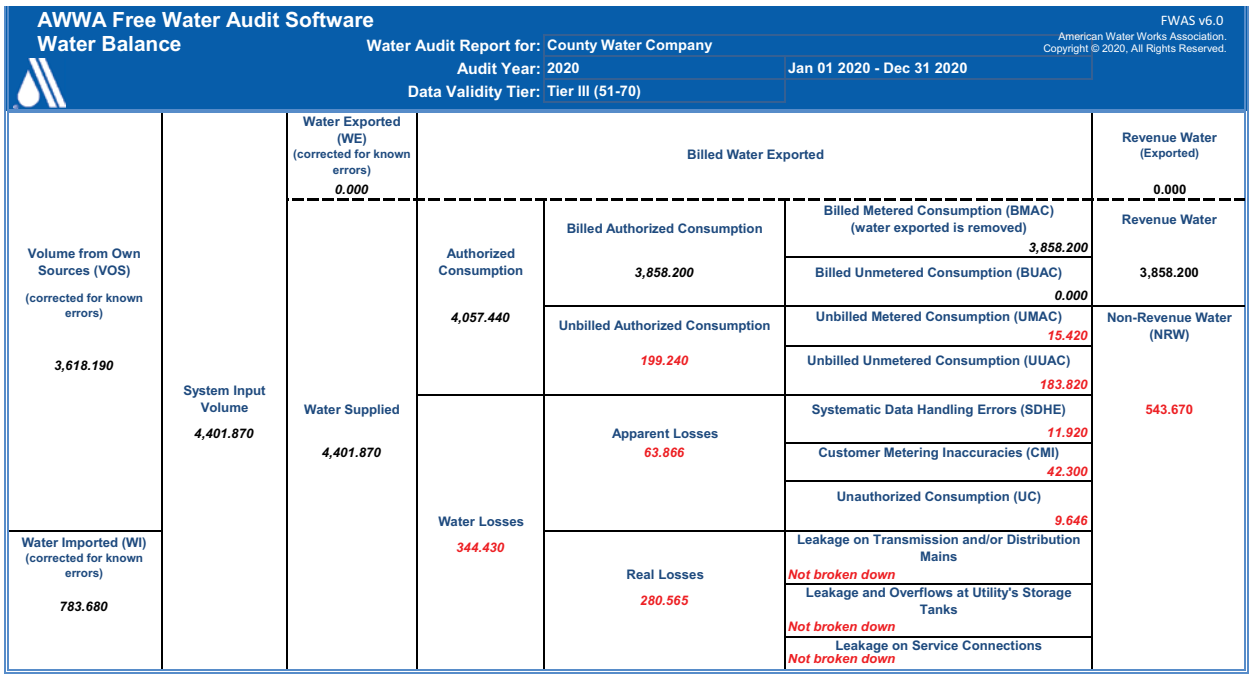


Figure 2-2 Water balance for County Water Company—2020 calendar year using the AWWA Free Water Audit Software (Version 6.0)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Since 2003, AWWA has recommended that water utilities, state agencies, and drinking water stakeholders avoid use of the poorly defined and imprecise term *unaccounted-for water* (Kunkel et al. 2003). More recently, research and development led by the AWWA WLCC (Kunkel et al. 2020a) formally discontinued support for any percentage-based performance indicators (such as volumetric "unaccounted-for" water percentages), instead employing a suite of key performance indicators (KPIs) to communicate water audit results in terms of the validity of the data, volumes of water consumption and losses, and value or cost impacts from losses. Stakeholders should employ the term nonrevenue water (NRW) and apply it as specifically defined in Table 2-1 and performance indicators as calculated by the Audit Software and discussed later in this chapter (see Task 7—Performance Indicators).

The performance indicators give a useful assessment of water loss standing from water resources management, financial, and operational perspectives. They are effective in evaluating current standing, for loss reduction target-setting, and for benchmarking with other utilities.

Table 2-1 Water balance terms and definitions

Water Balance Term	Definition
Volume from Own Sources	The volume of water withdrawn from water resources (rivers, lakes, streams, wells, etc.) controlled by the water utility for distribution. Volume shown in water balance includes any Error Adjustment input on the Audit Software Worksheet.
Water Imported	The volume of bulk water purchased to become part of the Water Supplied volume. Typically, this is water purchased from a neighboring water utility, wholesale water supplier, or regional water authority. Volume shown in water balance includes any Error Adjustment input on the Audit Software Worksheet.
System Input Volume	The annual volume input to the water supply system. This equals the Volume from Own Sources plus the Water Imported volume.
Water Supplied	The annual volume of water delivered to the distribution system. This equals System Input Volume minus the Water Exported volume.
Water Exported	The volume of bulk water conveyed by the water utility to neighboring water systems that exists outside of their service area or neighboring zones outside of the water audit boundary. Volume shown in water balance includes any Error Adjustment input on the Audit Software Worksheet.
Authorized Consumption	The volume of metered and/or unmetered water taken by registered customers, the water supplier, and others who are authorized to do so.
Water Losses	The difference between the Water Supplied Volume and Authorized Consumption, consisting of Apparent Losses plus Real Losses.
Apparent Losses	Nonphysical (revenue) losses from systematic data-handling errors (in the customer billing process), all types of customer metering inaccuracies, and unauthorized consumption.
Real Losses	The physical volumes lost through all types of failures (leaks, breaks, bursts) on water mains and service connections up to the point of custody transfer to customer metering, along with overflows from distribution system water storage tanks and reservoirs.
Revenue Water	All components of the System Input Volume that are billed and produce revenue.
Nonrevenue Water	The sum of Unbilled Authorized Consumption, Apparent Losses, and Real Losses. This can also be derived by calculating the difference between Water Supplied Volume and Billed Authorized Consumption.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

COMPILING THE TOP-DOWN WATER AUDIT DATA

This section provides step-by-step instructions for compiling the top-down water audit. Major tasks are listed, as well as individual steps under these tasks.

Before Starting the Water Audit

At the outset of the water audit, it is important to establish the scope of the audit by determining its key parameters.

Identify the system boundaries. The auditor must clearly define the system boundaries for the audit, noting where water enters the supply and where it leaves. Three distinct water supply configurations are shown in Figures 2-3a through 2-3c. Figure 2-3a illustrates a configuration for an untreated water transmission (wholesale) system. Figure 2-3b shows a treated water distribution (retail) system. Figure 2-3c displays a discreet subsector of a retail distribution system, such as a pressure district or district metered area (DMA). A water audit using the audit software can be conducted on any of the three configurations, as well as other system types such as a reclaimed water distribution system or a raw water distribution system. The audit software will accommodate reclaimed (also called recycled) water distribution and raw water distribution systems. Performing annual water audits is generally recommended to understand and cost-effectively manage losses for these types of systems. Water audits are most commonly performed on individual treated (potable) water distribution systems (Figure 2-3b), and the example given in this chapter follows this configuration. It is important that the system type and boundaries be clearly identified to match the justification put forward for compiling the water audit. Appendix B discusses water resources considerations that might justify expanding or isolating the audit to include raw water transmission systems, water use/loss through water treatment plants, or more detailed evaluations of customer consumption. When identifying the system

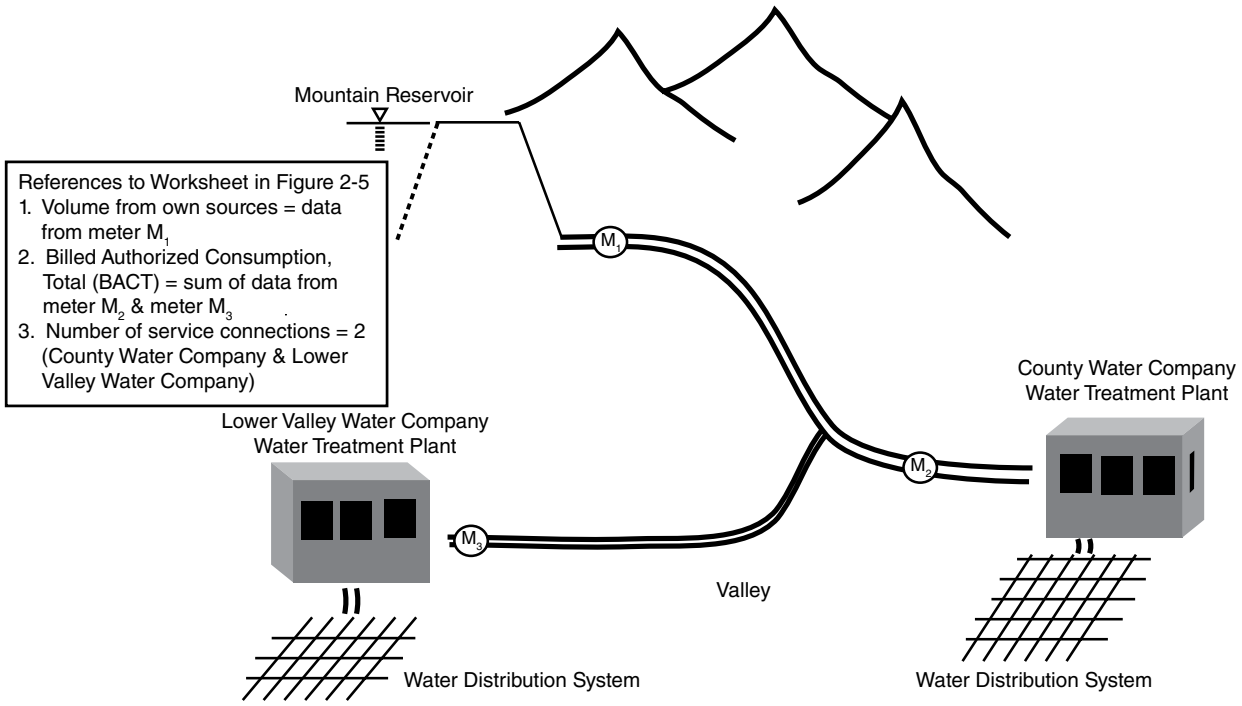


Figure 2-3a Identifying system boundaries for a water audit conducted on a wholesale transmission water supply system

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

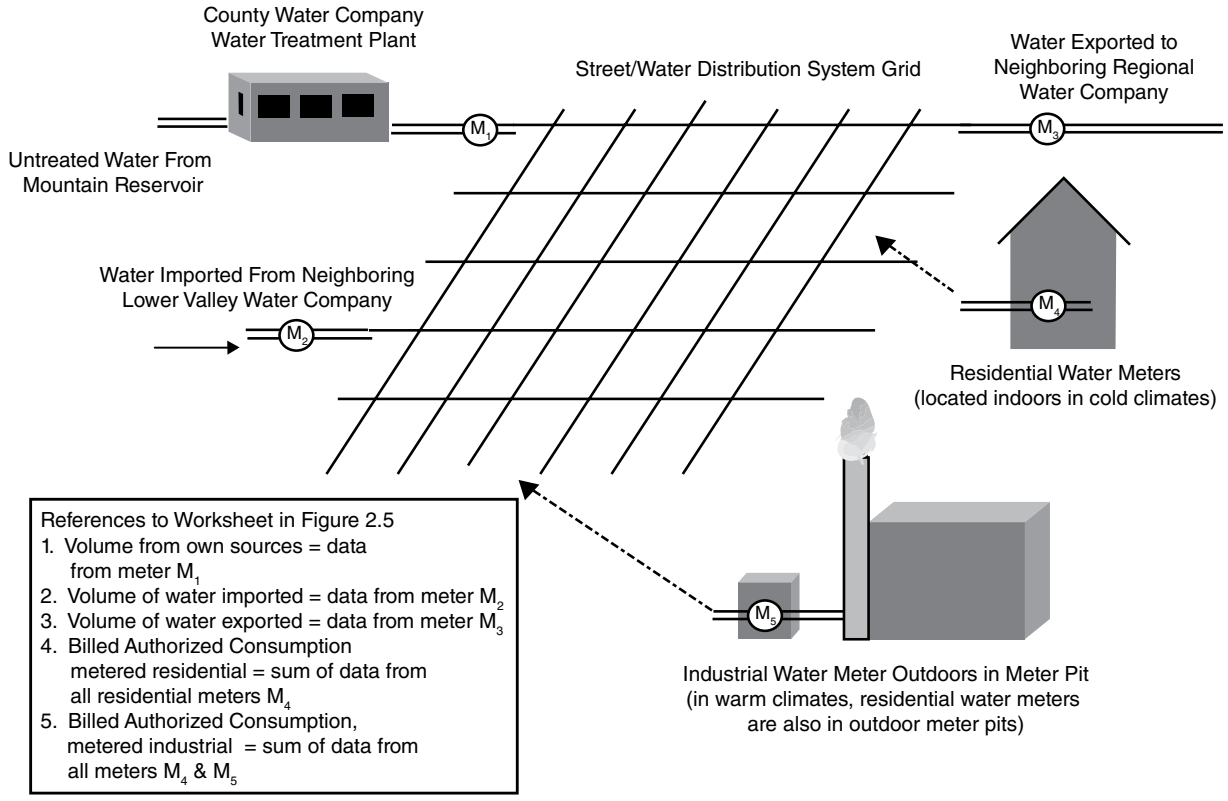


Figure 2-3b Identifying system boundaries for a treated water distribution system

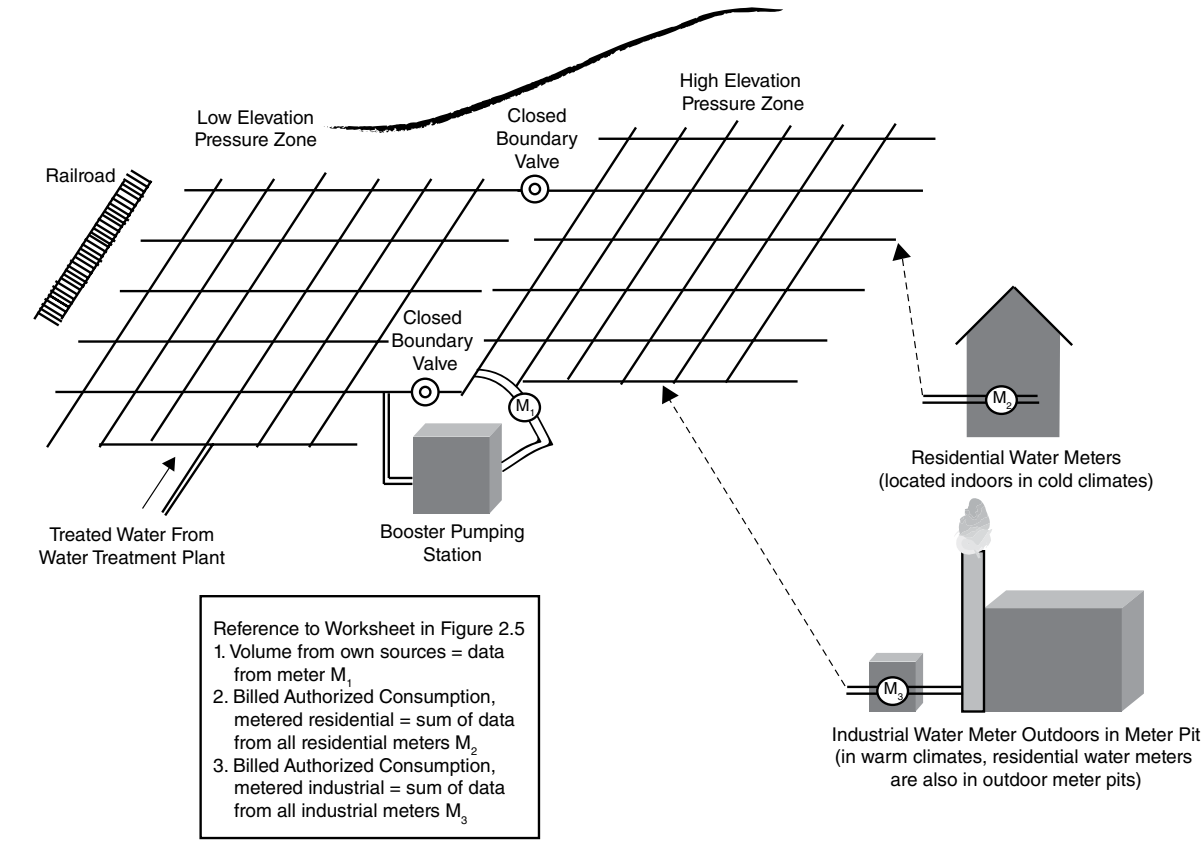


Figure 2-3c Identifying system boundaries for a discrete pressure zone or DMA

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-2 Metering locations in drinking water supply systems

Location	Function
Water source (untreated water)	Measure withdrawal of water from rivers, lakes, wells, or other raw water sources
Treatment plant or works	Process metering at water treatment plants; metering may exist at the influent, effluent, and/or locations intermediate in the process
Distribution system input volume	Water supplied at the entry point of water distribution systems; commonly at treatment plant, treated water reservoir, or well effluent locations but may also start at raw water sources depending on the type of system being audited
Distribution system pressure zones	Zonal metering into portions of the distribution system being supplied different pressure, also includes metering at major distribution facilities such as booster pumping stations, tanks, and reservoirs
District metered area (DMA)	Discrete areas of several hundred to several thousand properties used to analyze the daily diurnal flow variation and infer leakage rates from minimum-hour flow rates
Customers	Consumption meters at the point of custody transfer
Bulk supply	Import/Export meters to measure bulk purchases or sales
Miscellaneous	Capture use of water from fire hydrants, tank trucks, or other intermittent use

boundaries, it is important that accurate measurement of the water input is obtainable from existing installed meters, portable or temporary meters, or new meters proposed for installation at the input location.

The boundary limits should be defined by points of metering of the water supply. Typical metering locations for drinking water supply and distribution are given in Table 2-2. A water audit of the raw (untreated) water system (Figure 2-3a) utilizes metering data of the source water withdrawals as the System Input Volume and the water metered at the treatment plant influent or effluent (where the water improves in quality and value) as the "customer consumption" endpoint. If raw water end users exist in addition to the treatment plant, the raw water audit boundary would extend to the point of custody transfer for those end users.

For water audits conducted on treated water distribution systems (shown in Figure 2-3b and the typical example in this manual), the metered flow of finished water leaving the water treatment plant is taken as the starting point for system input and customer metered consumption is the endpoint.

Set a time period. A water audit is a detailed study over time. A duration of one month or even six months is too short to give an overall picture of water flow through the system. A 12-month audit period is recommended as it is long enough to include seasonal variations and reduces the effects of lag time in customer meter reading. Advanced metering infrastructure (AMI) technology can collect all customer meter readings on a daily (or more frequent) basis; thus, groups of customer meter data do not lag each other in data collection and the billed authorized consumption volume can be synchronized with the water Supplied volume in a reliable manner without causing any billing lag.

Most utility records are kept by the calendar or business (fiscal) year; either schedule makes 12 months of data available. The calendar year is illustrated in this chapter. The water audit draws upon water supply and consumption data, as well as cost data, production and customer meter testing data, investigations of UC, and other data. Most of the data are routinely assembled by water utilities on an annual basis as part of their business year reporting. Thus, the water audit is a yearly reporting endeavor. Water audits are not generally compiled on a more frequent basis since most of the data needed for the water audit is quite detailed and usually only assembled and reconciled on a yearly basis. Water utilities, however, can compile a cursory measure of the water supplied volume and

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

compare it to the Billed Authorized Consumption volume on a more frequent basis. See Table 7-2 in Chapter 7 for an example of such reporting. However, it is important that such comparisons be conducted using rolling 12-month data for the previous 12-month period. Comparing a single-month water supplied volume to a single-month Billed Authorized Consumption volume is unreliable and misleading because it fails to consider seasonal variations, customer meter reading lag issues, and other short-term effects. Water loss assessments should always be conducted using data derived from a 12-month period, whether formally compiled data of the annual water audit or the cursory monthly overview of Water Supplied volume compared to Billed Authorized Consumption volume.

Select the units of measure. The units of measure must also be chosen and standardized so that supply and customer consumption units are the same. In many water utilities, treatment and distribution operations use one unit of measure (e.g., gallons), while metering and billing systems often use a different unit (e.g., cubic feet). Moreover, different multiples are often found in different types of utility records, such as hundreds of cubic feet in billing records, millions of gallons per day in water production reports, or thousands of acre-feet per year in annual financial statements. Although a variety of units are used by North American water utilities (million gallons, acre-feet, cubic feet, megaliters), gallons will be used in the examples in this manual. Because the audit period is one year, the unit of measure (million gallons) is presented as a volume for the year. Unit conversion is a common source of error, requiring continual vigilance by the auditor. Units for the water audit are selected on the Start Page worksheet as shown in Figure 2-4.

Assemble records and data. One of the auditor's greatest challenges is to assemble records and data from a wide variety of operations in the water utility. Information is required on production metering; leak detection and repair; detailed customer metering and billing data; authorized consumption from flushing, firefighting, and related activities; distribution system pressures; water conservation activities; financial information (water rates and production costs); infrastructure rehabilitation; and a host of related data. Distribution system maps or geographic information systems (GIS), customer billing systems, maintenance management information systems, and supervisory control and data acquisition (SCADA) systems are some of the information management systems that can be accessed to assemble much of the needed data. In most water utilities, a considerable degree of interdepartmental cooperation is needed to access and assemble the data. Thus, it is essential that the auditing process be supported at the outset by the utility's senior management, who should communicate the importance of the water audit and urge active participation from all groups in the water utility.

Establishing procedures and contacts for the routine annual collection of the data is an important function. The auditor should be cognizant during the auditing process of the quality of information sources. Who provides the data? What is the format of the data? What is the degree of confidence in or validity of the water audit data? If new information sources are uncovered during the auditing process, the new information streams should be documented so that the desired data are available for the water audit in subsequent years. Because similar data are gathered on a yearly basis, routine data-collection processes greatly ease the amount of work needed to assemble this information each year after the initial water audit is conducted. Thus, the completeness of the water audit and the validity of its input data should improve over time.

Through the auditing process, the individual or team responsible for conducting the audit will be required to answer a series of practice-related questions in the Audit Software which determine the validity of each audit input, a process referred to as interactive data grading (IDG). Data grades are strongly dependent on the integrity of both the source of the data and the process by which the audit input value is derived from that data. To assess the validity of each audit input, the auditor must be diligent in the discovery process,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

including documentation of not only sources of data, but pertinent utility staff interviews regarding practices and procedures as well. The data grading for each input is automatically calculated by the Audit Software once all visible questions for that input have been answered (in the Interactive Data Grading tab). Throughout this chapter, guidance is offered about the factors to consider when assessing data quality and answering IDG questions for each of the data inputs of the water audit. The reader is also encouraged to reference the *Level 1 Water Audit Validation Guidance Manual*, 2nd Edition for additional guidance on quality control using the Audit Software (Sturm et al. 2021).

Using the Audit Software to Compile the Standard Water Audit

The Audit Software allows the user to easily compile the standard AWWA water audit detailed in this manual. It is standard spreadsheet software that includes eleven tabs. The four primary worksheets of the Audit Software are shown in Figures 2-4 to 2-7, including the Start Page, the Worksheet, the Interactive Data GradingIDG worksheet, and the Dashboard, respectively. Data are input on only the first three of these tabs. Figures 2-4 to 2-7 include input data and calculated performance indicators for CWC, the primary water utility example referenced throughout this manual.

Water utilities may desire to create their own spreadsheets or other software to assist the compilation of the standard water audit, particularly if they would like to include greater detail, such as breaking down the loss components into subcomponents. This could also have advantages since certain subcomponents might be valued at different cost rates (e.g., residential water rates vs. commercial/industrial water rates for CMI). However, the

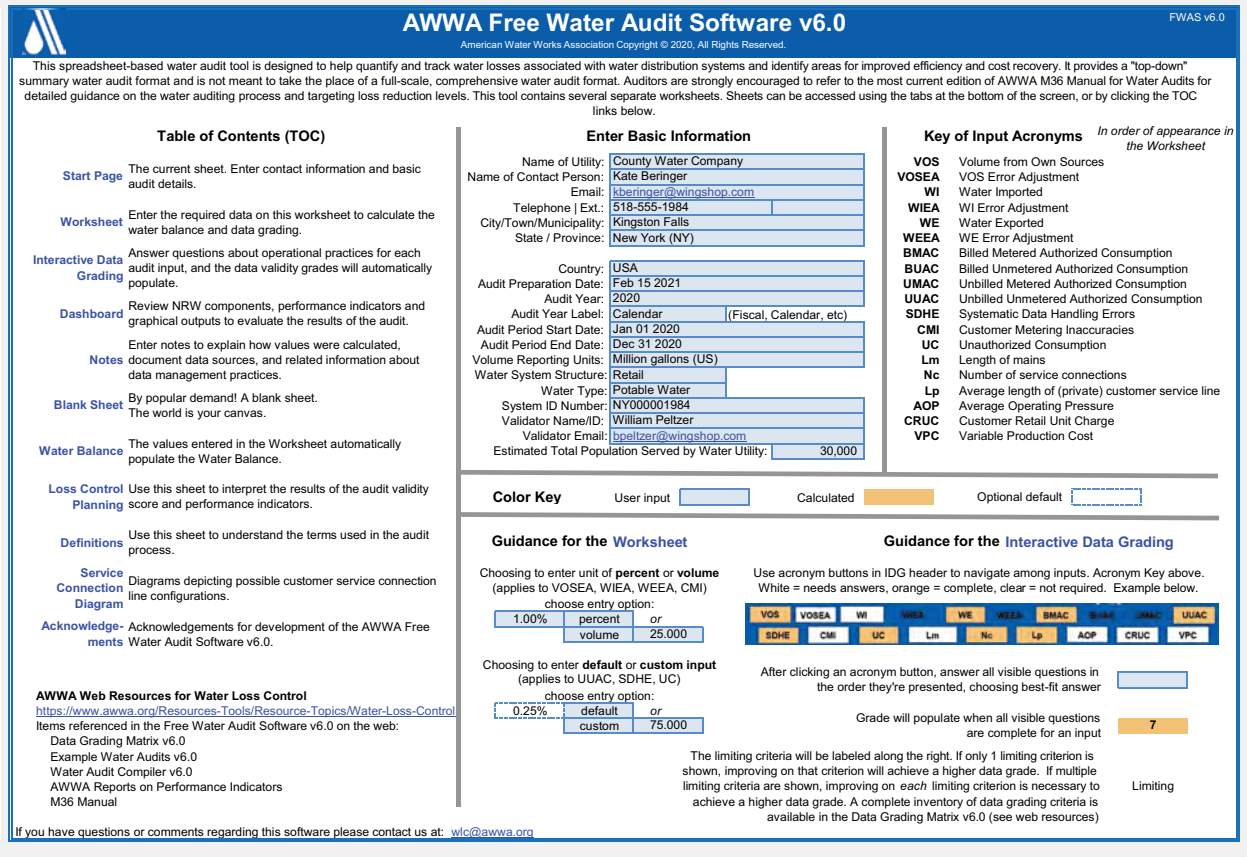
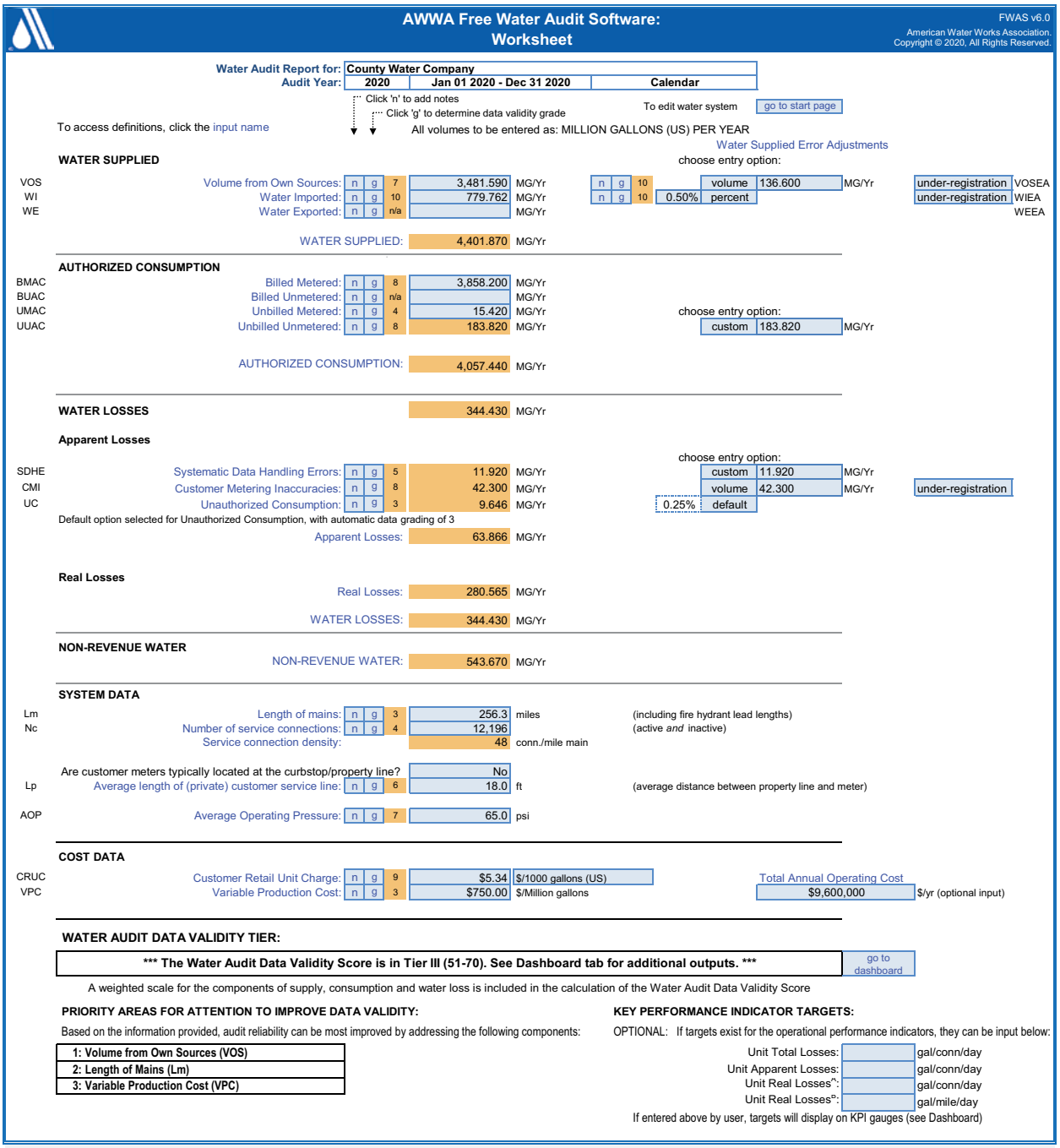


Figure 2-4 AWWA Free Water Audit Software—Start Page completed for County Water Company

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



AWWA Free Water Audit Software: Worksheet

FWAS v6.0

American Water Works Association. Copyright © 2020, All Rights Reserved.

Water Audit Report for: County Water Company
Audit Year: 2020 | Jan 01 2020 - Dec 31 2020 | Calendar

Click 'n' to add notes
Click 'g' to determine data validity grade
To edit water system: go to start page
To access definitions, click the input name
All volumes to be entered as: MILLION GALLONS (US) PER YEAR

								Water Supplied Error Adjustments					
	WATER SUPPLIED							choose entry option:					
VOS	Volume from Own Sources:	n	g	7	3,481.590	MG/Yr	n g 10		volume	136.600	MG/Yr	under-registration	VOSEA
WI	Water Imported:	n	g	10	779.762	MG/Yr	n g 10	0.50%	percent			under-registration	WIEA
WE	Water Exported:	n	g	n/a		MG/Yr							WEEA
	WATER SUPPLIED:				4,401.870	MG/Yr							
	AUTHORIZED CONSUMPTION												
BMAC	Billed Metered:	n	g	8	3,858.200	MG/Yr							
BUAC	Billed Unmetered:	n	g	n/a		MG/Yr							
UMAC	Unbilled Metered:	n	g	4	15.420	MG/Yr			choose entry option:				
UUAC	Unbilled Unmetered:	n	g	8	183.820	MG/Yr			custom	183.820	MG/Yr		
	AUTHORIZED CONSUMPTION:				4,057.440	MG/Yr							
	WATER LOSSES				344.430	MG/Yr							
	Apparent Losses								choose entry option:				
SDHE	Systematic Data Handling Errors:	n	g	5	11.920	MG/Yr			custom	11.920	MG/Yr		
CMI	Customer Metering Inaccuracies:	n	g	8	42.300	MG/Yr			volume	42.300	MG/Yr	under-registration	
UC	Unauthorized Consumption:	n	g	3	9.646	MG/Yr		0.25%	default				
	Default option selected for Unauthorized Consumption, with automatic data grading of 3												
	Apparent Losses:				63.866	MG/Yr							
	Real Losses												
	Real Losses:				280.565	MG/Yr							
	WATER LOSSES:				344.430	MG/Yr							
	NON-REVENUE WATER												
	NON-REVENUE WATER:				543.670	MG/Yr							
	SYSTEM DATA												
Lm	Length of mains:	n	g	3	256.3	miles	(including fire hydrant lead lengths)						
Nc	Number of service connections:	n	g	4	12,196		(active *and* inactive)						
	Service connection density:				48	conn./mile main							
	Are customer meters typically located at the curbstop/property line?				No								
Lp	Average length of (private) customer service line:	n	g	6	18.0	ft	(average distance between property line and meter)						
AOP	Average Operating Pressure:	n	g	7	65.0	psi							
	COST DATA												
CRUC	Customer Retail Unit Charge:	n	g	9	$5.34	$/1000 gallons (US)	Total Annual Operating Cost						
VPC	Variable Production Cost:	n	g	3	$750.00	$/Million gallons	$9,600,000	$/yr (optional input)					

WATER AUDIT DATA VALIDITY TIER:

*** The Water Audit Data Validity Score is in Tier III (51-70). See Dashboard tab for additional outputs. *** go to dashboard

A weighted scale for the components of supply, consumption and water loss is included in the calculation of the Water Audit Data Validity Score

PRIORITY AREAS FOR ATTENTION TO IMPROVE DATA VALIDITY:

Based on the information provided, audit reliability can be most improved by addressing the following components:

1: Volume from Own Sources (VOS)
2: Length of Mains (Lm)
3: Variable Production Cost (VPC)

KEY PERFORMANCE INDICATOR TARGETS:

OPTIONAL: If targets exist for the operational performance indicators, they can be input below:

Unit Total Losses: gal/conn/day
Unit Apparent Losses: gal/conn/day
Unit Real Losses^: gal/conn/day
Unit Real Losses^: gal/mile/day

If entered above by user, targets will display on KPI gauges (see Dashboard)

Figure 2-5 AWWA Free Water Audit Software—Worksheet completed for County Water Company

Audit Software is easy to use and widely recognized as the industry standard water audit format, so it is the recommended medium for North American water utilities.

The means to compile the water audit are explained in this chapter and referenced in other chapters using CWC data. The Audit Software also references https://www.awwa.org/Resources-Tools/Resource-Topics/Water-Loss-Control, where an example water audit for the City of Asheville, NC, United States, can be downloaded. The Audit Software makes use of many acronyms, and these are shown in Table 2-3 as well as the Start Page worksheet of the Audit Software.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

go to input | **Variable Production Cost (VPC) - Data Grading Criteria** | go to notes

vpc	Criteria Question	Select Best-Fit Answers to All Visible Questions	
vpc.1	Choose the option that best describes how the input was derived	Multiple sources of water exist, and a volume-weighted average was calculated for all sources	
vpc.2	Choose the option that best describes which short-run marginal costs have been included in the input, using the definitions below for reference. Short-run marginal costs can include the following: - chemicals + power for treatment, typically applicable if the utility is producing/treating water - power for distribution, typically applicable if pumps exist in the distribution network - water acquisition costs, typically applicable if the utility is purchasing water or incurs any extraction costs for withdrawing from a source Some short-run marginal costs may not be applicable. The auditor should analyze the system characteristics to determine which costs are applicable for inclusion in the VPC input derivation. See also the latest AWWA M36 Manual for further guidance.	Some but not all applicable short-run marginal costs are included	Limiting
vpc.3	Choose the option that best describes which long-run marginal costs have been included in the input, using the definitions below for reference. Long-run marginal costs can include the following: - water treatment residuals management, typically applicable if solids are produced from water treatment process - accelerated wear & tear on dynamic equipment, typically applicable if pumps exist for treatment and/or distribution, or any other equipment exists that wears out as a function of use instead of time (i.e. filter media, chemical dosing pumps, uv disinfection bulbs, etc) - payouts for damage claims from main and service line breaks, typically applicable if damage claims are paid by the utility - accelerated expansion of supply capacity, typically applicable if the utility is at or nearing supply capacity, or scarecity costs in water scarce areas - full cost pricing that includes all lifecycle costs and externalities (internalized or not) Some long-run marginal costs may not be applicable. The auditor should analyze the system characteristics to determine which costs are applicable for inclusion in the VPC input derivation. See also the latest AWWA M36 Manual for further guidance.	Long-run marginal costs have not been evaluated for applicability, and are not included	
vpc.4	Has the input derivation been reviewed by someone with expert knowledge in the M36 methodology?	No	
	FINAL DATA GRADE FOR THIS AUDIT INPUT:	**3**	

Figure 2-6 AWWA Free Water Audit Software—Interactive Data Grading for Volume from Own Sources (VOS) completed for County Water Company

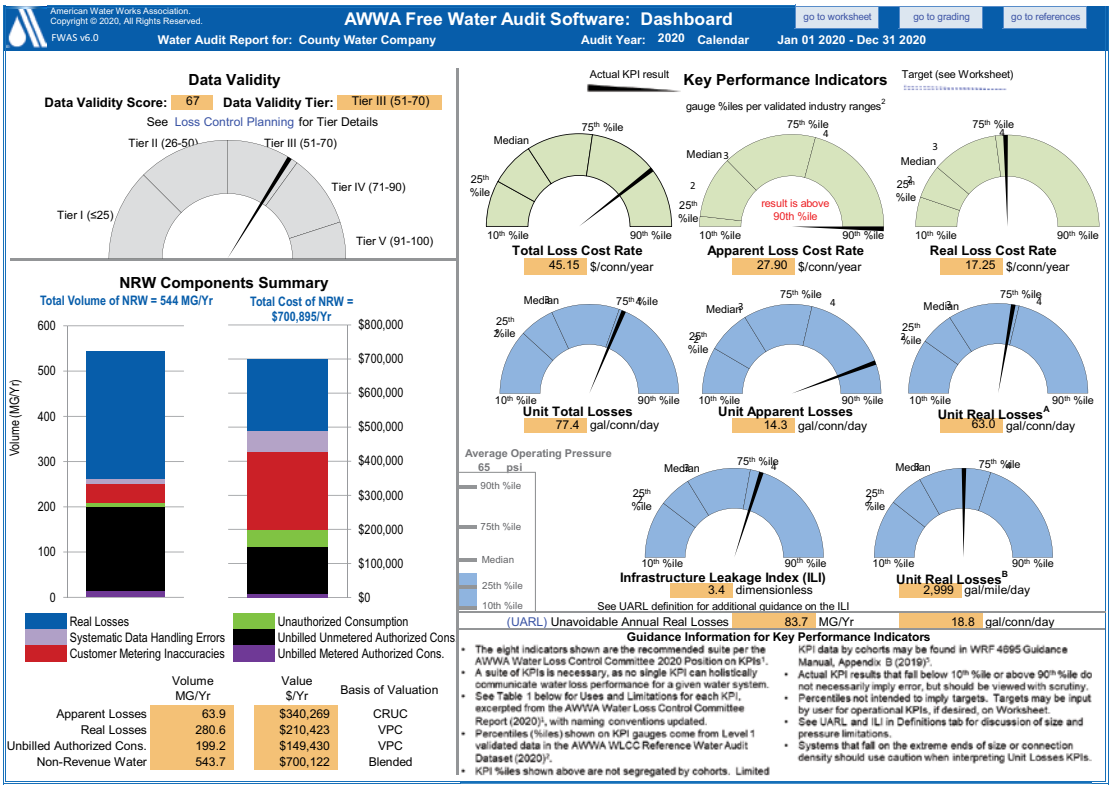


Figure 2-7 AWWA Free Water Audit Software—Dashboard for County Water Company

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-3 Acronym key for terms in the AWWA Free Water Audit Software

AOP	Average operating pressure	**Nc**	Number of service connections
AMI	Advanced metering infrastructure	**NRW**	Nonrevenue water
AMR	Automatic meter reading	**SCADA**	Supervisory control and data acquisition
AWWA	American Water Works Association	**SDHE**	Systematic data-handling errors
BMAC	Billed metered authorized consumption	**UC**	Unauthorized consumption
BUAC	Billed unmetered authorized consumption	**UMAC**	Unbilled metered authorized consumption
CMI	Customer metering inaccuracies	**UUAC**	Unbilled unmetered authorized consumption
CRUC	Customer retail unit charge	**VOS**	Volume from own sources
DVS	Data validity score	**VOSEA**	Volume from own sources error adjustment
DVT	Data validity tier	**VPC**	Variable production cost
GIS	Geographic information system	**WE**	Water exported
IDG	Interactive data grading	**WEEA**	Water exported error adjustment
Lm	Length of mains	**WI**	Water imported
Lp	Average length of (private) customer service line	**WIEA**	Water imported error adjustment

Data inputs to the AWWA Free Water Audit Software. As shown in Figure 2-5, the auditor is prompted to enter the primary water audit data into the Worksheet and Interactive Data Grading (IDG) of the Audit Software. These inputs fall into three categories: volume inputs, system/cost data, and IDG questions. Volume inputs include the quantities of water supplied, the various components of authorized consumption, the components of apparent losses, and error adjustments for various components. These are the inputs used to create the water balance and they are the primary inputs to the water audit. The system/cost data inputs describe the water supply operation and are used to calculate the performance indicators shown in Figure 2-7. IDG questions are practice-based questions that assess the validity—or trustworthiness—of the individual volumetric and system/cost data inputs. By answering IDG questions for each of the volumetric and system/cost data inputs, a composite Data Validity Score (DVS) from 1 to 100 is calculated and presented by the Audit Software to fall within one of the five Data Validity Tiers (DVTs) from 1-100. The DVT provides a measure of the overall validity of the water audit and provides context for the next steps that the water utility can consider. The DVT is given at the bottom of the Worksheet and is based on the cumulative data grading of the audit components. Data gradings are automatically calculated when all visible IDG questions are answered for a given input. Each IDG question has a menu of answer options, from which the auditor chooses the best-fit answer for the utility's operating conditions during the audit period.

Data validity. The top-down audit is highly useful, particularly for water utilities doing a first-time water audit, because it can be assembled using readily available data or rough estimates. The auditor can promptly complete the water audit and assess the values of the performance indicators for the water utility. The disadvantage of the top-down approach is that, for many first-time auditors, the quality and completeness of available data are often questionable. Although the audit can be completed and the performance indicators calculated quickly, how confident can the water utility manager be in those

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

results if it is believed that much of the data entered into the water audit are of marginal quality? This is the essence of the question of data validity.

This guidance manual and the Audit Software provide water utilities with a robust and reliable structure for water auditing. However, the quality of the outputs (performance indicators) of the water audit is only as reliable as the quality of the data inputs to the water audit.

No water utility has perfect data, and all data are subject to some degree of error. Data of low validity exist if the quantification method applied is cursory (such as use of rough estimates). However, even data that are quantified in a robust manner can be low validity if the quantification is not kept up to date. For example, the length of water mains in the system is an important data input. This value can be of low validity if poorly organized paper records are used to tally the total. Conversely, deriving this length from a well-maintained GIS will produce a value of high validity. However, if the auditor applies the same value year after year instead of accessing updated GIS data, the audit data will have less validity, particularly if the water system is adding considerable new piping every year and the length of mains is constantly increasing.

Water utilities can improve data validity over time by using bottom-up activities to improve the quality, completeness, and timeliness of the data (which often results directly in operational improvements and improved loss control). For example, a rough estimate of the metering (CMI) input can be improved by conducting meter accuracy tests on samples of meters, with the results providing a more trustworthy measure of the CMI input.

Several methods have been employed to assess the quality of data in water audits. Some consultants use proprietary water audit software that assigns statistical confidence levels to each component of the water audit. A composite degree of error can then be stated for the audit. The AWWA Free Water Audit Software includes a data grading capability and the calculation of the composite DVS and DVT to assess the validity of the water audit data. Rather than applying statistics, it uses a process-based approach that also provides specific guidance for water utilities on ways to incrementally improve their data and their operations. The DVT provides the auditor with context for next steps to improve the accuracy of the audit, as well as next steps for water loss control planning, target-setting, and benchmarking.

It is important that water utilities assess both the water audit outputs and the validity of the input data. The higher the level of confidence or validity of the input data in the water audit, the greater the level of confidence in the resultant performance indicators and the development of loss reduction strategies.

Assessing the Data Validity Tier of the Audit Software. The Audit Software includes guidance for water utilities to interpret the DVT. The Water Loss Control Planning Guide worksheet of the Audit Software features the Water Loss Control Planning Guide, which is shown in Figure 2-8. The guide groups DVTs based on actions that correspond to the level of validity of the water audit. Five DVTs exist across the full range of 1–100, and these are labeled as Tier I through Tier V. Additionally, a list of five functional focus areas is included in the rows of the guide. These focus areas are audit data collection, short-term loss control, long-term loss control, target-setting, and benchmarking.

The structure of the Water Loss Control Planning Guide provides that the higher-level functional focus areas (long-term loss control, target-setting, and benchmarking) should only be conducted once the water audit data have achieved a level of validity of Tier III and higher. Note that these activities are not recommended for water utilities whose DVS places them in Tier I or II. At these low tiers of data validity, the water utility should concentrate heavily on improving data generation and collection and not risk decision-making on higher functional focus areas when the integrity of the data exists at a relatively crude level. The Water Loss Control Planning Guide allows water utilities to reliably focus their attention on the needs most pertinent to their current water auditing and operational standing regarding water losses and loss control operations.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

AWWA Free Water Audit Software: Determining Water Loss Standing

FWAS v6.0

American Water Works Association. Copyright © 2020, All Rights Reserved.

Water Audit Report for: County Water Company
Audit Year: 2020 Jan 01 2020 - Dec 31 2020
Data Validity Tier: Tier III (51-70)

Water Loss Control Planning Guide

Functional Focus Area	Water Audit Data Validity Tier (Score Range)				
	Tier I (1-25)	Tier II (26-50)	Tier III (51-70)	Tier IV (71-90)	Tier V (91-100)
Audit Data Collection	Launch auditing and loss control team; address supply metering deficiencies	Analyze business process for customer metering and billing functions and water supply operations; Identify data gaps; improve supply metering	Establish/revise policies and procedures for data collection	Refine data collection practices and establish as routine business process	Annual water audit is a reliable gauge of year-to-year water efficiency standing
Short-term loss control	Research information on leak detection programs; Begin flowcharting analysis of customer billing system	Conduct loss assessment investigations on a sample portion of the system: customer meter testing, leak survey, unauthorized consumption, etc	Establish ongoing mechanisms for customer meter accuracy testing, active leakage control and infrastructure monitoring	Refine, enhance or expand ongoing programs based upon economic justification	Stay abreast of improvements in metering, meter reading, billing, leakage management and infrastructure rehabilitation
Long-term loss control		Begin to assess long-term needs requiring large expenditure: customer meter replacement, water main replacement program, new customer billing system or AMR/AMI system	Begin to assemble economic business case for long-term needs based upon improved data becoming available through the water audit process	Conduct detailed planning, budgeting and launch of comprehensive improvements for metering, billing or infrastructure management	Continue incremental improvements in short-term and long-term loss control interventions
0.000 Target-setting			Establish long-term apparent and real loss reduction goals (+10 year horizon)	Establish mid-range (5 year horizon) apparent and real loss reduction goals	Evaluate and refine loss control goals on a yearly basis
3,858.200 Benchmarking			Preliminary Comparisons - can begin to rely upon with PIs for performance comparisons for real losses	Performance Benchmarking with PIs is meaningful in comparing real loss standing	Identify Best Practices/ Best in class; PIs are very reliable as real loss performance indicators for best in class service

For validity scores of 50 or below, the shaded blocks should not be focus areas until better data validity is achieved.

Figure 2-8 AWWA Free Water Audit Software—Water Loss Control Planning Guide Worksheet completed for County Water Company

The auditor can use the guide to identify the applicable DVT and then refer to the corresponding guidance for each of the rows in the functional focus areas. Figure 2-8 highlights Tier III for the results from CWC because its DVS of 67 falls within the Tier III range of 51–70.

Within Tier III, the CWC auditor can read down the column for the five functional focus areas to obtain guidance on the most appropriate means to launch or refine specific loss control efforts. Because Tier III is in the midrange of data validity, CWC can assess that its overall water audit data exist with reasonable validity, but efforts should continue to improve its data to a higher level of confidence. Thus, the guidance provided for Tier III is that the utility should continue improving its data validity and furthering short-term loss control efforts but only begin assessing long-term loss control planning, target-setting, and benchmarking with other water utilities.

Transparently stating the reliability of water audit results has become very important as water auditing becomes standard practice in the North American drinking water industry. This is critical when communicating audit results to decision-makers, customers, and other stakeholders and comparing results with other utilities (e.g., benchmarking). The best course of action is for water utilities to compile regular annual water audits and consistently improve their data via the bottom-up approaches discussed in this chapter and detailed further in Chapters 4 and 6. For most water utilities, it will take several years of top-down and bottom-up water auditing as a standard business practice before reliable performance benchmarking against other water utilities can or should take place. For this reason, the water utility's initial focus should be to strive for sufficient data validity to establish reliable performance indicators. After that, the utility can then benchmark performance against itself for several years and finally, begin to look at long-term loss control planning and benchmarking with similar utilities.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

AWWA Water Audit Compiler. In addition to the Audit Software, AWWA's WLCC created and maintains a companion spreadsheet tool—the Water Audit Compiler—which is used to quickly assemble and analyze data from multiple water audits. Data from water audits from many utilities can be easily transferred into this spreadsheet, or the water audit data from a single water utility for a multiyear period can be assembled with trends identified.

Starting the water audit. On the Start Page, the name of the person compiling the audit (auditor) should be listed, as well as the reference period that the audit covers, along with the other required information. This is also the page on which the Volume Reporting Units are selected. Instructions on the use of the Audit Software are provided on the Start Page, as well as a description and link for each of the subsequent tabs in the tool.

All remaining data input occurs on the Worksheet, including volumes of Water Supplied, Authorized Consumption, Apparent Losses, System Data, and Cost Data. The auditor selects answers on operational practices on the Interactive Data Grading tab, and data validity grades are automatically assigned when all visible questions have been answered for a given input. Once all data are entered and gradings assigned, audit results including KPIs are calculated and displayed on the Dashboard. These steps are described in detail in the sections (tasks) that follow, and the reader can also reference the Audit Software while reviewing these steps.

Task 1—Quantify the Volume of Water Supplied to the Distribution System

Proceed to the section of the Worksheet to document Water Supplied, as shown in Figure 2-5. The Water Supplied volume is calculated automatically by the Audit Software after the individual inputs of VOS, Water Imported, and Water Exported, and their respective Error Adjustments are quantified and input by the auditor. This task demonstrates how much water enters the treated water distribution system and from where it originates.

Step 1-1. Compile the data for VOS. All water sources that are owned or managed by the water utility to supply the water for treatment and the water that goes into the distribution system should be identified. Such sources can include wells, rivers, streams, lakes, reservoirs, or aqueduct turnouts (withdrawal points from an aqueduct). Most water audits (including this example for CWC) are performed on the potable, retail water distribution system (see Figure 2-3b). In this case, the source is often the location where finished, treated water enters the distribution system. The finished water supplied from a water treatment plant is a primary example. This also represents the point where the water increases in value by virtue of being treated to meet water quality standards and pressurized for delivery. Ideally, production flowmeters (meters that measure bulk flow rates) will be located on the effluent piping leaving the water treatment plant and measure the treated water leaving the plant. However, this configuration is not always the case. For some systems, flowmeters are located on pipelines conveying untreated (raw) water into the water treatment facility, and no flowmeters exist on piping leaving the treatment plant. With flowmeters only at the raw water source, an additional adjustment may be needed to account for water used in the water treatment process, water lost to leakage in the raw water piping, and flowmeter error in measuring the untreated water.

The VOS quantity should be metered with routine flow accuracy testing and electronic calibration conducted so that volumes of water taken from all sources are registered accurately. Information about the flowmeters can be compiled in a table like Table 2-4. Data from these flowmeters should also be archived in an electronic format that allows for easy retrieval and analysis. Data should be available on a daily, weekly, and monthly basis to compile into an annual volume of water supplied from each source. Monthly water production volumes should be assembled in a table like Table 2-5.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-4 Source water measuring devices for CWC

	Volume from Own Sources		Water Imported
Characteristics	Source 1 Mountain Creek Treatment Plant	Source 2 Well Field	Source 3 City Interconnection
Type of measuring device (flowmeter)	Venturi	Propeller	Electromagnetic
Identification number (may be serial number)	0000278-A	8759	OC-16
Frequency of reading	Daily	Weekly	Daily
Type of recording register	Dial	Dial	Encoder
Units registers indicate	100,000 gal	Gal	Cubic feet
Multiplier (if any)	1.0	1.0	10.0
Date of installation	1998	1988	2009
Size of flowmeter	24 in.	8 in.	12 in.
Frequency of testing	Annual	Every 2 years	Every 6 months
Date of most recent testing	4/1/2020	8/21/2020	1/15/2021
Frequency of calibration	Annual	None	Every 6 months
Date of most recent calibration	4/1/2020	n/a	1/15/2021

Table 2-5 Total water supply in million gallons for CWC (uncorrected)

2020 by Month	Source 1 Mountain Creek Treatment Plant	Source 2 Well Field	Subtotal Own Sources (unadjusted)	Source 3 City Interconnection (water imported)	Total for All Sources 1, 2, and 3 (unadjusted)
January	126.50	0	126.50	104.27	230.77
February	162.40	0	162.40	65.17	227.57
March	146.83	87.24	234.07	0	234.07
April	280.18	125.94	406.12	0	406.12
May	274.83	132.50	407.33	0	407.33
June	260.62	172.51	433.13	81.46	514.59
July	242.64	190.68	433.32	84.72	518.04
August	283.89	104.93	388.82	89.61	478.43
September	343.42	34.06	377.48	32.59	410.07
October	161.28	31.95	193.23	98.76	291.99
November	160.09	0	160.09	126.42	286.51
December	158.27	0	158.27	96.76	255.03
Annual Total	2,600.95	879.81	3,480.76	779.76	4,260.52
Daily Average, million gallons/day					11.67

Accurately measured source or production flows are critical to the efficient operations of water utilities and wise resource management as overseen by regulatory agencies and other stakeholder groups. Therefore, utility managers and regulators should give high priority to the use of accurate metering at all sources and reliable compilation and balancing of water production and supply data. All water sources should include flowmeters that are technologically current, accurate, reliable, well-maintained,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

and—ideally—continuously monitored by a SCADA system or similar monitoring system. Detailed guidance is given in Appendix A on production flowmeter management and the means to balance flows across the distribution system, leading to an accurate quantity for the annual Water Supplied volume including VOS.

Occasionally one or more water sources of a water utility are unmetered or have flowmeters that are not routinely monitored. Because of the critical importance of the Water Supplied volume in the structure of the water audit, it is strongly recommended that new or replacement source flowmeters be installed as soon as possible, particularly if unmetered sources represent more than 10 percent of the utility's total volume entering the distribution system. If metered data are lacking in this manner on 5 percent or less of the supply, the following approaches can be taken.

- *No meters at a water source.* A portable meter should be used or the flow estimated. Portable meters can be insertion or strap-on types and can be installed on source piping just downstream of the treatment plant effluent or other source. Ideally, a minimum of 24 hours of continuous metering should be obtained, but even several hours of data will be useful. If portable metering is not feasible, one way to estimate flow is to utilize treated effluent water pumping records. If the water pump performance characteristics are known, a volume estimate can be derived by multiplying the number of hours that the pump was operated during the year by the average pumping rate. If water is taken from a large reservoir, an estimate of the withdrawal can be formulated by accounting for the drawdown of the reservoir level, adjusted by the amount of inflow from streams and rainfall and losses due to evaporation. These methods give an approximate volume measurement, but ultimately, unmetered sources should be designated for metering where possible.
- *Source water meters have not been routinely calibrated and flow verified for accuracy*. An inspection of the source flowmeter installation should be conducted. The type of metering device that exists should be noted (e.g., Venturi flowmeter, magnetic flowmeter, ultrasonic flowmeter, Parshall flume, weir, or stream gauge). Basic information about the measuring device should be noted: type, identification number, frequency of reading, type of recording register, unit of measure (and conversion factor, if necessary), multiplier, date of installation, size of pipe or conduit, frequency of flow testing, and date of last test. Using that information, a table like Table 2-4 should be constructed.

A record should be obtained about how much water was produced by each source during the period of the audit. Most meters have some type of register or totaling device. Older registers are round reading or direct reading. Round-reading registers have a series of small dials with pointers, registering cubic feet or gallons in tens, hundreds, thousands, and tens of thousands. Direct-reading registers have a large sweep hand for testing, and a direct-reading dial that shows total units of volume. Electronic encoder registers display and record flow data with a high degree of precision. If the meter has not been routinely read, tested for accuracy, or its secondary instrumentation calibrated, work should be initiated to flow test the meter, calibrate its secondary instrumentation, and institute routine reading of the meter. Currently, many drinking water utilities link source meters with SCADA systems that convey data in real time to centralized computers, where the flow data are totaled and archived for easy retrieval. See Appendix A for further guidance.

Tables 2-4 and 2-5 illustrate the example of CWC with flowmeters on the two sources that it owns, "Mountain Creek" and "Well Field," as well as water imported (purchased) from a neighboring water utility, "City Interconnection." These tables illustrate how source flowmeter and flow data can be arranged and adjusted for the water audit period.

Step 1-2. Adjust figures for total supply. Once a volume is established for each source for the audit year, the measured amounts should be reviewed and corrected for

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

known systematic or random errors that may exist in the metering data. Figures for the total water supply, based on readings from source meters and measuring devices, are raw data. The raw data must be adjusted for several factors, including

- changes in treated water reservoir and storage tank levels,
- any other adjustments such as losses that occur before water reaches the distribution system, and
- corrections due to data gaps or data corruption.

Step 1-2A. Adjust reservoir and tank storage. If source meters are located upstream of treated water reservoirs and storage tanks, treated water in storage must be accounted for in the water audit. Generally, water flowing out of storage is replaced; as the replacement water flows from the source into storage, it is measured as supply in the system. If the reservoirs have more water at the end of the audit period than at the beginning, the increased storage is measured by the source flowmeters but not delivered to consumers. Such increases in storage should be subtracted from the VOS quantity. Conversely, if there is a net reduction in storage, then the decreased amount of stored water should be added to the VOS quantity. Table 2-6 shows how to compute the change in storage volume using this general approach. Although this method suffices as a concise example of the adjustment for reservoir storage, a more accurate quantification of adjustment due to reservoir and tank storage can be obtained by monitoring storage levels and making adjustments on a day-to-day basis. An example of this more accurate method to quantify this adjustment is given in Appendix A under the "Example of the Mass Balance Technique" subsection.

It should be noted again that decreases in storage are added to the supply; storage increases are subtracted from the supply.

For CWC, this is +825,580 gal or +0.83 mil gal, as shown in Table 2-6.

Large, open reservoirs may require volume adjustments because of the effects of evaporation (water lost) and rainfall (water gained) and should be considered by the auditor.

For CWC, no such adjustments exist, so no adjustment is required.

Step 1-2B. Other adjustments. Some water suppliers may be subject to other types of contributions or losses. One example would be losses incurred during the treatment process (filter backwashing, etc.) if the source meter is located at the point of entry to a water treatment plant. In another case, there may be an additional source that enters the water system between the source meter and the finished water system. This could result from infiltration into an open channel. Likewise, losses may be introduced through an unlined or open channel. The auditor should document these additions or losses.

For CWC, no such adjustments are listed in this example.

Step 1-2C. Corrections due to data gaps or data corruption. As discussed in Appendix A, gaps in flow recordings may occur because of instrument or SCADA equipment failure. Data archives may also become corrupted periodically. The auditor

Table 2-6 Changes in reservoir/tank storage for CWC

Reservoir/Tank	Total Volume (gal)	Start Volume (gal)	End Volume (gal)	Change in Volume (gal)
Apple Hill	50,000	32,350	36,270	+3,920
Cedar Ridge	300,000	278,100	240,600	–37,500
Monument Road	1,000,000	978,400	318,400	–660,000
Davis	250,000	187,300	55,300	–132,000
Total change in reservoir storage				–825,580

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

should attempt to reconstruct the data archive with representative values. This type of surveillance and correction is best accomplished by monitoring the data daily.

Step 1-2D. Determine the adjusted VOS due to reservoir/tank storage changes, other adjustments, and data gaps. Only reservoir/tank storage changes have a quantity to enter for this category of adjustments. This adjustment is calculated as follows:

> *The value of "other adjustments" and "data gaps" is zero. Thus, the total adjustments made to the raw VOS quantity from Table 2-5 of 3,480.76 mil gal is calculated by adding the annual volume that was removed from reservoir/tank storage of +0.83 mil gal to result in the adjusted VOS quantity of 3,481.59 mil gal entered in the Worksheet shown in Figure 2-5.*

Step 1-2E. Answer IDG criteria questions to determine the data grade for VOS. Key factors affecting the data grading for this input include the percentage of treated water sources that are metered, the frequency of flow accuracy testing and electronic calibration, and management of the flow data. CWC is proactive in metering all three of its water sources. As noted in Table 2-4, CWC conducts regular meter accuracy testing on the flowmeters of its own two sources: Mountain Creek and the Well Field. However, its Well Field flowmeter is only tested for accuracy every two years. Collectively for those meters that measure >90% of the finished water volume, the most representative answer to the IDG question regarding frequency of flow accuracy testing is "Less than annual but within last 5 years." The CWC auditor answered all visible IDG questions for the VOS input in accordance with the utility's practices, resulting in automatic calculation of the data grade, which is 7 for CWC. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting" on the IDG worksheet of the Audit Software.

Step 1-2F. Calculate the VOS Error Adjustment (VOSEA). Step 1-2D reflects the potential for adjustment to the VOS input due to changes in distribution storage, data gaps, and/or other adjustments. As an alternative, these adjustments may be reflected in the VOSEA. The auditor should closely analyze the specific situation for a given supply infrastructure configuration and determine which potential sources of error exist and the most appropriate way to reflect this in the water audit. Regardless, the auditor should ensure all calculations and assumptions for Steps 1-2D and 1-2F are clearly documented. The guidance presented below assumes the auditor has determined to reflect only the meter error in the VOSEA and to reflect any other adjustments in the VOS.

Throughout the water supply industry, a wide variety of flow measuring devices exist, some of which are more suited to untreated or raw water supply and others better suited for sediment-free potable drinking water. Although most water utility source flows are measured by meters, some are measured by other devices, such as Parshall flumes or weirs, particularly if the flow is untreated water being supplied to a water treatment facility. Water supply data (like those used in Table 2-5) are based on readings of these measuring devices that exist with the utility's water supply system. No meter or flow measuring device is 100 percent accurate, and some degree of error always exists. An assessment of the likely degree of error in the measuring devices should be undertaken and included in the VOSEA input in the water audit. However, because error can exist in either direction (over- or underregistration), caution is advised against arbitrarily inputting an error amount not based on testing activity or other reliable evaluation of the supply meter accuracy.

To determine the accuracy of the production flowmeters, the results of accuracy tests should be compared to applicable AWWA standards and guidance manuals. If meter error exceeds the standard for its category, the meter and its related instrumentation should be repaired, recalibrated, or replaced to function within standard limits. If the meter has not been tested for accuracy within the past 12 months, the meter should be tested as soon

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

as possible. See Appendix A for detailed discussion of production flowmeter verification practices.

Possible causes of production flowmeter error. Normal wear is not the only cause of inaccurate meter readings. Inspect the flowmeter in its field installation to determine if the flowmeter is the right type and size for the application and that it is installed correctly. See AWWA Manual M33, *Flowmeters in Water Supply* (AWWA 2018), and Appendix A for guidance on typical flowmeter types and applications. The meter should be installed according to manufacturers' recommendations. As feasible, the flowmeter can be removed from service and inspected to see if hard-water encrustation is interfering with the measurement. Also, it should be verified that the proper registers were selected and installed correctly. Finally, the register should be read to see that the signal from the meter is properly transmitting to the SCADA system. A person familiar with metering instrumentation (third-party testing service or a knowledgeable employee) should perform the calibration of the instrument and should make a special reading of the source meter. That person might also accompany the meter reader to verify sample readings. It should be verified that the meter is read and recorded correctly and that the correct conversion factor is used.

Checking Venturi flowmeters. Venturi flowmeters are one of the most common metering devices in use as production flowmeters in the drinking water industry. These meters should only be used for sediment-free water and should periodically be checked for blockages in the throats of the meters or in the sensing lines. The primary device should be regularly tested by comparing it with a measurement taken from a portable insertion-type or strap on–type meter at an appropriate location downstream on the supply pipeline. Testing the meter with an insertion meter shows whether the installation is adequate for flows with low turbulence. The meter's primary device should be tested at different flow ranges. If pressure differential for appropriate flows is adjusted without checking the Venturi itself, the meter may still record flows, but the data will be erroneous.

Testing flowmeters for accuracy. Flow accuracy testing is defined in the Audit Software as "a process that confirms the flow measuring accuracy of the primary device (the flowmeter), in its installed location, using an independent reference volume." Water utilities often conduct flow accuracy testing as well as electronic calibration of related instrumentation such as differential pressure cells connected to Venturi flowmeters. Unfortunately, electronic calibration does not confirm the flow measuring accuracy of the primary metering element. Many utilities conduct regular calibration but do not conduct flow accuracy testing. Regular flow accuracy testing is necessary for water utilities to ascertain the degree of error in the flowmeter installation and to yield a higher data grade in the Audit Software. In-situ test methods are generally recommended over test methods involving meter removal; however, all test methods can have limitations, and factors affecting uncertainty must be assessed with all test methods. See Appendix A for further discussion on testing of supply flowmeters. Five general ways that production flowmeters may be tested are listed below.

1. Test the meters in place.
2. Conduct a clearwell or storage tank drop test, if feasible.
3. Compare flowmeter readings with readings of a calibrated meter installed in series with the host meter.
4. Record flowmeter readings for a given flow over a specified period. Remove the meter and replace it with a calibrated meter. Record readings from the calibrated meter using the same flow rate for the same duration; compare the readings.
5. Remove the host flowmeter and test it at a meter testing facility (not practical for larger flowmeters).

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

These methods are discussed in detail in Appendix A, and it is recommended that the reader review this information to determine the most appropriate testing options for flowmeters in their system. Flow accuracy testing may be conducted by trained utility personnel, specialized consultants, meter manufacturers, and meter-testing laboratories for meters that can be removed and shipped to the test facility.

The monthly and annual supply data shown in Table 2-5 should be reviewed and the VOSEA quantified. The VOSEA volume can be taken as the difference of the uncorrected metered volume minus the corrected meter volume. The calculation to determine the corrected metered volume is given in Equation 2-1. The uncorrected metered volume should be divided by the measured accuracy of the meter (a percentage expressed as a decimal) and subtracted from the uncorrected metered volume as follows:

$$\text{Uncorrected metered volume} - \frac{\text{uncorrected metered volume}}{\text{percent accuracy}} = \text{VOSEA volume} \qquad \text{(Eq 2-1)}$$

If the auditor wishes to calculate the VOSEA as a percent instead of a volume, the Audit Software includes additional guidance for this in the Water Supplied Error Adjustments definition.

Table 2-7 shows how to adjust the supply totals from Table 2-5 to yield the adjusted measurements.

Enter the net VOSEA in the Audit Software Worksheet as shown in Figure 2-5. For CWC, this is a quantity of –136.60 mil gal, which represents underregistration of the meter. The volume should be entered as a positive number in the Audit Software and the "underregistration" option selected to the right of the input.

Step 1-2G. Answer IDG criteria questions to determine the data grade for VOSEA. Key factors affecting the data grading for this input include monitoring of storage volumes and incorporation of accuracy test results into the VOSEA input. The CWC auditor answered all visible IDG questions for the VOSEA input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 10 by the Audit Software. CWC's data grade of 10 for this component reflects that it is currently performing

Table 2-7 Volume from Own Sources in million gallons for CWC (adjusted for meter error)

Source	Yearly Total: Uncorrected Metered Volume*	Meter Error (%)	Percent Accuracy (%)	(Uncorrected Metered Volume) – (Uncorrected Metered Volume/Percent Accuracy†) See Equation 2-1	Meter Error (mil gal) ‡	Adjusted Metered Volume (mil gal) §
1. Mountain Creek	2,600.95	–3	97	2,600.95 – (2,600.95/0.97)	–80.44	2,681.39
2. Well field	879.81	–6	94	879.81 – (879.81/0.94)	–56.16	935.97
Total	3,480.76				–136.60	3,617.36

* Based on Table 2-5.
† A percentage, written as a decimal (95 percent = 0.95).
‡ Negative (–) meter error represents underregistration. Positive (+) meter error represents overregistration.
§ The total corrected meter volume for sources 1 and 2 is 3,617.36 mil gal; note that this is 136.60 mil gal greater than the total supply given for these sources in Table 2-5. This is a way to double-check the arithmetic.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

at the highest level of flowmeter management. CWC's focus should then be to sustain these strong practices without backsliding.

Step 1-3. Compile the volume of Water Imported (WI) from outside sources or purchased from other water utilities. Tables 2-4 and 2-5 include Source 3, which is an interconnection flowmeter on the City Interconnection. This flowmeter registers water purchased by CWC from a neighboring water utility. Interconnections between water utilities are considered custody transfers of water, and the pipelines supplying these flows typically include flowmeters that are carefully maintained and monitored because the metered data provides the basis for billing large water volumes. Both the water utility supplying the water and the system purchasing the water have a strong motivation to keep this bulk measurement accurate because significant costs are at stake for each water utility. For this reason, it is common practice that custody transfer flowmeters are flow accuracy tested and calibrated on an annual basis or more frequently. As with the data from VOS, the data derived from "import" flowmeters should be carefully managed and adjusted accordingly during the water audit, based on the validation methods like those noted in Step 1-2 and described in detail in Appendix A.

Enter the WI volume on the Worksheet shown in Figure 2-5. From Table 2-5, obtain the uncorrected value of 779.76 mil gal for the City Interconnection imported volume to CWC and enter it as shown in the WI cell.

Step 1-3A. Answer IDG criteria questions to determine the data grade for WI. Key factors affecting the data grading for this input include the percentage of treated water sources that are metered, the frequency of flow accuracy testing and electronic calibration, and management of the flow data. CWC and its imported water supplier coordinate to conduct meter accuracy testing and flowmeter maintenance every six months. Flow accuracy test results are not listed for the City Interconnection in Table 2-4; however, they are known to be a composite underregistration of 0.50 percent inaccuracy (99.5 percent accurate) for calendar year 2020. The CWC auditor answered all visible IDG questions for the WI input in accordance with the utility's practices, resulting in automatic calculation of the data grade of 10. CWC's data grade of 10 for this component reflects that it is currently performing at the highest level of flowmeter management.

Step 1-3B. Calculate the WI Error Adjustment (WIEA). Table 2-4 shows that flows from the City Interconnection are measured via a 12-in electromagnetic (mag) flowmeter that is flow-accuracy tested against an insertion flowmeter inserted downstream of the flowmeter every six months. This flowmeter installation is well maintained by the water utility that supplies the water to CWC. Calibration of the mag meter is also conducted every six months using diagnostics software built into the flowmeter. These actions have resulted in superior meter accuracy test results of a composite 0.50 percent inaccuracy (accuracy of 99.5 percent) for the City Interconnection flowmeter.

The WIEA is entered (as shown in Figure 2-5) into the water audit in the WIEA cell, with "percent" selected as the entry option. The average inaccuracy of one half of 1 percent, or 0.5 percent, is entered. This inaccuracy value, which translates to –3.90 mil gal, is calculated but not displayed by the Audit Software. The percentage should be entered as a positive number, then the "underregistration" option selected to the right of the input.

Note: If a water utility does not import any water, then the Audit Software does not display the WIEA cell.

Step 1-3C. Answer IDG criteria questions to determine the data grade for WIEA. Key factors affecting the data grading for this input include parameters of the agreement between importer and exporter, incorporation of accuracy test results into the WIEA input, and access to flow data. CWC and its importer water utility maintain a very

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

high level of rigor in the management, testing, and data archival of the water imported supply. The CWC auditor answered all visible IDG questions for the WIEA input in accordance with the utility's practices, resulting in automatic calculation of the data grade of 10. CWC's data grade of 10 for this component reflects that it is currently performing at the highest level of flowmeter management. CWC's focus should then become sustaining these strong practices without backsliding.

Step 1-4. Compile the Water Exported (WE) volume delivered to outside water utilities or jurisdictions. The WE volume is water sold to wholesale customers who are typically charged a wholesale rate that is different than rates charged to retail customers existing within the service area. Many state regulatory agencies require that the WE volume be reported to them as a quantity separate and distinct from the retail customer billed consumption. For these and other reasons, the WE volume is always quantified separately from Billed Authorized Consumption in the standard water audit. Be certain not to "double-count" this quantity by including it in both the WE box and as part of the BMAC cell of the water audit worksheet. This volume should be included only in the WE cell. The WE volume should be monitored and adjusted with the same scrutiny given to imported water, because the same revenue implications exist. As with the WI volume, a separate entry for Error Adjustment is included on the worksheet in Figure 2-5.

> *Enter the volume of WE in the appropriate cell in the worksheet shown in Figure 2-5. In this example, CWC exports no water to neighboring water utilities, so no value is entered in this cell.*

Step 1-4A. Answer IDG criteria questions to determine the data grade for WE. Like IDG questions for WI, the auditor must consider the extent of metering of the exported supply pipeline and flowmeter accuracy test frequency and results. Because CWC does not export any water, the auditor answers the first IDG question accordingly, and the data grade automatically calculates as "n/a" (not available) because CWC exports no water.

Step 1-4B. Determine the Water Exported Error Adjustment (WEEA). Like the WIEA, the auditor should consider the extent of metering of the exported supply pipeline and flowmeter accuracy test frequency and results. The test results should provide data for the auditor to quantify the WE volume for the audit year. If a water utility does not export any water, the Audit Software does not display the WEEA cell. Thus, the CWC auditor will not encounter this cell as shown in Figure 2-5.

Step 1-4C. Answer IDG criteria questions to determine the data grade for WEEA. Like the IDG questions for WIEA, the auditor must consider parameters of the agreement between importer and exporter, incorporation of accuracy test results into the WEEA input, and access to flow data. Because CWC does not export any water, and the data grade for WE is "n/a," the data grade for WEEA is also designated as "n/a."'

Step 1-5. Quantify the Water Supplied volume into the distribution system. The Water Supplied volume is automatically calculated by the Audit Software after the inputs for VOS, WI and WE are entered into the water audit.

> *The Water Supplied volume to the distribution system is calculated as the sum of VOS, plus WI minus WE and includes the Error Adjustments of these three components. The Water Supplied volume for CWC is calculated to be 4,401.87 mil gal for the water audit period of calendar year 2020.*

Task 2—Quantify Authorized Consumption

Authorized consumption is any water delivered for consumptive purposes that are approved by the water utility as a benefit to the community. Steps 2-1 through 2-4 describe how to quantify authorized consumption. Steps 2-1 and 2-2 deal with billed authorized

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

consumption, while Step 2-3 details unbilled authorized consumption. Step 2-4 sums up all authorized consumption.

Billed authorized consumption represents the collective amounts of water delivered to individual customers that have accounts in a customer billing system. Billed authorized consumption is the basis for revenue generation for the water utility. Billed accounts are customer properties served by permanent customer service connection piping. Most of the water supplied to the distribution system is attributed to this type of consumption. In North America, most water utilities and utility commissions require customer meters on service connections and bill based on metered consumption on a monthly, bimonthly, or quarterly basis. Metered water can be categorized as residential, industrial, commercial, agricultural, governmental, or other uses. Some water utilities do not require meters for a portion or all their customers; instead they charge a flat billing rate per consumption period or based on property or other characteristics. Therefore, billed authorized consumption may be metered or unmetered. AWWA recommends that all customers with permanent service connection piping be metered with billing based at least in part on measured consumption.

Some utilities also have temporary meters for construction and other uses that should be added to billed authorized consumption totals. These meters are sometimes attached to fire hydrants or allowed to travel with contractors or other service providers to capture water consumption from hydrants. It is most helpful to obtain readings from these meters at the same intervals as other fixed meters, and readings at the beginning and the end of the audit period will be required in any event. Ultimately, temporary meters should be replaced with permanent meters to the extent practical for any network access locations with continuous use.

Unbilled authorized consumption is water that is provided to customers without a charge being assessed for the water use. Unbilled consumption also occurs as water used by the water utility (flushing fire hydrants) or community services (firefighting, street cleaning). Unbilled consumption may be metered or unmetered. Unbilled metered consumption of permanent customers allows their water consumption to be captured even if the water utility policy allows the service to go uncharged. A frequent example of this is unbilled water supplied to government buildings by municipally owned water utilities. By metering these buildings and routinely reading the meters, water consumption is monitored even though the property is issued a "no charge" bill. Unbilled unmetered consumption includes water taken irregularly in a variety of manners from nonaccount connections, such as fire hydrants that typically do not supply permanent structures. Withdrawing water from fire hydrants and water used for firefighting are the most common examples of such nonaccount consumption.

Remember: To be accurate, the water audit period must be consistent across all water audit components. Be sure to use the same 12-month audit period and the same units of measure as was used to quantify the Water Supplied volume when evaluating authorized consumption.

Step 2-1. Compile the volume of Billed Metered Authorized Consumption (BMAC). Modern metering, automatic meter reading (AMR), AMI, and customer billing management technologies offer outstanding capabilities for water utilities to gather and utilize accurate customer consumption and billing data. It is strongly recommended that water utilities measure individual customer consumption via water meters and use computerized customer billing systems to store customer account data. AMR and AMI systems are being implemented by a growing number of water suppliers because of their cost-effectiveness in gathering metered consumption data and ability to provide enhanced customer service. For water utilities that utilize these technologies, consumption data are typically accessed via a variety of reports from the customer billing system. Examples of typical reports are shown in Tables 2-8 and 2-9, where consumption is summarized by

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-8 Number of customer accounts and metered consumption by meter size for County Water Company: Jan. 1–Dec. 31, 2020

Meter Size (in)	Number of Accounts	Percent of Total Accounts	Percent of Metered Consumption
⅝	11,480	94.10	71.2
¾	10	0.08	0.1
1	338	2.80	2.8
1½	124	1.00	2.8
2	216	1.80	11.7
3	15	0.12	6.6
4	7	0.05	2.2
6	6	0.05	2.6
Total	12,196	100.00	100.0

Table 2-9 Total metered water consumption by category for County Water Company (uncorrected)

2020 by Month	Residential (mil gal)	Industrial (mil gal)	Commercial (mil gal)	Metered Agriculture (mil gal)	Total for All Meters (mil gal)
January	156.6	36.8	7.4	0	200.8
February	171.9	37.8	7.6	0	217.3
March	170.9	37.8	7.7	5.10	221.5
April	286.2	40.1	8.2	48.3	382.8
May	303.8	40.4	8.4	50.1	402.7
June	320.2	42.9	8.5	55.9	427.5
July	323.4	44.9	8.7	57.0	434.0
August	335.4	46.9	8.6	71.8	462.7
September	283.2	45.6	8.2	60.0	397.0
October	181.7	40.8	8.1	5.20	235.8
November	193.9	37.8	8.0	0	239.7
December	191.6	36.8	7.8	0	236.2
Annual Total (mil gal)	2,918.8	488.6	97.2	353.4	3,858.0
Daily Average, mgd	7.97	1.33	0.27	0.97	10.54

meter size and customer consumption category, respectively. Note that the consumption shown in Table 2-9 is unadjusted and a minor adjustment will be made to these data.

Chapters 3 and 4 discuss the sometimes unintended impacts to the integrity of consumption data that may occur in customer billing system operations. The auditor should develop a sound understanding of the workings of the customer billing system to ascertain the true amount of customer consumption and identify any billing system functions that unduly reduce the actual volume of customer consumption.

Step 2-1A. Maintain customer accounts data. If computerized billing records or reports do not exist, the auditor must assemble customer account information from available records. Start by identifying all customer users with permanent structures that should

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

have meters. Accounts should be identified by several descriptors such as account number, property street address, meter size, meter serial number, connection size, assessor's parcel number, and the name and address of the property owner as well as any tenants that receive bills in their own name. To track customer consumption patterns and water conservation impacts, it is important to list the consumption category for each account: residential, industrial, commercial, agricultural, governmental, and so on.

Step 2-1B. Maintain customer meter and AMR/AMI data. All active accounts should include the meter identification number, meter size, meter type, and installation date. If an AMR or AMI system exists, the reading transmission device number and meter-reading route number should also be included in the customer billing system, along with any other pertinent information. If the AMR or AMI system is compatible, readings should be collected from connected meters at times that coincide with the beginning and end of the water audit.

Step 2-1C. Compile metered consumption volumes for the water audit period. First, assemble the total (uncorrected) water consumption volumes for all accounts and connections for each size of meter by month (or other billing period) and for the entire audit period, as shown in Table 2-9. The same unit of measure for supply should be used; this may require performing a conversion, for example, from cubic feet to million gallons.

Step 2-1D. Adjust for lag time in meter readings. Corrections must be made to metered consumption data when the production flowmeter reading dates and the customer meter-reading dates do not coincide with the beginning and ending dates of billing cycles and the water audit period. AMI systems can eliminate this issue because customer meter consumption data are automatically compiled each day. If the auditor obtains AMI billed consumption data, then the data can be synchronized with the production flowmeter data and the auditor can proceed directly to Step 2-2. If the AMI system will accommodate the practice, it is best to align consumption meter data read timing exactly with supply meter data read timing, down to the day and hour. Note, however, that many utilities with AMI systems do not change their traditional billing cycles; thus the meter-reading software could be used to acquire the volumes. In this process, the utility should be aware of the possibility of reporting duplicate volumes such as master meters, sanitary sewer deduction, or irrigation meters measuring nonpotable volumes.

Adjusting for lag time for meter routes read on the same day. For this example, consider a utility auditing one calendar year, January 1 through December 31, with source production flowmeters read on the first day of each month and customers' meters read on the 10th day of each month. The goal is to calculate the amount of water supplied and consumed for the calendar year:

- *Source production flowmeters.* No lag time correction is made for source flowmeters because their reading usually occurs on the days that the water audit period begins and ends. If the last reading (December 31) was a day late (January 1), then the water supplied for January 1 should be subtracted from the total water supply reading.
- *Customer meters.* For utilities that do not use an AMI system, customer meter readings usually do not coincide neatly with the study period; thus, a correction must be made. The best way to account for changes in the number of customers and in consumption patterns is to prorate water consumption for the first and last billing periods within the water audit period. As an example, consider a system that has a first billing period with only 10 days that occur in the water audit period that started on January 1, yet the billing information represents 31 days of consumption. If consumption for December 11 through January 10 is 33.204 mil gal, the amount applicable to the water audit period for January (January 1-10) is

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

$$33.204 \text{ mil gal} \times \frac{10 \text{ days}}{31 \text{ days}} = 10.711 \text{ mil gal} \quad \text{(Eq 2-2)}$$

Thus, 10.711 mil gal of the consumption read on January 10 applies to the water audit period that started in January.

At the end of the water audit period, there are 21 days not included in the billing data collected on December 10 (December 11–31). Consumption for the last 21 days in December is obtained from the following month's billing (January of the next year). If consumption recorded on January 10 of the next year was 36.66 mil gal, the amount applicable to the water audit period is

$$36.66 \text{ mil gal} \times \frac{21 \text{ days}}{31 \text{ days}} = 24.83 \text{ mil gal} \quad \text{(Eq 2-3)}$$

Thus, 24.83 mil gal is added to the consumption read on December 10 of the water audit year. The total adjusted volume of customer consumption added is (10.711 + 24.83) mil gal = 35.541 mil gal.

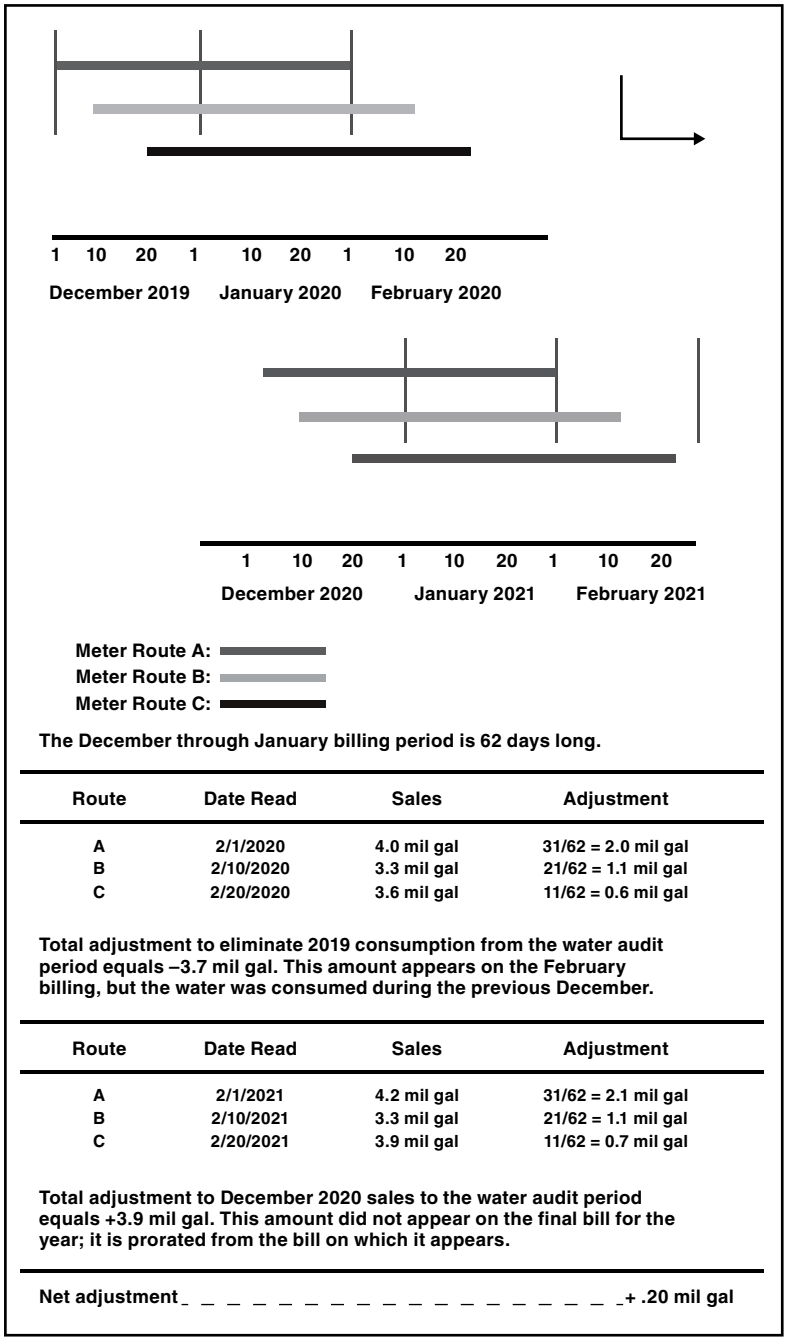
Adjusting for meters read on varying dates through the billing period. The preceding discussion describes the basic method for correcting lag time in meter reading when all customers' meters are read on the same day. That seldom happens, however, in utilities that do not employ an AMI system. Usually, meters are assigned to different routes and read on different days. Therefore, a meter lag correction should be used for each meter reading route, particularly if each customer's meter is read on the same day each month. Figure 2-9 gives an example of this situation.

A meter lag correction can involve several steps. In the example shown in Figure 2-9, CWC has three meter-reading routes, each with its own reading date. The water audit period is one calendar year (2020), and the consumption is prorated for each meter route or book. Meters are read monthly: route A on the 1st of the month, route B on the 10th, and route C on the 20th.

The uncorrected metered water consumption of 3,858.0 mil gal shown in Table 2-8 is based on bills issued during the water audit period. However, because of the monthly billing schedule, these bills do not include all water consumed during the year. Some water shown as used in the first bills issued in February 2020 occurred in December 2019. The last set of bills, issued in November and December 2020, do not include all water consumed in December 2020 because water used in the last days of December were included in bills issued in January 2021. Two corrections need to be made. First, water consumed in the month preceding the water audit period (December 2019) must be subtracted from consumption figures. Second, water consumed in the final month of the water audit period must be added. The more frequent (monthly as opposed to bimonthly or quarterly) the readings, the smaller the adjustment and the less likely the estimated use will be prone to error.

Additional considerations in making lag-time adjustments. It is important to acknowledge that prorating customer meter readings carries the inherent assumption of equal daily usage between meter read dates. The auditor should be on alert for any consumption anomalies that can occur at the end of an audit period (for example, holiday season uses at end of a calendar year or weather-based swings in irrigation uses at end of a fiscal year). If such anomalies are present, the auditor should consider custom adjustments for such accounts rather than an assumption of equal daily usage as described above. The auditor should strive to be consistent with the lag-time adjustment method utilized from one year to the next to minimize variability from the derivation itself.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



The December through January billing period is 62 days long.

Route	Date Read	Sales	Adjustment
A	2/1/2020	4.0 mil gal	31/62 = 2.0 mil gal
B	2/10/2020	3.3 mil gal	21/62 = 1.1 mil gal
C	2/20/2020	3.6 mil gal	11/62 = 0.6 mil gal

Total adjustment to eliminate 2019 consumption from the water audit period equals –3.7 mil gal. This amount appears on the February billing, but the water was consumed during the previous December.

Route	Date Read	Sales	Adjustment
A	2/1/2021	4.2 mil gal	31/62 = 2.1 mil gal
B	2/10/2021	3.3 mil gal	21/62 = 1.1 mil gal
C	2/20/2021	3.9 mil gal	11/62 = 0.7 mil gal

Total adjustment to December 2020 sales to the water audit period equals +3.9 mil gal. This amount did not appear on the final bill for the year; it is prorated from the bill on which it appears.

Net adjustment + .20 mil gal

Figure 2-9 Detailed meter lag correction

The auditor should also be aware that lag-time adjusted totals for BMAC will not match the unadjusted BMAC totals present in other places where this volume is tracked and reported. Care should be taken to communicate as needed to any stakeholders with access to this information about why different totals are observed and the importance of lag-time adjustments for reducing error in the water audit.

Figure 2-9 shows how to adjust consumption figures for meter lag time. Many utilities combine accounting and billing procedures into a computerized format to make this procedure easier and quicker.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The auditor for CWC prorates the water consumption figures to adjust for lag time in meter reading. As shown in Figure 2-9, the net adjustment of +0.20 mil gal was calculated. This adjustment was added to the BMAC value of 3,858.0 mil gal to give the revised BMAC value of 3,858.20 mil gal. The revised number is entered as the BMAC input in the Worksheet shown in Figure 2-5.

Step 2-1E. Answer IDG criteria questions to determine the data grade for BMAC. Key factors affecting the data grading for this input include the percentage of customers with volume-based billing from meter readings, meter-reading frequency, incorporation of lag-time adjustments, and data review practices. The CWC auditor answered all visible IDG questions for the BMAC input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 8. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 2-2. Compile the volume of Billed Unmetered Authorized Consumption (BUAC). Most North American drinking water utilities have policies to universally meter their customers and bill based on measured consumption. This is standard practice recommended by AWWA. However, not all utilities meter all their customers; instead, some water utilities bill customers a flat fee per billing period. Others meter a portion of their customer accounts. This latter scenario can occur if

- the utility is in transition to a fully metered customer population;
- larger commercial and industrial accounts are metered whereas smaller accounts are not;
- utility policies dictate that certain accounts, such as municipal properties or fire connections, need not be metered; or
- some of the meters are known to be nonfunctional or highly inaccurate or readings are unobtainable, in which case estimates of consumption are used in place of measured consumption. This position, however, reflects that—by policy—the water utility requires water meters but has difficulty in keeping all meters functional or accessible.

In a system that is completely or largely unmetered, the auditor must devise an estimate of the water consumed by unmetered customers. One way to accomplish this is to install water meters in a small sample of accounts (50-100) based on consumption category or meter size. Data from these meters could be used to develop average consumption trends that could be inferred for the entire population in each category such as residential, industrial, and so forth. Any estimating process that is developed should be fully documented and based on current conditions. Estimation invariably interjects a degree of error into the measure of customer consumption. For this reason, it is recommended that all customers be metered with routine meter readings and archived consumption data, thereby keeping BUAC at zero or a minimal quantity.

Include the total estimate of BUAC in the Worksheet shown in Figure 2-5. For CWC, this value is zero because the company meters and regularly reads all accounts, so this cell is left blank.

Step 2-2A. Answer IDG criteria questions to determine the data grade for BUAC. Key factors affecting the data grading for this input include the percentage of billed accounts that are unmetered and method for estimating unmetered customers' consumption. CWC meters its entire customer population and reads meters manually monthly, and the volume entered into the water audit for this component is zero. Thus, the auditor answered the first IDG question accordingly, and the data grade automatically calculates as "n/a."

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Step 2-3. Compile the volume of Unbilled Consumption. As discussed previously, unbilled authorized consumption describes water that is provided to customers without a charge being assessed for the water use. Unbilled metered authorized consumption (UMAC) of permanent customers allows their water consumption to be captured even if the water utility policy allows the service to go uncharged. These may include municipal facilities such as city hall, police and fire stations, libraries, public parks, and fountains. Unbilled unmetered authorized consumption (UUAC) includes water taken for sundry purposes, including water taken from fire hydrants for firefighting as their primary purpose, flushing, testing, street cleaning, and other public purposes. Rarely is such consumption metered or directly billed, although sometimes revenue is recovered via flat fees paid by fire departments or other users. These uses should include backflow prevention protection with usage policies in force to protect water quality and public safety. Water utilities often utilize water from the distribution system at their own treatment plants and facilities in uses that include backwash water to clean treatment plant filters, internal building use, and sampling; these uses may fall under the category of unbilled unmetered consumption.

Unfortunately, many water utilities do not employ clearly written policies that include procedures for unbilled water uses. Similarly, good accounting often does not exist for the types and volumes of such consumption occurring throughout the year. It is recommended that the auditing process review utility policies and practices and improve them as needed to expand the use of temporary/portable meters where practical to convert unbilled consumption to billed authorized consumption. Where this is not possible, the utility should ensure that unbilled water consumption is not unsafe or wasteful and can be accounted for to the extent practical.

The UUAC volume is usually a small portion of the total volume of Authorized Consumption; therefore the auditor may choose to use the default value in the Audit Software for UUAC when validated data for unbilled unmetered activities is unavailable. However, auditors are encouraged to collect data and quantify unbilled unmetered use to the extent possible. Under conditions such as severe drought, publicly visible use of water for flushing, or other operations could generate negative public perceptions for the water utility. In such cases, auditing should review all instances of unbilled authorized consumption and ensure that they are efficiently managed.

Step 2-3A. Compile the volume of UMAC. Any unbilled consumption that is metered can be quantified by obtaining meter readings at the beginning and end of the consumption period(s) throughout the year of the water audit. If a permanent meter exists and supplies a permanent structure (such as a municipal building or a water treatment plant), it is best if the property is eventually assigned an account in the customer billing system and is read regularly, just as all other water meters are read. The consumption of unbilled accounts must be tracked separately from water that is billed, and any metered Authorized Consumption that does not generate revenue for the utility, even if it exists in the billing system, should be included as UMAC. In addition, activities that can be metered using portable hydrant meters in the field or measured at bulk water fill stations should be identified and policies established to explicitly measure consumption and tally the consumption each billing period. Water use should be metered to the greatest extent possible, whether the use is billed or unbilled.

> *Include the total of all UMAC documented on the Worksheet shown in Figure 2-5. For illustration, the auditor of CWC obtained meter readings on a sporadic basis for known unbilled metered water users and tabulated a total of 15.42 mil gal for UMAC for the audit year.*

Step 2-3B. Answer IDG criteria questions to determine the data grade for UMAC. Key factors affecting the data grading for this input include the status of utility policies for billing exemptions, extent of unmetered accounts, and frequency of reading

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

and review of unbilled-metered consumption data. CWC meters its entire customer population and reads billed metered accounts manually monthly. However, an unknown number of unbilled metered water users exists. CWC has documentation on several of the high-profile water users, including the water treatment plant, the wastewater treatment plant, and the municipal office building. For these accounts, meter readings were gathered quarterly throughout the audit year.

Unfortunately, CWC is also aware that an unknown number of other metered water users exists, including at maintenance sheds, park and playground facilities, the municipal airport, and police and fire stations. Written policy documents permitting such water-using facilities to go unbilled are incomplete and dated. CWC is aware that it has not accounted for all unbilled metered water consumption in its system. The CWC auditor answers all visible IDG questions for the UMAC input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 4. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria that are flagged as "limiting."

Step 2-3C. Compile the volume of UUAC. The most common occurrences of UUAC include

- firefighting and training;
- flushing water mains, storm inlets, culverts, and sewers;
- fire flow tests;
- street cleaning;
- landscaping/irrigation in public areas, landscaped highway medians, and similar areas;
- decorative water facilities;
- swimming pools;
- construction sites: water for mixing concrete, dust control, or trench setting; and
- water consumption at public buildings (plants, municipal buildings, etc.) where metered data are unavailable due to the absence of a meter or collection of consumption data.

Some of the items listed above are included as UUAC due to challenges associated with metering the activity using portable hydrant meters, equipment meters, and metered bulk water fill stations. Sometimes activities that can be metered are simply unmetered due to an administrative decision not to meter the activity (e.g., filling municipal equipment such as street sweepers and other equipment at an unmetered hydrant).

Activities such as fire flow tests conducted by the water utility are often measured using portable instruments (e.g., pitot gage) that provide discharge and pressure data but do not explicitly meter the volume used. In such cases, the flow should be averaged over the time period that the fire hydrant was open. Volumes of water from such tests should be totaled for the entire water audit period.

Water consumed in water supply operations, such as water quality testing, filling tanks and reservoirs, and loading water mains, may also be classified as UUAC. If extracted from the distribution system after the finished water meter(s), process water at treatment plants should be metered and quantified because water treatment plants are permanent structures. If unmetered, then an estimate of this volume must be included in this category of the water audit period based upon available operations data.

The UUAC volume is typically a relatively small portion of the total Authorized Consumption, but it is an important component of consumption and should be quantified using portable field instruments to measure flow or estimated using tank truck volumes or similar means. If data are unavailable, the default value in the Audit Software may be selected (see the "History of the Unbilled Unmetered Authorized Consumption (UUAC)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Default Values" sidebar). If the default is used, the auditor is encouraged to develop a list of the activities currently assumed to be included under UUAC. This can form the foundation for future data collection efforts. If validated data are available, then this data should be used to derive a system-specific UUAC volume which should be input in the Audit Software instead of selecting the default value.

The auditor for CWC developed a comprehensive tracking system for the various types of UUAC volumes which are shown in Table 2-10. The individual volumes were summed up, the entry option "custom" for UUAC was selected on the Worksheet, and then the volume of 183.82 mil gal was entered.

History of the UUAC Default Values

Versions 4.2 (2010) and 5.0 (2014) of the AWWA's Free Water Audit Software (Audit Software) included a default value for UUAC equal to 1.25 percent of a system's Water Supplied volume. This approach was based upon data collected from 2002 to 2003 from 23 privately managed water companies in the United Kingdom. These data represented the best available information at the time because reliable North American data were not yet available.

Version 6.0 (2020) of the Audit Software includes an update to the default option for UUAC, equal to 0.25 percent of a system's Billed Authorized Consumption (BMAC + BUAC). Collecting and reviewing many years of validated water audit data in North America made it evident that the data from the United Kingdom did not accurately represent North American systems. This is largely due to differences in system management, regulatory requirements, and implementation of conservation practices. In general, the data indicate that North American systems have less volume associated with unbilled unmetered activities than the volumes indicated by the data from the United Kingdom. Deriving this default as a percentage of Billed Authorized Consumption rather than a percentage of Water Supplied also avoids inappropriately exaggerating the volume for high leakage systems.

Additional guidance for typical methods to estimate components of UUAC is available at www.awwa.org/waterlosscontrol.

The following are several important additional insights regarding UUAC.

Table 2-10 Sum of individual estimates of Unbilled Unmetered Authorized Consumption

Item Description	Volume (mil gal)
Firefighting and training	9.70
Flushing water mains, storm inlets, culverts, and sewers	2.55
Fire flow tests	0.00
Street cleaning	1.75
Landscaping irrigation in large public areas	162.89
Decorative water facilities	1.75
Swimming pools	0.42
Construction sites	0.56
Water quality and other testing	1.20
Water consumption at public buildings not included in the customer billing system	2.15
Other	0.85
Total Unbilled Unmetered Authorized Consumption (UUAC)	183.82

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- Careful policy considerations should be employed regarding water withdrawn from fire hydrants (see "Fire Hydrant Usage Policy: Does the Utility Have Control of Its Fire Hydrants?" sidebar).
- The utility should strive to determine how UUAC instances can eventually become metered and billed accounts. Over time, water utility managers should attempt to establish permanent metering at unmetered sites, particularly if they are permanent structures or facilities, such as municipal buildings, public parks, or irrigation within street rights-of-way.
- While unbilled activities do not generate revenue for the water utility, they are an authorized activity; therefore, the volume should be tracked over the audit period. If a meter is used, meter readings should be reliably collected. If the activity is not metered, then data should be collected, and the estimating procedures described within this section should be utilized to quantify the volume of Unbilled Unmetered Consumption. In this way, water consumption will be tracked to ensure that it is not wasteful.
- There should be consideration for how water efficiency improvements (the need for which often becomes evident once meters are installed) could be implemented to ensure that no more water is going toward these uses than needed.

Fire Hydrant Usage Policy: Does the Utility Have Control of Its Fire Hydrants?

The primary purposes of fire hydrants are firefighting and water distribution system testing and maintenance, including flushing water mains. In many water utilities, however, the use of fire hydrants—for both authorized and unauthorized purposes—goes far beyond these basic functions. UC from fire hydrants, which is classified under Apparent Losses, occurs when hydrants are illegally used to fill tank trucks for landscaping or construction purposes, to wash cars, or for recreational purposes such as for personal cooling in hot weather. Many water utilities have policies that permit water to be drawn from fire hydrants for a variety of community-spirited purposes and therefore are considered authorized consumption. This water is frequently unmetered, and the user is not billed, so it falls under UUAC in the water audit. This may include, but is not limited to, a variety of activities such as street cleaning, filling public swimming pools, providing transient supplies (such as nonpotable supply to a traveling circus), watering community gardens, and for use in construction sites. Some communities allow hot weather cooling relief from fire hydrants using spray caps. These varied uses of fire hydrants pose potential problems for water utilities and customers for the following reasons.

- Water taken from fire hydrants is often unmetered. The more hydrants that are opened, the greater the amount of water that must be estimated in the water audit.
- Water taken continuously from fire hydrants should include backflow protection to prevent contaminants from entering the distribution system during a negative pressure event. Often no backflow protection is used.
- Water drawn from a fire hydrant could pose a health risk if used for human consumption because water quality degradation can occur as the water passes through the barrel of the hydrant.
- Using the spray of a fire hydrant to cool off is a significant safety risk because fire hydrants are usually configured to face the street. The public (often children) can be pushed by water under high pressure into the roadway in the way of vehicular traffic.
- Widespread unauthorized openings of fire hydrants can result in greatly reduced pressure in the distribution system, crippling firefighting capability and greatly increasing the risk of backflow contamination.
- Allowing multiple uses of fire hydrants sends a poor public relations message that water is free for the taking to those who can manage to open a hydrant. This is a precarious position particularly because of the need to secure drinking water systems.

For these reasons, it is recommended that water utilities keep the number of permitted uses of fire hydrants to a minimum. Utility managers should maintain strong control of fire hydrants

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

and resist requests for sundry uses of hydrants. It is important that utility managers establish a sound policy for fire hydrant usage that is supported by fire departments and elected officials. Procedures for permitting and tracking allowable uses should be put in place and enforced. Many water utilities are establishing bulk water sales stations to supply tank trucks rather than allowing the use of fire hydrants. This is one step of a good policy on fire hydrant use. Water utility managers should work to educate public officials, contractors, customers, the media, and other stakeholders on the need to maintain strict utility control over fire hydrants.

Step 2-3D. Answer IDG criteria questions to determine the data grade for UUAC. Key factors affecting the data grading for this input include record keeping and estimation methods. If the auditor employs the use of the default value, a data grade of 3 is automatically assigned by the Audit Software. The CWC auditor has chosen to input a custom value based on the tracking and estimations reflected in Table 2-9. The CWC auditor answers all visible IDG questions for the UUAC input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 8. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 2-4. Quantify the Authorized Consumption volume for the audit year. The Authorized Consumption volume is calculated by the Audit Software as the sum of the four subcomponents of BMAC, BUAC, UMAC, and UUAC.

The worksheet in Figure 2-5 calculates the volume of Authorized Consumption to be 3,858.20 + 0.0 + 15.42 + 183.82 = 4,057.44 mil gal.

Task 3—Calculate Water Losses and Quantify Apparent and Real Losses

Water losses are made up of apparent and real losses and are calculated for the audit year as described in Step 3-1. Apparent Losses are quantified as described in Step 3-2. The Real Losses quantity is calculated as the remaining loss volume by the Audit Software and is described in Step 3-3. A rigorous means of directly quantifying real losses is described in the "Leakage Component Analysis" section in Chapter 5.

Step 3-1. Quantify total water losses. Water Losses are autocalculated and shown in Figure 2-5 as the Water Supplied volume minus the Authorized Consumption volume.

The worksheet in Figure 2-5 calculates the volume of water losses as Water Losses = Water Supplied – Authorized Consumption. For CWC, Water Losses = 4,401.87 – 4,057.44 = 344.43 mil gal.

Step 3-2. Quantify apparent losses. Apparent losses are the nonphysical losses that occur when water is successfully delivered to the customer but is not measured or recorded accurately. Apparent losses have several undesirable effects on utility operations. They distort customer consumption data and cost water utilities revenue when accounts are underbilled. Conversely, reductions in apparent losses tend to increase water system revenue from a given water rate structure and improve equity among customers. Apparent losses are comprised of SDHE, CMI, and UC.

The top-down approach relies on the auditor to devise estimates or measures of apparent losses to include in the water audit. Methods to quantify apparent losses are given in the following steps.

Step 3-2A. Quantify SDHE. For water utilities that meter customer consumption, integrity must exist not just with the accuracy of the water meter but also with the process to read, transmit, archive, and report customer consumption totals as derived from the meter population. An error at any point in this process potentially represents an apparent loss by distorting the ultimate documented value of customer consumption, causing a portion of the consumption to be understated and possibly missing a portion of revenue.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Systematic data-handling errors can therefore occur anywhere from the time that the meter reading is registered to the final reporting and use of the consumption data. SDHE can occur in three broad manners: systematic data transfer errors, systematic data analysis errors, and policy and procedures shortcomings. The means to quantify losses in these classes of error are discussed in the next steps. SDHE in the context of the entire customer population is reasonably expected to be a loss to the water utility, though a gain to the water utility is possible in a fringe scenario. The reader is directed to Chapters 3 and 4 for additional information on systematic data-handling errors.

Step 3-2A1. Systematic data transfer errors in customer meter reading. Considerable error can occur in the customer meter-reading process. Meters are typically read in two manners: manual (visual) meter reading or automatic meter reading via an AMR or AMI system. Manual meter reading, with meter-reading personnel visiting individual meters to collect readings, is a traditional approach still used by many water utilities in North America. In many systems, however, AMR or AMI systems are supplanting manual reading.

AMR is usually more accurate, less labor intensive, safer, and often (but not always) more cost-effective than manual meter reading. AMR/AMI systems have a strong history in the electric and gas utility industries, with implementation in the water industry growing rapidly since the late 1990s. Many successful case studies in water utility AMR/AMI systems have been documented, an example of which is given in the "Benefits of AMR/AMI Systems" sidebar. AMR/AMI systems have greatly reduced the accessibility and safety problems that have plagued manual meter reading programs. These systems transmit the current meter reading to a device or communications network outside of the building or meter pit in which the meter is located. With mobile AMR systems, readings can be collected by meter readers with handheld devices or, more economically, via vehicles patrolling scheduled meter-reading routes in which multiple readings are gathered almost simultaneously. Fixed communication networks with AMI systems have emerged as the more comprehensive and effective means of data collection. Fixed AMI networks typically include permanently installed data collector units or another communication mode such as cellular covering the service area. While the traditional AMR systems gather single meter readings every 30, 60, or 90 days, AMI systems generate detailed customer consumption profiles by obtaining readings as frequently as every 15 minutes. By collecting more granular data in this manner, AMI systems can utilize capabilities to reduce and more quickly resolve customer billing complaints, identify plumbing leaks, and assist water conservation and loss control efforts. Thus, AMI systems also bring about significant customer relations benefits. In addition to the above capabilities, AMI includes functions such detection of tampering, leaks, and reverse flow, and more endpoint capabilities are likely to be developed as these systems see wider use in the industry.

While AMR/AMI is less susceptible to data transfer error than manual meter reading, all forms of meter reading can incur errors. Meter-reading attempts can fail for many reasons. Manual meter reading has encountered a growing number of pitfalls, particularly in gaining access to meters located inside customer premises, the typical location of water meters in colder climates of North America. Many properties may have no one at home during business hours to let a meter reader into the house. Indoor water meters are often located in hard-to-reach corners of basements, boiler rooms, or other subterranean areas. Often, owners store items that block access to the meters. Outdoor meters in meter pits can have access difficulties, such as flooding and snow cover in colder climates. Meter readers entering private properties often encounter safety risks from aggressive dogs, dark or poorly maintained spaces, or hostile customers. For these reasons, manual meter-reading success rates have declined in recent years for many water utilities. AMR/AMI meter reading can fail due to a malfunction of the reading device from causes such as battery failure. Billing system analysts should evaluate billing data to detect accounts with successive cycles of "zero consumption" to identify potential AMR failure or possible

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

tampering of metering or meter reading equipment. Modern AMR/AMI systems typically feature self-diagnostic alerts to the utility when meter-reading devices fail or communication links are interrupted.

Benefits of AMR and AMI Systems

Prior to the start of AMR installation in 1997, Philadelphia's Water Department and Water Revenue Bureau had such poor meter reading that only one of every seven water bills issued was based on an actual meter reading; six were based on estimates. With the installation of more than 425,000 residential AMR units by 2000, the city witnessed a meter-reading success rate of over 98 percent in its monthly billing process. A system of mostly estimates was replaced with a system of mostly actual meter readings. This greatly improved the confidence of customer consumption data, lessened the number of customer billing complaints, and aided the detection of systematic data-handling errors and UC in Philadelphia. In 2019, the city entered a contract to upgrade its AMR system to AMI to continue its progress in efficiently obtaining customer meter- reading data along with new capabilities to efficiently manage the water supply and improve customer service.

Since the late 1990s, many North American water utilities have installed AMR or AMI systems. DC Water, providing service to the District of Columbia, was one of the first large-scale AMI systems and was installed in 2003. In addition to improving customer meter reading and billing operations, this AMI system has helped bring about a much improved environment keyed on customer engagement and service.

Many large- and medium-sized water utilities have moved from manual reading to AMR, given that this approach is usually more cost-effective for systems that must collect data from 10,000 or more customer meters on a regular basis. For small North American water utilities that read only several hundred or several thousand meters once every 60 or 90 days, a productive business case for AMR/AMI systems may not exist based solely on meter-reading success. AMI systems, however, offer a host of other benefits, including improved customer relations, leak detection, and more efficient water supply operations. Hence, water utilities of all sizes can investigate improvements in their operations by employing an AMR/AMI system.

When a meter-reading attempt is unsuccessful in obtaining actual meter data, most water utilities bill customers based on an estimated volume that reflects the customer's consumption based on their recent history. While this is a reasonable approach, multiple cycles of meter readings without an actual reading greatly increase the prospect of inaccurate estimates. Over time, buildings are sold, and new owners with different water consumption habits may become the permanent occupants. An estimate generated for a household of two persons may be fine until the house is sold to a family of seven. Water consumption could triple, but understated billings based on the outdated estimate could continue for some time. When an actual meter reading is eventually obtained, a large billing adjustment will confront the new property owner, a scenario that commonly creates customer ill will toward the water utility. Clearly, obtaining routine, accurate meter readings is key in maintaining sound oversight of customer consumption patterns and maintaining stable billing and good customer relations.

The auditor should review records to gain a general sense of the meter-reading success rate for both residential and industrial/commercial categories of accounts. The number of estimates assigned should also be tracked, and an approximation of the error due to poor estimation should be attempted. Accounts that register zero consumption for several successive meter-reading cycles should be sampled and investigated to determine if the zero consumption is valid (which should only occur in unoccupied or underutilized buildings) or whether AMR/AMI failure or tampering has occurred. Building alerts into the billing system for zero consumption events is a recommended practice and a standard feature in many billing applications. Other sources of systematic data transfer errors can exist in any given water utility. Depending on the time and resources available to the auditor, investigations can be conducted to assess any errors that are unique to the utility.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The auditor should attempt to quantify the major components of apparent loss due to data transfer error and include them in the water audit.

The quantity attributed to data transfer errors for CWC is related to several meter data collection functions, including meter-reading error, estimating error, and computer programming error. The auditor estimates the total of errors identified in these areas to be 4.21 mil gal. This quantity must be added to the systematic data analysis errors quantity and the policy and procedures shortcomings quantity to get a total volume for SDHE before inputting the total volume into the Audit Software Worksheet.

Step 3-2A2. Systematic data analysis errors. Typically, meter readings are transferred to customer billing systems where they are used to calculate the volume of customer consumption that has occurred since the previous reading. In the United States, consumption is most often recorded in units of cubic feet or thousand gallons. Billing systems often include programming algorithms that assign estimates of consumption if an actual meter reading cannot be obtained. These algorithms often base the estimate on the recent trend of customer consumption. If a poor or outdated estimation algorithm exists in the customer billing system, underestimation or overestimation of customer consumption can occur, either of which could distort consumption data needed for operational purposes. The auditor should understand the method used to estimate consumption and consider programming refinements if it is determined that the existing method creates inaccuracies. A quantity representing the amount of missed or overstated customer consumption should be included in the water audit.

A significant error can also occur by billing adjustments that distort consumption data registered by water meters and collected in the customer billing system. An important question is this: Are billing adjustments triggered by modifying actual consumption volumes? As described in the "Using the Customer Billing System" sidebar and discussed in the "Hypothetical Example" section that follows, billing systems designed with the intention of good revenue collection may corrupt the operational integrity of customer consumption volumes when generating a credit.

Distortions in customer consumption because of billing adjustments can occur when billing systems do not distinguish between registered consumption (from meter readings) and billed consumption listed on the customer bill and archived in the billing records. Billed consumption can differ from registered consumption when the customer is due a monetary credit. If the billing system creates the credit (negative revenue to the utility) by creating negative consumption values, actual consumption data becomes distorted. Billing systems that include separate fields for registered and billed consumption avoid this problem. Financial managers can track data stored in the billed consumption field, while operators and engineers can separately track the registered consumption.

In determining the amount of data analysis error occurring in billing system operations, the auditor should determine how billing adjustments are calculated. If adjustments are triggered by changes in consumption, then an approximation of the number of adjustments—both overstating and understating actual consumption—should be attempted.

The quantity attributed to systematic data analysis errors for CWC is estimated to be 5.11 mil gal. This quantity must be combined with the systematic data transfer errors quantity and the policy and procedures shortcomings quantity to get a total volume for SDHE before inputting the total volume into the Audit Software Worksheet.

Using the Customer Billing System to Extract Customer Water Consumption

The customer billing system is a standard feature of most drinking water utilities. Revenue is generated via billing to customers, typically on a monthly, bimonthly, or quarterly basis. For utilities that meter their customers, the billing system stores customer account data as well as

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

routine customer meter readings from which consumption volumes are calculated. These systems historically have been designed with a primarily financial purpose—to generate bills that result in revenue collection.

It has become evident in recent years that the value of customer consumption data goes beyond serving as the basis for billing. Customer consumption data are also relied on for a variety of engineering purposes. Consumption data are needed for demand forecasting which determines capital investments needed for new supplies and to evaluate water conservation practices. Data are required to realistically size customer meters and service connection piping on an individual basis and to size water supply infrastructure on a community basis. Consumption data are necessary to develop accurate hydraulic models and to assist water loss control programs by separating components of authorized consumption from components of loss. Unfortunately, many billing systems were designed with only the financial function in mind, and water utilities that now also use billing system data for engineering purposes may be doing so without knowing whether adequate controls exist to ensure the engineering integrity of customer consumption data.

It is important that water utility managers understand the workings of the customer billing system regarding consumption data integrity. Many billing systems, although configured with sound billing intentions, may unknowingly corrupt the engineering integrity of water consumption data. Some systems, when generating a credit to the customer, backcalculate the adjustment by changing the actual meter readings or consumption. A monetary credit to the customer is thereby triggered by reducing, eliminating, or creating negative consumption values for the period in question. Frequent adjustments in this manner can greatly distort the true amount of consumption for individual customers or whole communities. Other programming features in customer billing systems might unintentionally corrupt consumption data in an engineering sense, even though they were created with good financial intention.

It is recommended that sufficient controls be designed into the customer billing system if the system is to be used for both billing (financial) and operational (engineering) purposes. This will protect customer consumption data integrity while providing proper billing functions. The primary function of most existing customer billing systems is to accurately account for the revenue received by the utility for services rendered to individual customers. Utility operators embarking on conservation, hydraulic modeling, or water loss control programs should undertake a careful review of the billing system function and configuration to ascertain that the actual consumption volumes are not unintentionally modified by billing operations and that the customer consumption amounts recorded as output of the billing system are unchanged from the data generated by customer water meters. The utility should undertake a flowcharting or similar investigative exercises of the billing process to identify any impacts to customer consumption integrity and any apparent loss components from the data-handling process. If consumption data are found to be unduly modified by billing operations, the utility manager should consider reprogramming the billing system to record both the registered consumption and billed consumption as separate fields, thus ensuring that the accuracy of billing functions and customer consumption data are preserved. Until this process is implemented, an estimate of the impact of such adjustment activity should be included as a component of the apparent losses.

Hypothetical example. Table 2-11 gives an example of a hypothetical residential customer account that incurred estimates for a 23-month period, during which time the property was temporarily vacant and then sold to a new owner who consumed less water than his predecessors. The auditor reviewing billing data to compile the 2020 water audit discovered a problematic billing adjustment that affected the 2020 consumption data. Beginning in October 2018, the water utility was unable to obtain a reliable meter reading at this property. This may have been caused by blocked access to the meter or a failure of AMR equipment. Unfortunately, the water utility was unable to correct this condition and obtain an accurate meter reading until August 2020. During the period without meter readings, the water utility assigned an estimate of the consumption based on the customer's recent history—in this case, 885 cubic feet per month (ft^3/month). This estimate, shown in Column D, closely matched the actual consumption (shown in Column G for illustrative purposes) until April 2019, when the property was vacated and placed for sale.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-11 Distorted customer consumption data due to customer billing adjustments triggered by using negative consumption values (example data for a ⅝-in residential meter account)

A Year	B Month	C Meter Reading (estimates shown in gray)	D Billed Consumption (current minus previous meter reading; estimated consumption shown in gray) (ft^3)	E Cumulative Billed Water Consumption (per year) (ft^3)	F Actual Meter Reading	G Actual Consumption (ft^3)	H Cumulative Actual Consumption (ft^3)
2017	Dec	15004			15004		
2018	Jan	15838	834	834	15838	834	834
	Feb	16654	816	1,650	16654	816	1,650
	Mar	17496	842	2,492	17496	842	2,492
	Apr	18304	808	3,300	18304	808	3,300
	May	19220	916	4,216	19220	916	4,216
	Jun	20162	942	5,158	20162	942	5,158
	Jul	21130	968	6,126	21130	968	6,126
	Aug	22105	975	7,101	22105	975	7,101
	Sep	23007	902	8,003	23007	902	8,003
	Oct	23892	885	8,888	23867	860	8,863
	Nov	24777	885	9,773	24722	855	9,718
	Dec	25662	885	10,658	25535	813	10,531
2019	Jan	26547	885	885	26360	825	825
	Feb	27432	885	1,770	27184	824	1,649
	Mar	28317	885	2,655	28021	837	2,486
	Apr	29202	885	3,540	28433	412	2,898
	May	30087	885	4,425	28513	80	2,978
	Jun	30972	885	5,310	28578	65	3,043
	Jul	31857	885	6,195	28633	55	3,098
	Aug	32742	885	7,080	29255	622	3,720
	Sep	33627	885	7,965	30059	804	4,524
	Oct	34512	885	8,850	30836	777	5,301
	Nov	35397	885	9,735	31592	756	6,057
	Dec	36282	885	10,620	32315	723	6,780
2020	Jan	37167	885	885	33032	717	717
	Feb	38052	885	1,770	33740	708	1,425
	Mar	38937	885	2,655	34462	722	2,147
	Apr	39822	885	3,540	35150	688	2,835
	May	40707	885	4,425	35884	734	3,569
	Jun	41592	885	5,310	36686	802	4,371
	Jul	42477	885	6,195	37520	834	5,205
	Aug	38345	–4,132	2,063	38345	825	6,030

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-11 Distorted customer consumption data due to customer billing adjustments triggered by using negative consumption values (example data for a ⅝-in residential meter account) *(Continued)*

A Year	B Month	C Meter Reading (estimates shown in gray)	D Billed Consumption (current minus previous meter reading; estimated consumption shown in gray) (ft^3)	E Cumulative Billed Water Consumption (per year) (ft^3)	F Actual Meter Reading	G Actual Consumption (ft^3)	H Cumulative Actual Consumption (ft^3)
	Sep	39113	768	2,831	39113	768	6,798
	Oct	39811	698	3,529	39811	698	7,496
	Nov	40515	704	4,233	40515	704	8,200
	Dec	41230	715	4,948	41230	715	8,915
2021	Jan	41951	721	721	41951	721	721

The property was vacant until August 2019 and experienced minimal water consumption during periodic caretaker visits from April to August 2019. Upon sale to a new owner in August 2019, a regular pattern of water consumption resumed but at a slightly lower rate than that of the previous owner.

Between April 2019 and August 2020 (17 months), the assigned estimate (885 ft^3) notably overestimated the consumption for this account. When the water utility was once again able to gain an accurate meter reading, it found that its estimate of the July 2020 meter reading (42477) was overstated by a total of 4,132 ft^3 since the last accurate meter reading in September 2018. This resulting cumulative overestimation error was compounded by the

- lengthy duration (23 months) of the period with no meter readings,
- four-month period of vacancy of the property, and
- lower water-consuming habits of the new property owner.

When an accurate meter reading was obtained in August 2020, an adjustment of *negative* 4,132 ft^3 was necessary, and a credit due to the customer in the dollar amount commensurate with the volume of adjusted consumption.

How the customer billing system awards this credit has bearing on both the billing (financial) and registered (engineering) functions of the system. While money can flow to and from the drinking water utility via charges and credits, respectively, water flows in only one direction, supplied by the utility to the customer. If the billing system contains only a single field for customer consumption, the billed consumption value for August 2020 is *negative* 4,132 ft^3.

While a negative consumption number is acceptable for use for billing (financial) reasons because it translates into a monetary credit, a negative consumption number is unacceptable for operational (engineering) purposes because the actual consumption for the month of August 2020 was 825 ft^3 (Column G), not *negative* 4,132 ft^3 as shown in Column D.

The distortion of the consumption data is further reflected in the estimated vs. actual consumption based on yearly periods. Water utility analysts reviewing the account data shown in Table 2-11 for conservation or loss control purposes would be in error by 3,840 ft^3 (10,620 – 6,780) over the actual consumption in 2019. Conversely, the analysis would be understated for this account by 3,967 ft^3 (8,915 – 4,948) in 2020. Some may reason that the periods of estimation and adjustment ultimately balance with no net difference over the long term; therefore, using a single consumption value is acceptable. However, many analytical and reporting functions are performed over the course of a calendar or

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

business year. If a given account has been poorly estimated for many years, the use of a huge multiyear adjustment in the last year will greatly distort the consumption for that final year. Additionally, in any given drinking water utility, many hundreds or thousands of accounts could utilize estimates for varying periods of time. Reliably estimating the net impact of the aggregate overestimation or underestimation of these accounts in a given year is unnecessarily complex. Clearly, while a negative consumption value can be acceptable for billing (financial) purposes, it is quite harmful to the integrity of the data for operational (engineering) purposes.

For the reasons previously explained, it is recommended that water utility customer billing systems include two fields for customer consumption: one for registered consumption and a separate field for billed consumption. Using the same data from the example in Table 2-11, the form of the data with separate fields is shown in Table 2-12.

Table 2-12 includes separate columns for billed consumption (Column D) and registered consumption (Column G). When actual meter readings resumed in August 2020, the consumption adjustment of –4,132 ft^3 appears as billed consumption in Column D and is used to generate the monetary credit to the customer. However, Column G reflects the revised estimate of consumption for the prior 30-day period, which is based on the difference between the two most recent actual meter readings (September 2018 and August 2020). This one-time estimate is determined as

$$\left(38{,}345\ ft^3 - 23{,}007\ ft^3\right) / 23\ \text{months} = 667\ ft^3 \qquad \text{(Eq 2-4)}$$

By September 2020, when the second consecutive actual monthly meter reading was obtained, estimates were no longer utilized, and billed consumption once again matched registered consumption. The benefit to the operational integrity of data using separate billed and registered consumption fields is shown by comparing the cumulative consumption for 2020 in Column E and Column H, or 4,948 ft^3 and 9,747 ft^3, respectively

Table 2-12 Utilizing separate fields for registered and billed consumption in the customer billing system (example data for a ⅝-in residential water meter account; see Table 2-11)

A Year	B Month	C Meter Reading (estimates shown in gray)	D *Billed Consumption* (current minus previous meter reading, estimated consumption shown in gray) (ft^3)	E Cumulative Billed Water Consumption (per year) (ft^3)	F Actual Meter Reading	G *Registered (actual) Consumption* (ft^3)	H Cumulative Registered (actual) Consumption (ft^3)
2017	Dec	15004			15004		
2018	Jan	15838	834	834	15838	834	834
	Feb	16654	816	1,650	16654	816	1,650
	Mar	17496	842	2,492	17496	842	2,492
	Apr	18304	808	3,300	18304	808	3,300
	May	19220	916	4,216	19220	916	4,216
	Jun	20162	942	5,158	20162	942	5,518
	Jul	21130	968	6,126	21130	968	6,126
	Aug	22105	975	7,101	22105	975	7,101
	Sep	23007	902	8,003	23007	902	8,003

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-12 Utilizing separate fields for registered and billed consumption in the customer billing system (example data for a ⅝-in residential water meter account; see Table 2-11) *(Continued)*

A Year	B Month	C Meter Reading (estimates shown in gray)	D *Billed Consumption* (current minus previous meter reading, estimated consumption shown in gray) (ft^3)	E Cumulative Billed Water Consumption (per year) (ft^3)	F Actual Meter Reading	G *Registered (actual) Consumption* (ft^3)	H Cumulative Registered (actual) Consumption (ft^3)
	Oct	23892	885	8,888		885	8,888
	Nov	24777	885	9,773		885	9,773
	Dec	25662	885	10,658		885	10,658
2019	Jan	26547	885	885		885	885
	Feb	27432	885	1,770		885	1,770
	Mar	28317	885	2,655		885	2,655
	Apr	29202	885	3,540		885	3,540
	May	30087	885	4,425		885	4,425
	Jun	30972	885	5,310		885	5,310
	Jul	31857	885	6,195		885	6,195
	Aug	32742	885	7,080		885	7,080
	Sep	33627	885	7,965		885	7,965
	Oct	34512	885	8,850		885	8,850
	Nov	35397	885	9,735		885	9,735
	Dec	36282	885	10,620		885	10,620
2020	Jan	37167	885	885		885	885
	Feb	38052	885	1,770		885	1,770
	Mar	38937	885	2,655		885	2,655
	Apr	39822	885	3,540		885	3,540
	May	40707	885	4,425		885	4,425
	Jun	41592	885	5,310		885	5,310
	Jul	42477	885	6,195		885	6,195
	Aug	38345	–4,132	2,063	38345	667	6,862
	Sep	39113	768	2,831	39113	768	7,630
	Oct	39811	698	3,529	39811	698	8,328
	Nov	40515	704	4,233	40515	704	9,032
	Dec	41230	715	4,948	41230	715	9,747
2021	Jan	41951	721	721	41951	721	721

Unknown, No Readings

(Table 2-12). If only a single field is used for consumption, the billed value of 4,948 ft^3 greatly understates the actual consumption for the year. The registered consumption value of 9,747 ft^3 is a much more representative value of the water consumed by this account during the water audit year of 2020.

Step 3-2A3. Policy and procedure shortcomings. Apparent losses can occur because of policies and procedures that are shortsighted, outdated, or poorly designed, implemented, or managed. Such occurrences can be subtle and numerous. Chapter 4 illustrates how flowcharting the customer billing process with a focus on impacts to customer consumption values gives insight to the likelihood of these types of apparent losses. Some of the common occurrences to consider are

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- ignoring the installation of meters in certain customer classes despite utility policy to meter all customers;
- municipally owned buildings in water utilities run by local governments;
- provisions allowing customer accounts to enter unbilled status, a potential loophole often exploited by fraud or poor management;
- bureaucratic regulations or inefficiencies that cause delays in permitting, metering, or billing operations; and
- poor customer account management (accounts lost, uninitiated, or transferred erroneously).

The degree to which such shortcomings in billing account management exist is largely dependent on the accountability culture that exists in the water utility. If accountability is only casually emphasized, it is likely that numerous opportunities for missed consumption exist. If sound accountability is trumpeted by the utility's leaders and managed at all levels of staff, then such occurrences are likely to be isolated and/or of minor significance. The auditor should consider including an estimate of apparent loss that represents the collective policy and procedural shortcomings of the water utility. During the top-down audit, perhaps only a rough approximation can be attempted. During subsequent audits, bottom-up investigations can give greater insight into such problems, and corrections can be identified.

The quantity attributed to policy and procedure shortcomings for CWC is estimated to be 2.60 mil gal. This number is added to the quantity of systematic data transfer errors of 4.21 mil gal and systematic data analysis error of 5.11 mil gal to give a total SDHE quantity of 11.92 mil gal for the audit year. This quantity is entered into the Audit Software Worksheet in Figure 2-5 in the row for SDHE in the cell, with "custom" selected for entry option. Alternatively, if the auditor is not able to undertake a detailed quantification of this SDHE, the default value may be selected. If the default option is used, then the Audit Software calculates a SDHE quantity equal to 0.25 percent of Billed Authorized Consumption (BMAC + BUAC).

Step 3-2A4. Answer IDG criteria questions to determine the data grade for SDHE. Key factors affecting the data grading for this input include the extent of investigation into billing software multipliers, rigor of new service account policies to avoid billing lapses, extent of billing data auditing, and the method for input derivation. If the auditor employs the use of the default value, a data grade of 3 is automatically assigned by the Audit Software. The CWC auditor has chosen to input a custom value based on the tracking and estimations described in Steps 3-2A1 through 3-2A3. The CWC auditor answers all visible IDG questions for the SDHE input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 5. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 3-2B. Estimate CMI. Most drinking water utilities in North America provide meters on all or most of their customer service connection piping to measure customer consumption. This is good industry practice supported by AWWA. However, meters typically are not 100 percent accurate in all situations. Meters are subject to wear and loss of accuracy with continued use. Another common source of meter inaccuracy occurs when meters are oversized or undersized for the flow profile that they encounter. Many meter types fail to accurately measure very low flow rates which often represent leakage within the customer premises; therefore, meters frequently experiencing low flows will be less accurate than appropriately sized meters. Chapters 3 and 4 also provide guidance on meter management and the means to maintain a high degree of accuracy in the customer meter population. For water utilities with an unmetered customer population, there is no amount of apparent loss caused by CMI; therefore, this component does not apply.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Meter rightsizing. Historically, meter sizing calculations have been based on conservative techniques, basing the size of the meter on the peak flow it might encounter, even though peak flows occur rarely and for short durations. Meters sized in this way are usually larger than they need to be, resulting in substantial meter inaccuracy at low-flow rates. This practice has resulted in a significant number of oversized meters in many water utilities. Changing building uses, such as a factory converted to office space, can result in an oversized meter if the original meter that passed high flows remains in place after the low-flow office setting is established. Leakage within customer premises can generate low-flow rates that are commonly unrecorded or underrecorded, even by new meters. The degree of inaccuracy in the meter population at any point in time depends on the meter types that were specified, the amount of cumulative flow that meters have registered, whether the meters are appropriately sized and installed, the aggressiveness of the water in creating internal corrosion, and the degree of upkeep in the meter population by the water utility management. Taking these factors into account, the auditor can determine an estimate of the composite amount of water loss attributable to the inaccuracy of customer meters. Meter rightsizing programs can recoup much of that loss with significant gains in billed consumption. A flow recorder (data logger), like the one shown in Figure 2-10, can provide accurate flow-rate data to assist meter rightsizing decisions.

Because there are typically many hundreds or thousands of customer meters in any drinking water utility, it is not practical to inspect and test every meter each year. Instead, periodic inspections and testing should be considered for all high-revenue meters, along with a random sample of smaller meters. As a minimum, it is important to ensure that the meters serving the largest users are sized properly and tested on a regular basis.

Step 3-2B1. Check for proper meter installation. The utility's practices on meter selection, sizing, and installation should be reviewed to determine whether current practices result in a high accuracy customer meter population. If they do not, the practices should be revised as necessary so that meters will operate correctly. The reader is advised to refer to AWWA Manual M6, *Water Meters—Selection, Installation, Testing, and Maintenance*



Courtesy of F.S. Brainard and Co.

Figure 2-10 Customer meter flow recorder

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

(2012), and AWWA Manual M22, *Sizing Water Service Lines and Meters* (2023) for additional information.

Large industrial, commercial, and agricultural meters typically produce a much larger share of revenue per account than do residential meters. These accounts should be inspected for proper selection, sizing, and installation. In addition, best practice dictates that meters be inspected and tested before they are used. Not all new meters are sufficiently accurate. The results of a large-scale project for testing of residential meters can be found in the research report *Accuracy of In-Service Water Meters at Low and High Flow Rates* (WRF 2011).

Step 3-2B2. Test residential meters for accuracy. In most cases, a random sample of in-service residential meters should be tested on a regular basis; as few as 50 to 100 may be sufficient, but the optimal number to be tested depends on the size of the customer meter population, the degree of confidence required in the test results, and the variance in the actual test results observed. Residential meters may be tested on a utility-owned test bench or sent to the factory or a testing service contractor for testing. More information on meter accuracy testing is given in Chapter 4 of this manual and in the AWWA M6 publication.

Meter testing and replacement programs. Many utilities operate ongoing meter testing and replacement programs. Particularly for small meters, it has become more cost-effective to replace meters than to repair them. Random or targeted testing to determine the accuracy of installed customer meters can be conducted to monitor the wear of meters. A representative sample of newly purchased residential meters should also be tested to confirm the acceptability of the delivered meters. This test data represent a good source of information to infer the overall degree of inaccuracy existing in the customer meter population. Large meter replacement programs offer an excellent opportunity to ensure that older meters are replaced with new meters of the correct type and size. Flow recorders (data loggers) can assist with this selection process by recording the daily variation in flow and ensuring that low-flow regimens are identified and included in the meter-sizing determination. Traditional mechanical meters such as compound and turbine meters offer advantages for specific flow profiles. However, the potential inaccuracy of older meters and any flow data recorded from them should be considered if they have not been maintained properly. Electromagnetic and ultrasonic meters with no moving parts are growing in popularity within the drinking water industry and offer a variety of benefits that are discussed in Chapter 4.

Conducting meter accuracy testing. A water meter–testing apparatus (test bench) is precision instrumentation and must be operated and maintained properly to provide representative accuracy results for meters being tested. Proper testing procedures must also be followed and the results carefully documented. It is important that skilled, trained technicians are employed as the testing operators; this job should not be assigned to unskilled workers. If the testing program fails to follow the best practices noted here, unrepresentative data will be obtained, wasting the value of the investment in testing and potentially misguiding decisions on meter replacement goals. The investment for a water utility to purchase a meter-testing apparatus and assign trained technicians as operators can be a considerable one. If the water utility cannot provide sufficient financial and human resources to conduct meter testing properly, then it is better to utilize a qualified meter-testing service to conduct the accuracy tests. This is also an appropriate approach if the customer meter population is small and the number of meters to be tested each year is few.

The meter accuracy testing protocols included in manual M6 call for testing customer meters at three flow rates: minimum or low, intermediate or medium, and maximum or high. When testing meters taken from service, always test the minimum (low) flow rate

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

first to avoid disturbing any particulate matter in the meter that may be washed away if the meter is tested at the maximum (high) flow rate first. Accuracy measurements can be obtained for each flow rate. It is also important to know the relative portion of the water supply to meters flowing through customer meters that occurs in the minimum, intermediate, and maximum flow regimens. Tables 2-13 through 2-16 illustrate how accuracy test results are used to calculate a composite customer meter inaccuracy value for the meter population or subset of the population, such as residential meters. The calculated inaccuracy value is then input into the Audit Software.

The optimal determination of % volume weighting for each flow range should come from sample flow logging of residential accounts for a given utility. See the "Customer Consumption Data Collection" sidebar on Austin Water.

The "Customer Consumption Data Collection" sidebar provides an account of small residential meter accuracy testing and data-logging work conducted in Austin, Texas. The testing work allowed Austin Water to determine the composite accuracy for ⅝-in water meters. Data-logging work allowed the utility to determine the composite amount of flow occurring at the low-, medium-, and high-flow regimens, which were found to be 13.1 percent, 52.5 percent, and 34.4 percent, respectively. These composite flow profiles are representative of ⅝-in meter customers in the Austin service area and cannot be taken to be levels representative of other utilities. Ideally, water utilities can undertake their own assessment of these levels, as Austin Water did. Realistically, though, not all utilities have the resources and time to conduct this work. There is no "default" set of percentages for allocating test results to low-, medium-, and high-flow regimens. However, the "% Volume" values presented in Table 2-15 can represent a reasonable temporary starting point for a water utility to use until system-specific data can be obtained.

Step 3-2B3. Calculate the composite CMI. Water usage varies among user categories including single-family residential, multifamily residential, industrial, commercial, agricultural, and others. In general, single-family residential meters are often considered small meters. Most other uses—multifamily housing, commercial, industrial, agricultural—are often referred to as mid-sized and large meters. Because the water usage patterns of small, mid-sized, and large meter accounts are different and the types and function of small and large meters are different, it is worthwhile to quantify inaccuracies in each of these types of accounts and add them to obtain a combined, or composite, quantity of loss due to CMI.

Customer Consumption Data Collection and Meter Accuracy Testing in Austin, Texas

Knowing your meter accuracy is critical for calculating apparent losses, and it will vary for every utility. Factors such as meter age and total flows recorded, water quality, manufacturer and model of meters, and customer usage patterns are going to affect accuracy, particularly in the low- and high-flow ranges. Consequently, utilities should provide for testing to determine the accuracy of their customer meter population.

Soon after stepping up its water loss control efforts, Austin Water in Texas decided to improve the accuracy of its apparent loss calculations. First, the utility pulled a random sample of meters of up to size 2 in (meters 3 in and larger are tested regularly and brought to within AWWA standards). The meters were tested on a test bench, and each meter size's test results were averaged together, yielding accuracy rates of between 86 percent and 98 percent for minimum (low) flows, 98 percent and 99 percent on intermediate (medium) flows, and 98 percent and 100 percent on maximum (high) flows.

However, to combine those average accuracy rates into a single rate for each meter size, the utility had to know which percentages of customer water moved through the meters at each of the three flow rates. The utility installed data loggers on the accounts where the tested meters had been pulled and tracked flows for a week. The data logging yielded flow percentages of 8 percent to 28 percent for minimum flows, 42 percent to 52 percent for intermediate flows, and 24 percent to 41 percent for maximum flows. By multiplying the accuracy rates by the flow percentages for the

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

three flow rates and summing them up, Austin Water determined that its customer meters were between 97.32 percent and 98.90 percent accurate, depending on the size of the meter.

Sample meter accuracy for ⅝-in meters*

Flow Rates†	Tested Accuracy (%)	Portion of Total Flows (%)	Proportion of Flow Times Accuracy (%)
Minimum (low) (<0.7 gpm)	86.04	13.1	11.27
Intermediate (medium) (0.7–5.45 gpm)	99.47	52.5	52.22
Maximum (high) (>5.45 gpm)	99.43	34.4	34.20
Total accuracy:			**97.69%**

* Tested accuracy rates and portion of total flows were measured by Austin Water for its own specific meter population and operating conditions.

† Flow rates are for classifying flows as data-logged through customer meters. Tested flow rates were taken from AWWA Manual M6, *Water Meters—Selection, Installation, Testing, and Maintenance* (2012). For ⅝-in meters, the test rates used were 0.25 gpm for minimum (low) flows, 2 gpm for intermediate (medium) flows, and 15 gpm for maximum (high) flows, as per M6.

Calculate residential (small) meter error for CWC. Water meters should be tested at minimum, intermediate, and maximum flows. The results, shown as a percentage of accuracy, are used to calculate the total meter error at average flow rates. Tables 2-13 through 2-16 demonstrate how to use existing meter test data to calculate composite total residential meter errors. The data in these tables are based on the audit year customer consumption for CWC shown in Table 2-9. Table 2-13 lists the percentage of typical residential flows that occur in CWC at the low, medium, and high flow rates. This information was calculated by CWC after data-logging 100 residential customer accounts and performing analysis such as that conducted by Austin Water in the "Customer Consumption Data Collection" sidebar. Meter accuracy testing was also conducted by CWC throughout the audit year by selecting a mix of 50 randomly selected meters and high cumulative consumption meters. The results of this testing are shown in Table 2-14.

CWC found that the minimum flow accuracy of the tested residential meters was 97.9 percent. Even at this high level of accuracy, it is likely that very low flows occurring in some CWC properties, such as toilet leaks, are not registered at all. The accuracy at intermediate and maximum flow rates was better, at 99.8 percent and 98.3 percent, respectively. Table 2-15 shows that the composite or weighted accuracy of this test group of meters is 99.0 percent, a high level of accuracy.

Table 2-16 shows the calculation used to determine the volume of water occurring as an apparent loss due to residential CMI. For CWC, this is 30.21 mil gal.

Table 2-13 Example of weighting factors for flow rates related to volume percentages for ⅝-in water meters in County Water Company

Flow	Range (gpm)	% Volume*
Minimum (low)	<0.75	15
Intermediate (medium)	0.75–5.5	50
Maximum (high)	>5.5	35

* Percent volume refers to the proportion of water consumed at the specified flow rate, as compared to the total volume consumed at all rates. In this example, only 15 percent of the total water consumed occurs at the minimum (low) flow range of less than 0.75 gpm.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-14 Meter testing data from a random sample of 50 meters for County Water Company

Test Flow Rates (gpm)	Mean Registration (%)
Minimum (low) (0.25)	97.85
Intermediate (medium) (2.0)	99.80
Maximum (high) (15.0)	98.30

Table 2-15 Sample calculation of residential water-meter error

Flow Rates	Tested Accuracy (%)	% Volume	Weighted Accuracy (%)
Minimum (low)	97.85	15	14.7
Intermediate (medium)	99.80	50	49.9
Maximum (high)	98.30	35	34.4
Weighted meter accuracy			99.0

Table 2-16 Calculation of residential water meter error

Percent Volume* (%V)	Total Sales Volume† (Vt) (mil gal)	Volume at Flow Rate (Vf) (%V × Vt) (mil gal)	Meter Registration (R)‡ (%)	Meter Error (ME) ME= Vf/ (0.01R) – Vf (mil gal)
15	2,918.80	437.82	97.85	[(437.82/0.9785) − 437.82] = **9.62**
50	2,918.80	1,459.40	99.80	[(1,459.40/0.998) − 1,459.40] = **2.92**
35	2,918.80	1,021.58	98.30	[(1,021.58/0.983) − 1,021.58] = **17.67**
Composite residential meter inaccuracy volume, mil gal				**30.21**

* From Table 2-13.
† Based on residential water sales data in Table 2-9.
‡ From Table 2-14.

Calculate industrial/commercial (large) meter error. Tables 2-17 through 2-19 show how meter test data are used to calculate the composite meter inaccuracy for the large customer meter population. The mean registration data in Table 2-17 are used to calculate the meter inaccuracy for large meters. Table 2-17 lists the percentage of flow occurring in the minimum, intermediate, and maximum ranges, derived by conducting flow recordings (data logging) on 25 large meters during the audit year. Table 2-18 shows the individual meter accuracy test results for five large meters tested by CWC during the audit year and the mean registration for these meters. Table 2-19 shows the calculations used to quantify the composite large meter inaccuracy volume for CWC. The accuracy test data shown in Table 2-18 are used in the calculations shown in Table 2-19. As determined from Table 2-8, CWC has 706 large water meters of size 1-in and larger in its system. With this size population, the test sample of only five large meters may be too small to be representative to calculate the composite inaccuracy volume of the large meter population. CWC would likely obtain more representative results by testing 15 to 25 large meters during the next audit year.

One of the benefits of a water audit is the potential increase in revenue resulting from testing, replacing, or rightsizing large meters. While testing can be used to better quantify composite loss volumes for the water audit and to improve data validity, the utility can also directly act on meters found to be sorely inaccurate, thereby realizing a fast recoup of

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-17 Example volume percentages for large meters for County Water Company*

Flow Rates	% Volume Delivered
Minimum (low)	10
Intermediate (medium)	65
Maximum (high)	25

* CWC conducted flow recordings for 24-hour periods in July and February for 25 large meter accounts to indicate the percentage of volume delivered by large meters at low, medium, and high flow rates.

Table 2-18 Meter test data for large meters for County Water Company

Meter ID Number	Size (in.)	Meter Type	Date of Installation	Manufacturer	Test Date	Mean Registration at Various Flow Rates (designated as percentage of registration)		
						Min	Inter	Max
XYZ001	3	Turbine	June 1998	Sensus	Apr 2020	97.0	99.0	100.0
X00ZAA	3	Turbine	June 1999	Sensus	Apr 2020	96.0	98.2	99.5
NB123	4	Turbine	July 2008	Neptune	Apr 2020	95.0	99.0	102.0
NB456	6	Compound	Sept 2012	Badger	Oct 2020	98.0	94.6	102.0
AA002	6	Magnetic	May 2017	ABB	Oct 2020	98.0	99.0	104.0
					Sum of mean registrations	484.0	489.8	507.5
					Mean registration for five meters tested	96.8	98.0	101.5

Table 2-19 Calculation of large water meter error

Percent Volume* (%V)	Total Sales Volume† (Vt) (mil gal)	Volume at Flow Rate (Vf) (%V × Vt) (mil gal)	Meter Registration (R)‡ (%)	Meter Error (ME) = Vf/ (0.01R) – Vf (mil gal)
10	939.20	93.92	96.8	[(93.92/0.968) − 93.92] = **3.10**
65	939.20	610.48	98.0	[(610.48/0.98) − 610.48 = **12.46**
25	939.20	234.8	101.5	[(234.8/1.015) − 234.8] = **-3.47**
Composite large meter inaccuracy volume for large meters				**12.09**

* Data from Table 2-17.
† Data from Table 2-9 sum of industrial, commercial, and agricultural metered consumption.
‡ Data from Table 2-18.

missing revenue. The auditor can estimate the amount of revenue to be gained by improving the function of large meters by applying the appropriate cost factor.

The resulting commercial/industrial meter error from Table 2-19 for CWC is 12.09 mil gal. This number is added to the residential meter error of 30.21 mil gal, and a total volume of CMI of 42.3 mil gal is quantified and input into the Worksheet in Figure 2-5 with "volume" selected as the entry option. The "underregistration" designation is selected to the right. An auditor has the option to designate the aggregate CMI input as "overregistration," but this is not typically recommended and would be highly unusual to encounter since mechanical meters, which still make up the majority of meters in many utilities, underregister as they wear from cumulative flow.

Note that the audit software also gives the auditor the option to enter the composite CMI as a percentage (instead of a volume) for the customer meter population. This can be done

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

if the auditor has not been able to undertake calculations as shown in Table 2-16 and Table 2-19. In such a case, the audit software calculates a volume based on the percent inaccuracy provided by the auditor multiplied by the volume of Metered Authorized Consumption (BMAC + UMAC).

Step 3-2B4. Answer IDG criteria questions to determine the data grade for CMI. Key factors affecting the data grading for this input include the type and extent of customer meter testing and replacement, and the method for input derivation. CWC tests a small but generally representative number of customer meters each year and replaces meters found to be inaccurate or having exceeded an expected life. The CWC auditor answers all visible IDG questions for the CMI input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 8. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 3-2C. Estimate UC. UC includes water that is taken against the policies of the water utility and can include

- illegal connections,
- open bypasses,
- buried or otherwise obscured meters,
- misuse of fire hydrants (see the "Fire Hydrant Usage Policy" sidebar earlier in this chapter and the discussion in chapter 4 on fire hydrant policy and management),
- unauthorized connections to dedicated firefighting systems piping (unmetered fire lines),
- vandalized or bypassed consumption meters (meter tampering),
- tampering of meter reading equipment,
- unauthorized opening of intentionally closed valves or curb stops on customer service piping that has been discontinued or shut off for nonpayment, and
- unauthorized opening of intentionally closed valves to neighboring water distribution systems designed for emergency or special use.

Water utilities sometimes allow a spacer pipe to be installed in place of a water meter in new building construction with the intention to install a water meter later in the occupancy process. Unfortunately, water utilities sometimes neglect to install the meter, and although the customer may be aware that they are not being billed for water use, they continue to consume water without notifying the water utility. Policies that allow water service to be established in this manner without a meter are discouraged. However, if such a policy is required, a periodic audit should be conducted to verify that each property has a meter and that occupied buildings show positive water consumption.

The potential for UC exists in any drinking water utility but varies from system to system. For large- and medium-sized systems, the total annual volume of water lost to UC may be a small portion of the utility's water supply volume, but this can vary. For expediency during the top-down water audit, the auditor may choose to use the default value of 0.25 percent of the Billed Authorized Consumption (BMAC + BUAC). The default is available for use until the utility is ready and able to conduct sufficient investigation into UC to develop a system-specific estimate. In this case, the Audit Software calculates the volume of water lost to UC using the default option.

For CWC, the auditor determines that there are not sufficient resources available to fully investigate the occurrence of UC, although it is known that a certain amount of such consumption occurs. In this case, the auditor selects the default value in the Audit Software, which is calculated as (Billed Authorized Consumption) × (.0025) = (3,858.20 + 0.0) × (.0025) = 9.646 mil gal. The "default" is selected as the entry option.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

For small systems, the occurrence of UC may be a larger portion of the Water Supplied volume than that of large systems. If the auditor believes that UC is significant and has the time and resources to investigate, the auditor can conduct work to examine the UC occurrences and obtain quantities for these components. This work can be tedious, however, and auditors should use their own judgment to determine whether the extra effort to obtain specific estimates of UC is worthwhile compared to merely applying the default value.

Step 3-2C1. Answer IDG criteria questions to determine the data grade for UC. Key factors affecting the data grading for this input include the extent of tracking and oversight for UC and the method of input derivation. Since the CWC auditor employs the use of the default value, a data grade of 3 is automatically calculated by the Audit Software. Alternatively, if the auditor inputs a specific UC volume, then all visible UC questions in the IDG worksheet should be answered, and the Audit Software will determine the data grading that is assigned.

Step 3-2D. Quantify the Apparent Losses volume. The total apparent losses are determined automatically by adding the apparent loss components for SDHE, CMI, and UC.

The Reporting Worksheet in Figure 2-5 calculates the apparent loss volume for the audit year. For CWC, the apparent loss is calculated as SDHE (11.920) + CMI (42.300) + UC (9.646) = 63.866 mil gal.

Step 3-3. Quantify the Current Annual Real Losses volume. Water losses consist of the apparent losses plus the real losses occurring in the drinking water utility operations and management. The Audit Software mathematically autocalculates Current Annual Real Losses (CARL) simply as water losses (see Step 3-1) minus apparent losses.

The Worksheet in Figure 2-5 calculates the volume of CARL or "Real Losses" as Real Losses = Water Losses – Apparent Losses, or (344.430 – 63.866) mil gal = 280.565 mil gal.

While the straightforward approach of the Audit Software makes the Real Losses calculation easy to determine mathematically, care should be taken in the interpretation of the volume of real losses determined in this manner. In using the top-down calculation of the Audit Software, real (leakage) losses are a "catch-all" quantity, basically the amount of water left over after authorized consumption and apparent losses have been quantified. The quantification of the amount of leakage losses is therefore only approximate because

- the accumulated inaccuracies in the other components will directly affect the estimate of real losses;
- the catch-all nature of this estimate of leakage losses gives no indication of the breakdown of individual leakage components, particularly unreported leaks and background losses; and
- a water balance normally covers a completed (retrospective) 12-month period, so it has limited value as an early warning system for identifying new leaks.

The annual top-down water audit using the Audit Software is a vital step in understanding real losses. As a subsequent step, for the reasons noted above, real (leakage) losses should also be assessed in more detail by bottom-up activities, namely,

- leakage component analysis of real losses and
- quantification of leakage components via field measurements and minimum-hour flow analysis, as in DMAs.

These methods are discussed in detail in Chapters 5 and 6.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Task 4—Calculate NRW

NRW is the portion of the water that a utility places into the distribution system that is not billed and, therefore, cannot generate revenue for the utility. NRW consists of the sum of UMAC (Step 2-3A in this chapter) and UUAC (Step 2-3C), Apparent Losses, and Real Losses. NRW can also be calculated as the Water Supplied volume minus the sum of BMAC (Step 2-1C) and BUAC (Step 2-2).

Step 4-1. Quantify NRW. The Worksheet in Figure 2-5 autocalculates NRW as the volume of Water Supplied minus the sum of the BMAC and BUAC. For CWC, this is determined as 4,401.87 – (3,858.20 + 0.0) = 543.67 mil gal.

Task 5—Collect Distribution System and Cost Data

This section of the Worksheet in Figure 2-5 provides for the entry of pertinent water distribution system data that are necessary to describe the utility and calculate the performance indicators. The information is provided under two headings: System Data and Cost Data. The water audit approach assumes that the utility distribution system is operated 365 days per year and is continually pressurized during these operations. This is true for North American systems; however, in many developing countries, intermittent supply systems are typical, providing pressurized water supply for only a portion of each day or only for certain days of the week.

Step 5-1. Calculate the length of mains (Lm). Lm is the length of all pipelines (except service connections) in the system starting from the point of system input metering (e.g., at the effluent of the water treatment plant). It is also recommended to include in this measure the total length of fire hydrant lead pipe. Hydrant lead pipe is the pipe branching from the water main to the fire hydrant. Fire hydrant leads are typically of a sufficiently large size that is more representative of a pipeline than a service connection. The average length of hydrant leads across the entire system can be assumed if not known and multiplied by the number of fire hydrants in the system, which can also be assumed if not known. This value can then be added to the total pipeline length. Total Lm can therefore be calculated as

$$Lm\,(miles) = \begin{array}{c}(total\ \ pipeline\ length, miles) + [\{(average\ \ fire\ hydrant\ lead\ length, ft)\\ \times (number\ of\ \ fire\ hydrants)\} / 5,280\ ft / mi]\end{array} \quad \text{(Eq 2-5)}$$

Enter the total Lm on the Worksheet in Figure 2-5. For CWC, the auditor enters the quantity of 256.3 miles of pipeline. The units for this length will be in miles unless the auditor selects metric units on the Start Page, in which case, the length will be in kilometers.

Step 5-1A. Answer IDG criteria questions to determine the data grade for Lm. Key factors affecting the data grading for this input include mapping inventory management practices and method of input derivation. CWC has a system of paper maps of its water distribution system. While it has a strong policy for upkeep of records, the management of this function has slipped in the past five years, and it is believed that the records of new piping installed in several housing developments has not yet been recorded in the mapping system. The CWC auditor answers all visible IDG questions for the Lm input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 3. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 5-2. Calculate the Number of Active and Inactive Service Connections (Nc). Nc represents the number of pressurized customer service connections extending

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

from the water main to supply water to a customer. This includes the actual number of distinct piping connections, including fire service connections, whether active or inactive. Nc does not include the pipeline leads to fire hydrants. Inactive connections are included as they can experience leakage and are relevant to the calculation of KPIs. An inactive service connection may be present at vacant, abandoned, or even demolished building locations, and a pressurized customer service connection may remain even where a meter has been removed.

Nc may differ substantially from the number of customers, number of billing system accounts, or number of meters. The derivation of Nc typically should come from some form of services inventory, which can include billing systems and GIS. Nc should be premise based (tied to a physical location for the service connection) rather than nonpremise based (tied to a customer or meter). Typical sources of error encountered when deriving Nc from nonpremise-based data include single service connections that serve multiple meters, a single customer who owns multiple properties, and inactive service lines serving a location that does not have a meter installed. The auditor should examine available service inventory data to identify if available counts are premise based or nonpremise based, adjusting as necessary. The inactive service connection count should be based on what is reasonably documented, and any estimates of this count should be properly substantiated.

> *Enter the Nc on the Worksheet in Figure 2-5. For CWC, the auditor enters the quantity of 12,196 active and abandoned customer service lines (see Table 2-8). After this number is entered, the Audit Software automatically calculates the Service Connection Density by dividing this number by the length of mains. For CWC, the Service Connection Density is calculated and displayed as 48 connections per mile of pipeline.*

Step 5-2A. Answer IDG criteria questions to determine the data grade for Number of Active and Inactive Service Connections (Nc). Key factors affecting the data grading for this input include services inventory management practices and method of input derivation. CWC has a robust electronic customer billing system that manages its customer accounts; however, it uses a system of paper maps of its water distribution system. Some of the same issues in keeping good track of the Lm parameter also plague the count of customer services. Notably, it is believed that the records of new piping installed in several housing developments have not yet been recorded in the mapping system; thus, the new service connections are also not recorded. The CWC auditor answers all visible IDG questions for the Nc input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 4. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 5-3. Calculate the Average Length of Customer Service Line (Lp). Lp represents the average length of a service connection piping under customer responsibility. Worldwide, the majority of leaks occur on customer service piping and not on water mains and appurtenances. Many water utilities have policies that require the customer to own and maintain a portion or all of the customer service connection piping. When leaks on service connections are detected on the portion of the piping that falls under the responsibility of the customer, leak run times are inherently much longer than leaks on the utility-owned piping. Customers are not as versed in arranging for leak repairs as the water utility, and many customers balk at the expense of this work, particularly with losses that occur upstream of the customer's water meter. The AWWA Water Audit Method reflects this difference in leakage management of customer- and utility-owned service piping.

This value is determined based on the water utility's policy for leak repair responsibility and the delineation point of this responsibility, such as the curb stop or customer water meter. Figures 2-11 through 2-13 show the definition of this value in various customer service connection piping and metering configurations. Policies that require the utility to implement repairs result in faster repair times and shorter leak run times than repairs

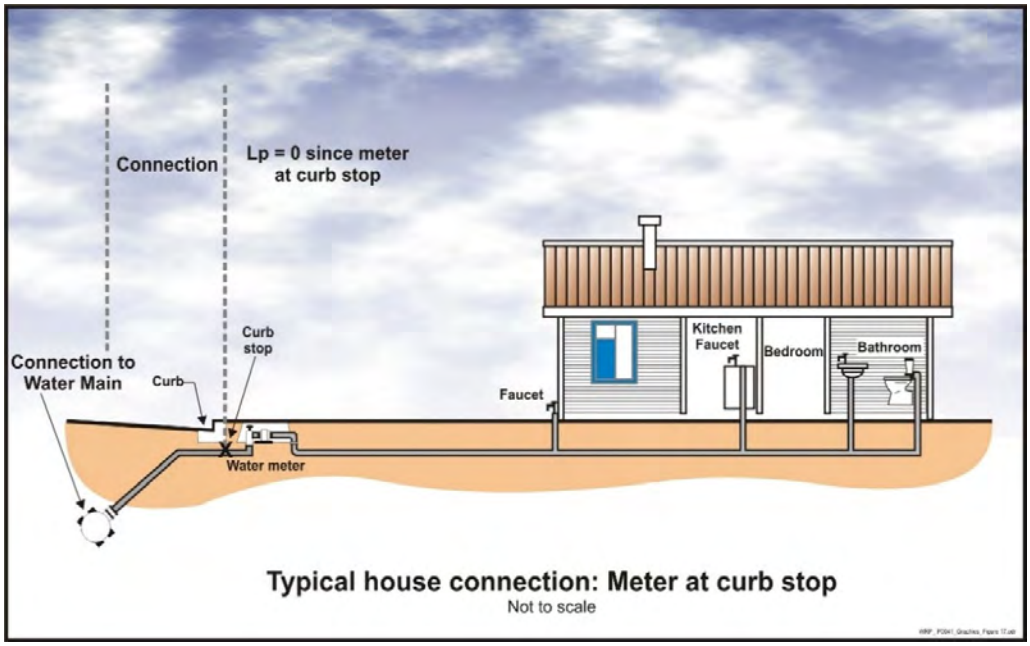
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

arranged by customers using contractors or plumbers. The average length needed for this parameter can be approximated if not known. If the utility's policy and practice are to place the customer water meter at the customer curb stop or property line as the point of ownership change, this value is zero since the water utility owns the service connection piping upstream of the customer water meter that serves as the customer endpoint in the water audit. For customer service connection piping configurations as shown in Figure 2-11, Lp is zero and the data grading automatically calculates as a 10. The Audit Software includes a question asking whether the customer water meter is located next to the curb stop or property boundary. If the auditor selects "Yes" in the dropdown box, the Audit Software automatically enters a value of zero, with data grade of 10.

> *Enter the Lp on the Worksheet in Figure 2-5. In CWC, most of the customer water meters are located inside customer building premises. Based on measurements of 12 to 15 locations, the CWC auditor estimates that the average value for this length across the water distribution system is 18 ft.*

Step 5-3A. Answer IDG criteria questions to determine the data grade for Lp. Key factors affecting the data grading for this input include private service line inventory management practices and method of input derivation. CWC has a robust electronic customer billing system that manages its customer accounts; however, it uses a system of paper maps of its water distribution system. CWC has not yet had the opportunity to do more than 12 to 15 field checks at varying building configurations that represent a range of service connection lengths. The CWC auditor answers all visible IDG questions for the Lp input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 6. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

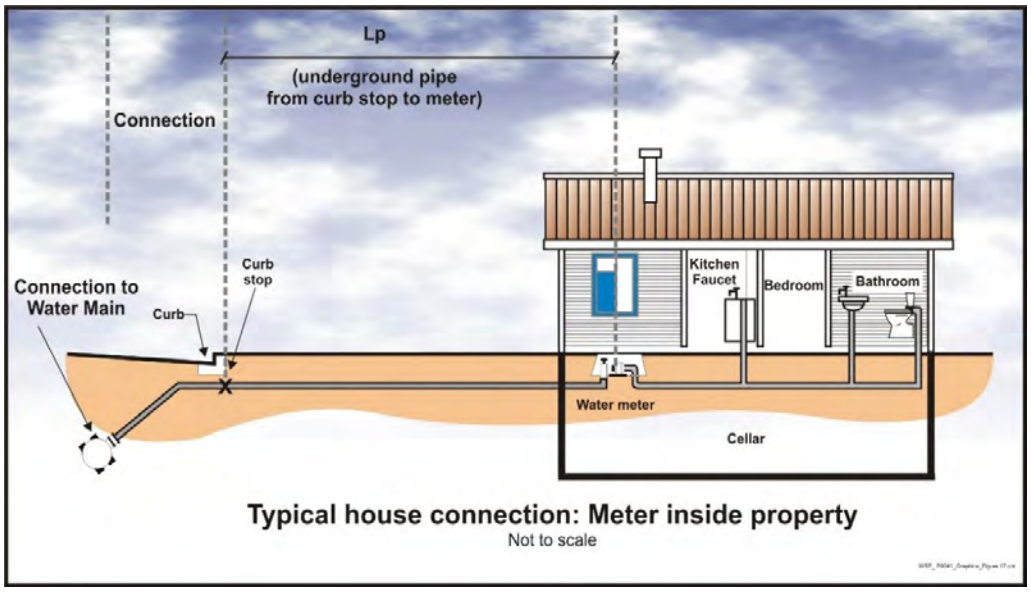
Step 5-4. Calculate the Average Operating Pressure (AOP). This is the average operating pressure in the distribution system that is the subject of the water audit. Many water utilities have a calibrated hydraulic model of their water distribution system that



Courtesy of Ronnie McKenzie, WRP Pty Ltd.

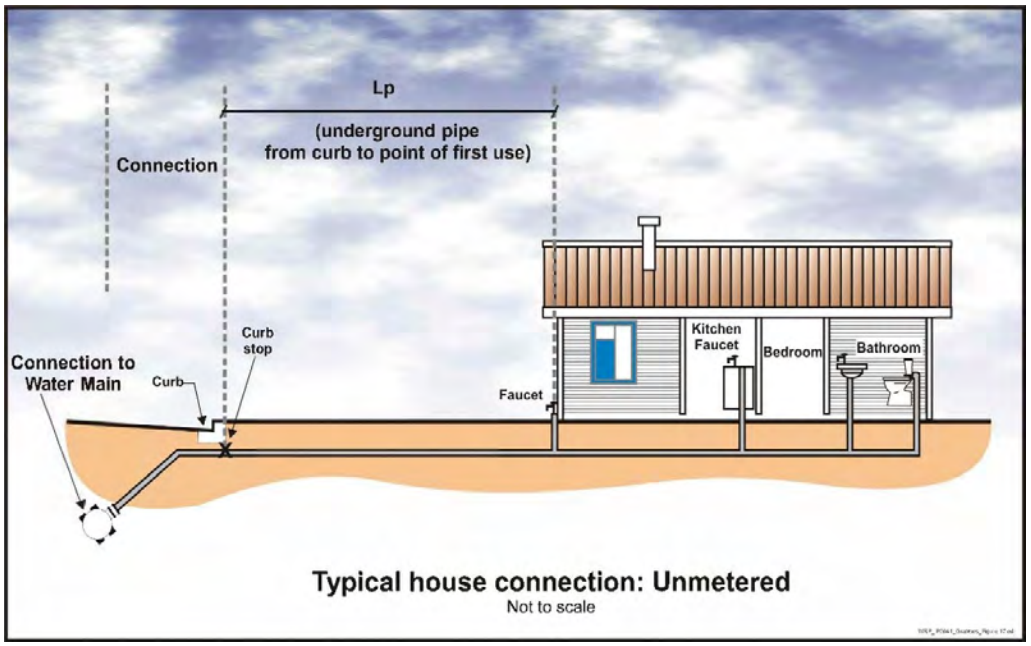
Figure 2-11 Determining the average length of customer service line, Lp, for customer meter located at the curb stop, where line ownership changes

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Ronnie McKenzie, WRP Pty Ltd.

Figure 2-12 Determining the average length of customer service line, Lp, for customer meter located inside customer premise



Courtesy of Ronnie McKenzie, WRP Pty Ltd.

Figure 2-13 Determining the average length of customer service line, Lp, for unmetered customer properties

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

can be used to obtain a very accurate value of AOP. In the absence of a hydraulic model, the AOP may be approximated by obtaining readings of static water pressure from a representative sample of fire hydrants or other system access points evenly located across the system. A weighted average of the pressure can be assembled but be sure to consider the elevation of the fire hydrants, which typically exist several feet higher than the level of buried water pipelines. If the water utility is compiling the water audit for the first time, the average pressure can be approximated, but a low data grading will be assigned by the Audit Software. In subsequent years of auditing, effort should be made to improve the accuracy of this quantity and qualify for a higher data grading. The "Determining Average Operating Pressure in a Water Utility Distribution System" sidebar, Tables 2-20 and 2-21, and Figure 2-14 provide guidance on calculating the average operating pressure. Care should be taken to derive the AOP value accurately, as some of the audit performance indicators are highly sensitive to error that exists in this number.

A well-calibrated hydraulic model of a distribution system includes detailed information on piping configurations and the ability to perform numerous calculations quickly to obtain the AOP. Hydraulic models are comprised of data for pipe segments and connection points (nodes). Calculating the average pressure of all nodes in the system model is not necessarily the most accurate way to calculate the AOP, especially if there are significant pressure differences throughout the system, and the nodes are not evenly distributed throughout the distribution system. The most accurate calculation captures the average pressure that each pipe model segment experiences. To obtain this, the model should calculate the pressure at each end of the pipe model segment, take the average of those two values, and multiply that number by the length of that pipe model segment. This operation is conducted for all pipe segments, and the products are summed up for the entire distribution system. The sum is divided by the total pipe length to calculate the AOP. This is essentially a weighted average of pressure over the total pipe length.

Table 2-20 Determining average operating pressure in a water utility distribution system with varying topography

Zone Reference	Length of Mains (Lm) (miles)	Number of Customer Service Connections (Ns)	Connection Density per Mile/Main	Average Zone Pressure (Pav)* (psi)	Ns × Pav	Lm × Pav
A	253.9	8,124	32.0	52.5	426,510	13,330
B	153.0	5,760	37.6	38.1	219,456	5,829
C	175.1	5,204	29.7	61.0	317,444	10,681
D	135.3	4,483	33.1	43.4	194,562	5,872
E	110.7	2,722	24.6	62.0	168,764	6,863
F	54.8	2,332	42.6	55.1	128,493	3,019
G	60.0	2,162	36.0	48.7	105,289	2,922
Column totals	943	30,787	—	Column totals =	1,560,518	48,516
Number of zones	7	7	—	Divide by	Ns = 30,787	Lm = 943
Zone average	134.7	4,398	32.7†	System Pav estimate =	50.7	51.4
†System density is >32/mile, so best estimate of system Pav =					50.7 psi	

* Average daily pressure measured at the location of the average elevation in the zone
† Weighted average, using Length of Mains as basis of weighting

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-21 Determining average system pressure in a section of a water utility distribution system (see Figure 2-14 for reference)

County Water Company—Downtown Region Listing of Fire Hydrants and Ground-Level Elevation					
Street	Cross Street	Elevation	Street	Cross Street	Elevation
Washington	1st	850.0	Washington	W. of 3rd	865.0
1st	N. of Adams	854.0	3rd	N. of Adams	872.5
1st	N. of Jefferson	861.5	Adams	W. of 3rd	873.0
1st	N. of Madison	869.0	3rd	N. of Jefferson	879.5
Madison	1st	872.5	Jefferson	E. of 3rd	882.0
1st	N. of Monroe	877.5	Madison	W. of 3rd	885.0
Monroe	1st	879.5	Madison	E. of 3rd	888.5
1st	N. of Jackson	883.0	3rd	N. of Monroe	892.5
Jackson	1st	886.0	3rd	N. of Jackson	899.0
2nd	N. of Washington	854.5	Jackson	E. of 3rd	902.0
2nd	N. of Adams	863.0	Washington	W. of 4th	874.5
Adams	W. of 2nd	862.0	Adams	E. of 4th	883.0
2nd	N. of Jefferson	871.0	Adams	W. of 4th	882.0
Jefferson	W. of 2nd	871.0	4th	N. of Jefferson	887.0
Madison	W. of 2nd	879.0	4th	N. of Madison	893.0
2nd	N. of Monroe	885.0	Madison	E. of 4th	898.0
Monroe	W. of 2nd	884.5	4th	N. of Monroe	902.0
Jackson	W. of 2nd	890.5	4th	N. of Jackson	909.5
2nd	S. of Jackson	893.5	Jackson	W. of 4th	910.0

Weighted Average Calculations					
Lower Limit	Upper Limit	Mid-point	Hydrant Count	Count Times Mid-Point	
850	860	855	3	2,565	Weighted average ground elevation = 33,480/38 = 881.0 ft
860	870	865	5	4,325	Nearest location of average zone point: 881.0 ft Adams, W. of 4th: 882.0 ft Measured pressure at this fire hydrant = 58 psi; for underground piping, take as 59 psi
870	880	875	10	8,750	
880	890	885	10	8,850	
890	900	895	6	5,370	Nearest location of zone critical point: 910.0 ft Jackson, W. of 4th: 910.0 ft Measured pressure at this fire hydrant = 45 psi; for underground piping, take as 46 psi
900	910	905	4	3,620	
—	—	Total	38	33,480	

Notes: The average zone point (AZP) in a zone is defined as the location of the average static water pressure. The critical point (CP) in a zone is defined as the location of the lowest static water pressure. The AZP and CP are important design parameters in the application of pressure management techniques which are discussed in Chapter 6. The auditor is cautioned not to assume that the AZP is the average of the zone inlet and CP pressures. In this example, the AZP and CP are taken as the location of the average and highest elevations, respectively. It is recognized that the locations of the AZP and CP are influenced by both elevation and the level of head loss in the distribution system. Identifying these locations is therefore most accurate when using a hydraulic model. However, the method shown in this example gives a reliable way to identify the AZP and CP with limited data collection needs.

Many water utilities do not have a hydraulic model of their water distribution system. An example calculation for determining average zone pressure from fire hydrant pressure data in a given zone is shown in Table 2-21. The example focuses on one region of CWC's service area: the downtown region. The water piping grid for this region is also displayed in Figure 2-14 in which fire hydrant locations are shown as well as ground-elevation contours at 10-ft intervals. The ground elevation of this region varies from 850 ft to

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

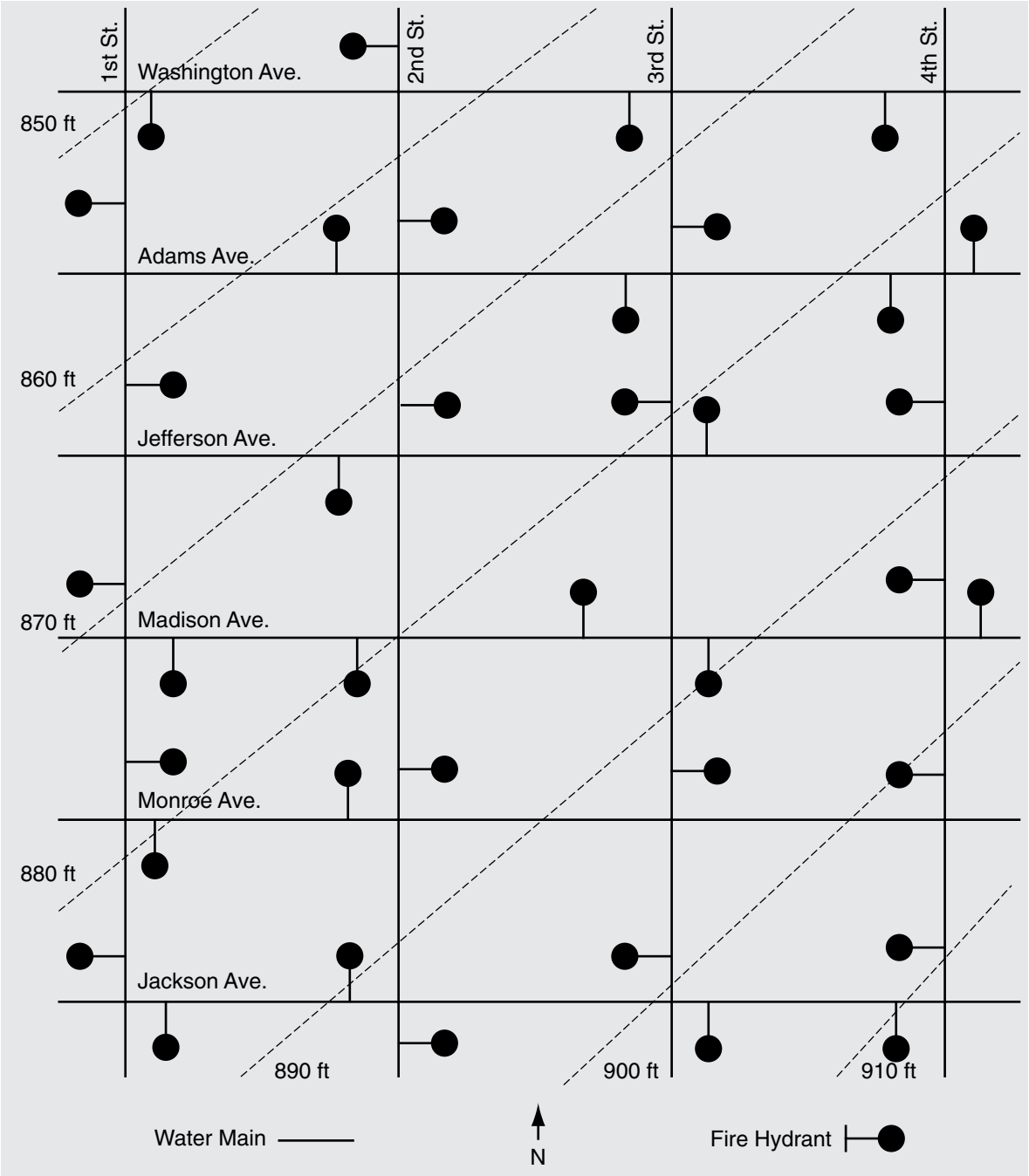


Figure 2-14 Map of fire hydrants in a section of a water utility distribution system (used to assist the calculation of average operating pressure)

more than 910 ft above sea level. In calculating the average operating pressure in this zone, this value can also serve as the average zone pressure (AZP) point if the utility is applying pressure management techniques. The example also shows the means to determine the location of the critical point (CP) and its value. See Chapter 6 for the specifics of pressure management design.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Enter the AOP on the Worksheet in Figure 2-5. CWC is largely located in an area of relatively flat terrain, but a small portion of the system exists in an elevated area where water pressures are lower than average, and another section of the system exists in a valley where higher water pressures prevail. CWC does not have a calibrated hydraulic model of its system and relies on mapping of the system and topography along with pressure data from six locations monitored by its SCADA system to obtain a value of the average operating pressure for the entire system, which is determined to be 65 psi. The auditor for CWC enters the value of 65 psi on the Worksheet shown in Figure 2-5 for the AOP.

Determining Average Operating Pressure in a Water Utility Distribution System

Water utility managers need to understand the variation of water pressure across their distribution systems to assess the potential for improved pressure management and to calculate the Unavoidable Annual Real Losses (UARL) parameter using Eq. 2-6 (later in this chapter). The UARL is typically calculated for the entire water distribution system, and the average operating pressure across the network is one of the inputs into Eq. 2-6. Although a mathematical average of the pressure throughout the water distribution system can be calculated, pressures can vary considerably from one part of the system to another, particularly if the system exists in hilly or mountainous terrain. In such cases, the utility manager should become familiar with those regions where static system pressures are notably lower or higher than the average level, and the impact of these regional pressures on leakage rates and levels of customer service. Chapters 5 and 6 discuss pressure management.

Calculating Average Pressure Across a Water Distribution System

Several means exist to determine the average system pressure with sufficient accuracy to calculate the UARL.

- A calibrated hydraulic model can be used, which can provide pressures at nodes across the water distribution system under various water demand conditions. The average pressure in each zone can be readily calculated by the data from this model and a weighted average pressure among zones calculated for the system, weighted by Lm or Nc (see Table 2-20). In the absence of a calibrated hydraulic model, one of the following methods may be selected to approximate the average pressure.
- For water distribution systems existing across a relatively flat service area, the average pressure can be determined by gathering static pressure readings from approximately 30 fire hydrants spaced proportionally across the system. The mathematical average of these readings should be calculated. Because fire hydrants in North America are typically located aboveground, water pressure in the underground pipelines is slightly higher (1-2 psi, depending on depth) than the level measured at the fire hydrant.
- For water utilities with a distribution system that extends across hilly or mountainous terrain, the system should be sectioned into several distinct zones that represent different pressure regimens. In each zone, topographical data (ground-level elevations) should be gathered, and a weighted average technique should be used to determine the location of the average elevation. Water pressure can be measured for 24 hours at the average elevation site from a fire hydrant or other system appurtenance to give a good approximation of the average pressure in the specific zone. The average pressure values from individual zones can then be used to calculate the weighted average pressure among zones for the system (see Table 2-20).

In calculating weighted average pressure among zones, the auditor should first determine miles of main and the number of connections per zone, then calculate a weighted average based on both parameters. The overall weighted density of connections should then be used to determine which weighting basis to use:

- if greater than 32 connections/mile of main overall weighted density, use the number of connections as the basis for weighting, or
- if less than or equal to 32 connections/mile of main overall weighted density, use miles of main as the basis for weighting.

Overall density of connections should be weighted using length of mains. Additionally, the

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

auditor is cautioned not to use "number of billed properties" in lieu of "number of connections" because these two values are generally not the same. An example table depicting the calculation of weighted average pressure among zones is presented in Table 2-20.

Step 5-4A. Answer IDG criteria questions to determine the data grade for AOP. Key factors affecting the data grading for this input include the extent of pressure data collection and the method of input derivation. The availability of a calibrated hydraulic model is also a strong factor. CWC operates three pressure zones, but the integrity of the valving that forms boundaries between the zones is known to be suspect; it is believed that several valves allow water to pass from one zone to another. Although CWC does have a reliable SCADA system that provides pressure at several locations, the auditor did not have the time or resources to gather additional pressure data from a sample of fire hydrants across the system. The calculation of average operating pressure is based on very limited data. The CWC auditor answers all visible IDG questions for the AOP input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 7. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 5-5. Total annual cost of operating water system, optional input. The audit software version 6.0 has designated this input as optional, as it is no longer required in the calculation of KPIs. The input was retained as optional for those utilities that wish to continue to track this figure as part of their annual water auditing process. To remain consistent with the cost data collected in software versions prior to 6.0, the costs collected should include those for operations, maintenance, and any annually incurred costs for long-term upkeep of the drinking water supply and distribution system only. These costs typically include employee salaries and benefits, materials, equipment, insurance, fees, administrative costs, and all other costs to sustain the drinking water supply distribution system only and not waste water, stormwater, or biosolids systems. Depending on utility accounting procedures or regulatory requirements, it may be appropriate to include depreciation in this cost. This parameter was an input in the Audit Software in versions prior to version 6.0. No data grading is associated with this optional input. CWC input the cost value of $9,600,000 as an optional input for its water audit.

Step 5-6. Calculate the Customer Retail Unit Charge (CRUC). The CRUC represents the volumetric charge that customers pay for water service. This unit charge (sometimes called a commodity charge) is applied routinely to the components of Apparent Loss, since these losses represent volumes of water reaching customers but not fully accounted for and thus not (fully) paid for. Many water utilities have a rate structure that includes fixed charges as well as charges per unit of water, but such fixed charges are collected regardless of metered usage and thus not pertinent to the valuation of Apparent Loss. Also, many utilities have unit charges that differ based on class of customer and other factors, so a volume-weighted average of these different individual charges by customer class can be calculated to determine a single overall composite cost that should be entered into this cell. Where tiered rates exist within a customer class, as with an inclining block structure to promote water conservation, the sum of all volumetric revenues received from that customer class can be divided by the total deliveries to that class during the audit period to yield an average rate for deliveries to that customer class. Special surcharges, such as for customers in certain elevation zones, can be averaged over all customers within a class.

Importantly, the CRUC should also include additional charges for sewer, stormwater, or biosolids processing, but only to the extent that these charges are based on the volume of potable water consumed. In many communities, the charges for sanitary sewer service are as high or higher than the charges for water service, so it is critical for an accurate valuation of Apparent Loss that any sewer charges and any other charges that are assessed based on metered water readings be included in the CRUC. In the event that

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

sewer charges are based on water meter readings but the sewer revenue is collected by an entity separate from the water utility, sewer charges should be evaluated and included in the CRUC if there is opportunity for the water utility and sewer entity to collaborate on Apparent Loss mitigation efforts.

Enter the CRUC on the Worksheet shown in Figure 2-5. Note: The assumed currency in these examples is US dollars. The auditor should select the appropriate units that are charged to customers ($/1,000 gallons, $/1,000 liters, or $/hundred cubic feet) and the Audit Software automatically converts these units to the units that appear in the "Water Supplied" box.

The auditor for CWC uses two approaches to derive a volume-weighted average of $3.95/1,000 gallons for the CRUC input.

Approach 1: Billed volumes by billing category and billing rate

In this approach, CWC uses their rates and volumes billed by category to determine a volume-weighted average for CRUC. Note: CWC bills a portion of its customer population for sewer services. This billed volume is 1,354.62 mg for the audit year; it is part of the total billings in Table 2-22.

Approach 2: Total billed volume and total revenue

In this approach, CWC can extract from their billing system the total billed volume for water services for the audit year, as well as the total volumetric revenue for water and sewer services, even though not all water customers are assessed a sewer charge. They were able to exclude fees not related to volumetric water or sewer billings, such as connection or activation fees and late charges. They used this to determine the volume-eighted average CRUC in Table 2-23.

Step 5-6A. Answer IDG criteria questions to determine the data grade for CRUC. Key factors affecting the data grading for this input include the extent of applicable rates included in the input derivation and the type of the averaging method (simple vs. volume-weighted) used among distinct customer classes. Also, if other charges to the customer, such as sewer, are based on the water meter readings, they should be applied. CWC determined a volume-weighted average for this input. The CWC auditor answers all visible IDG questions for the CRUC input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 9. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Step 5-7. Calculate the Variable Production Cost (VPC). Lastly, real (leakage) losses should be valued at an appropriate rate. The cost rate, which depends on the local economic and water resource considerations of the utility, can include short-term, direct variable water treatment and pumping power costs, the unit cost of bulk purchased (imported) water, indirect costs of new or upgraded system infrastructure to meet higher demands which can be "avoided" if leakage reductions are achieved (avoided costs), and other indirect environmental or social costs.

Assessing costs for real losses can be complex, but the methods included in this manual allow the auditor to keep the evaluation simple. Real losses include water that has been extracted from a water source, treated, energized, and transported a distance before being lost from the distribution system. Because these quantities of loss occur in addition to the water successfully supplied to customers, real losses effectively impose on the water utility excess extraction, treatment, and delivery charges, and/or excess imported water purchase charges.

The simplest costs of supply are the short-run marginal costs which include but are not limited to the basic costs to provide the next volume unit (e.g., mil gal) of water,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-22 Volume-weighted average calculation of CRUC for County Water Company using billed volumes by billing category and billing rate

	Unit Charge Rate ($/1,000 gal)	*Volume Billed (mil gal)*	*Volumetric Revenue by Rate ($) (Unit Charge Rate × Volume Billed)*
Residential water	*$ 2.49*	*2,919.00*	*$ 7,268,310*
Industrial water	*$ 3.33*	*488.60*	*$ 1,627,038*
Commercial water	*$ 2.78*	*97.20*	*$ 270,216*
Agricultural water	*$ 3.48*	*353.40*	*$ 1,229,832*
Any class sewer	*$ 3.57*	*1,354.62*	*$ 4,835,993*
		Total Volumetric Revenue	*$ 15,231,389*
Total BMAC (mil gal)		*3,858.20*	
Volume-Weighted Average CRUC = ($15,231,389 / 3,858.20 mil gal) / 1,000 gal/mil gal = $3.95 / 1,000 gal			

Table 2-23 Volume-weighted average calculation of CRUC for County Water Authority starting with total volumetric revenue

Total BMAC (mil gal)	*3,858.20 mil gal*
Total volumetric water and sewer revenue (excluding connection fees, late charges, etc.)	*$15,231,389*
Volume-weighted average CRUC = ($15,231,389 / 3,858.20 mil gal) / 1,000 gal/mil gal =	***$3.95 / 1,000 gal***

typically the operating costs for water extraction, water treatment, and power for pumping to convey the water throughout the distribution system. If water is purchased (imported) from another water utility, the unit purchase cost is used, along with any distribution pumping and additional treatment operating costs. Many systems supply a combination of water from their own sources and some volume of imported water. Most drinking water utilities routinely compile these direct operating costs, and the data are readily available.

Long-run marginal costs may additionally include those for water treatment residuals management, capital depreciation from wear and tear on dynamic equipment, and liability for damages from leakage incidents. These long-run marginal costs may also include capital costs such as expansions of capacity of water treatment or pumping facilities to meet growing water demand. A utility's capital improvement plan can be consulted to check for any impending expansion of supply or treatment capacity, which may justify a substantial avoided cost value to be assigned to the reduction of real losses, especially in a water-scarce location or condition. More information on the derivation of long-run, indirect costs can be found in "*Appendix A: Costing Methods* of the Alliance for Water Efficiency Report: *Building Better Water Rates for an Uncertain World"* (2014) available at www.financingsustainablewater.org. If full-cost pricing has been determined and implemented, components from this can be used to identify these long-run indirect costs, including lifecycle costs and externalities. For example, to practically apply the impact of wear and tear, the annual depreciation costs for the dynamic equipment could be determined and then added to the total cost components.

Additionally, because real losses represent volumes of water taken from a source that do not generate a benefit, these losses could also be assessed indirect costs relating to

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

their environmental, economic, and social impacts. Reducing leakage could mean smaller withdrawals from a river, which could improve instream flows, benefiting aquatic life, recreation (boating, fishing), or economic development (waterfront amenities). Clearly, in the long term, such impacts exist. Because these impacts are difficult to quantify and are system specific, methods to derive them are not included in this manual. If any costs are missing, an estimate can be used until a separate cost assessment can be performed later.

For a water utility using only a single own source (i.e., one treatment plant, no imports), it is recommended to add the annualized costs for all applicable short-run and long-run marginal costs described above and divide this total cost by the VOS (defined in Task 1—Including Error Adjustments) in volume units, yielding an answer in units of cost (i.e., $) per volume unit (i.e., mil gal). Similarly, a water utility using only a single import source (i.e., purchasing from a single supplier) should add the annualized costs for all applicable short-run and long-run marginal costs described above and divide this total cost by the WI (defined in Task 1—Including Error Adjustments). In this scenario, the water utility would answer "Only one source of water exists, which was the basis for the input derivation" for IDG question vpc.1 in the Audit Software.

For a water utility with multiple sources of water (i.e., more than one treatment plant or a combination of own sources and imported), it is important to first determine how leakage reduction anywhere in the distribution network would translate to reduction in water supply volumes from the different sources. If reduction in leakage anywhere in the distribution network would be indiscriminate with regards to which water supply would correspondingly be reduced, then the volume-weighted average method is recommended to determine VPC. The volume-weighted average method simply sums up all applicable short-run and long-run marginal costs for all sources and for distribution system pumping and divides by the sum of VOS and WI (defined in Task 1 – Including Error Adjustments). In this scenario, the water utility would answer "Multiple sources of water exist, and a volume-weighted average was calculated for all sources" for IDG question vpc.1 in the Audit Software.

However, if reduction in leakage, irrespective of where this reduction occurs in the distribution network, directly translates to a reduction in water supply required from the most expensive source, then the highest-cost method is recommended to determine VPC.

For the highest-cost method, the source-specific VPC for each source (i.e., each treatment plant and each import source) should be determined by summing up all applicable short-run and long-run marginal costs and dividing by the sum of supply volume from each source. The source-specific VPCs from all sources should then be arranged from highest to lowest cost. If the volume of real losses is less than the volume supplied by the highest-cost source, then the highest-cost source VPC plus any distribution pumping costs should be used as VPC in the water audit. If the volume of real losses exceeds the volume supplied by the highest-cost source, then a weighted-average should be determined based on the volumes supplied by source relative to volume of real losses with the distribution pumping costs added in. An illustration of this is provided below.

Highest-Cost Method Illustration 1

VOS volume	= 900 mil gal	source-specific VPC = $500 / mil gal
WI volume	= 200 mil gal	source-specific VPC = $3,650 / mil gal
Real Losses volume	= 350 mil gal	

Distribution pumping short- and long-run marginal costs = $225 / mil gal

Portion of Real Losses @ highest-cost source = 200 mil gal (WI volume)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Portion of Real Losses @ next highest-cost source = 150 mil gal (Remaining Real Losses volume)

Highest-cost method VPC =
($3,650/mil gal × (200/350)) + ($500/mil gal × (150/350)) + $225/mil gal = $2,525 / mil gal

Highest-Cost Method Illustration 2

VOS volume	= 900 mil gal	source-specific VPC = $500 / mil gal
WI volume	= 200 mil gal	source-specific VPC = $3,650 / mil gal
Real Losses volume	= 150 mil gal	

Distribution pumping short- and long-run marginal costs = $225 / mil gal

Real Losses volume of 150 mil gal < highest-cost source (WI) volume of 200 mil gal

Highest-cost method VPC = $3,650 / mil gal + $225 / mil gal = $3,875 / mil gal.

In these highest-cost method scenarios, the water utility would answer "Unit costs for the most expensive source utilized based on utility's discretion" for IDG question vpc.1 in the Audit Software.

One exception to the highest-cost method is noted. Where purchased water is provided under a "take or pay" contract (i.e., the same total charges apply whether the utility accepts the full volume of purchased water to which it is entitled), the cost of the purchased water takes on the quality of a fixed charge and can't be used as a truly variable production cost, particularly if such arrangements are present in long-term contracts. If such contracts are subject to revision or renewal on an annual basis, however, the costs of such purchases are tantamount to variable production costs and can be used in the audit.

Another situation for consideration is that of a water utility facing extremely constrained water resources with water-use restrictions in effect. In this case, real losses might be valued at the CRUC rate (same as apparent losses) because the reduction of these losses could result in the sale of like volumes of water to customers, thereby allowing new development to occur without increasing water withdrawals. A caveat that should be considered in this approach concerns systems with rate structures that do not proportionately recover variable operating costs from variable operating revenue; in such cases, utilization of the CRUC as the VPC may risk under- or overvaluation of the VPC. In this scenario, the water utility would answer "The VPC was entered using the CRUC value, based on the utility's discretion" for IDG question vpc.1 in the Audit Software.

> *The assumed currency in this example is US dollars. Enter the VPC on the Worksheet shown in Figure 2-5. The auditor for CWC obtains this value from the accounting unit of the company, which is $750/mil gal.*

Step 5-7A. Answer IDG criteria questions to determine the data grade for VPC. Key factors affecting the data grading for this input include the extent of applicable short-run and long-run marginal costs included in the input derivation and the nature of the averaging method (simple vs. volume-weighted) used among distinct supply sources. The CWC auditor answers all visible IDG questions for the VPC input in accordance with the utility's practices, resulting in automatic calculation of a data grade of 3. CWC can increase its data validity in subsequent water audits by focusing its next steps on any criteria flagged as "limiting."

Task 6—Determine System Attributes

System attributes quantify features that are specific to the individual water utility being audited as discussed below. Performance Indicators, detailed in Task 7, present data from

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

the water audit in a way that allows for comparison with other water utilities (benchmarking) and tracking a water system's performance over time.

Step 6-1. Calculating Unavoidable Annual Real Losses (UARL). UARL represents the minimum level of leakage that is modeled in a system-specific manner for a water utility. It represents the theoretical low limit of leakage that could be achieved in a system that is well-managed and in good condition at the current average pressure level.* UARL is a reference value and does not refer to a specific type or volume of leakage occurring in the water distribution system. A calculation exists for the UARL value, which accounts for the specific characteristics of an individual water system.

The derivation of the UARL calculation is given in Tables 2-24 and 2-25. Equation 2-6 is the calculation for the UARL for an individual water distribution system. The data needed to calculate the UARL are typically available to water utility staff and include the Lm, AOP, Nc, and total length of private service connection piping maintained by the water customer (Lp), taken as Lp (see Step 5-3) × Nc (see Step 5-2).

The UARL calculation was devised by the IWA Water Loss Specialist Group (formerly the Water Loss Task Force) during its development of the water audit method. In conducting work to develop a reliable benchmarking performance indicator (the Infrastructure Leakage Index or ILI), the Water Loss Specialist Group determined to devise a means to evaluate the technical low limit of leakage that could be expected to be achieved in a given water distribution system. It is recognized that leakage in any water distribution system can never be totally eliminated, and there is no reasonable expectation that such is possible. However, several water utilities have been successful in driving leakage down to extremely low levels and maintaining very low-loss operations.

The Water Loss Specialist Group obtained data from dozens of water utilities with strong leakage control practices and observed the rate at which new leaks arise despite having comprehensive leakage controls in place. From this information, data allowances were created for various leak types according to response times typical of strong leakage management operations (see Table 2-24). The allowances were developed for the three leak types: background leakage, reported leakage, and unreported leakage. These leakage types are defined and discussed in Chapters 5 and 6. An allowance for each leakage type was assigned for key infrastructure components, such as water mains, customer service connection piping maintained by the water utility, and customer service piping typically maintained by the customer.

The equivalent leakage rates that occur under the conditions in Table 2-24 are shown in Table 2-25. The Audit Software Worksheet shown in Figure 2-5 shows data for CWC (miles of water main, average pressure, average length of customer service line, and number of customer active and inactive service connections) that is used to calculate the UARL value.

$$\text{UARL (gal)} = \left(5.4L_m + 0.15Nc + 7.5Lc\right) \times P \times 365 \text{ d/year} \qquad \text{(Eq 2-6)}$$

Where:

L_m = length of water mains (miles; including hydrant lead length)

Nc = number of active and inactive customer service connections

Lc = total length of customer service connection line (miles) (Nc × Lp)/5,280 ft/mi, where Lp = average length of customer service line (ft, see Figures 2-11 through 2-13)

* Because pressure management is an important part of many loss reduction programs, the term "unavoidable" in UARL may be misleading. Nevertheless, UARL can serve as a useful reference level of leakage that is likely to prevail in even the most well-managed systems. The water auditor should be aware that this "unavoidable" level is not necessarily permanent since the implementation of pressure management interventions can push UARL to lower levels.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-24 Component values of the UARL calculation

Infrastructure Component	Background (acoustically undetectable) Leakage	Reported Leaks and Breaks	Unreported Leaks and Breaks
Mains or pipelines	8.5 gal/mi/hr	0.20 breaks/mi/year at 50 gpm for 3 days' duration	0.01 breaks/mi/year at 25 gpm for 50 days' duration
Service connections, main to curb stop	0.33 gal/service connection/hr	2.25 leaks/1,000 service connections at 7 gpm for 8 days' duration	0.75 leaks/1,000 service connections at 7 gpm for 100 days' duration
Service connections, curb stop to meter or property line (for 50 ft average length)	0.13 gal/service connection/hr	1.5 leaks/1,000 service connections at 7 gpm for 9 days' duration	0.50 leaks/1,000 service connections at 7 gpm for 101 days' duration

NOTE: All flow rates are specified at a reference pressure of 70 psi.
Source: Lambert et al. 1999

Table 2-25 Standard unit values used for the UARL calculation

Infrastructure Component	Background Leakage	Reported Leaks and Breaks	Unreported Leaks and Breaks	UARL Total*	Units
Mains, gal/mi of main/d/psi	2.87	1.75	0.77	5.4	Gal/mi of main/d/psi
Service connections, main to curb stop, gal/service connection/d/psi	0.112	0.007	0.030	0.15	Gal/service connection/d/psi
Service connections, curb stop to meter, gal/mi of service connection/d/psi	4.78	0.57	2.12	7.5	Gal/mi of service connection/d/psi

* The UARL is calculated via Eq 2-6.
Source: Lambert et al. 1999

P = average operating pressure in the system (psi). In Eq. 2-6, 365 d/year is included given that water distribution systems in developed countries operate with continuous service for all 365 days in the audit year. For systems that operate with intermittent service, the number of days that the system was operated should be used instead of 365 days.

The Audit Software (version 6.0) presents the calculated UARL and ILI for systems of all sizes and all pressures. Some published research is now available on predicting how UARL is likely to be modified when modeling low leakage limits in systems that are very small (less than 3,000 service connections), have very low average pressures (less than 35 psi), or have very high pressures (greater than 125 psi). These are the boundary cases. Inherent over- or underestimation of the UARL volume may exist in these boundary cases, as they operate at or near the limits of the UARL modeling assumptions. While it is not perfect, UARL remains the best available method for system-specific estimation of technical minimum leakage level. More widespread application and understanding of system-specific corrections to the UARL model in these boundary cases is now likely to occur, but corrections of this nature were not included in the Audit Software (version 6.0) at the time of its publication (2020). Caution is advised when using the standard UARL modeled value (and subsequently the ILI) for boundary cases. In boundary cases, the ILI may still

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

be considered a general Performance Indicator but not used as an absolute performance measurement or for benchmark comparisons.

For CWC, the UARL is calculated by the Audit Software to be 83.7 mil gal (18.8 gal/conn/day) for the audit year. This is shown on the Dashboard (Figure 2-7).

Step 6-2. Calculating the cost impact of Apparent and Real Losses. The process of compiling a water audit is effective in tracing the water supplied by a drinking water utility to its various destinations. Such an audit captures data primarily on customers but also includes losses. Equally important, however, the method detailed in this publication also assesses the cost impact of all water audit components. Water utilities, like any business entity, cannot operate efficiently without knowing their costs and impacts on budgeting, operations, revenue collection, capital financing, and all other financial aspects of utility management. The Audit Software provides a means to calculate costs for each of the pertinent components in the water audit calculated on the Dashboard (Figure 2-7).

The nature of the valuation process of the water audit is compelling in the stark difference between apparent and real losses. Because apparent losses are quantified by the amount of water improperly recorded at the customer's delivery point, this water is valued at the retail cost that is charged to the customer. Apparent losses cost water utilities a portion of their revenue. Usually, the cost impact of apparent losses is higher than that of real losses, which are typically valued at the variable production costs to acquire, treat, and deliver the water. For most water suppliers, the retail rate charged to customers is notably higher than the variable production costs to provide the water. Therefore, apparent losses can have a dramatic financial impact to the water utility's revenue stream.

Step 6-2A. Cost impact of Apparent Loss components. To determine the cost impact of apparent losses, the Audit Software multiplies the Apparent Losses volume by the CRUC (which is converted to $/mil gal by the Audit Software if gallons units are selected on the Start Page by the auditor.)

The Dashboard shown in Figure 2-7 displays the calculated Annual Cost of Apparent Losses, which for CWC is $251,730 for the audit year.

Step 6-2B. Cost impact of Real Loss components. To determine the total cost of real losses, the Audit Software multiplies the Real Losses volume by the VPC.

The Dashboard shown in Figure 2-7 displays the calculated Annual Cost of Real Losses, which for CWC is $210,423 for the audit year.

Step 6-2C. Cost impact of Unbilled Authorized Consumption. To determine the total cost of this third component of NRW, the Audit Software multiplies the Unbilled Authorized Consumption volume by the VPC.

The Dashboard shown in Figure 2-7 displays the calculated Annual Cost of Unbilled Authorized Consumption, which for CWC is $149,430 for the audit year.

Step 6-2D. Total cost impact of NRW components. To determine the total cost of NRW, the Audit Software adds the cost impacts of Apparent Losses, Real Losses, and Unbilled Authorized Consumption.

The Dashboard shown in Figure 2-7 displays the calculated Annual Cost of NRW, which for CWC is $611,583 for the audit year.

Task 7—Performance Indicators

Performance Indicators present water audit data in a way that reveals the utility's water loss standing while letting it reliably track its performance over time; communicate

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

performance to decision-makers, customers, and other stakeholders; and compare its standing with other water utilities (via benchmarking. The performance indicators calculated by the Audit Software are discussed below. System attributes, detailed in Task 6, are numeric features that are specific to the individual water utility being audited and are not suitable to use for comparisons.

The performance indicators published by the International Water Association (IWA) in the publication *Performance Indicators for Water Supply Services* (Alegre et al. 2000, Alegre et al. 2016) provided the foundation for an array of performance indicators included in the Audit Software. More recently, the AWWA Water Loss Control Committee (WLCC) conducted research and development work on KPIs, culminating in the publication of a technical report (Trachtman et al. 2019) and a committee report (Kunkel et al. 2020). These landmark reports present the AWWA "WLCC 2020 position" on recommended key KPIs that are automatically calculated by the version 6.0 Audit Software and presented in the Dashboard worksheet (Figure 2-7 and repeated as Figure 2-15). In the AWWA water audit methodology, multiple indicators of varying detail are available to water utilities, allowing a realistic assessment of water loss standing in particular functional areas, along with the capability for reliable performance tracking and benchmarking. Collectively, the AWWA performance indicators give a realistic, objective assessment of overall loss standing in the water utility and are viewed as the current best practice means for these assessments in water utilities.

Of particular importance, the "WLCC 2020 position" also formally advocates against the use of percentage-based performance indicators, and percentage indicators are no longer included in the Audit Software (as of version 6.0) or in the M36 publication (as of this 5th Edition).

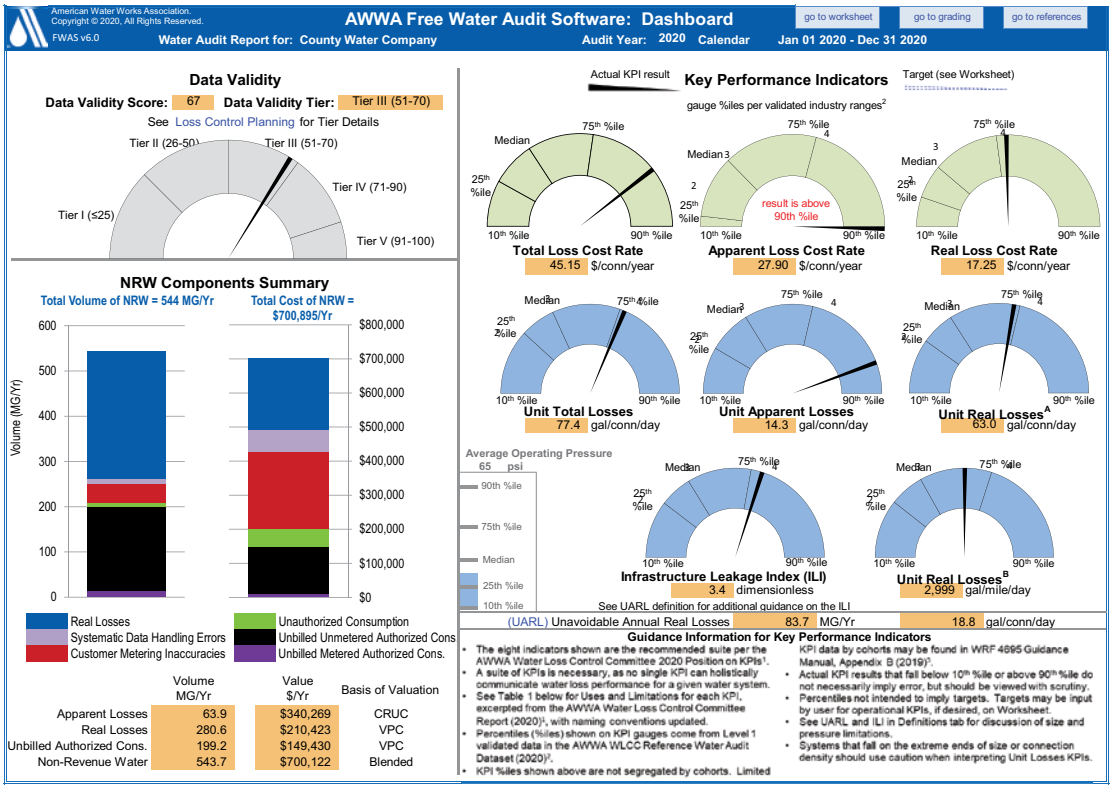


Figure 2-15 AWWA Free Water Audit Software—Dashboard for County Water Company

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The WLCC 2020 position introduced a new operational indicator and two new financial indicators. Table 2-26 lists the full array of performance indicators included in the version 6.0 Audit Software along with their descriptions, purpose, and uses. AWWA stresses that water utilities make use of the full suite of performance indicators, as no single performance indicator can holistically communicate water loss performance for a given water system. The *volume* and cost *value* of water consumption and losses and the *validity* of the water audit data must be considered in concert to assure cost-effective NRW management program design and decision-making. Thinking in terms of these "three Vs" (volume, value, validity) gives water utilities the holistic perspective needed to assess NRW objectively and plan NRW reduction strategies effectively. Several KPIs are detailed in the following section.

Step 7-1. Volume-based performance indicators. As listed in Table 2-26, the five volume performance indicators include four unit rate indicators (loss volume divided by the number of service connections) and the ILI. The unit rate indicators include a single unit rate for apparent losses, two versions for real losses (denominator of service connections and length of mains), and a total loss (apparent plus real) unit indicator (a new performance indicator as of 2020). Table 2-26 shows that the first three of the unit indicators are useful for all five purposes: assessment, benchmarking, target-setting, planning, and tracking. The unit total losses indicator should only be used for assessment and tracking purposes; the ILI may also be appropriate for these two purposes, but it is most useful as a benchmarking indicator.

In versions of the Audit Software prior to version 6.0, filters were in place for the Unit Real Losses indicator and the ILI. In the earlier versions, for small and/or low service connection density water systems, the ILI was not calculated in some cases, and only one form of Unit Real Losses was calculated. In the version 6.0 of the Audit Software, the threshold filters have been removed, and an ILI value is calculated for all utility water audits. Additionally, both Unit Real Losses indicators, Unit Real Losses^{A} (in vol/conn/day) and Unit Real Losses^{B} (in vol/pipeline length/day), are calculated and displayed for systems regardless of their size and service connection density.

As noted previously, the ILI is a performance indicator best used for benchmarking leakage standing among water utilities. The ILI is a dimensionless indicator that represents the ratio of the Current Annual Real Losses (CARL over the Unavoidable Annual Real Losses (UARL) or

$$\text{ILI} = \text{CARL} / \text{UARI} \qquad \text{(Eq 2-7)}$$

Water utilities can use the ILI to compare their real loss standing with other systems. However, it is best that utilities have in place an established leakage and pressure management program and to compare their ILI with those utilities that also have established programs in place. Because average operating pressure is a parameter that is included in the UARL and ILI calculations, changes in pressure (as might be performed in pressure management strategies) will alter the UARL and ILI. In tracking the ILI during periods of changing pressure, the water utility might observe leakage reductions via improved pressure management only to find that the ILI remains unchanged or may even rise. The ILI is most reliable for utilities that have implemented pressure management and have achieved new stable pressure regimens across their distribution system. Work continues to refine the ILI particularly to account for the influences on the UARL calculation from low average operating pressure and small system size (low number of service connections, typically less than 5,000). Lambert et al. (2020) have developed a system correction factor (SCF) that can be included as a coefficient in the UARL equation and is useful for small systems or individual DMAs. The SCF is calculated from several additional inputs that

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Copyright © 2024 by American Water Works Association. All Rights Reserved

Table 2-26 System Attributes and Performance Indicators of the AWWA water audit method (from Trachtman et al. 2019 and Kunkel et al. 2020)

Type	Indicator	Description	Suitable Purposes					Uses and Limitations	Principal Users
			Assessment	Benchmarking	Target Setting	Planning	Tracking		
Attribute	Apparent Loss Volume	Calculated by Free Water Audit Soware	✓	—	—	—	✓	Assess loss level	Utility, Regulators
	Apparent Loss Cost	Calculated by Free Water Audit Soware	✓	—	—	—	✓	Assess cost loss level	Utility, Regulators
	Real Loss Volume	Calculated by Free Water Audit Soware	✓	—	—	—	✓	Assess loss level	Utility, Regulators
	Real Loss Cost	Calculated by Free Water Audit Soware	✓	—	—	—	✓	Assess loss cost level	Utility, Regulators
	Unavoidable Annual Real Loss (UARL)	Calculated by Free Water Audit Soware	✓	—	—	—	✓	Reveal theorecal technical low level of leakage	Utility, Regulators
Volume	Unit Apparent Losses (vol/conn/day)	Strong and understandable indicator for mulple users.	✓	✓	✓	✓	✓	Used for performance tracking and target-setting	Utility, Regulators
	Unit Real Losses (vol/conn/day)	Strong and understandable indicator for mutiple users.	✓	✓	✓	✓	✓	Used for performance tracking and target-setting	Utility, Regulators, Policy Makers
	Unit Real Losses (vol/pipeline length/ day)	Strong and understandable indicator for use by utilities with low connection density.	✓	✓	✓	✓	✓	Data collection and assessment of systems with "low" connection density	Utility, Regulators, Policy Makers
	Unit Total Losses (vol/conn/day) **New KPI**	Strong and understandable indicator, suitable for high-level performance measurement.	✓	—	—	—	✓	High level indicator for trending analysis. Not appropriate for target-setting or benchmarking	Utilities, Customers
	Infrastructure Leakage Index (ILI)	Robust, specialized ratio KPI; can be influenced by pressure and connection density.	✓	✓	—	—	✓	Benchmarking after pressure management is implemented	Utilities
Value	Apparent Loss Cost Rate (value/conn/year) **New KPI**	Indicators with sufficient technical rigor. Provide the unit financial value of each type of loss, which is useful for planning and assessment of cost efficiency of water loss reducon and control interventions and programs.	✓	—	—	✓	✓	Data collection and assessment on AWWA indicators or contextual parameters to use in conjunction with Loss Cost Rates	Utilities, Regulators, Customers
	Real Loss Cost Rate (value/conn/year) **New KPI**		✓	—	—	✓	✓		Utilities, Regulators, Customers
Validity	Data Validity Tier (DVT)	Strong indicator of water loss audit data quality, if data has been validated. Tier provides guidance on priority areas of activity.	✓	✓	—	✓	✓	Assess caliber of data inputs of the water audit	Regulators, Utilities

include the percentage of rigid (metal) piping in the system compared to flexible (plastic) piping in the system, along with several other inputs. Varying pressure effects occur on the different pipe materials. By applying the SCF to the traditional UARL equation, water utilities with conditions that lie at the extreme values of the UARL inputs can be moderated so that the UARL (and therefore ILI) are not unduly hindered by these values. This work is allowing the traditional limitations in use of the UARL, and therefore the ILI, to be reduced.

Step 7-2. Value-based performance indicators. AWWA's 2020 position eliminated the use of the two percentage performance indicators that existed in the audit software through version 5.0. Version 6.0 of the Audit Software software includes the Loss Cost Rate (LCR) indicator, with a form for apparent losses and one for real losses. The LCR is expressed in units of value/service connection/year or US$/service connection/year for US water utilities. The LCR is a unit indicator with a conversion of the volume of loss to value of loss, expressed on a yearly basis rather than daily basis. As an example, for apparent losses, a water utility with an apparent loss unit rate of 5.5 gal/connection/day and a CRUC of $4.00/1,000 gal ($4.00/thousand gal or kgal) gives a calculated apparent LCR of

$$LCR_{apparent} = \frac{(5.5\ \text{gal/conn/day})\ (\$4.00/\text{kgal})\ (365\ \text{days/yr})}{(1{,}000\ \text{gal/kgal})} = \mathbf{\$8.03\ /\ conn\ /\ year} \qquad \text{(Eq 2-8)}$$

As an example, for real losses, a water utility with a real loss unit rate of 35.0 gal/connection/day and a VPC of $800.00/mil gal gives a calculated real LCR of

$$LCR_{real} = \frac{(35.0\ \text{gal/conn/day})\ (\$800.00/\text{mil gal})\ (365\ \text{days/yr})}{(1{,}000{,}000\ \text{gal/mil gal})} = \mathbf{\$10.22\ /\ conn\ /\ year} \qquad \text{(Eq 2-9)}$$

The LCR indicators marry the volume of loss with the value of the loss into a single indicator, thus the technical and financial aspects of a given loss are represented in a single value. This makes the LCR a powerful indicator. The highest LCR values result from utilities that experience a high loss rate and a high cost of water. Conversely, some utilities have a high unit loss rate but only a moderate cost. In this case, the LCR may be moderate or low even if the loss rate is high and may suggest that only a moderate loss reduction is cost-effective. In addition to the apparent and real loss forms of the LCR listed in Table 2-26, the total LCR is calculated by the audit software version 6.0, and it is included in Table 2-27.

The LCR offers utilities, regulatory agencies, and other stakeholders a more comprehensive perspective on the apparent and real loss impacts to the water utility. Using the volume indicators along with the value indicators (LCR) gives water stakeholders a highly effective means of assessing the NRW standing of a water utility and great insight into the NRW management priorities.

Step 7-3. Validity of the water audit data. Data validity is determined by the Interactive Data Grading (IDG) capability of the Audit Software in assigning a grading to each of the input components. After all data are entered, the Audit Software calculates the Data Validity Score (DVS) and assigns the Data Validity Tier in which the water audit data resides. As stated previously, guidance is included in the Loss Control Planning worksheet of the Audit Software (Figure 2-8) on the interpretation and use of the Data Validity Tier. This guidance informs utility staff with respect to the quality of their data and how it can be used in an objective manner. The utility auditor should take note of the tier of the water audit and follow the guidance of the Loss Control Planning worksheet to assist their planning for NRW management activities.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 2-27 Range of Values of the AWWA Performance Indicators in the AWWA Water Audit Reference Dataset (WARD) that are embedded in the "Gauge" or "Speedometer" Charts in the Dashboard of the Audit Software (US dollars)

Statistic	Total Loss Cost Rate, $/conn /year	Apparent Loss Cost Rate, $/conn/year	Real Loss Cost Rate, $/conn/year	Unit Total Losses, gal/conn/day	Unit Apparent Losses, gal/conn/day	Unit Real LossA, gal/conn/day	Infrastructure Leakage Index (ILI, Dimensionless	Unit Real LossB, gal/mile/day
Min value	-$0.23*	-$0.60*	$0.00	9.325	-0.51*	7.926	0.7	266.97
10th %ile	$5.08	$0.27	$1.90	21.095	1.186	16.278	0.9	879.72
25th %ile	$9.33	$0.87	$3.73	29.481	2.753	22.722	1.2	1,289.37
Median (50th %ile)	$18.28	$6.15	$7.95	45.389	5.300	36.775	1.8	2,048.93
75th %ile	$31.58	$14.13	$16.29	76.232	9.355	66.379	3.2	3,639.61
90th %ile	$57.80	$24.23	$35.55	125.151	16.321	115.433	5.7	6,074.13
Max value	$480.55	$216.69	$479.66	689.518	200.968	657.597	18.8	26,060.14

*One system input CMI as overregistration, leading to a negative apparent loss volume

Step 7-4. Interpreting the Audit Software Dashboard. Performance indicators are presented on the Audit Software Dashboard shown in Figure 2-15, in gauge or speedometer format against ranges of data in the WARD (Kunkel et al. 2021). The WARD is a filtered dataset of calendar year 2018 water audit data of 1,124 water systems in the US states of Georgia and California and the province of Quebec, Canada. These states and this province have progressive water loss policy that includes a formal data validation process, making the data the most trustworthy of data available in North America at this time. The Audit Software gauges for each performance indicator span the 10th to 90th percentiles from the WARD data. A percentile separates the data into groups based upon the percentage delineation. A performance indicator value that exists at the 10th percentile level means that only 10% of the 1,124 values, or 112, are less than that value and 90%, or 1,012, are greater than that value. A performance indicator value that exists at the 75th percentile level means that 75% of the 1,124 values are less than that value and 25% are greater than that value. The median value is the 50th percentile value, meaning half of the values are less and half of the values are greater. It is noted that for most performance indicators, a lower value represents better performance than a higher value. The exception is the DVS, where a higher value represents better performance.

The range of data by percentiles for each performance indicator is shown in Table 2-27. The minimum and maximum value of each performance indicator is also shown. Referencing Figure 2-15 and Table 2-27 allows the reader to assess the performance indicators of CWC, the example water utility shown throughout this chapter.

As seen in Figure 2-15, six of the eight performance indicators for CWC are greater than the 75th percentile values of the WARD. CWC's $LCR_{apparent}$ of $20.64/conn/yr is between the 75th percentile value of $14.13/conn/yr and the 90th percentile value of $24.23/conn/yr shown in Table 2-27. This correlates well with CWC's similarly high Unit Loss Rate of apparent losses of 14.3 gal/conn/day, which approaches the 90th percentile value of 16.3 gal/conn/day in Table 2-27. CWC might set an incremental goal to reduce its apparent loss volume to achieve a Unit Apparent Losses designation equal to the 75th percentile value

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

of 9.3 gal/conn/day in the next 2 to 3 years. Since its $LCR_{apparent}$ is quite high, it is likely to be cost-effective to set this initial, incremental goal. Chapter 4 discusses the means to control apparent losses and methods to set effective targets for apparent loss reduction. These methods can be applied to the CWC example after the initial, incremental loss reduction goals are established and met.

CWC's LCR_{real} of $17.25/conn/yr is slightly greater than the WARD 75th percentile value of $16.29/conn/yr—less than its standing for apparent losses but still a high loss cost rate. CWC's Unit Real Loss values of 63.0 gal/conn/day and 2,999 gal/mile of pipeline/day are slightly less than the 75th percentile values of 66.3 gal/conn/day and 3,639.6 gal/mile of pipeline/day. Because CWC's LCR_{real} value is high, CWC might set an incremental goal to reduce its real loss volume to bring its Unit Real Losses of 63.0 gal/conn/day down toward the median or 50th percentile value of 36.7 gal/conn/day in the next two to three years. If this initial target is deemed to be too aggressive, an interim level such as a 60th percentile level might be established as the initial goal. Chapter 5 discusses the means to set effective targets for real loss reduction. These methods can be applied to the CWC example after the initial incremental loss reduction goals are established and met. Note that the higher-level functional focus areas, including long-term loss control, target-setting, and benchmarking, should only be conducted once the water audit data have achieved a Tier III level of validity or higher (see the Assess the DVS of the Audit Software section on page 2-18 and Task 9).

The AWWA performance indicators give a detailed means to assess NRW standing and set targeted goals for loss control, apparent and real. Distinct performance indicators exist for apparent losses and real losses, and the cost impacts of each. Thus considerations for reducing the volumes of losses can consider the projected cost savings of the planned reductions. These savings can be compared to the proposed costs to undertake the loss control, so an intelligent economic assessment can be conducted. The performance indicators adopted under AWWA's 2020 position, which excludes percentage performance indicators, represent a significant advancement in assessment methodology that gives water utilities outstanding means to monitor their NRW standing and implement cost-effective NRW management strategies.

The auditor can also utilize the AWWA Water Audit Compiler (Sayers et al. 2021) to assemble the most recent water audit with prior year water audits. This can be a useful tool for analyzing both audit inputs and outputs over time and to discern trends in supply, consumption, and water loss performance.

Task 8—Compile the Water Balance

After all data are entered into the Audit Software, water supply, consumption, and loss volumes are shown on the Audit Software Water Balance worksheet. The completed water balance for CWC is shown in Figure 2-2 and is automatically populated in the Audit Software. The water managed by CWC is tracked from left to right across the water balance, and each column volume sums up to 4,401.87 mil gal; hence all flows "balance." The water balance reflects that all water managed by the drinking water utility is accounted for in the various categories of consumption and loss.

It is recognized that by quantifying the volume of real losses as the residual volume that is left after subtracting authorized consumption and apparent losses from the water supplied volume, the data are forced to balance. The discussion under Step 3-3 notes that this does not necessarily represent a wholly accurate quantification of the real losses because inaccuracies in the quantities reported for water supplied, authorized consumption, or apparent losses could induce a degree of error in the real loss value. The reader is referred to the "Leakage Component Analysis" section in Chapter 5 to learn of the methods and tools to rigorously quantify and assess leakage volumes in a systematic way that guides strategic leakage management planning.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Task 9—Assess the DVS

After all data are entered into the Audit Software and KPis are reviewed, the auditor should review the DVS calculated for the water audit, the corresponding DVT, and reference the Water Loss Control Planning Guide worksheet for high-level guidance on next steps for utility, based on the DVT.

For CWC, the DVS q*of 65 is shown in Figure 2-15, which places CWC in Tier III of the Water Loss Control Planning Guide shown in Figure 2-8.*

With a DVS placing CWC in Tier III of the Water Loss Control Planning Guide, CWC has an intermediate level of data validity. Thus, it can place sufficient trust in its data and begin to strategize and launch loss control interventions in specific areas, such as customer meter testing, increased leak detection, and better assessments of UC. At Tier III, it can also begin to reliably use the KPIs to track its ongoing loss control performance from year to year and may begin to compare its data with other water utilities, preferably utilities that have characteristics similar to CWC. On the Interactive Data Grading tab, each input shows which question or questions (i.e., operational criteria) are limiting and therefore which specific areas the utility can focus on to improve its data validity.

At this stage, CWC has completed its water audit for the audit year. The manager for CWC can now proceed to plan and implement needed loss control interventions, particularly as described in Chapters 3 through 6 of this manual.

CONSIDERATIONS FOR WHOLESALE WATER SYSTEMS

Wholesale water systems and retail water systems serve different customers and employ distinct methods of distributing, metering, and billing for water. However, the principle of mass balancing to systematically account for all volumes of supply, consumption, and water loss applies to both wholesale and retail systems. Whether the wholesaler supplies large (bulk) customers or a combination of large (bulk) customers and some retail (end use) customers, it is possible for wholesalers in either scenario to compile annual water audits, and they are encouraged to do so.

All water systems, whether retail or wholesale, experience Apparent Losses and Real Losses. To manage water losses cost-effectively, wholesale systems and retail systems should first evaluate water loss volumes through a water audit.

As a best practice, the AWWA Audit Software version 6.0 can be adapted for use by wholesale water systems. As a result, completing a water audit with the Audit Software can provide a wholesale water system with the fundamental information necessary to proactively evaluate and improve system efficiency. An important additional benefit of regular water auditing by wholesale systems is the generation of data that will improve the quality of the annual water audits of the retail water systems served by the wholesaler. When a retail utility's import meter is under the exclusive control of a wholesale water supplier, both systems benefit from the wholesaler's assessment of flow meter accuracy and the documentation of any meter error adjustments, which can be quite impactful for the data validity of the audits of both the wholesaler and the retailer.

In order for the Audit Software to be standardized and methodologically sound when used by a wholesale system, these particular areas need to be given special consideration:

- how system boundaries and volumes inputs should be considered in the Audit Software,
- how default options should be handled in the Audit Software,
- how Interactive Data Grading questions should be interpreted to apply to wholesale systems, and

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- which audit outputs provide insight into wholesale distribution system performance.

System Boundaries

Regardless of whether water volumes recorded in the water audit were treated water, it is essential that distribution system boundaries and corresponding flowmeters are identified before beginning a wholesale water audit. This may require that site visits to key boundary flowmeters are conducted or that a single flowmeter from a series of redundant inline flowmeters is selected to bound and inform the audit. Additionally, the location of reservoirs, treatment processes, and treatment process flowmeters relative to the water audit must be assessed. A modified water balance graphic which includes untreated water supplies is given in Appendix B, which may provide additional insight into the process to set boundaries for wholesale water suppliers.

Volume Inputs

Like retail systems, wholesale systems acquire water from various sources, including surface water, groundwater, and imports from other systems. All sources of supply should be accounted for in the Water Supplied section of the Audit Software. Like a retail system water audit, all water that a wholesale system treats or conveys should be considered VOS. All treated water that a wholesale system imports should be catalogued as WI. Most wholesale systems should not allocate any volume to WE. Instead, metered exports and other authorized sales should be treated as BMAC. Like a retail system water audit, any known metering error or inaccuracy in registered VOS or WI should be incorporated as an Error Adjustment, either VOSEA or WIEA. See Appendix A for more information on assessing supply metering error. Because error can exist in either direction (over- or underregistration), caution is advised against arbitrarily inputting an error amount not based on testing activity or other reliable evaluation of the supply meter accuracy.

Categorization of Authorized Consumption for a wholesale system water audit will generally follow the same approach as a retail system water audit, inasmuch as a wholesaler must inventory forms of Authorized Consumption that are either metered or unmetered, billed or unbilled. Typically, a wholesale system will have a much smaller number of customer connections than a retail system. This can make the effort to compile these volumes relatively accessible. Any water metered and sold in bulk to another utility with its own customers typically should be included in BMAC.

Wholesale systems typically deliver water to customers through large flowmeters that are as likely to overregister as to underregister. Overregistering flowmeters may produce negative Apparent Losses, indicating that the volume of water registered and billed is greater than actual throughput. To quantify wholesale CMI, it is important to test flowmeters at their typical operating flow rates. Test results can be incorporated in the Audit Software as a CMI input (percentage or volume), with an accompanying designation of over- or underregistration. In the absence of test results, wholesale CMI can be difficult to estimate. Therefore, it is strongly recommended that wholesale systems institute meter accuracy testing for its flowmeters, as described in Appendix A. Wholesale systems that do not have test results are recommended to leave CMI blank in the Audit Software until a meter-testing program can be established where feasible.

Default Options

The Audit Software provides default options for UUAC, SDHE, and UC wholesale systems, the amount of physical infrastructure and count of customers is typically much smaller than retail systems relative to the amount of water delivered. Thus for most wholesalers,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

the true volumes for these three categories will be either zero, near-zero, or an amount considerably below the default-calculation (0.25% x Billed Authorized Consumption). It is recommended that wholesalers endeavor to produce system-specific estimates for these three inputs. In the absence of utility-specific estimates for any of the three default options, wholesale systems are advised to use a zero placeholder. As the Audit Software does not allow a true zero entry in either of these three inputs, this can be accomplished by entering a near-zero volume (i.e., 0.0001).

IDG Questions

IDG questions in the Audit Software are structured with prepopulated answer options, and the user is prompted to select the best-fit answer to each visible question. This premise remains true for a wholesale water audit. If none of the answer options aligns with the operational practices at the wholesale system, the user should simply select that answer that most closely aligns.

Generally, the IDG questions and answer options for most inputs are expected to be applicable and align with the realm of most wholesale utility operations. Considerations for two inputs are highlighted here that may be expected not to align with typical wholesale operations.

- CMI: The concept of CMI is applicable in a wholesale system. However, because the customer meter sizes in a wholesale system tend to be larger, IDG questions cmi.2 and cmi.3 which deal with CMI and small-size customers may not be applicable. Wholesale systems which do not have any residential meters are recommended to answer cmi.2 as "No testing conducted, but at least 10% of meter stock has been replaced within two years of the audit period." With this answer, cmi.3 will no longer be visible and will render cmi.2 and cmi.3 to be functionally irrelevant to the data-grading assessment.
- Lp: Generally, this concept is not applicable in a wholesale system. Additionally, the Lp input is used only in the derivation of the Infrastructure Leakage Index (as part of the UARL denominator), which is not a recommended performance measure for wholesale systems. Thus, wholesale systems are advised on the Audit Software Worksheet to indicate "yes" to the question "Are customer meters typically located at the curb stop/property line?" on the Audit Software Worksheet. This will render Lp functionally irrelevant to the data-grading assessment.

Water Audit Outputs

Many of the KPIs in the Audit Software provide limited insight for wholesale systems because the scale of infrastructure relative to the amount of water delivered is fundamentally different from retail systems. Additionally, a large dataset of validated benchmarks for wholesale systems is not yet available in North America, so comparisons of any metrics among different wholesalers will have very limited value.

Presented below are those measurements that can be useful for wholesale systems as they assess water loss standing and track performance over time.

- DVT is generally useful for high-level planning (per Loss Control Planning worksheet in the Audit Software), and limiting criteria flagged on the IDG questions are useful for mapping out specific next steps to improve data validity.
- Annual volume of Apparent Losses can be tracked over time to observe trends, progress toward goal, and effectiveness of interventions employed to reduce or optimize Apparent Losses.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- Annual volume of Real Losses can be tracked over time to observe trends, progress toward goal, and effectiveness of interventions employed to reduce or optimize Real Losses.
- Annual volume of Unbilled Authorized Consumption can be tracked over time to observe trends, progress toward goal, and effectiveness of interventions employed to more accurately quantify or reduce Unbilled Authorized Consumption
- Unit Apparent Losses (volume/connection/day), Unit Real Losses (volume/connection/day and volume/length/day), Real Loss Cost Rate ($/connection/ year) and Apparent Loss Cost Rate ($/connection/year) can provide limited value for tracking performance over time, with an understanding of the impact of Real and Apparent Losses at the unit level (connection or mile); however, comparisons of these KPIs against those of retail systems is not recommended.

SUMMARY

Water utility managers can assemble the top-down water audit by gathering records, data, and procedures from various operations routinely occurring in their provision of a water supply. The top-down water audit is largely a desktop exercise, with minimal field testing or investigations required. The advantage is that the top-down audit can be assembled relatively quickly and give a reasonable sense of the nature and extent of a utility's losses. It is extremely important that the water utility verify the accuracy of its source flowmeters and correct any gross malfunctions of these devices as part of the top-down process (see Appendix A). To refine the top-down water audit and formulate strategies to cut losses, work should then shift to the bottom-up approach. Over time, bottom-up activities should be pursued to better audit and control apparent losses (described in Chapters 3 and 4) and real losses (described in Chapters 5 and 6).

REFERENCES

Alegre, H.; Hirner, W.; Baptista, J.; & Parena, R. 2000. *Performance Indicators for Water Supply Services*. 1st Edition. Manual of Best Practice series. London: IWA Publishing.

Alegre, H.; Baptista, J.; & Cabrera Jr., E. 2016. *Performance Indicators for Water Supply Services*. 3rd Edition. Manual of Best Practice series. London: IWA Publishing.

Alliance for Water Efficiency Report: *Building Better Water Rates for an Uncertain World* (2014) available at www.financingsustainablewater.org Appendix A: Costing Methods

AWWA (American Water Works Association). 2019. Assessment of Performance Indicators for Non-Revenue Water Target-Setting and Progress Tracking. Denver, CO: AWWA

AWWA (American Water Works Association). 2018. Manual M33. *Flowmeters in Water Supply, 3rd Edition*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2012. Manual M6. *Water Meters—Selection, Installation, Testing, and Maintenance*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2023. Manual M22. *Sizing Water Service Lines and Meters*. Denver, Colo.: AWWA.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

AWWA (American Water Works Association). 2021. Water Loss Control Resource Community, https://www.awwa.org/Resources-Tools/Resource-Topics/Water-Loss-Control Accessed Mar 12, 2021.

Chastain-Howley, A. et al. 2014. *AWWA Free Water Audit Software (version 5.0).* Microsoft Excel. Denver: American Water Works Association.

Jernigan, W. et al. 2020. *AWWA Free Water Audit Software (version 6.0).* Microsoft Excel. Denver: American Water Works Association.

Kunkel, G., et al. 2003. Water Loss Control Committee report: Applying worldwide BMPs in water loss control. *Journal AWWA*, 95(8):65–79.

Kunkel, G.; Trachtman, G.; Wyatt, A.; & Jernigan, W. 2020. Water Loss Control Committee Report: Key Performance Indicators for Nonrevenue Water—AWWA's 2020 Position.

Journal AWWA. https://awwa.onlinelibrary.wiley.com/doi/full/10.1002/awwa.1428

Kunkel, G.; Jernigan, W.; & Trachtman, G. 2021. AWWA WLCC Water Audit Reference Dataset (WARD). AWWA. Denver, CO. https://www.awwa.org/Resources-Tools/Resource-Topics/Water-Loss-Control/Free-Water-Audit-Software

Lambert, A. 2020. *Low ILIs and Small Systems*. https://www.leakssuitelibrary.com/low-ilis-and-small-systems

Lambert, A.; Brown, T.G.; Takizawa, M.; & Weimer, D. 1999. A review of performance indicators for real losses from water supply systems. *Aqua*, 48(6):227.

Sayers, D.; Jernigan, W.; Chastain-Howley, A.; Kunkel, G. 2021. *AWWA Water Audit Compiler (version 6.0).* Microsoft Excel. Denver: American Water Works Association.

Sturm, R.; Jernigan, W.; Stief, C.; & Blackwell, D. 2021. *Level 1 Water Audit Validation Guidance Manual, 2nd Edition. Project #5057*. Denver: Water Research Foundation.

WRF (Water Research Foundation). 2011. *Accuracy of In-Service Meters at Low and High Flow Rates*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 3

The Occurrence and Impacts of Apparent Losses

Water losses include the water volumes that do not achieve beneficial use or that cost utilities a portion of the revenue to which they could otherwise recover. Water losses in drinking water utilities occur as two distinct types. *Real losses* are the physical losses from distribution systems and include leakage and overflows from storage reservoirs/tanks. *Apparent losses*, the subject of this chapter, are the nonphysical losses that occur when water is successfully delivered to a water user but for various reasons is not measured or recorded accurately. This inaccuracy introduces a degree of error in the amount of actual customer consumption. When apparent losses occur systematically in an appreciable volume, the aggregate measure of water consumption can be greatly distorted and cause significant revenue loss. Accurately quantifying and reducing apparent losses will improve results from the water audit.

Apparent losses are considered nonphysical losses in that no water is physically lost from the water distribution system due to pipeline, equipment, or operator failure. However, the metering, accounting, and data-handling inefficiencies that constitute apparent losses can have a significant impact. Apparent losses consist of three primary components:

1. systematic data-handling errors in tracking customer consumption, particularly in customer billing systems;*

* This component was established by the AWWA Water Loss Control Committee and does not explicitly exist in the definition of apparent losses established by the International Water Association (IWA) Water Loss Task Force in the IWA publication Manual of Best Practice, *Performance Indicators for Water Supply Services* (Alegre et al. 2000).

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

2. customer metering inaccuracies caused by faulty meters, improperly sized water meters, the use of the wrong type of meter for the application, or poorly designed meter settings that adversely affect meter accuracy; and
3. unauthorized consumption.

This chapter explains the causes of apparent losses and describes the significant impacts that these losses exert on consumption data integrity and revenue recovery in systems with metered customers. Insight into quantifying apparent losses is given in Chapter 2, while strategies and programs to economically control apparent losses are detailed in Chapter 4.

HOW APPARENT LOSSES OCCUR

Apparent losses occur because of inefficiencies in the measurement, recording, archiving, and accounting operations used to track water volumes in a water utility. The causes of apparent water loss can be both subtle or obvious depending on the experience of the utility since various assumptions must be made to quantify these occurrences. These assumptions can be verified as bottom-up investigations are conducted and the water loss control strategy develops. These inefficiencies can be caused by inaccurate meters, improperly sized or typed customer meters, poorly designed meter settings, poor meter-reading practices, and errors in the customer billing system.

Unauthorized consumption is another source of apparent loss. Common examples of unauthorized consumption include customers (or perhaps others) tampering with meters or meter-reading devices, illegally drawing water from fire hydrants, and unauthorized usage and/or creation of bypass lines. Unauthorized consumption may be negligible in some water utilities, but it could cause a significant volume of loss for others. For any type of apparent loss, it is incumbent on utility managers and operators to realistically assess metering and billing inconsistencies and then to develop both internal policies and periodic auditing programs to economically minimize these inefficiencies. It is also important to clearly communicate with customers, governing bodies of the utility and municipalities, financing agencies, and the media about the problem of apparent losses and the need to minimize them.

The specific ways in which apparent losses occur vary greatly and, particularly with unauthorized consumption, are always changing. Individuals who purposely choose not to pay for water do so for many reasons. Some may believe that water is a basic human necessity and therefore should be free of charge. Some may not necessarily have the financial resources to pay for the service, and others are always thinking of new ways to "beat the system."

The water utility must therefore be vigilant in its effort to manage its product (water) via effective meter management and rational billing, collection, and enforcement policies to realize projected levels of revenue from consumption of water and to maintain accurate measures of the water that it supplies.

Apparent losses may indicate errors in the billing process of a water utility: some of the water that reaches a user may be underbilled or perhaps not billed at all, and this error obviously will affect the utility's revenue. In many water utilities, a portion of the issued billings is not collected in a timely fashion—or perhaps at all—because some customers are either delinquent or even resistant to paying. The *collections rate* is a financial performance indicator that gives the rate at which customers pay their water bills. The collected payments are measured as a percentage of the money billed each cycle for the utility's services. Collections rates at the 30-, 60-, and 90-day delinquency milestones are often tracked to portray the customer payment record.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

While the collections rate is important to utility revenue, collections are not included in the AWWA water audit, which has as its terminal boundary the customer meter and billed water consumption. The AWWA water audit focuses upon utility billing efficiency. The collections process is affected by payment efficiency, however, which is beyond the scope of this manual. Water utilities are urged to track their collections rate and institute policies that maximize collections, but these processes are not discussed in this manual.

Systematic Data-Handling Errors

Not all water utilities meter the consumption of their customers. In North America, however, the use of water meters is standard practice for most water utilities. Meters are used to provide an accurate measure of the water passing through the customer service line. While the meter must provide an accurate measure of the water flow, the subsequent processes—including those to obtain and transfer meter readings (manually or automatically) to billing systems, data processing, and manipulation by billing systems, as well as archival and reporting operations—must also be handled accurately; otherwise, the actual customer consumption may be distorted or lost entirely for certain accounts. In many water utilities, it is not uncommon to find accurate meter data transposed erroneously, adjusted improperly, or incorrectly archived or reported. If any part of the data path lacks integrity, it is easy to misinterpret apparent losses solely as meter inaccuracy. If loss control decisions are based on a faulty assumption of meter inaccuracy, there are potentially costly consequences, such as replacing large numbers of accurate meters.

Because water utilities input metered consumption volumes into their customer billing system, the data and reports available from the billing system provide utility managers the information needed to assess the consumption patterns of the customer population. Thus, the customer billing system is the recommended starting point for systems to assess their overall water consumption accountability.

The reading generated by a customer water meter is the first step in an occasionally complicated trail that ultimately generates a large amount of customer consumption data. Because most water utilities manage data for many thousands of customers, systematic data inaccuracies can easily be masked by the sheer volume of the bulk data. Figure 3-1

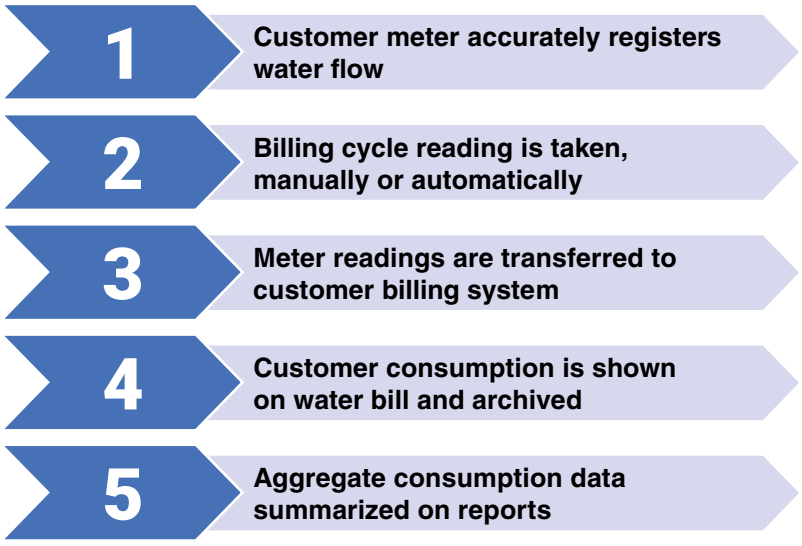


Figure 3-1 Metered consumption data archival path

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

gives an overview of the typical steps existing in the data trail from meter to historical archive. Errors in the data collection (transfer), billing, or archival and reporting processes can result in distortions in the summary data that are ultimately documented as customer consumption.

Obtaining water meter readings. Although the customer meter may record customer consumption accurately, the manner in which the meter displays and transfers meter-reading data can be prone to error. Many water utilities have a variety of meter types in their system with many made by different manufacturers. The same type and model of meter from a single manufacturer may have undergone design changes over several years, so even water utilities with a relatively homogeneous meter population may find important differences between individual meters. Meters used in most North American water utilities typically measure the volume of consumption in units of gallons or cubic feet, while metric units (cubic meters) are common in Canada and other parts of the world. In many instances, the unit of measure of the meter is not the same as the unit employed in the customer billing system. For example, a meter may measure the flow of water in gallons and totalize the flow in hundreds of gallons with billing to the customer shown in hundreds of cubic feet.

The granularity of the consumption data refers to the unit of measure employed by the meter. A smaller unit of measure is considered more granular; hence, using the unit of gallons is considered more granular than using cubic feet because one gallon of water is a smaller volume than one cubic foot of water (1 gal = 0.134 ft^3). The smaller, more granular volume unit is considered more accurate for water loss control purposes than a larger, coarser volume unit of measure. Residential static (electromagnetic and ultrasonic) water meters may be ordered to read in units as small as hundredths of a gallon or thousandths of a cubic foot.

The volume displayed or recorded by the meter may not be the volume that is transferred to the customer billing system by an automatic meter reading (AMR) or advanced metering infrastructure (AMI) system. AMR and AMI are technologies that provide electronic readings of customer water meters and transfer the data to a central location. These technologies offer more capabilities and efficiencies than manual meter reading. Either mechanical or electrical failures in older mechanical meter registers could generate errors, and some of the original AMR systems were not configured to capture the data from every digit on the meter register. Typical mechanical meter registers contain six-digit totalizers, but older reading technology may have recorded only the first four of the six digits. The precise totalizer digits of a given meter being read can vary considerably, depending on meter size, meter-reading technology, and meter manufacturer. Current production registers for mechanical meters have virtually eliminated those sources of both mechanical and electrical errors, and they have been proven to be very accurate instruments. Static meters transmit the usage data directly from their onboard computer. Moreover, many current production registers for residential mechanical meters offer up to eight-digit output, and many residential static meters offer nine-digit output.

Water utilities use a variety of methods to read meters. These can include manual (visual) reading, reading by touchpad or handheld recording device, mobile (drive-by) radio reading, and a variety of mobile AMR and fixed-network AMI technologies, with fixed (communication) network features of AMI technology being the most comprehensive data collection method. Many water utilities have in place a mix of the above methods, as they either transition from one technology to another or target specific regions, such as pilot project areas, or by deploying a hybrid network with an AMI system for urban and suburban areas with their higher population density and deploying a mobile AMR system to read meters in rural areas.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Historically, water utilities conducted manual meter reading with readings visually gathered and then written on paper cards or forms. Mobile AMR technology became both affordable and reliable by the 1990s, and many water utilities deployed fixed-network AMI systems by the middle of the first decade of the 21st century. For those utilities that still employ paper meter-reading forms, however, the opportunity for transposition errors still exists. Utilities using AMR/AMI technology typically have software gathering the data from the meters with a programmed interface uploading the data to the customer billing system.

Errors can still occur in AMR/AMI systems if programming errors exist in the data transfer interface, in the billing system, or in the programming of the water endpoint. Very common errors include entering the wrong meter size, entering the wrong multiplier for billing, and reading only one register on either a compound meter or a fire service meter with two registers. Another common error is assigning a meter reading to the wrong customer premises. Also, AMR/AMI equipment can fail, and the resulting estimated readings for billing purposes can be a potential source of error. Weak procedures for data continuity during meter replacement can also create errors. Additional information on the features and considerations in using AMR/AMI systems is provided in Chapter 4.

Data transfer errors occurring in the meter-reading process. The process of obtaining readings from the meters and then transferring them into the customer billing system can be rife with the potential for creating apparent loss if absent strong written procedures and careful oversight. Some of the ways in which the integrity of customer consumption data may be compromised from poor data transfer practices include the following:

- *Manual meter-reading misread.* This can occur due to poor lighting, flooded meter vaults or pits, dirt/debris obscuring the meter register, damaged meter registers, and poorly trained meter readers. Utilities that have a wide variety of meters (and registers) from different manufacturers may cause confusion for meter readers, even when the meter register is clearly visible. When a manual reading is taken to check the validity of data in the billing system, field personnel may perpetuate past incorrect assumptions of the units or totalizer dials to read based upon a printed history of incorrect readings from the billing system. Manual meter-reading errors can also occur when all dials on a meter totalizer are not read, particularly when the dials being read vary among meters in the same system. Figure 3-2 shows a water meter register that cannot be read manually because of moisture that has accumulated inside the meter register. Finally, it is not uncommon for



Figure 3-2 Accumulation of moisture inside a water meter register obscuring the dial and preventing visual meter readings

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

meter readers to enter an approximate meter reading rather than gather an actual reading. This might occur when the meter reader is avoiding entry to a hard-to-access meter location or is reluctant to deal with harsh weather, excessive traffic, or other challenges of the manual meter-reading process.

- *Manual meter-reading data entry and transposition errors.* An accurate reading can be transposed incorrectly, particularly if readings are handwritten on paper cards or forms. Errors can also occur when the correct numbers are entered in the wrong columns. This often happens in systems that do not record every dial on meter totalizers. Readings may also be entered incorrectly into handheld devices, although handheld devices that read directly from the meter/AMR equipment avoid this problem. Sometimes a correct meter reading is entered, but an incorrect code is entered for the account (such as mistakenly entering a "property vacant" code).
- *Manual meter-reading masquerading as an automatic meter-reading.* Utilities employing plug-in type automated meter reading technology or drive-by radio read technology can experience this type of error. The technology used may require or allow field employees to read the data on a display and later enter the displayed data into a field data logger or directly into the customer billing system. Meter readers may keep a handwritten list of displayed reads and enter these at the end of their shift, incorrectly believing that this practice increases efficiency. Aside from the possibility of errors in entering the numbers, this is also a possibility of entering a valid reading into the wrong customer account. Office staff may or may not be aware of this practice, and the coding in the customer billing system will indicate that the reading was obtained automatically, making these types of errors particularly difficult to detect.
- *Automatic meter-reading error or equipment failure.* Most AMR/AMI systems and equipment have been proven in the field to be very accurate, but none is infallible. Meter-reading devices connected to customer meters periodically fail. Water utility managers and AMR/AMI equipment providers should ensure that these systems have alerts that immediately inform personnel when equipment failure has occurred. Similarly, utilities must have procedures in place to ensure a timely investigation and either repair or replace defective AMR/AMI system equipment.
- *Improper meter register units or truncation factors erroneously programmed into meter-reading equipment.* Utility managers should be very careful to ensure that the proper units and factors are entered into meter-reading equipment. This is particularly important if many different meter types and brands exist in the meter population.
- *Errors due to mixed meter-reading type, frequency, and granularity.* Many water utilities, particularly those piloting AMR/AMI technology, have a mix of manual and automatic meter reading processes in place. In such cases, procedures should exist to ensure that meter readings are entered into the customer billing system accurately and according to the appropriate billing cycle. If a portion of accounts are read and billed on different cycles (some monthly, some quarterly), the potential exists for data transfer errors. The granularity of meter-reading volumes can also create the potential for data distortion. Meters reading in gallons but billed in units of cubic feet risk the potential for data disruption if sufficient controls are not in place to ensure that the correct unit conversion programming is in place. The greater the variety in meter-reading and billing mechanisms, the greater the need for careful management of metering and billing data.
- *Procedural/data entry errors during meter replacements.* When a meter is replaced, the reading on the new meter's register should be used to initiate a new billing

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

sequence. If the final reading from the former meter and the reading from the new meter are not properly reconciled at this point, then a distorted consumption value may result.

Data analysis errors in the archived data or in management reports. Once valid data on customer consumption are reliably transferred into the customer billing system, the potential remains for additional systematic data-handling errors to be introduced into the data path. Some of the ways in which the integrity of customer consumption data may be compromised from data analysis errors include the following.

- *Use of poorly estimated volumes in lieu of meter readings.* When a meter reading cannot be obtained or the gathered reading is in obvious error, it is common practice to enter a reading value that provides a representative estimate of the customer's consumption. The procedure for estimating meter readings is typically based on either an average of historic consumption or a similar period from the historic database. When estimates occur on multiple billing periods for the same billing account, the likelihood of error in billed consumption compared to registered consumption can grow.
- *Customer billing adjustments granted by manipulating actual metered consumption data.* Customer billing systems are typically designed with a focus on the financial aspects of the customer relationship—issuing bills and maintaining cash flow to the utility. The dual role of the consumption volume as an operational or engineering parameter may or may not have been considered in the billing system design. As part of standard financial transactions, water utilities need the ability to periodically issue a monetary credit to a customer just as they have the ability to issue a charge. A credit is typically issued when a valid, actual meter reading is obtained after several periods of billed consumption based on poorly estimated volumes. Some billing systems implement credits by adjusting meter readings or consumption volumes to create a negative charge. A large one-time adjustment to a customer account, however, can cause a distortion of the actual consumption. This is particularly disruptive if the adjustment accounts for a long period that spans reporting years. See Step 3-2A2 in Chapter 2 for a detailed example of this issue.
- *Poor customer account management.* Weak procedures for closing accounts and for transferring ownership of meters and premises can lead to accounts not being properly activated and meters not being read.
- *Aggregate data error in the archival and/or reporting processes.* The annual water audit relies on metered consumption reported from the customer billing system for aggregated groups of customers in the audited system. For most water utilities (perhaps other than those employing fixed-network AMI technology), meter readings are collected on different days throughout each meter-reading cycle. A reporting procedure is typically used to collect and manipulate the actual meter-reading data from different days to aggregate it across the billing period. The data are further aggregated into quarterly and annualized data in the archives and shown on various management reports. The frequency of meter readings can affect the accuracy of annualized data, as the monthly data will need less manipulation than quarterly data. The granularity of the meter readings can also affect the accuracy of annualized data, as readings taken in the smaller measure of gallons are more accurate than volume based on billings in the larger measure of cubic feet.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Policy and procedure shortcomings. Poorly conceived or implemented policies can allow apparent losses to occur, often without the impact being realized. Longstanding practices ("That's the way we've always done it!") are often prevalent instead of structured procedures that include accountability and transparency. Such shortcomings may be illustrated by the following examples.

- Despite policies for universal customer metering, certain customers are intentionally left unmeasured or unread—common for municipally owned buildings in water utilities run by local governments.
- Provisions that allow certain customer accounts to enter "unbilled" status, which often becomes a loophole that is exploited by customer fraud or becomes a source of unintended loss if the water utility is not rigorous in its oversight and auditing of these accounts.
- Billing adjustment policies that do not take into account preservation of actual customer consumption lead to apparent losses.
- Bureaucratic regulations, performance lapses, or organizational silos/tensions can cause delays in permitting new accounts, installation of water meters, or timely activation of billing functions.
- Poor preparation, procedures, and accountability affect losses, especially during the transition to a new customer billing system and software.

The above list provides but a few of the major data-handling problems that might be encountered in a water utility. It is not exhaustive, however, and any utility might identify an apparent loss situation that is unique to the organization. Any action that unduly reduces the actual volume of customer consumption can be considered as contributing to apparent loss.

Customer billing system. The customer billing system is a standard feature of water utilities. The implications of the use of customer meter reading and billing data on the financial and engineering functions of the water utility are detailed in the "Using the Customer Billing System to Extract Customer Water Consumption" sidebar in Chapter 2, and the reader is urged to carefully review this discussion.

The customer consumption data-gathering process, from meter installation and meter-reading through billing, can be flowcharted (see Chapter 4) and confirmed by the utility to determine the potential for apparent losses from systematic data-handling errors. This type of error can be subtle and detectable only after appreciable investigation. However, corrections can often be implemented quickly and inexpensively, sometimes requiring only minor procedural or programming changes in the billing system software. In pursuing these types of apparent loss recoveries, a fast and cost-effective payback can often be attained from the resulting additional revenue recovery. Addressing systematic data-handling errors early in the water loss control program also creates a foundation of data integrity that is essential as the loss control program matures.

Customer Meter Inaccuracy

Customer metering in North American water utilities. AWWA's policy statement on metering and accountability supports metering of water production flows and customer consumption, and this discussion is in the context of utilities having fully metered customer populations. Metering provides water consumption data that is used for long-term planning and to evaluate nonrevenue water (NRW) management and conservation programs. It also serves to elevate the value of water in the mind of the consumer by linking a price with a volume. The growing use of AMR and AMI systems is generating consider-

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

able customer consumption data, giving utilities greater insight into the use and waste of water occurring in their operations.

Detailed guidance on customer meters is given in AWWA publications, including Manual M6, *Water Meters—Selection, Installation, Testing, and Maintenance,* which covers customer meter management (AWWA 2022). AWWA Manual M22, *Sizing Water Service Lines and Meters* explains customer-demand profiling and sizing criteria that are critical for meter accuracy (AWWA 2022). The Water Research Foundation publication titled *Advanced Metering Infrastructure: Best Practices for Water Utilities* covers planning, installation, and use of AMR and AMI systems (Schlenger et al. 2011).

Not all water utilities meter their customers. When auditing an unmetered utility, meter inaccuracy cannot be evaluated as an apparent loss, and the utility must employ other methods to quantify the annual volume of customer consumption and separate this volume from authorized consumption and water losses. One approach is to estimate customer consumption by metering and data-logging a representative sample of customer accounts and statistically evaluating the results to infer general customer consumption trends.

Customer meter inaccuracies can be a major source of apparent loss in drinking water systems. Generally, meter accuracy is influenced in four principal ways: the physical accuracy of the meter as a flow measuring device, the appropriate sizing of the meter to fit the customer's consumption profile, the appropriate type of meter to best record the variations in flow, and proper installation of the meter.

Physical accuracy. To assess whether meters are functioning properly, it is recommended that the water utility own, install, and test customer water meters as part of an ongoing program. As volumes of water pass through a mechanical meter, it will eventually wear out and lose accuracy—some more quickly than others. Water quality also has a direct impact upon mechanical meter accuracy. Therefore, mechanical water meters must be tested and replaced with new or refurbished meters. Static water meters tend to hold their accuracy throughout the service life of the meter, but it is still prudent to test a statistically valid, random sample of static water meters annually. All meters—even meters newly received from the manufacturer—can be expected to register flow with varying degrees of accuracy. The suggested best practice per AWWA Manual M6 is to test new meters as complete assemblies before installation in a customer service line (AWWA 2022).

Appropriate meter sizing. Approaches to meter functionality and management have advanced in ways that promote greater accuracy of customer consumption measurements. In the past, it was common to size customer service connections and meters based upon the peak flow rates that the meter was expected to encounter. Because peak flows occur only rarely, most of the time, meters register flows at the low end of their design range. Many meter types—especially mechanical meters—are less accurate at the low end of their flow range, with very low flow usage rates not accurately captured at all. Current design philosophy focuses on the flow range most usually encountered and not on seldom-occurring peak flows.

Many water utilities have recovered considerable water revenue and reduced apparent water loss from rightsizing oversized customer meters. In the early 1990s, for example, the Boston Water and Sewer Commission's meter-downsizing program recovered more than 100,000 ft^3 of additional water per day in apparent loss, which translated into millions of dollars in subsequent additional revenue (Sullivan and Speranza 1991).

Conversely, meters that are significantly undersized for the flow profile that they serve will pass much higher flows of water than they are designed to accommodate, and the result is often a rapid degradation of meter accuracy for mechanical meters and no known long-term effects for static meters. Utility managers should monitor their billing data to flag billing accounts with unusually high consumption for the existing sized meter

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

and test these meters with additional frequency. Meters that are confirmed to be undersized should be replaced with a larger meter.

Appropriate type of meter. Many meter types exist to measure flows in a wide variety of settings. Displacement meters historically have been most common for smaller residential service, although multijet and single-jet meters are also used in this application. Compound, turbine, single-jet, and propeller meters are examples of the types of meters employed to serve large commercial or industrial connections of greater than 1 in. The use of ultrasonic and electromagnetic flowmeters is increasing in all types of customer applications. Fire service meters have unique features for this particular type of service. Metering technology is constantly advancing, with new types of meters and metering capabilities being developed.

Utility managers should carefully evaluate the performance specifications of water meters that are being considered for particular applications and select those with a range of accuracy that best meets the flow profile of the proposed use. For instance, an irrigation-only account that serves a large area could use a meter with a low-flow threshold that is relatively high and still provides accurate registration. This would be the case since this type of account typically consumes water at a relatively high rate of flow for nearly 100 percent of the time that water is used. Conversely, a large residential property that has high periodic irrigation demands for its landscaping needs might also encounter numerous times when only relatively low domestic flows are encountered. This type of account requires a type of meter that is highly accurate over a wide range of flows, from low flow to high flow. The cost of the meter tends to rise as the range of accuracy expands, but the long-term recovery of apparent loss costs generally justifies the additional expense.

Correct installation. Meter installation practices directly affect meter accuracy, especially for inferential type meters (e.g., multijet, single jet, and turbine). All meter manufacturers provide specifications and guidance for the proper installation of their water meters, and water utilities should follow this guidance. Most water meters should be installed in the horizontal position, with adequate upstream and downstream spacing and appurtenances (strainers) as required. A water meter should be installed in a space free from the extremes of weather and other stresses. Figures 3-3 and 3-4 show water meters that have not been properly installed and therefore may suffer compromised accuracy and performance. Meters should be inspected upon receipt of delivery to ensure that any damage in transit is identified and these meters are not installed in a customer site.



Figure 3-3 A water meter installed upside down is an example of a poorly installed meter; water meters should be installed upright and usually in the horizontal position

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Figure 3-4 A bank of water meters showing numerous meters improperly installed at an angle instead of upright and horizontal

MONITORING METER PERFORMANCE

Although most commercially available brands of water meters are highly accurate and reliable, it is incumbent upon the water utility to regularly monitor customer consumption data to flag potential problems from meter malfunction or outright failure. Meter performance may be compromised in several ways. Here are two common examples.

- *Frozen* or *stuck* meters fail to advance, which means that zero consumption is recorded for the meter-reading period. This condition usually reflects a complete failure of the meter (unless tampering with the meter is confirmed as the cause).
- *Intermittent performance* of some meters results in underregistration from the meter running "slow" or in interrupted fashion. This can happen due to cold temperatures, sediment in the water, or malfunction of the meter apparatus. Meter readings from intermittently performing meters typically show a pattern of low consumption rather than zero consumption. The reading history in the billing system may show a remarkably consistent low-consumption level from period to period, and poorly designed meter consumption "flags" or alerts in the customer billing system may not identify these underregistering meters.

The use of water meters to measure customer consumption is a cost-effective way for water utilities to track the end use of the water supply and properly bill customers based on the volume consumed. However, water utility managers should be proactive in ensuring that meters are properly specified, sized, installed, and monitored to provide this service to the water utility. Lapses in this process result in apparent losses due to customer meter inaccuracies for the water utility.

Unauthorized Consumption

Unauthorized consumption can occur in any drinking water utility. It often occurs through the deliberate actions of customers or other persons who take water from the system without appropriately metering the use and paying for the water. The nature and extent of unauthorized consumption in a system will depend on the combination of

- the economic health of the community;
- the value the community accords to water as a resource, often as a function of the relative abundance or scarcity of water in the region;
- the retail cost charged to customers for water service;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- the strength and consistency of the enforcement policies and practices existing in the water utility; and
- the political will of water utility management and public officials to enact and enforce effective policies to thwart unauthorized consumption.

Establishing features of a good NRW management program—water auditing being foremost—will inevitably uncover situations where unauthorized consumption is occurring. Unauthorized consumption might occur as

- illegal connections;
- open bypasses around large customer meters;
- illegal bypasses around residential meters;
- buried or otherwise obscured meters;
- misuse of fire hydrants and firefighting systems (tapping connections to unmetered fire lines);
- vandalism of consumption meters (meter tampering);
- tampering with meter-reading equipment;
- illegally opening intentionally closed valves or curb stops on customer service piping that has been shut off for nonpayment;
- illegally opening intentionally closed valves to neighboring water distribution systems designed for emergency or special use;
- failure to notify the water utility to activate a billing account after water use has been initiated;
- fabricated meter-reading data from a meter reader acting in bad faith to generate an illicit discount for a customer, perhaps in return for an illegal payment or "kickback"; and
- inadvertent tapping by one system of an adjacent water system's main(s).

The water audit should quantify the component of unauthorized consumption occurring in the utility. The water utility auditor has the opportunity to quantify a volume of unauthorized consumption directly or to apply the default value in the AWWA Free Water Audit Software. For water utilities with well-established water audits or those believing that unauthorized consumption is excessive, the extent and nature of unauthorized consumption should be investigated through inspection of suspect customer accounts. Policies and practices that may unwittingly create opportunities to manipulate metering equipment to reduce or avoid payment should also be researched. As an example, the utility manager could institute a routine auditing function to investigate a sample of active customer accounts that register zero consumption for consecutive billing cycles. Accounts with meters being tampered might be identified during these inspections. As another example, the utility manager might periodically inspect fire connections in large commercial/industrial buildings within the service area. These inspections could detect the presence of illegal piping tapped into an unmetered fire line. Further ways to detect and control unauthorized consumption are discussed in Chapter 4.

Recognizing that a portion of customers in any region live with real economic hardship, the water utility may choose to operate programs offering affordable rate structures, discounts, grants, or similar services to qualified customers to keep essential water service affordable. Having such a program working in tandem with persistent unauthorized consumption enforcement is the best policy. It is appropriate that water utilities recognize the limitations of certain customers in verifiable need and offer them an avenue to legitimately purchase water service at affordable rates. Some utilities have established a separate tier of "life-line" water rates to apply to this subpopulation of customers.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

PROBLEMS CREATED BY APPARENT LOSSES

Because apparent losses underrecord the volume of customer consumption, they generate two major impacts on water resources management:

1. Apparent losses introduce a degree of error into the quantification of customer water demand, thereby impacting the decision-making processes used to determine needed source water withdrawals, calculate the appropriate capacities of water supply infrastructure, and evaluate conservation practices.
2. Apparent losses cause water utilities to underbill a portion of the customer population and suffer a loss of revenue potential.

Both impacts can be significant. If a high level of apparent loss exists in a water utility, its recorded volume of customer consumption could be subject to a significant degree of error. Consider a water utility that documents customer consumption of 3.65 billion gallons of water in a year (10 million gallons per day [MGD]). If routine water auditing found apparent losses equal to 1 MGD, actual customer consumption during the year being audited was 4.015 billion gallons, an increase of 365 million gallons. Such a loss creates a distortion of the true customer consumption volume, in this case understating it by 365 million gallons. Activities that rely on accurate customer data are compromised by this degree of error. These can include efforts to evaluate the success of water conservation programs, using consumption data to assign demands in hydraulic models, and evaluating community drinking water requirements needed for regional water resource plans. Apparent losses, therefore, represent a degree of error that is interjected into a wide range of analytical and decision-making processes regarding water resources management. Figure 3-5 shows schematically how uncontrolled apparent losses create a growing error in the volume of water needed to supply customers. Given that the water industry in the United States is highly fragmented with many different-sized water utilities existing in any given region, the degree of error from apparent losses can be compounded by the varying errors existing in many disparate water utilities. Gauging true customer needs on a regional basis can be difficult without a reasonable assessment of the apparent losses existing in the region's water utilities.

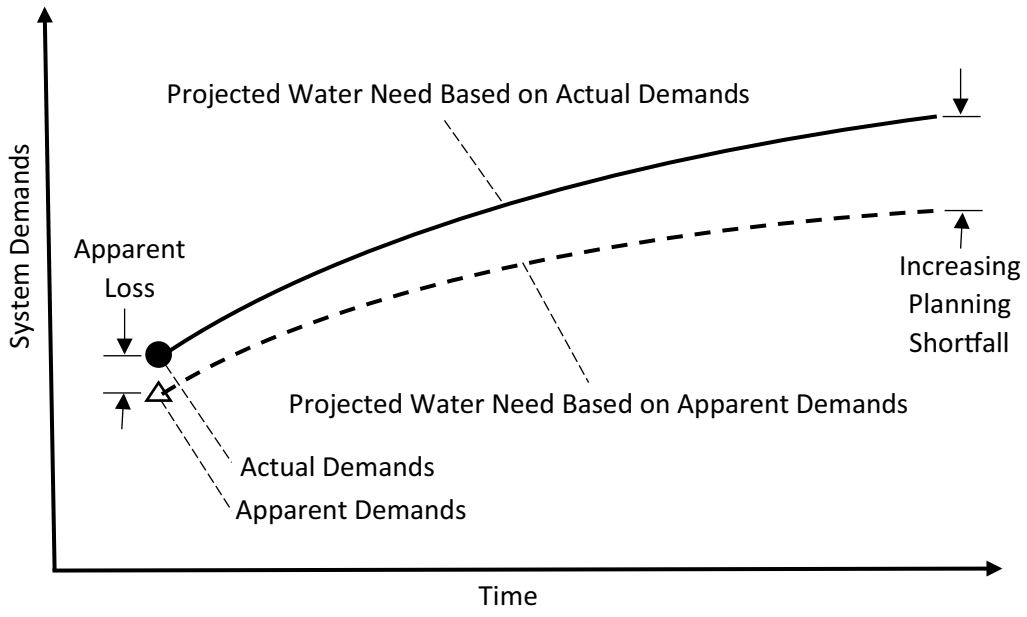


Figure 3-5 The impact of uncontrolled apparent losses in a water utility over time

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

From a financial perspective, apparent losses can exert a tremendous impact on the water utility's bottom line. The AWWA Water Audit Reference Dataset (WARD) includes validated water audit data from 2018. The WARD data from 466 water utilities from the states of Georgia and California found a collective apparent loss cost impact of over $174 million for these systems. This represents significant revenue loss for these systems that stand to gain from the revenue recovery of apparent loss control. Because apparent losses occur at the customer's delivery point, this water is valued at the retail cost that is charged to the customer. Water rates in many utilities include a water volume-based wastewater charge in addition to the volume charge for water supply, thus elevating the importance of accurate billing. The absolute costs of apparent losses are frequently higher than the absolute costs of real losses in many utilities when real losses are valuated based upon the variable production cost. For most water suppliers, the retail rate charged to customers is higher than the variable production costs. Therefore, apparent losses can have a dramatic financial impact on the water utility's revenue stream and overall financial condition.

The revenue impact from apparent losses also creates a problem of equity for the community. When apparent losses occur, a portion of the customer population obtains discounted or free water service. This means that the paying customer population effectively subsidizes those customers who are underpaying or not paying. This situation is particularly troubling as water utilities encounter pressure to raise water rates with paying customers shouldering an even greater financial burden for the entire water-using community.

Reducing apparent losses and recovering missed revenue can reduce the frequency of water rate increases by identifying underpaying and nonpaying customers and adding them to the active billing rolls. The recovery of apparent losses can create a direct financial improvement to the water utility, and many apparent losses can be recovered with relatively little effort and expense. This is important in terms of seeking early success and payback to the water loss control program. Funds recovered early in the program in this manner can serve to seed further activities in the long-term water loss control effort.

SUMMARY

An assessment of the volume of apparent losses gives a more realistic picture of the actual customer demand of the community and a preliminary measure of the distribution system efficiency. The reliability of the estimate of apparent losses has bearing on all quantitative aspects of accountability and the loss control program. Additionally, apparent losses exert a strong financial impact to the water utility by inhibiting revenue capture for a portion of the customer population. Addressing apparent losses, particularly early in the water loss control program, creates the potential for strong payback and a springboard of success in expanding the water loss control program to address real losses through improved leakage management. Chapter 4 discusses the specific means to economically control apparent losses.

REFERENCES

Alegre, H.; Hirner, W.; Baptista, J.; & Parena, R. 2000. *Performance Indicators for Water Supply Services*. Manual of Best Practice series. London: IWA Publishing.

AWWA (American Water Works Association). 2022. Manual M6, *Water Meters—Selection, Installation, Testing, and Maintenance*. Denver, Colo.: AWWA.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

AWWA (American Water Works Association). 2022. Manual M22, *Sizing Water Service Lines and Meters*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2021. Water Audit Reference Dataset (WARD). Denver, Colo.: AWWA.

Schlenger, D.L.; Hughes, D.M.; & Green, A. 2011. *Advanced Metering Infrastructure: Best Practices for Water Utilities*. Denver, Colo.: Water Research Foundation.

Sullivan, J.P., & Speranza, E.M. 1991. Proper meter sizing for increased accountability and revenues. In *Proceedings, AWWA Annual Conference and Exposition*. Denver, Colo.: American Water Works Association.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 4

Controlling Apparent Losses: Optimized Revenue Capture and Customer Data Integrity

Apparent losses occur when water is delivered to a water user but for various reasons is not measured or recorded accurately, thereby inducing a degree of error in the amount of actual customer consumption. When apparent losses occur systematically in an appreciable volume, the aggregate measure of water consumption can be significantly understated, leading to under-billings and resulting in revenue loss. Apparent losses consist of three primary components:

1. Systematic data handling errors (SDHE) in tracking customer consumption, particularly in customer billing systems or customer information systems (CIS);
2. customer metering inaccuracies (CMI); and
3. unauthorized consumption (UC).

The nature and causes of apparent losses are explained in detail in Chapter 3. Chapter 4 takes the utility to the next level by defining the cost-effective means to address apparent losses and control them.

There is a tendency for many to assume that their system's apparent losses are caused solely by customer metering inaccuracies and that the replacement of large numbers of supposedly faulty meters is the best remedy. It is an expensive and possibly unproductive proposition to implement a customer meter replacement project if the bulk of the apparent losses is caused by billing system errors or unauthorized consumption.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

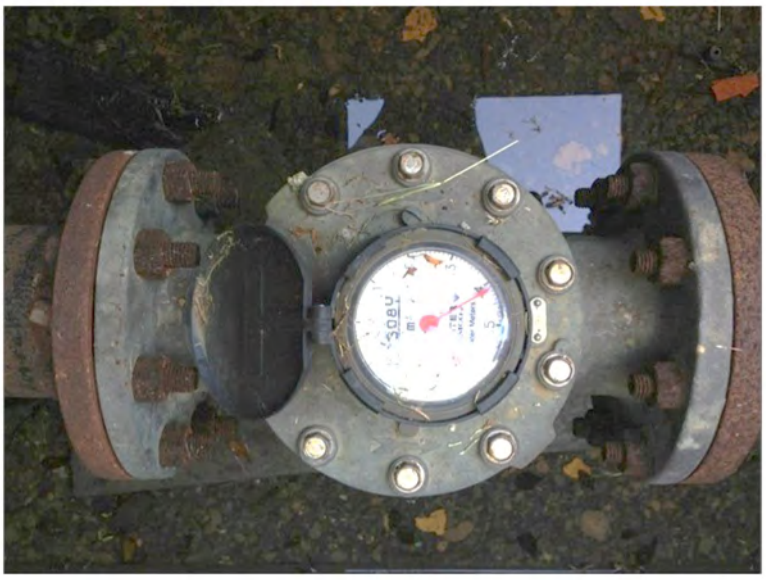


Figure 4-1 This meter was listed as a 3-in meter in the utility billing system software, but a quick field audit revealed it to be a 4-in turbine meter (identified by the eight flange bolts). There is also a small brass tag with "4-in" riveted to the body next to the register. The total monthly base charges for a 3-in meter at this utility are $598.49, but the total monthly base charges for a 4-in meter are $1,042.33. Thus, the utility was losing $5,326.08 a year in unbilled base charge revenue from this one meter.

Before setting an apparent loss control strategy, the water utility should first undertake the water audit process detailed in Chapter 2 that quantifies the three components of apparent losses and their cost impacts. The auditor should compile a representative water audit and identify the nature, quantity, and costs of the apparent loss components and only then develop a rational loss control strategy. Planning the apparent loss control strategy according to the results of the water audit is always the best way to proceed.

Investigating and fully understanding the customer meter-reading and billing process is the recommended first step in the apparent loss control strategy once the water audit has been completed. A billing system evaluation gives insight to the type and extent of apparent losses occurring in the utility. Data-handling error, such as the miscoded meter size shown in Figure 4-1, is an example of apparent losses that can be uncovered during billing system analysis. Data-handling errors revealed in the meter-reading and billing process evaluation are often quickly addressed by relatively inexpensive computer programming or procedural improvements. In this way, a quick payback can be earned by additional revenue recovery.

QUANTIFYING APPARENT LOSSES IN THE WATER AUDIT

Chapter 2 details the process used to assemble the best practice water audit. Once the water audit is assembled, the utility manager has a good initial sense of the quantities of apparent loss components. The approximate total volume of apparent losses and their effects on revenue shortcomings can be used as the basis to begin to develop an apparent loss control strategy. To refine this strategy, however, the auditor should begin to perform more detailed investigations of the source data and metering/billing activities to validate

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

the preliminary data and obtain a more accurate picture of the apparent losses. These "bottom-up" auditing practices involve launching more detailed investigative auditing work, like the detailed financial audits that accountants perform. The distinctions between "top-down" and "bottom-up" auditing approaches are explained in the "Water Auditing: Top-Down vs. Bottom-Up Approach" sidebar in Chapter 2.

This chapter focuses on the bottom-up water auditing and targeted loss control functions that utilities should employ to address apparent losses. The water utility manager can consider undertaking the steps listed in Figure 4-2 as the primary bottom-up activities to expose apparent losses in great detail.

Additional activities that can be conducted in each of the steps listed in Figure 4-2 are described in this chapter. Again, it is recommended that the billing system analysis always be performed as the first step because gaps in this process could affect the data that are evaluated in the other steps. Figure 4-3 provides the aggregate results of meter-reading and billing system evaluations conducted for 32 water utilities by a consulting firm over 25 years. Figure 4-3 displays estimated apparent water losses in gallons (gal) for the 32 water audits broken down into categories of SDHE (lost accounts, meter-reading error, and reclaimed water billing error) and UC (open bypass valves around meters of large water customers).

Figure 4-4 shows one type of unauthorized consumption that can be confirmed in field inspections of customer water meters. As shown, a customer water meter has been removed in an unauthorized manner and replaced with a straight pipe to prevent water consumption from being measured at a customer property.

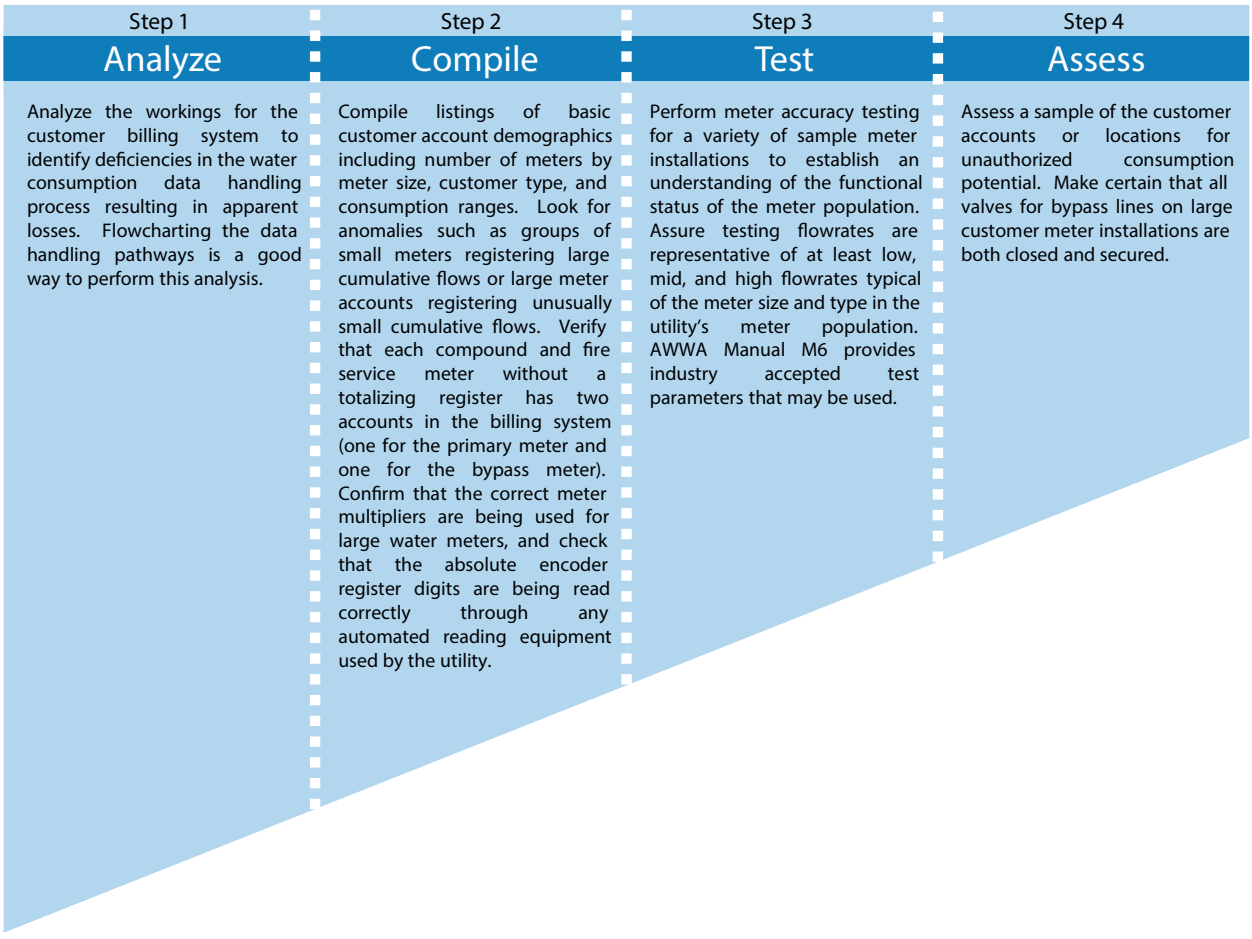


Figure 4-2 Primary activities for reducing apparent water loss

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

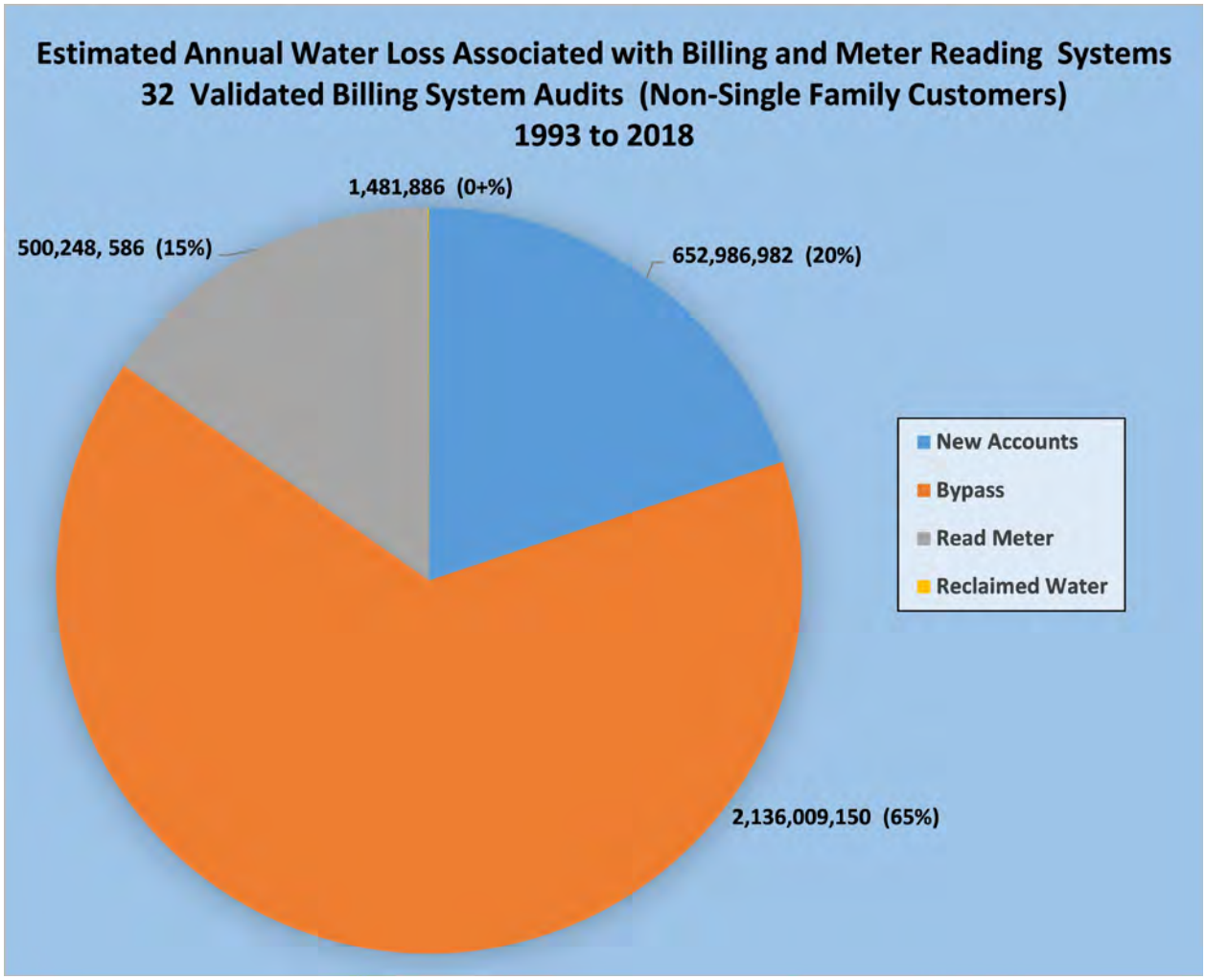


Figure 4-3 Apparent water loss in gallons from 32 validated billing system audits over 25 years



Figure 4-4 A "cheater" or "jumper" pipe (top) that was illegally used to replace the top meter in the meter setter: in the lower part of this meter pit is a typical ⅝-in residential meter

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

CUSTOMER BILLING SYSTEM

Most North American water utilities meter their customers and store customer meter readings in a computerized billing system that calculates water consumption, determines the charges, and bills the customer. For these utilities, the customer billing system serves as the source of all customer attribute data in addition to the water consumption history. Revenue is generated via billings to customers for water consumption, traditionally on either a monthly, bimonthly, or quarterly basis. It is important to realize that most customer billing systems historically have been designed with a primarily financial purpose—to generate bills that result in revenue collection. However, water utilities have also come to rely on customer usage data for a variety of engineering purposes, including tracking water conservation trends, water loss studies, and assigning water demands in distribution system hydraulic models.

It is important that water utility managers understand the workings of the CIS regarding consumption data integrity. While many billing systems are configured with sound billing intentions, they may unknowingly corrupt the engineering integrity of water consumption data. Frequent billing adjustments may change the customer consumption generated by meter readings. The implications of the use of customer meter-reading and billing data on the financial and engineering functions of the water utility are detailed in the "Using the Customer Billing System to Extract Customer Water Consumption" sidebar in Chapter 2, and the reader is urged to carefully review this discussion.

For the above reason, the utility should undertake an investigation of the billing process to identify any impacts to customer consumption integrity, as well as to identify any apparent loss components from the data-handling process. If consumption data are found to be modified by billing operations, the utility manager should consider reprogramming the billing system to record both the registered or actual consumption as the engineering value and the billed consumption as the financial value in separate fields in the computer record. This will help ensure that the accuracy of billing functions and customer consumption data are preserved. Until this procedure is implemented, an estimate of the impact of such adjustment activity should be included as an SDHE, a component of apparent losses. Advanced meter reading often has its own records of consumption in high detail that can assist in this separation of financial and consumption applications.

AUTOMATIC METER READING AND ADVANCED METERING INFRASTRUCTURE SYSTEMS

Automatic meter reading (AMR) and advanced metering infrastructure (AMI) systems are two proven technologies that can provide numerous benefits to water utilities in controlling apparent losses, as well as improving operational efficiency, financial stability, and customer relations. AMR systems use a mobile collector to obtain customer meter readings at virtually any time, but billing data collection is made every 30, 60, or 90 days. AMI systems use a fixed communication network to obtain at least hourly or more frequent interval meter readings, alerts, and alarms. Some AMR/AMI systems also provide water temperature and pressure data, support remote disconnect/reconnect valves, transmit leak noise logger data to detect leaks, and may even support internet of things (IoT) devices.

A complete review of AMR/AMI technology is beyond the scope of this manual, and the reader is referred to two valuable references. The Water Research Foundation (WRF) report titled *Advanced Metering Infrastructure: Best Practices for Water Utilities* (Schlenger et al. 2011) gives detailed information on metering, AMR, and AMI technologies. The publication *Advanced Metering Infrastructure: A Guidance Manual for Water Utilities* (Schlenger 2019) provides updated information on AMI technology and gives guidance

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

on development of the business case for AMI, along with the steps to consider in the procurement, installation, and commissioning of AMI systems.

Because the use of customer water meters is common in the North American drinking water industry, management of customer meter data is a critical part of utility operations, as well as the utility revenue stream. Before the development of AMR systems, water utilities typically read customer meters manually (visually) and then billed customers every 60 (bimonthly), 90 days (quarterly billing), or 180 days (biannually). AMR/AMI systems have enabled many water utilities to migrate to monthly meter reading and billing, which greatly benefits both the utility and the account holders. Customer payments are more likely to be made in full and on time in a 30-day billing interval compared to longer periods. Monthly data also allow customers to better manage their water consumption.

Moving away from manual meter reading reduces the potential for human errors that adversely affect the billing process. Meter-reading and billing activities occur at a cost to the water utility, and manual meter reading is no longer cost-efficient for many water utilities. AMR and AMI system technologies can address the many challenges in the meter-reading and billing processes, as well as offer other efficiency enhancements.

AMR is still a viable meter-reading technology and remains a cost-effective and practical meter-reading solution for many utilities for the foreseeable future. Many AMR system endpoints (reading device connected to the meter) available commercially support hourly data logging for at least 35 days. Some utilities with AMR systems activate the hourly or daily data logging feature on their customer endpoint devices and then choose to read the meters as frequently as needed to keep continuous data. This enables the utility to capture hourly usage data at a safe-driving speed, and therefore, the utility has the key benefit of an AMI system—hourly usage data—but without the expense of the communication network infrastructure.

AMI, however, provides many capabilities in addition to meter reading. AMI allows water meter readings to be gathered at short intervals (hourly readings are the most common format for the water industry), and some AMI systems offer on-demand readings. For systems that collect data at the top of the hour in closed metered zones like district metered areas (DMAs), comparisons can be made with aggregate customer use and the zone meter on an hourly basis. AMI's high-resolution customer consumption data, combined with alerts and alarms from its meter and data analytics software, can reveal individual customer consumption profiles and flag anomalies quickly. AMI systems can create many cost savings and operational efficiencies for the utility and facilitate exceptional levels of customer service.

Examples of the enhanced customer service features (benefits) of AMI include the following.

- Proactive detection of high or continuous water consumption due to plumbing or toilet leaks can alert the customer before considerable time has passed and a "high bill" is issued. Many utilities have a policy of issuing refunds that in effect shifts billed water to nonrevenue water (NRW), and these incidents can be curtailed by having this capability.
- High-bill complaints that arise can be investigated and often resolved by customer service representatives in the office by referring to hourly data that can often pinpoint the day (and time) that high water usage commenced.
- Substantial reduction of estimated bills.
- Customer engagement portals allow account holders to monitor their water usage, be alerted to potential issues via email and/or text messages, and compare their usage to both historical data from their account and peer groups of customers. The customer engagement portal may provide account holders with tips for reducing water usage and for finding leaks.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- Use of this technology reduces visits by utility staff to customer sites (i.e., fewer "truck rolls"), which reduces vehicle travel costs. Visits are no longer needed when AMI can provide on-demand "move-out/move-in" readings when customer property ownership changes. Some AMI systems even include remote disconnect capability in the customer endpoint, which results in fewer truck rolls as service discontinuation and reactivation can occur via commands issued from the utility's office. Customers spend less time waiting for the arrival of utility staff.

Compared to AMR systems, AMI systems offer many additional benefits for the utility, including

- detailed flow analysis for better meter selection and management;
- monitoring for water consumption at inactive accounts and zero usage at active accounts;
- the ability to quantify customer consumption in zones or DMAs during minimum hours, a component that assists leakage management and allows a water balance to be assembled on a zonal basis;
- the ability to collect readings of all customer meters on the same day (e.g., December 31) to better match water consumption data with water production data, facilitate water auditing, and simplify billing;
- usage of leak noise loggers to provide alerts through the AMI fixed network; and
- support for pressure transducers, water temperature sensors, water quality sensors, remote disconnect shutoff valves, and IoT devices.

These capabilities of AMI systems can be quantified with cost savings based upon the specific projections of their use by the water utility.

It is important to assess the benefit–cost of an AMR or AMI system over the life of the system (which may reach 20 years) rather than focusing solely upon the upfront installation costs. It is also important to include both the expected cost savings of the benefits in addition to the cost expenditures to install and maintain the system.

In terms of upfront installation costs, an AMI system costs more than an AMR system installation for a given water utility, but AMI offers more benefits. While the costs to install equipment at the customer endpoint are similar for both systems, the installation cost of the fixed, two-way communications network (i.e., collectors/routers, possibly repeaters, or a cellular system) for AMI results in a higher initial cost. There are also initial and annual fees for the network software and the data analytics software that enable the utility to take full advantage of the hourly usage data. Some AMI system collectors have batteries that require periodic replacement, an another cost incurred. The annual maintenance fees for an AMI system are also typically higher than those for an AMR system. The ultimate costs will be determined when the system requirements are established and the system installation schedule is developed. Water utilities serving large, high-density populations in a relatively flat topography will have notably different system needs than utilities serving small, low-density populations that are scattered across a wide geographical area of undulating terrain. Water utilities should assemble a rational business case for the pursuit of a new or replacement AMR or AMI system, and a rational economic assessment of both the costs and benefits should be detailed in the business case.

Utility staffing changes, and process changes must be considered when implementing either an AMR or AMI system. If manual meter reading is replaced by an AMR system, the meter-reading staff can either be reduced or reassigned as the meter-reading function becomes automated. The move to an AMI system can create additional changes in staffing and department processes. Hosted AMI networks (software as a service or SaaS) and customer engagement portals have greatly reduced the need for a utility to hire additional information technology staff, but staffing roles may need to change in the following two areas:

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

1. **Meter data management:** One or more staff members may be needed to monitor the AMI system health and performance. The status of the fixed network must be monitored because equipment such as collectors, routers, and repeaters can fail or be disrupted by storms, vandalism, or other factors. Likewise, endpoint units periodically fail or are damaged. Staff must be ready to detect and respond promptly to these problems to keep the system at peak performance.
2. **Data analytics:** By gathering highly granular customer consumption data, endpoint alerts, and status data, AMI systems generate massive amounts of data. A large water utility might generate more than a billion datapoints per year. Managing such a large volume of data is a significant undertaking. The use of data analytics software can provide the utility with the tools to quickly analyze large volumes of data to detect trends and anomalies. While data analytics software from an AMI system can provide great insight into the customer population, the utility must have the staffing in place to take advantage of the metering and billing trends, alarms, alerts, and anomalies in the data. This need is greatly aided by sophisticated meter data management (MDM) systems available through either the AMI system provider or other MDM companies. AMI MDM analytics can reveal a host of trends and alerts to the water utility.

Many of these work functions can be provided on a contract basis with the AMI system provider. These providers are continuing to expand their offerings of "as a service" work functions, including software as a service, metering as a service, and other capabilities. This can be an attractive option for many utilities that would prefer to outsource these roles.

Accurate water usage data from AMR and AMI systems greatly aid in responding to and solving customer questions and complaints, which improves customer confidence in the utility and the services it provides. The true value of an AMI system is only realized if the information that it generates is translated into meaningful action, improved outcomes, and increased customer satisfaction. AMR and AMI systems can provide many benefits to the water loss control program to reduce both apparent and real loss. AMR/AMI systems improve overall water accountability by ensuring a high degree of success and accuracy in obtaining actual customer meter readings. The number of estimated bills is greatly reduced, water usage at inactive accounts is quickly revealed, suspicious consumption patterns are flagged, accounts with poor metering are more easily identified, and many other benefits might be realized to better control apparent losses.

Real losses can also be better managed by employing AMR and particularly AMI technology. By providing accurate assessments of customer consumption, leakage losses can be more reliably quantified in pressure zones or DMAs. Leak noise loggers installed as part of an AMR or AMI system can readily detect potential leak noises as new leaks emerge, providing the water utility with the opportunity to provide a quick response and keep leakage losses to a minimum. Pressure transducers connected to an AMI system endpoint might also be used to find areas with high hidden leakage. AMR and AMI systems give water utilities strong tools to improve both their overall accountability and their efficiency in controlling and achieving minimum levels of water loss.

Since these systems also enhance customer service and provide new data to improve operational efficiency, it can be beneficial for any water utility to investigate their use and explore means of funding to procure, install, and operate them. Funding should be sufficient to ensure that the positive benefits to the utility are realized over the life of the investment. It is important for the water utility to develop a sound business case for either an AMR or an AMI system based upon the lifecycle evaluation of both costs and benefits to ensure that the enhanced capabilities targeted for the system are delivered in a cost-effective manner.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

SDHE

The water utility operator should develop a detailed understanding of the ways in which consumption data are managed in the utility's customer billing system. Using flowcharts to assess various subprocesses of billing operations allows the auditor to confirm the billing functions that are working properly and identify gaps that cause customer consumption to be understated and the utility to underbill their customers. Any such deficiencies that allow customers to exist without billing accounts, accurate metering, or continual meter reading can create apparent losses. Apparent losses may also be caused by systems that allow metered consumption data to be unduly modified for engineering purposes. Figure 4-5 below illustrates sources of SDHE.

A flowchart is a picture of the separate steps of a process in sequential order. It is a generic tool that can be adapted for a wide variety of purposes and can be used to describe

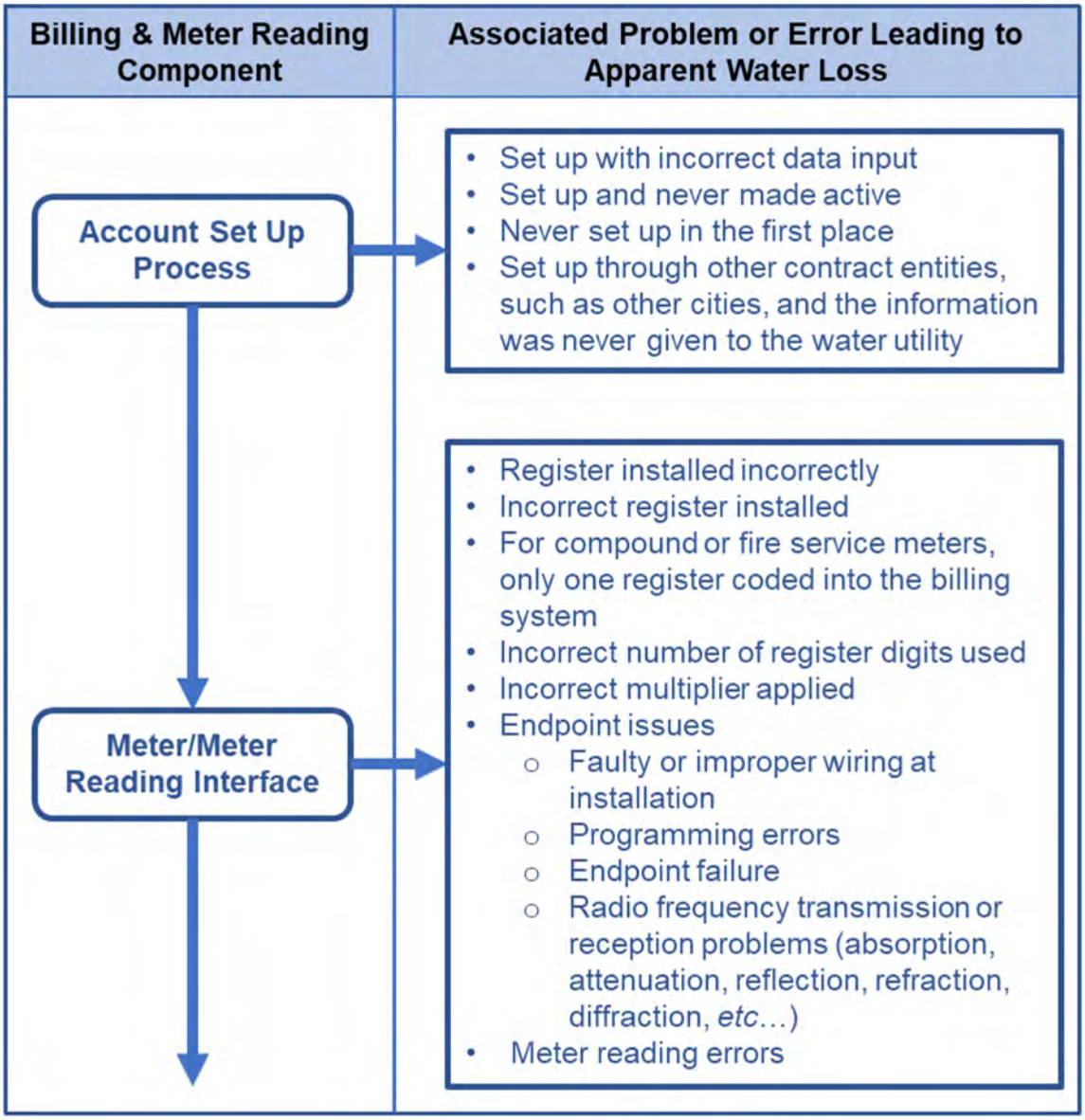


Figure 4-5 Sources of SDHE

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

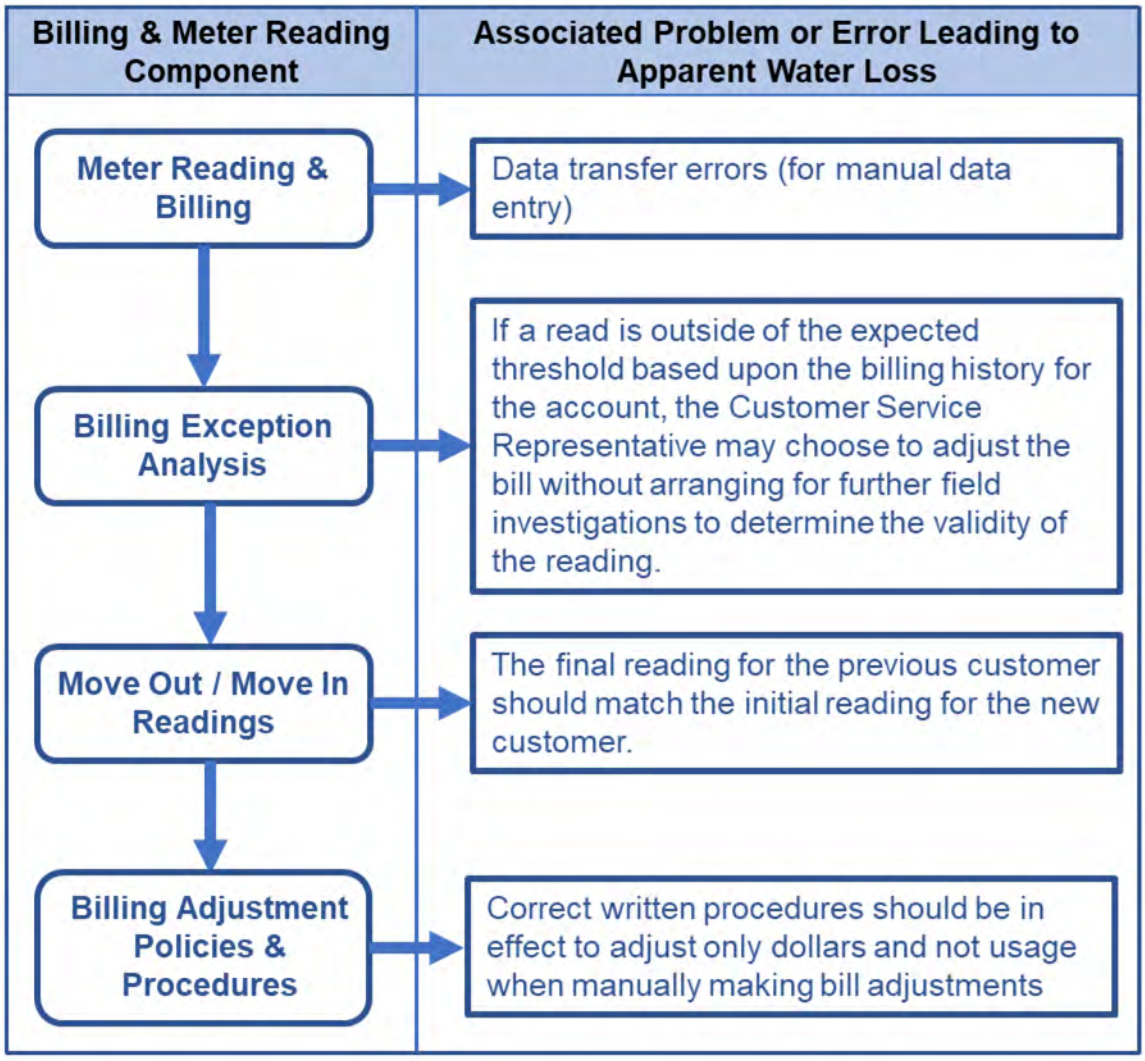
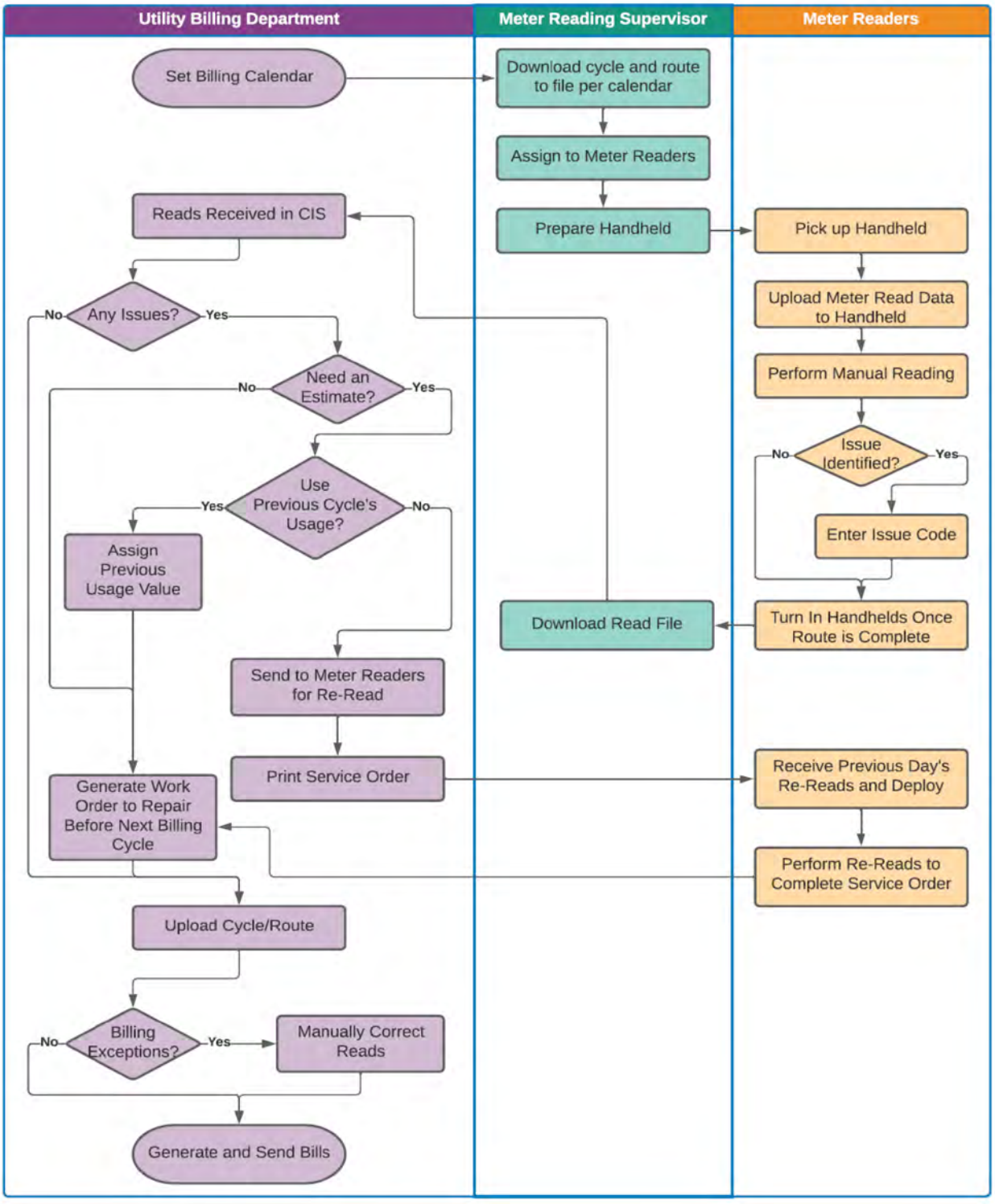


Figure 4-5 Cont'd Sources of Systematic Data Handling Errors

various processes, administrative services, or a project plan. Figures 4-6 to 4-8 represent customer billing system flowcharts for a water utility with manual meter reading, AMR meter reading, and AMI meter reading, respectively. The billing system flowcharts shown in Figures 4-6 to 4-8 are given for illustrative purposes only and do not apply for all utilities. Every water utility has a customer meter-reading and billing process with features that are unique to their organization; therefore, each utility should generate flowcharts that reflect their individual processes. For example, it is not unusual for water utilities to employ AMR in a portion of their service area while manually reading meters in the remaining areas. Likewise, some utilities with AMI over most of their service area may use a mobile AMR system to read meters in outlying areas.

It may be advisable to analyze a small sample of several dozen to several hundred customer accounts in various categories to determine if any loss impacts exist. The auditor should analyze samples of accounts in any special billing categories (municipal properties, unbilled accounts) as well as a sample of the largest water consumers to reveal likely and potentially significant occurrences of apparent losses. In analyzing customer billing system operations and billing data, the auditor should consider policies, procedures, and practices:

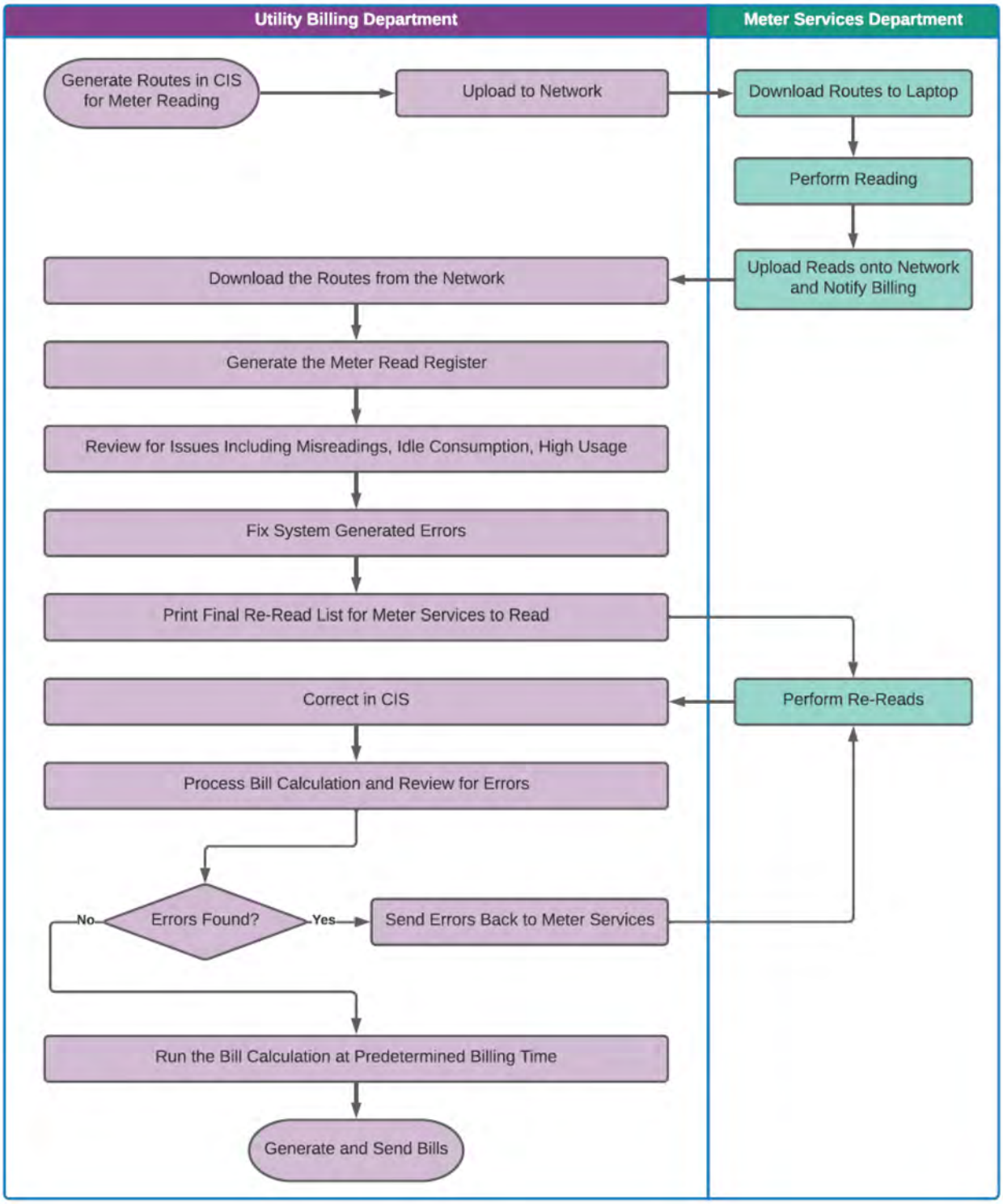
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



(Courtesy of C. Mastic, MARS Company, adapted from flow chart provided by SL-servo, Inc.)

Figure 4-6 Billing process chart for a utility with manual meter reading

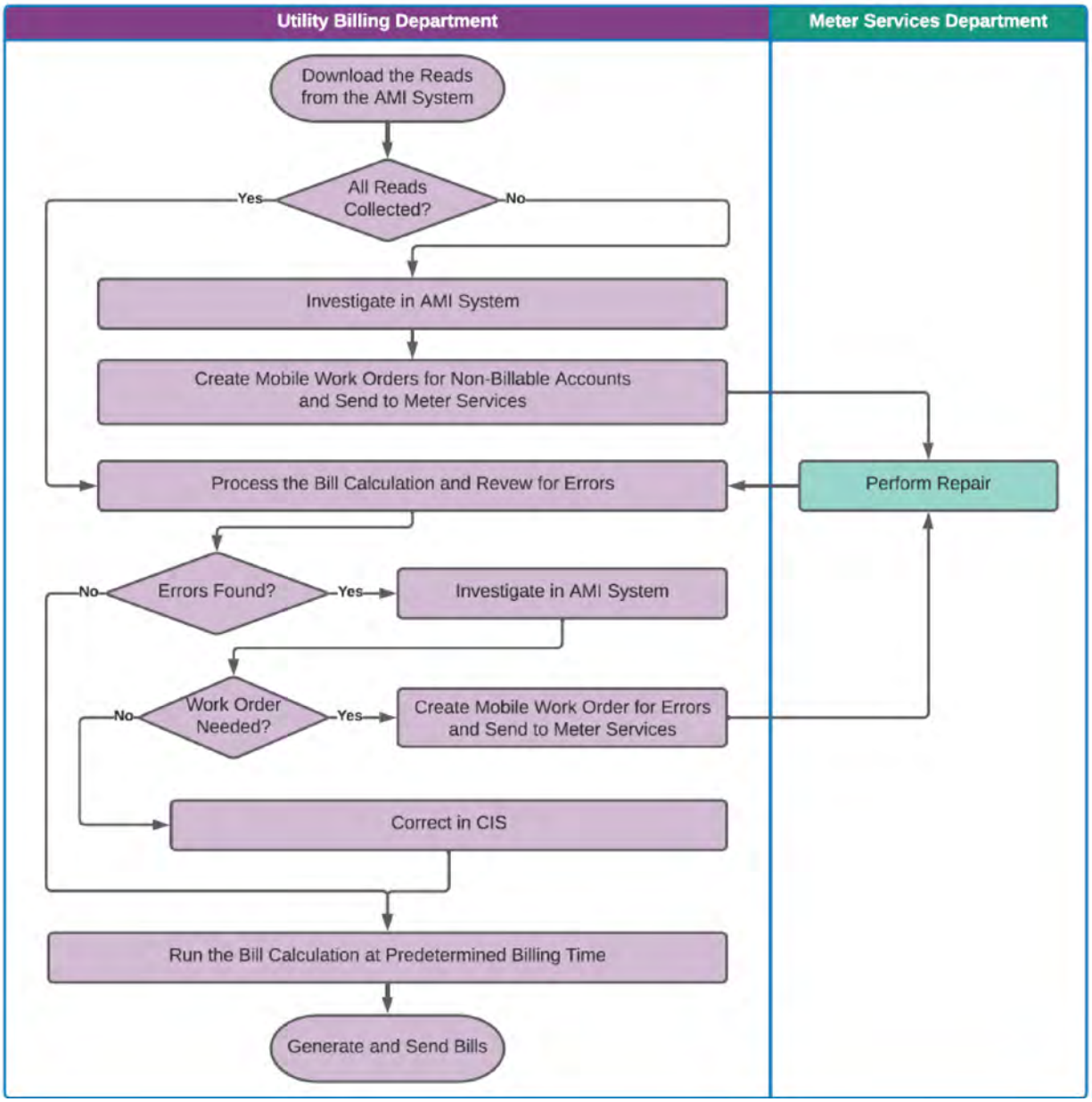
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



(Courtesy of C. Mastic, MARS Company, adapted from flow chart provided by SL-serco, Inc.)

Figure 4-7 Billing process chart for a utility with AMR meter reading

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



(Courtesy of C. Mastic, MARS Company, adapted from flow chart provided by SL-serco, Inc.)

Figure 4-8 Billing process chart for a utility with AMI meter reading

- Are **policies** regarding customer metering, billing, water rates, customer service connection responsibilities, etc., rational, consistent, codified, and well communicated?
- Do written **procedures** exist? Are procedures used to ensure that consistent metering, meter-reading, and billing functions are employed for all customers? Are checks and balances built into the system to flag breakdowns or gaps in the process?
- Do the actual **practices** follow the details of the written procedures? Are meter readers, billing clerks, or similar employees given the proper training and

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

evaluation to maintain or improve upon their skills to minimize human error or malfeasance in transferring water consumption data and to ensure that policies and procedures are being followed? Are annual refresher courses being offered to maintain employee skills and knowledge? Are new employees given the proper training needed to minimize errors?

Additionally, the following questions should be answered in an analysis of billing systems:

- Are certain classes of customers, such as municipal properties, exempt from metering and billing? If so, how is their water consumption accounted for by the water utility?
- Are there potential instances of customer classification miscoding, which may be resulting in misapplied billing rates to certain customer accounts? Is each customer being billed at the correct rate? For example, is a customer who lives outside the city limits being billed at the inside city limits rate? Is a customer who is connected to the sanitary sewer system not being billed for sewage fees?
- Are there potential instances of meter multiplier miscoding, which may be resulting in miscalculated water consumption values on the bill? (See the "Communication Error Results in Underbilling Customers by a Factor of 10" sidebar as an example.)
- Can customers be placed in a unbilled status due to property vacancy, delinquency, or shutoff account? If so, are these accounts routinely monitored?
- Are estimates of customer consumption employed if meter readings are not available? If so, how accurately does the estimate reflect actual consumption? Do checks exist to validate or periodically update the estimates?
- Does a policy exist for enforcement to deter unauthorized consumption? Can customers have service terminated for nonpayment? If so, are significant numbers of customers illegally reactivating their service? Is there a mechanism to detect and thwart this activity?
- Do programming algorithms incorporate billing adjustments that create a variance between "volume registered" and "volume billed," such as shown in Tables 2-11 and 2-12 in Chapter 2? This may reflect unintentional modification of legitimate consumption volumes in the billing adjustment algorithms.
- Are metering, meter-reading, and billing functions actively tracked and monitored by the issuance of routine management reports that are structured to summarize performance, identify trends, and flag anomalies?
- Do reliable procedures exist to ensure that accurate consumption volumes are determined during the meter replacement process? Proper closing readings must be gathered from the outgoing meter and a new initial reading input from the incoming meter. Do billing system operations maintain the integrity of the actual customer consumption during this transition?
- How many meters of size 1-in and larger are a misapplied size or type of meter, which may produce billing patterns indicative of consumption underregistration and underbilling? (This topic is discussed further in the next section.)
- Are customer consumption and billing trends evaluated on a regular basis to discern specific and overall trends in consumption and loss patterns in response to water conservation efforts, NRW management programs, or demographic trends such as growth in the industrial sector?
- How are bill adjustments processed so that changes in consumption values do not adversely affect the water audit and the volume of NRW calculations?

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

These are just some of the questions that might be posed during bottom-up auditing of the data-handling process. For every water utility, unique processes exist and should be scrutinized by the auditor. As an exercise, customer billing system flowcharting can be revealing, and process flaws or misunderstandings between different groups in the water utility are often discovered. For some of the system shortcomings, corrections can be implemented by relatively inexpensive computer programming changes or improved coordination between metering, meter-reading, and billing groups in the utility. These corrections are usually implemented by a change in procedure, additional auditing, modification of meter-reading cycles, billing processes, or revisions in computer output reports. These are all operational functions that can be executed in the office. They do not necessarily require new equipment or significant staffing changes. In some cases, the findings of the flowcharting analysis may point auditors to a questionable process that needs to be further investigated to confirm whether water volumes are being lost. In some cases, new policy may need to be established by the utility management.

Communication Error Results in Underbilling Customers by a Factor of 10

A water utility located in a coastal area of the United States has large underground meter vaults that are flooded throughout the year. The utility installed a touch-read AMR system in order to read those large water meters. Based upon the results of its AWWA water audit, the utility suspected that it was incurring high levels of apparent water loss and billing much lower water and sewer charges than expected, possibly due to errors in the utility billing system. The utility attempted to find sources of the apparent loss and with the assistance of a consulting firm was eventually able to identify the mysterious source of apparent loss.

The consultant discovered that the purchasing department had ordered many large water meters with only the first five digits of the six-digit encoder register as readable digits. No one had informed the meter readers of this fact, however. When reading these meters, the touch-read probe registered only five digits instead of the complete six-digit reading. Figure 4-9 shows both a meter register of one of the meters and the resulting display of the touch-read system. The utility was consistently billing these customers only one-tenth of their actual water consumption since the touch-read probe could not read the sixth digit of the water meter register. What is best practice for all water utilities? Physically inspect large meter accounts at least annually, and audit the complete meter-reading process.



Figure 4-9 Meter register showing active six digits on the meter, along with erroneous AMR touch-probe meter reading of only five digits

Once the customer billing process has been mapped via flowcharts, the auditor can determine whether to undertake a more detailed investigation of billing operations and the customer consumption database. If the database is extracted in digital format, the analysis can be performed in a semiautomatic manner, leveraging the functionality of database software tools such as MS Excel or appropriate data analytics software. This typically requires extraction of the database from the customer billing system and is sometimes referred to as a data-mining analysis. This is a more involved but worthwhile

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

undertaking. Although there is no standard approach to this type of analysis, a utility may consider developing a process in-house or using a consultant to perform the task. From a data-mining analysis, anomalies of all types can be flagged for further investigation, including the extent of accounts registering zero consumption for consecutive billing cycles, accounts registering negative consumption (possibly due to errant billing adjustment algorithms), accounts registering unusually high or low consumption volumes, accounts registering repeated consumption volumes, and many other possible trends of unusual activity.

Customer Billing Data-Mining Analysis Case Study

The city of Wylie, Tex. located northeast of Dallas, has over 13,350 water accounts, and the city is growing at annual rate of 2.08%. Until the fall of 2020, Wylie used a mobile AMR system to read meters monthly, and their customers were billed for water usage monthly. The mobile AMR system was approaching the end of its useful life, however, and the city chose to replace it with a fixed-network AMI system. As part of a comprehensive water loss optimization project, the city hired a consultant to perform a data-mining analysis of its water-billing data from fiscal year (FY) 2019 (October 1, 2018, to September 30, 2019). The consultant tested a statistically valid, random sample of the ¾-in water meter population at a certified independent laboratory. The city had standardized using a ¾-in meter for typical single-family residences. The overall weighted average accuracy of the ¾-in meter population was 99.8%.

Billing records for the ¾-in accounts were analyzed during the data-mining study. It should be noted that the city requires a separate meter on irrigation systems on customer properties and that irrigation meters provide consumption data that is billed in an account separate from the domestic water service account. Therefore, irrigation consumption was not included in the billing data-mining analysis.

Some of the important findings of the data-mining analysis of the ¾-in meter population are listed in Tables 4-1 and 4-2 and shown graphically in Figures 4-10 and 4-11. Table 4-1 and Figure 4-10 display the number of ¾-in accounts that fell into various annual consumption ranges. Table 4-2 and Figure 4-11 show the total annual water consumption volumes in gallons of ¾-in accounts broken down into consumption ranges. In FY 2019, there were 11,839 active ¾-in residential accounts out of a total of 12,959 active accounts, meaning the ¾-in single-family residential meters comprised 91.36% of the total number of active accounts.

Table 4-1 shows that the total population of ¾-in meter accounts consumed an average of 78,440 gal during the year or 215 gallons per day (gpd), and the median consumption was 70,960 gal for the year or 194 gpd. Table 4-1 and Figure 4-10 reveal that 6,216, or 52.5% of all ¾-in meter accounts, consumed water in the four consumption midranges between 121 and 300 gpd (or between 43,801–109,500 gal/year). Table 4-1 also reveals that 2,624, or 22.2% of all ¾-in meter accounts, recorded consumption in the seven ranges above 300 gpd or much higher consumption volumes than would be expected for residential accounts with a ¾-in meter. Additionally, a total of 764 accounts (6.45%) had annual consumption volumes at least twice the average volume, and 100 accounts (0.84%) were billed annual consumption volumes at least three times that of the average annual volume.

Although inordinately high consumption volumes generate high billings, they should be investigated to confirm whether excessive water waste (as from internal plumbing leaks) may be the cause of the unusually high water consumption. AMI systems usually have continuous flow alerts to flag high flow rates when they emerge. Additionally, Table 4-1 shows that 41 active residential accounts with ¾-in meters registered zero consumption for the entirety of FY 2019, and another 1,582 accounts, or 13.4% of all ¾-in accounts, consumed less than 80 gpd during FY2019. Many of these low-consumption accounts may

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 4-1 Single-family residential (¾-in meter) water consumption for FY 2019 in Wylie, Tex.

Annual Consumption Range (gal)	Range in gpd	Number of Accounts	% of Total
0	0	41	0.35%
1–29,200	> 0–80	1,582	13.36%
29,201–36,500	81–100	635	5.36%
36,501–43,800	101–120	741	6.26%
43,801–58,400	121–160	1,556	13.14%
58,401–73,000	161–200	1,585	13.39%
73,001–87,600	201–240	1,447	12.22%
87,601–109,500	241–300	1,628	13.75%
109,501–116,800	301–320	434	3.67%
116,801–131,400	321–360	688	5.81%
131,401–146,000	361–400	479	4.05%
146,001–175,200	401–480	574	4.85%
175,201–182,500	481–500	75	0.63%
182,501–219,000	501–600	235	1.98%
> 219,001	> 600	139	1.17%
	Total	**11,839**	**100.00%**
78,440	Average annual consumption (gal)		
215	Average annual consumption (gpd)		
70,960	Median annual consumption (gal)		
194	Median annual consumption (gpd)		

Table 4-2 Annual water consumption in gallons by range of water consumption in Wylie, Tex.

Annual Consumption Range (gal)	Range in gpd	Total Annual Consumption (gal)	% of Total
0	0	0	0.00%
1–29,200	> 0–80	25,777,957	2.79%
29,201–36,500	81–100	20,989,689	2.27%
36,501–43,800	101–120	29,832,395	3.22%
43,801–58,400	121–160	79,497,579	8.59%
58,401–73,000	161–200	104,060,182	11.24%
73,001–87,600	201–240	116,033,310	12.54%
87,601–109,500	241–300	159,316,891	17.22%
109,501–116,800	301–320	49,110,667	5.31%
116,801–131,400	321–360	85,288,860	9.22%
131,401–146,000	361–400	66,280,620	7.16%
146,001–175,200	401–480	91,176,059	9.85%
175,201–182,500	481–500	13,384,630	1.45%
182,501–219,000	501–600	46,552,575	5.03%
> 219,001	> 600	38,132,185	4.12%
	Total	**925,433,599**	**100.00%**

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

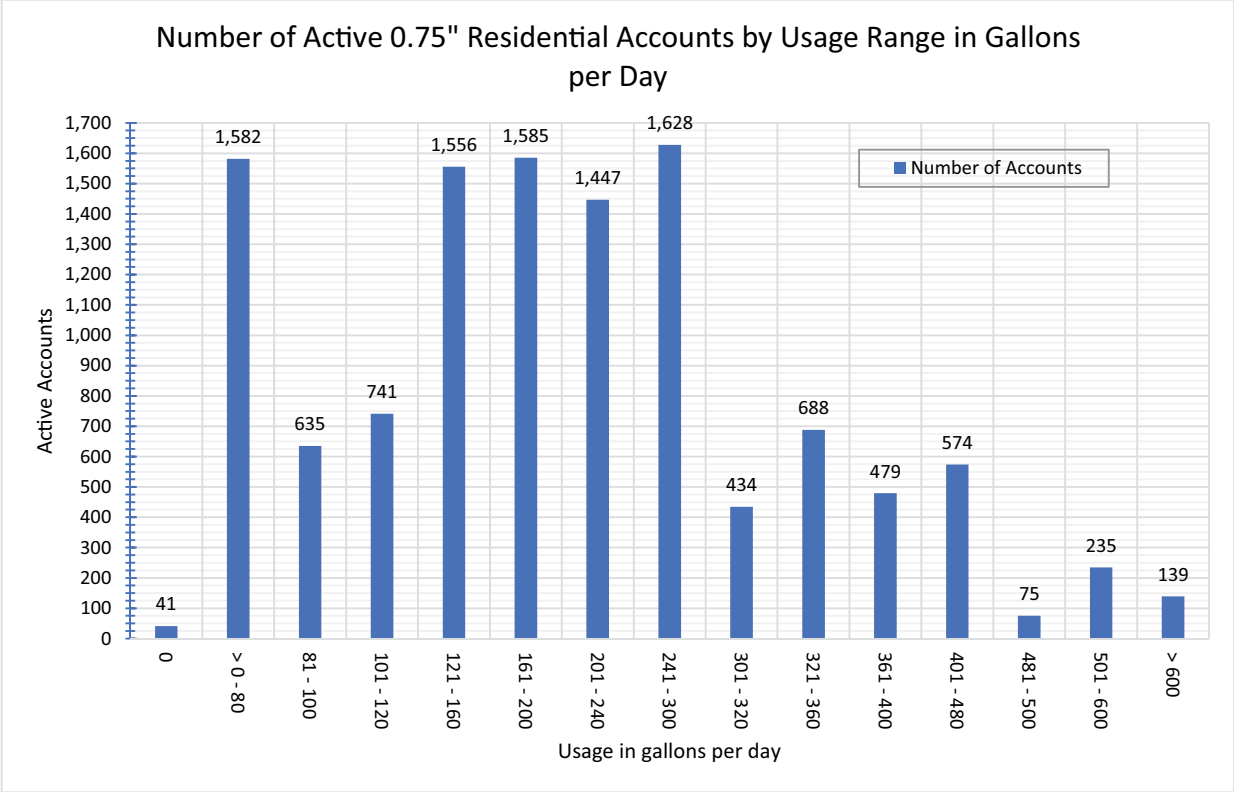


Figure 4-10 Data-mining analysis chart for the City of Wylie, Tex., showing the number of active ¾-in meter single-family residential accounts in water consumption ranges

have been vacant for part or all of FY2019; however, a portion may be accounts with meter and/or meter-reading equipment failure or UC that occurred due to tampering of meters and/or meter-reading equipment. It is often worthwhile to inspect a portion or all zero and low consumption accounts for the possibility of metering equipment failure or unauthorized consumption.

Table 4-2 and Figure 4-11 provide data for the active residential ¾-in meter population but display the annual billed consumption volumes that occurred in the ranges of consumption. While residential ¾-in meter accounts represent 91.36% of all accounts by number, the ¾-in residential meter accounts make up just 64.5% of the total billed consumption for all Wylie billed accounts in FY 2019. The data indicate that residences that used between 161 and 300 gpd consumed only 41% of all water used by residential customers with ¾-in meters. Conversely, 42.1% of consumption in these accounts occurred in the ranges above 300 gpd, which again provokes suspicion that some high consumption accounts may have incurred water waste due to internal plumbing leaks.

While Wylie had no accounts with a negative consumption in FY 2019, water utilities should be aware of the potential for negative consumption values and consider modifications to billing algorithms to prevent negative values from corrupting the tally of customer water consumption. Readers should review the discussion around Tables 2-11 and 2-12 in Chapter 2 to determine how to maintain the integrity of customer consumption values for both billing and water accounting activities.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

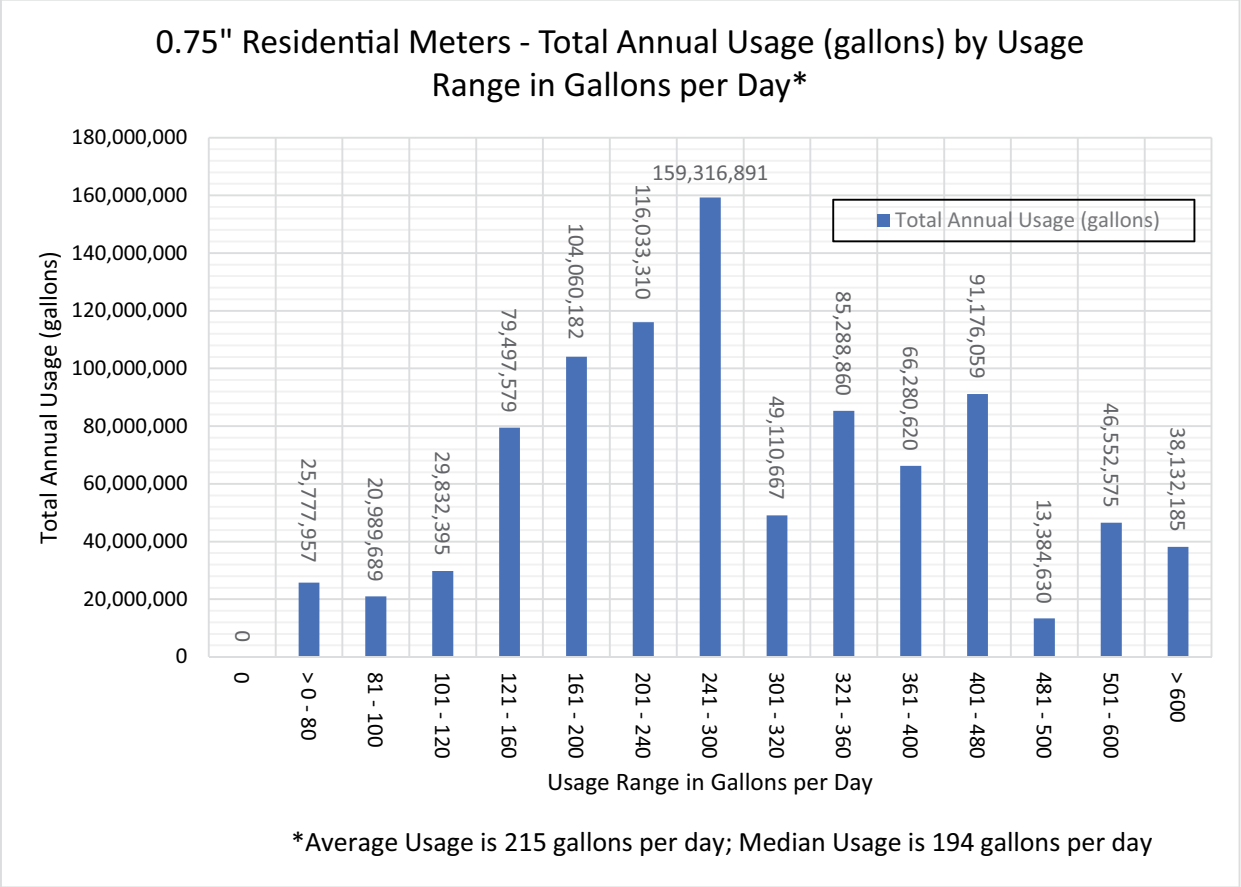


Figure 4-11 Data-mining analysis chart for the City of Wylie, Tex. showing annual billed water consumption for ¾-in residential meter customer accounts in various annual water consumption billing ranges

Table 4-3 Annual water consumption for accounts with 3-in meters in Wylie, Tex.

Account	Sum of Consumption	Overall Weighted Average Accuracy
Movie theater	1,231,000	99.37%
Hotel	1,708,549	Not Tested
Middle school	566,430	Not Tested
High school football stadium	155,450	99.93%
Municipal workshop	82,600	99.80%
Office	449,990	100.07%
Car wash	1,874,700	84.83%

Table 4-3 and Figure 4-12 show similar data for the accounts with 3-in meters in Wylie, a class of accounts that exists for the largest of water users, typically commercial and industrial buildings. Hence, it would be expected to find most of these accounts residing in the higher consumption ranges. Two of the seven 3-in meters in Wylie were not tested for accuracy because they lacked both outlet valves and test ports. Five 3-in meters

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

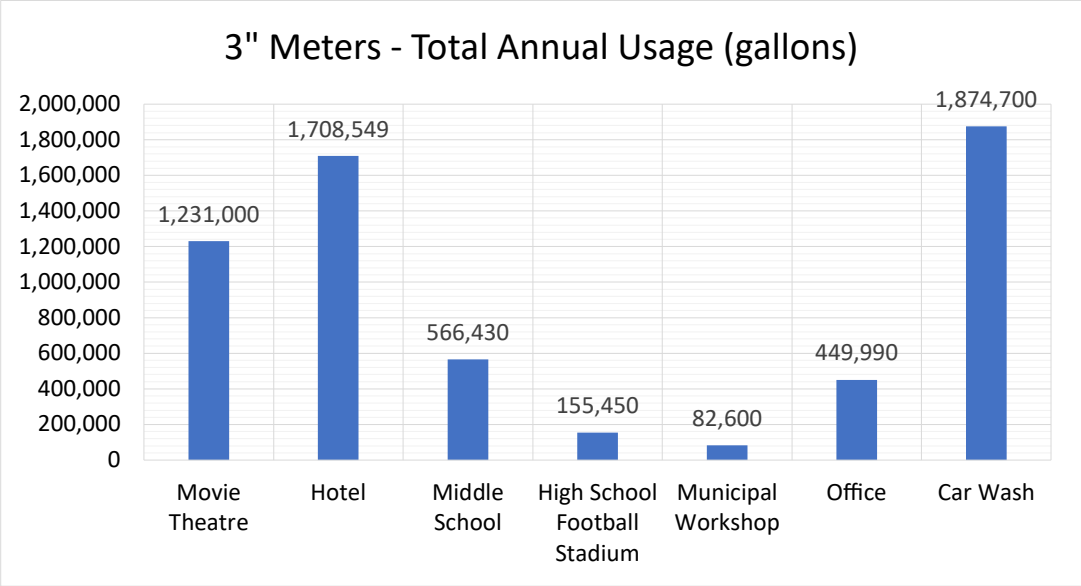


Figure 4-12 Data-mining analysis chart for the 3-in meters in Wylie, Tex.

were tested. Four of the five found superior accuracy close to 100%. However, the meter for the car wash, the largest water user in Wylie, tested at 84.83% accuracy, essentially under-registering water by 15.17%; this is a very high degree of inaccuracy. Wylie can recover a significant volume of missed water consumption and billings by improving the car wash's metering installation to a high level of accuracy.

Summarizing the steps to evaluate SDHE

By flowcharting the customer meter-reading and billing processes and then by conducting a data-mining analysis, the water utility may identify classes of accounts, as well as potentially numerous individual accounts, with suspicious billing patterns. Armed with this knowledge, the utility can begin to conduct physical inspections of customer premises and metering installations to identify the cause of unusual consumption patterns. These inspections represent the detailed "bottom-up" auditing that is most valuable to the water utility. Such inspections do require staff time to conduct, but these visits can lead to direct revenue recovery in many instances and can reveal additional information to the water utility.

In large water systems, it is not uncommon to find that many accounts with suspicious billing patterns exist. The utility should attempt to inspect portions of these accounts to the extent possible. Priority should be given to those accounts that appear to offer the greatest revenue payback potential or that seem to have abnormal metering or billing data. For example, the Boston Water and Sewer Commission examined hotels in their service area and evaluated the ratio of gallons used per room. A high ratio for a given hotel revealed a likelihood of internal plumbing leaks, and a low ratio revealed open meter bypass lines and meter accuracy issues.

As the water utility matures in making refinements to its metering and billing operations, standard procedures to flag anomalies and to initiate inspections can be developed to keep the water utility at a high level of billing efficiency and to maximize revenue capture. AMI system data analytics software both automates and greatly simplifies this type of auditing. Data analytics software often makes the utility aware of metering issues

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

within one day, giving it the opportunity for a quick response and limiting the duration, volume, and costs of apparent water losses.

Bottom-up auditing investigations of suspect accounts should be carefully documented so that volumes of water and revenue recoveries can be reliably tracked. Recovered water volumes should be tallied each year in subcategories of SDHE and included in the annual water audit. Recaptured revenue should also be tallied each year to use in a benefit–cost analysis of the bottom-up auditing and loss control activities.

By investigating SDHE first, water utility managers can accelerate revenue recoveries and ensure that customer consumption data are accurately recorded. This activity is both critical to the success of the water loss control program and a significant opportunity for water utilities to maximize revenue capture into their operations.

CMI

It is common in most—but not all—water utilities to use water meters to measure the amount of water passing through water service piping into customer premises. When properly installed, sized, and typed for the specific application, metering technology is accurate in recording flow across a range of flow rates, and water utilities have the potential to reliably record the collective water consumption for their customer population by regularly reading or collecting data from them. However, utility managers are challenged to select meters of the proper size and type for the wide variety of customer usage patterns that typically exist and may change over time. This presents a challenge in keeping the meter population current and functioning with a high level of accuracy.

Wear of moving parts, debris, and sediment deposits may cause mechanical water meters to lose low-flow accuracy over time, and it is low-flow accuracy that is needed to capture flow from toilet leaks and other low flow–rate leaks in customer premises. Water quality is an important factor if mechanical water meters are employed by the water utility. Acceptable water quality can be reasonably described as water with a minimum of debris and chemistry that includes hardness less than 200 mg per liter (mg/L) of calcium carbonate. Utilities that are concerned with the amount of lead and copper in the water should use only the amount of sequestration agents (polyphosphates and orthophosphates) needed to meet the minimum standards for lead and copper content because high concentrations of sequestration agents may adversely affect meter accuracy. Electromagnetic and ultrasonic meters, with no moving parts, typically maintain strong accuracy over an expected life cycle if the meter is installed correctly and used in water of reasonable quality. Tuberculation is an encrustation that may form inside either bronze or ductile-iron meter bodies if the meter is used in soft water with a bicarbonate alkalinity greater than 240 mg/L or water with concentrations of sulfate greater than 12 mg/L and chloride greater than 64.9 mg/L.

With water utilities that manage thousands, tens of thousands, or even hundreds of thousands of water meters, the sheer scope of keeping high meter accuracy is a great challenge. Thus, virtually all water utilities that employ water meters can have meter-accuracy issues with some portion of their customer population. AWWA provides detailed guidance on all aspects of customer meter management in its Manual M6 publication titled *Water Meters—Selection, Installation, Testing, and Maintenance* (6th edition, 2012). Utility managers responsible for meter management can familiarize themselves with the best practices presented in this publication to approach meter management in a proactive manner.

The water auditor should quantify the volume of apparent losses attributed to the collective customer metering inaccuracies in the water system. (See Chapter 2, Task 3, Step 3-2B for information on estimating apparent losses caused by CMI.) Meter accuracy is dependent on several components, and for those water utilities that meter their customers,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

the consumption data trail begins at the customer meter with the flow measured by this device. For a water meter to accurately register a volume of water passing through it, the meter must first be appropriately sized for the application. Second, the proper type of meter must also be selected. Third the meter must be installed correctly at each application. Meter installation has a direct bearing on accuracy and useful life, particularly for mechanical meters. Finally, the meter should be maintained as needed and tested for accuracy as determined by an ongoing meter test program.

Customer meters must work properly and be appropriately sized and typed for the customer demand pattern if they are to avoid underregistering flows, which causes underbilling, loss of revenue potential, and increased apparent water loss. Water utilities that employ best management practices for meter management likely have a thorough understanding of their customer meter demographics and the accuracy of the different types of meters in their system. Many water utilities, unfortunately, do not have current information about the status of their meter population or current information about the accuracy of specific types and makes of meters. It is not uncommon for an incoming water utility manager to inherit a meter population that was installed 15, 20, or 25 years ago but has not experienced ongoing meter testing, replacement, rightsizing, or right typing. In such cases, the size, type, make, and performance of the meter population are usually poorly documented and understood. It is important to conduct bottom-up data gathering and investigations to establish the basic customer meter demographics and determine the accuracy levels of the meter population.

Meter Demographics

The word *demographics* refers to the gathering statistics on a population, and meter demographics refers to statistics on the customer meter population. If the meter population characteristics are not known, the auditor can conduct research using meter purchase and installation records, billing records, customer complaint histories, and meter accuracy test results to compile information on the sizes, types, manufacturers, years of installation, and cumulative consumption levels of customer meters. Many manufacturers can provide an approximate year that the meter was manufactured based upon the serial number. Physical inspection of various customer premises (possibly an interview with larger customers) may also be necessary for some properties to verify the size, type, and make of the existing water meter. Table 4-4 is an expanded version of Table 2-7 in Chapter 2 for the fictitious County Water Company (CWC), which serves as the illustrative example throughout this publication. This table lists the summary demographics of the CWC customer meter population at the close of its calendar water audit year.

It is recommended that customer meter demographics be integrated into the customer account and consumption data system to allow for frequent systematic analysis of meter performance and the overall accuracy of the customer meter population.

Because meter technology is always improving, new types and models of meters are frequently introduced into the commercial water market. Many water utilities purchase meters in lots during a competitive bidding process and over long periods of time may gradually install a variety of makes and models in their system, particularly in the large customer meter classes. It is important that the auditor have a reasonable sense of the meter population demographics to establish a sound meter testing, rightsizing, right typing, and replacement strategy.

In the past, it was common for water utilities to periodically retrieve small (residential) meters from customer locations, repair and test them, and place them back into service, typically in a different customer account. This "rotation" process has largely given way in recent years to a meter "replacement" or change-out process, with an older meter retrieved and scrapped and a new replacement meter installed. The economics of small meter repair

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 4-4 Customer meter population demographics and metered consumption for CWC as of Dec. 31, 2020

Meter Size	Number of Meters	Percent of Total Meters	Meter Type (No.)	Meter Manufacturer (No.)	Average Age (years)	Percent of Metered Consumption
5/8"	11,480	94.1	Positive Displacement (11,480)	Badger Meter (6,123)	13	71.2
				Neptune (4,682)		
				Sensus (675)		
¾"	10	0.08	Positive Displacement (10)	Neptune (10)	26	0.1
1"	338	2.8	Positive Displacement (338)	Badger Meter (250)	18	2.8
				Neptune (88)	11	
1.5"	124	1.0	Positive Displacement (124)	Badger Meter (18)	18	2.8
				Neptune (106)	9	
2"	216	1.8	Positive Displacement (216)	Sensus (54)	12	11.7
				Badger Meter (146)	22	
				Neptune (16)	20	
3"	15	0.12	Turbine (15)	Sensus (15)	15	6.6
4"	7	0.05	Compound (2)	Neptune (2)	15	2.2
			Turbine (5)	Badger Meter (5)	26	
6"	6	0.05	Turbine (2)	Badger Meter (2)	15	2.6
			Compound (2)	Neptune (2)	9	
			Electromagnetic (2)	Honeywell-Elster (2)	4	
Total	**12,196**	**100.00**				**100.0**

and reinstallation are no longer justifiable for most water utilities, and many newer meter types simply cannot be repaired by water utility staff. Most large mechanical meters allow for the measuring chamber and the register to be replaced, and most large static meters allow for either battery pack replacement or for complete measuring chamber and battery pack replacement.

Customer Meter Testing

To determine the physical accuracy of the meter population, many water utilities conduct regular meter accuracy testing to identify meters with declining performance, particularly large meters. By obtaining meter accuracy test results of groups of water meters, the water utility benefits by

- calculating a quantity of the collective water consumption that is underregistered in the customer meter population throughout the audit year, thereby improving the validity of the water audit and
- identifying individual water meters with poor accuracy that can be replaced (small- and intermediate-sized meters) or repaired (large meters); and cohorts of meters where more investigation into meter inaccuracies may be warranted. This serves as an intervention to reduce apparent loss from CMI.

AWWA's guidance manuals on meters give excellent instruction on meter accuracy testing. The AWWA Manual M6 includes an entire chapter on testing titled "Testing of Meters—Test Procedures and Equipment" (AWWA 2012). Readers are referred to this

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

chapter to obtain detailed instruction on the methods and structure of the meter-testing program. In addition, WRF has sponsored a project that tested a significant number of new water meters, and the results can be instructive about how various types of meters perform over the long run. The project, described in *Accuracy of In-Service Water Meters at Low and High Flow Rates* (WRF 2011), included testing of hundreds of small water meters from a wide variety of manufacturers and offers great insight to the meter accuracy testing process and accuracy findings for many meters.

Generally, accuracy tests should be conducted at low-, intermediate-, and high-flow rates. Mechanical meters that have been in service should always be tested first at the low-flow rate because this is the most "sensitive" test. If mechanical meters are tested at the high-flow rates first, the high velocity of the water may remove scale or other debris in the measuring chamber of the meter that may have impaired low-flow accuracy while the meter was in service. Conducting a low-flow test after the high-flow test in a mechanical meter may give an accuracy value that is higher than the actual low-flow accuracy for the meter as it performed at the customer premises, which is not a representative test result.

Many water utilities operate a customer meter accuracy testing program and have their own test facility and equipment to perform ongoing accuracy testing of meters. Water utilities that do not have their own meter-testing facilities can outsource their testing to specialty companies. For water utilities that operate their own test equipment, water utility managers should review the procedures and testing practices that are employed by the water utility personnel. The meter test apparatus is a precision instrument and should be operated and maintained by skilled, trained personnel who follow a standard procedure for testing and documentation of the test results. As it is a precision instrument, the meter test apparatus should also be regularly calibrated to assure that it is reporting accurately and repeatably within acceptable tolerances.

Different sizes of "test benches" exist for testing the smallest meters of size ⅝ in through 2 in. Larger-sized test benches can handle meters of size 3 in up to 12 in or larger. A water meter test bench is shown in Figure 4-13.

While the full spectrum of customer meter sizes can be tested on test benches, many water utility managers prefer to test large water meters (e.g., 3 in and larger) in situ at the site of installation. For this purpose, portable test equipment is available to test these large meters in situ testing of the meter as it is installed and avoids the need to remove and transport large and heavy water meters to the meter shop for testing. Meter installation has a direct influence on both accuracy and useful life, and such testing can be used to confirm if a meter is either overregistering or underregistering usage because of a disturbed flow profile.

Customer meters that are installed close to backflow preventers, pressure-reducing valves, and other features that can disturb the flow of water into the measuring chamber of the meter are likely to suffer compromised accuracy. Figures 4-14 and 4-15 show examples of water meters improperly installed close to devices that caused the meter to overregister. In Figure 4-14, a pressure-reducing valve installed immediately upstream of the meter causes the meter to overregister by 2.40%. In Figure 4-15, a 90° elbow installed immediately upstream of a compound meter without a strainer impacts the flow profile and allows debris to clog the low-flow measuring element of the meter. This meter tested poorly, with no flow registered in most of the low-flow testing regimens. Testing these meters in situ provides the data that result from these pipeline features. If these meters had been removed from service and tested in a test laboratory, the meters might perform in with greater accuracy, misleading the water utility into thinking that the meter was functioning properly where it was installed. To the extent possible, utility operators should inspect meter installation sites—particularly for meters of size 1 in and larger—to determine if pipeline features exist that can impair meter accuracy. An example of a successful large

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



(Courtesy of MARS Company)

Figure 4-13 Meter test bench for testing small water meters

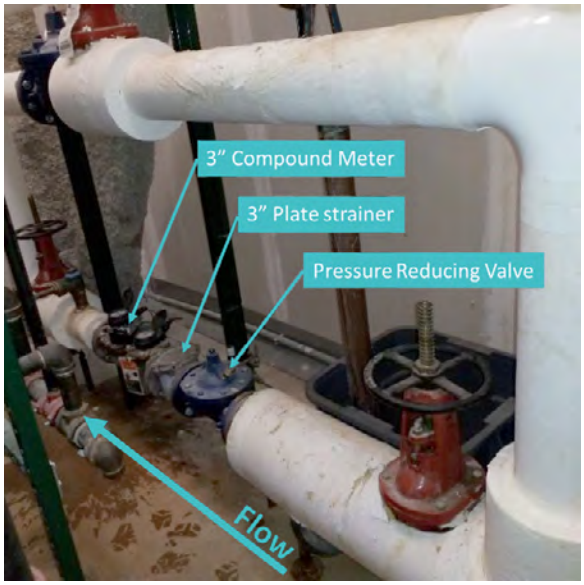


Figure 4-14 The pressure-reducing valve installed immediately upstream of this compound meter disturbed the flow profile entering the meter despite the presence of the plate strainer, causing the meter to overregister usage (102.4% overall weighted average accuracy)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

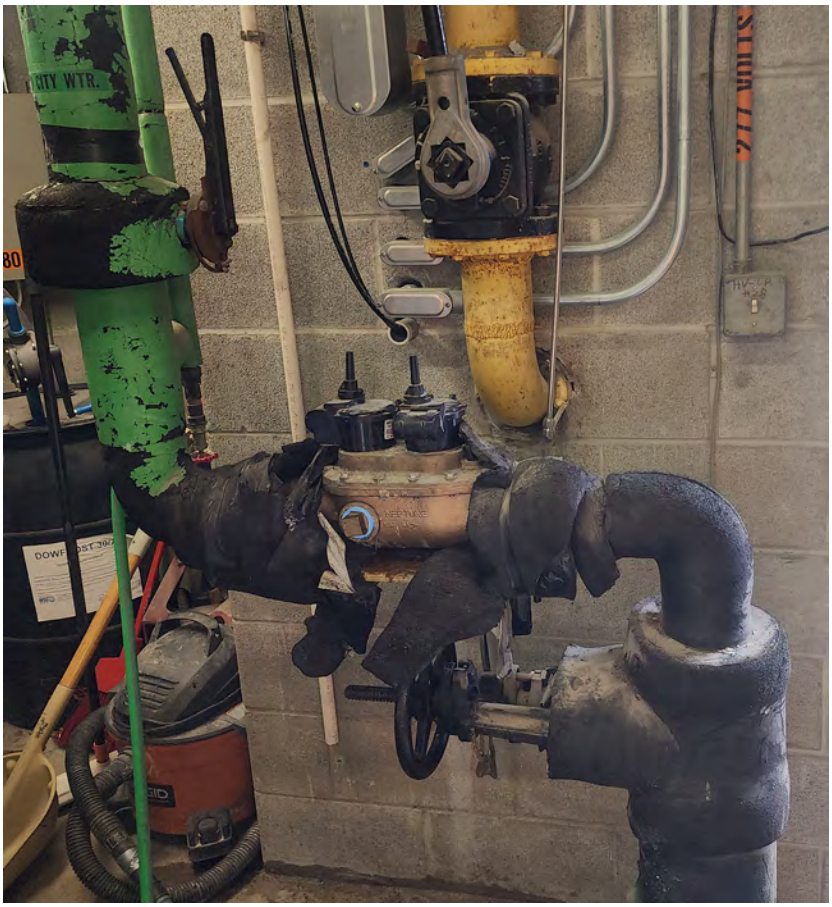


Figure 4-15 The 90° elbow just upstream of this compound meter disturbed the flow profile entering the measuring chamber, and the absence of a strainer allowed debris to clog the bypass meter measuring chamber. This meter tested at 74.7% accuracy at the maximum flow rate through the main meter and 68.1% accurate at the intermediate flow rate through the main meter. Low-flow rate testing registered zero flow (0.0% accuracy) for three test regimens: minimum flow rate through the bypass meter, maximum flow rate through the bypass meter (below changeover), and during the changeover. It registered 16.3% accuracy above the changeover level. On a composite basis, the meter performed very poorly.

meter test program is given in the "Large Customer Meter Accuracy Testing Program" sidebar.

For small (residential) meters, several regimens of testing should be established (see AWWA Manual M6 for guidance [AWWA 2012]). Random samples of newly received water meters should be tested to confirm that manufacturers are meeting performance specifications. Other sample groups of meters can be tested, including those with high-bill complaints, low or zero consumption, or for random sampling purposes. A separate test regimen of mechanical meters with high cumulative consumption should also be tested since these meters might be starting to show wear and have the potential for reduced accuracy. While volume should not affect static meter accuracy, these meter types should still be tested regularly to make certain that accuracy is being maintained.

Results of longevity testing of high cumulative consumption mechanical meters can help to develop a long-term meter replacement strategy based upon the identified level of cumulative consumption when accuracy begins to decline. Select small, intermediate, and large mechanical meters should also be identified for testing and/or replacement,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

including 1-in through 4-in meters, a midrange that is sometimes overlooked by utilities. These larger meters typically register high volumes of water and contribute large proportions of revenue, and large meter testing will confirm the ability of the meters to accurately register usage. Utilities with test equipment should annually test a statistically valid random sample of each meter size to gain a true sense of the collective accuracy of the customer meter population.

Large Customer Meter Accuracy Testing Program in the Louisville Water Company

The Louisville Water Company (LWC) is a proactive water utility that conducts customer meter accuracy testing of small and large meters supplying customers in Louisville, Ky, and the surrounding communities that it serves. LWC owns and operates a meter-testing apparatus at its facilities, which is used mostly for testing small water meters. LWC also owns meter test equipment for testing large customer meters in situ at their sites of installation. A truck is outfitted with equipment to allow testing of these large meters, which typically exist in a meter pit adjacent to the building that is supplied water. Figures 4-16 through 4-19 illustrate this equipment in use. LWC operates with a policy to test the water meters of the top 100 water-using customers on an annual basis and to test the water meters of the top 50 users on a semiannual basis. In this way, LWC keeps a strong level of surveillance of the metering of the largest customers and revenue producers and can act quickly to address those meters that begin to display declining meter accuracy.

Switching to the AWWA data for CWC, Tables 4-5a and 4-5b give example of large meter test results as described in Chapter 2. Table 4-5a lists the test results of compound water meters, and Table 4-5b shows results for turbine meters. It is always best practice to address large meters individually because of their distinctly different usage patterns, although one may attempt to calculate a composite accuracy value for a group of meters that are of the same size, manufacturer, age, and model for both lifecycle analysis and for completing the annual AWWA Water Audit. If a composite accuracy value is calculated for each meter size, then the degree of inaccuracy for the meter population can be determined as shown in the methodology of Tables 2-17 through 2-19 in Chapter 2.



Courtesy of Louisville Water Company

Figure 4-16 Truck equipped with large meter testing apparatus for in situ meter testing at LWC

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Louisville Water Company

Figure 4-17 Hose connected to large meter test port in underground meter pit and ready for testing



Courtesy of Louisville Water Company

Figure 4-18 Interior of large meter test truck showing calibrated test meter



Courtesy of Louisville Water Company

Figure 4-19 Water being discharged to waste during large meter testing operation. Utilities should be certain to follow applicable state and local requirements for the safe discharge of drinking water to surfaces.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 4-5a Compound meter test data for large meters for CWC

Meter Size	Meter Mfg.	Meter Model	Min. Accuracy	Below Change Over Accuracy	Change Over Accuracy	Above Change Over Accuracy	Inter-mediate Accuracy	Maximum Accuracy	AWWA M6 Overall Weighted Average Accuracy
4	Neptune	TruFlo	100.5%	98.9%	98.8%	101.2%	100.9%	101.4%	100.27%
4	Neptune	TruFlo	100.3%	100.7%	94.7%	100.9%	99.5%	100.1%	100.37%
6	Neptune	TruFlo	99.2%	99.4%	99.1%	100.1%	100.6%	100.7%	99.77%
6	Neptune	TruFlo	100.3%	100.8%	96.7%	99.1%	100.8%	101.5%	100.87%

Table 4-5b Turbine meter test data for large meters for CWC

Meter Size	Meter MFG	Meter Model	Minimum Flow Rate Accuracy	Inter-mediate Flow Rate Accuracy	Maximum Flow Rate Accuracy	AWWA M6 Overall Weighted Average Accuracy
3	Sensus	Omni C^2	100.9%	98.9%	100.9%	99.60%
4	Badger Meter	TSM 1000	98.6%	99.1%	99.5%	99.15%
4	Badger Meter	TSM 1000	98.8%	98.8%	99.4%	98.95%
6	Honeywell-Elster	evoQ4	98.9%	99.2%	100.1%	99.40%

Installation and upkeep of meters should be included as part of the ongoing functions of the water utility; therefore, funds should be budgeted to accommodate regular testing and replacement of customer meters. Implementing a program that routinely tests groups of customer meters can be an efficient and economical way to provide guidance to keep a meter population current, and the program can provide essential data to develop a rational long-term meter replacement plan for the customer meter population.

In addition to the normal low-, intermediate-, and high-flow rate tests, it is advisable for three additional tests to be conducted for compound water meters. A compound water meter consists of three primary components: a small measuring chamber (bypass meter) used for measuring consumption at low-flow rates, a main (turbine) meter for measuring consumption at both intermediate- and high-flow rates, and a crossover valve assembly that keeps flow out of the main meter until the minimum normal-flow rate for the main meter is reached. Utilities may determine if the crossover valve is operating properly by testing the meter (1) just below the flow rate at which the crossover valve is designed to open, (2) during the opening of the crossover valve, and (3) the flow rate just after the crossover valve moves to the fully open position.

Customer Meter Selection

In selecting meters for single-family residential service, utility managers should recognize that customer-side leakage is commonplace and that leakage flows often occur below the "low-flow" test level in the meter manufacturers' specifications. As previously noted, the cost of delivering water that is either unrecorded or underrecorded by the customer meter is ultimately recovered from all paying customers, and individual customers with hidden leaks may be unaware of this unintended consumption if it is not reflected in their water bill. Utility managers may consider investigating the use of higher performance water meters that are accurate at flow rates as low as ⅛ gpm, a flow rate that is typical of household leakage, particularly toilet leaks.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

For many water utilities, more than 50 percent of revenue is received from fewer than 20 percent of customer accounts classified as commercial, multifamily, or industrial that use meters of size 1 in and larger. It is, therefore, critical that these accounts be systematically reviewed to ensure that they are metered and billed correctly. If large meters have been in service for many years, current customer flows may not match the water demands that existed at the time the meter was installed. Customers may have become more water efficient and/or building use may have changed from a highly water-intensive use (as in a factory) to a low water use application (as in a warehouse). In such cases, low flows may not be registered accurately by old, large meters. Utilities with an AMI system may use the data analytics software to assist staff in identifying these accounts, and data-logging can be used to justify either downsizing the existing meter to an appropriate size or selecting a meter type that is better suited for the application.

Turbine meters are designed to register water consumption at continuous intermediate and high flows, but if the user experiences periodic or frequent consumption at low flows, a considerable portion of this water consumption may not be registered. Variations in flow rates from low to high can be measured by compound, electromagnetic (mag), single-jet, ultrasonic, and certain models of horizontal turbine meters (e.g., "floating ball technology"). Manufacturers publish the low-flow accuracy limits of the meters that they produce, so utility managers should make certain that meters they are considering will accurately register the expected low flows at a given customer site.

Data logging to obtain the customer water consumption profile. Data logging is the process of employing instruments that record the flow registered by water meters at very short time intervals, thus revealing the detailed water consumption pattern or flow profile. Data logging is often conducted for large water meter customers but can be carried out for any size meter. The water utility can select a representative sample of large meter accounts for data logging to confirm the customer water consumption profile and to determine which meters are properly sized for existing customers.

There are two ways to conduct data logging. One method is to reprogram the AMR/AMI system endpoint to its highest (most frequent) reading rate (i.e., every five minutes or every 15 minutes) and then download the meter-reading data for analysis before the nonvolatile memory in the endpoint is full. It is imperative that the endpoint be reset to its default reading interval setting as soon as the data-logging exercise is complete to preserve the life of the endpoint battery.

Utilities without an AMI system and data analytics software may use portable data-logging devices attached to the meter to record the flow registered by the meter. This method is useful for certain mechanical water meters such as positive displacement, turbine, and compound meters. The same loggers can also be used on both static water meters and mechanical water meters with a pulse output. This method involves either temporarily attaching the sensor for the data-logging device to the meter and recording the rate of change in the magnetic flux around the register drive magnets or splicing a special cable from the data logger to the pulse output cable from the meter register and recording the electronic pulses. These pulses can be converted into flow values and a detailed flow profile developed for the customer consumption for the period of the logging. The high, low, and average consumption values can be evaluated to determine whether a water meter is sized correctly for the actual water usage pattern. If flow through a customer meter is mostly occurring at either the low end or high end of the specified range of the meter, then the meter is likely improperly sized for the application.

When analyzing compound water meters, it is imperative that the amount of time that the flow resides within the crossover range for that specific meter (e.g., when the crossover valve is opening that enables flow to be registered by both the bypass meter and the main meter) is kept to a minimum. If the customer consumption profile includes flow rates that frequently transition between high and low rates of flow, some of the flow—

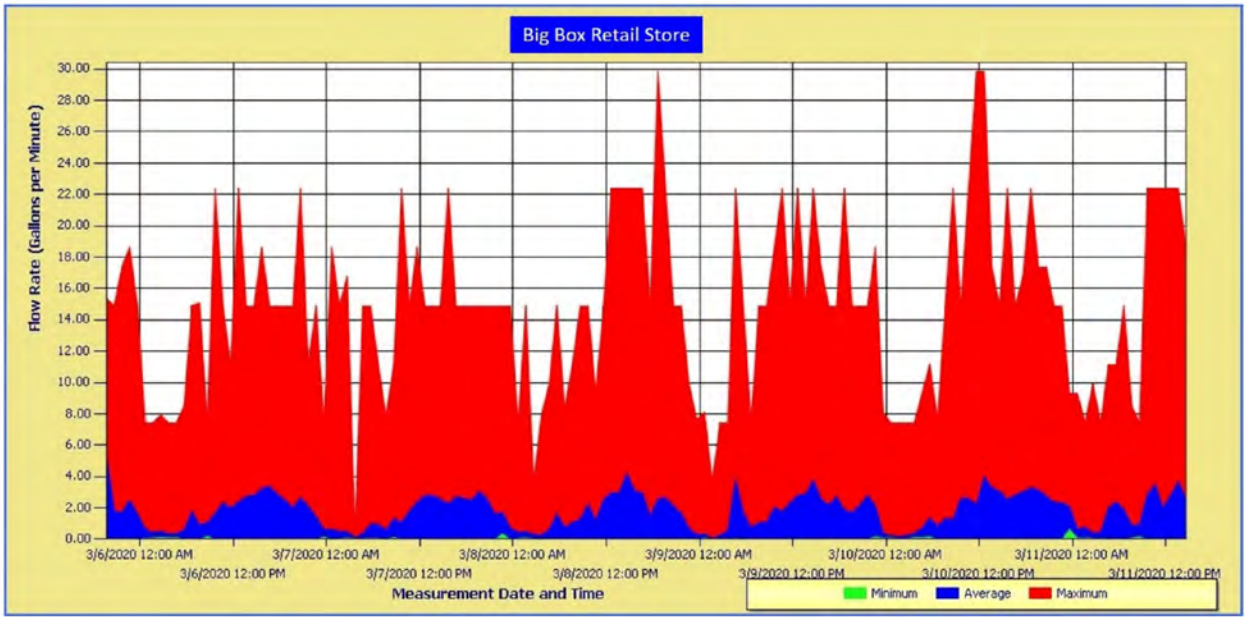
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

occurring at the transition flow rate—may not be accurately registered by the compound meter. Knowing the crossover range for a given compound meter and the customer water consumption profile is necessary to reliably specify the appropriate size of compound meter for a large water-consuming customer.

Data-logging technology provides the means to obtain detailed customer consumption profiles in a range of time increments: from hourly logging to establishing a basic consumption profile to every few seconds using high-resolution data loggers to capture transient peak flow rates. Some AMR/AMI system endpoints can transmit data continuously at intervals ranging from every one to 15 minutes. By obtaining and analyzing detailed logged data, the most appropriate size and type of meter can be specified for a given application.

The AWWA Manual M22, *Sizing Water Service Lines and Meters*, provides good guidance for the use of logged data to specify meter sizing for specific flow applications (AWWA 2024). Reliable data logging for meter sizing is dependent upon the resolution of the data. When using high-resolution data logging to capture peak flow rates, the time interval should be as short as possible (e.g., 10-sec intervals) while still registering appreciable non-zero flow data so that actual flow rates are recorded, as opposed to just a collection of average flow rates that may not accurately reflect the flow rate.

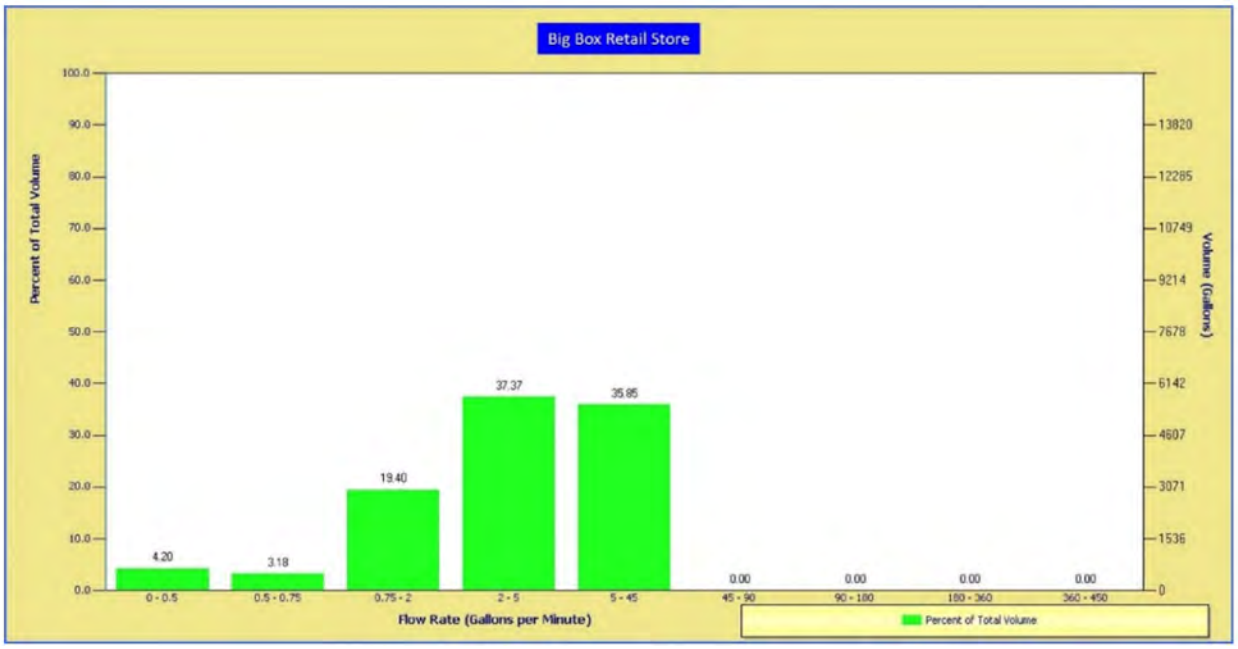
An additional example from Wylie, Tex., illustrates the value that data logging provides to the utility. In this case, a large meter survey identified a 3-in horizontal turbine meter serving a "big box" retail store, and this meter appeared to be improperly sized and typed for the application. The meter manufacturer's specification sheet stated that the normal flow range for this meter was between 5 and 450 gpm, and the accuracy curve for that meter indicated that 97% accuracy would not be realized until the flow rate exceeded approximately 2 gpm. A consulting engineer installed a data logger on the meter for seven consecutive days during a period of normal usage, and the data logger was set for a 10-sec sampling interval. The results of the data logging are presented graphically in Figures 4-20 and 4-21.



Courtesy of F.S. Brainard and Co. and the City of Wylie

Figure 4-20 Graph produced from Wylie, Tex., big box retail store consumption meter data logging showing minimum/average/maximum flow rates

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of F.S. Brainard and Co. and the City of Wylie

Figure 4-21 Graph produced from Wylie, Tex. big box retail store customer consumption meter data logging showing percentage of flow in given flow ranges

The data recordings found that a significant portion of the recorded flow occurred below 5 gpm and the maximum flow rate during the logging period was 29.85 gpm, which is far below the meter's maximum rate flow. These findings confirmed that the 3-in meter was oversized for the application. Figure 4-21 shows that approximately 26.78% of the flow during the logging period was at rates of less than 2 gpm, and 37.37% of the flow was between 2 gpm and 5 gpm. The meter analysis software estimated that the existing 3-in horizontal turbine meter could only account for 48.65% of the flow through the meter, and only 35.85% of the flow was within the normal range of the meter.

The analysis software recommended a 2-in meter with compound meter-like performance for this application, but there was insufficient room inside the meter vault to downsize the meter. The consulting engineer recommended replacing the existing 3-in turbine meter with a 3-in ultrasonic meter with the same lay length. A 3-in ultrasonic meter, with an accuracy of 100.0% ± 3.0% at a flow rate of 0.50 gpm and a normal flow range of 0.75 to 500 gpm was assessed to be able to account for 96.98% of the flow through the meter, and 90.15% of the flow was within the normal range of the meter. Using a water rate structure with tiers of $6.42 USD per 1,000 gallons (kgal) and $3.25 USD per kgal, the City of Wylie was projected to realize at least $4,432 USD per year in increased water and sewer revenue because of this large meter retyping project.

As another example, an energy services company (ESCO) developed a water loss optimization project for a large water utility in Illinois in 2018. This water utility had a total of 404 large water meters of size 3 in and larger. All these water meters were of the traditional horizontal turbine type. The utility had chosen to install this type of meter because of its lower purchase cost and lower maintenance costs compared to a traditional compound meter, and these meters were installed regardless of the water use profile of individual customers. Recall that traditional horizontal turbine meters are designed to accurately measure consistent moderate- to high-flow rates; applications that experience

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

low-flow rates will likely not be registered by this type of meter. Installing traditional horizontal turbine meters in applications with low-flow rates has been a common occurrence of applying the wrong "type" of meter in many water utilities. The utility did not consider the potential for increased NRW and uncaptured revenue at the time (~2009) of this purchasing decision.

Shortly before commencing the water loss optimization project, the utility replaced the measuring chamber assembly (the "turbine" unit that measures the flow) of 43 large horizontal turbine meters with a new chamber that was designed to accurately measure usage at low-, intermediate-, and high-flow rates. Engineers at the ESCO identified an additional 124 large horizontal turbine meters that appeared to be improperly typed for the application. The engineers installed flow recorders to each of those services and then analyzed the data. Findings from the loggings of the 124 horizontal turbine meters included

- eight of the meters were the proper type of meter for their application,
- 18 of the meters were both improperly typed and grossly oversized for their application, and
- 98 of the meters were improperly typed for their application.

In the case of the 18 grossly oversized meters, one option to consider is replacing the existing meter with a smaller, correctly sized and type meter. However, many water utilities include in their water rate structure a fixed or base charge that is based upon the size of meter, with higher charges for larger meters. In replacing large horizontal turbine meters with smaller meters, a loss of revenue will occur from the decrease in the base charge, and usually this is significant. In many cases such a loss of revenue will not be offset by the increased revenue due to the greater flow registration of the replacement meter. In such cases a water utility might consider replacing the less accurate turbine meter with a more accurate meter (mag, ultrasonic, other) of the same size to gain better registration without a reduction in the base charge.

Regarding the 98 improperly typed meters, the assessment found that replacing the measuring chamber assembly would be a cost-effective solution. At a water rate of $2.97 USD per 100 cubic ft, the utility would realize an annual revenue increase of more than $105,000 USD from replacing the existing horizontal turbine measuring chamber assembly with a new measuring chamber assembly designed to improve accuracy. Had the utility provided sewer treatment included as part of the water charges, the total annual revenue benefit from these large water meter improvements would be more than double that amount.

Table 4-6 presents the data logging results and analysis for 20 of the 4-in horizontal turbine meters in this study. Flows were recorded in units of cubic feet per minute (cfm), and the maximum flow is listed along with the percentages of time that the flow spent across five ranges of flow. The percentage of the flow measured by the meter is shown along with the percentage of flow occurring in the normal range for the meter. The projected performance levels of improved meters are shown under the heading of "Proposed Meters." Finally, the annual financial benefits of the proposed meters are shown in the column on the right.

In Table 4-6, the percentage of flow between 0 and 0.1003 cfm (0 and 0.75 gpm) is the volume of logged consumption that is below the extended low-flow accuracy limit (95%–101.5%) for the proposed meter. The percentage of flow between 0.1003 and 0.2005 cfm (0.75 and 1.5 gpm) is the volume of logged consumption that lies between the extended low-flow accuracy limit of 0.75 gpm for the proposed meter and the lower limit for normal flow (100% ± 1.5%) for the proposed meter. The percentage of flow between 0.2005 and 1.0694 cfm (1.5 and 8 gpm) is the volume of logged consumption that lies between the lower limit for normal flow for the proposed meter and the extended low-flow accuracy limit (97%) for the existing 4-in horizontal turbine meter. The percentage of flow

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 4-6 Data logging results for 4-in horizontal turbine meters for an Illinois water utility

#	Meter Mfg.	Meter Model	Max. Flow Rate (cfm)	% of Flow within Range [cfm = cubic feet per minute]					Horizontal Turbine Meter		Proposed Meter		Annual Benefit from Meter Typing
				0 – 0.1003 cfm	0.1003 – 0.2005 cfm	0.2005 – 1.0694 cfm	1.0694 – 1.3368 cfm	1.3368 + cfm	% Metered Flow	% Flow In-Normal Range	% Metered Flow	% Flow In-Normal Range	
1	Badger Meter	TSM 1000	9.98	0.11%	0.36%	25.29%	9.45%	64.78%	91.78%	85.22%	98.47%	99.52%	$1,230
2	Badger Meter	TSM 1000	8.67	0.18%	0.48%	18.37%	8.05%	72.91%	92.51%	88.60%	98.43%	99.34%	$1,220
3	Badger Meter	TSM 1000	6.28	0.58%	1.68%	47.09%	12.14%	38.50%	76.90%	67.21%	98.33%	97.74%	$2,305
4	Badger Meter	TSM 1000	3.68	3.59%	6.02%	62.25%	14.57%	13.58%	58.98%	40.41%	96.30%	90.40%	$1,128
5	Badger Meter	TSM 1000	6.74	0.05%	0.61%	34.26%	15.67%	49.41%	87.99%	79.18%	98.49%	99.35%	$1,580
6	Badger Meter	TSM 1000	2.90	1.15%	3.90%	38.90%	14.70%	41.35%	80.29%	65.65%	97.90%	94.95%	$875
7	Badger Meter	TSM 1000	3.02	1.62%	3.76%	54.80%	16.46%	23.36%	64.73%	39.82%	97.99%	94.63%	$1,381
8	Badger Meter	TSM 1000	6.42	0.32%	0.83%	35.65%	19.58%	43.62%	89.63%	82.14%	98.36%	98.85%	$1,195
9	Badger Meter	TSM 1000	4.60	1.05%	9.57%	52.95%	8.91%	27.52%	65.89%	48.15%	98.35%	89.38%	$2,281
10	Badger Meter	TSM 1000	1.54	13.73%	23.55%	42.17%	20.26%	0.29%	30.61%	22.20%	91.21%	62.73%	$546
11	Badger Meter	TSM 1000	2.89	1.24%	6.52%	82.46%	5.65%	4.13%	46.14%	22.28%	98.29%	92.24%	$2,813
12	Badger Meter	TSM 1000	2.44	6.73%	14.53%	72.41%	2.70%	3.63%	35.66%	18.01%	95.24%	78.73%	$745
13	Badger Meter	TSM 1000	5.89	1.89%	4.06%	42.66%	9.29%	42.10%	72.36%	62.32%	97.60%	94.05%	$1,763
14	Badger Meter	TSM 1000	0.46	45.25%	34.21%	20.53%	0.00%	0.00%	0.23%	0.00%	81.66%	20.54%	$583
15	Badger Meter	TSM 1000	58.36	1.89%	2.89%	10.02%	1.23%	83.97%	88.42%	87.04%	97.90%	95.23%	$1,459
16	Badger Meter	TSM 1000	6.10	0.00%	0.36%	44.97%	16.83%	37.84%	84.79%	72.85%	98.50%	99.64%	$1,861
17	Badger Meter	TSM 1000	7.23	1.09%	2.69%	57.58%	8.93%	29.72%	67.79%	53.78%	98.14%	96.22%	$2,396
18	Badger Meter	TSM 1000	16.33	0.00%	0.00%	49.43%	10.33%	40.24%	88.41%	72.70%	98.50%	100.00%	$1,457
19	Badger Meter	TSM 1000	1.62	23.45%	21.28%	52.30%	2.16%	0.81%	25.19%	6.57%	82.67%	55.26%	$155
20	Badger Meter	TSM 1000	4.17	0.00%	0.00%	8.22%	29.30%	62.48%	80.30%	50.15%	98.52%	99.99%	$2,099

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

between 1.0694 and 1.3368 cfm (8 and 10 gpm) is the volume of logged consumption that lies between the extended low-flow accuracy limit and the lower limit for normal flow for the existing 4-in horizontal turbine meter. The percentage of flow greater than 1.3368 cfm (10 gpm) is the volume of logged consumption that is within the normal flow range of the existing 4-in horizontal turbine meter. All but one of the 4-in meters were found to have a positive financial benefit from the proposed metering improvement. The assessment method shown in Table 4-6 allows for improved metering solutions to be identified along with the financial benefit. Data logging consumption measured by large meters is an approach that can be effectively employed by water utilities to reliably assess their large meter populations and make strategic decisions to control apparent losses from CMI.

Meter Replacement

Traditionally, water utilities set mechanical meter replacement schedules based upon years of service and meter size, with the largest of meters changed out as often as every 2 years and smaller residential meters extended to every 20 years. Alternatively, for small residential mechanical water meters, utility managers might consider meter-testing and replacement criteria based upon the cumulative flows passing through the meter rather than a fixed time interval. Replacement decisions for ultrasonic and electromagnetic meters are typically a function of battery life. Most of these meters in small sizes have a 20-year design life. For larger ultrasonic and electromagnetic meters, either the measuring chamber assembly or the battery pack typically needs replacing every five to 10 years.

Targeting mechanical meter replacements based upon cumulative measured volume is similar to automobile maintenance, where the 5,000-mile oil and oil filter change occurs not at any set time but only when the next 5,000-mile increment is reached on the vehicle odometer. By analyzing metered consumption data and conducting meter accuracy testing at certain levels of consumption, it may be possible to define—for a given mechanical meter size and type—the high cumulative consumption value of the meter, beyond which meter accuracy begins to decline notably. The methodology for this calculation for a small system in Arizona has been documented in a technical paper (Davis 2005).

This approach can be more efficient than a time-based cycle because heavily used mechanical meters will be replaced on a timely basis that will ensure accuracy is maintained, while lightly used mechanical meters will not waste resources by replacing meters too soon while they still maintain high accuracy. The ultimate mechanical meter replacement schedule, however, can be based around the cumulative consumption target in conjunction with crew deployment and scheduling realities. It may be logistically advantageous to have crews replace multiple meters in a given area at the same time, even if some of the meters have not yet reached their cumulative target. Additional discussion on meter replacement life and economics is found in the WRF report titled *Advanced Metering Infrastructure: Best Practices for Water Utilities* (Schlenger et al. 2011).

Using Midsize Turbine Meters for Low-Water-Consuming Buildings

It is advisable for all water utilities to review billing records and initiate data logging or other investigative actions for customer accounts in the midrange sizes of 3 in, 4 in, or 6 in that employ a traditional horizontal turbine meter with a relatively high low-flow threshold specified. Particularly for low-water-using building occupancies, there is a good likelihood that in many such metering installations, the turbine meter is failing to register a considerable portion of the flow, which is often below the meter's low-flow accuracy threshold. This may be causing considerable apparent losses and reduced revenue at these sites. Replacing these meters with an appropriate smaller meter or different type of meter with improved accuracy at low flow rates can improve accountability and revenue capture at many of these sites.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Based upon the methods discussed above, it is evident that there are many aspects of customer meter management that water utility managers must consider. By conducting meter accuracy testing and data logging as needed, the manager can obtain data to quantify the collective accuracy of the customer meter population. This includes how a utility is viewed in terms of completing its water audit. If these activities are undertaken, a high grading of the CMI component in the AWWA Free Water Audit Software will be assigned. If the utility does not conduct regular meter accuracy testing, data logging, or meter replacement, the quantity of meter inaccuracy entered into the water audit will be assigned a low grading. (See Chapter 2 for specific guidance on data grading in the audit software.) By following the steps outlined in this chapter, the manager can take steps to improve the accuracy of the customer population and will be able to monitor this improvement objectively. Good composite accuracy of the customer meter population can be maintained by

- replacing mechanical meters according to a reasonable schedule based upon cumulative flow registration, age of meter, or a combination of these criteria;
- having the means to closely monitor customer consumption trends and use meter-accuracy test results to flag mechanical meters for replacement close to the time when they begin to appreciably lose accuracy from wear;
- using available meter data analytics to look for unexplained drops in water consumption and other anomalous meter flags;
- identifying oversized meters and replacing them with a similar meter of the appropriate size;
- identifying meters of the wrong type for the application and replacing them with a more appropriate type of meter; and
- staying abreast of innovative metering technology and applying it to the benefit of the overall meter population and revenue capture.

Managing a population of customer meters requires knowledge of meters and meter-reading equipment as well as billing policies and customer relations. Policy and procedures regarding the sizing and installation of customer meters also play a role in water supply efficiency, and these should be reviewed to ensure that inappropriate meters are not installed inadvertently because of policy shortcomings. The benefits of accurate customer metering, however, continue to evolve as consumption data are recognized as critical to evaluate revenue protection programs, water loss control, and water conservation programs.

Metering Fire Services

Water supply for fire protection service is routinely provided to buildings of all sizes from large commercial and industrial buildings, multifamily residential buildings, and one- and two-family residential buildings. Fire flow supply is typically provided via separate, dedicated fire connection piping in larger multifamily residential, commercial, and industrial buildings. These are connections typically size 3 in and larger. Fire supply to smaller buildings is supplied by either dedicated fire connection piping in addition to the domestic supply connection or by a single connection that supplies both domestic and fire supply. Guidance on the sizing of fire connection piping and meters, types of meters, and related criteria including backflow protection, valving, rates and charges, and other considerations can be found in a variety of references. Specific requirements are often established by local jurisdictions and can be defined by fire, building, plumbing, and water service codes and ordinances. The National Fire Protection Association (NFPA) provides guidance in three of its standards: NFPA 13, 13R, and 13D. AWWA manuals M6, M22, and M31 give

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

guidance on water meters, sizing service connections and meters, and fire protection in water supply, respectively. AWWA Standards exist for fire service meters and residential fire service meters. Detailed guidance is given for residential fire sprinkler systems in the AWWA white paper *Residential Fire Sprinkler Systems: Guidance for Water Utilities* (2018). The specifics of metering of fire services are beyond the scope of this manual, and the reader is referred to these references.

UC

No water utility is immune to the occurrence of UC. UC occurs in all water utilities; only the extent of the occurrence varies (Thornton 2002). UC can occur from weak policies, practices, and oversight by the water utility, coupled with deliberate actions by a segment of the customer population set on avoiding paying for water service. UC can be significant for some water utilities while negligible in others; thus, utility managers should conduct investigations to estimate the general likelihood of the occurrence of UC and project the amount of attention that the issue requires in their operations.

Water utilities can exert control over the occurrence of UC by focusing on the following:

- Having coherent, workable policies that clearly identify which uses of water are authorized vs. those that are not authorized. Policies should exist to cover the wide spectrum of activities that occur in providing water service.
- Having good detection capabilities to become aware of the various UC events that can occur in the utility, close to the time in which the use begins.
- Having the enforcement means to halt such consumption and invoke appropriate penalties.

Having mechanisms in place to detect trends of UC is most important. Flowcharting the processes of the customer billing system as illustrated in Figures 4-6 through 4-8 gives the auditor insight into any loopholes that may exist that allow unauthorized consumption to occur and go unnoticed by the water utility. Once identified, loopholes can often be expeditiously closed by procedural, programming, or permitting corrections, perhaps resulting in a return of additional revenue. Water utilities should also create routine output reports that present data from the customer billing system in a way that reveals unusual consumption trends and flags data anomalies that suggest suspicious activities. Such reports should be run and reviewed at the same frequency as the billing cycle. The water utility should assign appropriate staff time to this function.

An example of such a tracking report is a zero-consumption report that lists active accounts with unchanged meter readings (hence, zero consumption recorded) for two or more consecutive billing cycles. Customer accounts in active billing status that register unchanged meter readings for ongoing billing cycles might be indicative of vacant properties with no ongoing water consumption. This status does not represent UC or a specific problem for the water utility. Zero consumption can also occur because of meter failure (stopped meter) or AMR/AMI device failure. This incidence is also not unauthorized consumption but an apparent loss occurring as CMI. The third possible cause for zero consumption billings is tampering of the customer meter and/or AMR/AMI device, an activity that indeed is UC. Customer meter tampering might be controlled by locking devices that secure the covers of outdoor meter pits and bypass valves. AMR/AMI systems have a variety of alerts that flag tamper events on the meter or meter-reading equipment, such as empty pipe alarms and reverse flow alarms. These give the water utility immediate indication of unauthorized activity.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of New Jersey American Water Company

Figure 4-22 Public relations flyer cautioning communities about the impacts of unauthorized water consumption

If the water utility has a significant number of accounts that register zero consumption for more than two consecutive billing cycles, then inspections of a sample of the individual customer premises should be undertaken to uncover the reason for the unchanged water meter readings. These physical inspections are bottom-up auditing activities that locate apparent losses in an account-by-account manner. By physically inspecting a representative sample of these accounts, the utility manager can gather data to determine the percentage of zero consumption accounts occurring because of vacancy, meter/AMR/AMI system failure, or tampering.

Similar bottom-up loss control activities can include the following.

- **Collecting data on field reports of illegally opened fire hydrants.** The auditor should review opportunities for the unauthorized use of fire hydrants and ensure that a rational policy regarding fire hydrant use exists. The Loudoun County Sanitation Authority in Loudoun County, Va., developed a comprehensive policy and detailed procedures for fire hydrant usage that has allowed them to better balance the need for access to water supply vs. protection of the water distribution system and water quality. Their efforts were well documented in an article in AWWA's *Opflow* publication (Villegas 2006). Field data can show the extent to which these policies are violated. Actions to control such unauthorized usage may

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

include the installation of fire hydrant locking devices, increased public education functions, and the enactment of stricter penalties for unauthorized use of fire hydrants. Figure 4-22 illustrates an effective public relations flyer to communicate with customers the negative impacts of unauthorized consumption and utility policy. Recently, some manufacturers have begun to produce "smart fire hydrants" with leak detection and pressure sensors that could detect hydrant tampering in addition to other functions.

- **Conducting random inspections after terminating service.** If utility policy allows customer water service to be terminated because of payment delinquency, follow-up random inspections should be conducted to ensure that customers have not reactivated their service illegally. If the water utility halts water service by closing a "curb stop" or valve in an outdoor meter pit or at the building property line, some unscrupulous customers may gain access to these valves and reopen them. Locking devices can deter these activities. AMI systems will register forward water usage at the time that this type of illegal activity occurs, allowing the water utility to respond quickly to this unauthorized consumption.
- **Identifying meter/AMR/AMI device tampering.** Unauthorized consumption occurs in all utility/services sectors, including electric, cable TV, water, gas, and other services. In virtually any water utility, there are certain customers who will tamper with meters and AMR/AMI system devices to create an erroneously low measure of water consumption on their water bill. Tampering can be identified by review of zero consumption accounts, as described previously. Shrewd individuals, however, may create a low consumption bill rather than a zero-consumption bill, making it less likely to draw attention to the consumption history. Meters may be removed and installed backward for several days or weeks each month to reverse the consumption accumulation. Meters may be removed and replaced with a straight pipe ("jumper" or "cheater") for several weeks each month to obtain unmetered water for a portion of the billing period. Figures 4-23 through 4-26 show examples of unauthorized consumption that occur due to meter installation tampering. Careful inspection of billing records can identify suspicious consumption patterns that may identify tampering, and mag and ultrasonic meters



Figure 4-23 The open bypass line valve on this 4-in compound meter setting was costing the utility approximately $127,100 USD annually in unbilled water and sewer revenue.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

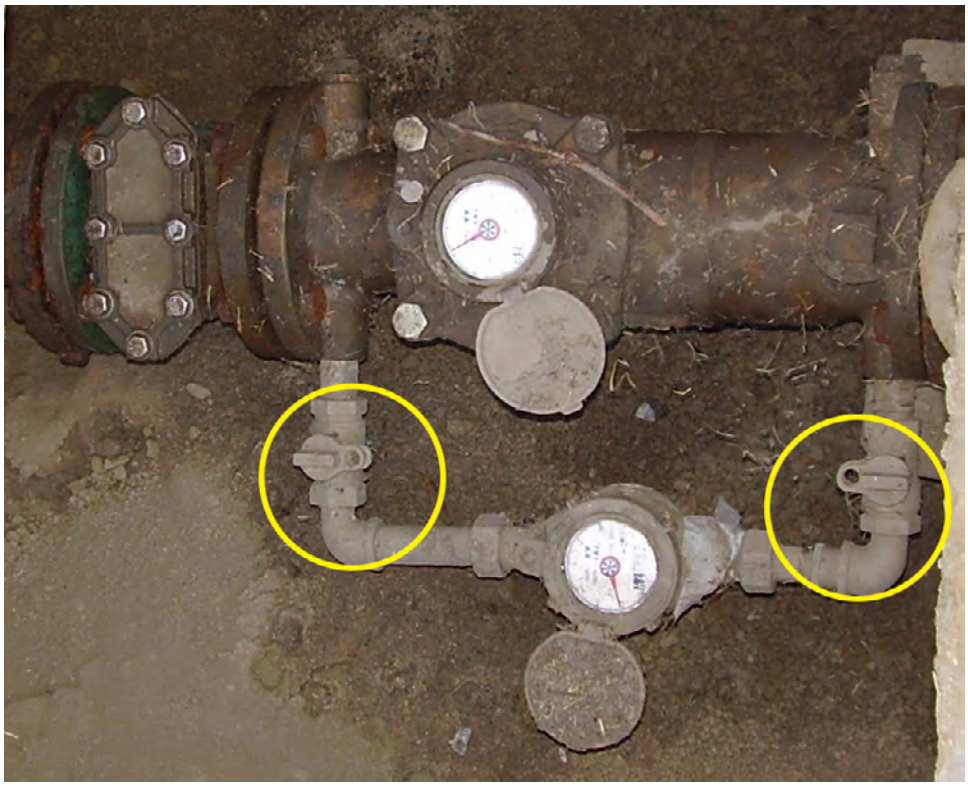


Figure 4-24 The bypass meter isolation valves on this 3-in dual-body compound meter have been closed, which renders the meter incapable of registering any usage below 8.5 gpm.

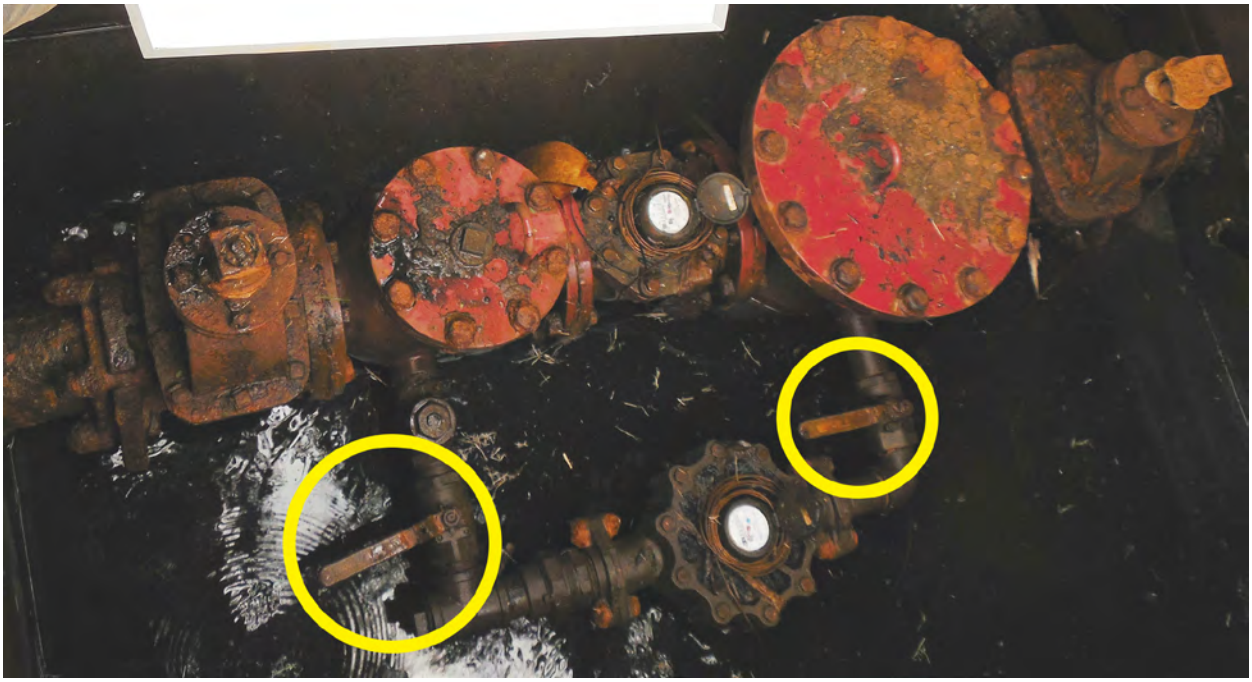


Figure 4-25 The bypass meter isolation valves on this 6-in fire service meter serving a condominium have been closed, which renders the meter incapable of registering any usage below 45 gpm.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Figure 4-26 A finishing nail has been driven through the meter register between the face and the flow indicator, impeding the ability of the impeller to rotate on the spindle of this single-jet meter.

can generate an "empty pipe alarm" if removed. Utility managers can schedule inspections of customer metering installations on a periodic basis to attempt to identify tampered meters. AMR and AMI systems with tamper alerts also provide evidence of equipment tampering and suspicious consumption and may be the most reliable source of information on these activities.

- **Conducting periodic inspections for illegal bypass piping.** Many customers with large meter settings include unmetered bypass piping around the meter that can be used to sustain water service while the meter is periodically taken out of service for maintenance or replacement. By simply closing and opening certain valves, the customer can route water flow through the unmetered bypass line with little or no flow going through the metered line. Water utilities should schedule large meter installations for periodic inspections to identify cases of illegally opened bypass piping. It is best practice to lock valves in the closed position on bypass piping to prohibit tampering. The use and monitoring of water consumption output reports may show significant changes in water consumption patterns for given accounts, which can indicate that a bypass has been illegally opened. Imposing significant fines and penalties may also convince customers to avoid this illegal activity.
- **Conducting inspections to identify illegal connections.** Unfortunately, illegally installed connections into water supply piping are common throughout the world. In some countries, it is not uncommon for water pipelines to be flagrantly excavated and tapped illegally with multiple lines. More often, illegal lines are not connected in such a blatant manner but exist more surreptitiously. It is not unusual

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

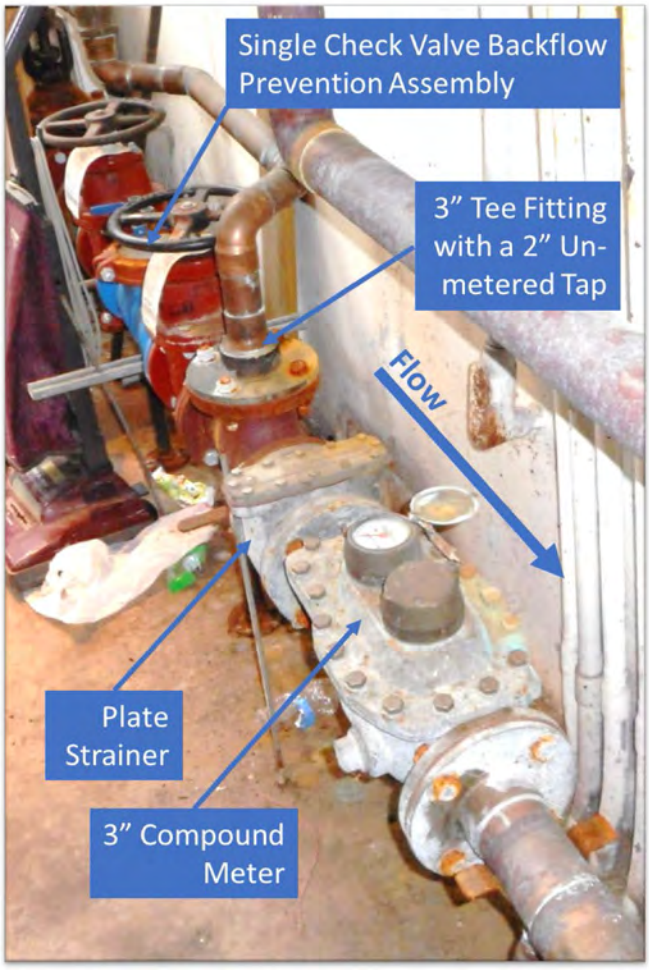


Figure 4-27 Unmetered 2-in tap upstream of a 3-in compound meter. There also is a single check valve backflow prevention device upstream of the meter. Backflow prevention devices should never be installed upstream of a water meter.

for unmetered fire connection piping to be tapped inside building premises. Sometimes this occurs inadvertently when contractors misidentify fire connection piping as the domestic water supply line. However, many such illegal service connections are installed purely with intent to obtain water without paying for the service. Since these illegal connections are visible inside of building premises, water utilities can schedule periodic internal inspections of building supply line and plumbing to identify such connections. These inspections might be combined with existing inspection activities, such as leak detection or cross-connection control inspections. This would minimize the need for additional resources to conduct the inspections. Figures 4-27 and 4-28 show examples of illegal water lines tapped into existing piping systems to gain unmetered water.

- **Inspecting valves on interconnecting water systems.** Like bypasses around meters, valves on standby or emergency unmetered water supply pipelines to neighboring water systems might be occasionally opened inadvertently or unscrupulously to cause UC. Water utility interconnections with neighboring water utilities are common. Many of these pipelines supply water on a continuous basis,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Figure 4-28 Illegal bypass at a residential meter service discovered during a water meter and mobile AMR system project

but some interconnections exist for standby or emergency supply purposes. These latter connections normally exist with supply line valves closed until authorized for temporary use during emergencies or special conditions. If valves on these lines are inappropriately opened, water supply may be obtained by the receiving water utility in an unauthorized manner. It is not unusual for such connections to be unmetered since they are not intended for continuous water supply. Valves on such interconnection pipelines should be inspected periodically to ensure that they are in the proper position. If UC is occurring on these interconnections, the water utility selling the water should consult its contract with the receiving water utility and determine what penalties might apply. Cutting in a water meter and then connecting that meter to either an AMR or AMI system will provide documentation of the water usage which could then be billed.

The "Criminal Prosecution of Unauthorized Consumption in the Kansas City Water Services Department" sidebar describes the case of an illegal connection used to supply irrigation water to a golf course in Kansas City, Mo. In cases of hidden illegal connections, water utility managers may be able to identify suspicious supply if the consumption pattern for a water user appears to be inordinately low for the class of service. This was the case in Kansas City. Once illegal connections are identified, the water utility must rely on its enforcement policies to invoke appropriate warnings and penalties to offending property owners to eliminate the illegal connection and pay back charges that are due. Legal and/or criminal remedies may also be needed.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

These are some of the actions that are typical of the bottom-up procedures utilities can undertake to quantify and control unauthorized consumption. Every water utility is unique, however, and certain circumstances may exist in each system that allow for the possibility of UC. Therefore, it is worthwhile for utility managers to conduct a reasonable level of bottom-up auditing investigations to identify opportunities that might allow unauthorized consumption to occur in their system.

For control of UC on a long-term basis, the water utility should employ effective policies and enforcement capabilities. This may require changes in existing regulations, statutes, or codes or the creation of new ones. Implementing changes in these instruments can be politically sensitive and requires skilled effort over potentially long periods of time; however, a strong legal framework will ultimately allow the water utility to operate with sufficient enforcement powers to keep UC to an economic minimum.

Criminal Prosecution of Unauthorized Consumption in the Kansas City Water Services Department

The Kansas City Water Services Department (KCWSD) is a large water utility that is active in controlling water losses and has long emphasized monitoring its operations for apparent losses. When proactively assessing the billing records of its large consumers, utility auditors took note of the billing records of a particular golf course as very suspicious. Golf courses can rely on large volumes of water for irrigation. While some golf courses are supplied recycled water to use for irrigation, KCWSD supplied this golf course from its potable water distribution system. This facility was expected to register high volumes of consumption during the irrigation season, but it routinely registered unusually small, metered water consumption volumes. An inspection of the service line and metering installation found that an illegal connection, with a line valve, had been tapped into the primary water service connection piping to divert water—in unmetered fashion—to a pond on the golf course property. Water from the pond was used to irrigate the golf course.

Not only did KCWSD then work to halt the illegal use of water, but it followed with legal action that ultimately led to the owner of the golf club pleading guilty and being placed on four years of probation under the conditions that he repays the city $251,400 in four installments. The judge in the case left open the option to sentence the owner up to seven years in prison if the debt went unpaid. The golf course owner admitted that between 2003 and 2008, golf course employees routinely opened the valve on the illegal line to divert water supply to the pond. KCWSD initially placed a lien on the golf course and billed the owner $1.6 million in back charges for water services and fees. As part of the court rulings, KCWSD agreed to settle the claim against the golf course operator for the $251,400 amount.

This incident testifies to the significant extent to which water customers will go to take water unlawfully, and to the large volumes of water and revenue that can be lost. The volume of unbilled water taken over a period of five years was considerable and resulted in uncaptured revenue of several hundred thousand dollars per year to KCWSD. And this was just from a single customer! This event underscores the fact that water utilities are vulnerable to occurrences of unauthorized consumption and—like KCWSD—should take proactive steps to identify suspicious customer behavior and control the occurrence of unauthorized consumption to economic levels.

*Club and Resort Business 2009.

DEVELOPING THE APPARENT LOSS CONTROL STRATEGY

Figure 4-29 is a graphic that represents the four-pillar approach to loss control interventions applied to apparent losses. The center boxes represent three levels of apparent losses, as defined here.

- The outer box perimeter represents the current annual volume of apparent losses listed in the water audit.
- The perimeter of the middle box represents the utility-specific target level for apparent losses. Conceptually, this is the Economic Level of Apparent Losses

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

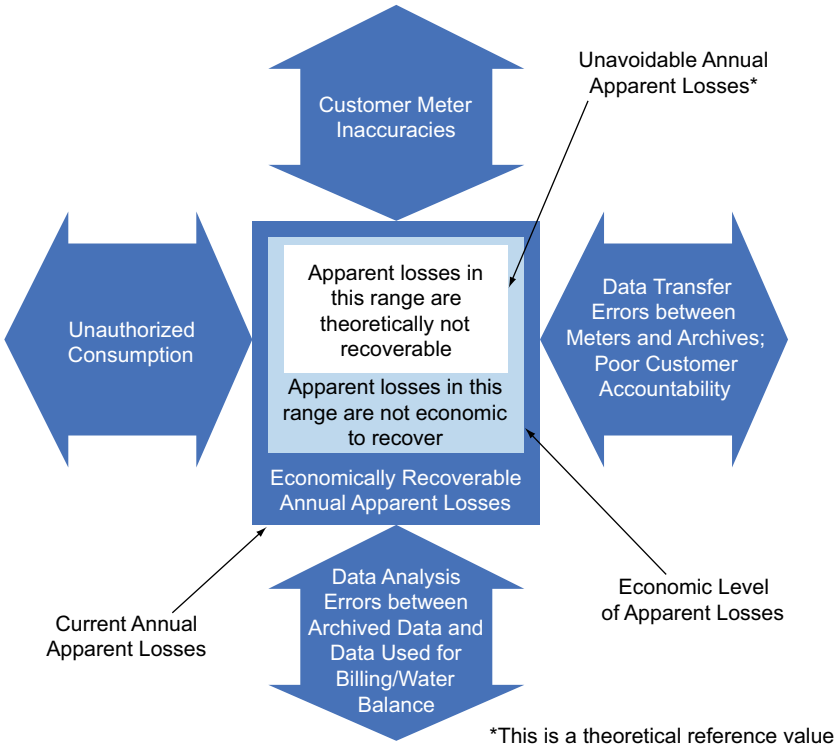


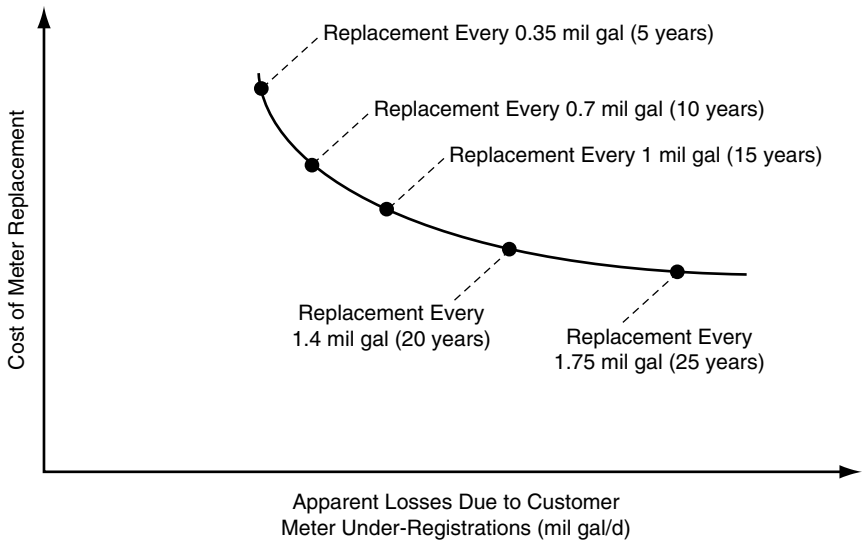
Figure 4-29 Four-pillar approach to the control of apparent losses

(ELAL) or the level at which the cost of the loss control efforts equals the savings garnered from the loss recovery. The ELAL represents a breakeven point, beyond which the effort to control apparent losses costs more than the likely recoveries of revenue.

- The inner box is defined by the Unavoidable Annual Apparent Losses (UAAL) as the quantity representing its perimeter. This is a conceptual level of loss representing the lowest level that could be attained if all possible loss controls could be exerted. Unlike the unavoidable annual real losses (UARL) that have an established calculation, an established formula or reference value for the UAAL does not currently exist. Discussion on the means to develop a calculation for the UAAL has occurred at different times (Rizzo et al. 2007).
- The four arrows represent the mechanisms by which apparent losses occur and the intervention actions needed to reduce them. The dual directions of the arrows reflect the fact that by exerting control in each component of loss, the total annual volume of losses (outer box) can be reduced. The arrows also reflect that lack of control of these components results in the total volume of apparent loss increasing.

Controlling losses in almost any field of endeavor is an effort of diminishing returns, as many losses can never be completely eliminated. When losses are extensive, relatively large reductions can often be gained early in a loss control program; this is known as picking the low-hanging fruit. However, further loss reduction requires ever greater cost and effort to recoup ever-decreasing returns. Figure 4-30 provides a cost curve for mechanical water meter replacement with moving parts that eventually wear out and cause underregistration. Points plotted on the curve represent meter replacement frequency (in years) and

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Source: WRF 2007

Figure 4-30 Cost curve for a mechanical water meter replacement program

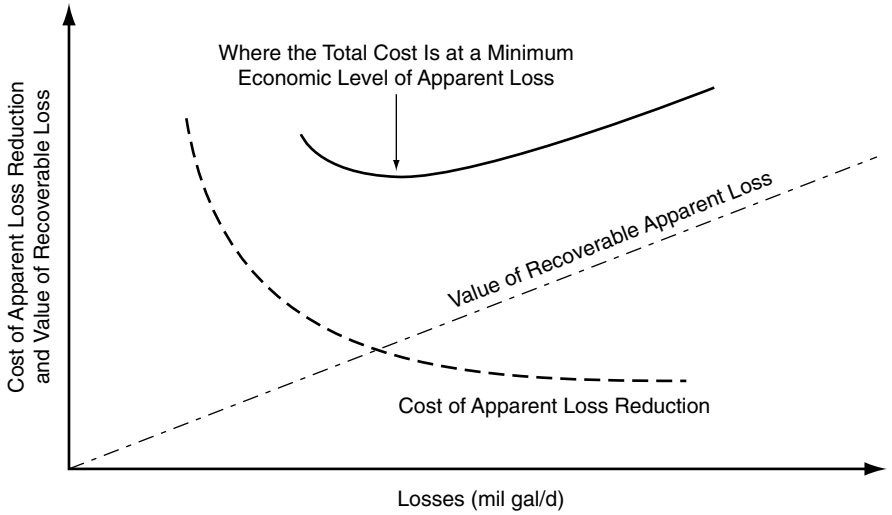


Figure 4-31 Economic balance for an apparent loss reduction by mechanical meter replacement

average cumulative consumption passed through the meters (million gallons). Replacing mechanical meters at a high frequency results in less apparent loss since a younger, more accurate meter population will exist. However, a high-replacement frequency requires higher replacement costs. Replacing meters less frequently saves costs for water meters but allows apparent losses to grow and run longer.

When setting an apparent loss reduction target, there exists a breakeven point beyond which the effort to control the losses costs more than the benefits derived from the recoveries; thus, further loss control effort is not an economical pursuit. This is the ELAL or the economically ideal target of apparent losses to seek. The ELAL for a mechanical meter component of apparent losses can be derived via the approach shown graphically in Figure 4-31. In this figure, the mechanical water meter replacement cost curve from Figure 4-30 is matched against the cost recovery line, which reflects the savings generated

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

by apparent loss recovery. A third curve is generated at the top of the graph by adding the two values and plotting the added values, giving a total annual apparent loss cost curve. The ELAL for apparent loss caused by mechanical meter inaccuracy is found by taking the level of loss at the minimum point of this curve, as shown in Figure 4-31. The optimum level of apparent loss reduction at the ELAL is determined by reading back off the apparent loss reduction cost curve. For apparent losses caused by CMI, the optimum frequency of mechanical water meter replacement is determined by selecting the point on the mechanical water meter replacement cost curve that matches the minimum point of the total cost curve in Figure 4-31.

A benefit–cost approach can be used to assess the subcomponents of apparent loss. The benefit–cost analysis should account for all costs, including administrative and billing personnel expenses, to manage errors, refunds, and the cost to verify readings.

In generating a particular curve, the economic analysis should start with determining the volume and cost value of the most significant sources of apparent loss. For each apparent loss component, it is necessary to analyze the problem and determine why these errors are occurring. It is then possible to consider various solutions to reduce these losses. Possible solutions might range from improved auditing, new reports to identify these errors, or better training as low-cost endeavors to full AMR/AMI system implementation or a new customer billing system at the opposite end of the cost spectrum. Chapter 7 provides further discussion on planning loss control activities.

As an example, solutions to reducing apparent losses caused by meter-reading errors may range from better training for meter readers, improved auditing of meter readings, and improved software on handheld meter-reading computers to the implementation of a complete AMR/AMI system. The cost of each of these alternative solutions should be compared to the projected revenue recovery from the reduction in apparent loss and the solutions ranked in terms of benefit–cost ratio. Savings from related benefits such as reduced visits (truck rolls) to customer premises in the AMI environment should also be included in the analysis. Only those solutions with a sufficiently attractive benefit–cost ratio payback period should be included in the apparent loss control plan. Clearly, the scale and the shape of the cost curve for solutions to the various components of apparent loss could be very different and will vary from utility to utility. Until further research has been undertaken and standardized models or spreadsheets are available, it is up to each water utility to develop appropriate utility-specific costs and cost curves for the various apparent loss components that they perceive to be significant.

The previous example illustrates two limitations in the current status of apparent loss target-setting. First, in applying the method using cost curves, considerable data on meter accuracy at varying meter consumption levels must be generated. This can be a complex and time-consuming undertaking. Second, separate cost curves must be developed for each component (and subcomponent) of apparent losses that are deemed significant: one for CMI, one for mechanical positive displacement meter replacement, one for turbine meter replacement, one for failed (stopped) meters, and so on. Similar curves are needed for subcomponents of SDHE and UC. Unfortunately, there is no single, composite ELAL for a water utility. Instead, there is an economic level for each apparent loss subcomponent control solution considered, and the overall ELAL for the utility is the sum of the individual solutions that are selected. Therefore, the present means of rigorously developing the ELAL is a demanding task that cannot be executed without considerable data. AWWA and the water industry continue to work to develop a simpler, more straightforward method of obtaining the ELAL.

While the current approach to identify the overall ELAL is time and resource intensive, apparent loss recovery can often generate considerable "new" revenue. Until a simpler method to calculate the ELAL is available, water utilities should undertake a cursory

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

analysis of their apparent losses and identify approximate levels of desired apparent loss reduction. If a water utility is only beginning to audit its water supply, it is likely that appreciable apparent (and real) losses exist, and it will be economical to recover a notable volume of losses. In lieu of a complex apparent loss analysis, the following recommendations are reiterated as standard starting points for water utilities in apparent loss control.

1. Flowchart the customer meter-reading and billing process. Understanding this process and identifying any lapses or loopholes that allow apparent losses to occur is fundamental to the management of all apparent loss components. This exercise can be conducted largely in an office setting with limited resources and at modest cost and may identify several loss components that can be quickly and inexpensively corrected by policy, procedural, or computer programming improvements.
2. Unless the customer meter population is young and well documented, collect data on water meters and create meter demographics reports and statistics. Perform annual meter accuracy tests on a statistically valid, random sample of customer meters, ideally as part of a formal meter test program. Data from this testing will reveal the current accuracy status of the existing meter population and the yearly trend will ultimately reveal the points at which mechanical meters lose accuracy significantly from high-cumulative volumes passed through the meter.
3. Review customer billing records and consider data logging 5 to 10 meters of the highest use customers to identify large-water-using accounts that are underbilled because of inappropriate meter sizes or types. Utilities with AMI systems may take advantage of the analytics software to identify those accounts. Target traditional turbine meters sized 3 in, 4 in, and 6 in for investigation, since these meters often do not capture low flows that are occurring.

The above first steps are manageable in terms of effort and expense and can provide good data and possible revenue recoveries that can get apparent loss control efforts started. Once water auditing has been performed for several years, additional bottom-up data will be available, and a more robust assessment of apparent losses can be undertaken.

Figure 4-32 identifies a sequence of steps for executing the apparent loss control strategy after the initial top-down water audit has been compiled and bottom-up activities

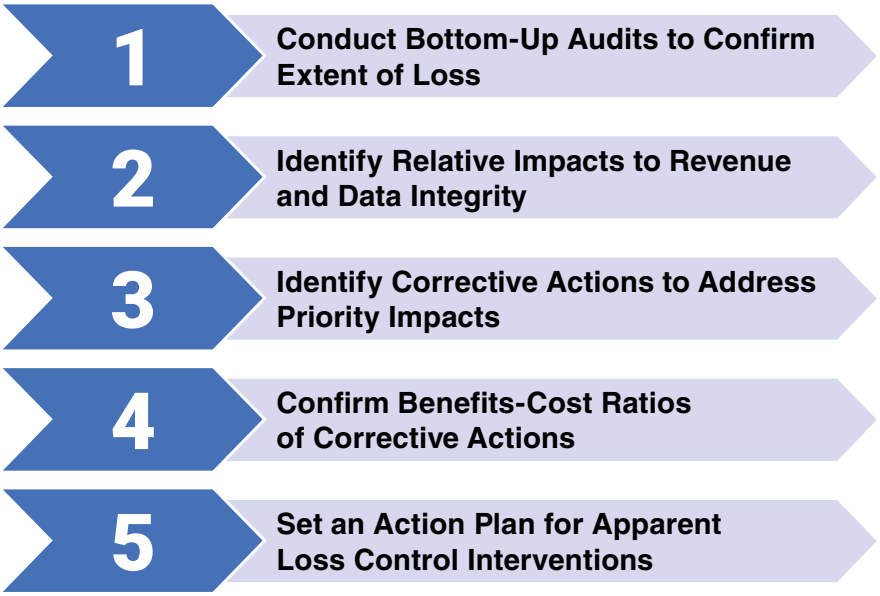


Figure 4-32 Steps to execute an apparent loss control strategy

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

are launched. These steps, starting with the bottom-up auditing procedure, should be followed in sequence to ensure that intervention actions are economically justified, well planned and executed, and documented.

DEVELOPING A REVENUE PROTECTION PLAN TO CONTROL APPARENT LOSSES

The most significant impact of apparent losses for water utility managers is uncaptured revenue. The term *revenue protection program* is used to identify the host of procedures put in place to protect the utility's revenue base by controlling apparent losses. As previously noted, several distinct components and subcomponents of apparent losses occur in water utilities; therefore, a revenue protection program must be tailored to the individual apparent losses of the water utility. The "Sample Revenue Protection Plan" sidebar shows an example revenue protection plan for CWC based upon their water audit for calendar year 2020. Revenue protection plans should be developed by considering each of the three major components of apparent losses: SDHE, CMI, and UA, and potentially subcomponents of each of these. Data from the water audit should be evaluated to assess the relative value that each component offers the water utility. In the CWC sidebar example, the volume of UC is the lowest of the three apparent loss components, and CWC determined to not include it for specific revenue protection interventions at this time. Work in subsequent years can include investigation of unauthorized consumption for action in the revenue protection plan.

The example, expressed in US dollars, shows that the cost impact of potentially uncaptured revenue to CWC caused by apparent losses is $251,730 for the 2020 audit year. In following with the previous recommendations, the manager determines to launch a revenue protection program that will analyze the customer billing process and institute annual customer meter accuracy testing.

The billing process analysis (flowcharting) is envisioned as a two-month project costing $24,024. This cost includes the analysis and any apparent loss corrections that can be immediately incorporated into the process. CWC conducted accuracy testing of a sample of customer meters during the compilation of its initial water audit and determined that it should continue testing a sample of meters on an annual basis to track the accuracy of the customer meter population and monitor degradation of accuracy over time. The projected annual cost of this effort is $7,140 to test 50 residential meters and five large meters.

The total first-year cost of the two-component revenue protection program is estimated at $24,024 + $7,140 = $31,164. By applying its equivalent composite customer retail billing rate of $3,945/mil gal, CWC need only recoup 7.90 mil gal (31,164/3,945) of apparent loss to break even during the first year of program operation. This is only 12.4 percent of the total apparent loss volume of 63.866 mil gal quantified in the water audit. If each residential customer consumes 42,000 gal/year (3,500 gal/mon) of water, then the equivalent of recovering 188 missing accounts from the billing roles would meet the cost-effective breakpoint of 7.90 mil gal recovered. This is only 1.6 percent of the total of 11,490 residential customer (5/8-in and 3/4-in meter sizes) accounts in the customer billing system. It is evident that recovering losses valued at the customer retail rate can potentially offer a swift and strong payback.

As the program matures, the water utility can consider more extensive improvements to control the subtler forms of apparent loss occurring in the system. Such efforts will require additional costs and may include replacement of large numbers of customer meters, installation of an AMR/AMI system, and data analytics software. Chapter 7 gives guidance on water loss control program planning with consideration of long-term upgrades to the major systems and processes of the water utility that impact on apparent loss control.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Sample Revenue Protection Plan

Name of Water Utility: *County Water Company* Date: *07/10/2021*

I. Revenue Protection Plan

After completing County Water Company's (CWC's) first annual water audit for calendar year 2020 (see Figures 2-4 through 2-7 in Chapter 2), the manager created an ongoing revenue protection program that identifies causes of the most significant apparent loss components and launches efforts to reduce these losses to economic levels. After initial gains are evaluated, additional less significant occurrences of apparent loss will be evaluated for reduction. All costs are expressed in US dollars.

The CWC manager broke down the apparent loss volume of 63.866 mil gal into subcomponents and their cost impacts, using the CRUC of $3.95 per 1,000 gallons of water billed. These include

• Systematic data transfer error	4.55 mil gal	$ 17,972
• Systematic data analysis error	3.16 mil gal	$ 12,482
• Data policy/procedure impacts	4.21 mil gal	$ 16,629
• Residential meter underregistration	29.26 mil gal	$115,037
• Industrial/commercial/agricultural meter underregistration	13.04 mil gal	$ 51,508
• Unauthorized consumption (default 0.25% of water supplied)	9.646 mil gal	$ 38,102
• Total Apparent Losses	63.866 mil gal	$251,730

From this summary, the financial impact of the three subcomponents of the SDHE is calculated as $17,972 + $12,482 + $16,629 = $47,083. The cost impact of CMI is $115,037 + $51,508 = $166,545. UC, which is an estimate derived from the default value of 0.25 percent of water supplied, resulted in the smallest cost impact to CWC of $38,102. From the results of the water audit, it was determined that the revenue protection plan should focus primarily on customer metering inaccuracies, with a secondary focus on SDHE. By following the recommended first step in addressing apparent losses, the manager of CWC plans to flowchart the workings of the customer billing system to ascertain the integrity of the customer consumption data and identify occurrences of SDHE.

II. Customer Billing Process Analysis

II-a. The manager assigns one CWC billing analyst to work part-time over a period of two months in conjunction with a billing system consultant to perform an initial analysis of the customer meter-reading and billing process. From the initial findings, any areas of apparent loss that are deemed to be readily correctable will be implemented. Such corrections are recognized as relatively minor procedural or programming changes, an example of which might be a programming lapse that inadvertently left a two-year-old housing development of 50 homes off the meter-reading/billing rolls. The cost of this effort is basically the cost of human resources to implement it.

II-b. Staffing costs, including wages and benefits for CWC personnel

Number of CWC Staff: 1	Cost, $/hr 33.50	$/day 268.00
Number of Consultant Staff: 1	Cost, $/hr 75.00	$/day 600.00

II-c. Duration

Days, per Project Task	Flowcharting/Analysis	Corrections	Total Days	Total Project Costs, $
CWC Staff	14.00	4.00	18.00	$4,824.00
Consultant	25.00	7.00	32.00	$19,200.00
Total				$24,024.00

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

III. Customer Meter Accuracy Testing

III-a. The water audit for CWC estimates that customer metering inaccuracies caused underregistered consumption worth $166,545 of revenue during the audit year. This amount represents most of the revenue recovery potential in CWC. During the water audit process, CWC undertook customer meter testing on a sample of meters: 50 random residential meters and 5 random large (industrial, commercial, and agricultural) meters. The analysis of the meter test results is shown in Tables 2.13–2.19 in Chapter 2. The findings of this meter testing were extrapolated to the entire meter population to determine an estimate of the entire apparent losses attributed to CMI. Based upon the value of this testing, the CWC manager continues such testing on an annual basis, both to continually gauge meter accuracy and to also observe the rate of long-term degradation in accuracy with increasing cumulative consumption. CWC does not have its own meter-testing facility; therefore, it uses contracted testing services. The metering supervisor and one staff person will also participate by identifying meters for testing, retrieving meters from customer properties, and performing the administrative and analysis work.

III-b. Staffing and testing service costs, including wages and benefits for CWC personnel Number of CWC Staff: 2

Supervisor:	Cost, $/hr 35.00	$/day 280.00	# of days 3	Cost,	$840.00
Service Worker:	Cost, $/hr 27.50	$/day 220.00	# of days 15	Cost,	$3,300.00
Total				CWC Staff Cost,	$4,140.00

III-c. Estimated costs of meter-testing program—55 annual meter tests

Meter Testing Services:	Cost, $/small meter 35.00	Cost for 50-meter tests, $1,750.00
Meter Testing Services:	Cost, $/large meter 250.00	Cost for 5-meter tests, $1,250.00
		Meter Testing Service Cost $3,000.00

III-d. Total cost for annual meter testing program, $7,140.00

IV. Revenue Protection Program Summary

IV-a. The total cost of the two components of the initial revenue protection program:

Customer Billing Process Analysis:	$24,024
Annual Meter Testing Program:	$7,140
Total Revenue Protection Program Cost:	$31,164

IV-b. Economic level of revenue recovery

During the first year of the new revenue protection program, CWC anticipates spending $31,164 to launch the program. To recover the cost of this program, CWC would need to recover revenue equal to this amount. By applying the composite customer retail billing rate (see Figure 2-5 in Chapter 2) of $3,945/mil gal of customer consumption, an equivalent volume of consumption can be determined:

$$\text{breakeven recovery volume} = \frac{\$31,164}{\$3,945 / \text{mil gal}} = 7.90 \text{ mil gal}$$

If CWC's initial revenue protection efforts recover merely 7.90 mil gal of consumption, the revenue protection program will have paid for itself in its first year of operation. This level is only 12.4 percent of the total apparent losses of 63.866 mil gal quantified in the water audit in Figure 2-5 in Chapter 2. Because apparent losses are valued at the customer retail rate, recovering these losses can be highly cost-effective. CWC has strong potential to more than recoup its first-year revenue protection program costs in its first year. If this level of revenue recovery is met or exceeded, CWC will be well on its way to creating a cost-effective apparent loss control and revenue enhancement program.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

SUMMARY

Apparent losses distort the measure of the volume of customer water consumption and cause water utilities to lose revenue. Apparent loss control results in more accurate consumption data for the service population and gives a better portrayal of community water demand. This is particularly helpful in regions that have limited water resources and/or are encountering drought or water shortage. Controlling apparent losses can also be very cost-effective because initial corrections may require relatively little work with potentially high payback. It is often advantageous to target apparent loss control early in the NRW management program to quickly generate recoveries that can seed further loss reduction activities, particularly real loss reduction. Loss control is an endeavor of diminishing returns, but it is likely that many water utilities have significant apparent losses, which can be recovered in a cost-effective manner to enhance the utility's revenue stream and further promote the water loss control program.

REFERENCES

AWWA (American Water Works Association). 2019. C703-19 *Standard for Cold-Water Meters—Fire-Service Type*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2012. Manual M6. *Water Meters—Selection, Installation, Testing, and Maintenance.* Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2013. C714-19 *Standard for Cold-Water Meters for Residential Fire Sprinklers in One- and Two-Family Dwellings and Manufactured Homes*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2023. Manual M22. 2024. *Sizing Water Service Lines and Meters*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2008. Manual M31. *Distribution System Requirements for Fire Protection.* Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2018. *Residential Fire Sprinkler Systems: Guidance for Water Utilities*. Denver, CO: AWWA.

Club and Resort Business. 2009. Staley Farms GC owner pleads guilty to stealing water. 2009. www.clubandresortbusiness.com/tag/staley-farms-golf-club/ (accessed Sept. 14, 2014).

Davis, S.E. 2005. Residential water meter replacement economics. In *Water Loss 2005 Conference Proceedings, Halifax, Nova Scotia, Canada.*

National Fire Protection Association, Codes and Standards: http://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards Accessed November 11, 2020.

Rizzo, A.; Vermersch, M.; St. John, S.G.; Micallef, G.; & Race, R. 2007. Apparent water loss control: The way forward. In *Water 21.* London: IWA Publishing.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Schlenger, D.L.; Hughes, D.M.; & Green, A. 2011. *Advanced Metering Infrastructure: Best Practices for Water Utilities*. Denver, Colo.: Water Research Foundation.

Schlenger, D.L., 2019. *Advanced Metering Infrastructure: A Guidance Manual for Water Utilities*. Ridgewood, NJ.: Don Schlenger & Associates, LLC

Thornton, J. 2002. Component analysis of apparent losses (non-physical losses). In *Proceedings, Technical Session on Promoting Progressive Approaches for Water Loss Control: AWWA, Distribution and Plant Operations Conference*. Denver, Colo.: American Water Works Association.

Villegas, S. 2006. Hydrant use: Balancing access and protection. *AWWA Opflow*, 32(10): 24–26.

WRF (Water Research Foundation). 2007. *Evaluating Water Loss and Planning Loss Reduction Strategies*. Denver, Colo.: AwwaRF.

WRF (Water Research Foundation). 2011. *Accuracy of In-Service Water Meters at Low and High Flow Rates*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 5

Understanding Real Losses: The Occurrence and Impacts of Leakage

Real water losses, as identified in the water balance illustration of Chapter 2 (Figure 2-1), are the difference between the annual water supply volume and the sum of authorized consumption (billed and unbilled) and apparent losses. Real losses described in this chapter are the physical losses of pressurized water from the distribution system and include leaks and breaks from water mains, appurtenances, and the portion of customer service lines managed by the water utility, as well as leaking or overflowing finished water tanks and reservoirs. This chapter explains how real losses occur in water distribution systems and why leakage occurs and provides an overview of approaches to contain them. In addition to describing the causes and influencing factors, the chapter describes the significant impacts that real losses exert on water utility operations and finances, potential damage to roadways and property, and the strain that they can place on water and energy resources of the community and region. This chapter also details efforts at categorizing and measuring leakage and provides an overview of planning a leakage management program, as well as the criteria for assessing its cost-effectiveness. The reader is advised to refer to Chapter 6 to learn about the various methods to control leakage in a cost-effective manner.

For most North American water utilities, leakage is the most significant portion of real losses. While finished water storage overflows (Figure 5-1) are included in the definition of real losses, these events are typically infrequent and often visible; therefore, they are less likely to run unattended for extended periods unless the storage is underground.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Pittsburg Tank & Tower Group

Figure 5-1 Tank leaks and overflows are a minor component of real losses

Regardless, observant water operators performing routine checks generally discover this form of loss quickly. Given this observation, the content of this chapter focuses on pipeline and service connection leakage, the primary sources of real losses. Leakage from pipe networks is inevitable. Pipe materials and appurtenances degrade over time, losing strength through various forms of corrosion or molecular change. Failures occur when stresses, including water pressure, break through a deteriorating element in the pipe network.

THE IMPACT OF LEAKAGE

Although leakage from water distribution systems is labeled lost water, it should be recognized that the water cycle is a continuous process. Leaked water is lost only in the sense that it has escaped from the pressurized water distribution system before it could be used as potable water. Water leakage, like rain, percolates into the ground or enters breaches in sewers, stormwater collector systems, or other underground conduits. The negative effects that leakage imparts on the cost of water operations and society are numerous. Here are some examples.

- High-leakage losses indirectly require water suppliers to extract, treat, and transport greater volumes of water than their customers require. Such losses waste water treatment chemicals and electric power and add to wear and tear on equipment. High-leakage losses are also a pertinent energy management issue.
- Leakage results in unnecessary withdrawals from watersheds, possibly contributing to adverse environmental impacts and adding stress on raw water storage supplies.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- High-leakage losses require greater infrastructure (storage, treatment, pumping, and pipe) capacity than needed to meet customer demand, a compelling factor because infrastructure rehabilitation and renewal are of great concern. Infrastructure condition assessments may include an evaluation of leakage losses to distinguish that portion of infrastructure capacity that provides water to end-user consumption vs. the portion of capacity that exists merely to overcome distribution system leakage. Upgrading the conveyance efficiency of distribution systems is an important part of any long-term water supply infrastructure planning and improvements.
- Leaks and breaks may cause damage to nearby infrastructure, roadways, and property, increasing liability for utilities usually in the form of higher insurance premiums. They may also lead to road closures that impact local traffic, residents, and businesses and elicit negative public reactions.
- Leaks and breaks result in repair and restoration costs including labor, equipment, and materials. Materials, including repair parts, are usually held in an expanded inventory for a wide variety of repair issues.
- Because leaks create water loss and can cause significant immediate damage from water ponding, ice, and sinkholes, utility workers/contractors must perform some repairs under adverse conditions at any time of year and any time of day. Emergency repairs may proceed without full knowledge about nearby infrastructure. This implies increased risk for workers when repairs are made immediately as opposed to performing work within the normal work schedule. It often means additional costs for more repair work and overtime for unplanned repairs.
- Investigating and repairing leaks and breaks can consume considerable utility personnel time. When operators are frequently responding to emergency leaks, routine maintenance is deferred, which can lead to other problems. Making repairs under emergency conditions results in a disruption to the operators' routine job duties. If they were not deployed to the emergency, the operator would be available to perform other important maintenance duties. These opportunity costs are difficult to measure and are often not considered when determining the cost associated with this type of water loss. The cost of performing an emergency repair goes beyond the construction costs. In addition, frustration and stress typically occur during emergencies that can potentially be prevented. It may be more difficult to assemble a proper crew to respond to a water main break at 2:00 on a Sunday morning in frigid weather. This type of situation can significantly affect work-life balance for workers and hinder employee morale.
- Leaks and breaks may compromise distribution system water quality from potential contamination during high-flow leak events, low-pressure periods, or during shutdowns for repairs. Good response and repair practice, such as maintaining positive pressure in pipes until the leak and pipe are exposed, mitigate water quality risks from many less severe events. However, severe breaks may force shutdowns and cause backflow risks or generate high velocities in the pipe that shake loose deposits and tuberculation into the supply.
- Significant volumes of leakage may drain into community wastewater or combined sewer collection systems with the leaked water treated at the local wastewater treatment plant, thereby adding costs of increased volume for treatment without ever providing any benefit (Thornton et al. 2008).
- Unnecessary water withdrawals caused by leakage may limit growth in a region because of restrictions on scarce available source water.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- Leaking treated water can drain into natural waterways and sensitive ecosystems and cause significant damage like fish kills.

THE HOW AND WHY OF LEAKAGE

Real losses represent the portion of water volumes that do not achieve beneficial use because they never reach the customer. Water that leaks from the water distribution system between source and customer occurs in all utilities—only the flow and duration vary. There are numerous types of leaks in water distribution systems, each with different typical ranges of flow. Even recently constructed pipe systems installed by best practice will typically experience some level of leakage. Leakage in water distribution systems can be attributed to any of the following causes (Lambert et al. 1998):

- Inferior or defective materials, especially the pipes and joints;
- Pipe breaks resulting from poor workmanship or materials handling in pipe-laying, unsupported lengths of pipe, stones in contact with pipes, nonadherence to required joint gaps, poor backfilling of trenches, excessive joint deflection, improper tapping for connections, plastic pipe exposed to sunlight during storage, and similar occurrences;
- Leaking fittings and appurtenances, e.g., valves, air valves, saddles, hydrants, leaking stuffing boxes, and drain or blow-off valves that are closed but passing water;
- Operational errors like excessive pressure and surge caused by filling pipelines too rapidly, closing valves and/or hydrants too rapidly, and incorrect operation (starting and stopping) of pumps (water hammer);
- Corrosion, either internal corrosion caused by aggressive water or external corrosion caused by insufficient protection of pipe materials (Metallic pipes are prone to corrosion from aggressive soils, groundwater, or stray electric current. Plastic pipes may be vulnerable to certain contaminants and care must be taken to avoid prolonged exposure to sunlight during storage.);
- Seasonally induced stresses, such as frost loading, soil expansion and contraction, and thermal effects of temperature changes, including pipe contraction due to winter cold water in the pipe from surface water sources (While groundwater maintains a more constant temperature throughout the seasons, the water temperature will be impacted with exposed elevated storage facilities or pipes located close to the frost line.);
- Poor quality of leak repair work;
- Accidental or deliberate damage to water mains, hydrants, or other appurtenances; heavy traffic loadings; or careless construction activity over shallow water mains. Traffic loading may have grown with higher demand and loading (truck traffic) on roadways, increasing stress on buried pipelines; and
- Changing stresses in the pipe environment: pipe installed in bygone eras met the design standards of their day. Many systems expanded with an increasing need for fire protection. As a result, as systems grew, flow rates and pressure levels often increased with larger pumps and the addition of elevated tanks, placing greater stress on piping systems. Rapidly changing pressures known as surge in transients also contribute to pipe stress.

The total volume of leakage losses occurring in a particular water distribution system over a given period depends on the following:

- the operating pressures in the piping distribution system;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- the different types of pipe materials and failures;
- the depth of pipe, geology, soil type, and road cover materials that delay the time that leakage water becomes visible at the surface;
- the speed, frequency, and scope of the water utility's active leakage control to detect, locate, and repair leaks; and
- the integrity of the piping infrastructure and its degree of upkeep via best-practice rehabilitation and renewal programs.

The extent of the occurrence of leakage within a water utility depends on the following:

- characteristics of the water distribution system, notably pipe materials and controls;
- importance attached to leakage control by the water utility;
- how the distribution system is operated and maintained; and
- level of expertise and technology available within the utility.

To summarize, the above factors can be aggregated into three primary categories:

1. the characteristics of the water distribution system;
2. the stresses produced in the local environment, e.g., weather extremes as well as aggressive soils, shifting ground, and heavy traffic loads; and
3. the level of proactive leakage management employed by the water utility.

How much control can a utility operator exert upon these factors? Water system operators can exert some change in the characteristics of the water distribution system, but major system changes (expansion, cathodic protection, and pipe renewal) occur slowly, owing to the size of systems and the cost of improvements that can only be accomplished on a long-term basis. Relatively little control over the environment can be gained; weather, geologic conditions, or even traffic loading cannot be controlled to any great extent. The last of these categories, proactive leakage management, is where the utility operator can exert the greatest day-to-day degree of control on the occurrence of leakage. While leakage in water distribution systems is inevitable, utility operators can employ various methods to limit the extent of leakage and the volume of leakage losses. The motivation of the utility to reduce leakage is likely influenced by the availability of water supply and the cost to secure, treat, and deliver drinking water to the customer. Some North American regulatory agencies have begun to consider setting leakage reduction targets for water utilities.

THE EFFECT OF TIME ON LEAKAGE LOSSES

The volume of leakage losses in a distribution system over a year depends on the number of leaks occurring, their magnitude, operating pressure, and—perhaps most importantly—the total time that the leaks are permitted to run. Modest undetected leaks that run for long periods often account for the greatest volume of leakage losses in a water distribution system. While large, dramatic water main breaks (see Figure 5-2) cause havoc and garner much attention, these events typically contribute comparatively small volumes of nonrevenue water (NRW) on an annual basis. Despite the high rate of water flow spewing from a severe water main rupture, the disruptive nature of such events usually prompts a quick response by the water utility and a relatively speedy shutdown of the broken section of pipe. Because the run time of the break is often limited to a period of hours, the total volume of lost water from the event is lower than a steady, low-flow leak running for weeks or months.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Hazen & Sawyer

Figure 5-2 Large water main break

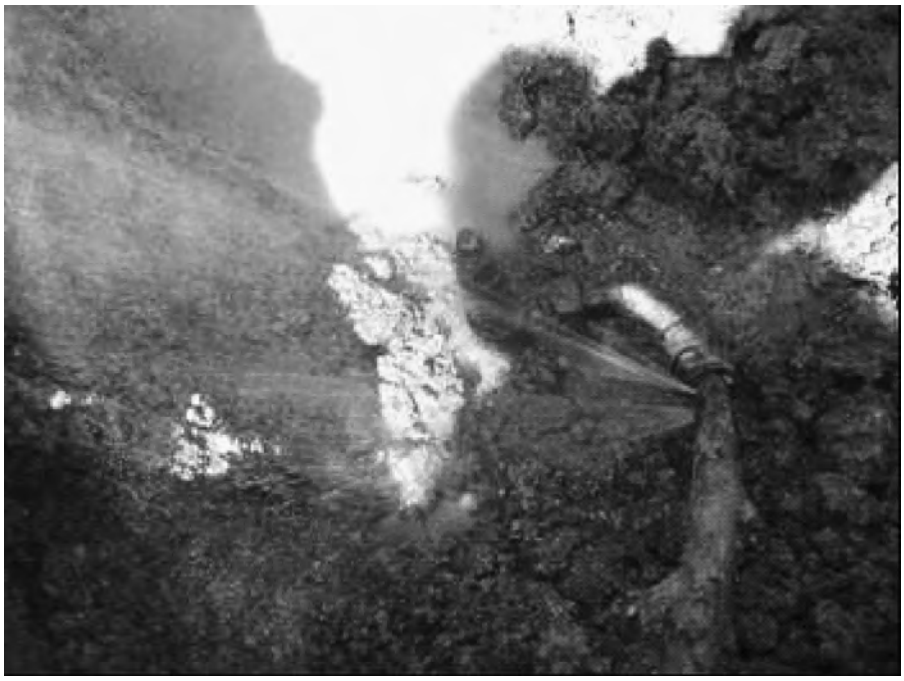
It is the numerous small, long-hidden leaks (see Figure 5-3) that account for the greatest overall volume of annual leakage losses in a distribution system. In well-maintained systems, the greatest annual volume of real losses occurs from long-running, small to medium-sized leaks on customer service lines, except in systems with very low densities of service connections (Brown et al. 2000).

While leaks can run undetected for a time, once suspected, utilities often work swiftly to limit their running time or the actionable leak duration. As depicted in Figure 5-4, the run time of discovered leaks comprises three periods:

1. **Awareness time** is the time needed for water operators to become aware that a leak exists, a parameter strongly influenced by the presence of an active leakage control program, its comprehensiveness, and its intensity.
2. **Location time** is the time taken to pinpoint the source of the leak once the operator is aware of its existence.
3. **Repair time** is the time to reduce and then halt the leakage flow once the leak position has been identified. This is not just the time of the shutoff of the leakage flow but all preceding time (while the leak is still running) needed to route the repair work order, schedule the repair, notify customers, notify other utilities, and secure street opening permits and other activities, which can take hours, days, or weeks depending on the policies of the water utility and the severity of disruption caused by the leakage.

The influence of total leak run time, as illustrated by example in Figure 5-5, is the primary factor in the volume of water lost to leakage over a year (WRF 2007a). In systems with no active leak detection programs, the run time of undetected, hidden leaks is continuous until the leaks are identified by the water utility or an external party such as a customer. Such detection occurs usually after a leak becomes evident, often through some form of damage or disruption that it is causing.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Modernizing Distribution

Figure 5-3 Small leak on customer service piping

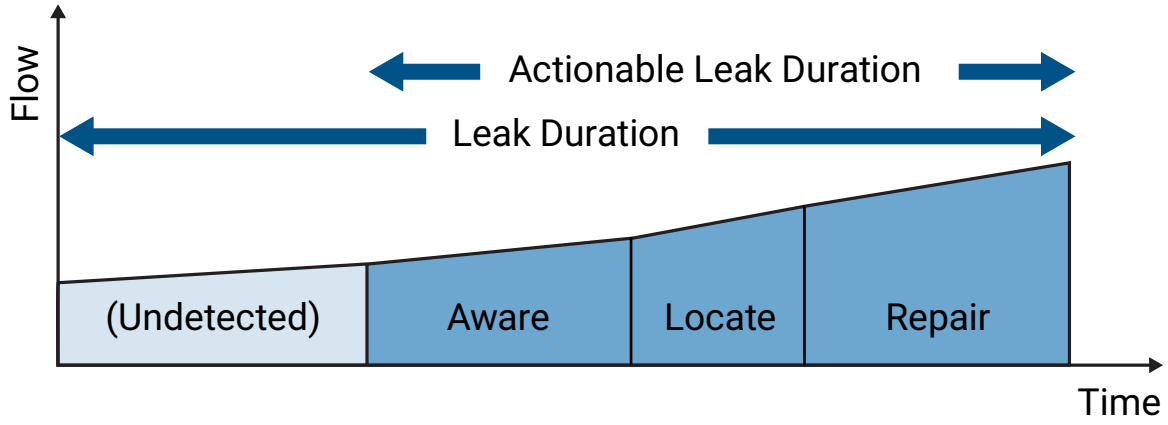


Figure 5-4 Time periods in the life of a leak

In many water systems, small leaks and breaks run for periods of weeks, months, or even years before they are discovered and repaired. Consequently, despite a low-flow rate from many such leaks, the annual volume of hidden leakage losses is usually a significant proportion of the total leakage volume and far exceeds the water lost in catastrophic, visible main break events.

For illustrative purposes, the schematic diagrams shown in Figures 5-4 and 5-5 have been drawn as slowly increasing leaks with distinct start and end times of leaks. This is to

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

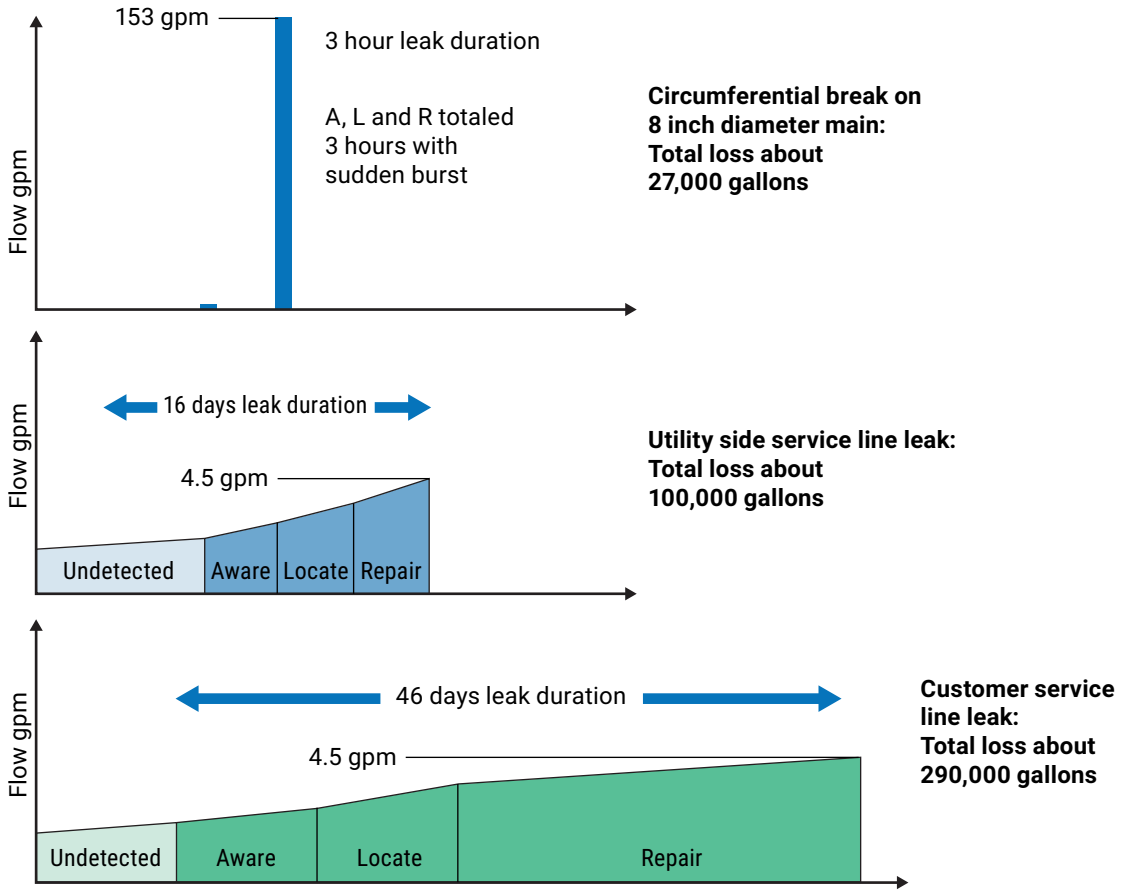


Figure 5-5 Example of various leakage types and impacts of time on the volume of loss

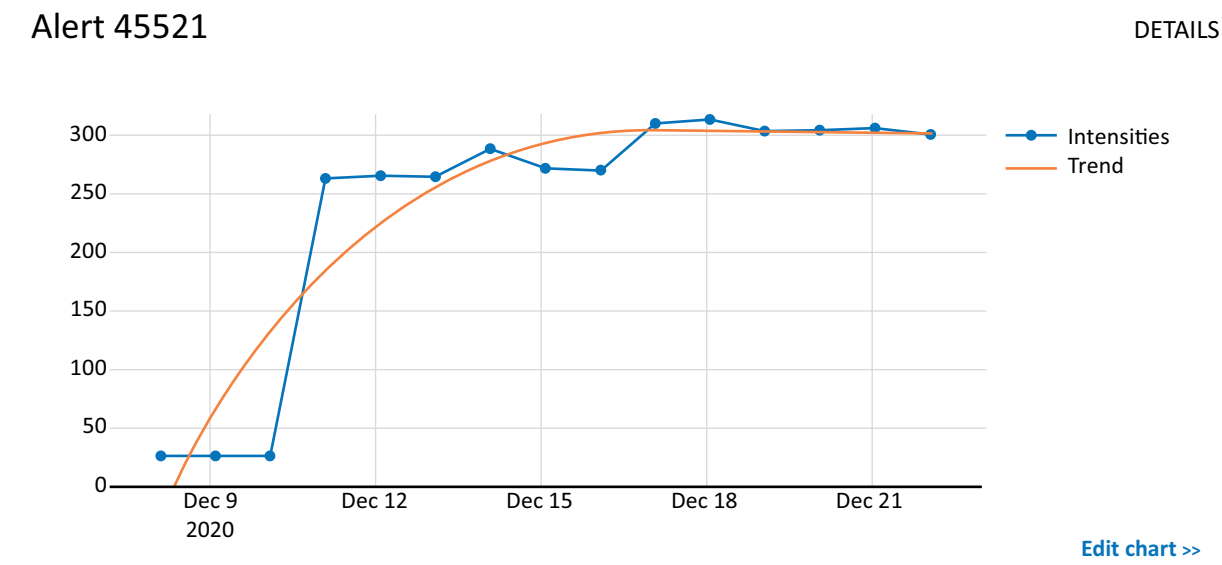
reflect the reality that the flow emanating from a leak varies over the life of the leak, usually accelerating over time, perhaps with a notable rupture after leaking for some period.

Figure 5-6 depicts a leak noise signature for a leak detected by a leak noise transmitter. The sound intensity corresponds well with the leakage flow rate and shows how the rate varies and generally increases over time. Analysis of acoustic data from many leaks suggests that some leaks maintain a steady flow while some jump in steps and still others can erupt suddenly. The leakage pattern shown in Figure 5-6, nonetheless, further suggests the value that can be gained in minimizing leak run time to optimize water loss reduction. Employing an efficient network of continuous acoustic monitors (CAMs) is a way to significantly reduce awareness and location time to a few days, allowing repair to be scheduled during normal working hours (WRF 2011).

CHARACTERIZING LEAKAGE EVENTS

Because leak run time is such a prominent factor in the occurrence of leakage losses, developing a strategy to minimize leakage run time is key to a successful leakage management program. It is just as important to the leakage management strategy to understand how leakage incidents become evident. Three different types of leakage occur in water distribution systems. These are defined below and illustrated in Figure 5-7.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



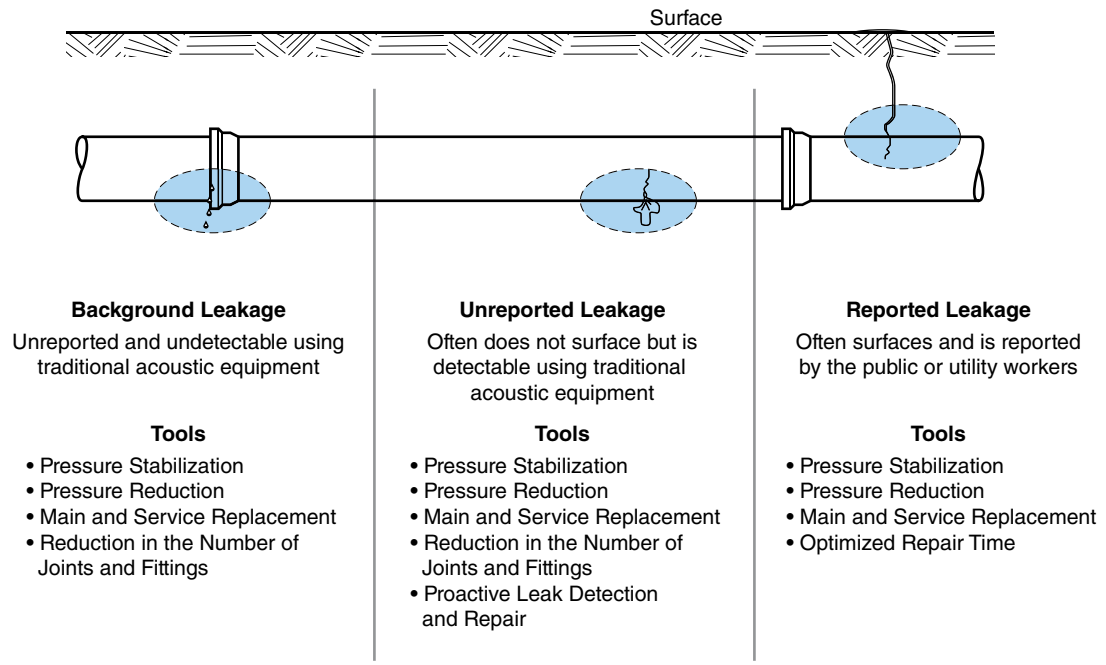
Courtesy of Aquarius Spectrum

Figure 5-6 Example leak noise signature suggesting increasing leak flow rate over time

1. **Reported leaks.** These are leaks that are reported by customers, traffic authorities, other utility staff, or any other outside party because of their visible and/or disruptive nature (such as sinkholes). Leaks detected by high flows and/or noticeable drops in water pressure in supervisory control and data acquisition (SCADA) systems may be categorized as reported leaks. Figure 5-2 is an illustration of a dramatic reported failure.
2. **Unreported leaks.** These leaks run hidden underground and escape public knowledge. They are only identified through the active leakage control work of the water utility. The acoustic leak detection survey (detailed later in this chapter and in Chapter 6) is the most common means currently used in North America to identify unreported leaks. Unfortunately, many water utilities do not regularly perform proactive leak detection. Figure 5-3 shows a low-flow rate customer service line leak, a common type of leak that often occurs as an unreported failure.
3. **Background leakage.** These are the collective weeps and seeps of water largely at joints and fittings that occur at very low-flow rates but may exist pervasively across the water distribution system, particularly if the piping and service connections are in poor condition. This type of leakage generates nearly no leak noise and is not acoustically detectable, so it cannot be addressed by conventional acoustic leak detection. However, background leakage is sensitive to changes in pressure levels and can often be mitigated economically by pressure management techniques that reduce the discharge of the leakage. Background leakage can also be abated by pipeline renewal, but such work is rarely cost justified based on leakage reduction alone. Background leakage in a system is estimated by using the pressure step testing technique in the distribution system (see Chapter 6).

Figure 5-7 illustrates the three components of real losses—background leakage, unreported leakage, and reported leakage—and the appropriate activities to control them. All drinking water utilities experience reported leaks. If utilities' only leakage control

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Source: *Tardelli Filho 2004*

Figure 5-7 Components of leakage and appropriate intervention tools

activity is to respond to reported leaks, they are operating a reactive leakage management program. Systems that additionally seek to identify unreported leaks and perhaps reduce excessive background leakage are operating a proactive leakage management program. Because many utilities respond rapidly to repair surfacing leaks, most of the leakage losses over the course of a year occur from unreported leaks and/or background leakage. Those utilities with a reactive leakage control policy are likely addressing only a minority of the leakage occurrences in their distribution system.

Background leakage, by definition, defies detection through conventional acoustic means. The tiny weeps and seeps of background leakage can be numerous and widespread in any given distribution system but are not readily "heard" or located. In the past, those leaks falling under the heading of background leakage may have been viewed as unavoidable leakage, in the sense that it was not practical to detect and repair them on an individual basis. However, if an individual pipe segment is suspected as a background leak source, non-acoustic techniques such as ground-penetrating radar or thermal imaging might be used to find these small leaks (usually less than 1 gal/min [gpm]). Even then, it is unlikely that excavation and repair of such small flow leaks will be cost-effective. It should be noted that where pressure levels are high and pipes are shallow, especially if soils are not porous, most system leakage may be reported leakage because leaks quickly surface and become visible.

The concepts of awareness, location, and repair (ALR) times led to the development of leakage component analysis (LCA) as a method for discerning leakage patterns in a specific water distribution system. This analysis recognizes reported and unreported leaks have different ALR times based on the nature of the leakage occurrences in the distribution system, and the attention that the water utility staff direct to them. The following are some examples.

- Many types of surfacing water main breaks are the most recognized form of reported leaks and typically have very short ALR times (see Figure 5-5). These

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

include pipes made of materials that split or fracture as well as circumferential breaks of iron pipe. Because of the disruptive nature of such high-flow events, they encounter a short awareness and location time, and a repair time of perhaps several hours to gain a shutdown of the broken section of pipe (recognizing that the actual pipe repair may be extensive and require additional time).

- Unreported or hidden leaks on underground water mains and customer service connections can have brief or lengthy awareness times depending on the leakage management program. Leaks that remain hidden commonly are corrosion holes or joint leaks on mains, fittings, or service connections. Reactive leakage management programs commonly have a longer awareness time, especially in areas where leaking water can effectively drain below ground. In arid regions or in regions where subsurface conditions do not readily accept the leaked water, these leaks can be discovered more quickly. Conversely, systems with a continuous acoustic monitoring (CAM) program may find hidden leaks at the time they become acoustically detectable (Illinois Sustainability Technology Center 2016). Utilities that conduct physical leak detection sounding surveys can have leaks that run short or long duration, as a function of the frequency and quality of the leak detection surveys. Once aware of a hidden leak, many utilities can locate and repair the leak quickly and see value in making prompt repairs to contain leak repair costs and limit damage.
- Unreported leaks on the customer-owned section of service lines may also have variable awareness times for the same reasons as stated above. A notable difference for these leaks is that they can also have variable repair time depending upon the utility's policies. Most utilities take responsibility for the portion of the service line in the public right of way and make timely repairs or line replacements to address leaks. However, many water utilities require their customers to arrange for repairs on sections or the entire length of the service connection piping which is owned by the customer. Because many customers respond slowly in arranging for such repairs, leaks can run at length, and leakage volumes can be significant on customer leaks. Some water utilities operate customer warranty or insurance programs that they offer to customers to implement repair or replacement of customer service lines as leaks are identified. Such programs assist customers by giving them an affordable repair option and a one-call action to initiate repairs while assisting the utility in keeping leak run times to a minimum.

A FURTHER WORD ON CUSTOMER SERVICE LINE PIPING LEAKAGE

In water utilities throughout the world, much of the leakage volume occurs on customer service line piping, and to a lesser degree, on the water main piping of the distribution system. There are several reasons for this. Distribution system piping often tends to be relatively uniform in materials, design, and construction. Utilities use quality materials and repair techniques intended to last for years. Service connection piping from the tapping point on the distribution system water main and into the property has more fittings that can fail and are often found to be much more variable in materials and installation practices. Different piping types have been employed over the years, from lead and galvanized iron in the past to copper and plastic pipes currently. Some service pipe materials, such as galvanized iron and polybutylene pipe, are prone to failure well before their water main material counterparts. Many utilities require work on distribution piping and utility-owned service pipe to be performed only by their personnel or a construction contractor selected and inspected by them. Conversely, they allow customers (which typically include

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

builders for new buildings) to hire independent contractors to install and repair service connection piping, typically without inspection. Drinking water utilities can reduce the risk of customer service piping failures by establishing uniform quality standards for this piping, as well as sound installation, repair, and quality assurance procedures.

The primary factor in the high volume of leakage losses occurring on customer service piping, however, is the repair policy employed by many water utilities as noted above. It is common for many North American water utilities to require customers to not only own their service connection piping but to arrange for repairs of leaks found on their portion of the pipe. When the customer's water meter is located at the property line in a meter pit, leakage on the downstream section of piping owned by the customer is metered. Customers are incentivized to initiate prompt leak repair action in this case since high water bills result from the leakage. However, repairs by customers of an unmetered hidden leak often result in a protracted loss of water.

There are arguments for utilities to aggressively address customer service line leaks. During a severe drought in the United Kingdom from 1995 to 1996, the regulator imposed a precedent-setting requirement on water companies to repair known private (largely unmetered) customer service piping leaks (Lambert et al. 1998). The reduction in leakage losses was so dramatic that the regulator implemented a permanent requirement for all water companies in England and Wales to institute a policy for utility-implemented repairs of private service piping leaks. The utilities found that after an initial backlog of leaks was addressed, repairing new customer leaks was a manageable proactive approach that saved water and money for the utility when compared to the prior practice of customer-implemented repairs.

WATER PRESSURE AND LEAKAGE

Water pressure levels in distribution systems vary widely throughout all countries. Regulatory requirements and design guidelines also vary considerably. In some parts of the developing world, very low pressures (20 psi or less) are common, and some regions lack 24-hour pressurized water service and only intermittent supply is provided. These conditions often create considerable leakage and infrastructure deterioration, as well as a strong likelihood of water quality compromise. Although this type of operation certainly merits attention for improvement, the following discussion applies to systems with a continuous water supply and constant positive pressure, which includes nearly all of the United States and Canada where many systems have high pressures, often in regions with a significant change in topography.

Pressure management, which typically includes reducing excessive pressures and controlling pressure surges, generally has two major benefits: 1) it reduces stress on pipes, thereby delaying failure events, and 2) if there are leaks, less water flows through flaws in piping under lower pressure. Consequently, pressure management is a well-supported practice and one of four pillars of leakage management. Additionally, the AWWA Partnership for Safe Water Distribution System Optimization Program (AWWA 2011) recognizes pressure management as one of its three focus areas, along with maintenance of acceptable chlorine residual and controls to limit the number of water main breaks.

Many North American water utilities encounter a wide range of pressures across their systems. Changing topography presents a pressure challenge for single-zone distribution systems with high elevation areas of lower pressure that may struggle to fully meet pressure needs at taller buildings or meet fire flow requirements, while pressures at lower elevations within the pressure zone may rise considerably. Consider water supply pumped from a treatment plant at a low elevation near a lake or stream to communities at higher elevations some distance from the plant. The elevation between the plant and customers may also be highly irregular with multiple hills and valleys that result in variations in

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

pressure. Plumbing codes require customers in systems having pressures notably above 75 to 80 psi to have and maintain pressure-reducing valves on their service connection piping to prevent damage to customer plumbing, hot water heaters, and other devices. Sections of water distribution systems with high-pressure levels have a greater opportunity for high leakage and failure rates, especially if piping was not installed using a robust class of pipe designed to withstand high pressure. Conversely, higher system pressures can also make some leaks visible more quickly, which reduces the time to detection.

Many water systems have sections of their water distribution systems with pressure above 80 psi. Pressure data for 1,124 water systems were collected by the AWWA Water Loss Control Committee from water audits from the states of Georgia and California and the province of Quebec in the Water Audit Reference Dataset (WARD) (AWWA WRF 2021). The findings are summarized in Table 5-1, with average operating pressure (AOP) values collected using the AWWA Free Water Audit Software (Audit Software). Twenty-five percent of the utilities reported an AOP over 80 psi, and 35 percent reported AOP over 75 psi. These data strongly suggest that water utilities with notably high pressure are common in some parts of North America, though not necessarily in all states and provinces. Quebec, Georgia, and California have more varied terrain, which may be the cause of higher operating pressures, than states with flat terrain like Delaware or Iowa. State and provincial water system guidelines do not always include provisions setting maximal pressure levels. Some regulatory agencies have established maximal pressure levels, but it is not uncommon for some water systems under their jurisdiction to operate with pressures above the maximal level.

Some water utilities with robust water distribution systems and high pressure may experience less loss from unreported and background leakage and only occasionally visible reported breaks as the pressure results in the quick surfacing of water. But many systems likely have a financially feasible potential to reduce excessive pressure levels for greater benefit. In water distribution systems that are particularly pressure sensitive, such as poor infrastructure condition and/or high background leakage, excessive pressures can exert a perceptible increase in leakage and more frequent water main breaks. There is evidence that higher pressures accelerate pipe deterioration and increase breakage over time. For some systems, maintaining a higher pressure requires more energy to pump water to higher levels. Separating zones to reduce pressure may present an opportunity to effectively monitor high flows from leakage in designed zones called district metered areas (DMAs), described later in this chapter.

Water pressure levels have a bearing on the amount of water escaping from a leak in a pressurized pipe. The higher the pressure, the greater the rate of water flow coming from the leak. The larger the leaking pipe opening, the greater the flow will be. However, not all pipe openings behave in the same manner. The theory of fixed and variable area discharge paths, or the FAVAD model, was developed in 1994 and has advanced the understanding of pressure–leakage relationships for water distribution systems (May 1994).

The FAVAD formula, given in Equation 5-1, expresses that a change in leakage rate L (volume/unit time) varies with a change pressure, P, to the power of an exponent, N1, as

Table 5-1 Summary of AOP in psi in the WARD*

State/province	Number of reporting utilities	Average AOP	Median AOP	Lowest AOP	Highest AOP
Georgia	198	76.7	70.0	41.0	165.0
California	268	76.0	73.4	45.0	120.0
Quebec	658	68.7	68.6	26.8	139.2

*AWWA WARD, AWWA (2021)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

$$L_1 / L_0 = (P_1 / P_0)^{N1} \quad \text{(Eq 5-1)}$$

Before the FAVAD model, it was assumed that the amount of leakage through a fixed-size hole in a pipe varied according to the square root power of the pressure, implying that a 10 percent change in pressure will produce only a 5 percent change in the velocity of water leaving the leak. The FAVAD model considers the fact that certain types of leaks, such as holes in a metal pipe, which do not change size with changing pressure, follow a fixed-path model and demonstrate an N1 exponent equal to the square root, or 0.5 power, relationship in the pressure–leakage calculation.

Certain other types of leaks follow a variable leakage path (e.g., cracks in plastic pipe, wherein the size of the crack also increases with increasing pressure). The pressure–leakage relationship can vary up to an N1 value of 2.5 in such cases. Background leakage typically has a FAVAD N1 exponent of 1.5. The higher the N1 exponent, the more sensitive existing leakage flow rates are to changes in pressure. Many distribution systems have a variety of leakage types occurring, and it is now common to assume an N1 of 1.0 rather than 0.5 for most systems.

The FAVAD Equation 5-1 allows for accurate forecasting of the increase or decrease of leakage loss rates in a piping system caused by changes in operating pressure. The development of this model serves as the foundation for pressure management applications.

High pressures can greatly exacerbate the rate of water escaping from active leaks. Studies on various pipe materials have been conducted to confirm the extent to which water main break and service leak frequency is accelerated with higher pressures (WRF et al. 2005, Lambert et al. 2013, WRF 2014c). Reducing excessive system pressure may also result in less damage to adjacent property and infrastructure during water main break events, provided that the failure is addressed in a reasonable time and does not result in extended erosion. Studying the effects of pressure on leakage rates and infrastructure condition is a relatively recent undertaking, and additional work is needed to provide comprehensive conclusions on the techniques and benefits of pressure management which include extending the useful life of active pipelines.

Because all water distribution systems are unique in their configuration and operation, each system must be assessed individually for the potential for improved pressure management. Many systems operate with pressure regimens that are dictated by water storage tank elevations, and pressure is defined. Where pressure refinement or reduction is implemented, it is essential to ensure that adequate minimal pressure exists to reliably meet customer water demands at all locations and that fire fighting flow and pressure requirements can be met.

Pressure management projects have been implemented in many water systems worldwide and are often installed within a DMA configuration, which is then known as a pressure managed area or PMA. Pressure-reducing valves (PRV) and other equipment are usually needed to establish improved pressure control. Some systems without elevated storage may be able to change pressures with affordable pump modifications. The costs of equipment set up to establish a PMA should be weighed against the savings from anticipated leakage and main break reduction. Benefits, constraints, and results will vary, and it is best to undertake a business case assessment of any proposed pressure management scheme before implementing it. The technique of pressure management is discussed in detail in Chapter 6.

In recognizing the pressure–leakage relationship and by employing new means to target its use for individual distribution systems, pressure management has become an effective tool in the control of leakage losses that are distinctly influenced by excessive pressure levels. Particularly in addressing background losses that are, by definition, undetectable by traditional acoustic means, pressure management has become a

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

highly cost-effective means to reduce leakage below what was previously viewed as an unattainable leakage reduction. Establishing pressure management as a strategy in the leakage management toolbox is one of the most effective innovations of recent years for utilities with excessive pressure and high leakage.

HOW TO QUANTIFY LEAKAGE

Water leaking from pipes, whether seeping or pouring out through flaws in the pipe system, is a challenge to accurately quantify. Two effective approaches for quantifying leakage flow rates are discussed in this section. First, the bulk leakage rate in a specific area of the water distribution system can be monitored by direct flow measurement as featured in a DMA and conducting minimum hour flow analysis. This is a reliable means to quantify the bulk leakage that is currently running across the DMA. For severe short-term breaks, a comparison of pump rates and storage between a normal day and the day of the break can aid in approximating the leakage loss volume. Secondly, if DMAs are not employed in a water utility, the water utility can refer to the section about estimating losses from discovered leaks later in this chapter. This method is typically used to quantify leaks that have been repaired and documented.

District Metered Areas (DMAs)

Perhaps, the most accurate means to quantify the individual components of leakage in a water distribution system is to measure water flow, along with pressure and customer consumption data, in DMAs. The DMA approach involves isolating a portion of the water distribution system into a separate zone by closing valves and installing flowmeters on one or more supply water mains entering (and possibly leaving) the DMA. The flowmeters continuously measure the flow into (and sometimes out of) the DMA. Since equipment and commissioning costs are incurred to install a DMA, the cost-effectiveness of a proposed DMA should be determined in advance as part of the planning process. The principles for the design and operation of DMAs are given in the publication *A Manual of DMA Practice* (UKWIR 1999) and are discussed below with modifications that consider the distribution system characteristics prevalent in North American water systems. This section examines DMA design and installation concepts, and principles of operation, while Chapter 6 addresses the operating practices for DMAs.

DMAs can provide many advantages in controlling water losses, managing pressure, and sustaining distribution infrastructure. Each DMA must be carefully engineered to serve its purpose while maintaining a reliable water supply for all customer uses including fire protection. A variety of factors must be considered to confirm that the DMA design will serve its objective economically. For example, distribution systems that exist in a grid and those with water storage tanks may present challenges in measuring leakage while preserving good hydraulics. A grid system may require the closure of many valves that create small, local dead ends that might create water quality issues. A careful design process should identify such potential problems and devise a design that addresses them. Thousands of DMAs have been successfully implemented in many countries around the world. Though not in widespread use across North America, the installation of DMAs is growing, and water utility managers are assisted by being aware of this method of leakage management as another tool in their arsenal of methods to control leakage.

Figure 5-8 shows schematically some configuration options for DMAs. The graphic illustrates that flow metering begins at the effluent of the water treatment plant (water supplied to distribution). Multiple DMA meters can serve large pressure or gravity zones

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

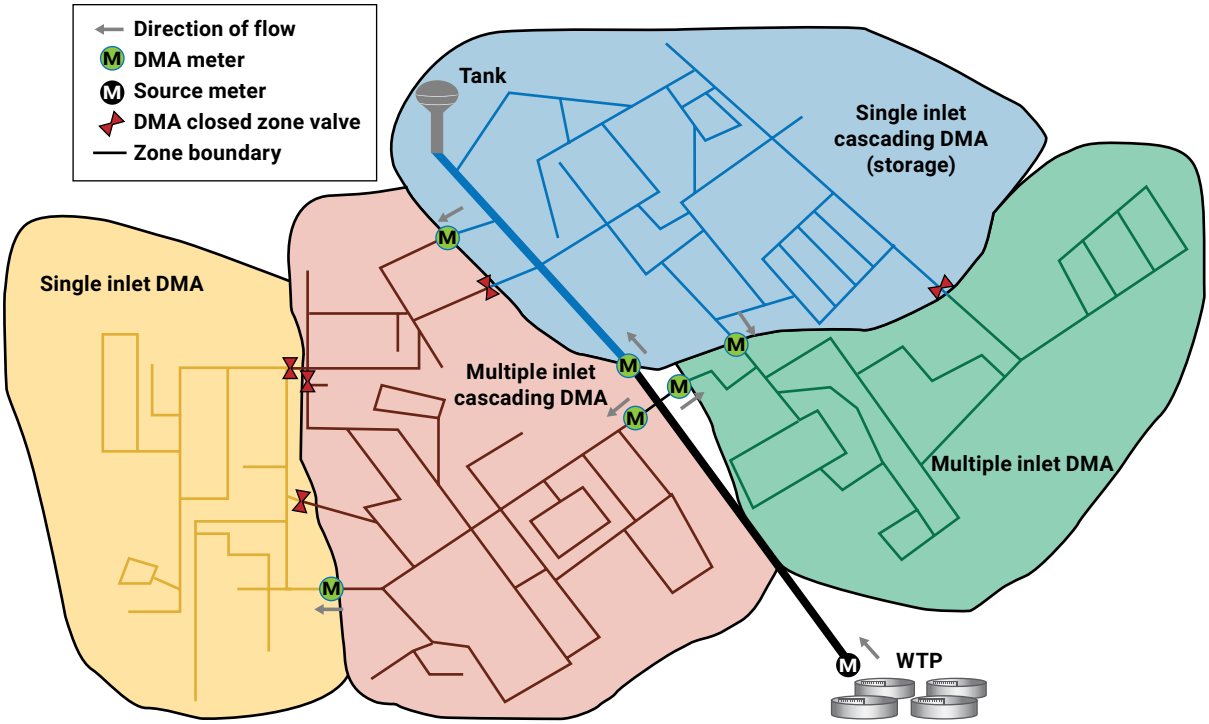


Figure 5-8 Distribution system flow metering and DMA design configuration options

placed into discrete DMAs, nominally sized to 1,000 – 3,000 customer service connections,* or even to subdistricts or sub-DMAs of 1,000 or fewer properties. DMAs are best suited for zones of the distribution system that supply residential customers with predictable consumption patterns. It is better to avoid areas with large 24-hour water users (industrial users) and large diameter transmission piping, as well as water storage tanks or reservoirs. These features can complicate the design and operations process and may incur additional risks. An individual DMA can be supplied via a single supply main or limited number of feeds. It can be a discrete isolated zone or can be interconnected with adjacent DMAs. Interconnections can be a simple pass-through or be equipped with control valves that open under extraordinary circumstances to provide supplemental emergency supply into the DMA.

Principles of DMA operations. The primary concept and advantage of DMA monitoring are to isolate and monitor a small area of the distribution system with supply feeds into the DMA of sufficient scale so that flows from leaks can be distinguished from customer consumption flow. Components of normal consumption and leakage rates are typically evaluated, especially during off-peak or minimum consumption times of the day. Although flow monitoring in DMAs does not provide the ability to pinpoint individual leaks, it offers the important capability of identifying emerging leaks (when flows rise) and quantifying the collective leakage rate occurring within the DMA. Well-managed DMAs provide an early warning of newly rising leakage and alert the operator when to optimally schedule leak detection work to pinpoint leaks. DMAs are useful in alerting the operators to rising flows that might indicate a leak once they become a detectable portion of minimum night flow. Pipelines made of plastic, operated at low pressures or located

* With many low-density water systems and some concentrations of industry, the DMA designer might consider miles of main and flow volumes when sizing DMAs.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

below heavily vegetated areas are typical of systems that produce little evidence of the presence of leaks other than the increased flows measured by the DMA and step testing. (See sidebar discussion in Chapter 6 on DMA use in Wilson County, TN.)

The technique of flow measurement to estimate leakage volumes requires metering and tracking of flows that supply sections of the water distribution system. The design of the DMA leakage monitoring approach seeks to achieve two primary goals:

1. to partition the distribution network into one or more appropriately sized zones with a defined and permanent boundary, appropriately sized so that flows can be regularly monitored and analyzed to detect rising flows from emerging leaks as distinguished from routine customer consumption and
2. to assess the pressure in zones so that the potential to optimize pressure levels and reduce background leakage is determined.

In cases where the DMA is also a PMA, the flowmeter installation will typically be supplemented with a PRV to reduce pressure into the DMA or modified pumping operations to better regulate pressure in the zone.

Analysis of DMA data typically relies on monitoring DMA inflows during minimum consumption hours. During these periods, which are often at night, the proportion of leakage (a relatively fixed flow throughout the day) to total inflow is at its greatest. Legitimate customer consumption during the minimum consumption hours should be measured or estimated. Advanced metering infrastructure (AMI) systems give the capability of gathering continuous detailed customer consumption data at short time intervals. Subtracting customer consumption from the total inflow gives a reliable estimate of the leakage in the DMA. If an AMI system does not exist, legitimate nighttime consumption can be estimated in residential DMA zones by allotting 1.5 gal per customer connection per hour. If the DMA includes large, continuous-use consumers, such as industrial plants, or if the DMA includes customers with irrigation systems operating during nighttime hours, minimum consumption will be difficult to estimate and may not occur during nighttime hours.

DMAs enable a water utility to quantify the current level of leakage in a discrete area and to subsequently prioritize its leak detection activities, sending leak detection personnel with acoustic leak detection equipment into those DMAs where leakage rates are rising appreciably and deferring investigation if leakage rates remain contained. By regularly monitoring DMA inflows into a well-managed zone, the operator can identify the occurrence of new leaks and breaks by the rise in water flow during the minimum hours of consumption. This information enables a utility to intervene and repair the leaks once the action level of leakage is reached and avoid expending leak detection crew time when the presence of excessive leakage is not indicated.

DMA planning and design considerations. Various factors should be considered when planning DMAs, including

- if the annual water audit reveals a high unit real loss rate that justifies additional leakage management steps beyond the utility's historic practices,
- if the annual cost of leakage indicates that a sufficient return on recovered leakage will exist to justify the expense of establishing one or more DMAs,
- if leakage is concentrated in one portion of the system, which necessitates the installation of a DMA as a pilot project, or
- if the DMA approach provides a more economical and effective solution than an alternate leakage control approach. (See sidebar discussion in Chapter 6 on DMA use in Wilson County, TN.)

Many water utilities divide at least some portions of their water distribution systems into distinct pressure zones. Some of these existing zones may be amenable to the creation

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

of DMAs with the relatively minimal work of installing a flowmeter and implementing routine monitoring. The size and configuration of an individual DMA will vary, depending on local factors and system characteristics, such as

- the level of economically targeted leakage reduction desired for the DMA;
- geographic/demographic/water use factors (e.g., urban, rural, or industrial);
- existing leakage control techniques used by the utility;
- individual water utility preference (e.g., identifying customer service connection leaks, ease of leak survey deployment);
- minimum flow and pressure required to meet customer demands, as well as fire flow requirements;
- the size and configuration of existing or desired pressure zones;
- variations in topography;
- looping and redundancy considerations of the piping grid;
- water storage tank and transmission main layout; and
- water quality and hydraulic considerations when closing system valves.

Many small North American water systems have piping configurations that can serve as a DMA without closing valves and with only a flowmeter installation required. For larger North America systems with widespread piping grids, creating a new DMA involves closing valves to form its boundary and reconfigure flow patterns to meet the objectives of the DMA flow monitoring. When closing valves, potential can exist for water quality degradation at dead-end piping along the DMA boundary. Care should be taken to design water quality safeguards into the DMA implementation. Water quality sampling and assessment should be conducted during the planning and implementation phases of the DMA. This will allow the utility operator to provide any needed water quality controls into the DMA design.

Innovative technology has reshaped DMA design elements. AMI can provide hourly readings for ready-made analysis between DMA metered flow and the aggregate of AMI readings of customer locations within each DMA (Schlenger and Hughes 2019). Improved leak pinpointing techniques provide both competitive and complementary options for DMAs. While DMAs alert the operator to emerging or increasing leakage, the utility must still deploy staff to localize and pinpoint the leak for repair. Several options assist the utility in leak pinpointing, including a variety of leak correlation methods, pressure monitoring, step testing (closing valves to subdivide the DMA and observing the change in flow and pressure, see Chapter 6), and internal flow monitoring.

During the DMA planning phase, calibrated hydraulic models are often used to simulate prospective DMAs and verify that pressures and flows will be adequate during routine, peak, and emergency conditions. Preliminary flow measurements can be gathered using portable flowmeters to determine which areas indicate high leakage levels. A pilot DMA employing permanent metering can be implemented to give a better indication of the feasibility of using DMAs on a wider scale across the distribution system.

DMA boundaries should be designed to cross as few water mains as possible, following the "line of least resistance" by using natural geographic and hydraulic boundaries, such as open spaces, highways, railroads, or existing pressure zone boundaries. This minimizes the cost of DMA installation, operation, and maintenance.

As noted earlier, transmission mains supplying water tanks should be avoided in larger systems because the additional balancing of flows noting the effect of changing storage volumes must be conducted and the fill/drain cycle of tank operations must be unhindered. Where the DMA boundary crosses a transmission main, a valve is closed or

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

a meter is installed so that all flow at the boundary crossing, either into the DMA or into an adjacent DMA, is continuously monitored. Most DMAs are sized to keep the number of needed flowmeters to a minimum, typically one or two flowmeters, with all other main connections isolated with verified valve closures. Existing valves designated to be closed and serve as boundary valves should be located on smaller mains.

Adequate supply must be available for emergencies such as a large water main break or a heavy fire flow drawn from fire hydrants. Solutions include a check valve that can be configured in place of a closed boundary valve with an adjacent pressure zone or a PRV set to a specific pressure to allow flow from an adjacent zone with higher pressure to augment supply to the DMA. Potential locations of unacceptably low pressure or flow are identified during the preliminary design phase, and adequate design adjustments must be made when the potential for reduced service is identified. Hydraulic models can be used to identify potential problems in advance of DMA construction.

In zones fed by PRVs, the PRV is set to maintain adequate pressure at critical locations within the zone where pressure is marginal. This is accomplished in several ways, including using the remote location to function as the pressure setting for the PRV or using flow through the PRV to adjust the pressure (low flows adjust the PRV to a lower pressure setting).

Any reconfiguration of the distribution network transmission mains, pressure control valves, pumping stations, and water tanks needs to be carefully assessed and included in the planning. It is necessary to size the primary components to accurately measure the routine daily flows, but the system must also include capabilities to meet peak and emergency flows. To provide accurate flow measurement during the minimum flow period while also reliably measuring peak flows, the routine supply feed can be configured as smaller piping. A bypass or normally closed interconnection can be equipped with a check valve or PRV so that additional supply can be triggered to provide high flows during an emergency event. Such an arrangement is schematically shown in Figure 5-9.

Determining DMA Cost-effectiveness. While DMAs provide effective monitoring of leakage rates, they need not be installed on a blanket basis across a distribution system, particularly if portions of the system include robust infrastructure and experience low leakage. Some small water utilities might be able to convert their entire distribution system into a single DMA with minimal cost and effort. Larger systems might justify the cost of

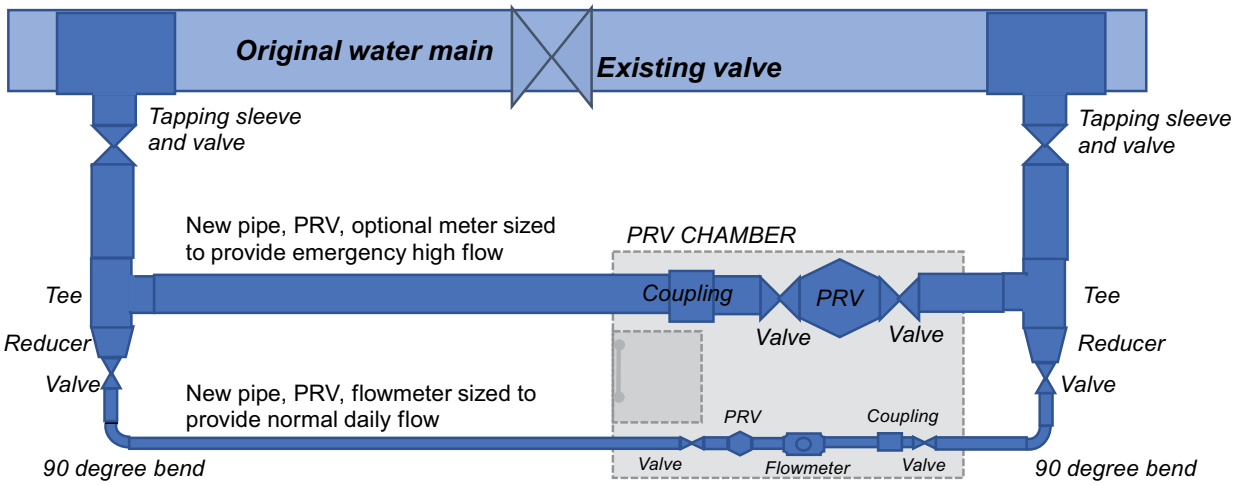


Figure 5-9 Preliminary design sketch for installation of a PRV chamber metered for daily monitoring with emergency bypass

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

DMA implementation for only a portion of the system, perhaps sections with older piping infrastructure, high leakage, high pressure, poor soils, or other aggravating factors. Larger utilities can assess the leakage trends in parts of their system and implement only those DMAs deemed to be cost-effective on an incremental basis to gradually improve their leakage management capabilities.

To determine if a prospective DMA is economically justified, the water utility should tabulate the projected costs of installation and the annual operation of the DMA, as well as the financial benefits of the expected leakage/main break reduction. This analysis should be conducted on a lifecycle basis, using the 15 to 25 years that represent the life of the primary DMA equipment such as flowmeters and PRVs. Service costs could include the design and testing effort, including the time utilized by utility staff and any consulting services employed. Equipment costs typically include flowmeters, PRVs, valves, chambers, sensors, SCADA equipment, electric power and communication equipment, and related necessities. Annual recurring costs for services and maintenance of equipment, electric power, software licenses, etc., should be tabulated and included in the lifecycle analysis.

The financial savings of benefits derived from the DMA should also be determined. The primary benefit and goal of a DMA is operating an area with a lower level of leakage than the level before DMA implementation. By being alerted quickly to newly emerging leaks, run time can be contained, and the volume of water lost to a leak can be less than the pre-DMA environment. The difference between the pre-DMA flow rate and the new reduced leakage flow rate can be converted to an annual volume, and the annual savings can be calculated. The estimated costs should be compared to the projected savings over the lifecycle of the DMA to determine cost-effectiveness.

DMA Construction. The utility operator should install pressure loggers at the critical point or location of lowest pressure in the DMA and the average zone pressure or locations with pressure closest to the average pressure across the DMA. Data should be collected for several days before closing the boundary valves to enclose the new DMA. Pressure should be transmitted or logged at locations near any critical customers, such as healthcare facilities. Comparing the data with the pressure values recorded after the DMA is isolated, and suggested modeling levels helps to identify any problem locations. Designers should notify customer service staff to be on the alert to any customer pressure or water quality complaints that might arise during the initial implementation period. If an unacceptable pressure reduction occurs in operating the DMA, operators should look for partially opened boundary valves or valves in the wrong position.

Pressure loggers can be installed inside and outside the DMA boundaries to verify isolation of the zone. A pressure drop test can be conducted to confirm that the DMA is fully isolated from the neighboring water distribution grid by slowly closing a valve on the piping that supplies input to the DMA. During the pressure drop test, pressure inside the DMA should drop further as the supply is reduced. The test should be conducted for only a short time to limit the period of low pressure. If the inflow supply is reduced and the DMA pressure fails to drop, it is likely that one or more boundary valves are open or not holding tightly and are allowing flow from a neighboring higher-pressure grid to pass into the DMA. Boundary valves must be checked, repaired, or replaced as needed and the DMA confirmed to be hydraulically tight before continuing with the DMA work.

After determining that the DMA boundary is hydraulically tight, the operator should verify that the DMA supply can meet peak demands. High-flow conditions can be created by opening a fire hydrant or boundary valve to a neighboring lower pressure zone or DMA, thereby creating an additional flow demand through the subject DMA. The utility should monitor pressures at locations of any sensitive customers during testing. If the pressure drops incurred during the peak flow conditions are unacceptable, the DMA design must be revisited. The DMA should be monitored over several days to determine the selection of an appropriately sized flowmeter and if specified, the PRV. Other equipment should

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

then be specified and procured. Once all equipment is procured, it can be installed and the DMA gradually commissioned into permanent operation.

Variations to Conventional DMAs. Other approaches to monitor for leakage using modified versions of DMAs are now being considered as data analysis and modeling tools improve and metering systems become more robust. Temporary DMAs and virtual DMAs are two variations considered for this manual.

Temporary DMAs. Some water systems have constructed temporary or "mobile" DMAs (Jenks and Papa 2020). Rather than permanently installing flowmeters and communications equipment and closing valves to isolate a portion of the system, these utilities created zones to temporarily examine minimum hour flows for short periods. The flowmeter and an optional PRV may be housed in a truck that measures flow into the system, minimizing equipment and commissioning costs needed for permanent DMAs. Systems with AMI can access their granular customer consumption history and flow patterns to estimate a nominal minimum hour flow. Without AMI, a utility can close valves and employ a portable insertion or strap-on flowmeter for several days to measure flows just as in a permanent DMA to obtain the minimum hour flow rate. If the minimum hour flow rate is sufficiently higher than expected, the utility can conduct a leak detection survey to identify active leaks for repair. After this work is complete, valves can be reopened to restore the area.

The temporary DMA can provide brief periodic assessments in leak-prone areas of a system without creating hydraulic issues or water quality issues at dead ends. Because a temporary DMA is assessed only periodically, however, unreported leakage that emerges between temporary DMA measurements can occur until detected by subsequent leak detection operations. Thus, a temporary DMA will be less effective in addressing newly emerging leakage than a permanent DMA.

Virtual DMAs. A virtual DMA is an analytic technique where flow is metered in and out of a zone at certain points so that the area is not physically isolated from the adjacent portions of the system. Pressure monitoring is also utilized in conjunction with software models to predict leakage without fully closed boundaries. (Note that virtual DMAs are only considered such if they do not have metering infrastructure at every inlet or outlet to the DMA, as a standard DMA does). The virtual DMA flows are examined in the aggregate, and a composite night flow is analyzed for significant net increases that suggest leakage has emerged.

Virtual DMAs must still be equipped with well-placed flowmeters and pressure sensors that generate data that are integrated with a calibrated hydraulic model of the system adjusted for water demands that vary diurnally, weekly, and seasonally. The presence of significant leakage can be identified in the virtual DMA when modeling suggests that flow rates are higher than expected and/or pressure is lower than expected in the model. Like a standard DMA, if integrated correctly, the virtual DMA has the potential to provide an early alert to newly emerging or increasing leakage, which allows leak detection crew deployment to occur strategically and leak run time to be minimized.

A virtual DMA acts as a digital twin (a digital representation) of the physical network and can provide the benefit of a DMA's early alert capability without the need to enclose the entire distribution system. However, because a virtual DMA also depends upon reliable field data, including a calibrated hydraulic all-pipe model and well-placed calibrated flow and pressure, measurements are essential. If sensors are not placed in representative locations and if field conditions like inadvertently closed valves are unknown, the analysis provided by the virtual DMA will be in error. Utility operators must also be aware of field operations. Virtual DMAs are likely to be unable to identify low-flow rate leaks such as customer service line leaks because pressure drops are minimal with many small leaks.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

ESTIMATING LOSSES FROM DISCOVERED LEAKS

Water utilities can obtain an approximate measure of leakage by multiplying the duration of the leak and the flow rates from a known distribution system failure by referring to the tables included in this section. In the absence of monitoring by DMA or acoustics, the duration can only be estimated. Some well-disciplined crews can take field measurements or use their experience to estimate leak flow, but it is more common to use tables to estimate leak size. The tables in this section provide estimated leakage flow rates broken down into categories, and this requires that water utility collects some field information on individual failures. The desired information should include the type of leak or break (main leak, service leak, etc.), size of the pipeline, and pressure at the site of the failure. With this information, an estimated leakage flow rate can be obtained from Table 5-2.

The reference leakage flow rates in Table 5-2 are based on a pressure of 70 psi. The leakage flow rate at pressures other than 70 psi can be estimated by taking the leakage from the appropriate type of leak listed in Table 5-2 and correcting for the different pressure that exists at the leak site. The leakage rate at the actual pressure P_a can be determined by applying Equation 5-2:

$$leakage\ rate\ at\ actual\ pressure\ P_a = \left(leakage\ rate\ at\ 70\ psi\right)\ [(P_a\ /70)^{0.5}] \qquad \text{(Eq 5-2)}$$

The exponent of 0.5 in Equation 5-2 relates to a leakage rate from metallic piping systems. Flexible (plastic) piping systems have a higher exponent value, typically 1.0 or 1.5 but could be as high as 2.5 to reflect higher leakage rates expected from plastic pipe

Table 5-2 Leakage flow rates for metallic piping systems

		Leakage Flow Rate at 70 psi				Leakage Flow Rate at 65 psi*			
		Unreported		Reported		Unreported		Reported	
Type of Leak or Break	Diameter	gpm	mgd	gpm	Mgd	gpm	mgd	gpm	mgd
Appurtenances									
Fire hydrant	—	3.5	0.005	3.5	0.005	3.37	0.0048	3.37	0.0048
Valve	—	6.9	0.010	6.9	0.010	6.65	0.0096	6.65	0.0096
Customer service connection piping leaks, all sizes	—	6.9	0.010	6.9	0.010	6.65	0.0096	6.65	0.0096
Water Mains									
Joint leak or repair band leak	6 in.	10.4	0.015	10.4	0.015	10.0	0.014	10.0	0.014
Joint leak or repair band leak	8 in.	17.3	0.025	17.3	0.025	16.7	0.024	16.7	0.024
Joint leak or repair band leak	10–48 in.	27.8	0.040	27.8	0.040	23.7	0.034	23.7	0.034
Circumferential crack	4 in.	34.7	0.050	69.4	0.100	33.4	0.048	66.9	0.096
Circumferential crack	6 in.	55.5	0.080	111.1	0.160	53.5	0.077	107.0	0.154
Circumferential crack	8 in.	76.3	0.110	152.6	0.220	73.5	0.106	147.0	0.212
Circumferential crack	10 in.	93.8	0.135	187.6	0.270	90.4	0.130	180.8	0.260
Circumferential crack	12 in.	111.1	0.160	222.2	0.320	107.0	0.154	214.1	0.308
A longitudinal crack or split bell	6 in.	69.4	0.100	138.9	0.200	66.9	0.096	133.8	0.193
A longitudinal crack or split bell	8 in.	93.8	0.135	187.6	0.270	90.4	0.130	180.8	0.260
A longitudinal crack or split bell	10 in.	111.1	0.160	222.2	0.320	107.0	0.154	214.1	0.308
A longitudinal crack or split bell	12 in.	138.9	0.200	277.8	0.400	133.8	0.193	267.7	0.385

Source: *Bristol Water Services 2001*
* Leakage rate at 65 psi = (Leakage rate at 70 psi) [(65/70)0.5]; mgd = million gallons per day

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

failures (which are known to split longitudinally and blow out). The second set of leakage flow rates in Table 5-2 lists leakage rates at 65 psi as an example, calculated based on Equation 5-2.

To improve on this estimate, the size and configuration of the leak (including holes as well as splits and cracks) can be gathered at the time of repair. An estimated leakage flow rate can be obtained from Table 5-3 for joint leaks and cracks in pipelines or Table 5-4 for circular holes. If actual leak measurements cannot be taken, the water utility operator can use the leakage rates listed in Table 5-2 by leak type and pressure.

It is possible to calculate the leakage flow rate directly from Greeley's formula Equation 5-3 if the size and nature of the leak opening and the pressure at the point of failure are known or reasonably estimated. This calculation is especially helpful for large leaks where the flow is too great to measure and the water main must be valved off quickly. This method also makes assumptions regarding the shape of the hole, which may introduce errors.

Table 5-3 Leakage losses for circular holes under different pressures*

Diameter of the hole (in.)	Area of the hole (in.^{2})	Leak Losses (gpm) at various pressures, in psi									
		20	40	60	80	100	120	140	160	180	200
0.1	0.008	1.068	1.510	1.849	2.135	2.387	2.615	2.824	3.019	3.203	3.376
0.2	0.031	4.270	6.039	7.396	8.540	9.548	10.460	11.298	12.078	12.811	13.504
0.3	0.071	9.608	13.588	16.641	19.216	21.484	23.535	25.420	27.175	28.824	30.383
0.4	0.126	17.081	24.156	29.585	34.162	38.194	41.839	45.192	48.312	51.243	54.014
0.5	0.196	26.689	37.744	46.226	53.378	59.678	65.374	70.612	75.487	80.066	84.397
0.6	0.283	38.432	54.351	66.566	76.864	85.936	94.139	101.681	108.702	115.296	121.532
0.7	0.385	52.310	73.978	90.604	104.620	116.969	128.133	138.399	147.955	156.930	165.419
0.8	0.503	68.323	96.624	118.340	136.647	152.776	167.357	180.767	193.248	204.970	216.057
0.9	0.636	86.472	122.290	149.773	172.943	193.357	211.812	228.783	244.579	259.415	273.448
1.0	0.785	106.755	150.975	184.905	213.510	238.712	261.496	282.448	301.949	320.266	337.590
1.1	0.950	129.174	182.679	223.736	258.348	288.841	316.410	341.762	365.359	387.521	408.484
1.2	1.131	153.728	217.404	266.264	307.455	343.745	376.554	406.725	434.807	461.183	486.129
1.3	1.327	180.416	255.147	312.490	360.833	403.423	441.928	477.337	510.294	541.249	570.527
1.4	1.539	209.240	295.910	362.415	418.481	467.875	512.532	553.598	591.821	627.721	661.676
1.5	1.767	240.199	339.693	416.037	480.399	537.102	588.366	635.508	679.386	720.598	759.577
1.6	2.011	273.293	386.495	473.358	546.587	611.103	669.429	723.066	772.990	819.880	864.230
1.7	2.270	308.523	436.317	534.377	617.045	689.878	755.723	816.274	872.634	925.568	975.634
1.8	2.545	345.887	489.158	599.094	691.774	773.427	847.247	915.131	978.316	1037.661	1093.791
1.9	2.835	385.386	545.019	667.509	770.773	861.750	944.000	1019.637	1090.037	1156.159	1218.699
2.0	3.142	427.021	603.899	739.622	854.042	954.848	1045.983	1129.791	1207.798	1281.063	1350.359

* Calculated using Greeley's formula, Eq. 5-3

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 5-4 Leakage losses for joints and cracks*

Area of Joint or Crack		Leak Losses (gpm)									
Length (in.)	Width (in.)	Water Pressure (psi)									
		20	40	60	80	100	120	140	160	180	200
1.0	1/32	3.2	4.5	5.5	6.4	7.1	7.8	8.4	9.0	9.6	10.1
1.0	1/16	6.4	9.0	11.0	12.7	14.2	15.6	16.9	18.0	19.1	20.1
1.0	⅛	12.7	18.0	22.1	25.5	28.5	31.2	33.7	36.0	38.2	40.3
1.0	¼	25.5	36.0	44.1	51.0	57.0	62.4	67.4	72.1	76.5	80.6

* For leaks from joints and cracked service (rigid) pipes, an orifice coefficient of 0.60 is used in Equation 5-4.

For losses from items such as broken taps, an orifice coefficient of 0.80 is assumed and the flow calculated in gallons per minute using Equation 5-3, which is applied in Table 5-3:

$$Q = \frac{43,767}{1,440} \times A \times P^{0.5} \quad \text{(Eq 5-3)}$$

Where:

Q = flow (gpm)
A = the cross-sectional area of the leak (in^2)
P = pressure (psi)

For Leaks from joints and cracked service (rigid) pipes, an orifice coefficient of 0.60 is used in Eq. 5-4.

$$Q = (22.796)\ (A)\left(P^{0.5}\right) \quad \text{(Eq 5-4)}$$

Where:

Q = flow (in gpm);
A = area (in $in.^2$);
P = pressure (in psi).

Many corrosion holes are close to circular, and the flow would be:

$$Q = (95.436)\ \left(r^2\right)\left(P^{0.5}\right) \quad \text{(Eq 5-5)}$$

Where:

Q = flow (gpm)
r = radius (in)
P = pressure (psi)

HOW TO LOCATE LEAKS

It has become essential for the utility operator to know where and how much water loss is occurring as leakage. Leaks have occurred in piped water systems for as long as these systems have been in existence. Historically, many water utility operators reacted to leaks only after they became visible, often causing disruption. In recent decades, however, technologies have been developed to allow the operator to address leakage proactively by detecting leaks while they are relatively minor and not evident from above ground. These techniques provide operators with accurate means to pinpoint leak sources and measure

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

quantities of water from leakage occurrences. An overview of these techniques is provided here. Refer to Chapter 6 for more detail about leakage control techniques.

Active leakage control methods employed by water utilities fall into four general categories:

1. Acoustic techniques. Water escaping from the pressurized system creates a sound that can be detected by sensitive listening devices. Leaks can be identified and pinpointed via these techniques, but the amount of water escaping from leaks cannot be quantified with high accuracy, so these techniques cannot detect background leaks. Background leaks are considered undetectable by sonic methods.
2. Flow and pressure measurement techniques. DMAs include equipment that measures the water flow rate into the area, and analysis can discern the portion of the flow that exists as leakage, but this method does not provide a capability to pinpoint individual leaks. Nonetheless, water flow within a distribution system can be measured between different points in the system and analyzed to infer and quantify the presence of a leak. For example, flow monitors installed on either side of a river or highway can confirm the presence of an otherwise undetected leak. With the advent of the digital twin hydraulic models and virtual DMAs, it is possible to project where a sizeable leak exists based upon where a continuous and discernible unexpected increase in demand is occurring.
3. Environmental detection techniques. Water from leaks causes disturbances in their surroundings that can be detected by various techniques. This evolving array of techniques scans from various locations: at ground level, from the air (plane, drone), and in space (satellite). Techniques include ground-penetrating radar to find subsurface water accumulations, thermal techniques to find leaking water by temperature differentiation, and subsurface profiling to identify liquid with characteristics of drinking water or shifting soils caused by erosion from leaks.
4. Predictive techniques. If the location of leaks can be predicted in advance, pipes can be strategically renewed before they fail or monitored, so the potential for breaks can be quickly anticipated and corrected to reduce operational stress. Condition assessment techniques now exist that determine remaining effective pipe wall thickness to varying degrees and estimate time to future failures (see AWWA 2019). Several predictive models have been developed that use pipe and environmental data to prioritize water mains for replacement, including identification of the consequence of pipe failure. More recently the application of machine learning has been applied to reliably predict the probability of failure (Hughes and Fitchett, 2020).

Effective leakage management relies most commonly on the use of the first two techniques, but all techniques continue to progress.

Acoustic Detection Techniques

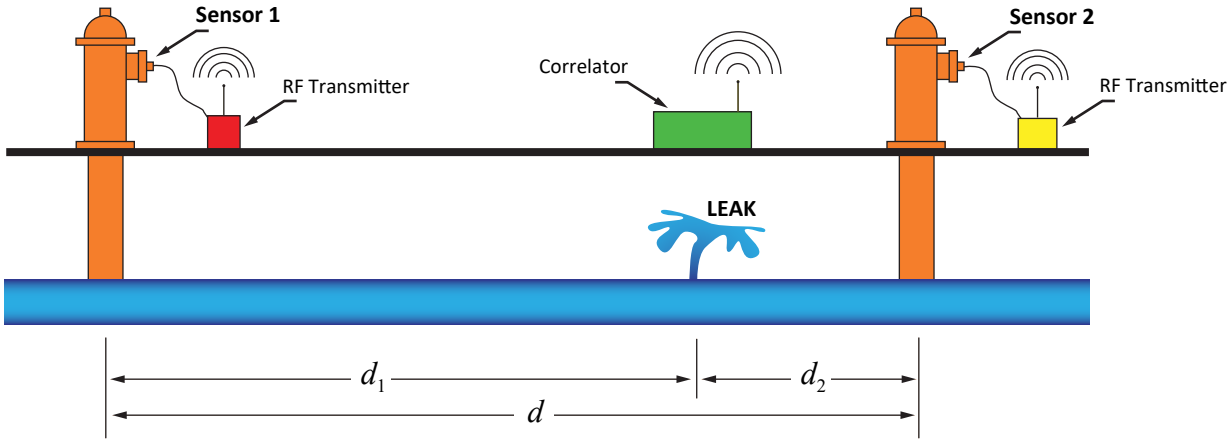
Acoustic devices have been used to detect and pinpoint leak noises for over 100 years. When water escapes from a pressurized pipeline, a distinct and recognizable leak sound is generated in the pipe at the point of leakage. Leak sounds can travel in the water and along the pipe in all directions and be detected at points away from the leak. Manual leak detection surveys feature acoustic devices as operators place sensors in contact with fire hydrants, valves, curb stops, or customer plumbing to listen for the presence of nearby leaks. Ground microphones are placed on the ground surface over the water main to pinpoint the location to excavate where the sound is loudest. Advanced techniques like CAM include leak sensors permanently deployed and units that work together to pinpoint an area for investigation.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

To understand the capability of the leak sensor options, it is important to understand how leak sound travels. Leak noise varies in frequency and distance as it travels. The vibration travels along the pipe as a function of pipe size and pipe material. Iron, steel, other metal pipes, and cement pipes transmit leak vibration in the pipe for greater distances than plastic materials. Leak noise in a smaller diameter pipe travels farther than leak noise in a larger pipe made of similar material. As noted previously, tiny leaks generate faint sounds, but once acoustically significant, smaller (pinhole) leaks generate audibly higher frequency noise than large flow rate ruptures, which produce lower frequency noise. Higher pressures generally generate louder sounds. Plastic pipe leaks vibrate at lower frequencies than comparable metal pipe leaks and tend to absorb vibration, making acoustic leak detection more challenging for polyethylene (PE) and polyvinyl chloride (PVC) pipe over a distance.

From early mechanical listening devices (sounding rods, geophones), leak noise detection advanced to devices using electronic sound amplification to better detect, filter, and break down leak noises. These methods are used largely to detect the presence of leaks. In the 1970s, leak noise correlators provided a significant advancement in the field by allowing utilities to pinpoint the exact location of the leak emerging from the piping. Leak correlation identifies the leak location by comparing simultaneous leak noise sound waves from two devices in contact with the pipe system on either side of the leak, as shown in Figure 5-10. The leak noise correlator has become a fundamental tool of a leak detection program for its capability to pinpoint leak locations.

More recently, leak noise monitors have been developed to record and analyze leak noise through data logging or transmission to the cloud where the vendor and the utility can "view" noise anomalies graphically. These monitors are designed to be deployed either permanently in fixed locations or rotated between areas of the distribution system (the "lift and shift" method). The monitors are usually programmed to monitor during normally low flow nighttime hours and record leak noises at the quietest time of the day. The daily recorded sounds can then be downloaded or transmitted and compared through a data management system to detect leak points of interest that can then be investigated. Leak noise monitors can be deployed relatively easily and require far less overall labor than traditional manual leak surveys. Figure 5-11 shows a geographic information system



Sound travels in the pipe at constant velocity V. Simultaneous leak signals arrive at Sensor 1 I time $t_1 = d_1/V$ and at sensor 2 in time $t_2 = d_2/V$. Time lag between signals $\Delta t = t_1 - t_2 = (d_1 - d_2)/V$. Given $d_2 = d - d_1$, $\Delta t = (d - 2d_1)/V$ and $d_1 = (d - V\Delta t)/2$

Figure 5-10 Leak correlation is used as a standard pinpointing tool of the leak detection squad in many water utilities

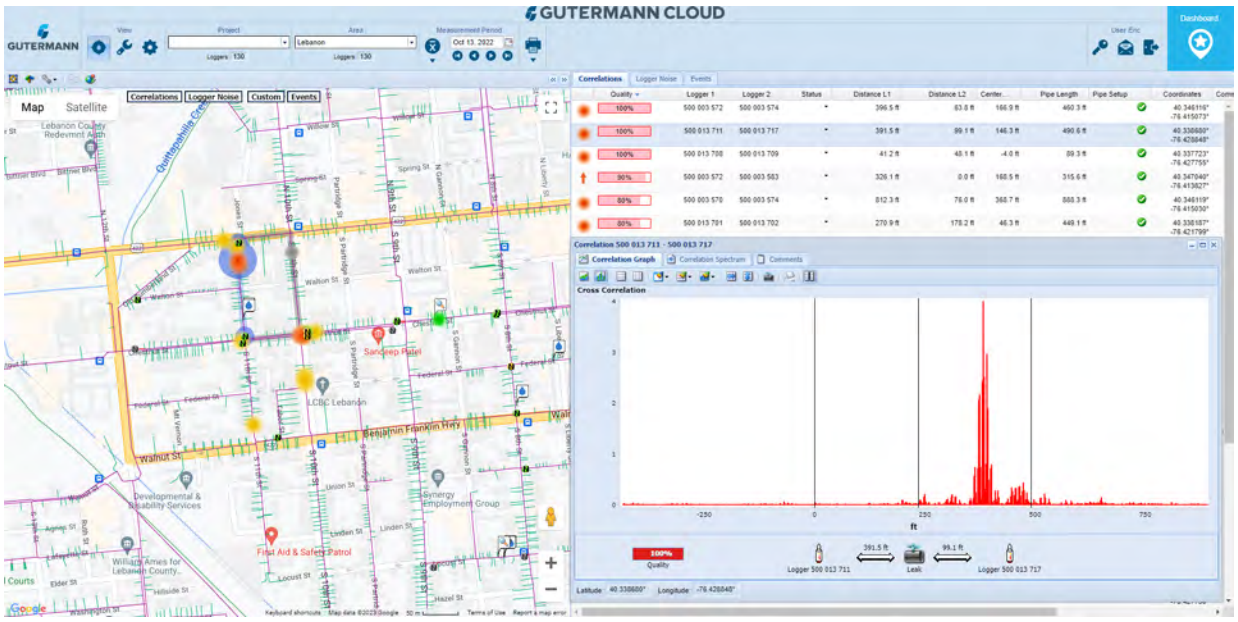
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

display map of leak noise loggers and a profile of an individual monitor identifying a potential leak.

Depending on the manufacturer, fixed acoustic monitors are designed to be attached to customer service connections, cling magnetically to gate valve operating nuts, or be incorporated to a fire hydrant as a sensor. At least one manufacturer offers "smart" customer meters that include acoustic monitoring capabilities to detect leaks on service lines and in the distribution system. An increasing number of fixed leak noise monitors can correlate with one another to detect leaks and narrow the area for field confirmation. Correlation between two monitors tends to reduce the number of false-positive alerts from other noise sources.

Leak noise monitors properly stationed throughout an area can find leaks at low-flow rates, literally as soon as they are acoustically detectable. Detecting leaks shortly after they emerge limits leak run time and reduces annual leakage volumes significantly and can keep the costs of repair, restoration, and damage low while improving worker safety. It is important to carefully develop the economic business case for the use of leak noise monitors since these "smart" devices require funding for purchase, maintenance, and replacement or upgrading at recurring intervals.

Leak monitors have evolved significantly recently to address challenges to using the technique. Leak correlation has helped to reduce false positives from nonleak noise sources. Nonetheless, initial fieldwork is usually required to screen out noise sources like pumps, throttled valves, and the off-peak customer uses. Units have become increasingly sensitive and some leaks flowing less than 0.5 gpm have been detected. A major challenge for acoustics is plastic pipe leaks where sound transmission is absorbed by the pipe material. Hydrophones are sensors that can be put in direct contact with the water to detect leak noise. Water contact can be achieved through connections to the pipe, curb stops, valves, or fire hydrants. These devices are decidedly more effective for finding leaks on plastic pipes or transmission mains than pipe contact acoustic sensors. Devices of this type may combine pressure/surge sensors. There is increasing development of hydrophone



Courtesy of Gutermann

Figure 5-11 Leak noise monitors help to automate the leak survey process and display leak candidates daily.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Digital Water Solutions

Figure 5-12 Hydrophone retrofitted to fire hydrant base

technology that listens to the vibration in the water using equipment attached to the water main as shown in Figure 5-12.

Another type of acoustic sensor is inserted into an active pipeline to make water contact. Sensors can be either tethered or free floating and pinpoint leaks by tracking acoustics as they travel through the pipe. These devices are generally used on large transmission pipelines and offer a high level of resolution and accuracy that can detect very small leaks that elude detection using traditional listening and correlating techniques. These technologies are typically more expensive to deploy than other methods and may require the pipe to be shut down to insert the monitoring tools.

Although effective acoustic leak detection devices are in common use, they do not directly measure the flow rate of water escaping from a leak or detect background leakage. It is helpful to know the flow rate of water lost from leaks to evaluate leakage control progress and reveal leakage patterns in the distribution system. Knowing leakage flow rates in the distribution system can help prioritize fieldwork. Permanently installed leak noise monitors can note changes in sound that reveal the rise in leakage from individual leaks over time.

Like all early leak alert methods, there is evidence that leak noise monitors not only help facilitate leakage reduction but also contain the damage that would likely occur if leaks were left to grow to disruptively high-flow rates. Additional cost efficiencies are gained when leaks are discovered early with low-flow rates. These leaks can be scheduled

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

for repair proactively during routine weekday hours, avoiding emergency repairs often done in off-hours and frequently in adverse weather, traffic congestion, and supply interruption conditions. The use of correlating units also narrows field investigation time. Several companies use sound transmission to provide condition assessment (pipe wall thickness information). Advances in machine learning are occurring including combining correlation which gives the distance from leak to sensor and leak noise frequencies to categorize the size of leaks.

The variety of acoustic monitoring approaches (as well as other options described below) provides the utility with the opportunity to pilot a technique before implementing it as a system-wide solution. Good pilot areas are generally those locations with historically problematic leakage (for example, a DMA where leakage flow rates can be measured) and the performance of the pilot technology in reducing leakage can be tracked. Pilot studies acquaint utility operators with the technique being evaluated and the level of support that utility staff must provide. Pilot projects reveal the performance and logistical requirements of the tested technique and serve as the basis for the decision on whether to extend the use of the technique to other areas of the distribution system.

Flow and Pressure Monitoring Techniques

By monitoring flow and pressure around the clock and linking it to hydraulic modeling analysis, operators can evaluate the potential for leakage by examining changes in minimum night flow. Pressure monitoring can be a tool to assist operators in identifying areas of the distribution system that are likely to be experiencing significant leakage, including large leaks, leak clusters, or undetected high customer consumption. Moderate- to large-volume leaks in water systems create a discernible localized drop in pressure on pipelines. Small leaks do not create significant pressure drops, but by employing accurate pressure sensors and a well-calibrated hydraulic model, analysis can be conducted to pinpoint areas of notable leakage. Details on the workings of DMAs and PMAs are provided in this chapter and Chapter 6.

Environmental Leakage Techniques

Considerable advances in monitoring tools and computing technology have been developed to look below the ground surface to find hidden leakage. These are technologies that evaluate the pipe environment to confirm existing water leakage. Leaks, by their nature, can cause underground erosion of adjacent soils, create accumulations of subsurface water pockets and water-soaked soil, and alter underground chemistry and temperature. Technologies that detect these symptoms include ground-penetrating radar, thermal sensors, radiometers, and energy pulsing tools. Measurements can be captured visually as snapshots or examined for changes over time. These are still evolving techniques and not yet mainstream methods. Finally, the use of canine leak detection (sniffer dogs) has found success in the United Kingdom, United States, and other countries. Trained dogs are proving to be very accurate in detecting the chlorine smell from hidden leaks, but these animals require an investment as well as training, handling, and care from owners.

Generally, acoustic leak surveys are a confirming or short-term leakage control process. Flow monitoring in DMAs and pressure management form part of medium- to long-term interventions, as do the environmental methods. Portable acoustic tools are generally employed to identify the location of a leak for excavation and repair.

Predictive Techniques

Historically, determining the time and location of high-flow rate-reported failures (main breaks) has been considered unpredictable. Nonetheless, aided by machine learning techniques, further development of risk management approaches to renewing pipes

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

is ongoing. Risk assessments have advanced, and a mathematical calculation has been developed for *risk*, which equals the calculated rating of the consequence of failure multiplied by the probability or likelihood of failure. Hydraulic modeling specialists have undertaken processes to assess the consequences of pipe failure anywhere in a distribution system. Consequences of large leaks and main breaks include damage to roadways and buildings, property loss, and interruptions of water service to customers with potential economic losses and additional liability for the water utility. Utility awareness of the potential costs of these consequences allows operations to focus monitoring and mitigation measures on pipes with the most consequential impacts. Several approaches exist to determine the probability of failure of pipe segments. Many methods employ cohort analysis, assess pipe stresses, or incorporate condition assessment. More recently, the artificial intelligence industry has applied machine learning techniques to evaluate a large data range looking at factors causing past breaks to predict future breaks on individual pipe segments. Results from these programs are encouraging, and combining them with the consequence of failure analysis allows utilities to focus their leak management resources on the pipes at highest risk. If a reliable risk model is generated, utilities can identify the pipe segments most in need of replacement while minimizing premature pipe replacement. They may also focus pressure and leakage control strategies on pipes that remain in service but are likely to incur periodic failures in the near future.

STARTING A LEAKAGE MANAGEMENT PROGRAM

All water distribution systems experience leakage, and utilities can choose to use a systematic approach to develop an appropriate leakage management strategy, especially if the utility is advancing from a reactive to a proactive approach. By compiling the annual water audit (see Chapter 2), the utility quantifies its leakage volume and cost impacts and can evaluate the real loss key performance indicators (KPIs) that are calculated by the audit software. To select the proper methods to manage leakage, a utility needs to understand the failures (types and frequencies) occurring in their water distribution system and the potential economic savings that can be realized from enhanced leakage controls. The level of effort needed to undertake the program planning varies from system to system as factors differ between individual water distribution systems and the variety of technologies available to mitigate leakage. The important first step is to assign a responsible person or team to take ownership of the program, regularly assess progress, and continually implement best management practices for leakage control. Table 5-5 outlines suggested steps that can be taken to prepare a program. Chapter 7 will detail planning guidance, including setting the leakage management strategy. A sample leakage analysis for the fictitious County Water Company (CWC) follows this process later in this chapter.

Most of the steps in Table 5-5 are described in detail in the next sections of this chapter, but steps 4 and 5 are described further in Chapter 6.

Establishing an Initial Leakage Reduction Goal

The water audit provides essential building blocks to make an initial projection of how much leakage can be reduced. The dashboard worksheet of the audit software provides charts that show where the calculated values of KPIs for a water utility fall relative to the data of 1,124 utilities in the WARD. Table 5-6 lists values of the unit real (leakage) losses in two forms: gallons per service connection per day and gallons per mile of pipeline per day.

The values are referenced to percentile levels, which reflect where the value lies within the dataset. For example, at the 90th percentile value of 115.43 gal/conn/day, 90% of the water audits in the WARD have a unit real losses rate below this level, with only 10% having a higher leakage rate. If the utility's unit real losses or the real loss cost rate exists

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 5-5 Steps for initiating a sustainable leakage management program

1. Identify a preliminary goal for real loss reduction, noting the cost savings projected from the water savings and any other major savings from leakage/break reductions as well as assessing the cost of leakage management options.
2. Quantify the volumes of leakage by its three components (reported leaks, unreported leaks, and background leakage) by applying an LCA technique (per the LCA section in this chapter). Alternatively, volumes can be estimated by analyzing minimum hour flow measurements in metered pilot zones or DMAs.
3. Assign costs to the volumes quantified in 2 above.
4. Consider a short-term plan for initial leakage reduction ("picking the low-hanging fruit") by identifying the leakage reduction methods and resources to achieve early success in meeting preliminary goals. Alternatively, consider planning a pilot program in a leak-prone portion of the system to test the approach and gather more information about the component breakdown of leaks.
5. Implement the short-term plan or pilot project and evaluate whether the leakage reductions that are achieved meet the levels expected for a cost-effective program.
6. Recalculate the LCA based on new findings.
7. Review results and confirm assumptions; expand or alter the plan as needed.
8. Set goals for medium- and long-term leak reduction, including revised methods and leakage reduction targets.

Table 5-6 Leakage unit rates from the WARD

Percentile Value	Unit Real Losses, gallons per service connection per day	Real Losses, gallons per mile of pipeline per day
10th percentile	16.28	879.7
25th percentile	22.72	1,289.4
50th percentile (median)	36.77	2,048.9
75th percentile	66.38	3,639.6
90th percentile	115.43	6,074.1

at a high level (over the 75th percentile value), the utility may have the potential to reduce leakage economically. The product of annual leakage volume and the variable production cost (VPC) gives the annual cost of leakage, which is an upper limit of excessive costs due to leakage. While it is not possible to eliminate all leakage, it is important to have a basic understanding of how much leakage can be feasibly reduced and which approaches can accomplish a meaningful reduction.

A utility with a unit real losses above the 75th percentile value might set a short-term goal of reducing its real (leakage) losses to the 75th percentile within the first two to three years. For example, a utility with unit real losses of 85 gal/conn/day might set a goal to reduce leakage to a level of 66 gal/conn/day within three years. This is a good first-step approach to obtain an initial target, but the utility should ultimately conduct an economic assessment as discussed later in this chapter to determine the level of leakage that is cost-effective or economical to target.

It is possible that a utility can fall into different percentile levels for the two components of loss per connection and loss per mile. If your system has a high connection density and leak-prone water mains, the loss per connection may rank lower than it would for loss per length of main. The reverse would be true in a low customer density system where losses per mile of main might look better than the loss per connection. A higher loss per connection may not necessarily mean high service line leaks. Other factors like inaccurate production flow metering need to be considered during the Water Audit process.

Leakage Reduction Approaches. Figure 5-13 shows the four-pillar approach to controlling leakage on an annual basis. The center box represents the annual volume of real

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

(leakage) losses from the water audit or current annual real losses (CARL). The perimeter of the smallest box within the CARL represents the unavoidable annual real losses (UARL) level; a reference level that is unique for each water system and is calculated by the audit software. Leakage that exists below the UARL is theoretically taken and not capable of being addressed, thus water utilities should generally not strive to reduce leakage below the level of the UARL. The perimeter of the second smallest box represents the economic level of leakage (ELL). Leakage cannot be economically controlled below this level because the cost of the leakage control measures will exceed the value of the savings from the additional leakage reductions that are achieved. The area between the CARL and ELL perimeters represents the portion of leakage that can be reduced economically. This is the ideal target range for leakage reduction for many water utilities. All water utilities should consider some level of activity in each of the four pillars in Figure 5-13 to contain leakage to economically low levels. Discussion in Chapter 2 defines the infrastructure leakage index (ILI) as the ratio of the CARL over the UARL. Guidance on the use and interpretation of the ILI is given in the sidebar "A Note About the ILI."

The four arrows shown in Figure 5-13 represent the four recognized water distribution management techniques that can sustain a low-leakage, water-efficient distribution system. The key is to assign the appropriate combination of these four primary leakage control methods, as defined below, to meet the leakage reduction goals of the water utility.

1. Active leakage control: identifying and locating existing leakage in the water utility transmission and distribution system
2. Optimized leak repair activities: making timely and lasting repair
3. Pressure management: optimizing distribution system pressures to minimize leakage and reduce the likelihood of disruptive rupture

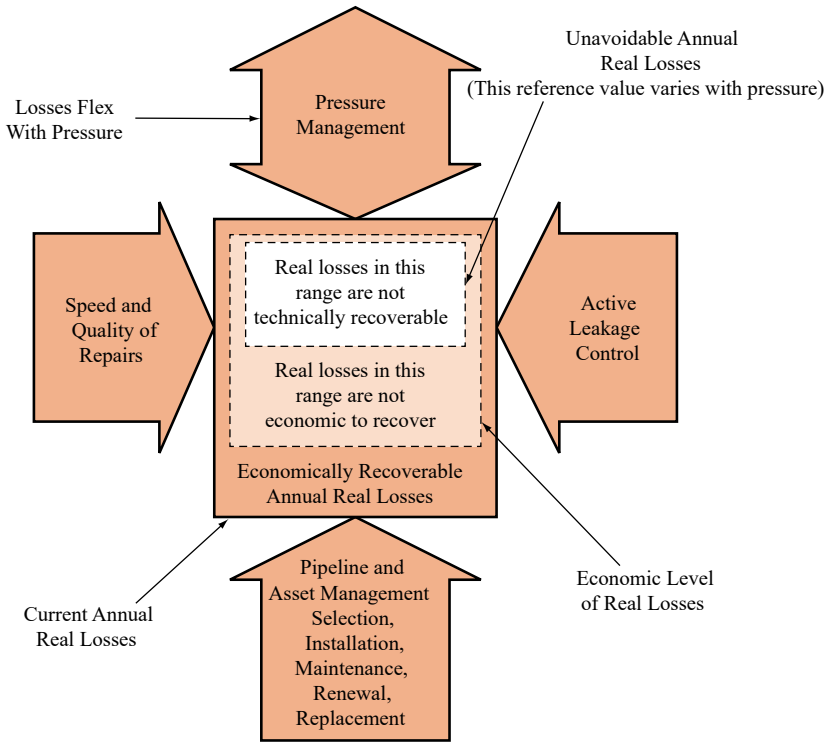


Figure 5-13 Four-pillar approach to the control of real (leakage) losses

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

4. Infrastructure rehabilitation and renewal: all pipeline assets eventually reach the end of their useful life and must be rehabilitated or replaced if they are to continue to provide service without structural failure

These four approaches can reduce leakage to different degrees depending upon the type of leakage occurring. New losses are continually occurring in a water distribution system; ideally, loss reduction activities should be designed to address and anticipate leaks to come and subsequently sustain a lower leakage level.

A Note About the ILI

The ILI is a dimensionless ratio used in many parts of the world to assess the leakage standing of a water utility. The ILI is the ratio of the CARL over the UARL. An equation has been developed by Lambert and others to estimate the UARL using inputs of the length of water mains, number of service connections, AOP, and average length of customer service lines maintained by customers (see Chapter 2). An ILI of 1.0 means that the CARL is equal to the UARL, thus a very low level of leakage exists. In theory, leakage is minimized when the ILI equals 1.0, but some systems with outstanding leakage control programs have achieved an ILI below 1.0, thus challenging the assumptions in the derivation of the UARL equation. As the ILI increases above 1.0, the more significant the leakage. Most North American water utilities should not strive for an ILI value of 1.0 since that will not likely be a cost-effective approach. A more effective approach is to understand the concept of the ELL. Data from large numbers of North American validated water audits, however, indicate that the UARL exists well below the CARL for many North American water utilities. Potential therefore exists for many of these water utilities to make significant incremental and cost-effective reductions in leakage by striving for an ILI of somewhat less than 1.0 if economically justified.

Leakage Components. The three types of leakage—reported leakage, unreported leakage, and background leakage—were previously defined along with the lifespan of a leakage event in terms of the ALR times, and together represent the leakage components. Knowing the degree to which the leakage types occur in a water distribution system helps the utility operator determine the best strategy to reduce leakage economically. For systems that have not managed leakage proactively, it may be best to establish short-, medium-, and long-term actions or interventions to achieve modest leakage reductions incrementally over time until the leakage components of the system are better understood.

Quantifying Leakage by Components

Setting goals for leakage reduction calls for identifying the value of leakage reduction by components. As discussed in Chapter 2, water auditing occurs at three levels of refinement, and these levels can be applied to the assessment of real (leakage) losses.

1. Top-down approach: the initial desktop process of gathering information from records, procedures, data, and other information systems.
2. Bottom-up approach: the method that validates the top-down results with actual field measurements and system data, such as leakage losses calculated from DMA minimum-hour flows or temporary deployment of acoustic monitors to evaluate the potential for leakage reduction. Background losses can be quantified in this approach by conducting pressure step testing in the small zones described in the DMA discussion in Chapter 6.
3. LCA: a technique that models leakage volumes based on the nature of reported leak occurrences and durations.

Top-down approach. The top-down approach relies on knowledge of the system including leak repair reports and general awareness of the types of leaks experienced in the water system. The experience and pipe environment knowledge of system operators can

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

offer important clues concerning the component breakdown of leakage for the individual utility. If pressures in the system are relatively high and pipes are shallow and/or off-road, it is likely that a significant portion of leakage surfaces quickly and would be reported as leakage. In such cases, much of the real losses can likely be quantified by adding up surfaced leaks (leak flow rate multiplied by duration). Conversely, if pressures are low and pipes are deteriorated, deeper, and found under paved surfaces sitting in pervious soils, then background leakage and unreported leaks would likely be the primary focus.

Operator experience with the nature of observed failures offers clues to leakage. A system may have an abundance of small breaks on service connections or water mains (like corrosion holes) as opposed to visible pipe ruptures. This suggests that significant quantities of leakage occur as unreported leaks. When leaks surface, crews may see evidence that the leaks have run for a long time: for example, evidence of sinkholes or underground streams from the leak flowing to nearby water sources or drainage areas. Continuous round-the-clock excessive water demand can signal significant unreported leakage running before a leak finally surfaces as a reported failure. Utility operators can be alerted to closely monitor the system water demands, noting the increased time to fill a water storage tank or discernible changes in the sound of running pumps at pump stations.

The top-down method of identifying real loss volumes is very useful for a quick, broad look at the entire system performance or volumes of real loss for distinct regions within a distribution system. The top-down approach is the best first step for water utilities just getting started in leakage management. However, the top-down approach does not produce a sufficiently detailed analysis of the separate components and volumes of real loss. Ultimately, the water auditor can better validate and improve the accuracy of the water audit when it is augmented by bottom-up field measurements, LCA, or both assessments.

Bottom-up approach: piloting. For utilities that recognize their water leakage demands immediate attention, the most viable approach often is to pilot a particular leakage control technology and evaluate its performance in their water distribution system. A variety of leakage control options exist, and research can be conducted to select the most appropriate. Regardless of which technology is piloted, it is important for the water utility to establish in advance the performance level and cost-effectiveness that will constitute a successful trial of the technology. The utility should carefully document the pilot results, evaluate technology performance, and gather new or additional data on the nature of system leaks that the technology may provide. The bottom-up approach is based on field activities that broaden the operator's knowledge of their system leakage and provide them a first-hand evaluation of a particular leakage control technology. Technologies that prove effective during a pilot phase can be considered for broader use as part of the utility's leakage management program.

LCA. LCA is a detailed method that analyzes the occurrence of leakage components in a water distribution system along with the costs of proposed leakage controls and other data to provide information on cost-effective leakage control strategies and frequencies. The Water Research Foundation (WRF) project *Real Loss Component Analysis: A Tool for Economic Water Loss Control* (WRF 2014a) provides a standard approach to conduct the LCA. This project created a free spreadsheet software tool (Leakage Component Analysis Model, or LCA model) that enables utilities to plan a cost-effective leakage control program. The LCA model is described in this chapter, with discussion modified to correspond with some alternative advanced leakage detection methods that have developed in recent years.

LCA requires more data than the top-down water audit approach and technical staff to execute the work, but it is still largely an in-office exercise and not as resource intensive as bottom-up assessments, which require flow measurement or leakage control equipment to be deployed in the water distribution system. LCA relies on the assessment of the average durations of detectable leaks, namely, the ALR periods. ALR periods were described earlier

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

and shown schematically in Figure 5-4 with an example given in Figure 5-5. Breaking down the water utility's CARL into its component quantities of reported, unreported, and background leakage and assessing their average ALR period is the fundamental approach of the LCA. Additionally, leakage events can be categorized as those occurring on water mains, mains fittings, and service connections to improve understanding of leakage control methods to reduce effectively.

Having good and comprehensive data on the number and types of failures occurring in the water distribution system is critical to obtaining reliable results from the LCA model. This is explained in the sidebar "The Importance of Reliable Distribution System Failure Data Collection" with example data collection tools shown in Figures 5-14 and 5-15.

LCA builds prudent estimates of leakage volumes by assessing the types of leakage events typically encountered in the system and analyzing them using specific leakage flow rates and response times to those events, along with average pressures and other readily available data. The sum of the estimates is then compared to the CARL identified in the top-down analysis, and the two models are calibrated.

The LCA method is rigorous and provides detailed results. It is meant for the technical staff of utilities to assemble. The LCA provides the full range of outputs to assemble the comprehensive leakage management plan. For utilities just getting started with quantifying leaks, staff may wish to proceed with top-down and/or bottom-up approaches and eventually conduct the LCA later.

The Importance of Reliable Distribution System Failure Data Collection

LCA model results are only as good as the quality and completeness of the water system failure data placed into it. It has been observed that terminology used by water utilities for pipe failures is inconsistent. The terms *leak*, *break*, and *burst* are used with different definitions and interpretations depending on the water utility. For consistency, WRF Project 4372a defined any pipeline or service deficiency that releases leaking water as "a failure" (WRF 2014a).

In addition to varied terminology, many water utilities fail to document all failures occurring in their systems. An accurate LCA requires that all failures on the transmission system, distribution system, and service connections need to be consistently documented. To meet this end, WRF Project 4372a also developed a companion spreadsheet—the *Leak Repair Data Collection Guide* (WRF 2014b)—to give water utilities a means for consistent documentation of water system failure events. This tool is also a free product available to water utilities to employ to ensure consistent data collection. Figure 5-14 lists the data selection options of dropdown menus built into this tool, and Figure 5-15 shows one of the data entry worksheets, in this case documenting failures on water mains. There are separate worksheets for mains fittings and customer service connections.

Additional field information on failures can include photos of excavated piping that is experiencing leakage. With the prevalence of mobile devices and Internet connectivity, great opportunity exists to photodocument leaking pipelines, fittings, and service connections as well as documenting excavations and shoring, repaired conditions before restoration, and other field conditions. These images can be employed to assist the evaluation of failure trends, serve as evidence in liability claims, and promote better maintenance and repair techniques.

LCA example. The detailed methodology of the standard LCA is conducted in the following steps and illustrated in the accompanying example for CWC given in a series of sidebars (see "CWC: Conducting an LCA to Quantify Individual Leak Types and Evaluate Economic Intervention Frequency" sidebars, Parts 1 through 6).

Step 1. Quantify reported leakage. The annual volume of reported leakage (visible leaks and breaks) can be assessed by adding the product of the flow rate of each reported break by the known or estimated run time of each leak. The flow rate of small leaks can be calculated at the time of repair as a function of the type of failure or size of the pipe opening and pressure. Tables 5-2, 5-3, and 5-4 can assist with this. The flow rate of larger leaks (that may reduce pressure significantly) can be derived from changes in the net water demand on the system, increased pumping output, and/or increased minimum-hour flows

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Mains - Drop Down Menu Sources For Macro					*Note: This page is linked to the macro to fill the drop down lists on the other worksheets*			WRF Project 4372
Failure Event Type	Network Category of Failure	Failure Event was reported/detected by:	Nature of Failure	Piping Material at Location of Failure	Suspected Cause of Failure	Soil Condition at Location of Failure	How was failure repaired?	
Reported	Transmission	Leak Detection Crew	Circumferential Crack	Cast Iron	Corrosion	Compacted Sand	Replaced section of pipe	
Unreported	Distribution	Leak Detection Service	Longitudinal Crack	Ductile Iron	Thermal effects (cold, heat)	Clay	Repair Clamp	
		Customer	Blow Out	Steel	Poor materials	Ash/Corrosive	Plug	
		Meter Reader	Pin Hole	Pressuized Concete Cyliner Pipe	Poor installation	Rubbish/Poor Quality Fill	Welded a patch	
		Sewer Inspection	Split Bell/Joint	PVC	Excessive pressure/surges	Flowable Fill (Cementious Material)	Rehabilitated pipe joint	
		Other	Joint Leak	HDPE Pipe	Improper operation	Rock	Other Repair	
			Other	Asbestos Cement	Third party damage	Other		
				other	Other			

Service Connection - Drop Down Menu Sources For Macro							
Failure Event Type	Service Connection Ownership Where Service Leak Occurred	Failure Event was reported/detected by:	Nature of Failure	Piping Material at Location of Failure	Suspected Cause of Failure	Soil Condition at Location of Failure	How was failure repaired?
Reported	Utility	Leak Detection Crew	Circumferential Crack	Copper	Corrosion	Compacted Sand	New Service Connection
Unreported	Customer	Leak Detection Service	Longitudinal Crack	PVC	Thermal effects (cold, heat)	Clay	Replace Section
		Customer	Service Connection Saddle Leaking	Galvinized Iron	Poor materials	Ash/Corrosive	Fixed Meter Leak
		Meter Reader	Pin Hole	Lead	Poor installation	Rubbish/Poor Quality Fill	Other Repair
		Sewer Inspection	Meter Leak	other	Excessive pressure/surges	Flowable Fill (Cementious Material)	
		Other	Joint Leak		Improper operation	Rock	
			Other		Third party damage	Other	
					Other		

Mains Fittings - Drop Down Menu Sources For Macro							
Failure Event Type	Type of Appurtenance in Failure	Failure Event was reported/detected by:	Nature of Failure	Piping Material at Location of Failure	Suspected Cause of Failure	Soil Condition at Location of Failure	How was failure repaired?
Reported	Blow-off	Leak Detection Crew	blow-off left (partially) open		Corrosion	Compacted Sand	Plug
Unreported	Hydrant	Leak Detection Service	packing leak on valve		Thermal effects (cold, heat)	Clay	Welded a patch
	Air Valve/Tap	Customer	leak at tapping point		Poor materials	Ash/Corrosive	Replaced fitting
	Valve	Meter Reader	leak at hydrant base		Poor installation	Rubbish/Poor Quality Fill	Leak repair (repacked, etc)
		Sewer Inspection	hydrant stripped open and running		Excessive pressure/surges	Flowable Fill (Cementious Material)	Other Repair
		Other	packing leak		Improper operation	Rock	
			Other		Third party damage	Other	
					Other		

Figure 5-14 WRF Project 4372-b, *Leak Repair Data Collection Guide* data selection options

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Water Main Failures			WRF Project 4372	
Water Distribution System Failure Tracking for:		*(Input Utility Name)*		
	Year or Period:			
		Unique ID# for failure or work order	Failure 1	Failure 2
Minimum Required Information	Failure Event Type	*Reported - from complaints Unreported - from proactive leak detection*		
	Network Category of Failure	*Distribution Systems Transmission System*		
	General Location of Failure Event	*For Example - Street Intersection*		
	Size Information	*Size of Main At Failure Location*		
	Failure Event Reported	*Date*		
		Time		
	Failure Event Pinpointed	*Date*		
		Time		
	Failure Event Contained/Valved-off/Repaired	*Date*		
		Time		
Additional Information for Reliable Leakage Component Analysis	Detailed Failure Event Information	*Failure Event was reported/detected by:*		
		Nature of failure		
		Piping Material at location of failure		
		Age of piping at location of failure		
		Average Pressure at Failure Location		
		Suspected cause of failure		
	Detailed Location Description for Failure	*Street Address*		
		Nearest House Number		
		GIS Coordinates (X)		
		GIS Coordinates (Y)		
	Additional Failure Event Information	*Soil condition at location of failure*		
		How was failure repaired?		
		Estimated Cost to repair failure (labor/materials/equipment/restoration of excavation and pavements)		
		Comments		
		Estimated Leak Flow Rate (using AWWA M36 recommended flow rates)		
		Estimated Leak Flow Rate (using utility specific estimations/measurements)		

Figure 5-15 WRF Project 4372-b, *Leak Repair Data Collection Guide* water main failures data worksheet

in DMAs. The calculation may need to factor variable flow rates over the duration that the leak is running.

In the LCA model, reported failures (breaks and larger leaks) are categorized as those occurring on the water main piping, on customer service piping, and mains fittings or appurtenances including fire hydrants, valves, and air valves. The "Reported Failures" section of the LCA model features a standardized calculation of the real loss volume stemming from failure events reported to the utility. An example of the calculation of the annual value of reported leakage for CWC is given in the "CWC: Part 1" sidebar.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

CWC: Part 1. Conducting an LCA to Quantify Individual Leak Types and Evaluate Economic Intervention Frequency

From the water audit data shown in Figures 2-5 and 2-7 in Chapter 2, CWC has CARL of 280.565 mil gal for the 2020 audit year. Its UARL is 83.7 mil gal, the theoretical low level of leakage that could be achieved.

For the CWC distribution system, it is likely that reductions can be achieved using all four approaches to leakage reduction (see Figure 5-13). To assist in targeting priorities, CWC performed an LCA utilizing the LCA model as follows.

Step 1. Quantify the current reported leakage. During the 2020 audit year, CWC experienced 185 reported failures (break/leak events). These events were recorded by CWC's work order management system. Utilizing the LCA model, the total volume of real losses stemming from these reported failures was calculated.

The reported failures are shown in Figures 5-16, 5-17, and 5-18 (graphics from the LCA model software tool): 80 failures on water mains, 3 failures on main fittings, and 102 failures on service connections. With average durations for ALR/shutoff of each failure approximated and entered, the LCA Model then calculated the total volume of real losses stemming from CWC's reported failure events.

For CWC in 2020 reported failures on mains accounted for 11.57 million gal (MG) (Figure 5-16), reported failures on mains fittings accounted for 0.43 MG (Figure 5-17), and reported failures on service connections accounted for 6.89 MG (Figure 5-18). Summing up these three volumes gives the total reported leakage of 18.89 MG for CWC in 2020, which is a small fraction of the CARL volume of 280.565 MG as calculated by the annual water audit (see Figures 2-5 and 2-7).

Mains by Size	Number of Leaks & Failures per Year	Length of Main (miles)	Failure Frequency (number / 100miles / yr)	Average Failure Flow Rate @ 70psi (gpm)	Average Pressure (psi)	N1 (Leakage-Pressure Exponent) Value	Average Failure Duration: Average Awareness Duration (days)	Average Failure Duration: Average Duration for Location and Repair/Shutoff (days)	Average Failure Duration: Total Duration (days)	Average Annual Loss per Failure (MG)	Total Annual Loss (MG)		Use Default Average Failure Flow Rate	Use Utility Specific Average Failure Flow Rate	Average Failure Flow Rate @ Average System Pressure (gpm)
Diameter 2"	1	-	-	13.90	65.0	1.00	1.00	1.00	2.00	0.04	0.04		⊙	○	-
Diameter 3"	-	-	-	13.90	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 4"	34	-	-	44.00	65.0	1.00	1.00	2.00	3.00	0.18	8.00		⊙	○	-
Diameter 6"	39	-	-	92.00	65.0	1.00	0.50	0.50	1.00	0.12	4.80		⊙	○	-
Diameter 8"	5	-	-	92.00	65.0	1.00	0.50	0.50	1.00	0.12	0.62		⊙	○	-
Diameter 10"	1	-	-	92.00	65.0	1.00	0.50	0.50	1.00	0.12	0.12		⊙	○	-
Diameter 12"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 14"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 16"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 18"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 20"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 24"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 30"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 36"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 42"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 48"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 54"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter 60"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Diameter >60"	-	-	-	222.00	65.0	1.00	-	-	-	-	-		⊙	○	-
Other Diameter	-	-	-	-	65.0	1.00	-	-	-	-	-		Enter avg flow rate		-
										SUB-TOTAL REPORTED FAILURES ON MAINS	11.57				

Figure 5-16 LCA model: leakage losses stemming from reported failures on primary water main piping in CWC (WRF 2014a)

System Appurtenances by Type	Number of Failures per Year	Total Number of Appurtenances	Failure Frequency (Number of Failures per 1000 Appurtenances)	Average Failure Flow Rate @ 70psi (gpm)	Average Pressure (psi)	N1 (Leakage-Pressure Exponent) Value	Average Failure Duration: Average Awareness Duration (days)	Average Failure Duration: Average Duration for Location and Repair/Shutoff Failure (days)	Average Failure Duration: Total Duration (days)	Average Annual Loss per Failure (MG)	Total Annual Loss (MG)		Use Default Average Failure Flow Rate	Use Utility Specific Average Failure Flow Rate	Average Failure Flow Rate @ Average System Pressure (gpm)
Hydrants	1	2,123	0	3.50	65.0	1.00	5.00	30.86	35.86	0.17	0.17		⊙	○	3.8
Valves	2	3,345	1	6.90	65.0	1.00	5.00	9.62	14.62	0.13	0.26		⊙	○	7.4
Meters	-	-	-	0.25	65.0	1.00	-	-	-	-	-		⊙	○	-
Other (e.g. Blow-offs, etc.)	-	-	-	-	65.0	1.00	-	-	-	-	-		Enter avg flow rate		-
										SUB-TOTAL REPORTED LEAKS ON SYSTEM APPURTENANCES	0.43				

Figure 5-17 LCA model: Leakage losses stemming from reported failures on mains fittings (fire hydrants, valves, meter, etc.) in CWC (WRF 2014a)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

STEP 3C: REPORTED FAILURES ON SERVICE CONNECTIONS

Instructions: Enter the total number of failures by service connection size (<1" or >=1") that were reported to and repaired the water utility during the audit period in cells C98 to C99. Next enter the total number of service connections for each service connection size group in cells D98 to D99 (Note: this information is not required for the real loss component analysis but serves to calculate failure frequencies by service connection size group). Next enter the average awareness duration for reported failure events by service connection size group in cells I98 to I99 (Awareness duration is the average time from the occurrence of the failure event until the water utility becomes aware of its existence). Next enter the average duration of locating and containing reported failure events by service connection size group in cells K89 to K99. Next select either the default average leak flow rate provided by the model in cells P98 to P99 or select and enter your own utility specific average leakage flow rates in cells Q98 to R99. Average failure flow rates are based on AWWA M36 recommended flow rates and flow rates collected by the research team.

Service Connections by Size	Number of Failures per Year	Total Number of Service Connections	Failure Frequency Number of Failures per 1000 Service Connections	Average Failure Flow Rate @ Avergage System Presure (gpm)	Average Pressure (psi)	N1 (Leakage-Pressure Exponent) Value	Average Failure Duration			Average Annual Loss per Failure (MG)	Total Annual Loss (MG)		Use Default Average Failure Flow Rate	Use Utility Specific Average Failure Flow Rate	Average Failure Flow Rate @ average system pressure (gpm)
							Average Awareness Duration (days)	Average Duration for Location and Repair/Shutoff Failure (days)	Total Duration (days)						
Services < 1"	-	-	-	6.90	65.0	1.00	-	-	-	-	-		⊙	○	-
Services >= 1"	102	-	-	13.40	65.0	1.00	1.00	2.50	3.50	0.07	6.89		○	⊙	13.4
										SUB-TOTAL REPORTED LEAKS ON SERVICE CONNECTIONS	6.89				
										GRAND TOTAL REPORTED FAILURES	18.89				

Figure 5-18 LCA model: Leakage losses stemming from reported failures on customer service connections in CWC (WRF 2014a)

Step 2. Quantify known unreported leakage. **T**he annual volume of unreported leakage can be assessed by adding the products of the flow rate of the unreported leaks found at the time of repair, as estimated using Tables 5-2, 5-3, or 5-4 and the known or estimated run time of each leak. The leakage volume of an individual failure can be increased from the estimate at repair if it is suspected that the leak "grew" significantly between the start time of detection and time of repair. Annual unreported leakage is calculated using the same principles used for calculating reported leakage. In the LCA model, when the utility does not undertake any proactive leak detection, the total volume of leakage stemming from unreported leaks is recorded as zero. This does not mean that the system is free of unreported leakage, just that there was no unreported leakage documented. Since CWC did not undertake any proactive leak detection during the 2020 water audit year, the annual volume of unreported leakage in this example is listed as zero. For water utilities with an active leak detection program, their data on the number, types, and flow rates of unreported leaks detected can be input into the LCA model. When one leakage reduction approach is being replaced by another, it is important to be consistent using total savings and cost or net change in savings and cost.

Step 3. Estimate the unavoidable background leakage and target background leakage. This is done in two tasks, detailed below:

- Task 1: Calculate the unavoidable background leakage (UBL) for the system. Equation 5-6 calculates the UBL (Lambert 2009).

$$UBL\left(thous\ gal/d\right) = \left[\left(0.20*Lm\right) + \left(0.008*Nc\right) + \left(0.34*L_c\right)\right] \times \left(Pav/70\right)^{N1} \quad \text{(Eq 5-6)}$$

Where:

Lm = total length of water mains (miles)
Nc = number of service connections (main to curb stop)
L_c = total length of private service connection piping owned by the customer; curb stop to customer meter (converted to miles) = Nc*Lp (see Chapter 2 Figures 2-11 to 2-13)
Pav = average system pressure (psi)

Equation 5.6 quantifies a minimal level of background leakage estimated to be "unavoidable" in the context of the significant (and likely not cost-effective) efforts needed to reduce below this level. The example unavoidable background leakage (UBL) calculation for CWC is shown in the Part 2 sidebar. The UBL value is indicative of well-maintained infrastructure, subject to intensive and efficient active leakage control. It serves as a point

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

of reference for calculating the target background leakage (TBL) level, as discussed in the text that follows.

CWC: Part 2. Conducting an LCA to Quantify Individual Leak Types and Evaluate Economic Intervention Frequency

For CWC, the UBL can be determined by using established values for three distribution system components: leaks on water mains, service connection sections (utility responsibility), and service connection sections (customer responsibility). See Tables 2-21 and 2-22 in Chapter 2 for the derivation of these values. Note: the standard values are set at 70 psi, so these values must be pressure corrected for CWC's average pressure level of 65 psi. The N1 exponent for equation 5.6 is taken as 1.5, reflecting that leakage rates are somewhat more likely to increase with increasing pressure (variable path) than a fixed path with an N1 of 1.0. (See the FAVAD discussion in this chapter.)

UBL (thousand gal/d) = $[(0.20 * Lm) + (0.008 * Nc) + (0.34 * Lc)] * (Pav/70)^{1.5}$ (5.6)

$= [(0.2 * 256) + (0.008 * 12{,}196) + (0.34 * 41.6)] \times (65/70)^{1.5}$

$= [51.2 + 97.6 + 14.1] \times (0.9286)^{1.5}$

$= 162.9 \times 0.895 = 145.8$ (thousand gal/d) or 53.2 mil gal/year

Note: Lc = [(12,196) × (18)]/5,280 feet/mile = 41.6 miles

- Task 2: Calculate the TBL for the utility by using a multiplier called the infrastructure condition factor (ICF). This factor is needed because not all systems have infrastructure that meets the assumed criteria for the UBL Equation 5-6. Different systems have varying characteristics including age, makeup of pipe materials and fittings, and pressure variability. The ICF expresses the ratio of the TBL to the UBL. The TBL is estimated by multiplying the calculated UBL by the ICF, where:

$$TBL\left(thousand\ gal/d\right) = ICF \times UBL \quad \text{(Eq 5-7)}$$

$$ICF = TBL/UBL \quad \text{(Eq 5-8)}$$

The ICF can be estimated or calculated in several ways. Methods include

1) measuring minimum hour flows in DMAs after completing comprehensive leak detection and repair;

2) performing a pressure step test by incrementally restricting flow to a DMA or discrete zone, measuring the flow and pressure reductions, and using these data as inputs to specific calculations;

3) using sensitivity analysis by estimating the best case/worst case values of the ICF from LCA or use the average; or

4) assuming that ICF will be equal to the ILI (as described in the Part 3 sidebar), an approximate method that is straightforward but may be less accurate (WRF 2007b).

CWC: Part 3. Conducting an LCA to Quantify Individual Leakage Types and Evaluate Economic Intervention Frequency

CWC lacks sufficient leakage repair data to use the refined methods of estimating its ICF to quantify background leakage. CWC, therefore, estimated the ICF as equal to the ILI. In CWC's case, the target ILI is assumed to be **4.0 or the next unit increment above the CWC ILI of 3.4 from their annual water audit and reported in Figure 2.7 of Chapter 2.**

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The LCA model provides a method of calculating the target/current background leakage volume for any given utility. The user can choose to select an ICF value based on average infrastructure age/condition or use one of the ICF estimation methods explained earlier. By basing its ICF value close to the ILI value, CWC's resulting total volume of background leakage for CWC is 210.93 MG as calculated by the LCA model (see Figure 5-19). CWC must update the ICF value as more leak detection and repair data become available or DMAs are established and measured minimum hour flow data are available to obtain a more precise ICF value.

If a utility is estimating the ICF, it could perform an informal comparison of the ICF factors involved in comparison to a typical utility. Categories and related questions and concerns could include the following:

- pipe components: Are the utility's water mains likely to have more small leaks as with corrosion of metallic pipe? The utility side service connection leaks are based upon the number of connections, so roadways wider than average could increase service line leakage. Newer systems may have issues such as pipes made of certain leaking plastics (e.g., polybutylene service lines) or atypical poor joint fusion with polyethylene (PE) pipe, complicated by the difficulty of detecting leaks on those materials.
- pressure component: Is the pressure so high and mains so shallow that even small failures will grow quickly and surface, decreasing the ICF? Or is the pressure low and pipe so deep that detecting leaks acoustically is more challenging, increasing the ICF?

The background leakage calculation totals for CWC are shown in Figure 5-19 for water mains and utility side and customer side service connections. As listed, CWC incurred 210.93 MG of background leakage in its system in 2020.

Step 4. Summarize the LCA model results. Calculating the targeted leakage requires subtracting the reported, unreported, and background leakage as detailed above from the water audit's calculated CARL, excluding reservoir leakage. It may be possible to quantify these factors between mains (with appurtenances) and service lines by assuming the ratio of reported leak volumes applies to the ratio of unreported and background leakage. For initial goal setting, the potential annual savings would be approximated as the product of the targeted leakage reduction volume and the variable production cost of water. Figure 5-20 shows the summary table of the real (leakage) loss component analysis, giving the structure and steps in the calculations of the leakage components. Figure 5-21 shows the summary table for CWC from the LCA model.

Step 2: BACKGROUND LEAKAGE ON MAINS AND SERVICES

Instructions: The background leakage volume from mains and service connections will be automatically calculated after selecting one of the default Infrastructure Condition Factors (ICF) based on infrastructure age or after a utility specific ICF has been selected and entered in cell P43. The level of background leakage tends to increase with infrastructure age and is higher for systems with higher pressure. The volume of background losses is generally a significant component of the total real loss volume, at times up to 50% of the total volume of real losses. Overestimating (ICF used in model is too high) or underestimating (ICF used in model is too low) the background leakage volume might lead to a real loss control strategy that is not truly optimized. Selecting the ICF based on general age of the distribution network should only be seen as a starting point. It is strongly recommended that over time a utility specific ICF is assessed and confirmed through field tests using one of the recommended approaches outlined in the AWWA M36 manual.

						Pressure Corrected Calculation	
System Component	Units	Quantity	Component UARL (g/unit/d/psi)	Infrastructure Condition Factor	Average Pressure (psi)	N1 (Leakage-Pressure Exponent) Value	Annual Volume (MG)
Mains	miles	256.3	2.87	4.0	65.0	1.5	67.45
Services - Main to Curb-Stop	number	12,196	0.112	4.0	65.0	1.5	125.26
Services - Curb-Stop to Meter	miles	41.6	4.78	4.0	65.0	1.5	18.22
					Total Background Leakage on Mains and Services		**210.93**

Select Infrastructure Condition Factor (ICF) based on age of distribution network

○ ICF 1 for System Age <50 years	1.0
○ ICF 1.5 for System Age 50 - 70 years	1.5
○ ICF 2.5 for System Age >70 years	2.5
OR	
⊙ Enter ICF values assessed through implementing the approaches outlined in the AWWA M36 manual	4.0

Figure 5-19 LCA model: background leakage volume calculation for CWC (WRF 2014a)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

System Component	Background Leakage (MGY)	Reported Failures (MGY)	Unreported Failures (MGY)	Total Leakage (MGY)	
Reservoirs	-	-	-	-	
Mains and appurtenances	step 3	step1	step 2	sum step 1-3	A
Service connections	step 3	step1	step 2	sum step 1-3	B
Total annual real loss	step 3	step1 A+B	step 2 A+B	A + B	C
Real losses as calculated by the water audit				water audit less reservoir leakage	D
Hidden losses/unreported leakage currently running undetected				D - C	

Figure 5-20 LCA model summary table showing the steps used to calculate the component totals

System Component	Background Leakage (MG)	Reported Failures (MG)	Unreported Failures (MG)	Total (MG)
Reservoirs	-	-	-	-
Mains and Appurtenances	67.45	12.00	-	**79.45**
Service Connections	143.48	6.89	-	**150.37**
Total Annual Real Loss	**210.93**	**18.89**	**-**	**229.82**
Real Losses as Calculated by Water Audit				**280.56**
Hidden Losses/Unreported Leakage Currently Running Undetected				**50.74**

Figure 5-21 LCA model: component analysis summary CWC (WRF 2014a)

CWC: Part 4. Conducting an LCA to Quantify Individual Leakage Types and Evaluate Economic Intervention Frequency

LCA Summary

The results of the LCA for CWC are summarized by the LCA model in a worksheet as shown in Figure 5-21. Out of the total real loss volume of 280.565 MG as calculated by the annual water audit (Chapter 2 Figures 2-5 and 2-7), the LCA reveals that about 210.93 MG occurs as background leakage and about 18.89 MG occurs from reported failures. (Reported leakage includes 11.57 MG on mains, 0.43 MG on mains fittings, and 6.89 MG on service connections.) The LCA summary results shown in Figure 5-21 indicate that about 50.745 MG are due to unreported failures that are currently running undetected. This total illustrates that, while reported leaks can be disruptive and garner much attention, they normally do not account for a major portion of the annual volume of real loss since they are quickly detected and repaired. Hidden unreported leaks that are left to run continuously and undetectable background leakage often account for a major portion of the leakage losses in many water distribution systems, and this is the case for CWC.

THE ECONOMICS OF LEAKAGE MANAGEMENT

Leakage represents inefficiency in a water utility's system that delivers water to its customers. Although limits exist below which leakage recovery is not cost-effective, many North American water utilities are likely operating with excessive levels of leakage that are cost-effective to recover. The challenge is to balance the expense of leakage control

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

activities against the cost of distribution system failures (leaks and main breaks) that can be mitigated or prevented.

Previously, we summarized the impacts of leakage, and many can relate to the tangible and intangible costs depending upon the specific leakage control remedy. In North America, real losses are typically valued at the short-term variable production cost to treat (chemical costs) and deliver (pumping power costs) the water. Other production costs such as water allocation fees, treatment residuals management, and mechanical equipment wear-and-tear may also be considered if significant. For utilities that purchase bulk water from wholesale water suppliers, which is typically more expensive, the unit purchase rate is used as the variable production cost.

The cost-effectiveness of the leakage management program should also consider long-term impacts. If the water utility's supply infrastructure is close to its capacity in meeting daily supply needs, leakage reduction may provide the added benefit of postponing expansions to a water treatment plant or pumping infrastructure. Such costs could be considerable and justify more significant leakage reductions than those suggested merely by short-term, direct variable production costs. A water supplier's capital improvement plan should be reviewed for scheduled investments in capacity expansion. Some of these projects might be deferred or downsized if leakage reductions are targeted as an alternative. When the scheduled and deferred investment scenarios are compared on a present-value basis, the difference can be added to the value of leakage reduction.

Indirect cost impacts vary from utility to utility, with unique issues possible in any water system. Social and economic costs like environmental damage, interruption of commerce and transportation, and limitations on regional economic growth may be difficult to quantify but may be relevant to the community. Reducing leakage significantly could mean lower withdrawals from a river that could improve instream flows, benefiting aquatic life, recreation (boating, fishing), or economic development (industrial, residential, waterfront amenities, etc.). There is increased recognition of the benefits in avoiding or mitigating disruptive main breaks that have high consequences in terms of damage and disruption.

Leakage costs can also vary with time and conditions. If a water shortage develops due to drought or other reasons, the unit savings from leakage reduction will likewise increase as the relative scarcity of the water increases. If water resources are severely limited, with strict water conservation measures in place and high rates charged to customers, it may be appropriate to value leakage at the customer retail unit charge, which makes the unit cost of leakage very high. In such cases, it can be argued that volumes of water from recovered leakage can be sold to current or future customers; therefore, the retail rate applies.

In some cases, especially where advanced leakage monitoring allows many leaks to be identified and repaired close to the time that they first become detectable, reduction of repair and restoration costs can be studied to determine if these savings are significant. Very early detection might prevent making a later repair that requires more extensive excavation and materials along with extra staff overtime hours under emergency conditions.

The Economic Level of Leakage (ELL)

The ELL identifies the amount of leakage that cannot be avoided without investing more in leakage management techniques than is recovered in further reduction of leakage. In short, it does not pay to spend another dollar on leakage management if you cannot save that dollar or more. The ELL can be used to set the economically ideal target level of leakage for the water utility. The ELL is typically based upon the short-term, direct costs of leakage but could include long-term, indirect costs if these are pertinent. The timeframe should correspond to the expected life of the leak detection methodologies. Approaches to real loss control should be developed with the ELL in mind. With an effective leakage management

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

program in place, reducing leakage eventually becomes an effort of diminishing returns; reducing more leakage usually costs increasingly more to achieve smaller reductions. When losses are excessive, relatively large reductions can occur early in a loss control program at relatively little cost. However, further loss reduction requires ever greater cost and effort to recoup ever smaller returns.

Determining the breakpoint beyond which the effort to control the losses costs more than the value of the recoveries is somewhat complex, owing to the variety of technologies that vary in cost and effectiveness. If a water utility has only just begun to audit its operations and has not employed active leakage control methods, significant leakage losses may exist, and it will be economically feasible to recover a notable portion of those losses. Initial leakage controls will likely generate considerable savings. A water utility should then project the additional loss recovery of further leakage controls, especially if initial programs (increased leak detection, repair procedures, pressure optimization, etc.) are successful in removing large flow rate leaks, with only lower flow rate and more elusive leaks remaining.

Several approaches have been developed to calculate the ELL. An example calculation of the ELL is not provided here, but the ELL and modified means of setting leakage reduction targets are discussed in the WRF report *Guidance on Implementing an Effective Water Loss Control Plan* (WRF 2019). Work continues within the water industry to analyze economic leakage control by improved leak detection and pressure management options that increase benefits in the reduction of leakage levels. Essentially, cost commitment for the array of approaches to leakage control compels utilities to evaluate approaches on a case-by-case basis. Regardless of the analysis, several basic elements are common to calculating the ELL.

The ELL is dependent on the incremental cost of water as well as the differential cost to pinpoint leaks. The higher the cost of leaking water, the lower the calculated ELL and the greater the level of leakage that will be economic to reduce. The more prone the system is to have unreported leaks, the more intense the leakage management effort is likely to be. Once initiated, an effective program that identifies unreported leaks usually means that more repairs will be made. The additional repair costs may be offset by the reduction of costs per repair and reduction of damage by repairing leaks when they are smaller and of shorter duration. Many leaks identified early are leaks that ultimately are repaired in the near term, avoiding higher costs for overtime and a higher risk of safety for workers and the public for higher volume reported failures that occur later.

It should be noted that, in some jurisdictions, water loss targets may be established by regulatory agencies. These regulatory targets may or may not match what the water utility management believes to be economical to achieve. In some cases, knowing its ELL may help the utility as it negotiates with the regulator to designate a leakage target that is technically feasible to meet while still economical for the utility.

Utilities that have not calculated their ELL but have completed a water audit using the AWWA audit software can use the KPIs found on the dashboard worksheet of the software to compare their utility with the validated data displayed on the "speedometer" charts of the dashboard. Incremental leakage reduction targets might be set against high percentile real loss levels shown in the charts.

LEAK DETECTION INTERVENTIONS

Leak Surveys

In addition to knowing how leak detection works (detailed in Chapter 6), it is useful to assemble a cost-effective means to define the size and functions of the traditional leak detection survey program. The costs to create an in-house leak detection program with

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

staffing or to contract leak detection services are a worthwhile investment but should be planned carefully. There is value in staff having improved knowledge about leakage detection. Increased use of leak detection technology may also help shorten the repair response time as leak pinpointing skills are likely to improve. Increased knowledge about unreported leaks may lead to improvements in selecting superior pipe materials for a replacement to inhibit leakage from occurring.

Most water loss financial models, including the ELL approach and the benefit–cost ratio approach (discussed later in this chapter), rely on the same assumptions regarding targeted leak detection work or a leak detection intervention. These assumptions are listed below.

1. Theoretically, an effective leak detection survey intervention should abate most unreported leaks and bring leakage down to the level where only background leakage remains. It should be noted that the intervention may run for an extended time as initial leak repairs may result in elevated system pressures causing more stress on pipes and a second wave of leaks to emerge, with follow-up efforts needed to detect, pinpoint, and repair them.
2. After full intervention as described above, the leakage rate is assumed to increase at a constant rate between interventions as new, unreported leaks gradually emerge. This is the rate of rise (RR) of leakage.
3. The cost of a leak detection intervention is assumed to be the same, regardless of the leakage level.
4. Shortening the leak detection interval between interventions means more leakage found and abated but more money spent on reducing that leakage. Lengthening the leak detection interval means less money spent on leak detection but more money spent treating and pumping water that ends up wasted as leakage. Designing a cost-effective program means finding the right balance point of spending for leakage reduction and savings from leakage abatement, essentially determining an economic level of leak detection.

Economic Intervention Frequency for Leak Detection Surveys

Active leakage control provides for proactive surveillance of unreported leaks that can be discovered through leak detection techniques detailed in this chapter and Chapter 6. In North America, acoustic leak detection is carried out by many water utilities that schedule crews to perform periodic leak soundings (leak surveys) of water distribution system infrastructure or placement of continuous acoustic monitors to identify unreported leaks.

The work to create a methodology and equations to calculate the economic intervention frequency (EIF) of leak surveys was published by Lambert et al. in several technical papers, including the paper by Lambert and Fantozzi (2005) titled "Recent Advances in Calculating Economic Intervention Frequency for Active Leakage Control, and Implications for Calculation of Economic Leakage Levels." This work is included in the following discussion. A preliminary schedule for acoustic leak detection surveys or zone measurement, including an appropriate annual budget for active leakage control interventions, can be assessed using three parameters:

1. The variable cost of real losses, CV (in dollars per thousand gallons, or $/thous gal). The utility derives this from the VPC of the water audit, with the conversion from MG to thous-gal units.
2. The cost of a leak detection survey intervention, CI (in dollars per mile of mains). Note that the repair costs have been excluded to keep the economic assessment simple and consistent with the leak survey frequency approach used in earlier

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

M36 editions.

3. The average change or RR of unreported leakage (thousand gal/mi of mains/d/year). NOTE: These units are more fully interpreted as "thousand gallons per day, per mile of mains, per year."

If no leak detection work has been conducted previously, an approximate estimate of the RR can be calculated from the rise in real losses from annual water audits conducted over several years. If the utility has previously conducted leak detection and repair work, the RR can be calculated using the number of leaks and flow rates of unreported breaks and leaks found, divided by the time between leak surveys.

The appropriate level of intervention to control unreported leakage can be evaluated in terms of determining the frequency at which leak surveys are economically effective. This is proportional to the square root of the CI and inversely proportional to the square root of the CV. With this relationship, the lower the cost of the leak detection survey work and the higher the cost of water, the shorter the survey frequency could be. It is also inversely proportional to the square root of the RR of unreported leakage. So, the more rapid the increase in unreported leakage between surveys, the shorter the survey EIF.

Changes in leakage due to leak detection interventions. When planning a leak detection program, it is important to project how leakage levels will reduce due to successful leak detection interventions or grow due to the lack of proactive leak detection. Periodic rounds of leak detection survey work throughout a distribution system should produce a reduction in leakage, as illustrated generally in the schematic graphic of Figure 5-22 with intervening times where unreported leakage levels grow over time represented by the increasing diagonal slope in the sawtooth triangles. Theoretically, unreported leakage can be brought down again to the background level of leakage after an effective leak detection survey and leak repairs, and this is shown in Figure 5-22 by the vertical line of the sawtooth triangles. However, in reality, many leak detection surveys do not remove all unreported leakage and reach the background leakage level, instead, reaching an "achievable" level of leakage based upon the quality and execution of the leak detection work. Still, for illustrative purposes, leakage reduction is taken as reaching the background leakage level in Figure 5-22.

It should also be noted that a constant RR of leakage may be an oversimplification of actual system performance. Numerous simultaneous breaks cause a drop in pressure and repairing many in close proximity may cause increased system pressure that can trigger an immediate rapid surge of leakage. After repairs, the leakage flows subsequently slow down, assuming the initial backlog of leakage is resolved. Pressure monitoring in areas of leak repairs initiated before and after repairs may be a prudent step to take, especially in areas that have not undergone leak surveys in the past. The methodology described below also assumes that leakage again rises to the previous level of leakage at a regular interval, but this pattern may not hold due to system changes and the effectiveness of other leakage control measures.

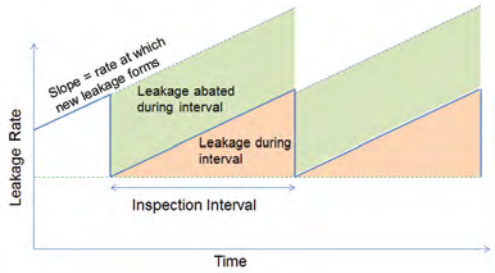


Figure 5-22 RR of leakage and changing leakage levels with an active leak detection program

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Under this sawtooth model shown in Figure 5-22, two quantities of leakage volumes during the leak detection interval exist:

1. The water volume that occurs as leakage during interval due to the emergence of detectable leaks during each cycle is the orange area of the triangles in Figure 5-22 across the inspection interval and can be calculated as ½ the base of the triangle multiplied by the height (leakage eliminated) of the triangle:

$$leakage\ during\ interval = ½\ inspection\ interval \times leakage\ eliminated \quad \text{(Eq 5-9)}$$

2. The leakage abated during the interval is the water volume that would have been lost due to detectable leaks during this cycle had the leak detection work at the beginning of the cycle not occurred. This volume of leakage abated during the interval is the amount of water that was recovered by this intervention and is the green area of the parallelograms in Figure 5-22 which can be calculated as the base times the height:

$$leakage\ abated\ during\ interval = inspection\ interval \times leakage\ eliminated \quad \text{(Eq 5-10)}$$

Although not shown here, the calculations can be expressed in costs per year and then solved to calculate the leak detection inspection interval at which these total costs per year are minimized, which is the EIF. To make this equation usable in a practical manner, particular units need to be applied to each variable and a unit conversion coefficient applied. This can be expressed mathematically as

$$EIF\ \ (months) = \left[0.789 * (CI / CV) / RR\right]^{0.5} \quad \text{(Eq 5-11)}$$

Where:

EIF = economic intervention frequency (measured in months)
CI: leak Intervention costs (measured in $/mile of mains surveyed for leakage)
CV: water value (measured in $/thous gal)
RR: rate of rise (measured in thous gal/day, per mile of mains, per year)

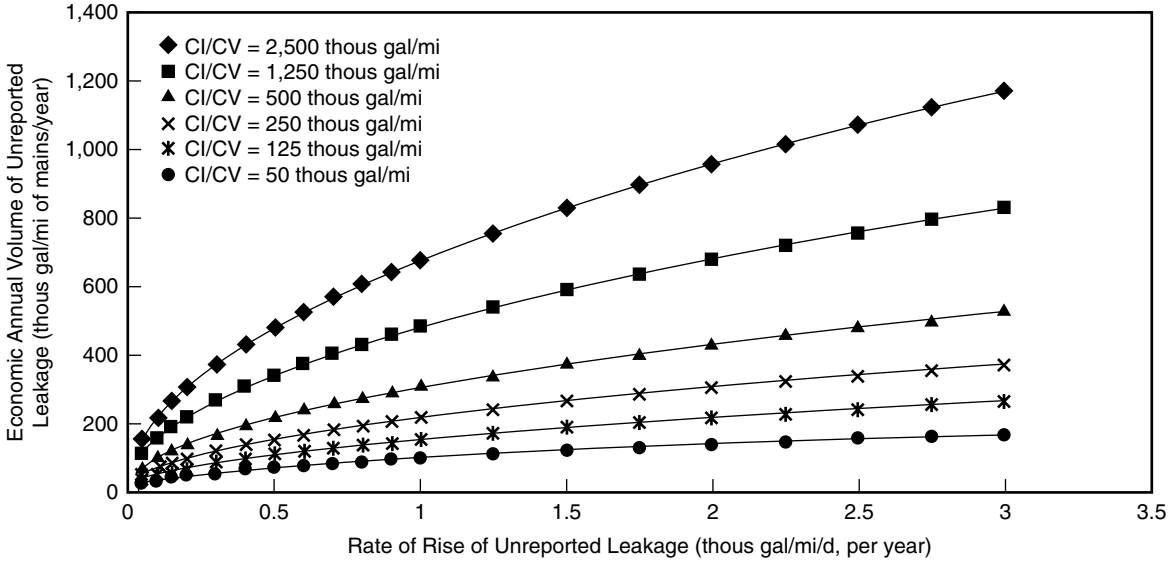
Running interventions less often than the EIF means that the utility is spending excessive money from water lost to leakage. Conversely, running interventions more often than the EIF indicates that the utility is spending more money on interventions than it can recover in reduced leakage.

The percentage of the system to be inspected annually (assuming a continuous program), known as EP%, is the inverse of the EIF expressed in years as follows:

$$EP\%\ of\ the\ system\ to\ be\ surveyed\ annually = (100\% * 12) / EIF \quad \text{(Eq 5-12)}$$

The appropriate annual budget for intervention (ABI) is the cost to survey the entire system for unreported leakage multiplied by the economic percentage of the system to be surveyed:

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Veritec Consulting Inc. and ILMSS Ltd. (2005)

Figure 5-23 Economic unreported real losses for regular survey

$$average\ ABI = cost\ of\ leak\ survey\ (CI) * EP\%\ of\ the\ system\ to\ be\ surveyed \quad \text{(Eq 5-13)}$$

The calculated economic unreported leakage (EUL) expressed in units of thousand gallons is the ratio of the annual budget for leak detection intervention to the variable cost of real losses:

$$EUL = average\ annual\ budget\ for\ leak\ detection\ intervention / CV \quad \text{(Eq 5-14)}$$

Figure 5-23 shows the relationship between RR of unreported leakage(x-axis), different ratios of the intervention to water costs (CI/CV curved lines), and the economic annual volume of unreported real losses, EUL(y-axis). To use this graph, the RR(x-axis) should be estimated, CI/CV should be calculated to identify the appropriate curved line, and the EUL annual volume can then be read on the y-axis. Alternatively, the LCA model can be used to determine these parameters (Lambert and Lalonde, 2005).

The calculations for the example of CWC are shown in the "CWC: Part 5" sidebar.

CWC: Part 5. Conducting an LCA to Quantify Individual Leakage Types and Evaluate Economic Intervention Frequency

CWC calculated the economic intervention frequency (EIF) for active leak detection utilizing the LCA model, which employs Equation 5-11. The three parameters necessary to calculate the EIF for CWC (or any given system) are as follows:

1. The variable cost of real losses, CV (in dollars per thousand gallons, or $/thous gal).

CV = $0.75/thous gal

2. The cost of a leak detection survey intervention, CI ($/mi of mains, excluding repair costs). The current cost for the leakage survey—conducted once every five years—is $70,400. Dividing this cost by 256 miles in the CWC system gives a survey cost rate of $275 per mile of mains.)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

CI = $275/mile of mains

3. The average RR of unreported leakage (thous gal/mi of mains/d/year). CWC does not operate an ongoing active leakage control program; instead, it hires a leak detection contractor to survey the entire water distribution system for leakage once every five years. Between leak surveys, CWC responds reactively to reported leaks and breaks as they are called in. The unreported leaks and breaks that are typically found during the interventions every five years total around 1 MGD. The average flow rate of unreported leaks locatable by survey in the five years is taken as one half this five-year finding, which is 0.5 MGD or 183 million gal/year. The economic intervention equations 5-11 to 5-14 can now be used to check if the frequency of the five-year survey and repair interventions are economic and to calculate the economic intervention parameters for this type of survey. The implied average RR of 0.2 MGD each year is 200,000/(256*1,000) miles = 0.78 thousand gallons/mile of mains/d/year.

RR = 0.78 thous gal/mi of mains/d/year

Figure 5-24 is a summary list of the calculations of the LCA model and indicates that the economic frequency of leak survey and repair intervention should include 32% or about one-third, of the system each year. This will require three years to complete the entire system instead of the five-year interval currently used by CWC. It would be preferable to increase the active leakage control annual budget (excluding repair costs) from an equivalent of $14,080 per year ($70,400/5) to $22,788 per year, and target around one-third of the system to be surveyed for leakage each year.

The economic intervention equations demonstrate that, as the variable cost of water increases, the EIF and the economic unreported leakage volume will decrease, indicating that more frequent leak survey and repair interventions are justified. Likewise, if the annual budget for intervention and economic percentage both increase, the average run time of unreported leaks and breaks reduces and the total volume of real losses for the year decreases.

WaterRF 4372: Real Loss Component Analysis: A Tool for Economic Water Loss Control

Water Audit: County Water Company , New York, United States, 2020
CALCULATION OF ECONOMIC INTERVENTION FREQUENCY FOR PROACTIVE LEAK DETECTION

	System Characteristics		
	Total Length of Mains	256.3	miles
	Number of Service Connections	12,196	service connections
	Service Connection Density	47.6	conn./mile main
	Average System Pressure	65.0	PSI
	Water Balance Results		
TBL	Current Annual Background Leakage	210.93	MG/Yr
CRL	Real Losses from Current Reported Leakage	18.89	MG/Yr
UL	Unreported Failures Identified Through Existing Proactive Leak Detection Program	-	MG/Yr
	Hidden Failures/Unreported Failures not Identified or Captured by Current Leakage Management Policy	50.74	MG/Yr
CARL	Current Annual Real Losses	280.56	MG/Yr
UARL	Unavoidable Annual Real Losses	83.93	MG/Yr
	Infrastructure Leakage Index (ILI) [CARL/UARL]	3.3	
	Variable Cost of Real Losses		
CV	Variable Production cost (applied to Real Losses):	0.75	$/per kgal
		750.00	$/MG
CI	Cost of comprehensive leak detection survey (excluding leak repair cost)	275.00	$/per mile
		70,483	$/for entire system
RR	Average Rate of Rise of Unreported Leakage	0.21	kgal/mile of mains/day in a year
		0.05	MG/day in a year
	CI/CV	366.7	kgal/mile
EIF	Economic Intervention Frequency [0.789 * (CI/CV)/RR] ^0.5	37.1	months
		1,129.0	days
	Economic Intervention Frequency - Average Leak Run Time	564.5	days
	Economic Percentage of System to be Surveyed per Year	32	%
ABI	Average Annual Budget for Intervention (Proactive Leak Detection)	22,788	$/year
EUL	Economic Unreported Real Losses	30,383	kgal/year
		30.4	MG/year
	Economic Infrastructure Leakage Index (ILI)	3.1	
PRL	Potentially Recoverable Leakage (CARL-CRL-EUL-TBL-UL)	20.4	MG/year

Source: Water Research Foundation Project 4372a Leakage Component Analysis Model

Figure 5-24 Economic intervention frequency for leak detection by survey for CWC

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The final calculations for the example of CWC are shown in the Part 6 sidebar and include the determination of the economic volume of unreported leakage reduced by conducting leak detection at the EIF.

CWC: Part 6. Conducting an LCA to Quantify Individual Leakage Types and Evaluate Economic Intervention Frequency

Based on the results of the LCA and calculation of economic intervention frequency (EIF), CWC estimates that around 20.4 million gallons of leakage losses (Figure 5-24) could be potentially recovered by proactively surveying about one-third of the system every year.

Currently, CWC's leakage management practices are largely reactive and focus almost entirely on reported leaks and main breaks. However, the initial component analysis shown in the Part 1 sidebar finds that the current reported leakage is only 18.89 million gallons/year, which is just 6.7 percent of the CARL of 280.56 million gallons. As stated in the Part 5 sidebar, the unreported leaks and breaks that are typically found during the leak surveys every five years total around 1 MGD. The average flow rate of unreported leaks locatable by survey in the five years is taken as one half this five-year finding, which is 0.5 MGD or 183 mil gal/year. However, the EIF analysis (Figure 5-24) shows that the EUL should be reduced to around 30.4 mil gallons/year if the leakage survey frequency was increased to approximately once every three years. The analysis estimates TBL (assuming an infrastructure condition factor, ICF, of 4) as 210.93 million gallons or around 75 percent of the CARL of 280.56 MG. This leaves 20.4 mil gallons/year as the first estimate of potentially recoverable leakage or about 7.3 percent of the CARL.

The economic intervention analysis shows that more frequent leak surveys are justified. However, utilizing CAM or providing flow monitoring in discrete zones or DMAs would improve the targeting of areas for active leakage control interventions. Continuous flow monitoring in DMAs can shorten the awareness time for unreported leaks and reduce the annual losses from unreported leakage. A more structured and continual active leakage control program will enable CWC to reduce and sustain an economically lower level of leakage. Many good active leakage control options exist for CWC to evaluate.

The significant quantities of background leakage suggest that CWC could also benefit from the optimization of distribution system pressures (discussed in Chapter 6), which will reduce flow rates of all existing leaks (particularly background leakage), in addition to reducing the frequency of new leaks, RRs of unreported leakage, and repair costs. Several pressure management improvements are relatively inexpensive and can be feasibly implemented on a short-term basis. Other techniques require the installation of pressure-reducing valves and are considered medium- to long-term improvements. Upgrading the distribution system via pipeline replacement or rehabilitation will also reduce background leakage. This option, however, requires the longest time to implement. It is also the most expensive option and therefore should ideally be pursued when water distribution piping assets have reached the end of their useful services lives.

CAM

There is an alternative to traditional acoustic leak detection surveys that uses CAM as opposed to periodic searching for leakage. The calculations described above and illustrated in the CWC example reflect the traditional leak survey approach. A continuous approach to leak detection employs monitors that are positioned in the system to detect leakage acoustically every day. The sensors have been proven to detect leaks flowing as little as 1 gpm, alerting distribution staff at the very early stages of a leak. Unlike leak surveys that are generally performed by technicians during quiet hours (often nighttime) expecting to monitor locations once or twice during a deployment, the CAM monitors provide longer-term night-after-night continuous monitoring. The analysis performed by CAM systems provides the technicians with the alarms and supporting data.

CAM systems have a different economic basis for analysis as leaks may be identified whenever they become acoustically significant and can be repaired quickly thereafter. With monitors placed throughout an area, the location of the leak is more readily identified and the time between surveys to arrest leakage is days or weeks instead of months or years.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

CAM Implementation

Whether CAM sensors are placed on valves, hydrants, or service lines, reliable communication of data is essential. Either cellular or radio frequency transmission options are available with many systems. Some CAM systems use the same network to transmit AMI data to the communication network. Transmission is generally received by the vendor-operated cloud interface that is made accessible to the utility. Data are typically updated daily at the quietest volume in an overnight window.

Depending upon the sophistication of the network, the general information provided is a map of the system showing sensors with indications of proper working status, a point of interest if detected, a tabular summary, and a history of acoustic sound sometimes showing frequencies. The more advanced CAM systems can correlate leak noises detected by multiple sensors and provide a reasonable approximation of the location of the leak. Note that correlated service line leaks will be located at the point of the service line connection on the main, not the specific point along the service line. Correlation serves to reduce false positives as well as reduce the area of investigation around the point of interest significantly.

The two primary issues with CAM are their effective range of detection and periodic identification of nonleak noise (false positives). As described earlier, sound traveling along a pipe varies according to pipe size and pipe material. Plastic pipes (service lines and mains) are poor conductors of noise, and the spacing of sensors might have to be more frequent than a single unit per block. Several manufacturers have sensors that can detect noise in the water column that carries sound farther. However, having effective contact with the water directly through subsurface main connections or hydrants can be expensive. Most older pipe materials, cementitious and ferrous products, are good conductors of sound, and the density of monitors may range from 4 to 15 monitors per mile of water main depending on the sensor and the system configuration.

False positives, especially after initial installation, can slow the process of finding actual leaks. Typically, leak noise correlators are used to confirm that a leak exists and pinpoint its location. Commonly found conflicting noise sources during pinpointing efforts have included throttled valves, air releases, nearby pumps, electric transformer vibration, and continuous water use sources such as industrial facilities or irrigation systems. Some CAM systems can effectively obscure these nonleak noises and identify actual leaks that occur in these areas. Locations of nonleak noises can be noted in the CAM software.

CAM Cost-Effectiveness

As with any technology, the expected costs and benefits of CAM systems should be evaluated before investing in such a system. Costs include upfront costs for sensors and the communication system and ongoing costs for staff assigned to monitor and evaluate the information provided by the CAM and perform upkeep. Costs and benefits should be assessed across a representative lifecycle that includes sensor replacement on a periodic basis, as well as software upgrades and other recurring system upkeep costs.

The benefits expected of the CAM can be projected before investing in the system. Since the onset of CAM installations, the benefits of early detection and repair of leaks have been demonstrated. In practice, an effective CAM system can minimize the time that a leak is categorized as unreported. This capability not only curtails leakage but has demonstrated that damage caused by leaks running longer and emergency repairs are reduced, which cuts overtime costs and improves working conditions for crews.

These types of benefits are demonstrated in a leak from an excavated water pipeline pictured in Figure 5-25. The excavated cast-iron pipe shows a crack propagating from the bell that has started to split as it traveled down the pipe. The crack was on the invert of the pipe; leaking water never surfaced, and the trench was dry when the top of the pipe

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Echologics

Figure 5-25 Excavated break detected by correlating acoustic monitoring

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

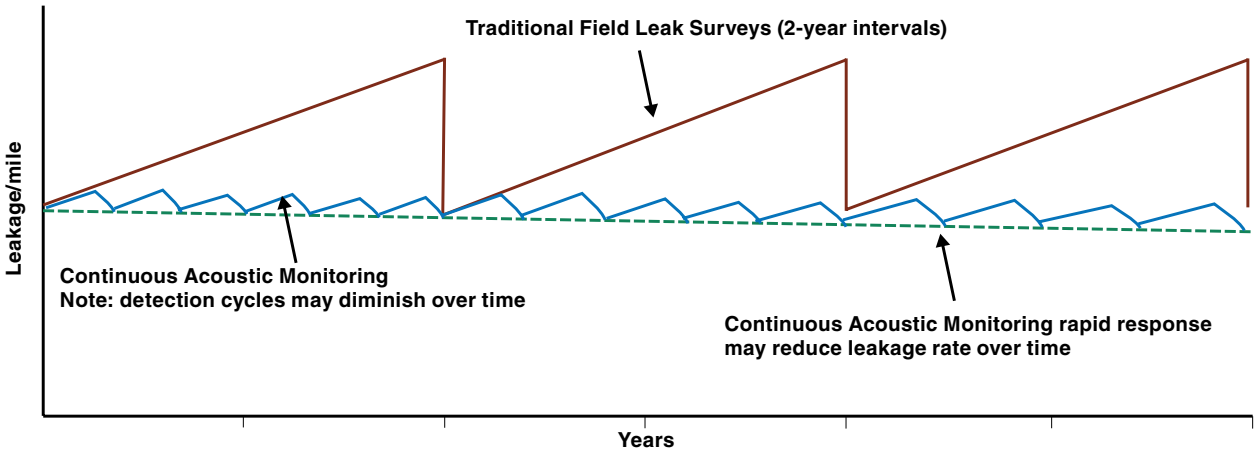
was exposed. The leaking water traveled below the pipe and likely found its way to a river about a quarter mile away. The leak was tracked for three weeks by a correlating CAM system that showed increasing acoustic noise over that time. Despite the wait, the repair was simpler at three weeks than it would have been had this pipe ruptured later. Damage around the water pipeline close to a residential street was minimal.

Comparing Traditional Leak Detection Surveys with CAM

Figure 5-26 is a chart similar to Figure 5-22 and compares the reduction and RR patterns incurred with the traditional leak detection survey approach and the use of a CAM system, with cycles of rising and falling unreported leakage making a "sawtooth" pattern. Using the traditional leak survey method, each rise in leakage continues during the intervening time between surveys when crews manually listen for leaks, allowing a relatively high leakage rate to occur before intervention. With the CAM system always in listening mode, leaks are identified and repaired soon after they emerge, thus the leakage rate and the run time are contained, which generates the much smaller sawtooth pattern for CAM. The difference in the areas of the sawtooth patterns represents the water savings from leakage that runs for longer times and at higher rates when using traditional leak detection surveys.

Experience with CAM systems in distribution networks with hidden leaks has revealed interesting characteristics about the nature of leaks and the networks they monitor. The approach has proven most effective in systems where leaks have the potential to run for months without surfacing, most notably areas with porous soils, river valleys, and concrete highways where water may not surface for a long time. Leaks have been identified flowing into drainage and sewer systems where surfacing would be highly unlikely.

One factor discovered in leak-prone systems is the rapid return to high leakage after the initial leakage repair program. After the first wave of repairs is made in response to suspected leaks, some distribution systems experience a brief increase in pressures resulting in more stress on pipe and a new wave of breaks following shortly thereafter. It has been the experience of users of these systems that eventually the system weak spots are repaired and the time between investigations increases. This is shown graphically in



Courtesy of Echologics

Figure 5-26 Impact of CAM on RR and leakage reduction

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Figure 5-26 which contrasts a typical leak survey cycle of water loss with that of a CAM system. It has been reported that CAM systems are generally more effective at finding leaks and come closer to reducing leakage to levels closer to the background leakage level.

If leaks are detected effectively by CAM as suggested in Figure 5-26, the results are a reduced level of leakage as small leaks are identified and repaired before they become acoustically significant. For many water utilities, this early response and cumulative leakage reduction through time suggests that a stronger business case can be made for CAM vs. the manual leak survey approach.

THE BENEFIT–COST RATIO APPROACH

Another approach applied to leakage economics for competing approaches is the ratio of benefits to costs, where the benefits are the leakage avoided due to the program and the costs are the leak detection program costs. Although this approach appears to have advantages of greater simplicity of use, it also has a notable limitation by restricting benefits to water loss only. This approach should not be used for utilities where reductions in repair costs and reduction in damage are significant. The benefit and cost parameters can be described according to the following equations using terms from Figure 5-22 which shows the RR of unreported leakage across intervals of acoustic leak detection activity:

$$costs\ per\ cycle = leak\ intervention\ costs \quad \text{(Eq 5-15)}$$

$$benefits\ per\ cycle = leakage\ abated\ during\ interval \times water\ value \quad \text{(Eq 5-16)}$$

It is important to ensure that the leakage abated during interval (which is the water that would have been lost had the intervention not taken place in Figure 5-22) be used as the benefit and not the leakage during interval or the unreported leakage that emerges after the leakage detection work. This is an easy error to make, as both can be referred to as "water lost"; however, total leakage during interval is actually a cost, and leakage abated during interval is a benefit.

The following equation is used to calculate the benefits per cycle:

$$benefits\ per\ cycle = inspection\ interval \times leakage\ abated\ during\ interval \times water\ value \quad \text{(Eq 5-17)}$$

The benefit-to-cost ratio is calculated as

$$benefits/costs = (inspection\ interval \times leakage\ abated\ during\ interval \times water\ value)/leakage\ intervention\ costs \quad \text{(Eq 5-18)}$$

A word of caution is needed when considering the use of the benefit–cost ratio. The benefit–cost ratio is useful in confirming that the benefits of a program at least match the costs. It cannot, however, be used to define the ideal program. Benefit-to-cost ratios will

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

always be increased by lengthening the inspection interval, but a higher benefit–cost ratio does not necessarily maximize net benefits. Consider the utility having two options:

1. Spending $10,000/year to yield a $50,000/year savings (a 5:1 benefit–cost ratio).
2. Spending $25,000/year to yield a $100,000/year savings (a 4:1 benefit–cost ratio).

The utility would see a better benefit–cost ratio under option 1 (5:1 vs. 4:1), but option 2 will yield a larger net annual benefit (and save twice as much water per year) to the water utility ($75,000 vs. $40,000).

The approach to calculating the EIF, as detailed earlier in this section, alternatively yields the intervention frequency (and associated leakage level) that yields the maximum net benefit to the utility.

More sophisticated cost–benefit analysis is usually required to evaluate increasingly advanced techniques. Such analysis could consider capital and operating costs, the time value of money and adjustments necessary for changing values including variations in the cost of water, the level of leakage anticipated, and changing labor costs.

REFERENCES

AWWA (American Water Works Association) Partnership for Safe Water. 2011. Self-*Assessment Guide for Distribution System Optimization*. Denver, Colo.: AWWA Partnership for Safe Water.

AWWA (American Water Works Association). 2021. Water Audit Reference Dataset (WARD). Denver, Colo.: https://www.awwa.org/Resources-Tools/Resource-Topics/Water-Loss-Control/Free-Water-Audit-Software (accessed June 19, 2021)

AWWA (American Water Works Association). 2019. Manual M77. *Condition Assessment of Water Mains*. Denver, Colo.: AWWA.

Bristol Water Services. 2001. *Leakage Management Assessment Project*. Final Report to the City of Philadelphia. Bath, UK: Bristol Water Services.

Brown, T.G.; Huntington, D.; & Lambert, A. 2000. Water loss management in North America—Just how good is it? In *Proceedings, Technical Session on Progressive Developments in Leakage and Water Loss Management, Distribution System Symposium*. Denver, Colo.: American Water Works Association.

Hughes, D., & Fitchett, J. How to Utilize Reliable Predictions of Main Break Locations from Machine Learning Tools, AWWA Water Quality and Finance Summit (virtual) 2020.

Illinois Sustainability Technology Center, 2016, Reduction of Non-Revenue Water through Continuous Acoustic Monitoring, Champaign, IL: Illinois Sustainable Technology Center.

Jenks, B., & Papa, F., Application of a Mobile DMA Testing Unit for Water Loss Management, AWWA Water Quality and Finance Summit (virtual) 2020

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Lambert, A.O. 2009. Ten years' experience in using the UARL formula to calculate infrastructure leakage index. In *Water Loss 2009 Conference Proceedings,* Cape Town, South Africa.

Lambert, A.O., & Fantozzi, M. 2005. Recent advances in calculating economic intervention frequency for active leakage control, and implications for the calculation of economic leakage levels. In *IWA International Conference on Economics, Statistics, and Finance*. Rethymno, Greece: IWA Publishing.

Lambert, A.; Fantozzi, M.; & Thornton, J. 2013. Practical approaches to modeling leakage and pressure management in distribution systems—Progress since 2005. Presented at CCWI 12th International Conference "Computing and Control in the Water Industry," Perugia, Italy, Sept. 24, 2013. Complimentary copies of the paper and presentation are available at http://www.leakssuite.com/wp-content/uploads/2012/11/CCWI_Sep2013paper_Pressure-burstsALMFJT-1-2003-2013K1.pdf.

Lambert, A.O., & Lalonde, A. 2005. Using practical predictions of economic intervention frequency to calculate short-run economic leakage level, with or without pressure management. In *IWA Leakage Conference*. London: IWA Publishing.

Lambert, A.O.; Myers, S.; & Trow, S. 1998. *Managing Water Leakage—Economic and Technical Issues*. London: Financial Times Energy.

May, J. 1994. Leakage, pressure, and control. https://www.leakssuitelibrary.com/wp-content/uploads/2021/04/JOHN-MAY-SEMINAL-1994-ARTICLE-4-16-April.pdf

Schlenger, D., & Hughes, D. Expert Guidance on Automatic Meter Reading (AMR) & Advanced Metering Infrastructure (AMI), North American Water Loss Conference workshop, 2019

Thornton, J.; Sturm, R.; & Kunkel, G. 2008. *Water Loss Control*. New York: McGraw-Hill.

UKWIR (United Kingdom Water Industry Research). 1999. *A Manual of DMA Practice*. Report Ref. No. 99/WM/08/23. UKWIR.

WRF (Water Research Foundation), Commonwealth Scientific & Industrial Research Organization (CSIRO), United Kingdom Water Industry Research (UKWIR). 2005. *Long-Term Performance Prediction for PVC Pipes. Denver*, Colo.: WRF.

WRF (Water Research Foundation). 2011. Continuous System Leak Monitoring-From Start to Repair, WRF #3183, Denver, Colo.: WRF.

WRF (Water Research Foundation). 2007a. *Evaluating Water Loss and Planning Loss Reduction Strategies*. Denver, Colo.: WRF.

WRF (Water Research Foundation). 2007b. *Leakage Management Technologies*. Denver, Colo.: WRF.

WRF (Water Research Foundation). 2014a. *Real Loss Component Analysis: A Tool for Economic Water Loss Control*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

WRF (Water Research Foundation). 2014b. *Real Loss Component Analysis: A Tool for Economic Water Loss Control: Leak Repair Data Collection Guide*. Denver, Colo.: WRF.

WRF (Water Research Foundation). 2014c. *Pressure Management: Industry Practice and Monitoring Procedures*. WRF #4321, Denver, Colo.: WRF

WRF (Water Research Foundation). 2019. *Guidance on Implementing an Effective Water Loss Control Plan*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 6

Approaches for Controlling Real Losses: Leakage and Pressure Management

This chapter offers information on how to economically control real (leakage) losses. It provides information on the most effective leakage and pressure management approaches available in a world of rapidly improving technology. Please refer to AWWA's Manual of Practice, M77, *Condition Assessment of Water Mains,* for coverage of leakage detection techniques in detail. This chapter limits information of those techniques covered in AWWA M77 but expands on others not covered in in the manual. The reader is urged to review Chapter 5 prior to reading this chapter. Chapter 5 explains the nature and impacts of leakage in water distribution systems; it also explains how to assess, quantify, and establish real loss reduction targets.

Water utilities are urged to complete the first three steps listed in Table 5-5 in Chapter 5. After these steps have been completed, the water utility will be poised to develop its leakage management program.

Figure 6-1 illustrates the four-pillar approach to controlling real losses, which is discussed in detail in Chapter 5. The center box represents the annual volume of real (leakage) losses from the water audit or current annual real losses (CARL). The perimeter of the smallest box within the CARL represents the unavoidable annual real loss (UARL) level, a level that is unique for each water system and is calculated by the AWWA Free

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Water Audit Software (audit software). Leakage that exists below the UARL is taken as undetectable with current technology and not capable of being addressed; thus water utilities should generally not strive to reduce leakage below the UARL. The perimeter of the second smallest box represents the economic level of leakage (ELL). Leakage cannot be economically controlled below this level because the cost of the leakage control measures will exceed the value of the savings from the additional leakage reductions that are achieved. The area between the CARL and ELL perimeters represents the portion of leakage that can be reduced economically. This is the ideal target leakage reduction that can be achieved in a cost-effective manner. All water utilities should consider some level of activity within each of the four pillars to contain leakage to economically low levels. Chapter 2 defines the infrastructure leakage index (ILI), a strong benchmarking indicator, as the ratio of the CARL over the UARL.

The four pillars of real loss control as shown by the four arrows in Figure 6-1 are as follows:

1. Active leakage control: identifying and quantifying existing leakage in the water utility transmission and distribution system. This includes conventional acoustic leak detection and flow monitoring in district metered areas (DMAs).
2. Improved leak repair time: ensuring timely and lasting repairs.
3. Pressure management: optimizing water distribution system pressures to minimize leakage and sustain water distribution infrastructure.
4. Infrastructure management: all pipeline assets eventually reach the end of their useful life and must be rehabilitated or replaced if they are to continue to provide service.

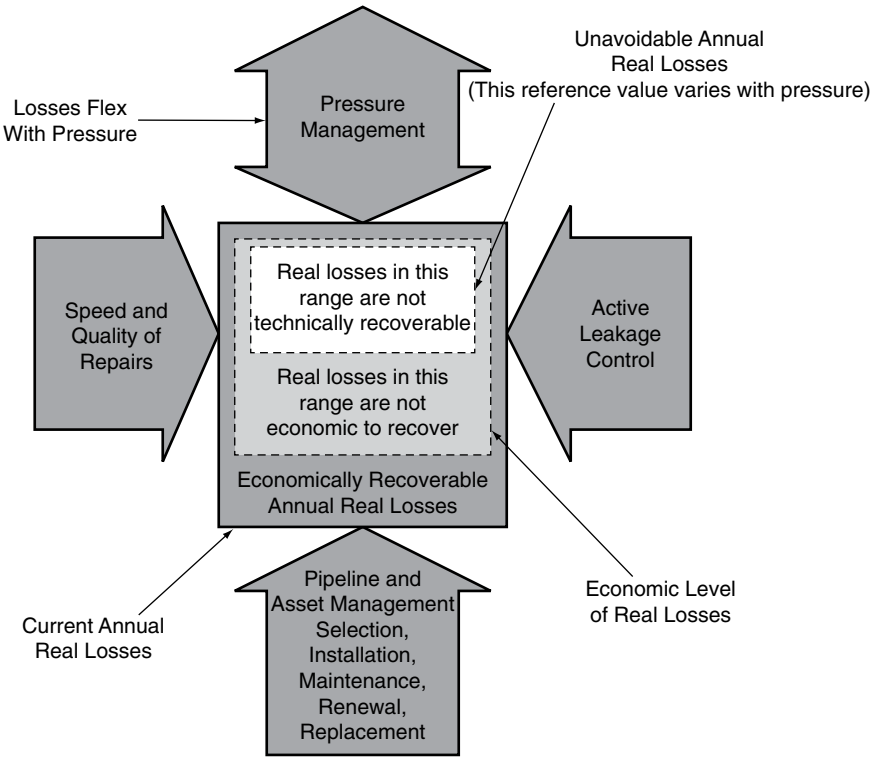


Figure 6-1 Four-pillar approach to the control of real (leakage) losses

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Utilities can follow the steps outlined in Chapter 5 to assess and quantify their real losses and establish a real loss reduction goal. This will help guide them in their approach to controlling real losses using the methodologies discussed in this chapter and to select the appropriate economic combination of the four primary leakage control methods. If utilities employ an appropriate level of activity in each of the four pillars, leakage can be maintained at economically low levels.

The remainder of Chapter 6 covers approaches and techniques for each of the four pillars.

ACTIVE LEAKAGE CONTROL

This section focuses on the various methodologies and techniques that can be used for proactive acoustic and non-acoustic leak detection and flow monitoring in DMAs. Utilities should evaluate the approaches and techniques discussed below based upon their system conditions. Utility-specific conditions such as pipe material, system pressure, staff skill level, or cost of water influence the success, effectiveness, and economics of the techniques used. What works for one utility may not work for another.

Leak Detection

Both acoustic and non-acoustic leak detection methods exist and are discussed in this section. Acoustic leak detection is the most common and affordable approach for proactive leak detection to identify and pinpoint unreported leaks in the water distribution system and on customer service lines. It uses mechanical and electronic listening equipment that can detect the sound that a leak generates. Depending on the type of leakage (background, unreported, or reported), water utilities may consider employing some form of regular acoustic leak detection using one of the technologies discussed below. It is best to have leak detection activities scheduled on a regular basis and available to confirm the location of unreported leaks and for pinpointing leaks. It should be cautioned that acoustic leak detection is not able to detect background leakage (small weeps and seeps that generate no sound). While reported (visible) leakage does not require acoustic leak detection for identification, the leak correlation process can assist in pinpointing the source location of reported leaks and breaks.

The traditional acoustic leak detection approach uses crews surveying portions of the distribution system at a set frequency. However, newer approaches described later in this section may be able to detect emerging unreported leaks sooner.

Leaks generate noise as pressurized water is forced through a failure in the pipe and loses energy to the pipe wall and to the surrounding soil area. This energy creates audible sound waves in the pipe and the water that can be sensed and amplified by electronic transducers or by simple mechanical devices in some cases. The sound waves are interpreted to determine the location of the leak. Most leak detection programs function by listening for leak sounds connected to the piping by placing sensors on valves, fire hydrants, water meters in outdoor pits, curb stops, or other accessible piping access points. Techniques also exist to sense leak noise sent through the less accessible water column. These are connected to the water through access points such as fire hydrant ports or by probes traveling inside active water piping.

Although acoustic leak detection is the most common way to detect unreported leaks, this technique has limitations. Acoustic leak detection is difficult in areas with interfering noise from pumps, electric transformers, and noisy equipment inside buildings. Depending on the time of day, water consumption and vehicular traffic can interfere with acoustic leak detection. Hence, many water utilities conduct leak surveys during quiet nighttime hours. Even at night, acoustic leak detection can also be compromised by noise

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

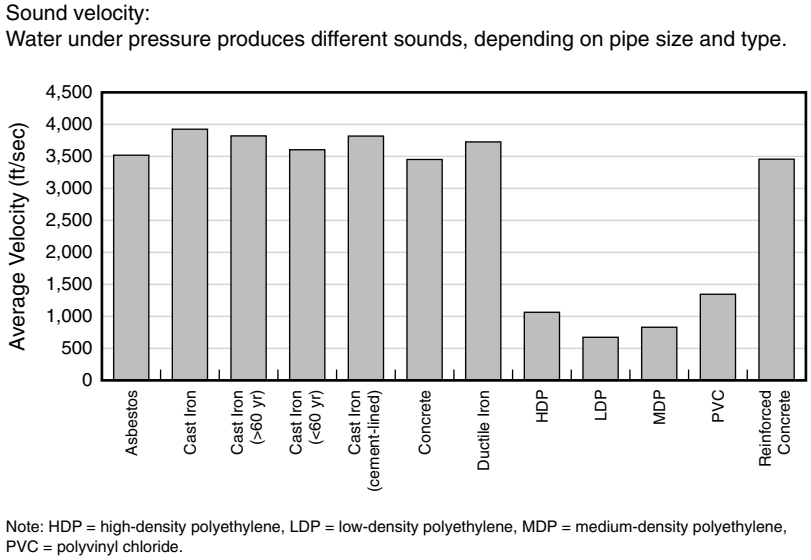
from continuous customer water use, air release activity, or water passing through nearly closed valves. Acoustic leak detection is complicated when multiple leaks exist within close proximity in a small area of the distribution grid. Repeat leak surveys are often needed after each leak repair is completed to confirm if additional leaks are still running. Without DMAs or other metering, acoustic leak detection provides limited ability to quantify leakage flow rates to a good degree of accuracy. Acoustic leak detection does not detect or quantify background leakage, the tiny weeps and seeps at pipeline joints. However, the lower limit of leak detection has been lowered with improved equipment to 1 gallon per minute (gpm) or less.

By using acoustic leak detection techniques, small unreported leaks can be detected early before they grow into large reported leaks. Eliminating those small hidden leaks may prevent undermining of the bedding soil support of nearby existing piping. Early detection can also limit repair and restoration costs by minimizing the size of the excavation and restoration area while potentially avoiding repairs during unanticipated nonworking hours.

Principles of Acoustic Leak Detection

The principles of acoustic leak detection must be understood to achieve success in pinpointing water system leaks. The following factors can affect leak sounds:

- **Pressure.** It is usually necessary to have a water pressure of at least 15 psi to employ acoustic leak detection successfully. Higher pressures generate more energy from escaping water and tend to make a stronger leak sound.
- **Pipe material and pipe size.** Acoustic techniques can be used on pipe and fittings of any material. However nonmetallic materials such as plastic tend to absorb noise energy and are much weaker sound conductors than metallic pipe. This requires acoustics to be located closer to the leak or placement of units to be at closer intervals when searching for leaks on nonmetallic pipe. See Figure 6-2 for typical sound velocities in pipelines of different materials. Larger pipes also tend



Source: Titus et al. 2013

Figure 6-2 Leak sound velocities in pipelines of different materials

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

to absorb noise energy, and, therefore, proximity to a leak is needed. Because many transmission pipes often lack points of contact, inserting probes or otherwise monitoring of the water is often employed.

- **Soil type.** The type of soil influences the amount of sound transmitted to the surface. Empirical observation indicates that sand is normally a better conductor of sound vs. clay.
- **Ground cover/surface type.** If the acoustic monitor is placed on the ground surface as opposed to direct contact with the pipe, the type of surface on which the sounding instrument is placed also influences how the sound travels. Grass or sod tends to insulate and muffle sounds, whereas asphalt and concrete are good resonators providing a uniform-sounding surface. Some ground surface monitors suggest placing a solid plate on softer surfaces to improve acoustic reception.

Types of Leaks Located by Acoustic Leak Detection

Water Mains. Detectable leaks on water mains range from less than 1 gpm to more than 1,000 gpm. Leaks on water mains can occur due to metallic pipe corrosion that usually originates as small leaks but can grow into large leaks. Splits at bells of piping can occur because of excessive pressure, improper installation, defective joint material, settlement, and overloading. Circumferential breaks are associated with stress from thermal changes and shifting soils. Joint leaks can occur because of corrosion, improper installation, improper materials, or overloading. Slow-developing main leaks (usually corrosion holes and joint leaks) have the potential to remain unreported because they can create subsurface paths for leaking water to travel away from the leak source. Alternatively, such leaks can undermine pipe support and lead to a larger pipe failure.

Customer Service Connections. Detectable customer service connection piping leaks range from a low of 0.5 gpm to more than 15 gpm and are caused by the same factors as main leaks. Customer service connection piping leaks account for the greatest number of unreported leaks in many water utilities and often account for the greatest volume of annual real losses. Many water utilities have policies that require their customers to arrange for leak repairs occurring on certain sections of their service connections within an extended time frame. Such policies tend to produce a delay in repairs and add to the quantity of current losses because many customers are unprepared to respond quickly, and service connection leaks run for excessive durations.

System Appurtenances. Detectable distribution system appurtenance leaks occur on valves, fire hydrants, air valves, and other system appurtenances and can range from less than 1 gpm to 500 gpm. Higher volume leaks or those visible on fire hydrants often become reported leaks, but many small leaks remain unreported leaks for long durations. Dry barrel hydrants that are just slightly open can waste water through the drain port unless inspected. Leaks may also occur in system appurtenances, such as pressure-reducing valves, pressure-sustaining valves, pressure relief valves, altitude-control valves, blow-offs, and related components of the distribution system. These leaks may occur due to malfunctions such as loose packing on valves or from operational problems that introduce hydraulic shock (also known as water hammer) to the system such as pressure surges and fire hydrants that are closed too quickly.

Customer Meter Leaks. Detectable customer meter leaks near the outdoor meter pit or box can range from less than ¼ gpm to the capacity of the service connection pipe. Leaks may be caused by loose spud nuts on the meter, loose packing nuts, damaged or broken angle stops, broken or damaged couplings, broken meters, or damaged or broken meter yokes. Some meters, when frozen, will break through the bottom (frost) plate to protect the mechanical parts.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Premise Plumbing. Detectable premise plumbing leaks on the customer side of the water meter can range from less than ¼ gpm to the capacity of the customer meter inside their premises. Current industry standards for customer water meters specify accuracy levels down to ¼ gpm, a flow rate too high to identify very small leaks such as slight toilet flapper valve leaks. These leaks may be caused by holes or breaks in customer service connection piping, ineffective hose-bib or shutoff valves, holes or breaks in interior plumbing lines, or leakage inside plumbing fixtures; toilet leaks are very common. Because many of these leaks occur downstream from customers' meters, this leakage may be metered and result in a higher bill to the customer. Very low-flow leaks may not register on customer meters, and this wasted water may go undetected until the leak increases to detectable limits. Note: Premises plumbing leaks are not included in the annual water audit because the customer meter is the terminus of the data tracked by the audit. However, the above information provides useful insight into the occurrence of such leaks because leak detection personnel will often detect the sounds of these leaks and must be able to distinguish them as being on customer-owned piping or utility-owned piping.

Acoustic Leak Detection Equipment

A variety of acoustic leak detection equipment exists for purchase or as part of service contracts in the commercial marketplace. The type of equipment that should be used depends wholly on the system that is being investigated. Determining the most effective type of acoustic leak detection technologies to use on a system is heavily influenced by the size and material of the water mains present coupled with the system age, failure history, soil properties (i.e., high corrosivity), water properties (high total dissolved solids could lead to high levels of tuberculation in mains), budget constraints (capital and operating costs), and criticality of water mains (i.e., risk of failure causing significant personal and/or property damage, nonredundant transmission main, etc.). A leak detection solution that works for one system may not work on another.

Ideally, every pipe segment in a system would have active leak monitoring coupled with periodic field surveys including noninvasive and invasive leak detection technologies. Such monitoring, however, is not economically or physically sustainable.

Mechanical listening equipment such as listening rods and geophones (that operate like a physician's stethoscope) are still in use, but the most effective tools are electronic listening and pinpointing devices like ground microphones, amplified listening devices, leak noise correlators, leak noise loggers, and inline sensors; most have visual readouts. Water utilities can choose to employ many of these tools in various combinations to develop an effective leak detection capability that best suits their situation. Leak detection consultants maintain a range of this equipment in their "toolbox." A description of some of the most notable equipment is given in the following subsections.

Simple Leak Noise Probes. The fundamental instrument for conducting leak noise surveys is an instrument that uses a probe that conveys sound to the user audibly or through a monitor. The original nonmechanical units were brought right to the ear to listen for the leak. Probes were brought into contact with part of the water system by direct contact if practical or indirectly using a valve key tool. Today's units convey sound electronically to devices that have amplifiers and may feature insulated headphones and filters to screen out selected frequencies. Many units have readout devices to provide a visual measure of the noise that can reach frequencies beyond human hearing abilities.

Ground Microphones. A variation of the probe is a ground microphone that is placed on a flat surface to carry sound without direct water system contact. These devices are often used to select the point to excavate where the loudest sound is often directly over the leak.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The objective of this method is to find the location of the loudest leak sound over the main or customer service connection piping. The first step is to determine the exact location of the piping. An electronic pipe locator can be used to locate the buried water pipe, and its location should be marked precisely on the roadway pavement or ground surface. Other nearby pipes from which the sound might be coming should also be located.

When using the ground microphone for pinpointing leaks, the volume should be set relatively low at the beginning, so loud sounds are not uncomfortable to the staff member listening. The volume adjustment should be kept at the same level throughout each pinpointing sequence. If sounds become uncomfortably loud, the volume can be reduced for safety, and the points should be surveyed again to locate the loudest leak sounds. The ground microphone should be used to listen for leak sounds every 5 to 10 ft. Notes should be taken on the sound intensity. If the equipment has a meter, then meter readings should be made. The strongest signal usually indicates the location of the leak. The setting of the volume or other controls should not be changed during this process. Where possible, comparing sounds at points with different surface and compaction characteristics should be avoided. If this is not possible, it should be noted that the same leak sound is quieter within a loosely compacted surface than in a dense one. After pinpointing the leak, its location should be verified by relistening using the ground microphone. The ground microphone is reliable in pinpointing many leaks but is limited by the existence of interfering noise, thickness of ground cover or pavement, and operator skill level. Ground microphones work best on flat, smooth surfaces; the ground should be prepared as best as practical, or a flat plate (thumb tack) should be used.

Leak Noise Correlator. This device is accurate in pinpointing many leak locations by analyzing leak sounds (including those that may be inaudible to the human ear) that travel along the pipe wall. These sounds can be sensed from aboveground by placing sounding sensors on valves, hydrants, and curb stops. Operators can also make direct contact with exposed mains or probe rods touching water mains through holes drilled in the street. Where taps are available, a hydrophone alternative that contacts the water directly offers improved sensitivity and greater range. This technology typically consists of a leak noise correlator unit (a receiver and processor) and two sensors (transducer or hydrophone depending on the application) with radio transmission capability. When the sensors are connected to pipe or water columns on either side of the leak, the correlator can display the leak location as a function of time of travel of leak sounds moving along the pipe wall or water column.

The primary leak noise correlator component is essentially a two-channel microprocessor that measures the time it takes for a leak noise registered at the two contact sensors on the water main. The characteristics of the leak sounds vary because of such factors as pipe materials and diameter; both should be known and are part of the input information entered into the correlator. Leak sounds are affected by the nature of the leak orifice, system pressure, ground conditions, and other factors. The leak sound velocity, the speed with which the leak sound travels along the pipe, remains constant. Correlation can also be accomplished using a network (more than two) of sensors.

To use the leak correlator, the leak sound must be detectable by a minimum of two contact sensors, and certain information must be entered into the correlator, including the distance along the pipe between the sensors, the pipe material, and size (diameter) of the pipeline. Two sensors, connected to and powered by portable transmitters, are attached to the selected contact points. The leak sound, picked up by the sensors, is then transmitted to the correlator unit by a radio.

By obtaining leak sounds at two points on either side of a suspected leak, the correlator analyzes the leak sounds and using main characteristics input provided by the operator, determines the exact location of the leak between the two sensors. Common noise interference, such as wind, traffic, and ambient system noise, has less of an impact

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

on the leak correlator than simple acoustic monitors. The depth of the main, type of cover, and surface conditions are generally not factors impacting results if contact between the sensors and the pipe system is solid. Correlation tends to eliminate some forms of false positives. Best practice is to use the leak noise correlator to get into close proximity of the leak and then employ the ground microphone to pinpoint the leak location as precisely as possible.

The distance between correlator sensors does have limitations. The ability for sound to travel long distances is attenuated in a large-diameter pipe; thus, contact points for correlators sensors may need to be spaced at smaller intervals than a specific pipeline alignment configuration may have available. Since contact points can have wide separation on transmission lines, excavation of the pipeline may be required to make direct contact with exposed piping. Alternatively, probe rods touching water mains through holes drilled in the street can be used by the sensors. Most leak noise correlators are limited to sensors spacings of approximately 1,500 ft often due to the need to be within radio transmission range of the sensor transmitter and the receiver. With the advent of cellular communications and improved hydrophone sensors, some manufacturers of models designed specifically to address the challenges of large-diameter piping quote allowable spacings of up to 5,000 ft. Air pockets within the area bracketed by two sensors detecting leak noise in the pipeline (more likely to occur in large-diameter piping) attenuate the acoustic noise caused by a leak and therefore may reduce the reliability of the correlation. Customer service connections, fire hydrant laterals, and connecting mains between sensors impact the location of the sensors. For example, the correlator with sensors connected to the main that detects a service line leak will correlate to the connection of the service line to the main and not the site of the leak on the service line.

If multiple leaks are present within the bracketed area, the correlator may only detect the leak creating the loudest noise. Some correlators allow the user to use frequency filters. This can allow for multiple leaks existing in different frequency domains to be found within the same correlation. There are also situations when a leak is repaired on a heavily corroded pipe, and the average pressure increases within that section of pipe which can cause one or more new leak(s) to develop close to the original leak location. This highlights the need to perform additional leak correlations after the initial leak is abated.

Advancements in correlation technology along with increased competition have resulted in the equipment becoming more affordable. Leak noise correlation has subsequently become a fundamental tool of the leak detection program.

Leak Noise Monitors. Acoustic leak detection equipment manufacturers have developed units that can be deployed in the field to take the place of sending staff to manually conduct leak surveys. Leak noise monitors have distinct advantages over conventional leak surveys. The units are programmed to provide daily output, listening to the quietest noise level of the night period vs. a leak survey crew listening only momentarily when some noisy activity may be occurring. The leak noise monitor units are strategically placed to ensure full coverage of the system, while a leak survey crew may struggle in the nighttime hours to access adequate points or listen inefficiently, at many more locations than might be necessary.

There are some drawbacks to using leak noise monitors. There may be other sources of noise that can resemble leaks, which may generate false positives. These can include legitimate water use, electrical transformers, nearby control valves, or other mechanical noises. Field staff may not always find an existing leak when dispatched to investigate. Initial reconnaissance usually identifies some noise sources that are not leaks. The noise monitors also do not provide a direct measure of the magnitude of the unreported leak, although some units provide frequency data that suggests leak magnitude. For leak noise monitors that rely on a network to collect data, it is important to ensure that there is a reliable and adequate network available for transmitting data.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Correlating leak noise monitors. Field correlation has recently been applied to leak noise monitoring systems. These systems function as leak noise monitors to find acoustic suspects but then supplement this finding by correlating either universally or in areas where suspects have been identified. This serves to reduce false positives and narrow down the area of interest to a much smaller area for field staff to investigate. Systems differ somewhat in their interpretation of correlation data; some systems can estimate the magnitude of a leak, while some provide further analysis of correlation to reduce false positives. Leak noise monitoring systems either log data or in most instances rely on a communication network to send acoustic data back to a server for analysis. Consideration should be given to ensure that there is adequate network coverage for the leak noise monitors to work effectively. The use of leak noise correlators by staff in the field is still generally required to confirm a leak conclusively, but the area of investigation is usually small, perhaps 500 ft from the monitor.

The type of acoustic equipment used to collect leak noise is broken into two categories related to how data are acquired: leak noise data loggers and leak noise transmitters, described below.

Leak noise data loggers (LNDLs). LNDLs sense and record sounds emanating from water distribution system piping, allowing operators to download the data and analyze sounds to detect and pinpoint leaks. LNDLs can conduct leak surveys by deploying them for periods of time at various locations within the distribution grid and setting them to "awaken" during night or low-noise times of the day to continuously listen and record leak sounds. The statistical variance of this noise is determined by the presence or absence of leakage. The noise signature obtained at each monitoring point confirms the presence or absence of nearby leakage points and indicates the relative location. In addition to use in leak survey work, LNDLs can also be deployed to "stand watch" over sensitive or hard-to-access locations. Some LNDLs can integrate with leak correlators and are thereby able to gather sounds from multiple loggers and correlate to pinpoint leak locations.

The development of LNDL technology was an important innovation in automating the leak detection process. LNDLs provide uniform listening, sound recording, and analysis, greatly reducing the human error associated with manual sounding methods. LNDLs can reduce the worker-intensive process of manually sounding many distribution system appurtenances. They can also greatly reduce the need for crews to work at night, at times in unsafe locations, to gather leak sounds during low-noise periods. LNDLs can be readily deployed in groups that are installed in valve boxes or other system access points at varying intervals up to approximately 1,500 ft. The area can be surveyed by gathering sounds, downloading the data, and correlating to pinpoint leaks. The LNDLs can then be relocated to the next area designated for survey. This method is commonly referred to as "lift and shift." In this way, a relatively small number of LNDLs can be purchased to outfit a leak survey team.

LNDLs can also be deployed on a permanent basis. LNDLs might be deployed permanently on critical infrastructure or other important/sensitive buildings to detect newly developing leaks quickly. Some water utilities have installed LNDLs on a permanent, widespread basis throughout major portions of their water distribution system.

LNDLs create cost efficiencies by reducing the labor involved in conducting leak surveys. Instead of a crew of two to four employees sounding individual appurtenances, one or two employees can install LNDLs relatively quickly in the same survey area and return the next day to download and analyze data. Leak noise correlator teams can then be dispatched to the areas that indicate leak sources. LNDLs also provide greater consistency in sounding for leaks. Whereas manual leak detection relies heavily on the sound detection capabilities of individual team members, LNDLs provide consistent sounding capabilities that can be assessed objectively. A slightly higher skill level may be required to analyze

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

data generated from LNDLs, so the labor trade-off may be somewhat offset by the need for additional training for the analysis of the LNDL findings. Still, the periodic downloading of the logged data remains a manual process.

Leak noise transmitters (LNTs). Several automatic meter reading (AMR) and advanced metering infrastructure (AMI) manufacturers provide (with or without connecting water meters) fixed network and mobile AMR/AMI systems to send leak noise data that can be received and analyzed with software in an office setting to identify potential leak locations. Fixed-network solutions offer the capability of next-day leak surveys, while AMI systems provide possible feedback through data logging on a periodic basis without going into the field. The leak noise candidates identified by LNDL and LNT do identify leak noise sources but in the absence of correlating can also identify locations that emit false positives. This requires periodic field visits to perform leak noise correlation at such locations, but the frequency of visits tends to diminish as a history of day-to-day leak noise is built and understood.

Economics of leak noise monitors. The economics for the use of leak noise monitors should be considered when planning the use of this method. A cost analysis between doing manual leak detection surveys and using leak noise monitors should be completed. If permanent deployment is considered, the economic return should be closely evaluated because dozens to thousands of devices might be deployed depending on the size of the system and the planned objectives. Large-scale deployment of leak noise monitors will require a large initial investment, as well as periodic recurring costs.

The decision to install many units across a system for an extended period should consider the useful life of the equipment and its effectiveness. Factors to consider include expected battery life, the robustness of the equipment in its working environment, and the frequency of needed hardware and software upgrades. Like many new technologies, the design of the equipment is evolving rapidly, and costs could drop in the future. If leakage is modest, the rate of rise of leakage is low, or leakage is believed to consist largely of background leakage or rapidly surfacing reported breaks, this would not be an appropriate technology to employ. For systems with high background leakage, plastic pipe under vegetated ground cover, high pressure, or frequent visible reported breaks, DMAs and pressure management (discussed later in this chapter) should be considered instead of systems using many leak noise monitors.

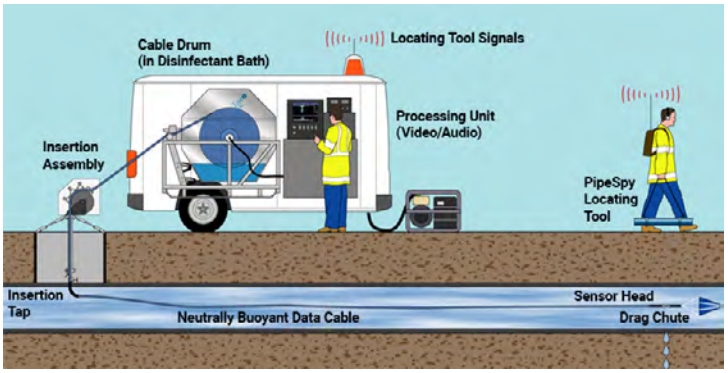
Inline leak detection sensor. In cases where external acoustic leak detection techniques are not practical, an inline acoustic survey can be conducted. Acoustic sensors have been developed to run through the in-service pipe, bringing the sensor to the leak sound rather than relying on the leak sound to find the sensor. This makes this technology relatively immune to external noise that can interfere with traditional acoustic leak detection. Inline surveys are often appropriate for large-diameter transmission mains, which are often poor at transmitting leak sounds and have limited access points to the pipe. However, some inline technologies have been developed to scan small-diameter piping. Some techniques use sensors tethered to an umbilical cable, and others use sensors that are free flowing in the pipeline. Both applications have been fully developed commercially and have proven usage histories in many utility situations. Inline methods can reliably identify single or multiple very small leaks on transmission mains with pinpoint precision, without requiring the water main to be taken out of service. This is particularly beneficial in assessing critical pipelines or those that run under rivers, major highways, or other areas where a failure would be particularly damaging.

Inline leak detection uses acoustic sensors that are typically inserted into tapping locations on the pipelines. Some systems allow insertion into 2-in-diameter valves (such as taps supporting air-release valves), while others require a 4-in-diameter opening.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Tethered Systems

In tethered systems, the acoustic sensor is attached via the umbilical cable to an aboveground monitoring station (typically in a vehicle large enough to house the inspection equipment). The cable can typically travel several thousand feet, depending on flow rates and the number of bends in the pipeline, but it cannot safely traverse butterfly valves or other inline obstructions. The sensor is propelled by the flow of water, and a trained operator in the aboveground monitoring station controls its movement and simultaneously listens to acoustic data recorded in real time while another operator tracks the sensor's progress aboveground, typically recording and marking any locations of interest such as leak locations and air pockets in the pipe. Tethered systems provide the operator with good control since the tether can be advanced or retracted as needed to better pinpoint leak noise and navigate the features of the pipeline. An example of a tethered system is shown in Figures 6-3 through 6-5.



Courtesy of Pure Technologies

Figure 6-3 The Sahara technology is a tethered inline leak detection system that is effective in pinpointing leaks on large-diameter piping.



Courtesy of Pure Technologies

Figure 6-4 Tethered inline leak detection insertion apparatus

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Philadelphia Water Department

Figure 6-5 Use of tethered inline leak detection technology in a 48-in water main



Courtesy of Pure Technologies

Figure 6-6 Free-flowing leak detection sensor inside the retrieval tool

Free-Flowing System

Free-flowing leak detection is typically employed by insertion of an acoustic sensor into the fully operational pipeline through check valves at pumping stations or tapping locations (such as those used for air valves), usually 4 in in diameter or larger depending on technology specifications. Prior to inserting the sensor, a suitable, predetermined retrieval location downstream is secured either by use of an inline retrieval tool (see Figure 6-6) or configuring removal through a fire hydrant or discharge into a reservoir. As the sensor travels through the pipeline, it constantly records acoustic data while intermittently

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

emitting a signal that allows its progress to be monitored at fittings aboveground by operators using tracking devices. The inspection concludes at the retrieval location where the sensor is removed from the pipe, and the recorded data are then downloaded for postprocessing and analysis. Along with acoustic recording capability, untethered sensors include a battery and onboard memory card and may typically include an accelerometer among other components depending on the provider's specific technology. The accelerometer records the sensor's movements throughout the inspection, allowing for calculation of its velocity, which is used in conjunction with the known tracking locations to accurately locate leaks.

Free-flowing sensors can navigate tight bends and travel up completely vertical pipes, provided flow velocity is sufficient.

Free-flowing sensors can travel much farther than tethered systems, but these devices cannot be retracted and re-sent in a single survey as with a tethered system; therefore, careful planning and preparation of routes is required, and the free-flowing device must be carefully tracked. Any branching mains from the transmission pipeline must have closed valves during the leak detection survey. Shorter spacing may be needed in pipelines with many bends. Sensors rely on a minimum water pressure of at least 5 psi so that leaks will generate an audible leak noise.

Because both tethered and free-flowing inline systems depend on the flow of water for propulsion, steps may need to be taken to adjust the flow to achieve the needed minimum velocity to propel the sensor such as opening valves and hydrants downstream or increasing the flow from pumps upstream.

Inline leak detection is generally offered as a service by specialized contractors, although equipment sales or leases may be available to utilities needing large volumes of surveys. the service may be expensive, but it can provide insight into locations that are inaccessible by traditional leak detection surveys.

Organizing an Acoustic Leak Detection Program

Leak detection is most often carried out with traditional leak surveys by manually sounding water system appurtenances such as valves, fire hydrants, service connection curb stops, or other accessible points on active piping. Water utility operators conduct a leak detection survey by systematically canvassing the water distribution system in such a manner. The deployment of lift-and-shift and permanent LNDLs allows a significant portion of the labor-intensive leak survey process to be automated. With new leaks constantly forming in water distribution systems, the optimum approach is to focus on areas where leakage is suspected. DMAs detecting high minimum-hour flow provide such a focus.

Analysis of historical leak records can also serve as a guide to predict areas of concern. However, some water utilities will schedule the distribution system for leak detection on a regular interval. For small water utilities that do not have the in-house resources, hiring a contractor to survey the entire distribution system on a frequency may be a cost-effective alternative. Larger utilities often have in-house leak detection crews that survey the system on an ongoing basis but may require three to five years to cover the entire system. The economic intervention frequency for a utility to conduct leak detection across its entire distribution system can be calculated using methods explained in Chapter 5. Leak surveys typically require two rounds of sounding to first identify leak noises and then confirm/pinpoint individual leak sources.

If the active leakage control program includes both DMA flow monitoring and acoustic leak detection surveys, leakage reduction can be conducted strategically, with leak detection staff deployed only where high minimum-hour flows indicate the presence of newly formed leaks.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Major considerations in creating an in-house leak detection program include the following tasks.

1. Developing program objectives: assessing the water audit's real loss volume and cost of leakage, project leakage reduction and cost savings from leak detection and repair (see Chapter 5).
2. Assessing the water distribution system characteristics, which includes
   a. Mains and customer service connections—materials, ages, sizes, joints, installation methods, inspections, leak histories, and operating pressures;
   b. Valves—accessibility (Are valve covers buried or stuck?), types, clockwise- or counterclockwise-turning, number of turns to exercise, frequency of exercising;
   c. Fire hydrants—types, sizes, locations, flushing frequencies, and unmetered usage;
   d. Customer water meter location—outdoor meter pit or indoors;
   e. Curb stops and customer service connection piping—typical locations, accessibility, mode of operation (quarter turn), and service pipe material (lead lines?);
   f. Blow-offs and air-release valves—locations, how often exercised;
   g. Pressure-reducing valves, pressure-sustaining valves, and pressure relief valves—locations and how often exercised/maintained; and
   h. Distribution system maps—What is the nature of the mapping (paper or electronic) and what is shown on maps (valves and other appurtenances), how current is the information, and how often is the information updated?
3. Evaluating and determining the use of manual leak detection, LNDLs, or a combination. Manual methods require more trained staff, while LNDLs require cost of equipment and training.
4. Projecting the needed level of staffing, equipment, training, and crew deployment.
5. Defining staffing/contracting needs. Effective leak detection teams can survey roughly two miles of pipeline per day.
6. Providing crew members with high-quality leak detection equipment, including sonic listening equipment with a high-frequency listening probe and a low-frequency ground microphone for pinpointing leaks. Providing standard tools and safety equipment.
7. Providing staff training, including instruction on the use of electronic leak detection equipment, available from equipment manufacturers, consultants, AWWA, or water operator organizations. Some regulatory agencies offer training and loaner equipment for utilities to undertake periodic leak detection work.
8. Considering the following when scheduling the leak detection survey
   a. Noise interference from electric transformers, underground transportation systems (subways), traffic, and other sources. Noise can also come from the water system, including pumps, throttled or nearly closed valves, air releases, and users who routinely consume water at night. Urban areas have more noise than rural areas.
   b. What time of day or night would be most effective to conduct the listening survey? Heavy daytime traffic noise in some areas may require leak sounding to be conducted during night hours.
   c. Define the sequence of survey and pinpointing work. Some utilities concentrate on the initial listening phase for several days and pinpoint leaks at the end of the week.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

d. If DMAs are in place, high minimum-hour flows will set leak detection priorities. If leak detection is scheduled periodically, historically leak-prone areas may warrant more frequent leak surveys than less leak-prone areas of the system.
e. Define the leak and repair data to be captured. See Chapter 5 discussion on leakage component analysis (LCA) and the "Sample Leak Management Plan" sidebar and sample "Leak Detection Survey Daily Log" sidebar. Leak detection and repair information should be part of the work order management system.
f. Define how leak detection crews communicate with repair crews to resolve "dry holes" that occur when repair crews excavate but find no leaks where the leak detection crew instructed them to dig.

Conducting Manual Acoustic Leak Detection Surveys

Water utility personnel often discover leaks fortuitously in the normal course of work, such as in valve exercising, fire hydrant flushing, and meter reading. Conducting an acoustic leak detection survey, however, means pursuing a systematic surveillance of the water distribution system to find hidden, unreported leaks. Many utilities survey their distribution systems according to zones or areas outlined on maps. Other utilities prioritize meter-reading routes that may minimize distances in covering the system. Many target high-leakage areas more often than low-leakage areas. It is important to recognize that leaks are continuously forming in water distribution systems, and although leak repair solves leakage, potential always exists for new leaks to form. A leak could occur the day after leak repairs are conducted, the day before the next survey starts, or at any time in between. Therefore, the average awareness time for leaks occurring between surveys is taken as one half of the time interval between the surveys. If leak detection and repair are conducted annually, the average run time for new leaks occurring is therefore one half of the year or 182.5 days. Knowing the average run times of individual leaks based on leak survey frequency is important when an LCA, a methodology used to set a system-specific leakage management strategy in a water system (detailed in Chapter 5). Leak surveys are typically conducted in a process that includes four phases:

1. Initial listening survey,
2. Relistening to suspect sounds,
3. Leak pinpointing, and
4. Leak repairs and confirmation of pinpointing.

The first three phases are detailed in the following sections with the fourth item discussed in the "Speed and Quality of Repairs: Optimized Leak Repair Functions" section later in this chapter.

Initial listening survey. During this phase, a trained operator conducts an initial listening survey of a large portion of or the entire distribution system, recording all suspect sounds. Leak detection is a process of discovery and elimination. The goal is to discover the contact points where leak noises can be heard and eliminate the contact points where leak sounds are not heard. A contact point is any accessible connection to the water main that transmits sound vibrations. This can be a fire hydrant, curb stop, valve, or probe rod touching a section of the water main. Building addresses should be noted of all locations where water use, meter sounds, or possible leak sounds exist. This initial search through each area of the system can usually be conducted relatively quickly. Prior to the start of the listening survey, a leak detection and repair plan should be prepared, like those shown in the "Sample Leakage Management Plan" and "Leak Detection Survey Daily Log" sidebars at the end of the chapter.

Sound travels a long distance on metallic mains, so listening at contact points allows the listener to hear the leak sounds along the length of the main between the points. Sound

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

travels roughly half the distance on nonmetallic mains, such as polyvinyl chloride (PVC), and additional effort is required during listening surveys on nonmetallic pipe. If sound does not carry the entire length of the pipe from one contact point to another and no other contact points can be found in between, the leak detection staff needs to listen over the main itself with a ground microphone. Figure 6-2 shows the wide variation of sound velocities in pipelines of different materials.

Several factors influence how far sound will travel along nonmetallic lines, including system pressure and pipe diameter. The sensitivity of listening equipment also limits the length of pipe along which sounds can be heard. To determine whether it is necessary to listen directly over mains in addition to contact points, perform the following test:

1. listen over the main with a ground microphone,
2. have a coworker turn on a hose bib at a customer's service, and then
3. determine how far along the main the sound of water escaping from the hose bib can be heard.

If the distance between contact points is greater than the distance that the sound travels along the main, the ground microphone should be used to listen over the main at appropriate intervals between 10 and 50 ft.

Many sounds can interfere with leak detection equipment. Sounds from customer consumption inside a dwelling include use of showers, toilets, washing machines, pumps, and meters. Even the sound of people talking may be picked up by electronic listening equipment. Sounds from outside a dwelling can be caused by aircraft, wind and rain, street traffic, interference from power lines or transformers, radio broadcasting, or lawn watering. Sounds from water noises usually come from adjacent leaks, valves, or turbulence. All of these sounds may be transmitted through leak detection equipment, making it difficult to isolate and identify leak noises. Faulty equipment, loose electrical connections, improper training, or system pressure less than 15 psi can also obscure or modify leak noises.

Relistening to suspect sounds. Because of variations in extraneous noise, even at night, it is often beneficial to revisit suspicious noise areas at a later time. The high-frequency contact microphone should be used to listen again for the sounds heard earlier. If the location is quiet, there is no leak. Where practical and where sounds are heard, the meter should be checked to see if it is running; a running meter indicates water consumption. If the meter cannot be accessed, it may be useful to return when the customer is present to view the meter or briefly shut off the service at the curb stop to determine if the source of noise is coming from the customer side of the curb stop. If sounds can still be heard when there is no water being consumed, a leak probably exists. That leak must be pinpointed.

Limitations of acoustic leak detection surveys. The use of acoustic listening instruments is a proven procedure for identifying and localizing hidden leakage. However, research organizations and practical experience have demonstrated that acoustic listening only on valves and hydrants or the ground surface leads to many unreported leaks being overlooked. Consequently, for effective leakage-reduction programs using acoustic surveys, soundings should also be performed on all service connections.

This technique has been practiced for many years, and while the acoustic monitoring equipment has improved, the technique does have some disadvantages. Surveys usually mean a labor-intensive process of walking the streets, especially if monitoring points like curb stops and valves are covered or difficult to access. Many utilities rely on experienced personnel to conduct the studies, drawing them away from other duties. Efficiency can be lost owing to the nature of the work conducted at night with factors like traffic, weather, and general safety a concern. As with other techniques, limited range of noise travel in plastic pipes and night flow add challenges to this approach.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Acoustic leak survey results can be optimized by using nighttime operations, uninterrupted listening, and extended listening periods. Nighttime operations add to the safety risk and cost of the work. Automated leak detection methods are an alternative to manual leak detection surveys and may improve the efficiency of the leak detection process.

There will likely be occasions where the field crew excavates and comes close but not exactly over the leak. The leak-pinpointing staff should be available to investigate this condition and determine whether a better location can be found and possible reasons why the leak location was not precise. There will also likely be times when the repair crew excavates and finds no evidence of a leak, just a dry hole. If there is no physical evidence of a leak in the area, the leak-pinpointing staff must be ready to respond immediately to investigate to prevent lost crew time. The excavated point provides an opportunity to listen directly on the pipe to determine if there is a leak noise in the area. These events, though undesirable, help the leak-pinpointing staff to improve their skill level in the future.

Non-acoustic Leak Detection

Although acoustic leak detection is the most common technique for leak detection in water distribution systems, it is only one means of detecting leaks in pressurized water piping systems. Several other techniques have been developed to identify leaks. While these techniques each have certain advantages, they may also have limitations. These techniques are currently in use commercially to varying degrees, while research continues to improve and enhance these and other new methods.

Satellite Leak Detection

Over the past number of years, several companies have been using satellites to help detect leaks. Satellites are programmed to take an image over a water utility's service area as it passes overhead. Water leaking from pipes has a certain signature/signal. The images from the satellite are filtered using proprietary algorithms that filter out unwanted signals leaving only the signal for leakage. The locations where possible leakage (points of interest) has been identified are plotted on a map for the utility.

Using this information, the utility can deploy a leak detection crew to the area to confirm and pinpoint the leak location. This approach to finding leaks is analogous to a doctor performing triage on a patient to determine where the most acute problems are located. Wide areas of the water distribution or the entire system can be scanned, and only the most likely leak locations are identified for further field inspection. These points of interest amount to a smaller percentage of the total length of pipe, therefore reducing the area that leak detection crews need to survey. Satellite leak detection may act more as a screening technology, rather than as a pinpointing tool. Further innovation is likely to occur in this technology as it has value in covering a wide area in a single scan.

Tracer Gas Method

Particularly for situations where leaks cannot be detected or pinpointed by traditional acoustic methods, tracer gas leak detection may be effective in identifying leaks. Tracer gas was initially used to find very small and hard-to-detect leaks after hydrostatic test failures on new pipelines awaiting commissioning after construction. Tracer gas has since been shown to be effective for detecting and pinpointing leaks in a wide variety of pipeline situations, both for dewatered pipelines and pressurized pipelines in service.

The tracer gas method uses either hydrogen or helium. The gas is injected into the pipeline through a standard tap or fire hydrant, and the gas travels with the flow of the water in pressurized pipelines or toward a vented outlet on dewatered pipelines. The gas will escape via locations where a leak is occurring. An operator will walk directly over

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

the section of pipe with a specialized instrument that can detect minor seepages of gas to atmosphere. When gas is detected at the surface, the instrument will verify and pinpoint the leak location. If the surface over the pipe is covered with asphalt or concrete or soil conditions include frost, it may be necessary to place test holes directly over pipe to allow the gas to vent to atmosphere. Helium gas detection may be less effective in clay or swampy soils.

Tracer gas leak detection technology is suited for any pipe material or any diameter and can identify leaks that do not generate audible leak noise. It can be used for very long runs of pipeline and is particularly suited for large-diameter mains, such as transmission mains where there are very few or distant access points for listening via acoustic leak detection methods.

For detection using hydrogen gas, it is not necessary to dewater the main because the mixture (less than 5 percent) is injected in a liquid form into the water. The gas mixture is a standard mixture of 5 percent hydrogen in nitrogen, purchased already mixed from a gas supplier. Exercise caution with this technique, however. The actual blending of hydrogen and nitrogen is a highly hazardous operation that should only be undertaken by the gas supplier. Do not handle hydrogen gas in any form other than ready-mixed to 5 percent hydrogen in nitrogen or less. Any hydrogen–nitrogen mixture containing less than 5.7 percent hydrogen is nonflammable (ISO 10156:2010). Helium gas leak detection should be obtained via certified service providers.

Ground-penetrating Radar

Ground-penetrating radar (GPR) can, in principle, be used to detect leaks in water pipes by detecting underground voids created by leaking water as it circulates near the pipe or by detecting anomalies in the pipe depth as measured by radar. Soil that is saturated by leaking water slows down radar waves and makes the pipe appear deeper than it is. GPR is similar in principle to seismic and ultrasound techniques. A transmitting antenna sends a short-duration pulse of high-frequency electromagnetic energy into the ground. The pulse is partially reflected back to the ground surface by buried objects or voids in the ground or by boundaries between soil layers that have different dielectric properties. Reflected radar signals are captured by a receiving antenna. The ground's interior is scanned with radar waves in a manner similar to that of ultrasound to obtain cross-sectional images (Hunaidi et al. 2002).

Because the method relies on detecting underground voids around leaks, soil conditions are a factor in the accuracy of the method. Impermeable clay soils may produce different leak-locating characteristics than sandy soils. Other limitations of the method include the requirement for sophisticated equipment and user skill. This technique is not widely used in North American water utilities; however, as research continues, it may find practical application as another effective leak detection tool. GPR may also prove useful in pinpointing the location of the leaking pipe if not the leak source.

Thermography

This technique detects thermal infrared radiation and displays it as visible images. In an infrared radiation image, the ground surface above a leak may appear cooler or warmer than the surface farther away from it. This temperature difference may reflect variations in the temperature of leaking water and the overlying soil; considerable heat may be transferred between leaking water and surface soil. Also, soil close to the leak becomes saturated by leaking water, which may change its thermal characteristics and make it a more effective heat sink relative to dry soil away from the leak. A thermographic survey of an

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

area uses a high-resolution commercial infrared camera system. The camera should be focused on the ground surface and will capture images over a period of time.

This technique, which also requires sophisticated equipment and user skill, may be affected by many variables, including ambient temperatures of air and soil, relative humidity, seasonal effects, and others (Hunaidi et al. 2002). Thermography is not yet available commercially in an affordable, user-friendly package.

Sniffer Dogs

Dogs trained to sniff chlorine from drinking water are working in the United Kingdom, United States, and other countries. These animals have been found to be effective in detecting and pinpointing water sources contributing to leakage. There is a cost associated with the training and care of the dog and often a designated handler must be assigned.

DISTRIBUTION SYSTEM KNOWLEDGE AND FIELD OPERATIONS SAVVY FOR DETECTING LEAKS

Acoustic and non-acoustic leak detection technologies offer sophisticated capabilities to leak detection staff and have greatly advanced the success of leak locating in the water industry. However, effective leak detection teams still must rely on knowledge of distribution system operation and develop field operations savvy to be highly successful. Staff should have good listening and observation skills to gather and analyze evidence in tracking sources of leakage. Staff should routinely operate valves, fire hydrants, customer hose bibs, and other appurtenances in activating and deactivating flow in attempt to identify which pipe segments are leaking. Distinguishing between water pooled in a basement or other subterranean space as potable drinking water, stormwater, ground water, or wastewater is an essential skill in confirming the presence of drinking water leaks.

Water systems that provide fluoride in the water treatment process may be able to use this chemical as a "tracer" to confirm a water sample as originating from the water distribution system. Unlike chlorine in drinking water which dissipates once the water is released into the ground, basement, or other area, fluoride remains detectable, and fluoride testing can be used as part of the leak detection process. Not all water utilities provide fluoride in the treatment process, so this capability is not available to all systems.

Distribution system knowledge is gained over time by staff and is important for a highly functional leak detection team. Water utilities can gain information on these practices from leak detection equipment manufacturers, consultants, and other sources.

FLOW MEASUREMENT AND ANALYSIS TO QUANTITY AND MANAGE LEAKAGE

District Metered Areas

An accurate means to quantify the individual components of leakage in a water distribution system is to measure water flow into DMAs. This approach involves subdividing a portion of the water distribution system into an isolated zone by closing valves and installing flowmeters on one or more supply water mains entering (and possibly leaving) the DMA. Flow patterns are routinely analyzed to discern when rising flows indicated the presence of newly emerging leakage. See Chapter 5 for guidance on the design and implementation of DMAs.

DMAs provide flow data that measure bulk leakage flows in a small zone and as an early alert to direct leak detection staff to areas of confirmed leakage. DMAs do not pin-

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

point the location of leaks but are used to make a utility aware that they have a potential leak, its magnitude, and a concise area to focus leak detection efforts.

Several analytic techniques available in the DMA environment are discussed below. Additionally, the sidebar titled "Managing Water Supply Costs by Focused Leakage Control in the Water and Wastewater Authority of Wilson County, Tenn." describes the successful use of DMAs on a system-wide basis to address the challenges of an all-plastic pipe distribution system that hampers acoustic leak detection.

Minimum-hour Flow Analysis

For utilities that have DMAs, leakage trends can be identified by analyzing DMA flow patterns. In most water distribution systems, flow follows a repetitive diurnal pattern, peaking at certain times of day and minimizing at other times (expect weekend and holiday patterns to differ from the five-day work week). During periods of relatively consistent customer consumption and leakage, a repetitive, characteristic pattern can be observed. As leakage rates rise, an increase in the minimum-hour flow conditions can be observed. A gradual rise in minimum-hour flows over periods of days or weeks gives a good indication that new leaks have developed. A target level of minimum-hour consumption can be established. When minimum-hour flows reach or exceed this level, leak detection personnel should be sent into the DMA to complete a leak survey to pinpoint leaks and arrange for repairs.

Minimum-hour flow analysis is an effective way to identify areas of rising unreported leakage before they grow into larger reported leaks that may cause additional damage and cost more to repair. For utilities that do not see most of their leaks coming to the surface, this approach helps to identify unreported leaks.

This approach may be hampered in DMAs that do not exhibit the repetitive diurnal curve. This could be the result of commercial/industrial customers who consume varying amounts of water throughout the entire day and customers who irrigate during the night. Additionally, if the net flow rate of the DMA is too large, smaller leaks may not be observed. Planning and design functions for a DMA should consider the customers and flow profile and variation expected to occur in the proposed DMA.

Step Testing Using DMAs

The term *step test* has been used to describe two types of testing techniques that can be conducted using DMAs. Each test method employs an incremental reduction, in a step fashion, in either the water supply to a DMA or the scale of the DMA by sectioning it to a smaller size by closing valves within the DMA. While the term *step test* is used for either test, these methods serve different purposes, and each has considerations in use.

Pressure step tests to quantify background leakage. This test is conducted by reducing the supply to the DMA by incrementally throttling the closure of a valve or pressure-reducing valve (PRV) on the sole supply main to the DMA in a similar manner to the pressure drop test described in the "Constructing the DMA" subsection in Chapter 5. Three or four incremental reductions in supply should be conducted with gaps of at least 15 min between step reductions. Pressure should be reduced approximately 10 to 15 psi (or as much as can be tolerated) in predetermined steps for each increment. Flow and pressure should be recorded at the start of the test and at each reduction. The profile of the graph created by these data points follows the shape of steps. The reduction in flow into the DMA relative to the drop in pressure is characteristic of the relation of background leakage to unreported leakage in the DMA. However, this test is valid for fully metallic

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

piping systems where leak detection and repair have been conducted to eliminate unreported leakage.

Pressure must be measured at the average zone pressure (AZP) during this test. In this way, the pressure step test is an important tool in setting the leakage management strategy because the amount of background leakage influences the degree to which pressure management should be employed. Pressure step tests often must be conducted at night or during other low-water demand periods because these tests require working pressures to be lowered notably such that customer service may be disrupted for approximately 1 hour. The pressure step test is a more rigorous version of the pressure drop test described in the "Constructing the DMA" subsection in Chapter 5.

Flow step tests to localize leakage. This technique attempts to section portions of a DMA to localize the subsection where significant leakage is occurring. If leakage is indicated by high minimum-hour flows, this method isolates the leakage by segmenting the DMA in a step fashion. Valves inside the DMA are closed to create a new temporary boundary that shrinks the DMA or zone by one-quarter or one-third of its size. If the flow drops notably by more than the reduction of the zone size, it can be surmised that much of the leakage exists in the area that was isolated from the DMA.

This test can be successful in narrowing the region in which leakage is occurring and is relatively straightforward to conduct in the DMA structure. However, flow step tests can be subject to an inherent flaw in that pressure also changes when the zone reduction is implemented. Without adjusting for the pressure change, the test might give misleading results. DMAs are useful in large part because flow and pressure patterns can be observed for seasonal variations and other supply fluctuations. Background leakage is very sensitive to pressure changes, and high pressure from the zone reductions may cause the leakage rates to change in the area being reduced. Often pressure increases as the DMA is further reduced in size, giving the indication that most of the leakage is located in the segment closest to the supply main.

Caution should be taken in conducting a flow step test by monitoring pressure at the supply point and other points in the DMA to see that undue pressure changes are not occurring, which could make the test results unreliable.

A major advantage of DMA technology is the ability to closely monitor a discrete, manageable area of the distribution system. By gathering data from the DMA on a continuous basis, the operator gains, over time, a solid understanding of the hydraulic performance of the zone. Deviations from normal flow patterns—caused by leakage, main breaks, fire flows, and so on—stand out and provide the operator with the capability to respond strategically to an event.

Managing Water Supply Costs by Focused Leakage Control in the Water and Wastewater Authority of Wilson County, Tenn.

The Water and Wastewater Authority of Wilson County (WWAWC), Tenn., is a small water utility located in the central area of Tennessee not far from Nashville. The WWAWC recognizes the value of employing best practices and uses the AWWA audit software to conduct a top-down water audit on an annual basis and has developed a very effective leakage management program.

The WWAWC operates with a staff of 15 employees (seven in field operations) and services a primarily residential population with 7,838 water service connections. The WWAWC purchases 100 percent of its potable drinking water supply as treated water from four different supplies through 15 meters that are read every five min via cellular telemetry and integrated into supervisory control and data acquisition (SCADA). The water purchased by the WWAWC is the single largest manageable expense in its annual budget. The WWAWC pays a notably high rate of $3.06 USD per 1,000 gal for this imported water supply, which is distributed in the eastern half of Wilson County through 348 miles of distribution piping, which is almost 100 percent PVC piping. System pressures vary from 25 to 140 psi, and the average pressure has been measured

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

at approximately 80 psi. Because of the significant expense of the imported supply, the WWAWC keys heavily on leakage management, as water lost to leaks causes a significant financial impact to the WWAWC. WWAWC has taken on the challenge of aggressive leakage management in a cost-effective manner. Its capital investment of $210,000 USD to improve metering and establish real-time communications on the 15 import metering sites, install metering at storage tanks, and establish additional DMAs yielding a total of 25 permanent DMAs has a simple payback period of nine months, a highly cost-effective result.

The WWAWC's primary leakage control strategy is to identify and abate leaks that tend to run for a long duration and to identify water main breaks before they affect the level of service to customers. The run time of a leak consists of awareness, location, and repair times. The WWAWC becomes aware of leakage from customer calls for low or no water pressure, surfacing water, delayed storage tank fill rates, routine sonic leak surveys, water auditing, SCADA alarms for high flows and low-pressure conditions, and minimum night flow measurements in DMAs that have been established throughout the water distribution system.

Once the WWAWC is aware of the leakage, then locating the individual leaks may consist of staff physically driving or walking the suspected area looking for surfacing water, partially closing main line valves and listening for water rushing through the valve, conducting a sonic leak survey by listening to the system appurtenances, using the ground-microphone technique with the water leak detector directly over the water main, analyzing minimum night flow measurements, and step testing in a DMA evidencing high minimum hour flow by temporarily limiting supply to a portion of the system and monitoring the change in flow.

To optimize repair time, the WWAWC repairs all main breaks/leaks immediately after they are located. Service leaks are typically repaired within four working days to allow for the other utilities to respond to a normal Tennessee One Call "underground utility locate" request. Service connection leaks that may affect the customer's water quality or quantity or cause property damage are repaired immediately after they are located. Large water main breaks, where the customer's service is interrupted, tend to surface and are relatively easy to locate using SCADA data to identify the location of the rupture. Therefore, these leaks tend to have a high-flow rate but run for a relatively short duration. Service connection leaks and small main line breaks may leak at a much lower rate, but if they are not discovered, they can run for long durations, and these leaks can be the major contributor to the annual real loss volume. Therefore, the WWAWC has established and implemented methodologies to reduce the awareness and location time for these leaks that would otherwise tend to run for a long duration and to identify the main breaks before they affect the level of service to the customers. This is not only a nonrevenue water issue but also a water quality issue. By becoming aware and locating main breaks before they blow out allows the main to be repaired under positive pressure with a repair clamp, eliminating the need to cut the pipe and depressurize the distribution system. If the system is depressurized, then a bacteriological sample must be gathered, and the depressurized section of the system may need to be disinfected. Repairing main breaks under pressure is extremely important in maintaining water quality and limiting service interruptions to customers.

The WWAWC's water distribution system exists in a rural area with a customer density of 23 connections per mile of main. Therefore, the average distance between listening points on services is 230 ft. In the more rural areas of the system, it is not unusual for the distance between listening points to be more than 2,000 ft, a distance too great to allow for effective sonic leak detection by direct contact. Listening to the service connections in these areas will tend to only identify service connection leaks. Additionally, the WWAWC's water mains are nearly 100 percent PVC pipes located out of the road right-of-way under soil and vegetation in fields and often behind fences. Using a ground microphone over the water main to listen for the impact and fountain sounds created by the leak works very well on a water main located under pavement or on the shoulder of the road. This process is difficult in WWAWC, where many water mains are under grass or soil, a condition that attenuates leak noise relative to leaks under a solid (paved) surface.

System pressures at higher elevations are as low as 25 psi. Low pressure produces a weak leak noise, making sonic leak detection much more difficult. In summary, on a rural PVC system like the WWAWC's, a sonic leak survey tends to identify service connection leaks but may not detect existing main line leaks. Therefore, the WWAWC implemented 25 DMAs covering 100% of the distribution system to identify and prioritize the areas where leakage exists or new leaks are emerging.

The DMAs were established based on existing locations of imported water supply flowmeters, storage tanks, pump stations, and pressure-reducing valves. Each DMA has a means

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

of measuring the input into the DMA by permanent supply meters with one gal resolution along with pressure monitors. The DMAs range in size from 0.8 miles of water main serving 114 service connections to 86 miles of water main serving 1,901 service connections. To reduce the run time of the leaks, the WWAWC monitors the minimum night flow (MNF) via SCADA at five-minute intervals. Given that the customer base is primarily rural residential, minimal consumption occurs during the early morning hours. During the summer months, some nighttime lawn irrigation may occur within housing subdivisions and for select individual customers. The minimum flow period in the WWAWC service area is typically between 1:30 a.m. and 3:30 a.m. when customer consumption is at its lowest level. For DMAs containing storage tanks, each night when pumping is shut down and the DMA is fed via gravity flow from the water storage tank, the tank level changes are converted to a volume over several hours to calculate the flow in gpm feeding the DMA. This process is a good verification to check the general accuracy of the flowmetering at the tank and is also a good way to measure MNF if there are no metering flows in and out of the tanks. DMA MNF analysis is conducted daily to determine if any leakage appears to be newly emerging within each DMA. To conduct a night flow analysis, the legitimate nighttime consumption (LNC) of customers must be measured or estimated.

Night flow analysis employs the equation: MNF – LNC = leakage.

If recoverable leakage exists, then the type of intervention to detect and pinpoint the leak(s) is determined based on the anticipated type and size of leak(s) and size of the DMA. Acoustic leak detection is typically performed in smaller DMAs with step testing first conducted in larger DMAs to localize the specific areas of the leakage within the DMA to focus acoustic leak detection activities.

If step testing and acoustic leak detection do not identify and locate the recoverable leakage, then a noninvasive portable ultrasonic flowmeter is temporarily installed at a strategic location in the subdivided DMA to measure flow. Data are typically logged at 30-sec intervals to conduct additional analysis to further diagnose the source of the leakage. This use of flowmeter and analysis methodology on a temporary basis is also a good strategy for water systems that do not have permanent DMAs.

The WWAWC's success in keeping leakage at minimal levels is reflected by its real loss performance indicators calculated by the audit software. Audit data from 2019 showed the annual leakage volume to be 75.6 mil gal, 595 gals/mi/day, and an ILI value near 1.0, a very low level of leakage. This is allowing the expensive cost of purchased imported water supply to be managed very effectively and to maintain a reliable level of service to the WWAWC's customers. The WWAWC encounters many of the challenges confronting small systems: a low density of connections, wide distances between pipeline-sounding locations, considerable piping installed in unpaved areas, hard-to-detect PVC piping, and limited finances and staffing. Yet by applying proactive leakage management techniques, the WWAWC has established a highly effective leakage management program. The WWAWC's experience proves that the leakage management techniques offered in this publication are effective when they are properly tailored to the specific conditions of the water utility, regardless of its size or general characteristics.

Pressure Management

Pressure management for leakage control is defined as the practice of managing system pressures to the optimum levels of service, ensuring sufficient and efficient supply of water for legitimate uses and consumers, while reducing unnecessary or excess pressures and minimizing transients, all of which cause the distribution system to leak unnecessarily. Pressure management offers many benefits in leakage control and distribution system sustainability and has become one of the most cost-effective leakage management strategies internationally.

Traditionally, the cost savings calculated in the benefit–cost ratio for introducing pressure management was the only recognized financial savings from the predicted reduction in leakage flow rates (leakage volume saved multiplied by the marginal cost of water). However, since 2006, members of the Pressure Management Team of the International Water Association (IWA) Water Loss Specialist Group have developed increasingly

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

reliable methods for quantitative predictions of reductions in break frequency on mains and services, which result in additional operational cost savings and a likely extension of residual infrastructure life. These and other additional benefits listed in Table 6-1 have transformed the economics of pressure management, which is now being implemented in some countries as a means of asset and energy management rather than simply for leakage control.

Various ways to manage pressure are available for use including:

- transient control to minimize surge,
- pressure sustaining or relief valving to avoid surge,
- altitude and level control in tanks and water storage facilities to avoid overflows,
- implementation of controlled pressure management areas (often in conjunction with DMAs), to reduce and stabilize pressure, and
- pump and motor controls for pressure stabilization and reduction.

Pressure stabilization and reduction strategies are probably the most widely used for leakage control and are considered proactive pressure management. However, all the above approaches can provide benefits of leakage control and infrastructure sustainability. It is important to know which tools to apply under specific conditions in the water distribution system.

Figure 6-7 displays a hydrograph of water and pressure supplied to a DMA in a North American water system. As is typical in most water distribution systems, the graph shows that this DMA experiences significant fluctuations in water demand throughout the day—peak consumption periods at the start of daytime activities and in the evening, coupled with off-peak periods of low demand often, but not always, during overnight periods. At the time of minimal customer consumption, typically overnight, head loss in the system may be near its lowest, and pressures might be approaching their maximum. Conversely, peak water demands coincide with periods of minimum pressure in many water distribution systems. This traditional water distribution supply pattern of ensuring that minimum pressures are maintained during maximum water demand periods results in higher pressure much of the remaining time in the day.

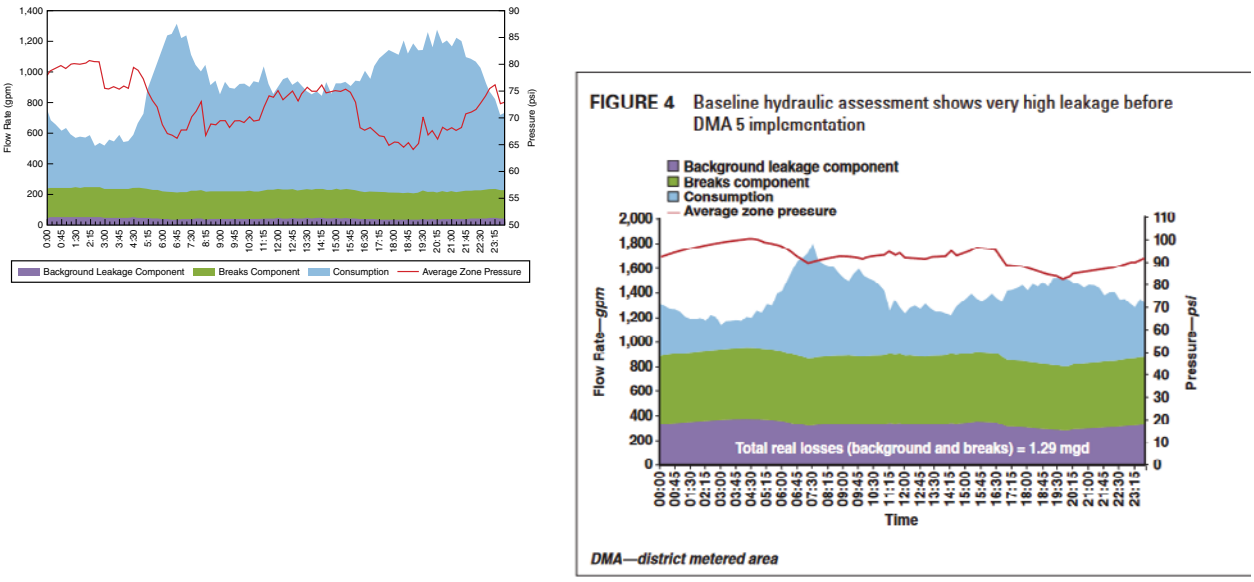
The risks of high pressure at minimum water demands are evident from the fact that major breaks in many water utilities tend to occur during the late evening and early morning hours when system pressures are at their highest, often due to full tanks and minimal friction loss from reduced flow or from transients that occur as a result of reduction (or increase) in pumping or rapid shutoff of system storage reservoirs or tanks that have completed daily filling.

Table 6-1 Multiple benefits of pressure management in water distribution systems

Pressure Management: Reduction of Excess Average and Maximum Pressures								
Conservation Benefits			Water Utility Benefits				Customer Benefits	
Reduced Flow Rates			Reduced Frequency of Breaks (Bursts) and Leaks					
Reduced excess or unwanted consumption	Reduced flow rates of leaks and breaks (bursts)	Reduced and more efficient use of energy	Reduced repair and reinstatement costs, mains, and services	Reduced liability costs and reduced bad publicity	Deferred renewals and extended asset life	Reduced cost of active leakage control	Fewer customer complaints	Fewer problems on customer plumbing and appliances

NOTE: Based on Australian WSAA PPS-3 Project 2008-11 (WSAA 2008) with addition of energy component

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Source: Thornton 2005a

Figure 6-7 Pressure and flow variations in a DMA without specific pressure management controls

Most of the roughly 50,000 community water utilities in North America perform basic methods of pressure management using booster pumping stations, level controls, and pressure zones. Refining their pressure management may be a feasible additional step for many of these utilities because system pressure is set by tank levels and existing pumps. Many of the same utilities likely do not have a full understanding of the significant range of financial and operational benefits that may be possible from improved leakage control and water main break prevention when employing optimized pressure management.

Benefits of optimized pressure management. The two primary objectives of pressure management for leakage control and infrastructure sustainability are to

1. reduce the frequency of new leaks and breaks occurring within a water distribution system by reducing the stress on the infrastructure due to unnecessarily high pressure and transient pressure and
2. reduce the flow rates of those leaks and breaks and background leakage by reducing excessive pressure.

Pressure management and infrastructure replacement/rehabilitation are the only real loss control methods that reduce background leakage flow rates. Background leakage is sensitive to water pressure. Infrastructure replacement/rehabilitation is the most comprehensive means to address leakage but also the most costly and is rarely justified economically based on leakage reduction alone. Alternatively, pressure management has a major advantage of often being a highly cost-effective leakage reduction tool for systems with high background leakage and high pressures.

Proactive pressure management cannot be applied universally across all water distribution systems; it is system specific in the degree to which it can be applied and the benefits derived. Changes in pressure generally cannot be employed where pressures are consistently at or near the low service-level requirements of the water utility for pressure and flow delivery. Pressure management may not provide cost-effective improvements where background leakage and break frequencies on both mains and services are low. Therefore, pressure at specific key points and repair frequencies on mains and services (separately) should always be assessed when considering a pressure management strategy.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Common questions raised by water utility managers with respect to pressure management techniques include suspicion that customer consumption and revenue will be reduced, fire flow capability will suffer, and other hydraulic limitations will appear. While such concerns are valid, they are usually addressed by sound engineering and hydraulic modeling at a cost that is more than offset by the benefits. As detailed later in this section, these questions can be predicted and addressed through a competent design process and seldom are a major impediment to optimized pressure management where this technique is otherwise deemed applicable.

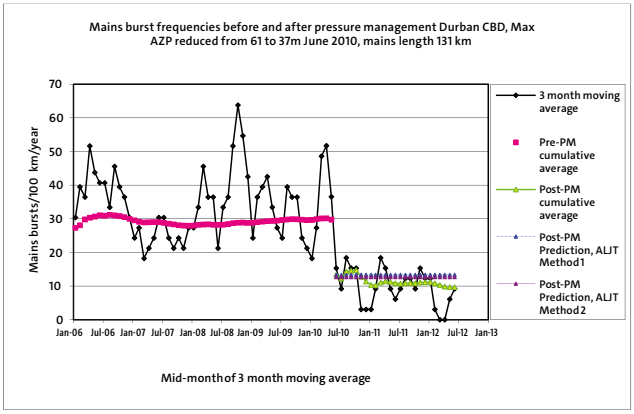
Reducing break frequencies through pressure management. The most reliable validation for a successful pressure management strategy is derived from analysis of "before and after" break frequencies in individual systems in which pressure management has been implemented. Members of the IWA Water Loss Specialist Group Pressure Management Team and utilities and consultants internationally have published many case studies where pressure management has produced immediate, significant, and sustained reductions in new break frequencies. Lambert, Fantozzi, and Thornton (2013) provide several such references. Methods have been developed and refined for predicting the relationships between maximum pressure and water main break frequency per 100 miles of pipeline per year (normally excluding fire hydrant and valve leaks) and service connection leak frequency per 1,000 service connections/year (excluding small leaks at meters and curb stops).

In many water distribution systems, the presence or absence of pressure surges is a major factor in the frequency of occurrence of water main and service pipe breaks and leaks. Surges or transients are brief but dramatic increases in pressure caused by pump activation and deactivation, control valves opening or closing too quickly, tank filling operations, or sudden change in water demands. Because they are usually very brief in nature, pressure transients can only be measured over very short time periods, of the order of one second or less. Precise data-logging instruments are readily available for measuring transients in supply zones, including those supplied by gravity. In developing the pressure and leakage management strategy, consideration should be given to launching an evaluation of the operations and function of pumps, control valves, and tanks and customers who have suddenly shifting water demand to determine if opportunity for harmful transients exists and if cost-effective controls can be incorporated into the strategy. In addition, operators of water systems (which includes firefighters) can be unintentionally inducing surges by opening and closing fire hydrants and valves too quickly.

Most breaks and leaks on water mains and service connections occur because of a combination of water pressure and pipe stresses in combination with environmental and local factors that causes pipe to weaken over time. Figure 6-8 shows how reduction of excess pressure in the Durban Central Business District (CBD) in Durban, South Africa, reduced not only the average main break frequency but also the seasonal variation. Excess pressure can be considered in simple terms as "the straw that breaks the camel's back" in exerting the additional stress to trigger many water main breaks. Similar reductions in service pipe break frequency were achieved in Durban CBD, and there are many such case study examples internationally.

As water pipes deteriorate over time from corrosion, traffic loadings, pressure transients, and other local and seasonal factors, the pressure at which failure occurs gradually reduces until at some point in time, breaks begin and break frequency starts to increase significantly. Pressure management rationale suggests that surges and excess pressures should be removed where possible to delay the operating pressure from attaining the point where the failure rate increases significantly, thus extending the life of the individual infrastructure components. Instead of the stress of high pressure causing a break, a lowered pressure (stress) will delay failures. Of course, the pipe will continue to deteriorate,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Source: Data from Ethekwini Water (South Africa)/JOAT/WLR and A Ltd

Figure 6-8 Pressure management changes water mains repair frequency in the Durban CBD in South Africa

and ultimately the lower pressure (stress) will be enough to cause failures if replacement or rehabilitation is not conducted prior to this time.

If a distribution system is "tight" with low background leakage that has undergone leak detection and repairs with a significant reduction of unreported leakage, it is likely more vulnerable to surges and excess pressure because the elimination of leaks, in effect, removes an unstructured form of surge relief that each leak in the past (openings in the pipe) provided. Without appropriate pressure control, a system with such a leakage reduction will operate as a tighter system and incur higher pressure, making it more vulnerable to surge. Therefore, it is important to consider pressure effects throughout the leakage management strategy, including a supplemental leak survey in the short term for vulnerable areas. In some leakage management projects, pressure controls are installed prior to the initial leak survey to keep pressures from rising after repairs are conducted.

The North American drinking water industry has experienced a growing concern in recent years about deteriorating water infrastructure and the looming high cost of renewing infrastructure to retain reliable water service in the future. In several prominent surveys conducted to gauge the extent of infrastructure needs, many of the water utilities based their condition assessments on the growing frequency of water main breaks as a primary factor in projecting near-term extensive replacement needs. An underlying assumption is that infrastructure replacement is the primary improvement option for long-term infrastructure sustainability. Pressure management can reduce main breaks and extend the life of existing infrastructure and defer pipe renewal, usually at a very modest cost. All pipelines eventually require renewal; however, pressure management has the benefit of ensuring that pipelines reach a full-service life with fewer failures before outright renewal is required.

Assessing the Potential for Leakage Reduction Through Optimized Pressure Management

The assessment process for proactive pressure management potential is like the process used in designing DMAs. In fact, in many cases, the design of DMAs with improved pressure management is conducted in a single comprehensive process where it makes sense to have the DMA boundaries coincide with the pressure managed area (PMA) boundaries. Several tasks should be undertaken to properly assess whether pressure reduction will be suitable for a DMA, zone, or distribution grid.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Desktop study to identify PMAs. Inspect maps or GIS records to identify areas of high/excessive pressure and lay out a preliminary configuration of a prospective PMA. The area must be controllable or easy to isolate from the general system naturally or by a series of valve closures. One or more PRVs may be used for controlled pressure reduction, especially if the emergency (fire flow) requirement is significant. At this stage, it is especially helpful to have input from the water utility field staff who operate the system daily because they are often aware of localized problems like poor fire flow, low carrying capacity, or partially closed valves, which need to be resolved prior to field measurements and analysis. Hydraulic modeling analysis is extremely useful to assess pressure impacts under various conditions expected to occur in the PMA.

Customer consumption analysis. Review customer consumption records to identify the categories of consumption: residential (indoor/outdoor), commercial, industrial, emergency, and so on. A primary focus of this analysis is to identify whether reduced pressures will negatively affect portions of customer consumption to negate the potential benefits of the pressure management to the level where it is no longer viable. A full year of consumption records should be reviewed to consider any significant seasonal variations in customer consumption such as irrigation and sprinkling functions.

Preliminary benefit–cost analysis. An initial estimated benefit–cost ratio should be calculated at this stage to identify the economic feasibility for real loss recovery in the pressure management scheme. Approximations relying on operator knowledge and hydraulic models can be used at this stage. In the model, fire hydrant flow can be simulated by adding a low friction pipe connected to a fictitious reservoir in which the hydraulic grade is set to a residual pressure requirement; the available fire flow would equal the total demand flowing from the object node to the fictitious reservoir.

A more detailed benefit–cost analysis can be conducted once field data are collected. A potential loss in revenue should only be included in the calculation if water conservation is not considered as an objective of the program.

Field flow and pressure measurements. If the desktop analysis indicates good potential for leakage reduction via pressure management, field measurements should be undertaken. It may be possible to simulate the zone by effecting valve closures and throttling the supply water main. Flow should be measured, and the daily, weekly, and when necessary, seasonal variations in water demand in the prospective PMA should be captured. Pressure should be recorded at the supply point to the potential district, the AZP point, and the point of lowest pressure or the critical point (CP). These measurements provide the data needed to perform the detailed benefit–cost calculation and serve as the design basis for the pressure management scheme. The data can be entered as inputs into hydraulic models or specialized pressure management models to predict loss control outcomes and benefits.

Enhancement of control methods and devices. The field data should be analyzed to decide on the type of controls and their configuration. The traditional pressure management control device is a PRV. PRVs are commonly placed in parallel on each supply feed to a PMA. One PRV is for normal service and a second on a bypass of large piping to provide higher flows under an emergency condition. In lieu of the bypass in the same location the second PRV can be installed at a location along the PMA boundary at the opposite side of the PMA. The second PRV is configured as a standby feed, set to open only when the outlet pressure drops to a preset low-level indicative of a high emergency flow condition. Once the control devices are identified, modeling can help confirm the best pressure control regime to obtain the optimum supply and leakage control conditions.

Final benefit–cost analysis. Once a proposed pressure management design has been assembled, the estimated costs of the project should be weighed against cost savings of the projected benefits. Often the main benefit is a direct cost savings from the reduction of lost water valued at the variable production cost (VPC). Usually, power and water treatment

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

chemicals are the main cost components; however, other costs may need to be considered. If a system is approaching maximum capacity, deferred or avoided costs to build a new pumping station or expand a treatment plant can be considered. The VPC may also be taken as the unit cost of imported water supply or a user fee if a utility purchases water from another utility or must pay for its water rights, respectively.

In addition to VPC savings, utilities find that break frequencies are often reduced after the implementation of pressure management. This can have a dramatic effect on infrastructure sustainability by extending pipeline life and containing infrastructure replacement costs. This will also provide a distinct direct benefit in reduced main break repair costs and damage to roadways and neighboring utility infrastructure. Deferred water main replacement or rehabilitation costs because of the extended life of the infrastructure can also be realized.

By following these steps, a rational planning approach to pressure management can be conducted, with anticipated levels of loss control benefits, costs, and impacts projected in advance of implementation.

Approaches to Optimized Pressure Management in Water Distribution Systems

Many approaches exist to incorporate optimized pressure management into water distribution operations, some of which are described below.

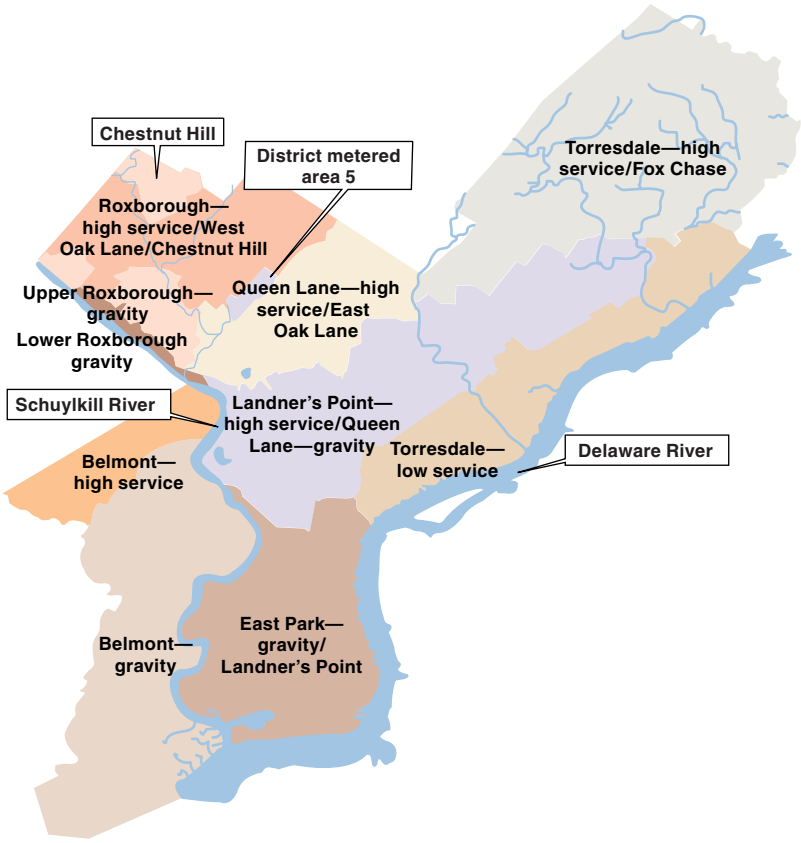
Pump controls. Pumps are common in water distribution systems and are typically activated and deactivated depending on system water demand that often includes maintaining appropriate elevated tank levels. Good pump control schemes incorporate a slow starting and slow stopping valve on the discharge side of the pump that inhibits the creation of transients in the distribution system, thereby minimizing risks of resultant leaks and breaks on system piping. Pumping systems employing variable-frequency drives (VFDs) can often meet widely varying water demands with fewer pump changes than systems with fixed speed drives. Improvements in hydraulic efficiency, such as use of VFDs, might also be accompanied by improved energy efficiency. It is likely that many water distribution systems experience daily surges related to pump activation and an opportunity for cost-effective refinement of pump operations exists in these systems.

Pressure relief systems. There are times when a sudden change occurs in the system outside normal operations. Power failures can cause pumps to stop suddenly along with disabling of electronic controls to regulate pressure. Malfunction of the control system can also occur occasionally. As a safeguard, the utility can install pressure relief valves that open to atmosphere to relieve excessive pressure that may occur, thereby inhibiting pressure surge.

Hydropneumatic tanks can be installed to dissipate surge from pump operations and other sudden increases in flow and pressure as well as to quickly provide extra water during a brief drop in pressure. Hydropneumatic tanks are connected at dead-ended branching pipes (including origin points like pump stations). They are pressurized with a combination of water and air. When pressure in the system drops, water flows from the base of the tank, and the air at the top expands. If the surge returns and sends water back to the tank, the air inside compresses to absorb the shock. The sizing of the tank requires careful design to be effective.

Pressure zones. As a result of variations in topography, pressure zones are established to ensure that minimum pressures can be provided to critical areas and to prevent over pressurization at lower elevations. Pressure zones represent the most basic and broadest level of sectorization of a water distribution system, with DMAs and PMAs being the finest level of sectorization. Figure 6-9 shows the pressure zone configuration for the Philadelphia (United States) Water Department's service area, with 11 pressure zones and

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



The Philadelphia (Pa.) Water Department tested advanced pressure management in district metered area 5, shown in this map of the utility's pressure zones, to determine how well low leakage levels could be achieved and maintained in its distribution system.

Courtesy of Philadelphia Water Department

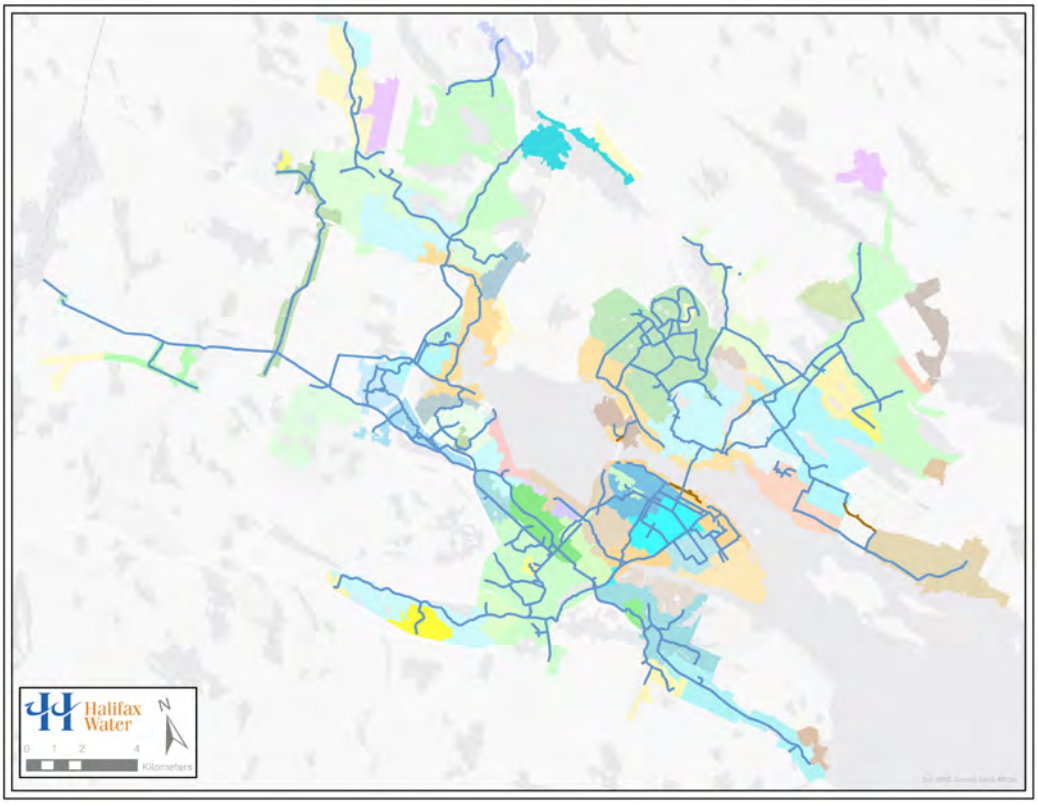
Figure 6-9 Pressure zones and DMA in the Philadelphia (United States) Water Department water service area

one DMA created within one of the zones. Figure 6-10 shows pressure zones in the water service area for the Ann Arbor, Michigan, United States, water system.

Systems with gravity feeds have zones usually configured based on ground elevations, and systems with pumped feeds have zones configured depending on the level of elevated tanks or storage reservoirs. Elevated tanks or ground-level tanks that set the pressure in a zone represent a challenge to adjusting pressure levels since they serve as useful storage for peak flows, fire flows, and emergencies such as significant water main breaks. The boundaries of existing pressure zones, storage tank operating ranges, and pressure variations within the zone should be well understood before undertaking a revised pressure management strategy.

Pressure-reducing valves. PRVs are commonly used in water distribution systems and other hydraulic applications. As featured in Figure 6-11, PRVs are designed to automatically reduce a higher inlet pressure to a designated lower outlet pressure and maintain the constant outlet pressure despite varying flows. This type of control is known as fixed outlet control. Separate electronic controllers or controls provided by manufacturers can be outfitted to PRVs to provide a range of additional control capabilities. Because topography can present significant challenges in providing consistent pressures in many

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of City of Ann Arbor, Michigan Public Works

Figure 6-10 Pressure zones in the water service area of Ann Arbor, Michigan, United States

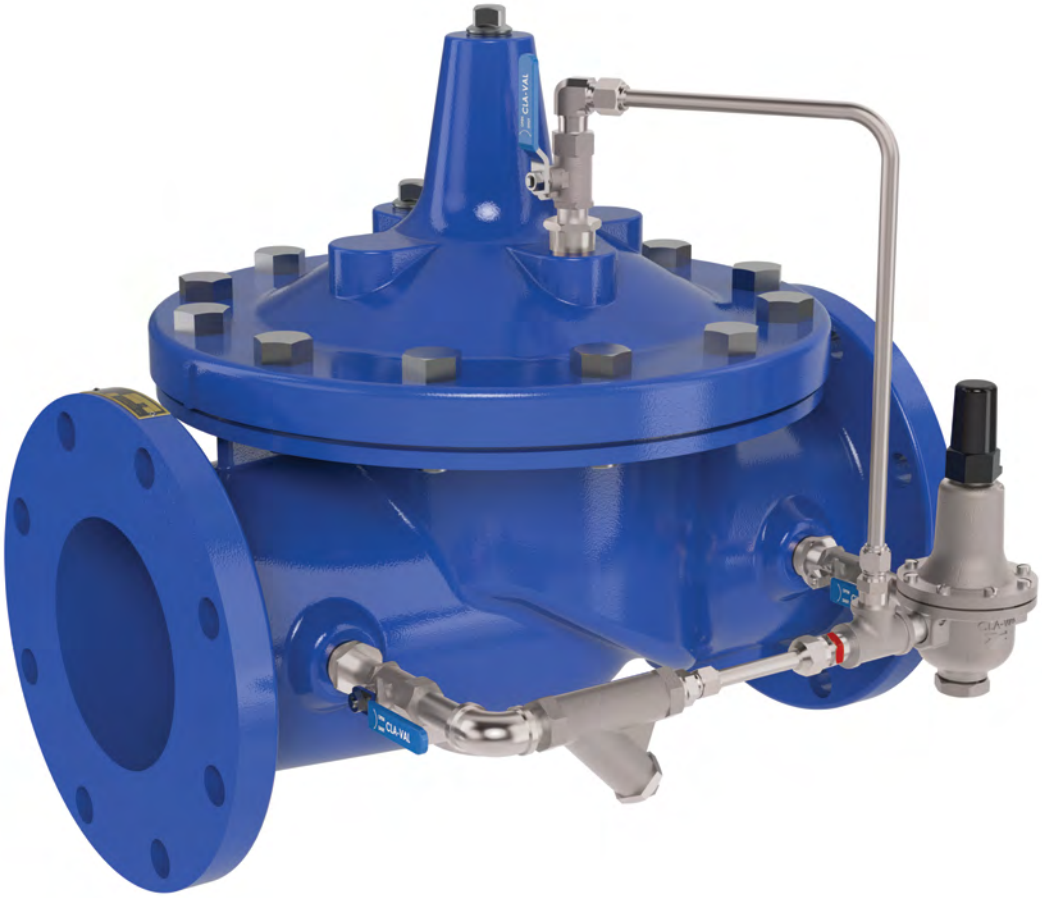
water distribution systems, PRVs are highly effective in reducing excessive pressures in certain sections of a distribution grid subject to widely varying pressure.

Different uses of PRVs for pressure management are covered in the following section.

Mechanisms for pressure reduction control using PRVs. Pressure reduction can be employed in various manners, each with advantages for certain applications. Selecting the appropriate level of sophistication usually depends on the distribution system condition, the components of loss, and the ability of the utility to maintain the equipment. Care should be taken when sizing a PRV or other control valve to check the potential head loss through the valve assembly, as well as gate valves, filter, meter, control valve, and pipe fittings. This is especially important when the pressure during the peak hours is low (as is often the case in systems with weak hydraulic capacity or small or corroded pipes) and modulated control is only desired during off-peak times. If care is not taken, supply may be constrained during peak hours, resulting in no-water or low-pressure complaints, and possible cavitation may occur. Traditional water distribution system operating schemes find, at times of minimal water demand, when the distribution system needs the least pressure, the pressure is usually higher. During peak demands, when the distribution system needs the most pressure to meet customer demands, the pressure is lower, creating an upside-down zone or district. Such a pressure-flow pattern is shown in Figure 6-12. However, a variety of pressure-flow control schemes exist to moderate or reverse this operating pattern, and these are discussed below.

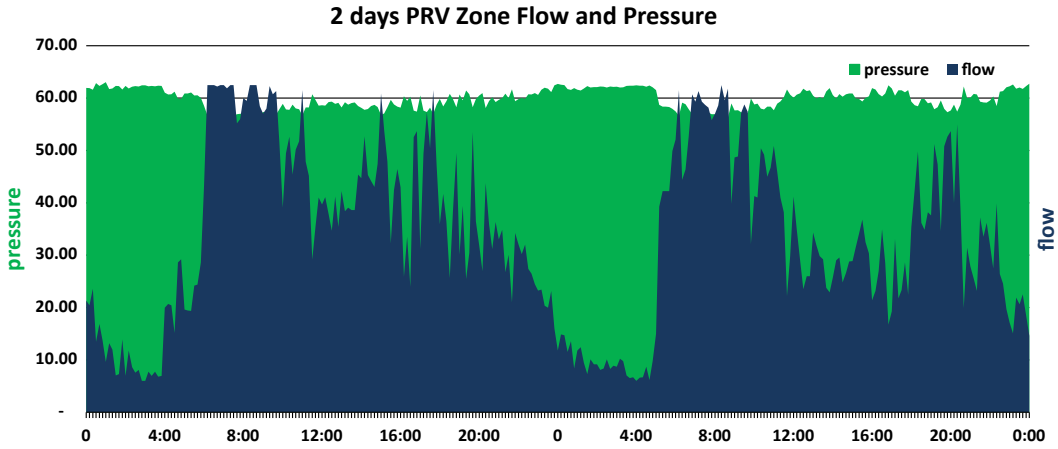
Fixed outlet pressure control. This is the traditional method of pressure control, typically using a hydraulically operated PRV or similar control valve. This method is effective in areas of uniform supply characteristics, pipelines with good flow-carrying capacity

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Cla-Val Company

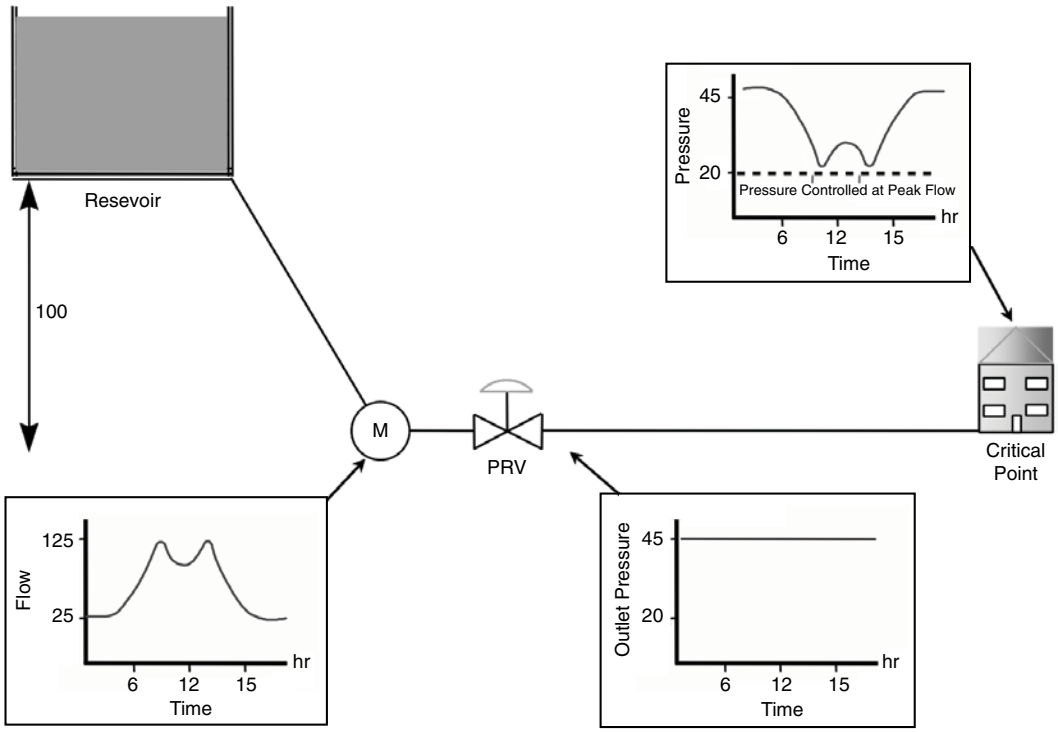
Figure 6-11 Pressure control devices, such as PRVs, provide consistent outlet pressure.



Source: Hughes 2014

Figure 6-12 Fixed outlet flow and pressure in zone

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Source: Thornton et al. 2005b

Figure 6-13 System schematic using fixed outlet control mode

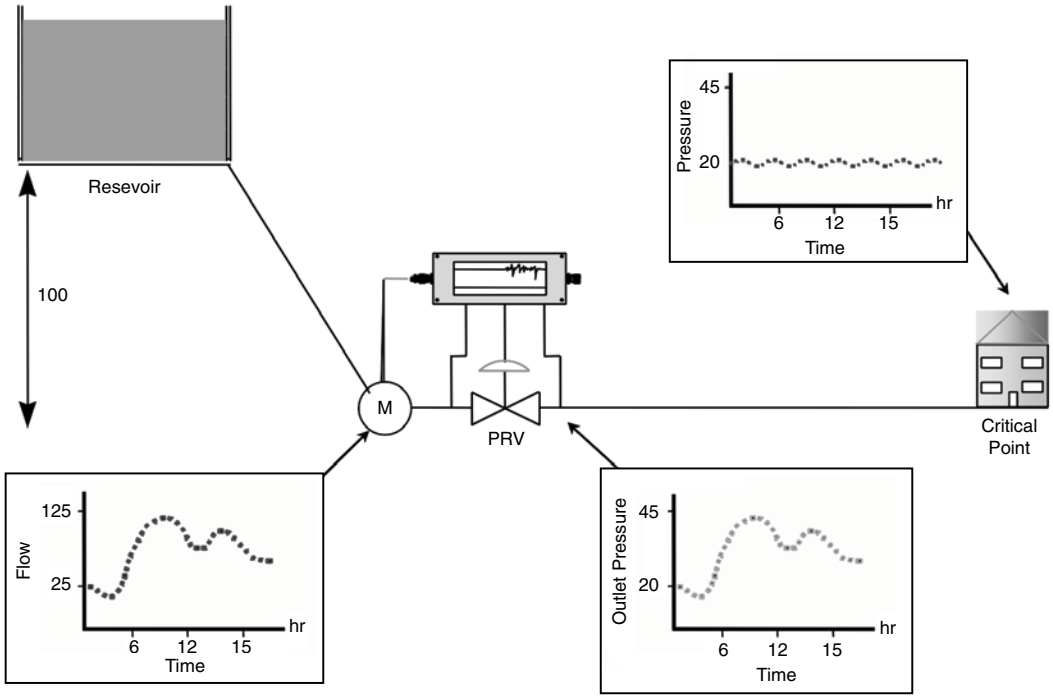
and low head losses, and water demands that do not vary greatly because of seasonal changes. This type of control is commonplace in North American utilities, albeit mostly because system designers and operators are generally not aware of the other control techniques and their benefits, as discussed below. The schematic shown in Figure 6-13 shows the effects of fixed outlet control as the PRV maintains a fixed outlet pressure (lower right) while the pressure at the CP (upper right) drops during the peak (daytime) period and rises during the off-peak (nighttime) period. This is largely due to lower frictional losses in pipes when flow is minimal. For systems operated in this manner, the minimal daytime pressure at the CP serves as the design minimal pressure level for the PMA, meaning notably higher pressure will occur at the CP and the remainder of the PMA for much of the time and pressure reduction gains throughout most of the PMA are minimal.

Time-based modulation. The pressure-regulating capabilities of a PRV can be modified by using a separate electronic controller with an internal timer connected to the PRV to regulate outlet pressure to preset levels at certain times of day. Control is affected in time bands in accordance with demand profiles. This methodology is very effective for areas with stable demand profiles and moderate pipeline head losses and is usually used where project cost containment is important but some advanced pressure management is desired. Time-based modulation controllers can be supplied with internal clocks or remote communication links to central control centers. Some manufacturers connect the controller to the pilot valve of the PRV and alter the set point of the pilot valve by introducing a force against the existing force of the pilot spring. Other manufacturers use a timer and a solenoid valve to reroute control through preset pilots. This type of control is not recommended for use with a single PRV as a sole means of advanced pressure control in North America because the timer will not respond to increased needs for high flows in an emergency

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

such as a sudden high fire flow demand. The use of a time-based controller can operate as the lead PRV with a second PRV at a lower pressure setting available to ensure flow can be delivered should pressure drop below that of the time-modulated unit.

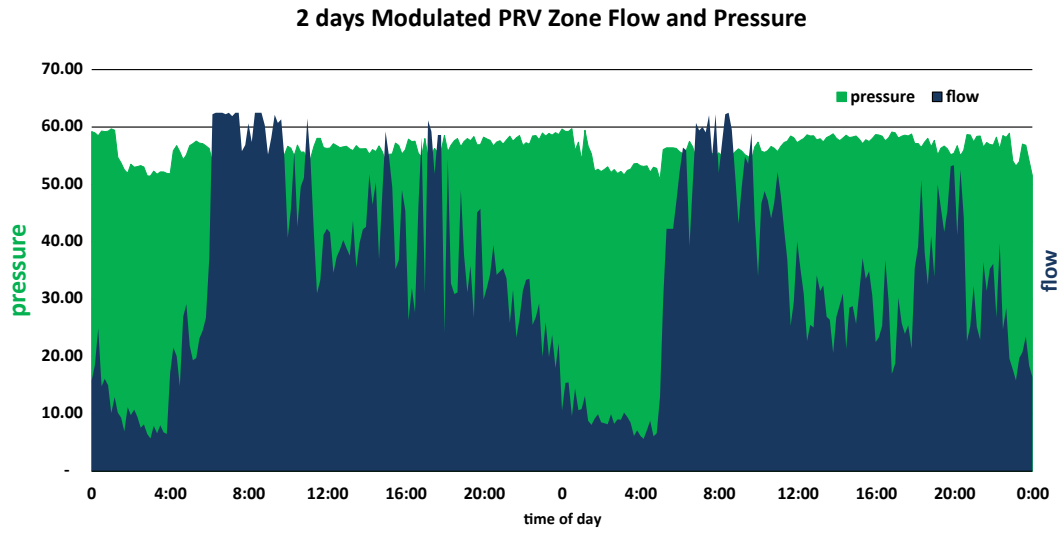
Flow-based dynamic modulation. This is a more efficient type of control for areas with changing conditions, pipelines with poor flow-carrying capacity and notable head loss, considerable fire flow requirements, and the need for advanced proactive pressure management to reduce leakage losses, conditions that exist in many North American water distribution systems. This type of control is implemented by controlling the outlet pressure in relation to the water demand in the PMA by connecting a separate electronic controller device to a metered signal output from a flowmeter measuring the water supply input to the PMA. As water demand increases, the controller increases outlet pressure. When water demand decreases, the controller reduces outlet pressure. Modulation of outlet pressure (within predetermined maximum and minimum settings) is achieved by altering the force against the regular hydraulic pilot spring of the PRV ensuring that, if the controller fails, the hydraulic pilot on the PRV will return the PRV to its highest hydraulic outlet pressure setting, thus providing a failsafe feature. The controller may be supplied with a local data logger or remote communications. Flow-based pressure modulation combats the effect of head loss in the system, ensuring that small diameter and structurally weaker pipelines and the CP receive a smooth constant lower pressure, as shown in the upper right of the system schematic depicted in Figure 6-14. Higher pressure is provided only when higher demand exists (routine and emergency), and lower pressure is provided when water demand is low and leakage is at a relatively high level. This control scheme converts the upside-down zone or district to a right-side-up PMA, and the pressure variation closely



Source: Thornton et al. 2005b

Figure 6-14 Flow-based dynamic modulation mode via a PRV combats high head loss in the distribution system and ensures a smooth pressure profile at the weaker points in the system.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Source: *Hughes* 2014

Figure 6-15 Using flow-based dynamic modulation mode results in PRV outlet pressure variation closely matching the water demand flow.

matches the flow variation as shown in Figure 6-15. The sidebar titled "Advance Pressure Management Using Flow-modulated Control" details the efforts of Halifax Water in Halifax, Nova Scotia, Canada, to successfully implement a flow-modulated pressure management scheme in an established DMA in its water distribution system.

Remote node control. This is implemented by controlling the outlet pressure of the PRV in conjunction with the pressure at a remote location or node in the area. The CP is often selected as the reference node. Alternatively, a location with a major customer or potentially large fire flow demand can be selected. This method requires the use of a communication link to continuously relay the pressure reading at the node or CP to the PRV site. This type of control is often affected with nonhydraulic electrically actuated valves of larger diameter.

An LCA shows that, in many cases, smaller-diameter mains and services that are usually found at the extremities of the system, often the CPs, have a higher failure frequency than the larger-diameter mains found at the entrance to most districts. In cases of very high break frequencies in small extremity mains, both the flow-based pressure modulation mode and remote node-based pressure modulation mode have the positive effect of reducing volumes of real loss and frequencies of new leaks and water main breaks.

A note of caution regarding manually throttled shutoff valves: Many system operators recognize the need for reducing system pressure at certain locations in their distribution systems. It is not uncommon for operators to throttle, or partially close, a gate or butterfly valve designed to serve as a shutoff valve to create a head loss and reduce pressure. This method is not recommended for gate valves, as the head loss created will change as system water demand changes and excessive wear can occur across the gate of the valve. Cavitation may also occur, a condition that risks additional damage to the valve.

Advanced Pressure Management Using Flow-modulated Control

Halifax Water (Nova Scotia, Canada) operates a large water system consisting of 932 miles of transmission and distribution mains that serve a population of 350,000. In 1999, as part of its drive to control leakage in the distribution system, the utility adopted the best practices for water loss control developed by the IWA and AWWA. Since then, Halifax Water has successfully transformed

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

55 pressure zones into 75 DMAs, incorporating approximately 110 pressure-control and flowmeter stations.

The utility has reduced its ILI from 9.0 to 2.5, which represents a reduction in real losses from 143 gal per service connection per day in 1999 to 44 gal per service connection per day in 2013. Annual system input volume reduced by 10.6 mil gal/day, reflecting a savings of $600,000 CAD per year in electricity and chemical costs related to water production.

More recently, Halifax Water has developed its pressure management strategy to take its water loss control program even further. As part of the Water Research Foundation Project Leakage *Management Technologies* (2007), Halifax Water utilized flow modulation to control pressure in the Dartmouth Central DMA. This successful strategy reduced water main breaks and leakage while maintaining fire flow and routine water service levels.

Dartmouth Central is a cascading DMA with two incoming feeds from a higher-pressure zone and three outgoing feeds to a lower pressure zone. The Dartmouth Central DMA has 37 miles of mains, 3,100 service connections, and 243 hydrants, and the average zone pressure is 80 psi.

The two PRV chambers feeding into the zone were configured for flow-modulated pressure control, and initial testing over several nights demonstrated that flow-modulated pressure control could reduce background leakage and that flow-modulated pressure reduction had little, if any, effect on customer consumption. Based on the initial tests, Halifax Water personnel decided to deploy flow-modulated pressure control in Dartmouth Central as a permanent pilot project.

Like many other utilities, Halifax Water's ability to deploy advanced pressure management was constrained by topography and the manner in which the water distribution system had developed over time. Halifax Water's systems are laid out in a series of cascading zones with unique hydraulic grade lines, stepping down through pressure-reducing facilities from the source to sea level. Also, because Halifax experiences low peaking factors and all piping is designed predominantly for fire flows, the hydraulic grade line is essentially flat across each pressure zone. Halifax Water's standard is to have minimum and maximum pressures under normal flow conditions of 40 psi and 100 psi, respectively.

Due to topography, within many of Halifax Water's DMAs pressures can be near the minimum service standard of 40 psi at the higher elevations and near maximum service standard of 100 psi at the lower elevations. This leaves very little excess pressure to reduce without customers noticing a difference. Therefore, the decision was made to reduce pressures in times of low flow, typically at night, and to allow pressures to fall below the minimum service standard while maintaining fire flows. Additionally, zones in which pressure was reduced had to be primarily residential or commercial customers, not industrial or institutional customers who use large volumes of water at night and might rely on a set service pressure. The Dartmouth Central zone fit the criteria, so the pressure was reduced by 11 psi in response to predetermined low-flow rates. The reduction was done in response to reduced flows and calibrated, so the pressure reduction generally started between 11 p.m. and midnight, with normal pressure resumed by 6 a.m.

The system operated in this manner for about one year and was mostly successful in achieving the intended objectives. Annual main breaks and service leaks were reduced by almost 50% (from 32 total breaks and leaks down to 17) in the year following advanced pressure management in Dartmouth Central, and minimum night flows were reduced from 2.8 ft^3 per hour to 2.5 ft^3 per hour. There were a few customer complaints related to lower nighttime pressure, but most of those who complained were amenable to the change when they understood the reasons for it.

One challenge with using flow-modulated pressure control on a DMA with two feeds was to feed the Dartmouth Central DMA simultaneously from two sources. In response to increased demand, one of the stations usually responded by increasing system pressure as flow through that station increased. The increase in system pressure was sensed by the second station, where a corresponding flow reduction occurred, which was interpreted by the PRV controller as a demand reduction. The second supply station responded to this perceived demand reduction by reducing system pressure, creating a cycle that eventually resulted in complete closure of the second, or lag-supply, station.

Various control settings were tested, but one supply station would eventually override the other. Programming the PRV controller at one supply station to function in time-based modulation with the other in flow modulation was also tried with limited success.

Following the trial period, Halifax Water staff identified the problem. In simple terms, the supply stations were connected hydraulically but were operating independently at the control level. In this mode, an increased flow through one supply station translated into a flow reduction in the other. To correct the problem, the controllers at each supply station would have to use the

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

same process variable: total DMA demand or total flow from both meters.

The solution was to develop a control algorithm that combined each meter's real-time flow into a single total DMA flow. The total flow value would be input into the PRV controllers at each station in place of the respective flowmeter values. In this way, controllers at each location would increase and decrease pressures together in response to total DMA demand. A peer-to-peer connection was established to allow required flow data transfer. The remote terminal unit at each site executes the algorithm and outputs total-flow demand to the PRV controllers. The algorithm incorporates fail-safes to accommodate possible communication losses and other technical scenarios with the fallback to fixed outlet pressure control.

Initial results have been positive. Communications are reliable, and flow updates to the controllers occur in less than four seconds. To accommodate local hydraulic conditions, controllers at each PRV can be configured at differing incremental levels. Since solving the control problem, operating results are more consistent than when the system ran initially in 2006.

Since permanently establishing this system in 2012, Halifax Water has been working to both refine the technology and evaluate other DMAs for implementation of advanced pressure management. This system has also relied on proprietary devices to hydraulically control the valves. While effective, the devices are expensive to replace and do not always provide the function required. Halifax Water developed its own hydraulic controller, which uses a programmable logic controller with control software developed by staff to operate solenoid valves on the control valve pilot piping.

This was initially tested at bench scale and is now installed in the Dartmouth Central DMA. This control strategy will allow settings to be downloaded remotely through the Halifax Water SCADA system.

Halifax Water initially thought this control methodology would be applicable to a small percentage of its pressure zones: those where water main breaks could be exacerbated by higher pressure and those with high background leakage. Since deploying this system in Dartmouth Central, Halifax Water has conducted additional research on the occurrence of water main breaks. This research indicates that in most DMAs, pressure does increase a small amount at night. This is either due to increased hydraulic grade line elevation because of lower flows and less friction head losses or alternatively (and more likely) from hydraulic grade line elevation increases because of PRVs that are unable to maintain a precise downstream pressure during very low nighttime flows. This research also revealed that approximately 80% of water main breaks occur at night when pressure creeps up. This suggests that the application of this pressure control methodology may have a much broader application, both within Halifax Water and to the water industry in general.

Installation of a Pressure Management System Using PRVs

PRVs have been used for many years to control the hydraulic condition of water utilities. With significant advances in control technology, PRVs have also become a very efficient means of reducing real losses in water distribution systems. As control options become more varied, however, it is important to ensure that proper sizing of PRVs and related equipment is conducted.

Care should be taken when sizing PRV or control valves so that the flows in the created PMA area do not fall below the minimum acceptable flow for the PRV at its operational settings after leakage has been reduced. It should be noted that diaphragm-type PRVs should operate within the valve manufacturer's recommended setpoints. If flows occur outside of this operating range, the PRV may have erratic control, with greater effects to the flow and pressure than when it is modulating in a nominal manner. This may result in either higher PRV maintenance costs or increased leakage. In situations of wide variation between high- and low-flow conditions, it is often common to install bypass piping and a second PRV around the main control valve to ensure smooth hydraulic control.

Once the PRV has been properly sized, the installation and type of control should be engineered to allow flexibility and ease of maintenance, ensuring that the investment will

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Halifax Water, Halifax, Nova Scotia, Canada

Figure 6-16 Typical pressure management and DMA supply installation

continue to pay off in the medium- to long-term future as well as providing a short-term benefit. Proper installation and startup of the pressure management installation are also critical to the success of the leakage reduction effort and sustainability of the results.

Important features of a good pressure management installation include the following items.

- A bypass piping arrangement or secondary supply feed to allow the primary supply feed/PRV to be feasibly taken out of service for maintenance work (If taking the PRV out of service would significantly increase pressure into the outlet zone, using a second PRV to sustain pressures during maintenance of the first should be considered [See Figure 6-16 in this chapter and Figure 5-9 in Chapter 5].)
- A mainline filter in the PRV to inhibit debris from entering the PRV and flowmeter if one exists
- A secondary filter on the PRV to protect the pilot assembly
- A flowmeter with suitable output signal (if flow-modulated control is desired)
- An air valve to release air from the PRV valve head
- Hand-operated ball valves on the PRV pilot circuit piping for easy isolation
- Gate valves on supply piping for isolation of mainline and bypass PRV feeds
- Inlet and outlet pressure gauges properly serviced for quick visual monitoring of inlet and outlet pressure

The PRV, piping, and related instrumentation should be housed in a secure, dry chamber protected from the elements that allows safe access for maintenance and calibration. The PRV may be installed on either the mainline or the bypass. However, designers of pressure management installations worldwide recommend installing the PRV on smaller-diameter bypass piping to supply routine flows and installing a second PRV on the larger-sized mainline piping to activate when needed to provide higher emergency flow (see Figure 5-9 in Chapter 5). This configuration often allows chamber access to be located off street thoroughfares, thereby facilitating easy access and safety for workers.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

If the installation is part of a DMA with a flowmeter, possibly all this equipment can be installed on the same bypass piping. The design concept that relies on two PRVs—one for smaller routine flows and one for larger emergency flows—can also be achieved with the PRVs on separate lines, perhaps at opposite ends of the PMA. The PMA configuration, water demands (particularly large water users), and system hydraulics will influence whether the two PRVs are best suited in a single location or split across two locations.

Special Design Consideration for Optimized Pressure Management

While the negative impacts of excessive pressures and transients are intuitively clear to utility operators, it is also common for the same operators to be apprehensive about reducing pressure, fearing that reduced pressures might generate customer complaints or impair firefighting capability. As discussed in the following sections, most of these concerns can be addressed by employing a competent pressure management design process. Such a design might include a gradual reduction in pressure by reducing the PRV setting in increments.

Regarding water distribution system pressure, how low is too low? From a reliability viewpoint, AWWA's Manual M31, *Distribution System Requirements for Fire Protection*, notes, "There is no such thing as a water system that is 100 percent reliable." Ruptures can occur at any time in any part of the system and reduce pressure. However, from a perspective of routine supply, Manual M31 also states that "a water system should be designed to provide some water at 20 psi," and 20 psi has become recognized as a customary, if not legally required, absolute minimum level of pressure needed in water distribution systems in the United States (AWWA 2008).

The primary concerns for water utilities in maintaining minimal water pressures are to (1) satisfactorily meet customers' varying water demands, (2) provide sufficient pressure for firefighting flows, and (3) to minimize the possibility of back siphonage of contaminants. The pressure determined to be the minimally designated service level requirement is ultimately determined in a case-by-case manner in individual distribution systems. By assessing these three design factors thoroughly, it is possible to define the low limits of the pressure reduction, below which system operation may be negatively impacted in some way. Following are some common questions raised by water utility managers when assessing pressure management and responses for competent design.

Will adequate fire flow capability exist in the pressure managed area?

Providing adequate water to fight fires is of utmost importance in North American water utilities. The design process should include a review of the types of buildings and potential fire risks existing in the area, as well as a review of prevailing national fire guidelines, like those from the National Fire Protection Association, and any state, provincial, or local building or fire safety codes that apply. Designing a PMA can provide the opportunity for utility operators to review fire risks in detail for perhaps the first time in many years.

The hydraulics of the PMA must also be carefully understood, and sufficient pressure and flow data should be gathered to understand the hydraulic conditions. Because routine water supply in a low-leakage distribution system encounters relatively low flows, and fire flows are generally quite high, the most common and effective design approach is to include multiple feeds controlled by separate PRVs. A larger feed can be installed to provide high-volume flows. This PRV is normally closed but will open when outlet pressure drops below a set point, representative of a large water demand on the system like a fire flow. Therefore, if there is a fire, the system has sufficient hydraulic capacity to maintain pressure and flow for firefighting. The PRVs will automatically regulate pressure as determined by the demand requirement plus the minimum safe

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

operating limit at residual conditions. The design should consider the amount of head loss occurring across the area served and ensure that the high-volume PRV supply can overcome the expected head loss to provide adequate pressure and flow across the zone. Head loss in pipelines can be significant, and the PRV must be specified and installed such that it will compensate adequately to supply distant locations of the PMA needing an emergency flow. Firefighters normally draw pressures down to 20 psi, but fire sprinkler systems resemble nonvolumetric demands. The design may place the emergency PRV in the same chamber as the routine supply PRV or as a separate supply point distant from the primary supply PRV if head loss is considerable or demands vary widely across the PMA.

From a design perspective, if adequate fire flow capability existed in the water distribution grid prior to the installation of a PMA, then adequate fire flow capacity can be engineered into the new configuration as well. It is merely a function of applying good research (of fire risks and regulations) and gathering sufficient hydraulic data to inform the PMA design. Sometimes improvements in the conveyance capability of the zone via enlarged or replacement water mains may be necessary to ensure that adequate flow and pressure can be maintained across the PMA under anticipated conditions.

Will customers consume less water at reduced pressures, thereby reducing revenue?

In residential buildings, more than one half of consumption typically occurs from uses that are volumetric, which means that water fills a tank or basin of a fixed volume so that the same amount of water is consumed regardless of the system pressure. Tank toilets, washing machines, bathtubs, pools, and other basins are common volumetric uses. Hence, reductions in customer consumption from reduced pressures are usually not nearly as significant as perceived. Water uses that are time based are affected by reduced pressure. Where outdoor water use for irrigation is a significant part of consumption, pressure reduction may have some impact on revenue. However, utilities with high rates of outdoor consumption are often located in areas where water is not a plentiful resource and reductions in irrigation use might be considered a desired conservation measure that is being matched with an appropriate water rate structure to moderate impacts on the revenue stream.

Many North American water utilities are developing water conservation programs and frequently tailor specific water rates as part of the effort. The cost of these programs incorporates the cost of lost revenue, which is usually less than the cost of development of new water resources and supply infrastructure. Pressure management can clearly assist a water conservation program by reducing distribution-side losses and direct pressure water use. Examples of pressure-influenced, nonvolumetric residential use include showers, dishwashers, and sink use that do not involve filling a basin. For water utilities with constrained water resources and water conservation programs, pressure management can be tailored to assist the reduction of water demands.

Systems with high leakage volumes will almost always see a positive benefit from reduced pressure, even when stacked against a potential loss of revenue, due to reduction of delivery pressure for metered consumption. This is also true for systems with lower losses and high costs to produce or purchase water. The trade-off in leakage reduction benefits gained vs. any reduction of revenue can be estimated and accounted for in the benefit–cost analysis of the leakage management strategy. In situations where a revenue loss is predicted and cannot be tolerated, pressure reduction can be limited to minimum consumption hours. During these times, legitimate consumption is at the lowest level, leakage is at the greatest proportion of the water demand, and system pressures are likely to be at their highest of the day.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Will hydraulic reliability suffer in pressure managed areas?

If good reliability exists in an "open" area of a water distribution system, adequate reliability can be designed into the PMA configuration. Typically, a primary supply feed providing routine flows coupled with a larger emergency feed should be adequate in most applications. A second emergency feed, possibly different from the PRV location (three feeds in total) can be added if circumstances dictate. Additional emergency feeds can be added in like manner if needed, but each additional feed brings with it the need for an additional PRV and increased complexity in the design. The utility can decide whether such seldom used emergency supplies merit a meter. If it appears that many feeds are needed to adequately supply a particular area, perhaps the proposed size of the area is too large, and the area can be subdivided into two or more areas. Pressure management or pressure reduction should be carefully designed when applied to large zones that include storage tanks or reservoirs or transmission mains that are responsible for transporting water through multiple zones. A calibrated hydraulic model is recommended to model the effects of pressure reduction on the system's ability to transport water from one point to another and make proper use of storage. Hydraulic models can be used to predict the function of an area under a pressure management scheme before it is put into use.

Can good water circulation and quality be maintained in the pressure managed configuration?

Water quality in any distribution system can be affected by a variety of factors, including the quality of the water leaving the treatment plant, piping materials, condition of the infrastructure, the status of valves, flow patterns and velocities, water temperature variations, storage tank turnover, and the use of flushing programs. A PMA may add concerns because the creation of a zone or DMA requires closing valves and creating boundaries. The reduction in leakage results in slower water velocities. Generally, water quality could suffer because of water age and poor circulation. Problems that might be encountered include red, rusty water in unlined ferrous pipe (where cast iron is common), loss of adequate chlorine residual, increased bacteria counts, and other impacts. Many water quality parameters are temperature dependent; good quality may be maintained during cool months of the year but may suffer during the warm summer months. Water quality data should therefore be gathered at different times throughout the year to determine the range of water quality variation in the PMA setting.

Each DMA/PMA configuration should be assessed on a case-by-case basis. However, several general steps can be taken to ensure that water quality is included as a primary factor in the design and operation of the PMA. First, the utility operator should gather baseline water samples and test for representative water quality parameters during the initial data-gathering phase of the project. Sampling can include gathering measures of chlorine residual along the proposed DMA/PMA boundary, both inside and outside of the zone. Additionally, several key sites within the proposed DMA/PMA should be selected for sampling, and a variety of water quality parameters gathered, including bacteria, turbidity, metals content, and other parameters typically collected by the water utility in its distribution system. One key site could be the location most distant from the primary supply source. Another might be a location with low water velocity or high water age. If problems exist in the distribution system, the utility may improve water quality by implementing pressure management. If good water quality exists in the open system, follow-up testing should be conducted after pressure management is implemented to determine whether any new water quality impacts emerged after the installation of the PMA.

Actions that minimize water quality problems might include configuring boundaries to minimize the number of dead ends. Utilities should install larger-diameter piping to serve as a supply "spine" through the central part of the area. They should also avoid creating areas of the grid with very low velocities. A flushing program may be considered if water quality problems are persistent. Small-diameter bleeder pipes can be installed as a

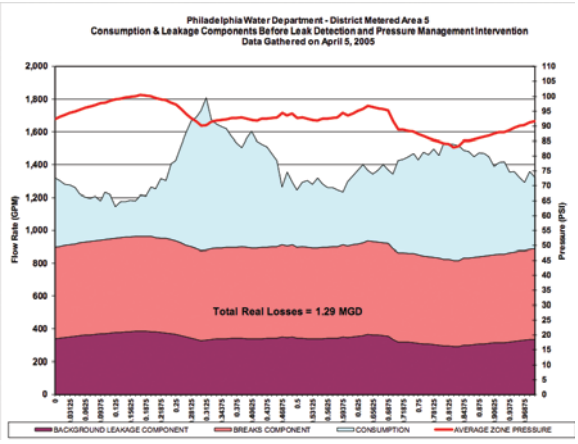
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

bypass around one or more boundary valves to eliminate dead ends, although this impacts accountability because some unmeasured flow leaves the area. If water quality is given sufficient attention at all stages of the project—planning, design, implementation, and operation—water quality is capable of being managed just as it is in an open distribution system. The reader is also referred to the discussion in the "DMA Planning and Design Considerations" subsection in Chapter 5.

Philadelphia's experience with pressure management. DMAs and PMAs can be successfully designed, installed, and operated in North American water utilities as confirmed by the experience of the Philadelphia Water Department (PWD) in the United States. PWD implemented such a system as part of the research project *Leakage Management Technologies* (WRF 2007) and followed with a detailed reporting in the *Journal AWWA* article "Piloting Proactive, Advanced Leakage Management Technologies" (Kunkel and Sturm 2011). PWD demonstrated the use of DMA and pressure management technologies in the North American setting and carefully completed and documented the steps necessary to achieve this kind of system. PWD identified an area of its water distribution system with average pressure over 100 psi that had a history of high leakage and water main break rates, the latter of which was driving a high level of water pipeline replacements in the area. After having previously studied four temporary DMAs, PWD established its first permanent DMA, known as DMA5.

The results of the work undertaken to establish DMA5 are shown over the course of time in Figures 6-17 through 6-20. The DMA resulted in an area of 12 miles of pipeline and approximately 2,250 customer service connections in a Philadelphia neighborhood that is largely residential and included a number of high-rise residential buildings. Figure 6-17 shows a hydrograph of baseline water supply conditions when the DMA was initially configured in 2005 and measurements gathered using portable instruments. The initial findings were startling; leakage within the DMA was approximately 1.3 mgd, roughly two-thirds the supply into the zone. This translated into a normalized leakage rate of 578 gal/conn/day.

Over the course of the next several months, a concerted leak detection survey was conducted, and the permanent DMA was implemented by constructing two chambers, one for a primary supply (with flowmeter and PRV) and one for an emergency supply at the opposite end of the DMA equipped only with a PRV set with a "trigger" pressure level to open only during an emergency condition of low pressure in the DMA. Once configured, an initial fixed-outlet reduction of 30 psi was implemented. Several customer service line leaks and two hidden water main breaks were uncovered during the leak survey work. A



Courtesy of Philadelphia Water Department

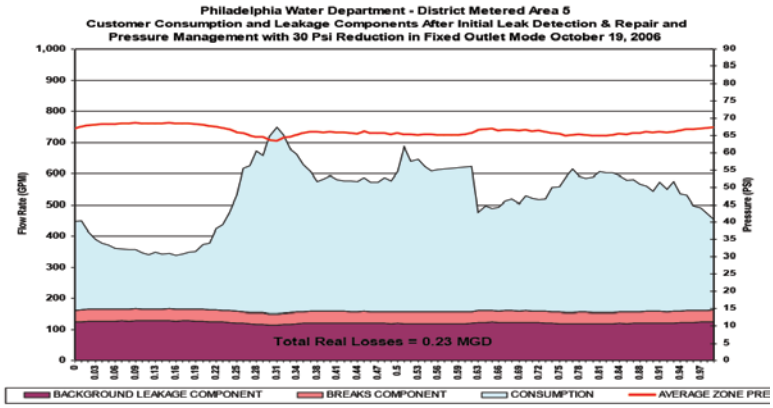
Figure 6-17 DMA 5 in Philadelphia baseline hydrograph of flow and pressure prior to DMA and pressure management

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

pronounced improvement in DMA conditions is shown in Figure 6-18 with supply into the zone greatly reduced. Further improvements are shown in Figure 6-19 once the final design adding a flow-modulated pressure management scheme was implemented in 2007. By 2009, after replacement of roughly one-half mile of piping was completed and another round of leak detection was undertaken, the DMA had reached what was judged to be its optimized state in terms of leakage and pressure management, shown graphically in Figure 6-20. At that time, the unit leakage rate had been reduced to 44 gal/conn/day, a reduction to less than 8% of the original leak rate. This project demonstrated that the use of leakage management technologies can produce meaningful and sustainable leakage reductions.

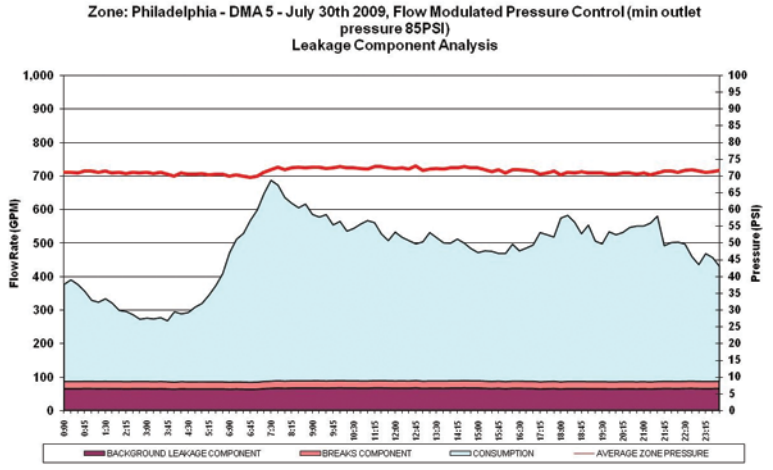
Infrastructure Renewal and Replacement

Even the best-maintained water distribution piping and infrastructure eventually serves its useful life and requires rehabilitation or replacement if it is to continue to provide reliable service. In managing water losses and maintaining infrastructure, water utility managers



Courtesy of Philadelphia Water Department

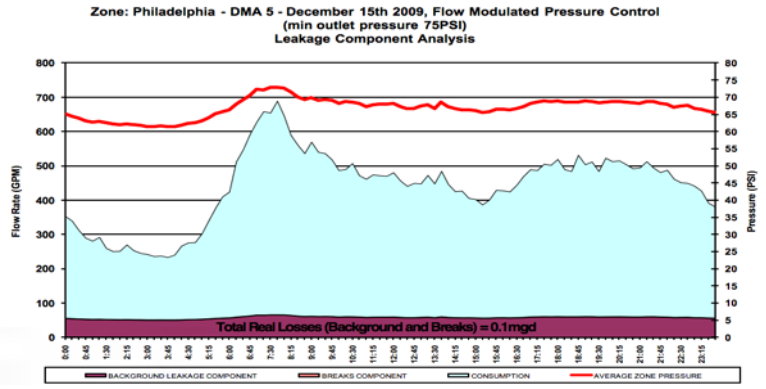
Figure 6-18 DMA 5 in Philadelphia after configuration into a DMA with basic pressure management and completion of leak detection survey



Courtesy of Philadelphia Water Department

Figure 6-19 DMA 5 in Philadelphia after refinement of pressure management strategy

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Philadelphia Water Department

Figure 6-20 DMA 5 in Philadelphia in an optimized state after final leak detection survey

can strive to ensure that infrastructure assets are maintained to attain their maximum life. Only then is the asset lifecycle optimized. Providing the appropriate balance of effective water loss control functions of active leakage control, pressure management, and optimized repairs will extend the life of piping assets to their ultimate range.

Many options exist for rehabilitation and renewal of water distribution system assets, and technology is rapidly advancing in this area. A detailed discussion of these methods is beyond the scope of this publication; however, AWWA offers excellent guidance on long-term infrastructure upkeep through a variety of publications in its Manual of Practice series. The reader is referred to the following AWWA manuals:

- *Concrete Pressure Pipe* (M9)
- *Steel Water Pipe: A Guide for Design and Installation* (M11)
- *PVC Pipe—Design and Installation* (M23)
- *External Corrosion Control for Infrastructure Sustainability* (M27)
- *Rehabilitation of Water Mains* (M28)
- *Ductile-Iron Pipe and Fittings* (M41)
- *Fiberglass Pipe Design* (M45)
- *PE Pipe—Design and Installation* (M55)
- *Condition Assessment of Water Mains (M77)*

Replacement and many structural rehabilitation techniques create renewed pipeline assets. Although this is ultimately necessary for all pipeline assets, it is also the most comprehensive, costly, and involved of the pipeline management options. Therefore, every effort should be made to extend the service life of piping to the ultimate level before renewing it. Historically, the only ultimate pipeline option was outright replacement with a new pipeline. This meant full trench excavation for new pipelines, disruption of traffic, noise, dust, and considerable disruption and nuisance to the surrounding area. Pipelining techniques have also been applied for many years, but these techniques typically only restore flow-carrying capability without providing structural rehabilitation. In recent years, many innovative techniques have evolved to provide structural renewal of existing pipelines in place. The advent of trenchless technologies has given water utility managers many more options in structural pipeline renewal.

Today's trenchless technologies include slip lining with structural and semistructural liners, pipe bursting, cured-in-place liners, and a variety of similar methods. The primary advantages of the trenchless methods include less excavation and aboveground

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

disruption. Also, because the existing pipeline is rehabilitated in place, less conflict arises with neighboring utilities as opposed to pipeline replacement, which usually requires installation of piping in a new location in the street or right-of-way while the existing piping is abandoned in place, meaning twice the lay length is consumed in the right-of-way. Trenchless methods may be the only practical means to rehabilitate pipelines existing in difficult locations like deep underground crossings of rivers, streambeds, railways, or interstate highways. The direct costs of trenchless methods can be more or less expensive than full pipe replacement, depending on the specific technique, the number of service connections to renew, logistics, and local restoration requirements. However, these techniques give utility operators greater versatility in designing the rehabilitation and renewal program, which is a necessary part of water utility management.

Many water utilities view water main replacement as the key to water loss reduction; however, in many cases, larger annual volumes of real loss are often recovered via customer service connection piping replacement programs than outright main replacement. The LCA will indicate where the largest real loss volumes can be recovered.

Rehabilitation and renewal form one of the four pillars to the successful control of real losses in water utilities, and it is essential that water utilities have a program to renew their infrastructure as it reaches the end of its service life.

Speed and Quality of Repairs: Optimized Leak Repair Functions

Active leakage control, which includes leak detection surveys and the ability to quantify leakage rates from continuous monitoring of minimum-hour flows in DMAs, alerts the water utility operator to the occurrence of leaks in the water distribution system. Once leakage is known to exist at a specific location or across a region of the water distribution system, interventions must be undertaken to abate or reduce the leakage. Leakage interventions should be

- **Timely**—Leak repairs should be implemented soon after the discovery of the leak to minimize leak run time and to contain disruptive effects of the leakage, thereby maintaining good customer relations and avoiding unnecessary liability. Where possible, repairs should be scheduled during favorable working conditions and during the normal working hours of staff.
- **Reliable**—Because of poor workmanship or inferior materials, many leaks recur at the site of previous leak repairs. Repair work should be executed with quality in mind, providing a lasting improvement to the water pipeline.
- **Cost-effective**—Leak abatement programs should be economical, with the annual costs of the program not exceeding the direct savings of the recovered leakage, along with indirect savings of less disruption, deferred infrastructure expansion, and similar savings. Additional factors, such as water quality, regulatory requirements, or reduced reliance on imported water supplies, should guide decisions to repair or replace pipelines.
- **Well documented**—The success of the leakage management strategy cannot be determined until leakage has been corrected and information on the nature of the leak obtained. The repair action is often the critical step in obtaining this information; therefore, a detailed, systematic documentation procedure should be employed to manage the important information to be collected.

Traditional leak repairs have several important steps, including excavation of the pipeline, physically repairing the leak, collecting information and documentation, and restoration of the street or ground cover above the pipeline. These steps are detailed in the following sections.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Excavating the leak. Before beginning any excavation, other utilities and/or the appropriate one-call underground service alert center must be contacted so that other underground infrastructure can be located and marked out by the owner/operator of that infrastructure.

As with any excavation, proper trench shoring techniques should be used at all times. When soil is saturated with water during a leak, it frequently becomes more unstable, so the protection of the crews working on the leak is paramount. Utilities are reminded to follow occupational health and safety regulations for trench shoring at all times.

Water distribution systems are composed of buried pipes, and unless leaks are visible in underground chambers or access hatches, leak repairs typically require excavation to expose the leaking section of pipe. The leak detection crew and the repair crew should work together to uncover the leak.

If the excavation is dry—meaning the pipe is not leaking at this location—the leak survey crew should sound the exposed pipe at this location and assist the repair crew in pinpointing the leak. Leaks emanating from the bottom of the pipe can be easily overlooked, and effort should be made to excavate around the full circumference of the pipe to confirm any such leakage. Sometimes a leak source can exist but give no visible sign of dampness. It should be noted that locations where water is visible or surfacing may not be the location of the leak source. Water from a leak may travel a significant distance from the leak source via underground conduits or undermined soil. Excavating a site based solely on the fact that it is the location of visible water can be a wasteful effort leading only to an intact pipeline. The location of the excavation should be based on the pinpointed leak source from the leak detection activity.

By working together, the leak detection and repair crews can share knowledge and experience that make locating the leak easier. Uncovering leaks requires careful excavation to avoid contacting neighboring underground utilities.

During the repair process, care should be taken to minimize the risk of contaminants like muddy water or soil from entering the pipeline. If a pipeline must be depressurized to perform the leak repair, the operators should maintain positive pressure in the pipeline until the excavation is complete and dewatering systems are installed in the excavation. Typically, a dewatering sump is created within the trench so that residual water from the pipeline can be pumped out of the excavation while the leak repairs are being completed in order to prevent any dirty water from entering the pipeline.

Local and state stormwater regulations should be adhered to which may require sediment control systems and/or dechlorination of water that leaves the leak site.

Measuring and estimating losses from discovered leaks. Obtaining a measure of the amount of water lost from leaks is important to gauge the success of the leakage management program and provide data for calculation of real losses in the LCA. For larger volume leaks or outright ruptures, distinct changes in metered flow may be registered in DMAs, on SCADA systems, or master meters at water treatment plants, tanks, pumping stations, PRVs, or other existing metering locations. Information from hydraulic models, transmission main flow gauging, and fire flow tests can be assessed to help distinguish routine flows from higher demand flows from main breaks or large ruptures. If DMA flow data are unavailable, flow rates can be estimated. Leakage rates from various leak types are shown in Tables 5.2 through 5.4 in Chapter 5.

Leak repair techniques. Pipe failures occur in many ways, and repair methods are equally numerous; thus, this discussion cannot offer an exhaustive account of all repair techniques that are available. Instead, a few of the most common repair techniques are mentioned. The water utility operator or manager is ultimately responsible to ascertain the appropriate repair technique for any given leak condition based on the nature of the leak, the pipeline materials and construction, how the pipeline is situated (heavily trafficked road, congested underground utilities, excessive depth of cover, etc.), and hydraulic priority. Utility personnel are urged to confer with pipe manufacturers, engineering

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of Halifax Water, Halifax, Nova Scotia, Canada

Figure 6-21 Repair clamps are commonly used to repair circumferential ruptures on distribution piping since they are quick to install and are highly durable.

consultants, AWWA, or other trade organizations to obtain information on the best repair technique and materials for a given repair project. The AWWA publication *Managing Water Main Breaks Field Guide* (2012) offers detailed guidance on repair methods and procedures.

The level of complexity of leak repair is usually commensurate with the severity of the failure. Following are several examples of typical repairs and considerations.

- Small leaks of a few drips per minute can occur from loose packing on a valve or pump. Simply tightening the bolts on a packing gland might quickly resolve this type of leak.
- Customer service connection leaks occur frequently in water utilities. Depending on repair policies, the water utility or the customer may arrange for a repair to a leak (replacing the damaged section of otherwise sound pipe) or outright replacement if there is evidence that the entire service is deteriorated.
- One of the most common repair techniques for small- to medium-sized pipelines is the use of repair clamps to repair breaks, as shown in Figure 6-21. These devices can be quickly installed to repair both reported and unreported breaks and are reliable for many years of service. Note that in higher pressure systems, repair clamps may be less effective, so they may be suitable for the temporary control of a leak while more permanent repair project is planned and executed.
- Pipeline joints are often the site of smaller leaks occurring because of worn joint materials, uneven settlement of pipe lengths, traffic loadings, and similar causes. The type of repair depends on the type of existing joint. For larger pipes, clamps specifically designed to encapsulate the bell are available. Some joints can be recalled, while others, such as split bell-ends of pipe, may need to be cut out and replaced by coupling in lengths of straight pipe.
- For larger ruptures or splits in pipelines, the effective repair may require cutting and extracting the damaged section of piping and installing one or more lengths of new pipe coupled or connected to the existing pipeline.
- Specific pipe materials require specific repair techniques. Many leak repairs on steel pipe, which is highly susceptible to corrosion, can be repaired by welding. Different plastic pipe materials are used in the water industry, and repairs require specific tools and equipment to complete them.

A variety of repair techniques are possible, and the potential for water utility personnel to innovate their own repair method is always present. Many unusual pipeline

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

configurations exist, particularly in older systems, so the rule to "expect the unexpected" applies. Weighing the costs associated with making multiple repairs on the same pipe segment against costs to replace that segment can assist the water utility in developing a sound basis for planning its leakage management program. Again, regardless of the repair method, leak repairs should be timely, reliable, and cost-effective to sustain the full benefits of the leakage management program.

Customer service connection piping leak repair policy. Usually, the water utility has responsibility for the connection piping from the water main to the meter (if outdoors in a meter pit) or at a curb stop or property line (if the meter is located indoors). A small percentage of water utilities assign customers ownership (and leak repair responsibility) of the entire customer service connection piping branching from the water main. The ability to contain leak run time is critical to an effective leakage management strategy. It is feasible for water utilities to operate customer service connection piping leak repair programs that efficiently implement repairs in two to four days after a leak is discovered. For most customer-arranged repairs, response time typically averages several weeks. The longer leaks run, the greater the leakage losses.

Leak repair information collection and documentation. During and after leak repairs, it is important that information is gathered and documented regarding the nature of the leak or break, the repair method, the underground conditions, street, weather, and costs. Information should be recorded on leak repair reports. Photos are helpful. This information is needed to conduct an LCA (see discussion in Chapter 5) to keep appropriate records for legal purposes, as well as to identify leakage trends and distribution system condition and to track the performance of utility staff. It is useful to determine the proportion of failures occurring on mains, fittings, and service connections. Additional information might be collected depending on local conditions and priorities. Possible parameters might include the time that a repair crew was called in, the time periods that water service was disrupted and restored, paving requirements, valve closures to execute shutdowns, fire hydrants operated, chlorine residual, and other useful information. In areas where external corrosion is a factor, measurements of pipe-to-soil galvanic potential as well as grab samples of soil for corrosivity lab testing can be helpful to inform a corrosion control program. Having data in an electronic database format allows for subsequent analysis of the types of pipes that fail, the possible causes, and their locations. Location data are extremely useful in making future decisions about pipe renewal priorities.

As an example, readers are urged to review the "Sample Leakage Management Plan," "Leak Detection Survey Daily Log," and "Leakage Management Program Cost-Effectiveness" sidebars following the summary of this chapter.

SUMMARY

All water distribution systems encounter leakage, with rates varying according to the conditions of the individual systems. For a water utility to be truly efficient in its operations, it must proactively manage leakage to contain it to economically low levels. Merely reacting to reported leaks when they become visible and disruptive means that a growing backlog of unreported and background leakage will plague the utility, wasting water resources and inflating production costs.

Many water utilities worldwide have moved beyond basic leak detection and repair to employ comprehensive and holistic leakage management programs. These programs not only seek to quickly identify and pinpoint existing leaks but also to better sustain water infrastructure by containing the rate of occurrence of new leaks and breaks. Such approaches are cost-effective for the water utility, fair to the customer ratepayers, and reflect good stewardship of water resources. Many innovative methods and technologies exist, and water utility managers have a great opportunity to create a new era of water-efficient operations in the North American water supply industry.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Sample Leakage Management Plan

Name of Water Utility: *County Water Company (CWC)* Date: *1/18/2020*

I. Leakage Management Approach

After completing CWC's first annual water audit and LCA, the CWC manager creates an ongoing leakage management program that (1) reduces the potentially recoverable leakage identified from the water audit and LCA and (2) sustains lower leakage conditions once initial leakage reductions are achieved.

The LCA conducted for CWC estimated that 20.4 mil gal of real losses exist as potentially recoverable leakage (see Chapter 5, sidebar "CWC: Part 6"). The initial economic intervention analysis concludes that, once the target leakage has been achieved, around one-third of the system should be checked each year on an ongoing basis.

The CWC manager also plans to improve pressure management in a portion of the CWC service area that is providing water pressures much higher than CWC's average level of 65 psi. Optimized pressure levels will be used to cut background leakage and reduce new water main break frequency in this area.

II. Leak Survey and Repair Plan

A. Leak Survey Area and Frequency

A-1. Based on records of previous leaks, type and age of piping, soil conditions, high pressure, and faulty installation practices, list the portion of the distribution system to be surveyed. List the survey frequency.

List the percentage of system to be surveyed each year: *33 percent*. Cover the entire system every three years. List frequency of surveys: *every year during spring to cover 33 percent of the distribution system.*

Describe each area to be surveyed under item B-2 of this plan.

A-2. Total miles of main to be surveyed: **(.333)(256) = 85.2**

When calculating pipeline mileage, include the total length of pipe and exclude customer service connection piping. If only a portion of the system is surveyed, calculate the benefit–cost ratio for only the portion surveyed.

A-3. Average number of miles surveyed per day: **2.0**

Typical survey crews can survey about two miles of main per day. Factors include distances between services, traffic/safety conditions, and availability of listening contact points. Explain if more than three miles per day are surveyed: *Assume 2.0 mi/d using a comprehensive mix of manual survey and leak noise loggers.*

A-4. Number of working days needed to complete survey (divide line A-2 by line A-3): **42.6, use 43**

A-5. Describe personnel deployment: *Two-person crew performs leak survey in spring each year and assists distribution repair crews during colder season of year when the number of leakage events increases.*

B. Procedures and Equipment

B-1. Describe the procedures and equipment for detecting leaks. The best results are obtained by a comprehensive leak survey technique: listening for leaks at all system contact points (such as pit water meters or curb stops, valves, hydrants, and blow-offs).

Purchase leak detection equipment, including electronic listening devices and eight leak noise loggers. Attend manufacturer training seminars and state training. Listen on all contact points except Downtown area.

B-2. Describe why the areas noted on the map in Step A-1 have the greatest recoverable leakage potential.

Year 1	*Year 2*	*Year 3*
The Valley District (area of high pressure)	*Remainder of Downtown area*	*Remainder of the system*
One quarter of Downtown (old ductile-iron mains)	*Steel mains more than 40 years old*	

B-3. If listening for leaks will not include all contact points, describe the plan for detecting leaks.

Rotate leak noise loggers in Downtown for nighttime listening; high traffic noise prevents daytime surveys.

B-4. Describe the procedures and equipment to be used to pinpoint the exact location of detected leaks.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Use low-frequency ground microphones to listen over pavement surfaces. Deploy leak noise loggers in Downtown area for night surveys. Use consultant with leak correlator for difficult leak pinpointing.

B-5. Describe how the leak detection team and the repair crew will work together. How will they resolve the problem of excavations of suspected leak locations that prove not to be the leak source (dry holes)?

The leak detection crew and the repair crew will jointly excavate all leaks for the first three weeks and resolve any dry holes thereafter.

B-6. Describe the methods to be used to determine the flow rates for excavated leaks of various sizes.

Use Table 5-2 (Chapter 5) to estimate leakage rates for all types of leaks. The pressure adjustment equation will be used to calculate leakage rates for the level of water pressure encountered at each leak location.

C. Staffing

C-1. How many utility staff members will be used? **2**

Staffing costs including wages and benefits: (NOTE: Night staff may require a higher wage.)

Person 1:	$/hr	*23.20*	$/d	*185.60*
Person 2:	$/hr	*15.75*	$/d	*126.00*
TOTAL			$/d	*311.60*

C-2. How many consultant staff members will be used? **1**

Cost of consultant staff:

Person 1:	$/hr	*75.00*	$/d	*600.00*
Person 2:	$/hr	*0*	$/d	*0*
TOTAL			$/d	*600.00*

D. Annual Leak Detection Survey Costs to Cover One Third of the Distribution System

Leak Detection Surveys	$/d	# of Days	Cost, $
D-1. Utility crew costs:	*311.60*	*41*	*12,776*
D-2. Consultant crew costs:	*600.00*	*16*	*9,600*
D-3. Vehicle costs:	*12.00*	*41*	*492*
D-4. Other:	—	—	*0*
D-5. Total survey costs:	—	—	*22,868*

E. Leak Detection Budget

E-1. Cost of leak detection equipment: *	$12,000	(Initial Cost)
E-2. Leak detection team training:	$3,000	(Initial Cost)
E-3. Leak detection survey costs:	$22,868	(Recurring Cost)
E-4. Total leak detection costs:	$37,868	(First-Year Cost)
	$22,868	(Year 2 and 3 Costs)

* Eight leak noise monitors and other electronic equipment

F. Leak Survey and Repair Schedule

Indicate realistic, practical dates:

F-1. When will the leak survey begin?	*Mar. 1, 2020*
F-2. When will the leak survey be completed?	*Aug. 6, 2020*
F-3. When will leak repairs begin?	*Mar. 15, 2020*
F-4. When will leak repairs be completed?	*Aug. 27, 2020*

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

III. Pressure Management/District Metered Area Plan

The average distribution system pressure for CWC is 65 psi; however, a lower elevation area known as the Valley District has an average pressure of 95 psi. The Valley District was developed 30 years ago and comprises 25 miles of pipeline or approximately 10 percent of CWC's total of 256 miles of pipeline. This area of the distribution system is supplied about 1 mgd on an average daily basis throughout the year, with a peak day of about 1.6 mgd. The Valley District also includes about 5 miles of plastic piping that CWC piloted 25 years ago. As detailed in the "Water Pressure and Leakage" section in Chapter 5, failures on plastic pipe follow a variable path failure mode with high N1 exponents, meaning that leakage rates change rapidly with pressure. With high pressure and plastic piping, the Valley District's infrastructure incurs significant background leakage that has recovery potential via optimized pressure management. The LCA shown in the Chapter 5 "CWC: Part 6" sidebar estimates 211 mil gal of background leakage and 20.4 mil gal of potentially recoverable leakage in the CWC system, with a proportionally larger volume of this leakage likely to exist in the Valley District. The CWC manager employs pressure management in this area in addition to ongoing leak detection surveys as the key component of the new CWC leakage management plan. The manager also reviews the break frequencies before and after pressure management to assess to what extent these may have been influenced by the pressure management strategy. The Valley District might ultimately require a different frequency of leak survey intervention than the other areas of the CWC water distribution system.

When considering a new pressure management program for the Valley District, the CWC manager arranges for an engineering assessment to be conducted via the use of a calibrated hydraulic model to evaluate performance implications in advance of conducting actual system changes.

The Valley District is partially isolated from the larger CWC service area by natural boundaries of state park land and a railroad. The Valley District is supplied via four distribution mains size 10 in, 8 in, and 6 in (2). By closing the two 6-in supply mains, the Valley District can be configured into a DMA, or more specifically, a PMA. PRVs will be installed on the 10-in and 8-in mains. With this configuration temporarily established, baseline flow and pressure data are gathered using portable instruments. Typical daily flow ranges suggest that the Valley District DMA can be served by the 8-in main routinely, with the 10-in main providing supplemental flows in case a higher flow is needed due to a fire or another emergency. The PRV on the 10-in main is set to open at a designated low-trigger pressure level indicative of a high-flow, low-pressure emergency event. A flowmeter will be installed on the 8-in main at an acceptable length downstream of the PRV. An electronic controller will be installed at this primary supply site to allow flow-modulated pressure control to be used for optimal pressure management. To keep startup costs reasonable, CWC determines to keep data collection local, with CWC technicians visiting the primary inflow site every two weeks to download data from the electronic controller that stores historical data. Minimum-hour flow data will be analyzed to determine leakage trends. Projected costs to establish the DMA with pressure management capabilities include

		Costs
Pressure-reducing valves:	8-in. diameter $2,200 10-in. diameter $4,100	$6,300
Flowmeter (electro magnetic)	8-in. diameter $7,000	$7,000
Electronic controller	$7,000	$7,000
Precast manholes (2)	$1,200/manhole	$2,400
Misc. piping and hardware	$500	$500
Construction: labor	3 workers, 5 d $24/hr × 3 workers, 8 hr/d × 5 days	$2,880
Equipment: truck	$125/d × 5 days	$625
	Total cost:	$26,705

IV. Leakage Management Plan Summary

Plan cost = leak survey cost + pressure management cost = $37,868 + $26,705 = $64,573, use $65,000

The LCA for CWC determined the economically recoverable unreported leakage of 20.4 mil gal. With a VPC of $750 per mil gal, the potential savings from the recovery of this volume of leakage will be $15,450 annually. The LCA quantified approximately 211 mil gal of targeted background leakage. With a majority of CWC's background leakage likely occurring in the Valley District where pressure management will be employed, it was calculated that background leakage can be reduced by one-half, or 105 mil gal, at a cost savings of $78,750. A total savings from leakage reduction is estimated at $15,300 + $78,750 = $94,050.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

CWC's leakage plan first-year cost of approximately $65,000 is well under the projected savings level, giving it a projected cost-effective payback of less than one year. After the first year, the leakage survey costs will reduce by approximately $15,000 per year as one-third of the system is subsequently checked each year, and the savings from the pressure management scheme will continue year to year without recurring costs, confirming that the plan is economical to undertake. It has been determined to move forward with this plan and reevaluate it after the first- and third-year intervals.

Prepared by: *C.M. Biggs, Manager* Date: *Jan. 18, 2020*

Leak Detection Survey Daily Log

Name of Water Utility: *CWC* Date: *Apr. 17, 2020*
Leak Detection Team Members: *Lloyd Williams and Raymond Smith*
Equipment Used: *Leak noise loggers and ground microphone*
Area Surveyed: *7* Map Reference: *Water Distribution Map*
Street and Block Numbers: *San Antonio, San Gabriel*
Page & Coordinates: *San Juan, San Carlos, San Luis, San Miguel 8600 Block*

Leak Number	Location or Address of Suspected Leak	Utility or Customer (U or C)	Leak Pin-pointed? (Y or N)	Leak to be Rechecked? (Y or N)	Confirmed a Leak? (Y or N)	Leak Re-paired? (Y or N)	Repair Code : Clamp, C Replace, R Weld, W Plug, P	Est Leak Run Time (Days)
51	8959 San Antonio	U	Y	N	Y	Y	C	2
52	NW Corner Firestone & San Gabriel	U	Y	N	Y	Y	R	3
53	SW Corner Firestone & San Gabriel	U	Y	N	Y	Y	C	12
54	SW Corner San Juan & Southern	U	Y	N	Y	Y	P	5
55	8990 San Antonio	U	Y	N	Y	Y	W	3
56	8996 San Carlos	U	Y	N	Y	Y	R	14
57	8921 San Luis	U	Y	N	Y	Y	C	6
58	8659 San Miguel	U	Y	N	Y	Y	C	3

	Meters/ Curb Stops	Hydrants	Valves	Test Rods	Other
Indicate Number of Manual Listening Points Used	483	43	88	0	0
Indicate Number of Leak Noise Logger Listening Points Used	0	0	12	0	0
Miles of Mains Surveyed	3.14	Survey time		16	Hours
Number of Leaks Suspected	8	To be rechecked		8	(Number)
Number of Leaks Pinpointed	0	Pinpointing time		0	Hours

Remarks

Found a 50/50 percentage between valve stem packing leaks and small service meter leaks. Also found two customer sprinkler system leaks; violation notices were delivered to each customer informing them that they are required to arrange for leak repairs within 10 days.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Calculating the Cost-effectiveness of CWC's Leakage Management Results

Name of Water Utility: *CWC*
Name of Report Preparer: *C.M. Biggs*
Date: *9/6/2020*

Leak Detection Survey

Total Number of Days Leak Surveys Were Conducted: 43

First Survey Date: 3/26/2020 Last Survey Date: 5/23/2020

Number of Listening Points:	Meters	Hydrants	Valves	Test Rods	Other
	4,025	862	1,605	0	17

Number of Suspected Leaks: 75
Survey Time: 344 hours
Pinpointing Time: 80 hours
Number of Pinpointed Leaks: 58
Miles of Main Surveyed: 82

$$\text{average survey rate} = \frac{\text{miles of main surveyed} \times 8\ \text{hr/d}}{\text{total srvey and pinpointing hours}} = 1.54\ \text{mi/d}$$

Total number of visible leaks reported since survey started, from other sources (not discovered during leak detection surveys): 0

Leak Repair Summary

First Leak Repair Made: 3/29/2020	Last Leak Repair Made: 6/28/2020	
Number of Repairs Needing Excavation: 37	Number of Repairs Not Needing Excavation: 21	Total Number of Repaired Leaks: 58
Total Water Losses from Excavated Leaks: 203.5 gpm	Total Water Losses from Nonexcavated Leaks: 78.9 gpm	Total Water Losses: 282.4 gpm

	Excavated Leak Repair Costs	Nonexcavated Leak Repair Costs	Total Repair Costs
Materials:	$699.36	$411.68	$1,111.04
Labor:	$4,377.39	$2,255.72	$6,633.11
Equipment:	$561.40	$248.75	$810.15
Other:	$35.00	$83.50	$118.50
Subtotal:	$5,673.15	$2,999.65	$8,672.80

A. Leak Survey and Repair Program

Step 1. Calculate the value of water recovered, Vwr, from all repaired leaks.

Vwr = (total leakage recovered in gpm)(average leak duration)(Variable production cost, VPC)

Ave leak duration = ½ of CWC's new 3-year leak survey interval = 547 days

Note: The cost-effectiveness for the 3-year interval will be reviewed to see that volumes recovered—once backlogs are removed—still warrant this survey interval.

VPC = Variable cost of water = $750/mil gal (See Chapter 2 Water Audit, Figure 2-5)

Vwr = 282.4 gpm × 1,440 min/d × 547 d × $750/mil gal × 1 mil gal/1,000,000 = $166,830

Step 2. Assemble Leak Survey Program First-Year Costs: from the "Sample Leakage Management Plan" sidebar, Section E. $37,868

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Step 3. Divide Vwr (from Step 1) by the total costs (calculated in Step 2).

$$\text{benefit/cost ratio (B:C)} = \frac{\text{value of water recovered}}{\text{total cost of leak detection survey}} = \frac{\$166,830}{\$37,868} = 4.40$$

For planning continuing leak detection efforts, calculate average survey costs per mile.

Step 4. Determine average survey costs per mile of main surveyed for 3-year cycle (C/mi).

$$\text{C/mi} = \frac{\text{3-year leak survey cost}}{\text{total number of miles surveyed}} = \frac{\$37,868 + \$22,868 + \$22,868}{256 \text{ mi}} = \$327/\text{mi}$$

At $327/mi the projected results are more expensive than the assumed value of $275/mi (see the Chapter 5 "CWC: Part 5" sidebar). Still, the program has a strong payback of $37,868/$166,830 = 0.23 year or just 2.7 months; a faster payback than was projected and very cost-effective.

B. Pressure Management Program

Step 1. Calculate the value of background leakage recovered, Vbr, from optimized pressures. Vbr—Assume that ½ of CWC's target background leakage (211 mil gal/year) and potentially recoverable leakage (20.4 mil gal/year) occurs in the Valley District. One half of (210 + 20.4) = 115.7 mil gal/year. Again, assume ½ of this volume, or 57.85 mil gal/year, is recovered.

Average leak duration: because the background leakage reduction occurs all year, the average background leak duration is 365 days.

Vbr = 57.85 mil gal × $750/mil gal = $43,387

Step 2. Assemble Pressure Management Program Costs: from the Pressure Management/District Metered Area Plan, Part III of the "Sample Leakage Management Plan" sidebar: $26,705

Step 3. Divide Vbr (from Step 1) by the total costs (calculated in Step 2).

$$\text{benefit/cost ration (B:C)} = \frac{\text{value of water recovered}}{\text{total cost of leak detection survey}} = \frac{\$43,387}{\$26,705} = 1.62$$

Step 4. Determine payback period for pressure control equipment $= \frac{\$26,705}{\$43,387} = 0.62$

The pressure control equipment has a life of many years, and payback occurs in 0.62 year or less than 8 months; thus, the pressure management program is projected to be cost-effective.

REFERENCES

AWWA (American Water Works Association). 2008. Manual M31. *Distribution System Requirements for Fire Protection*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2012. *Managing Water Main Breaks Field Guide*. Denver, Colo.: AWWA

Hughes, D. 2014. Improving System Performance with Flow Dependent Pressure Control. AWWA Infrastructure Management Conference. Denver, Colo.: American Water Works Association.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Hunaidi, O.; Chu, W.; Wang, A.; & Guan, W. 2002. Detecting leaks in plastic pipes. *Journal AWWA*, 92(2):82–94.

ISO 10156:2010. *Gases and Gas Mixtures—Determination of Fire Potential and Oxidizing Ability for the Selection of Cylinder Valve Outlets*. Geneva: International Organization for Standardization.

Kunkel, G., & Sturm, R. 2011. Piloting proactive, advanced leakage management technologies. *Journal AWWA*, 103(2):62–75.

Lambert, A.; Fantozzi, M.; & Thornton, J. 2013. Practical approaches to modeling leakage and pressure management in distribution systems—Progress since 2005. Presented at CCWI 12th International Conference "Computing and Control in the Water Industry," Perugia, Italy, Sept. 24, 2013. https://www.leakssuitelibrary.com/wp-content/uploads/2021/05/Practical-progress-in-modelling-leakage-and-pressure-2005-to-2013.pdf.

Thornton, J. 2005a. Metro Nashville Water Services Department uses new methods to reduce leakage efficiently. In *AWWA Distribution & Plant Operations Conference*. Denver, Colo.: American Water Works Association.

Thornton, J.; Shaw, M.; Aguiar, M.; & Liemberger, R. 2005b. How low can you go? A practical approach to pressure control in low pressure systems. In *IWA Leakage Conference Proceedings*. London: IWA Publishing.

Titus, R.; Meston, R.; Johnson, J.; Meschino, P.; & Moore, R. 2013. Leak detection programs identify and control water loss. *AWWA Opflow*, 39(11):8–12.

WRF (Water Research Foundation). 2007. *Leakage Management Technologies*. Denver, Colo.: WRF.

WSAA (Water Services Association of Australia). 2008. *Leakage and Pressure Management Stage 1, 2 and 3*. https://www.wsaa.asn.au/WSAAPublications/Documents/PPS3%20Leakage%20and%20Pressure%20Management%20Stage%201%20%202%20and%203.pdf.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 7

Planning and Sustaining the Nonrevenue Water Management Program

North American drinking water utilities should perform annual water auditing and would benefit from having specific programs in place to control water and revenue losses. A nonrevenue water (NRW) management program begins with gaining a true sense of the nature and extent of losses and their financial impact on the utility's operations. A good NRW plan should include these objectives:

- gaining awareness of the operational and revenue impacts that occur as a result of water that is inaccurately metered or billed, wasted, or stolen;
- verifying the accuracy of all production flowmeters and maintaining the integrity of the water supplied volume;
- putting in place mechanisms to ensure that unbilled water and associated revenue loss are monitored and contained; and
- employing proactive real loss management activities with appropriate combinations of leak detection and repair, flow monitoring, pressure management, and system rehabilitation.

Moving to an informed, proactive culture of routine water audits, active monitoring, and efficient loss control is a rational strategy to preserve water resources, minimize customer service disruption, keep water rates as affordable as possible, improve the financial bottom line of the water utility, and demonstrate accountability to its customers. This chapter discusses the planning considerations in establishing an NRW management program.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The first step for the utility is to compile a water audit on an annual basis as a standard business practice. The top-down water audit is straightforward to conduct and gives insight to areas needing the most attention, as well as initiatives offering a quick return on investment. The AWWA Free Water Audit Software (audit software) is an excellent tool to use for a first-time water audit because it can quickly give a water utility a preliminary sense of their water balance and functional areas where the accuracy of the audit by data collection can be improved. Progressively sophisticated activities should then be conducted over subsequent years as losses are better identified and quantified, and initial successes are achieved. All activities should be conducted in a manner that can be readily incorporated into the organization's culture and budget and sustained over the long-term planning horizon of the water utility. The major planning considerations for the NRW management program are detailed in the following sections.

IDENTIFYING DESIRED OUTCOMES AND BENEFITS OF AN NRW MANAGEMENT PROGRAM

A variety of potential benefits might be derived by the water utility from an NRW management program. Some of these benefits may impact the utility's performance, while others affect the community that is served.

- Program benefits to the water utility
  - Increased revenue
  - Reduced production costs
  - Reduced source water withdrawals
  - Reduced energy consumption
  - Deferred or reduced capital expenditures
  - Less infiltration into wastewater systems
  - Enhanced knowledge and operation of assets (piping, valves, meters)
  - Reduced liability
  - Regulatory compliance with water loss and efficiency standards
- Program benefits to the community served
  - Fewer disruptions in service
  - Equitable rate-setting process when all customers are metered and billed accurately
  - Better protection of the environment
  - Enhanced economic development opportunities
  - Less stress on underground utilities
  - Less traffic disruption and damage to streets
  - Enhanced utility preparedness in responding to distribution system emergencies
  - Improved appreciation for the value of efficient water service

The water utility can identify from the above lists the specific benefits that are desired outcomes of the program they are structuring. Clearly communicating that these benefits are targeted will help justify the program to decision-makers and other stakeholders.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

ESTABLISHING A CROSS-FUNCTIONAL NRW MANAGEMENT GROUP OR TASK FORCE

NRW can impact almost every facet of water utility operation; thus, water utilities must often bring together staff from across the organization to form a team to manage NRW effectively. These are functional areas affected by NRW and its implications.

- Source water and production metering: Water utilities should obtain accurate measures of water withdrawn from water resources and determine the volume of treated water supplied to the water distribution system.
- Production cost management: Efficiently maintaining energy, chemical, and residuals management costs can garner significant cost reductions.
- Water distribution system operations: Utility managers should employ flow monitoring, pressure management, and supervisory control and data acquisition systems to assist in system oversight and better control of leakage losses.
- Water distribution system maintenance: Utilities should conduct active leakage control activities and timely distribution system repair and employ maintenance management information systems to track these activities.
- Customer metering: Accurate metering of customer water consumption allows for accurate billing and a reliable measure of consumption by class of customer and the customer population in total.
- Customer meter reading: Automatic meter reading (AMR) or advanced metering infrastructure (AMI) systems provide outstanding capabilities to cost-effectively capture customer consumption data. While these systems can benefit most water utilities, for some water utilities, manual meter reading may remain the best option.
- Infrastructure management: Rehabilitation and replacement of the water infrastructure is part of a balanced strategy to ensure a low-loss system.
- Water quality: Addressing leakage more efficiently leads to less disruption to the integrity of the water distribution system. Optimized pressure management and effective renewal of distribution assets result in a lower likelihood of intrusion or backflow events resulting from pressure surges and ruptured water mains.
- Sewer collection systems: A portion of leakage often finds its way into the collection system where it adds to the flow burden in the sewer system and at the wastewater treatment plant.
- Water conservation: Utilities with a water conservation program should integrate their efforts with the water loss control program through a joint mission to manage water resources efficiently.
- Customer billing systems: Effectively managing large numbers of customer accounts, consumption data, billing actions, and revenue recovery is critical to program success.
- Water rate setting and finance: Pricing water to ensure full return on investment, efficient use of the resource, and displaying water-efficient operations elevates the stature of the water utility in the eyes of financial institutions.
- Customer service: Rapport with the customer is enhanced when the incidence of disruptive water main breaks and leaks, inaccurate meter readings, billing errors, or excessive water charges are minimized and an image of efficiency is exhibited by the water utility.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- Public relations: Efficient water management enhances the perception of the water utility in the communities that it serves by better protecting watersheds, keeping water rates affordable, valuing the resource in times of scarcity, and promoting economic development.
- External stakeholders: Interaction with fire departments is necessary to confirm fire hydrant usage policy and to estimate quantities of water used in firefighting and training. Most underground water piping lies below streets and highways, and activities must be coordinated with local or state street or highway departments. Coordination with other utilities (electric, gas, communications) is also an essential part of the operation of the water distribution system. Other governmental, community, business, or civic groups might also be important stakeholders to the program.
- Executive leadership of the water utility: Leadership provides the mandate for the listed groups to come together to focus on accountability improvements. Executives also provide guidance and resources to the effort and help communicate its success.

In planning for the new program, utility management should create an organizational structure that allows NRW management to be the central mission of its existence. As noted in the above list, water and revenue loss impacts the water utility's water distribution, metering, billing, customer service, water quality, renewal programming, and other groups. Therefore, the utility's organizational structure for NRW management activities should include participation by staff serving in these functional areas.

Large water utilities may be able to hire personnel dedicated to form a distinct NRW management group within the water utility. For larger water utilities, the head of the group might report directly to the chief executive of the utility, and the various stakeholders in the utility would be required to report water efficiency data and water loss reduction progress directly to the NRW manager. The advantage of this structure is that it assigns dedicated personnel to loss control activities in systems with many functions and large amounts of data. The challenge of this approach is that it requires dedicated funding to hire staff and maintain the program.

Another possibility is to form a team, committee, or task force of responsible individuals drawn from the above functional areas within the water utility. While still performing their normal duties, the members of this group can meet periodically to gather and review data, monitor progress, and plan loss control activities. This type of structure provides an opportunity for groups to learn how the various activities are handled by their peers within the water utility. Team dynamics often create the opportunity to quickly uncover water loss that exists merely because of misconceptions and gaps in procedure. A team-based approach also allows ideas to be communicated across the spectrum of the organization to obtain the buy-in of many employees. Improved employee cooperation and cohesiveness is therefore a strong side benefit of this approach. Good teamwork procedures need to be put in place to ensure that the team members perform their tasks and are accountable to each other and to the utility management. Systematic reporting procedures for the team and their communication to management are needed. The main advantage of this approach is that no additional personnel need be hired to launch the NRW management program. The primary challenges of the team approach are (1) to ensure that executive leadership provides a clear and concise mandate to all the involved groups about their role on the team and (2) for the current staff to incorporate the part-time requirements of new NRW management activities into their existing schedules.

Each utility can determine the best organizational structure for its needs. The key is to establish NRW management as part of the organization's overall mission and assign and communicate the goals, structure, and responsibilities to all personnel participating in the

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

program. Once the structure is set, the utility is ready to move forward knowing that the activities to control losses are well assigned and responsibilities are clearly defined.

IDENTIFYING RESOURCE NEEDS

The NRW management program can be launched or upgraded with little or no specific funding. Without earmarking specific funding for the program, executive management can launch a program simply by bringing together the appropriate personnel and assigning them meeting time to compile a top-down water audit and explore potential savings. The self-assessment nature of the water auditing process can often identify several quick, easy refinements, typically in billing system procedure or documentation. This can recoup revenue or create cost savings early in the program life. Designing such early funding recovery into the NRW management program provides swift payback that can "seed" further loss recovery activities.

To advance the NRW management program beyond its initial successes, the water utility should consider whether it can benefit from additional resources that can include the following.

- **Training.** Low- or no-cost training is available from a variety of sources, including drinking water regulatory agencies, water industry trade associations, and academic institutions. Instruction exists for water auditing, leakage and pressure management, meter management, and Automatic Meter Reading (AMR) and Advanced Metering Infrastructure (AMI). Many publications, applications, videos, webinars, and web resources exist in addition to this manual to provide training resources for operators embarking on NRW management.
- **Technology.** With rapid advancements in technology, considerable new equipment exists to assist water utilities in controlling losses and accounting for water. Permanent and portable metering equipment, leak noise loggers and correlators, data loggers, AMR/AMI systems, enhanced billing software, and countless other innovations are available at various cost ranges. The greatest challenge is to determine the most appropriate equipment from an array of many capable models.
- **Specialized services.** As the NRW management program matures, the water utility may consider advanced systems such as district metered areas (DMAs) for leakage management, AMR/AMI systems, hydraulic modeling of the water distribution system, and computerized information or control systems. Many water utilities find it helpful to utilize services from specialized providers to guide them in specifying and installing such advanced processes and systems. Specialized services are available from consultants and manufacturers in the drinking water industry and are sometimes available from regulatory agencies in the form of technical assistance for water utilities that qualify under specific assistance programs.

In launching or refining the NRW management program, it is quite natural, particularly for water utility board members, accountants, and managers to become alarmed about how much this program may cost. While a valid concern, the utility also should consider how much existing losses cost. No system is loss free, and losses impact water utilities with a variety of costs and liabilities. Unfortunately, these costs are often not quantified; instead, they are embedded in general operations, maintenance costs, and water rates, masking their impact to managers, executives, and customers. The water auditing process sets out to quantify both loss volumes and their cost impact to the utility. Planned expenditures for the NRW management program can only be viewed objectively

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

when compared to the costs of the losses currently affecting the utility and in the context of how the NRW management program meets the long-term goals of the water utility. Thus, if the program cost-effectiveness is questioned, the implementation team should state objectively that a certain program cost is necessary to obtain the targeted benefits, along with a financial benefit of a predicted amount of increased revenue and/or reduced production costs.

LAUNCHING THE NRW MANAGEMENT PROGRAM

Table 7-1 lists the primary activities that should be considered as part of a good NRW management program. This listing is neither exhaustive nor prescriptive; however, it is strongly recommended that of all primary activities, production flowmeters should be properly installed and managed (see Appendix A) as part of the top-down auditing process. Not all activities in Table 7-1 are needed by every water utility. The water auditing process will guide utility management in identifying the greatest loss occurrences and economic impacts. Selection of appropriate program activities—particularly those of long-term nature—should be made only when strategically and economically justified and in conjunction with the utility's long-term goals.

As shown in Table 7-1, once launched, several activities of the NRW management program can be conducted concurrently. For instance, by establishing pilot DMAs as discussed in Chapter 5, site-specific leakage data can be obtained to better direct leak detection crews. These data also provide leakage quantities as the basis for the leakage component analysis (LCA) that can accompany the water audit. Once established, many of the activities should be maintained in an ongoing manner. Water auditing should be conducted on a routine annual basis, just as regular financial auditing is conducted.

Table 7-1 also lists approximate time frames to initiate the activities of the NRW management program. These are designated as short-term (S), medium-term (M), long-term (L), and ongoing. No attempt is made to attach actual durations to these time horizons because the ability to enact activities varies greatly from one utility to another. Activities designated as long-term are more extensive, sophisticated, and costly in terms of scope and typically require long timelines for planning, funding, and execution. In most cases, however, utilities could implement the short-term activities in a matter of days, weeks, or months, as long as resources are properly assigned. Ongoing activities can be undertaken at any time once a sound basis for moving forward has been established.

Table 7-1 NRW management program activity planning matrix

Potential activities of an NRW management program (System-specific: Not all activities are necessary for every water utility.)					
		Intervention Activities			
Water Auditing		Apparent Loss Control		Real Loss Control	
Time*	Activity	Time	Activity	Time	Activity
S	Top-down water audit	S	Verify accuracy of production flowmeters (top priority)	S	Review maintenance records, gather and summarize statistics on water system failures (leaks and breaks). Establish this process as described in Chapter 5.

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 7-1 NRW management program activity planning matrix *(Continued)*

Potential activities of an NRW management program (System-specific: Not all activities are necessary for every water utility.)					
		Intervention Activities			
Water Auditing		Apparent Loss Control		Real Loss Control	
Time*	Activity	Time	Activity	Time	Activity
M	Start bottom-up water audit by launching field investigations into specific loss occurrences.	S	Flowchart the customer billing process; compile general statistics on the demographics of the customer/meter population.	S	Review policies for customer service connection piping ownership and maintenance, an opportunity to reduce customer service connection piping leakage durations.
Ongoing	Bottom-up water audit: Conduct detailed investigations of metering, meter-reading, and billing operations.	S	Perform meter accuracy testing on a statistically representative sample of customer meters. †	S	Conduct an initial leak detection survey as outlined in Chapter 6; consider use of leak noise monitors.
	Bottom-up water audit: Gather field measurement data and minimum-hour leakage analysis; conduct LCA	S	Audit billing records and inspect properties with suspect billing data to determine if unauthorized consumption is occurring.	S	Compile data on the variation of water pressure throughout the water distribution system. Identify areas of excessive pressure and evaluate potential for proactive pressure management.
		M	Install, upgrade, or replace production flowmeters. (See Appendix A for guidance.)	S	Establish a pilot DMA; perform minimum-hour leakage analysis.
		M	Review/implement policies and interventions to detect and thwart unauthorized consumption.	M	Create a leak detection squad, or hire a leak detection contractor, to regularly survey the distribution system for unreported leakage.
		M	Investigate the potential costs and savings of instituting an AMR or AMI system to reduce missing or erroneous customer meter readings.†	M	Install additional pressure management areas and/or deploy leak noise monitors as deemed feasible and cost-effective.
		L	Install an AMR/AMI system and institute monthly billing based on meter readings.†	L	Implement a maintenance management information system.
		L	Install a new computerized customer billing system.†	L	Create additional DMAs, if determined to be feasible.
		L	Replace most or all customer meters at an appropriate frequency to provide accurate measurements and keep metering technology current.†	L	Institute or expand capital replacement program for water main infrastructure.

*Time = time frame: S – short-term; M – medium-term; L – long-term; Ongoing – any time, assuming a sound basis exists for implementation.

†These initiatives assume that the customer population is metered. If not, the water utility should consider installing customer meters. While not a small undertaking in terms of scope and cost, this effort should be considered as a short- or medium-term initiative of high priority.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

For water utilities that are embarking on their initial effort to control losses, the top-down water audit approach is the recommended starting point. It is described in detail in Chapter 2, and descriptions of bottom-up approaches and component analysis for apparent losses are presented in Chapter 4. Bottom-up methods and LCA for real losses are presented in Chapter 5. While the water auditing process is not specifically a means to reduce losses, it is common for many utilities to realize loss reductions once they have instituted routine water auditing. This process can create an important shift in the organization's culture. The water audit also becomes the primary reporting mechanism to measure the success of the water loss control program from year to year. Once available, the findings of the water audit guide decision-making to determine which interventions are most appropriate and economically justified. With commitment, the water utility can build on successes incrementally to establish a culture of accountability and a reputation for water efficiency.

A solid planning approach that focuses on cost-effectiveness will ensure that the most appropriate loss control interventions are selected for the NRW management program. The water utility should develop a solid "business case" around the components of the program. Research has been conducted to investigate the key components of an NRW management plan and the means to evaluate cost-effectiveness. The Water Research Foundation Project No. 4695: *Guidance on Implementing an Effective Water Loss Control Plan* gives water utilities guidance on the ways to develop successful business strategies for managing NRW (WRF 2019).

REPORTING NRW MANAGEMENT PROGRESS WITHIN THE WATER UTILITY

The water audit provides detailed data and information on the utility's NRW standing and allows for tracking the success of the NRW management program. However, a water audit is conducted only once a year, and water utilities usually desire to also track their progress on a more frequent basis such as monthly or quarterly. It is possible to employ a "high-level" report as often as monthly, but only if a certain procedure such as a 12-month rolling average analysis is used to avoid misinterpreting the data.

By assembling the annual water audit, water utility personnel gather data that effectively measure performance, ultimately leading to the ability to better control and improve utility processes and operations. The standard water audit format advocated in this manual classifies and organizes data and calculates objective performance indicators that provide key information on the utility's water loss standing and the effectiveness of the intervention methods to better control losses.

In addition to a wide array of water quality information, water utilities frequently maintain data on water flow rates and pressures, infrastructure condition, maintenance activities, customer consumption and billing, leaks and breaks, water main replacement, and a host of other activities. It is important that the water loss control program identifies the vital information that is necessary to successfully operate the program and measure performance.

Reliable data collection and sound record-keeping functions should be a prominent feature of the working routine of the water loss control team. Quality and integrity of existing records and documentation during the development of the top-down water audit should also be evaluated. The team should carefully consider how data are generated, defined, categorized, stored, and reported, noting deficiencies in these processes and correcting them. The team should also make note of any data that are not regularly reported but should be. Information mishandling or omission alone can influence the success or failure of the program. Good information management identifies the nature, extent, and locations of losses and ably measures progress in controlling the losses to economic levels. It is likely that refinements to some of the existing record-keeping methods will be needed.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

At the start of am NRW management program, it is almost certain that gaps in the current data collection will be identified, resulting in the need for new documentation and reports. Employees in the water utility who serve as the "owner" of important data should be identified and given the responsibility for generating the data in the appropriate format for submission during the annual water audit data collection period.

A useful monthly report for water utilities, shown in Table 7-2, displays data from the fictitious County Water Company (CWC) featured in Chapter 2 and throughout this manual. The report gives a high-level, monthly overview of the water supplied volume, the billed authorized consumption volume, and the difference quoted as NRW. The number of active customer billed accounts is also listed because this number varies month to month and reflects the size of the customer population. Because of customer meter-reading lag time that is inherent in most customer billing systems and occasional swings in seasonal customer consumption patterns, both the water supplied volume and the billed authorized consumption volume are reported on a rolling 12-month basis. This minimizes the "noise" associated with variations in the underlying data due to short-term reporting intervals that can otherwise skew interpretation of existing conditions. Utilities must avoid reporting direct comparisons of a single month's water supplied volume vs. the same month's billed authorized consumption volume because such a comparison is often misleading. Only 12-month rolling average comparisons should be used for this purpose. Because this report is compiled and circulated every month, all stakeholders are regularly reminded of the general water efficiency standing of the water utility throughout the year.

The monthly report in Table 7-2 determines the volume of NRW as the difference of supplied and billed volumes, but it is not a water audit. It does not explain how the NRW volume breaks down to unbilled authorized consumption, apparent losses, and real losses. This breakdown is determined only in the detailed water audit process, which is conducted on an annual—not monthly—basis. Reports can also be created listing performance indicators such as main breaks per 100 miles of system pipeline per year, meter replacement rates, response time for reported leak repairs, and other statistics. Dashboard formats with features customized for water utility users are increasingly being used to present actionable information such as this. (See also the "Compiling a Monthly NRW Report" sidebar.)

Table 7-2 Monthly NRW Report for CWC

12-Month Period	Water Supplied Volume (mil gal)	Billed Authorized Consumption (mil gal)	NRW (mil gal)	Number of Commercial, Industrial Customer Accounts	Number of Residential Customer Accounts	Total Customer Accounts
February 2019–January 2020	4,242.83	3,712.90	529.93	693	11,391	12,084
March 2019–February 2020	4,241.28	3,728.40	512.88	694	11,398	12,092
April 2019–March 2020	4,241.57	3,733.80	507.77	694	11,401	12,095
May 2019–April 2020	4,250.46	3,792.90	457.56	694	11,422	12,116
June 2019–May 2020	4,256.08	3,829.50	426.58	696	11,439	12,135
July 2019–June 2020	4,267.95	3,835.30	432.65	696	11,437	12,133
August 2019–July 2020	4,280.82	3,840.90	439.92	696	11,439	12,135
September 2019–August 2020	4,293.01	3,846.60	446.41	698	11,452	12,150

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 7-2 Monthly NRW Report for CWC *(Continued)*

12-Month Period	Water Supplied Volume (mil gal)	Billed Authorized Consumption (mil gal)	NRW (mil gal)	Number of Commercial, Industrial Customer Accounts	Number of Residential Customer Accounts	Total Customer Accounts
October 2019–September 2020	4,289.15	3,829.20	459.95	701	11,464	12,165
November 2019–October 2020	4,276.26	3,748.90	527.36	703	11,461	12,164
December 2019–November 2020	4,268.05	3,754.60	513.45	709	11,478	12,187
January 2020–December 2020	4,402.16	3,858.20	543.96	706	11,490	12,196

Compiling a Monthly NRW Report

It is recommended that water utilities compile the standard water audit on an annual basis to detail the utility's water loss standing and measure its performance. However, water utility personnel need the ability to monitor system performance more often than annually, and the monthly reporting format in Table 7-2 is the recommended approach for providing a general monthly overview of water loss standing. This report provides stakeholders a high-level glimpse of the water loss standing of the water utility each month. There are two important details to remember in using this report:

1. This type of report is not a "water audit," given that it merely reports the total NRW volume and does not attempt to quantify the individual component volumes of unbilled authorized consumption, apparent losses, and real losses.
2. This report must be conducted using 12-month rolling average values for the water supplied volume and the billed authorized consumption volume. Attempting to use individual monthly quantities of these two volumes will invariably lead to a skewed data comparison at some point during the audit year and likely cause great confusion.

By employing a report such as the one shown in Table 7-2, water utilities can stay abreast of their water loss standing continuously throughout the year and use the annual compilation of the water audit to provide the detailed quantities of the NRW components.

New reports should be created when new technologies, such as DMAs, are implemented. The team should identify information sources and a contact person for each data source and identify internal stakeholders who have the need to know and should receive the report. Reporting is often a very straightforward way to promote success for the water loss control program.

COMMUNICATING WITH COMMUNITY STAKEHOLDERS—PUBLIC RELATIONS

Effective communication is key to sharing information, educating, and cooperating with consumers and a wide range of stakeholders. In undertaking an NRW management program, the water utility should implement communications that will announce program successes and educate stakeholders on the value of water efficiency to the community. Well-considered messages targeting stakeholders can

- promote the value of reliable water supply to the community;
- highlight the financial benefits of revenue recovery to stabilize water rates and build confidence in the water utility among the lending community;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- emphasize reduced service disruptions to customers and minimized damage to street or building infrastructure because of fewer water leaks and main breaks;
- tout the reduced impact on the environment from fewer withdrawals from rivers, lakes, or wells, as well as less energy consumption from optimized operations;
- delaying or foregoing new infrastructure construction—reservoirs, treatment plants, pumping stations, wells, and so forth—by better long-term management of existing water resources and infrastructure;
- establish reputation of reliability by sustaining water service during periods of drought;
- enhance customer perceptions by conveying visible accountability and operational efficiency improvements; and
- integrate the attributes of the program with messages delivered on water conservation and public education campaigns managed by the water utility or by partner organizations such as local watershed associations or conservation districts.

As part of the development of communication options, a wide variety of stakeholders should be considered. A list of pertinent stakeholders can be identified, with the appropriate messages and timing noted.

Improved communication is also part of the drinking water regulatory structure in the United States. The use of Consumer Confidence Reports is perhaps the most visible example of communication mechanisms that are required by regulatory agencies. Regulatory agencies, particularly in resource-limited areas, may consider increased reporting of water-efficiency practices, and water utilities should be aware of this possibility.

A good communication team knows that not all stakeholders need be targeted by all messages. An important approach of the communication plan is to consider who needs to know the information, what specifically they need to know, and when they should be informed. In this manner, the water utility manager or public relations official can build into the program the information or data that must be gathered and a schedule of communications to the appropriate groups. The best communications strategy will benefit the program by garnering support and gaining assistance and recognition.

SUSTAINING THE NRW MANAGEMENT PROGRAM

With a moderate level of forethought and organization, water utilities can create an NRW management program by forming a team of appropriate individuals from the organization and compiling the annual water audit. The AWWA audit software, described in Chapter 2, is a great tool to employ to get started on this endeavor.

The top-down water audit process will likely identify several initial loss reduction priorities to the group and enable them to capture the "low-hanging fruit" or relatively easy recoveries to launch the program toward success. Because loss control of any nature is an endeavor of diminishing returns, incrementally greater effort is required to recover incrementally smaller returns. It is important that the water loss control program include a realistic long-term vision to ensure that

- loss control efforts continue in a strategic, ongoing manner and a "backslide" does not occur and
- loss control efforts eventually define the economic breakpoint beyond which further loss reduction is not economically justified.

When creating an NRW management team, look for people who have demonstrated skill, perseverance, and innovation in their work history. Select a leader who will champion

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

the cause and motivate team members and outside stakeholders during the times when progress appears slow. Avoid adding personnel to the team who fear change, even if they are longstanding and experienced in their position in the utility.

Sustainability also hinges on the ability to implement long-term interventions to control losses. Water distribution system rehabilitation, AMR/AMI systems, customer billing systems, and extensive sectorization via DMAs are all significant undertakings that can require years to plan and implement, yet they hold the potential to generate enormous benefits. The NRW management group should show persistence over a long period of time to implement such improvements. Sustainability is a critical planning consideration because it protects against reversal of gains made early in the program and ensures that the ultimate desired outcomes are achieved.

With proper planning, water utilities can launch effective NRW management programs that will ensure efficient water supply operations and protection of natural resources.

REFERENCE

WRF (Water Research Foundation). 2019. *Guidance on Implementing an Effective Water Loss Control Plan*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Chapter 8

Considerations for Small Systems

Small drinking water systems (serving populations less than 10,000) can benefit from having a nonrevenue water (NRW) management program. While notable differences generally exist between large and small water systems in terms of institutional structure and resources, issues of water loss are largely the same: apparent losses exist due to customer metering and billing problems, and real (leakage) losses occur due to compromised infrastructure and from operational and maintenance limitations. Water audits are valuable for small systems to identify problem areas and consider possible solutions to control NRW. Small systems are diverse and can range from rural systems with widespread service areas to compact urban towns to seasonal mobile home communities. Small systems have challenges ranging from aging infrastructure, local policies and politics, and funding and staff limitations. This chapter addresses water audit and loss mitigation elements relevant to small utilities with an overall goal to minimize losses and optimize finances.

OBTAINING TECHNICAL AND FINANCIAL RESOURCES FOR WATER LOSS CONTROL

Perhaps the greatest difference between large water utilities and smaller ones is the availability of resources to implement water loss control activities. These resources include trained staff and funding to conduct water loss control activities such as acoustic leak detection and production and customer meter accuracy testing. Management and governing board support are critical to efforts to launch the water loss control program. Fortunately, resources are available to help smaller utilities with these needs.

In the United States, several programs exist to offer small systems technical and infrastructure funding, as well as guidance, training, and assistance. Some of the larger programs are offered by the US Department of Agriculture (USDA) through the Rural Utilities Service and the US Department of Housing and Urban Development Community

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Development Block Grant Program. USDA in particular offers both capital and technical assistance resources to rural water systems. In fact, USDA funding allowed AWWA to create the eLearning Program "Controlling Nonrevenue Water in Small Water Utilities" (2020), which is available to staff of small water utilities free of charge. The US Environmental Protection Agency provides funding through its Drinking Water State Revolving Fund administered through the states that can be used for training, technical assistance, and infrastructure improvements. The Rural Community Assistance Partnership and the National Rural Water Association's state affiliates also offer a wide variety of resources to small water utilities. Many states offer technical assistance as well.

Grant, loan, and training opportunities vary from year to year; therefore, small systems should work with their regional, state, and local water works associations, technical assistance providers, and regulatory primacy agencies for information on the most recent offerings and help with applications. Participation in these programs may provide additional insights into the financial and staffing challenges small systems often face. Loans from programs like the Drinking Water State Revolving Fund could be applied to such initiatives as pipeline renewal and customer meter upgrades, whereas grants may be focused on evaluations such as leak detection or district metered areas (DMAs). Technical guidance and promotion of best practices for small systems are also available from the Small Systems Divisions of AWWA.

IMPLEMENTING WATER LOSS CONTROL PROGRAMS IN SMALL SYSTEMS

Small water systems often face greater challenges than large water systems in implementing new programs or meeting new regulations. Utility staff reviewing the list of water loss control activities detailed in Table 7-1 in Chapter 7 may conclude that some of these activities are not practical for the typical small system. The following provides a road map that small systems can follow to identify and collect pertinent records and apply them to a water loss control plan.

Road Map for Getting Started

Developing a water loss control plan for a small system can be broken into two general phases: 1) using the water audit to determine what types of loss are most important to address in the utility and 2) prioritizing several key actions to better quantify or reduce the priority losses. Small utilities often have limited technical and administrative staff, and the initial phase of compiling a water audit can be challenging. The following steps provide a guide to starting the water audit process:

Step 1: Assign a person to coordinate the water audit. This person will coordinate with the other utility staff members who work with the data and information needed for the various inputs to the water audit. For example, this lead person could work with field staff to obtain estimates of unmetered water consumption and with the metering staff to obtain information pertaining to customer meter errors and inaccuracies. The coordinator should have a basic understanding of each of the various components of the water system included in the audit in order to properly manage the audit effort. The coordinator will input data into the water audit and provide this information to the utility leadership for review and approval.

Step 2: Gather data and information from existing records. Most utilities maintain the data and information needed to compile the annual water audit. Table 8-1 lists the typical data sources and reports used in small systems, along with tips for streamlining the data collection process so the annual water audit process goes smoothly. Although the water audit is designed to be compiled annually, the coordinator should gather and review data for seven key water audit inputs monthly. These inputs include

- **volume from own sources (VOS):** monthly water production reports (per flowmeter) and results of flowmeter accuracy testing;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- **water imported (WI):** monthly reports or invoices for bulk water imported or purchased from other utilities (per flowmeter) and results of flowmeter accuracy testing;
- **water exported (WE):** monthly reports or bills of bulk water sold to other suppliers (per flowmeter) and results of flowmeter accuracy testing;
- **billed metered consumption:** monthly billing summary reports (by customer classification) and results of customer meter accuracy testing;
- **billed unmetered consumption:** monthly billing summary reports listing estimates of unmetered consumption;
- **unbilled metered consumption:** monthly billing summary reports listing metered consumption that is not billed or separate records of meter readings for these customer accounts; and
- **unbilled unmetered consumption:** monthly summaries of the utility's unmetered water consumption (firefighting, water quality flushing, hydrant testing, etc.).

If these seven data inputs are tracked carefully each month—with any errors identified and corrected—the remaining data for the water audit can be collected at the end of the year.

Table 8-1 Example data sources for water volume inputs to the AWWA water audit

Water Audit Input	Potential Records	Tips and Best Practices
VOS	Water plant production monthly operating reports Flowmeter accuracy test results	• Add the production volumes from each water source or plant supplying the system. • Test each production flowmeter for accuracy annually and adjust volumes as needed.
WI	Invoices and operating reports from the wholesale utility Imported water accuracy test results	• Add the imported water volume from each interconnection supplying the system. • Test each imported flowmeter for accuracy annually and adjust volumes as needed.
WE	Billing records from exported water sold to utility customers Exported water accuracy test results	• Add the exported water volume from each interconnection supplying water to other utility customers. • Test each wholesale flowmeter for accuracy annually and adjust volumes as needed.
Authorized Consumption (Billed, metered)	Billing summary reports (monthly, quarterly, annual), customer water consumption data as gathered from manual meter reading, automatic meter reading (AMR), or advanced metering infrastructure (AMI)	• Add the billed metered consumption monthly values. • Monitor consumption based on meter size or customer classifications; note unusual or abrupt changes in routine consumption patterns. • Develop parameters for testing customer meters for accuracy and test a representative number of customer meters each year. • Billing cycles can sometimes run one or more months later than water production records. Align the data so billings reports match the water production time frames. • Perform lag time correction if data do not align with the water audit period. • Raw meter-reading data (like from AMI databases) do not always match the actual volumes billed to customers. Ensure the volumes are actual usage volumes.

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 8-1 Example data sources for water volume inputs to the AWWA water audit *(Continued)*

Water Audit Input	Potential Records	Tips and Best Practices
Authorized Consumption (billed, unmetered)	Lists of unmetered service lines or unmetered customers	• Estimate volumes used by unmetered customers based on similar metered customers. • Install and read meters on all customer service connections when possible.
Authorized Consumption (Unbilled, metered)	Meter readings for utility services (lift stations, wastewater treatment facilities, etc.), metered flushing points, metered hydrant testing, uses on portable hydrant meters, metered fire training, metered municipal uses	• Most metered uses can be reported from the utility billing system in many cases. An unbilled customer report can be created to track this consumption. • Perform lag time correction if data do not align with the water audit period. • Occasionally, unbilled meters are not read by meter-reading staff. Operational staff can record these meter readings or have the meters added to reading routes.
Authorized Consumption (Unbilled, unmetered)	Operations tracking logs for utility water uses in the distribution system (fire hydrant testing, flushing, etc.), run schedules and flow rates from automated flushing devices, fire department firefighting and training logs, fire flow estimations	• These volumes are typically estimated based on utility practices and typically include unmetered utility and municipal water use. • Volumes for utility uses can be tracked on monthly logs for hydrant testing, flushing, and other utility authorized system uses. • Annual water volumes for many uses are based on estimated flow rate and run time. • Fire flow volumes can be estimated based on the measured or estimated flow rate and the duration of the hydrant opening. If the system has a robust supervisory control and data acquisition (SCADA) system, fire flows can be derived from increased flow at pump stations or the water treatment plant during known fire events. • If trustworthy unbilled unmetered consumption data do not exist, the auditor can employ the default value in the audit software but should strive to obtain actual data or good estimates in following water audits.
Unauthorized Consumption (UC)	Annual logs or records of observed and documented unauthorized water consumption	• Water volumes should only be used in this entry if the utility has good quality estimates of unauthorized water consumption. • Ensure that strong policy states what fire hydrant uses are permitted and have in place a process to periodically inspect hydrants to confirm no unauthorized use. • Consider inspecting customer accounts with zero consumption for two or more billing cycles to ensure that meter tampering has not occurred. • If the utility has a large seasonal population, it may be beneficial to monitor for unbilled water consumption occurring on closed/ inactive accounts during the off-season.

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 8-1 Example data sources for water volume inputs to the AWWA water audit *(Continued)*

Water Audit Input	Potential Records	Tips and Best Practices
		• If trustworthy UC data do not exist, the auditor can employ the default value in the audit software but should strive to obtain actual data or good estimates in following water audits.
Customer Metering Inaccuracies (CMI)	Records of customer meter accuracy tests Logs of stopped/failed water meters Records of customer meters replaced, repaired, or rightsized (such as oversized meter replaced with one of a more appropriate size)	• Compile all customer meter accuracy tests for the audit year in a spreadsheet and determine the average meter accuracy. • Consider inspecting customer accounts with zero consumption for two or more billing cycles to ensure that meter failure has not occurred. • It can be beneficial to evaluate average meter accuracy by meter size or meter class and calculate a weighted average meter accuracy value. • If few meter accuracy tests have been conducted, multiple recent years of meter tests can be used as a baseline estimate of meter accuracy. • Perform customer meter accuracy testing during meter change-out and at regular intervals to establish trends in meter accuracy errors.
Systematic data handling errors (SDHE)	Reports of zero-consumption billings, billing adjustment reports, comparison of billings reports and raw meter readings or AMR/AMI water consumption data, billing system audit reports	• Zero consumption reports can identify vacant or miscoded properties. • Use summary reports of different steps in the billings process (e.g., meter-reading summaries vs. billed customer consumption) to identify missing consumption volumes. • Unusual seasonal NRW trends might be an indicator of customers missing from the billing roles in seasonal communities. • Incorrect meter multipliers are a common source of billing errors; inspect a sample of large meters that use multipliers to ensure accurate meter reading. • Volumes lost from inadvertently unmetered customers can be estimated based on monthly water use once such customers are identified and added to the billing system. • Raw customer meter-reading data (like from AMI databases) do not always match the actual volumes billed to customers. If there is a difference between metered customer consumption and volumes billed, this can be used as an estimate of volumes lost to data-handling errors. • If trustworthy SDHE data do not exist, the auditor can employ the default value in the audit software but should strive to obtain actual data or good estimates in following water audits.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This process should ensure that the annual compilation of the water audit occurs smoothly and without undue work burden. It is also helpful for the water audit coordinator to keep detailed notes on the data and information collection process in the comments worksheet of the AWWA Free Water Audit Software (audit software) and other spreadsheets.

Step 3: Identify data gaps. When compiling records for the water audit, the utility may find that records do not exist for certain types of information needed to create an accurate water balance. For example, consider a water utility that conducts significant fire hydrant flushing to maintain water quality but does not track the volume of water flushed for the water audit year. It may be beneficial to develop a methodology for estimating this water consumption by tracking a week's worth of flushing practices to determine a typical flow rate, time spent per flush, and number of flushes per day to provide a more accurate accounting of this part of the unbilled unmetered consumption component of the water audit. Developing reasonable estimates of the missing or roughly estimated data in Step 2 will provide value to the water audit as the water system develops more formal methods for collecting the required data. More accurate data also result in higher data gradings and a higher data validity score (DVS) in the AWWA water audit, ultimately yielding more realistic and defendable estimates of NRW and its component volumes.

Step 4: Complete the water audit using the audit software. Input as much data as are readily available from existing records or reasonably estimated. This starts with the water volume inputs to the system (water production, imported water, exported water), followed by the billed customer consumption, and all billed and unbilled, metered or unmetered consumption that exists in utility operations. It may be beneficial to keep a separate spreadsheet to tally the water volumes in the proper units prior to entry into the audit software.

Step 5: Meet and discuss the first draft of the water audit. Create a water loss team of key personnel from the utility, including staff overseeing water production and distribution operations and the people overseeing customer metering and billing as well as cost accounting. This team should meet to review the draft water audit and determine if the water balance volumes make sense based on staff experience. Staff may be able to provide additional data to improve the water audit. Reviewing the data gradings of the components can identify areas for improved data collection and a more representative water audit. Responsibilities should be assigned to people or groups that will implement the identified improvements prior to adjourning the meeting. The team should meet regularly throughout the year to discuss progress regarding the data collection and review effort.

Step 6: Update the water audit. Based on the team discussion and any new information obtained, update the water audit to include the most accurate information available.

Step 7: Assess the water audit results. Evaluate the loss volumes and cost impacts and compare them to the ranges encountered by other water utilities in the dashboard worksheet of the audit software. Which loss volumes and costs are the greatest? Are they high, above the 75th percentile values on the speedometer charts on the dashboard, relative to other utilities? Also, compare the water audit data from the utility's water audit data from previous years. Are the trends improving with losses being reduced, or are they regressing, with increasing losses and costs?

Step 8: Create an NRW management plan. Make a short-term (one to two years) NRW management plan with several priority action items identified to address the most significant data gaps that impact the water audit's trustworthiness or the most critical losses in the water system. This process will be discussed in detail in the next section.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

DEVELOPING A NONREVENUE MANAGEMENT PROGRAM IN A SMALL SYSTEM

Once a small utility has completed its initial water audit, it is suggested that it 1) improve tracking of its water audit data by creating or improving standard operating procedures (SOPs) for data management (refer to Table 8-2 and the audit software's interactive data grading capability and loss control worksheet for guidance on areas to focus on) and 2) identify priority action items to reduce losses of highest volume and/or cost as identified in the water audit. Many small systems have limited staff and budgets, and employees are often responsible for a wider array of functions than their counterparts in larger systems and challenged to prioritize their tasks. However, the AWWA water audit assigns costs to the components of NRW, so and the small water utility can assess which losses are the costliest to its operations. Absolute loss volumes (apparent and real) are also important considerations as is the data validity tier. Carefully evaluating these factors is a good approach to establishing priorities. Table 8-2 lists potential data tracking SOPs to consider, along with suggested action items to reduce NRW over time. The small water utility can select several key SOPs and several loss reduction strategies to implement based upon the priorities that they have assessed. SOPs should be placed in writing and regularly reviewed and updated to keep them current. Strategies to manage unbilled authorized consumption and control apparent and real losses are found in subsequent sections of this chapter.

Table 8-2 SOPs to track and reduce nonrevenue water

Nonrevenue Water Component	Loss Tracking Strategies and SOPs	Loss Reduction Strategies and Action Items
Unbilled Authorized Consumption	• Revise policy to meter and bill unbilled municipal buildings. • Meter as many unmetered utility water uses as practicable (treatment facilities, utility maintenance facility, etc.). • Track quantities used in fire hydrant flushing or automated flushing devices. • Track quantities used for firefighting, fire hydrant testing, and fire department training.	• Take actions to improve system water quality to reduce flushing needs (eliminate dead ends, adjust valves to improve flow in stagnant areas, use a unidirectional flushing program, etc.). • Provide an annual summary of unbilled municipal consumption with estimated costs and volumes for each customer to inform billing policymakers.
UC	• Keep monthly logs of observed unauthorized water consumption. • Review and update policy and SOPs on the use of fire hydrants • Create zero consumption report from the customer billing system. • Estimate water used by customers on zero-consumption reports confirmed as meter failure of UC, based on customer type or previous billings.	• Inspect zero-consumption customers and inactive accounts for water use. • Develop SOPs for periodic inspections for unauthorized connections and standby or emergency use interconnections with other water utilities. • Conduct an account audit in the customer billing system to ensure all addresses have a meter and an account in the billing system. • Implement SOPs for estimating and backbilling UC.
CMI	• Establish a report to track stopped/failed meters. • Implement a customer meter accuracy testing program to test a sample of customer meters annually and record meter accuracy test results for each meter size. • Consider breaking down meter accuracy by residential and commercial meters or similar customer classes.	• Use meter accuracy test data to identify inaccurate meters targeted for replacement. • Establish a meter replacement program with replacements guided by the volume of water loss, the financial impact, or remaining life of the meter population. • Use meter-testing data to determine at what age or cumulative consumption to proactively replace customer meters.

(continued)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table 8-2 SOPs to track and reduce nonrevenue water *(Continued)*

Nonrevenue Water Component	Loss Tracking Strategies and SOPs	Loss Reduction Strategies and Action Items
SDHE	• Review account initiation procedures to ensure that new buildings are not left off the billing roles. • Consider tracking volumes from billing adjustments in billing reports. • Check large meters to ensure meter multipliers are entered correctly.	• Implement SOPs to examine zero-consumption meters each billings cycle. • Consider an annual internal audit of customer billing records. • Consider an independent third-party audit of customer billing records and operations every three to five years.
Real Losses (leakage, breaks, etc.)	• Track leaks and breaks with estimated loss volumes. • Consider mapping repairs and completed work orders on a paper map of the system or in a GIS.	• Conduct periodic leak detection surveys to identify unreported leakage. • Prioritize and address leaks (repair/replace infrastructure). • Consider pressure management to reduce the negative impacts of excessive pressure including high background leakage. • Track time to repair leaks in the work order system to improve repair times.

AWWA promotes a common philosophy for water utilities of all sizes: a water audit should be compiled by the water utility on an annual basis as a standard business practice. Once a basic top-down water audit has been established, the water utility can progress to the use of bottom-up tracking SOPs every year to improve the quality of the water audit data and better confirm the source and extent of losses. For small utilities, water loss control begins with compiling existing records and information into an annual water audit so that losses can be examined on a systematic basis. Once the utility has a reasonable degree of confidence in its estimate of NRW, it can take small systematic steps each year to reduce NRW and capture additional revenue.

MANAGING UNBILLED AUTHORIZED CONSUMPTION

Measuring unbilled authorized consumption can be a significant challenge for small utilities. Some unbilled consumption is inevitable, but many small systems have policies in place that allow municipal or other "grandfathered" buildings to go unbilled and sometimes unmetered. Such arrangements can present a financial and political challenge to the utility wishing to increase revenue.

Unbilled authorized consumption is part of NRW, but this component is not considered a loss since the water utility holds the authority to determine what water uses are permitted to go unbilled. Hence, policy decisions can be a primary factor in managing unbilled authorized consumption. Understanding how unbilled authorized consumption factors into a utility's NRW is helpful in determining which strategies can be implemented to better manage NRW. Confidence in the volume of unbilled authorized consumption is critical to ensure that this volume is not under- or overstated in the water audit. This section provides tips for small utilities to track and better manage unbilled authorized consumption.

- **Meter and "bill" all water use in buildings.** To the extent practicable, all authorized water uses should be metered, and the water consumption should be billed even if the charge for water service is deemed to be reduced or zero (no charge). This policy will ensure that all water users have accounts in the customer

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

billing system, water meters are in place and are routinely read, and their water consumption is tracked, regardless of the rates charged for the water. With reliable consumption data, municipal managers are better armed to search for ways to improve water efficiency in their buildings. A typical example of an unbilled customer is a wastewater treatment facility owned by a municipality that operates the water utility. Such facilities often consume high volumes of potable water. With good water consumption data, the wastewater plant manager may find opportunities to reduce water consumption via efficiency improvements without compromising treatment operations.

- **Utilize portable meters on fire hydrants.** For construction projects, fire department training sessions, and authorized fire hydrant operations, the small utility can require that a portable meter is installed on the hydrant prior to the start of work. Meter readings should be taken at the beginning and end of each workday and recorded in a spreadsheet that tracks unbilled authorized consumption. Some utilities also have success with assigning one or more metered designated fill stations for filling of water tank trucks for construction, landscaping, and similar activities. By using fill stations, unmetered and unbilled uses can become metered and billed.
- **Develop reasonable estimates for unmetered water from field operations.** There are a variety of municipal water uses that may not be feasibly metered, either due to a limited number of portable meters or time limitations. Estimates could be developed by either utilizing a meter for a one-time event or measuring flow through alternative means (such as timing how long to fill a known volume bucket or measuring a velocity pressure with a pitot blade and pressure gauge) to determine a "typical" flow rate for unmetered water uses. Once a flow rate is determined, field operations could keep a record of dates and durations that unmetered water uses took place, and a volume could be calculated with the typical flow values. These uses are typically field operations and can include the following:
  - Water quality and water system flushing
  - Hydrant flow testing
  - Street sweeping, sewer and storm drain flushing, and other general system maintenance
- **Develop reasonable estimates for unbilled unmetered customers.** In situations where it is not feasible to install meters on unbilled customers, estimate consumption based on similar properties when possible. If no comparable properties exist, water consumption estimates could be made based on type of property, expected occupancy, and other factors based on type of customer. AWWA's M6, *Water Meters—Selection, Installation, Testing and Maintenance* (2012), can be helpful in determining the above-referenced factors.

ADDRESSING APPARENT LOSSES

Apparent losses are the nonphysical losses that occur because of SDHE in the customer billing system, CMI, and UC. Apparent losses result in underbilling to customers and the subsequent loss of revenue. A simple way to define apparent losses is water used by the utility customers that is not captured in the billing process. Apparent loss recovery is attractive for water utilities because it results in a higher billed volume and additional

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

revenue recovery. Considerations for apparent loss control that are practical for small systems to implement are presented here:

- Perform annual flowmeter testing and calibration on your source production/master meters. These meters provide the measure of source water volumes into the water audit and should be accurately maintained to provide a reliable water audit. See Appendix A for detailed guidance on the management of production flowmeters and water supply data. This testing regimen should also include flowmeters on any bulk water that is imported into or exported out of the system.
- Perform periodic review of customer billing system data, looking for patterns of strange, inconsistent, or incomplete data: for example, certain customer accounts that record zero or very low water consumption for multiple consecutive billing cycles. This consumption pattern may reflect customer accounts that are vacant properties but might also reveal stopped (nonfunctioning) meters or tampering with meters or meter-reading equipment.
- Make certain that the process flow of the billing system does not inadvertently underbill customers (see Chapter 4 for ways to address SDHE in the customer billing system). Consider drawing a flowchart of the meter-reading and billing process with billing staff and analyzing the procedures and actual practices conducted at each step. Are there steps that you believed were being conducted that are not actually occurring?
- Cross-reference local tax records or address lists of other utilities (electric, gas) to ensure that all occupied buildings in the service area have an account in the customer billing system. Sometimes a water-using building was inadvertently omitted from the billing rolls despite drawing water service.
- Review municipal or utility policy that permits some buildings to go unmetered and/or unbilled. Installing water meters and reading them routinely will track water consumption even if the building is billed a zero charge. High water waste from a toilet leak or other internal plumbing leaks can be discovered in this way.
- Metering is an essential tool for accountability, and all customers should be metered, including public buildings. Customer meter readers should be alerted to monitor for instances of UC, such as tampered meters or illegal connections, as well as meter malfunctions identified from suspicious data. For example, it is not unusual to find unauthorized service connections tapped into unmetered fire lines, and an annual inspection of all fire lines in the system can identify such occurrences. Accounts that consistently register unusually low consumption volumes are worth investigating for UC from meter or meter-reading equipment tampering.
- Initiate a customer meter testing program. Developing an active meter-testing program is an effective tool in combating apparent losses, especially for large water meters. Ensuring that the largest consumers are accurately metered and billed can provide a high level of revenue protection with relatively little effort. Focus on older meters sized 2 in and larger that serve commercial or industrial buildings. Older mechanical meters are susceptible to underregistration as they wear down or accumulate deposits from aggressive water quality. Traditional turbine meters are not designed to accurately register low flow rates, and oversized turbine meters may fail to register significant volumes of flow in some cases. These conditions cost the utility revenue when low flow rates frequently occur.
- Establish, communicate, and enforce a clear policy for the use of fire hydrants. A considerable amount of water is taken illegally from fire hydrants in many water utilities. In small systems in particular, customers, firefighters, and local public officials can assist by reporting illegal activity with regard to hydrant misuse.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Hydrant locks can also be a valuable tool.

- Automatic meter reading (AMR) or advanced metering infrastructure (AMI) may be effective, especially for systems with limited personnel. Effective, efficient meter reading, data capture, and billing are critical for all customer metered accounts, especially with large consumers. Large meter customers not only provide significant revenue to the water utility, but frequent variations in their consumption patterns can exert challenging operational demands on the supply system.

Many other approaches to apparent loss control, as described in Chapter 4, are viable methods for water utilities of all sizes. Because apparent loss recoveries translate to direct revenue recovery, the apparent loss control program can generate funding to pay for further (apparent and real) loss control activities.

ADDRESSING REAL LOSSES

Real losses include leakage from the water distribution system and customer service connections up to the point of customer metering, as well as water main breaks and overflows from treated water storage tanks or reservoirs. Leakage on the customer side of the meter is wasteful but is not considered to be a real loss since this water is usually billed and generates revenue. An active leakage management program will provide systems of any size the benefits of reducing real loss. It is important to note challenges and opportunities that small systems encounter relative to larger systems in managing leakage effectively.

Leakage can often exert a larger proportional impact on a small system than a larger system. A crack in a 6-in-diameter cast-iron pipe leaking at a moderate rate of 65 to 100 gpm may result in reduced pressures and compromised water service in a small system where the leaking pipeline is likely to be a primary supply line in the water distribution system. The same leak may go unnoticed in a larger system where the 6-in water main is one of many same-sized pipes in a grid of distribution mains. While leakage can exert significant impacts in a small system, it is often challenging for small systems to allocate resources to perform active leakage control. Effective leakage management may seem out of reach for many small systems, but there are some advantages that small systems have over their larger counterparts.

- Detecting and repairing one or only a few leaks in a small system could greatly improve its overall accountability, whereas many more leak repairs would be needed to generate the same benefit in a large system.
- Water systems with SCADA can monitor for sudden drops in water levels in storage tanks and in system pressure as a sign of a newly formed leak or break.
- Distribution piping in many small systems is configured independently and could serve as a DMA by monitoring flows closely. Techniques to measure leakage in DMAs could therefore be implemented with less cost than in large systems. Sometimes DMAs may encompass the system's transmission mains and water storage tanks, and design provisions must be undertaken to ensure that water levels in storage tanks can fluctuate within a desired range and still provide adequate pressure for routine supply and fire flow in the DMA. Tank levels should be monitored so that the net change in storage volume over short periods of time (similar calculation as shown in Figure A-15 in Appendix A) can be accounted for when quantifying supply flows to the DMA. (See Chapters 5 and 6 for detailed descriptions of DMAs.)
- The limited length of small water distribution systems means leak detection surveys can be conducted more quickly than in large systems. The lack of large urban

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

centers in most small systems means leak survey work does not necessarily have to be conducted at night to avoid noise from daytime activities and traffic common in cities. The unit costs of conducting leak survey work in small systems might be less than in large systems.

- Techniques like step testing are more easily employed in small systems. Step testing requires operation of selected system valves while monitoring for changes in system flows and pressures. Closing valves in a small area of the distribution piping with a significant leak can produce a noticeable change in pressure and flow into the area and help to localize the location of the leak.

It is also recognized that small systems can also encounter challenges that are not always present with large systems.

- Small systems often have piping existing under unpaved surfaces such as fields or lawns. Although leaks on such piping may surface more readily, locating underground piping and acoustic leak pinpointing is more difficult under these circumstances. Leak detection is also more difficult on nonmetallic (plastic) piping, and plastic piping buried under nonpaved surfaces presents some of the greatest challenges for acoustic leak detection.
- Many small systems lack accurate mapping of the underground piping network, and line location tracing is needed to locate the infrastructure.
- Plastic pipe has been installed in many small systems without tracer wire or tape, making line tracing and acoustic sounding difficult.
- Many small systems conduct leak survey work infrequently and often use contracted leak detection service providers to conduct the work. This makes it difficult to develop and retain in-house leak detection skills of existing water utility staff.
- The budgets for small water utilities are often very limited, and problems not appearing to need an immediate response (i.e., hidden leaks) are often deferred.
- The Water Research Foundation sponsored a project titled *Pipe Location and Leakage Management for Small Water Utilities* (Project 4144) that created technical guidance modules to assist small systems operators in locating their buried infrastructure, identify water loss sources, and locate leaks. The primary output of the project is not a written report but a series of presentation materials suitable for a workshop setting (WRF 2014).

Below are practical and affordable steps that small water utilities can take to establish and maintain reasonable leakage management practices in their water distribution systems.

- Maintain a means of water level monitoring or routine inspections to avoid storage tank or reservoir overflows. Inexpensive level-monitoring instruments and controls are available to monitor tank levels and avoid such events.
- Be watchful of tank levels that struggle to fill over an expected timeframe or begin to drop at an accelerated rate. In such cases, a newly emerging leak may be drawing water away from the tank. It is worthwhile to launch leak detection activity in the area served by the tank.
- Encourage customer meter readers and customers to report possible evidence of unreported leakage (leakage sounds, wet spots, damp cellars, flooded meter pits, etc.) on customer service connection piping or water mains.
- Look to reduce normally high water pressure that is common in the valley areas of systems in hilly or mountainous terrain. Excessive water pressure aggravates leakage and water main breaks. Do opportunities exist to reduce operating pressures?

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

(See Chapters 5 and 6 for discussions on pressure management.)

- Be mindful of the possibility of hidden leaks that may exist on water mains that cross under major highways, streams, rivers, railroad lines, or other major utilities. These are areas where water leaks without surfacing and escapes notice. Perform regular active acoustic listening surveys or inspections of these areas. Other considerations would be step testing, test shutdown, and checking upstream and downstream pressure changes, which may aid in detecting the leaks at such locations.
- Plan and budget for leak detection surveys of the entire water distribution system annually or as often as financial ability allows (up to every three to five years) or when water audits, high system flows, reduced pressure, or other signs suggest increasing leakage.
- Leak detection survey work also provides the opportunity to inspect, clean, or repair valves and fire hydrants in the system. Prior to leak detection surveys, this infrastructure should be inspected with pertinent asset management data collected to identify needed work on valves or fire hydrants. In addition to uncovering the required listening points for the equipment, issues with the infrastructure (such as a rounded valve nut) can be identified and logged for repair.
- SCADA system data assist monitoring of critical water infrastructure (such as tanks and pump stations) and can help water suppliers spot anomalies or leakage in the system.

Many of the approaches to leakage and pressure management as described in Chapter 6 are viable methods for water utilities of all sizes. In addition to cutting production costs, reducing excessive leakage may extend existing water resources and result in continued economic development in communities that are water limited and may avoid or defer the need to develop costly new water resources and treatment infrastructure.

ANALYZING AND PROCESSING WATER AUDIT RESULTS

As shown generically in Figure 8-1 and discussed in Chapter 7, the most basic way to track NRW is to track and compare volumes of water production, authorized consumption, and the calculated difference as NRW. Monitoring the trends in the three volumes across several years can assist utilities in better understanding the system operations and NRW trends. For example, if a utility is experiencing a trend of steady water production but declining authorized consumption (and revenue), increasing apparent loss is likely. This trend is exhibited in Figure 8-1, with a steady drop in authorized consumption in recent years, while the water production volume remains relatively constant. Perhaps customer water meters are losing accuracy from wear. In that case, comparing historical authorized consumption volumes to recent volumes can give insight to the growth of apparent losses. This assessment may serve as justification for increased customer meter accuracy testing to confirm the performance or decline of the meter population.

Conversely, if authorized consumption and the customer population is relatively steady, but the volume of water supplied and NRW are increasing, slowly rising leakage may be the likely culprit. Analysis should consider supply meter degradation or changes in data management.

The same three volumes may also be tracked using daily or monthly average values in a chart like Figure 8-2 with the same exceptions noted for Figure 8-1 where rolling averages may be more representative. Monthly trends can help utilities look at seasonal changes in water loss and consumption patterns that may not be obvious for annual summaries. Monthly trends may be most revealing for small utilities with very short

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

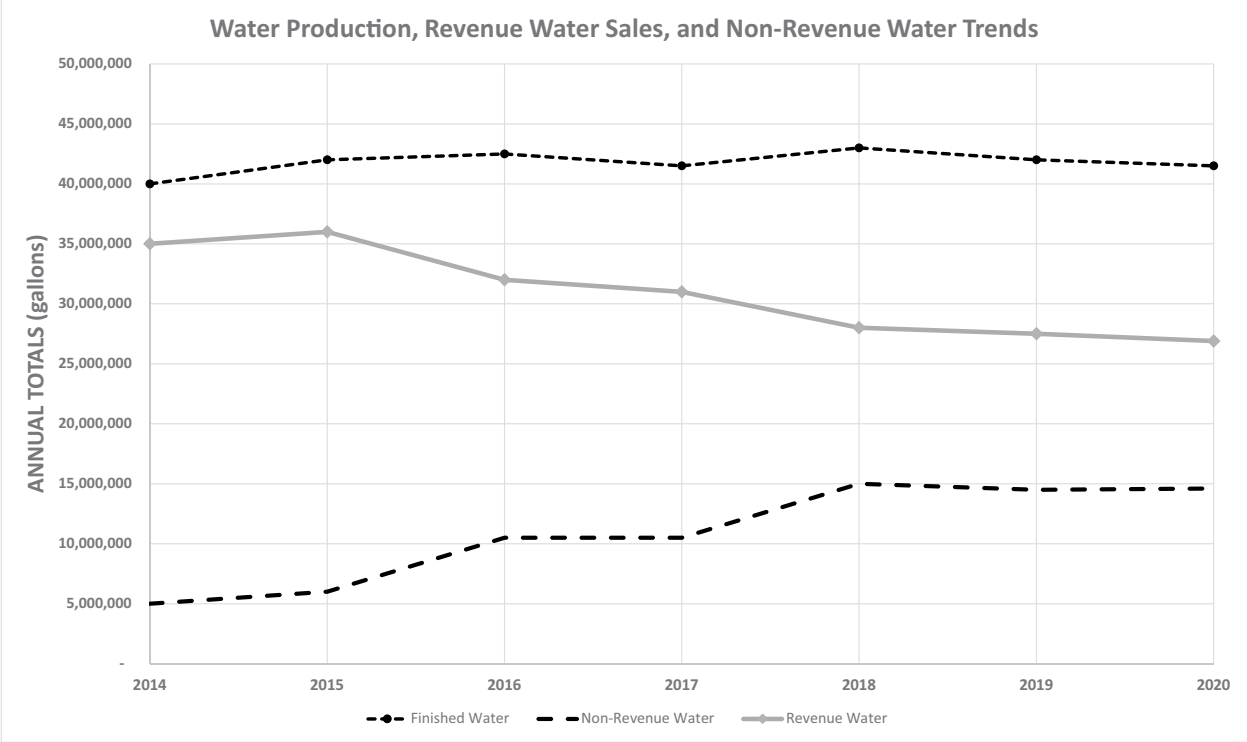


Figure 8-1 Annual volumes of water supplied, authorized consumption, and nonrevenue water

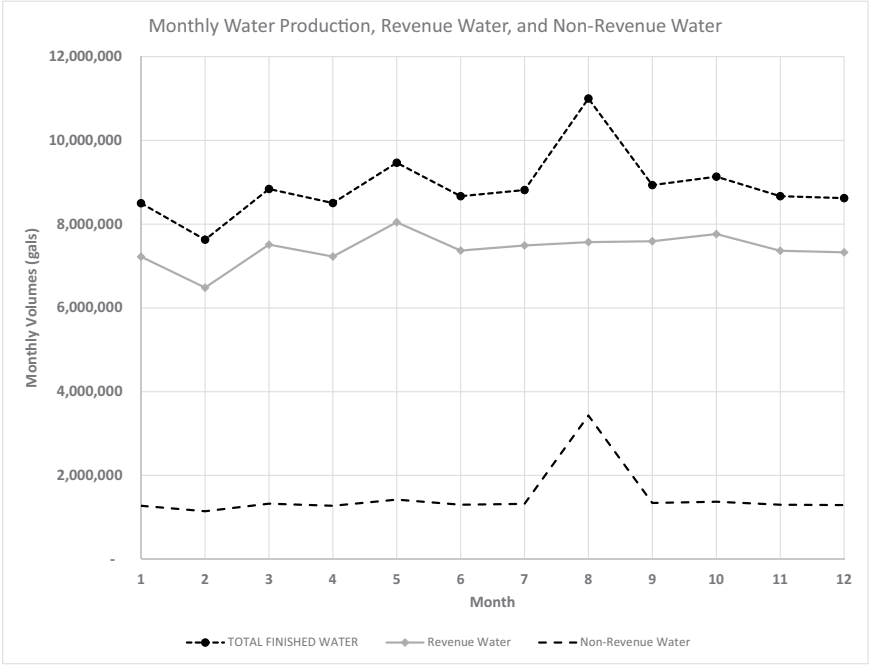


Figure 8-2 Monthly volumes of water supplied, authorized consumption, and nonrevenue water

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

meter-reading cycles. For water utilities that read customer meters throughout the billing cycle rather than over the course of a few days, the authorized consumption data may not align with the volumes of water production, and the billing cycle data might be misleading. For this reason, it is recommended that utilities also create rolling 12-month charts from the monthly data plotted in Figure 8-1. (This procedure is detailed in Chapter 7.)

Small systems often have a unique advantage in that their relatively lower water consumption volumes make it easier to diagnose certain water loss trends. In a large city, a high flow occurring to combat a major fire or loss from a broken water main may be difficult to detect in the data. But in a small-town water system, water consumed in a firefighting or main break incident can often be very visible in the monthly trending report. Figure 8-2 exhibits a notable spike in the monthly volume of water supplied and NRW during August, perhaps an indicator of a temporary large volume of water supplied for firefighting or a main break event. Understanding seasonal and monthly trends can be an important step for small utilities to examine the types of loss in the system or monitor the system for changes over time.

ACCOUNTABILITY IN SMALL WATER UTILITIES

Sound water accountability and an effective water loss control program are important for small water systems to operate efficiently. These are proactive steps that usually offer a financial payback as well as a boost in efficiency as the system ages. The improvement potential is often as strong—or stronger—in small utilities as it is in large systems. The possibility of a relatively quick boost in revenues or reduction in operating costs makes a loss control program particularly attractive for many small water systems.

The greatest challenge is often in getting started. Once an initial effort is launched by starting the water auditing process, opportunities for financial and operating improvements can quickly emerge. The methods and tools offered in this manual are designed to work for water utilities of all sizes but have been carefully structured with small water systems in mind. The AWWA free audit software (detailed in Chapter 2) was specifically designed to be straightforward and user friendly to allow utility staff members to input data and calculate performance indicators readily. With the tools and methods currently available, small water utilities have the potential to quickly assess their water and revenue loss standing and to begin to implement successful measures that will improve their operations while providing service to their customers and protection for valuable water resources. Various mechanisms exist to assist small systems with funding, training, or expertise from government-sponsored programs.

REFERENCES

AWWA, eLearning Program, Controlling Non-revenue Water in Small Water Utilities, (2020) https://www.awwa.org/Professional-Development/Small-Systems#125601037-controlling-non-revenue-water-in-small-water-utilities-courses-1-7

WRF (Water Research Foundation). 2014. *Pipe Location and Leakage Management for Small Water Utilities*. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

APPENDIX A

Validating Production Flowmeter Data and the Annual Water Supplied Volume

This Appendix provides detailed guidance on the methods for verifying the accuracy of production flowmeters and quantifying error in the data trail leading to the determination of the Water Supplied volume in the annual water audit. This discussion includes detailed instruction on accuracy testing for the larger flowmeters that measure water production flows, imported water flowmeters, and exported water flowmeters. It additionally highlights the need to properly balance supply flows across the water distribution system and make adjustments to correct data errors that inevitably occur from time to time.

IMPORTANCE OF PRODUCTION FLOWMETER MANAGEMENT

The term *production* is used to denote the bulk water volumes that constitute the overall water supply managed by the water utility, including the water audit components: *Volume From Own Sources*, *Water Imported volume*, and *Water Exported volume*. The Water Supplied volume is shown in the input Worksheet of the AWWA Free Water Audit Software (Audit Software) in Figure 2-5 in Chapter 2 and is calculated as

$$\text{Water Supplied volume} = \text{Volume From Own Sources} + \text{Imported Water volume} - \text{Exported Water volume} \quad \text{(A-1)}$$

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The Water Imported and Water Exported volumes are considered "custody transfer" volumes, because water is transferred from one water agency to another. The above three volumes typically represent the largest quantities measured by the water utility, usually by the largest flowmeters in the system. The accuracy of the annual water audit relies heavily on accurate measurement and data management of these volumes.

The Production Volumes Are the Most Important Quantities in Ensuring the Accuracy of the Annual Water Audit

The Volume From Own Sources component is the largest quantity that is input into the water audit by most water utilities. For water utilities that import all of their water supply, the Water Imported volume is the largest volume. **In each respective case, this volume is the critically important number in the water audit; and the reliability of all water audit calculations greatly depends on the accuracy of this bulk volume.** Because this is a large overall volume of water, even small inaccuracies translate into sizeable volumes of water in error, which can affect the accuracy of the remaining volume components input into the water audit. It is therefore critical that the water auditor give foremost attention to the management of flowmeters measuring this volume, as well as the data trail for these measured values.

In this Appendix, the term *verification* applies to the process of confirming the measurement accuracy of a flowmeter. *Calibration* refers to activities to ascertain proper function of instrumentation attached to a flowmeter. Version 6.0 of the AWWA Free Water Audit Software defines the desired validation function as onsite, or "in situ", testing of flowmeters to confirm their accuracy as installed.

In addition to confirming the accurate performance of flow metering equipment, the utility should identify and quantify the sources of error in the calculation of the Water Supplied volume. These errors occur in a data trail that can be affected by flowmeter inaccuracies, instrumentation error, data balancing flows in- and out- of water storage facilities, adjustments in computer archiving, and the final reporting that produces the Water Supplied volume.

The water utility's ability to maintain accurate measurement and reporting of water production volumes requires several important functions that should be conducted in ongoing fashion throughout the year for each flowmeter installation:

- Periodic, reliable verification of the measuring accuracy of the flowmeter that serves as the primary device generating the measured volume.
- Periodic calibration and correction of the secondary devices (instrumentation including differential pressure cells or chart recorders) that transfer, and may store, flow data.
- Ongoing surveillance of water production and supply data to detect and correct anomalies that can occur in the data. Such data are usually stored in a supervisory control and data acquisition (SCADA) system and/or other computerized archive, historian, or report generator.

Chapter 2 provides instruction on the data collection required to compile the water audit, starting with the production water supply volumes and adjustments that lead to an accurate quantification of the Water Supplied volume for the audit year. Task 1 in chapter 2 requires that the water auditor verify the accuracy of production flowmeters that generate these measured volumes and include necessary adjustments to the annual volume of production flows. The information provided in this Appendix gives detailed guidance on the methods available to accomplish this task. The reader is also referred to AWWA Manual M33, *Flowmeters in Water Supply* (AWWA 2018), for additional information on types of flowmeters, applications, and field-testing methodologies.

Described below are the means to ensure an accurate quantification of the production volume. Production water quantity data that is well managed according to these best practices will receive a high Data Grading, as calculated by the Audit Software, based on the information provided in the Interactive Data Grading worksheet.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

CONDUCTING FLOWMETER VERIFICATION AND SECONDARY DEVICE CALIBRATION

Flowmeters used to measure bulk supply flows come in a wide variety of sizes, types, and models. For the high flow-rate supplies managed by large water utilities, typical meters include Venturi meters, orifice plates, dall tubes, and magnetic flowmeters. Other meter types include turbine, propeller, and ultrasonic meters. Turbine, propeller, and positive displacement meters may be more common in small water utilities with lesser production rates than those encountered in large water utilities. Insertion flowmeters may offer greater ease of installation and possibly lower purchase cost (than other meter types in sizes larger than 12 inches). These devices can be installed in a standard tapping point in a pipeline, thereby occupying a smaller space within the pipeline. However, flowmeter types such as insertion magnetic meters must be custom sized to the installation to ensure correct positions of sensing electrodes to match the equally spaced concentric zones of flow. These meters also require regular maintenance to keep the sensing electrodes clean. Figure A-1 shows a large full-bore magnetic flowmeter being readied for installation. Figure A-2 shows an insertion flowmeter installed in a pipeline. Both meters use the same principle to measure flow, but they are notably different in size, installation requirements, and—for larger sizes—purchase cost.



Courtesy of the Philadelphia Water Department

Figure A-1 Magnetic flowmeter awaiting installation on a 60-in.-diameter raw water pipeline in Philadelphia, Pa.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of the Birmingham Water Works Board

Figure A-2 Insertion magnetic flowmeter in use on a 30-in. pipeline in Birmingham, Ala.

All meter types have certain advantages in use, accuracy, and cost, but *all* meter types also have limitations. Therefore, water utilities encounter trade-offs of cost, accuracy, ease of installation and testing, and other factors in determining which type of meter to install in an application. ***In any case, it is inherent that no meter is 100 percent accurate under all conditions, for all flow ranges, or for an infinite life.*** Thus, it is important that the water utility regularly verify the accuracy of the flowmeter on an annual—or more frequent—basis. Leading water utilities verify flowmeter accuracy on a semi-annual or quarterly basis. Others only perform verification based on local or state regulatory requirements, at intervals as infrequent as once every 5 to 10 years. Allowing more than one year to pass without verifying meter accuracy is not recommended because meter performance can degrade over time, flow patterns can change, and water consumption of the customer population can change dramatically, making the range of the flowmeter inappropriate to meet the current conditions. Also, data grading in the Audit Software for production components is dependent on how often flowmeters are tested and calibrated— the greater the frequency of these activities, the higher the grading.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

For example, internal corrosion and precipitation can slowly build on the throat of Venturi meters. Current Venturi designs include a ceramic coating to protect again this. However, the buildup of materials on the inside of the older Venturi meters can gradually impair accuracy. Similarly, buildup or sediment in Venturi meter pressure-sensing lines can also compromise the accuracy of the flowmeter measurements. As another example, accurate measurement of high constant flows that were achieved when a flowmeter was installed may not be registered by the same flowmeter when the water supply patterns shift to varying flows with frequent periods of low flow. Such flow-pattern changes are common when communities transition from large industrial economies to service-based businesses and residential developments.

Typically, water utilities install flowmeters for permanent service in supply pipelines to continuously measure and record water supply flows. The cross-sectional view of a typical orifice plate flowmeter is shown in Figure A-3. The metering element, which provides the capability to measure water flow rate, is known as the *primary device*. In most water utilities, production flowmeters are connected to a telemetry system, Meter Data Management (MDM) system and/or SCADA system to provide real-time data to the operator of the water supply system. In such installations, the signal or output generated by the flowmeter is connected to a *secondary device* such as a differential pressure transducer, or DP cell (Figure A-4), that converts the flowmeter measurement into a common electronic signal used in the telemetry or SCADA system. An electronic signal with a range of 4–20 mA (milliamperes) DC is standard in the industry. The signal from the primary device must be configured and processed to properly convert the flowmeter measurement units to flow rate volume units for the given application. Data are time-stamped and archived in the SCADA system and typically available to populate user-configured SCADA output reports. In the absence of a SCADA system, flowmeters may be connected to a circular chart recorder (Figure A-5), digital recorder, or other secondary output device where the

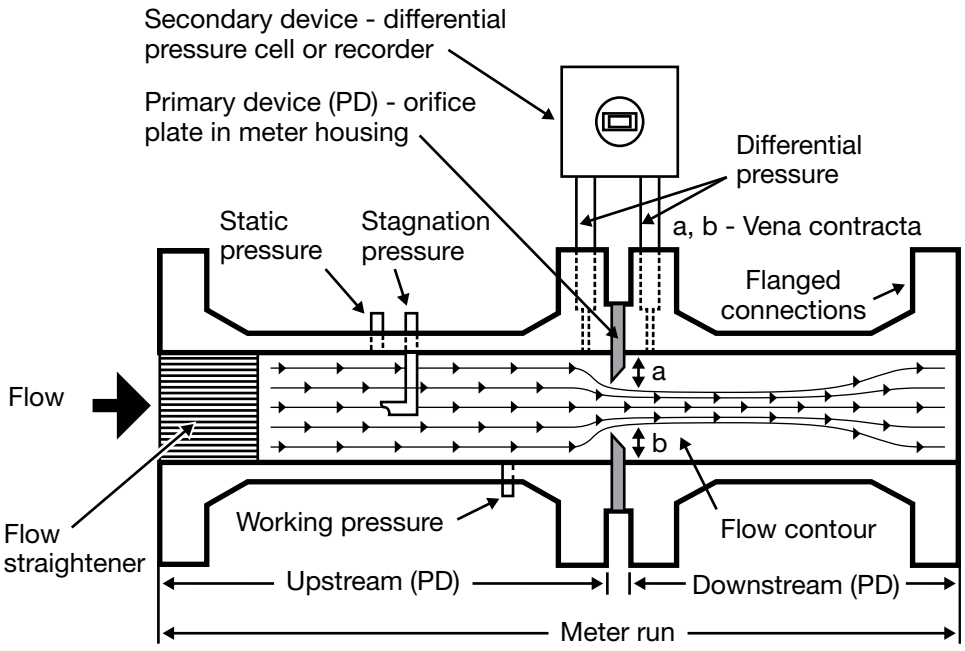


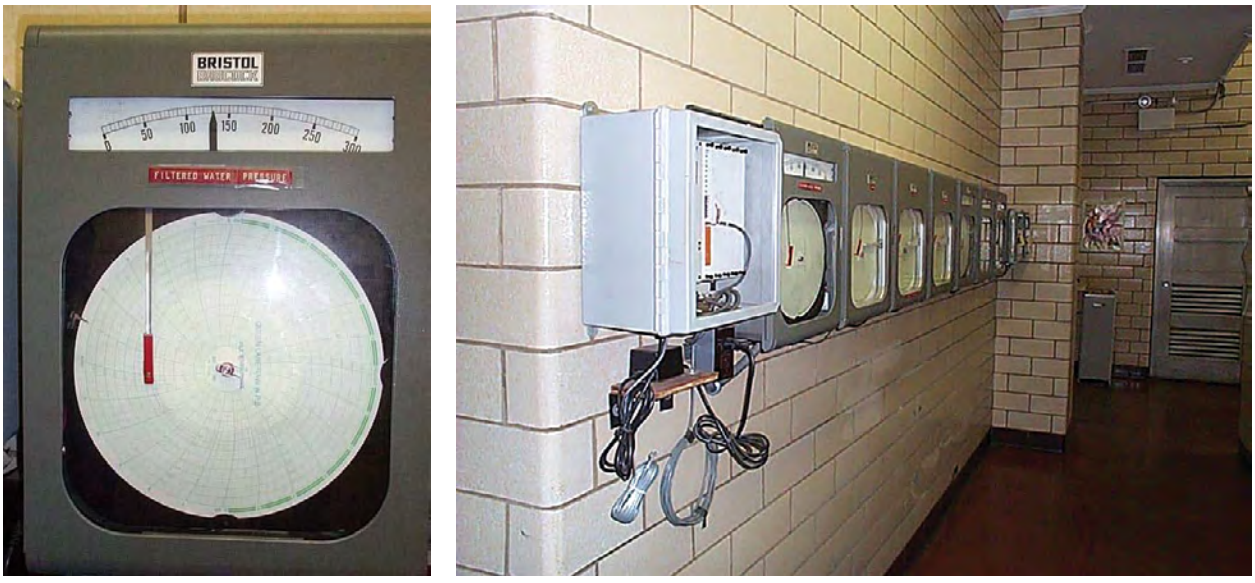
Figure A-3 Cross-sectional view of an orifice plate flowmeter installation

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Courtesy of the Louisville Water Company

Figure A-4 Bank of differential pressure transmitters (DP cells) at the effluent pumping facility of the Crescent Hill Water Treatment Plant, Louisville, Ky.



(a) **(b)**

Courtesy of the Philadelphia Water Department

Figure A-5 Chart recorder and bank of chart recorders at the Queen Lane Pumping Station in Philadelphia, Pa.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

measured value may be displayed. The data will also generally be stored on electronic data loggers for data archiving and analysis purposes.

To validate the annual volume generated by the flowmeter system, it is important to undertake each of the following validation steps:

- **Step 1.** Verify the flow-measuring accuracy of the primary device at least annually.
- **Step 2.** Calibrate the secondary electronic device on a routine basis, once every six months is recommended.
- **Step 3.** Confirm that the output reports of the telemetry system, MDM systems and/or SCADA system match the secondary device output and are accurately configured to balance flows across the supply system. The utility should also ensure that data errors and data gaps that occasionally occur in all systems are systematically identified and corrected on a frequent, routine basis.

Primary and Secondary Device Accuracy and Calibration

Be sure to verify the accuracy of the primary device—the metering element—as well as calibrating the secondary device—the electronic instrumentation transferring the flowmeter signal. Many water utilities are diligent in routinely calibrating secondary (electronic) devices such as differential pressure transmitters (DP cells) or other instrumentation. However, many of the same utilities fail to independently test the accuracy of the primary metering element of the flowmeter. Many flowmeters across the water industry have not been tested for years—or sometimes decades. If technicians "calibrate" the transmitters to produce a reading identical to the measurement from the flowmeter, then any existing error in the flowmeter measurement is carried through the transmitter to the SCADA system. It is essential that the flow-measuring capability of the flowmeter be confirmed with independent measurement on a periodic basis—at least annually—to properly manage these devices and to determine the quantity of Water Supplied Error Adjustment to include in the annual water audit. Technicians should inspect DP cells carefully before calibrating them to make certain that all connections are water-tight and leak-free. Water leaking from a Venturi or orifice meter pressure-sensing line to a DP cell can cause distorted readings.

It is important that the water utility manager undertake all of the above activities to ensure that supply volumes are accurately measured, collected, and stored for inputting to the annual water audit. It is very common for water utilities to conduct Step 2—calibrate the secondary device—as the sole meter verification activity and then assume that they have sufficiently validated the production flow data (see the "Primary and Secondary Device Accuracy and Calibration" sidebar discussion). This electronic calibration action alone does not confirm the flow-measuring capability of the primary device. Calibration also does not account for data configuration errors or data gaps that occasionally occur in the SCADA system data or archived data. It is essential to quantify any measurable inaccuracy in the flowmeter as the first and most important step and include this as a representative annual volume in the Water Supplied Error Adjustment components of the annual water audit.

In addition to verification of the primary device and calibration of the secondary device, it is essential that all telemetry, MDM, and/or SCADA system output reports are accurately configured to present a realistic reporting of production data. The data should be properly balanced across the water distribution system to account for changing water levels in storage tanks, flows transferring across pressure district boundaries, and other unique configuration features of the water distribution system. The data shown in these reports should be considered to obtain an accurate quantification of the annual Water Supplied volume, frequently monitoring and correcting data gaps or errors that periodically occur. Detailed information is provided in this Appendix, particularly on data gaps and other sources of error in reported data.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

SPECIFYING AND INSTALLING FLOWMETERS

In the design and construction of water treatment plants, pumping stations, and reservoirs, the specification and installation of flowmeters can often be viewed as minor details in the myriad of design features of the project. Quite often the task of specifying the flowmeter—a relatively small activity in an otherwise large project—becomes an afterthought, or entirely forgotten, until late in the design process. In such cases, the specification of the flowmeters is often rushed, and the flowmeter type, location, and installation logistics may be less than ideal, resulting in a poorly functioning metering system. Many flowmeters in use today suffer compromised performance because the wrong meter for the application was installed in too cramped a space with no means to verify the accuracy of the meter. Many engineers tend to specify oversized flowmeters based on perceived high, but rarely occurring, peak flows, resulting in a meter installation with built-in inaccuracy and limited means to verify its accuracy or improve its performance. Figure A-6 shows a meter installed in a poor configuration and Figure A-7 shows a Venturi flowmeter adversely affected by a throttled butterfly valve immediately downstream of the flowmeter. "The Problems with Poor Meter Installation Configurations" sidebar explains why these configurations are problematic in terms of compromised accuracy and the inability to test the meter.

Many of the flowmeters in use today are not properly sized, typed, or situated for the application in which they are intended to serve. A guidance document produced by the Department of Ecology of the State of Washington claims that up to 75 percent of installed flowmeters are not performing satisfactorily, and improper selection accounts for 90 percent of the problems users have with meters (Department of Ecology, State of Washington 2015). When embarking on a water infrastructure project, the water utility manager should check

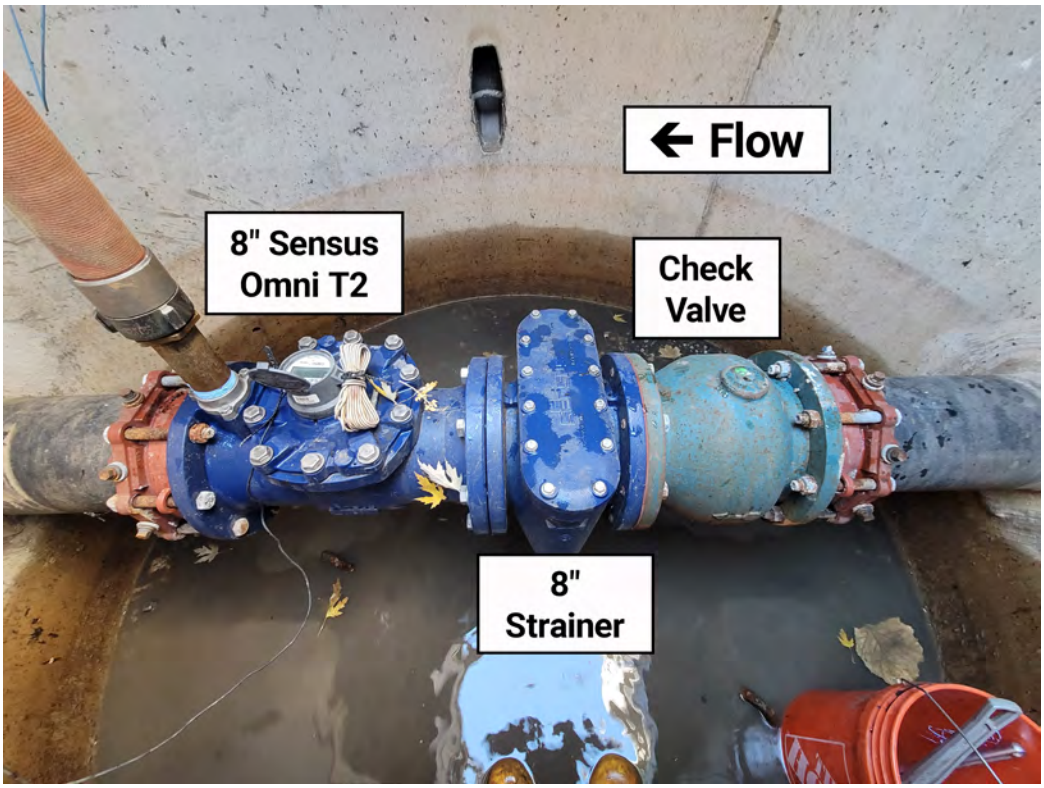


Figure A-6 Flowmeter installation with compression check valve located close to propeller flowmeter.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

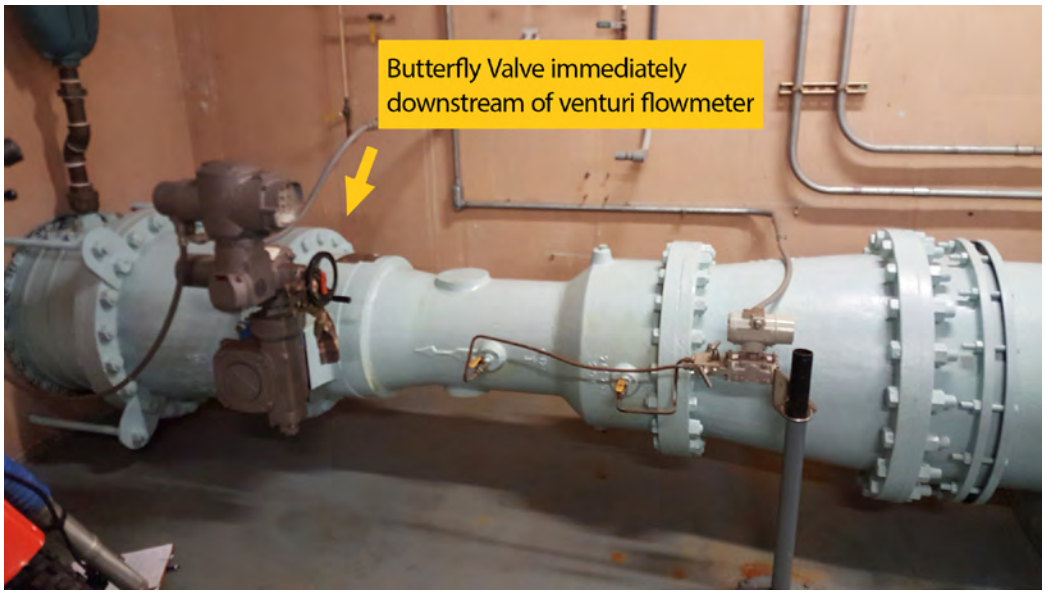


Figure A-7 Venturi flowmeter installation with butterfly valve connected to downstream flange of the flowmeter, and used to throttle flow.

to ensure that sufficient attention is given to the design of the flowmeter early in the facility design process. The design engineer must have a good understanding of the pipeline flow characteristics and issues that can cause distorted flow profiles that can eventually lead to inaccurate flow measurement by the flowmeter. Not all design engineers are well versed in flowmeter technology, and it is important that the designer obtain sufficient knowledge (through research or outsourcing the design) to assemble a competent flowmeter design. The design should also align with manufacturer specifications for flowmeter installation.

The Problems with Poor Flowmeter Installation Configurations

The installations shown in Figure A-6 and Figure A-7 are good examples of flowmeter installations that suffer from poor design, installation or operation, thus compromising the accuracy of the flowmeter. The 12-in. propeller meter installation shown in Figure A-6 is downstream from a compression check valve that is only 12 inches (one pipe diameter) away from the flowmeter. There is a great likelihood that the check value disturbs the velocity profile and could compromise the accurate registration by the flowmeter. Figure A-7 shows a Venturi flowmeter measuring flow leaving the water treatment plant serving a small town. A 24-inch butterfly valve is installed at the outlet side of the Venturi flowmeter. The butterfly valve disc might cause a disruption to the velocity profile and impair the accuracy of the flowmeter. However, in this case, the water utility uses the butterfly valve as a flow control mechanism by throttling the valve to varying valve positions. The partially closed positioning (throttling) of this valve very likely disrupts the accuracy of the flowmeter measurement.

Accuracy and repeatability are desired traits of a well-designed, specified, and constructed flowmeter installation. The engineer must specify the proper type of meter in the correct configuration to attain high accuracy and good repeatability. Ideally, flowmeters should be sited where the flow is relatively undisturbed. Flowmeters sited near bends in pipelines, butterfly valves, or similar devices can suffer compromised performance due to the skewed velocity profiles created by these pipeline features. Good practice requires that flowmeters be sited at an acceptable distance from pipeline features that skew the velocity profile. To achieve a streamlined velocity profile, a flowmeter can be installed at a location with appropriate lengths of straight, unobstructed pipe adjacent to the meter. Table A-1

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-1 Recommendations for desired upstream, unobstructed straight pipe length for common water utility flowmeters*

Flowmeter Type	Length of Recommended Straight Pipe (stated in terms of number of upstream pipe diameters for the application)
Venturi	4–10 diameters—depending on the type of any flow-disturbing obstruction in the pipeline
Orifice	5 diameters
Flow tube	4–10 diameters—depending on the type of any flow-disturbing obstruction in the pipeline
Pitot tube	10 diameters
Propeller	5 diameters
Turbine	10 diameters—assuming a flow-straightening element is used (otherwise 25 to 30 pipe diameters)
Magnetic	5 diameters
Ultrasonic (Doppler shift)	7–10 diameters
Ultrasonic (pulse transmission†)	7–10 diameters (and 5 diameters downstream)

*Information is based on engineering judgment and conservative best practices observed in the water industry by AWWA Water Loss Control Committee members
†Includes transit time flowmeters

includes guidelines for upstream clear distances for various types of flowmeters. Table 3-1 of the M33 Manual also provides spacing guidance for a variety of flowmeters. If it is not possible to allow for desired pipe spacing adjacent to the flowmeter, straightening vanes or strainers may be installed upstream of the flowmeter to condition the flow across the flowmeter. Some meter manufacturers advertise that they manufacture meters requiring little or no adjacent spacing. While technology continues to improve and reduced spacing requirements may be possible, caution should be exercised by the engineer to confirm the reliable performance of the particular meter type before making a purchase. The engineer should check with water utilities that employ the particular type of flowmeter to confirm its function.

Early in the design phase, the engineer should give attention to the flowmeter function, including the expected range and variation of flows, and site and space availability, allowing sufficient space for in-place accuracy testing, instrumentation, and the maintenance needs of the flowmeter system. Once constructed, the flowmeter should be tested for function and accuracy as part of the system commissioning, and startup problems should be immediately rectified. The water utility manager should require that all documentation, including product literature, specifications, and procedures, be submitted to the water utility by the manufacturer or installer. The water utility should keep these documents safely stored and refer to them regularly to help guide its maintenance and testing functions. All meter verification test records should be stored with this literature.

VERIFYING FLOWMETER ACCURACY

Several methods exist for verifying the accuracy of the primary measuring device used in generating a volume of water supplied through a production pipeline. The methods are somewhat varied, but they ultimately provide an assessment of the entire flowmeter system, including the primary metering device, secondary instrumentation, and SCADA system data handling. Refer to AWWA Manuals M33, *Flowmeters in Water Supply* and M6,

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Water Meters—Selection, Installation, Testing, and Maintenance, for further guidance on meter testing options and procedures (AWWA 2022)

The general methods available to quantify the accuracy of production flowmeters are more or less feasible depending on the size, type, and configuration of the flowmeter installation. These methods include the following options:

1. Conduct a water storage clearwell or tank drop test. This method makes use of the availability of the rate of change in level of a known volume in water stored in a clearwell, tank, or reservoir to serve as a comparative measure of flow through a flowmeter. Use of this method relies on having a specific piping-flowmeter-tank configuration and it cannot be employed for all flowmeters.
2. Test the meter in-place by comparing the meter-generated volume to the volume generated by a second, portable calibrated metering device. This can be accomplished by using a trailer-mounted meter test apparatus for production meters in the size ranges of 3-in. to 8-in.; or for larger meters, use of an inline meter (insertion or strap-on) installed temporarily on the same pipeline as the primary flowmeter.
3. Use a meter testing facility. Remove the meter and test it at a meter testing facility. This is generally practical only for meters up to 20 inches in diameter. Some utilities keep two meters in stock, rotating one out-of-service for testing while the second meter is installed in the production pipeline to keep the water supply metered with only a brief interruption to swap the flowmeters. Bear in mind that the conditions at the test facility may not mirror the conditions that the flowmeter experiences when in-service, and test facility accuracy results may be more favorable than the in-situ performance of the flowmeter.
4. Compare volume measurements from the primary meter to other permanently installed flowmeters existing in series within the same supply pipeline or system. By employing a *mass balance technique* to compare this data, unusual variations in flow patterns can be discerned. This technique does not serve as a direct indicator of the level of accuracy of a particular flowmeter, but instead serves to denote if the flow trend recorded by two or more flowmeters in series is in, or out, of control. It therefore can be helpful in indicating when a given flowmeter begins to lose accuracy.

The goal of the above methods is to obtain an assessment of the accuracy of the flowmeter at flow ranges that are representative of actual operating conditions. Accuracy is stated as a percentage representing disagreement of data within a range (AWWA 2018). To obtain a numerical determination of the accuracy of a flowmeter, an appropriate meter testing procedure should be followed. Each of the above methods has certain advantages and limitations in assessing flowmeter accuracy. The basis for selection of the most appropriate of these methods is usually dictated by the attributes of the flowmeter installation and scheduling logistics. A detailed description of each of the above approaches is given below.

Conduct a Clearwell or Water Storage Tank Drop Test

This approach compares a measure of the volume of water leaving a water storage facility (clearwell or tank) over a fixed period of time to the volume of water measured by a production flowmeter for the same period of time. The term *clearwell* is used to denote a treated water storage structure that exists immediately downstream of the treatment works, and upstream of the production flowmeter on the pipeline supplying water to the distribution system. This is a very typical configuration found in many water utilities; however, many varied configurations also occur in the water industry and may or may not

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

lend themselves to conducting a drop test. Like the mass balance technique, this method can only be applied in water systems with a supply configuration that allows the water being supplied to emanate from the storage facility and pass through the flowmeter.

Example of a drop test for County Water Company. An acceptable configuration for a drop test is shown in Figure A-8 with a clearwell at the effluent of a water treatment plant for County Water Company (CWC), the fictitious water system audited in chapter 2. Treated water enters the clearwell where it is briefly stored before flowing out of the clearwell, into the service area pumping station, and then into the supply transmission pipeline, which is equipped with a flowmeter (flowmeter B). In this case, a single flowmeter exists on the supply pipeline; however, it is not uncommon to have two or more supply pipelines—each with a flowmeter, or a single flowmeter on a common header for all of the pipelines—emanating from a clearwell.

A clearwell drop test can be conducted on CWC's system shown in Figure A-8 by halting the flow of water into the clearwell for a limited period of time, generally 2–4 hours. Drop tests are often conducted during night hours when water demand is low and water treatment rates can be halted temporarily, or the supply redirected to other service areas that the water treatment plant might supply. In this way, halting flow for several hours has minimal impact on normal operations. Executing the drop test during the low-demand night hours is also a good way to determine the accuracy of the flowmeter at the low flows that occur during nighttime. This is important since large flowmeters tend to be less accurate at low flows than medium or high rates of flow. As stated previously, however, it is essential that the assessment of flowmeter accuracy include flow ranges that are representative of actual operating conditions. As such, a group of drop tests at various flow ranges should be considered. For many small water systems, treatment is halted for several hours every day based upon distribution system storage tank fill-drain cycles, thus a drop test could be scheduled for a routine treatment stoppage period without modifying normal operating practices.

At the start of the drop test treatment effluent flow must be halted and valves on the influent piping supplying water to the clearwell must be fully closed. Thus, during the

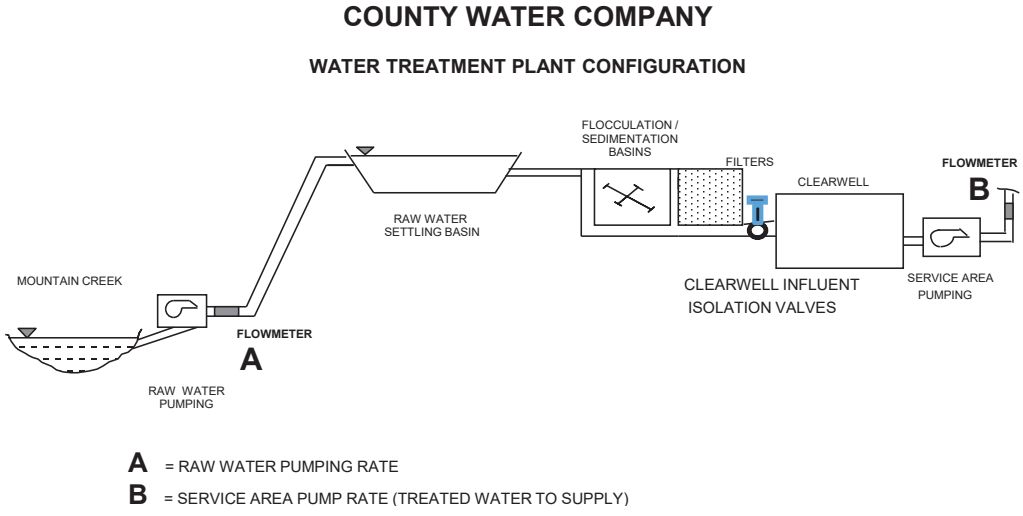


Figure A-8 Schematic of the treated water clearwell and water supply piping configuration for County Water Company's treatment works from the Mountain Creek Source

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

drop test, water only flows out of the clearwell; there is no incoming flow to the clearwell from the treatment works. Hence, the storage level in the clearwell "drops" as water is sent into the water distribution system. Charting the change in water level and knowing the geometry of the clearwell, the water utility manager can calculate the volume of water that exited the clearwell during the test period. This calculated volume is divided by the total time of the test to obtain an average test flow rate. The test flow rate is then compared to the average flow rate generated by the production meter over the same period of time.

The results of the drop test for the primary source of CWC (treatment plant at Mountain Creek) are shown in Table A-2. In this case, the volume of water supplied to the distribution system was determined to be 315,970 gallons over the 3-hour test period. Dividing this volume by 180 minutes of the test period gives an average test flow rate of 1,755.4 gpm. While the test was being conducted, the instantaneous flow rate from the production meter was recorded at each 15-minute interval. The resulting average value

Table A-2 Example tabulations of a drop test conducted at County Water Company's water treatment plant at Mountain Creek

County Water Company **Clearwell Drop Test Analysis for Production Meter Accuracy Testing**				
Date of test: Mar. 31, 2020		Tank height: 18 ft		
Source 1: Mountain Creek		Tank volume: 511,000 gal		
Cylindrical concrete tank diameter: 70 ft		Tank geometry: 28,389 gal/ft		
Time	Clearwell Elevation (ft)	Volume in Clearwell at Given Water Level (gal)	Clearwell Volume Reduction in 15-Minute Period	Production Flowmeter Reading at Each 15-Minute Increment (gpm)
2:00:00 a.m.	17.82	505,892		
2:15:00 a.m.	16.55	469,838	36,054	2,325
2:30:00 a.m.	15.44	438,326	31,512	2,042
2:45:00 a.m.	14.48	411,073	27,253	1,771
3:00:00 a.m.	13.58	385,523	25,550	1,655
3:15:00 a.m.	12.73	361,392	24,131	1,548
3:30:00 a.m.	11.94	338,965	22,427	1,432
3:45:00 a.m.	11.18	317,389	21,576	1,391
4:00:00 a.m.	10.46	296,946	20,443	1,302
4:15:00 a.m.	9.68	274,806	22,140	1,417
4:30:00 a.m.	8.82	250,391	24,415	1,561
4:45:00 a.m.	7.83	222,286	28,105	1,802
5:00:00 a.m.	6.69	189,922	32,364	2,091
Change over 3-hour period	17.82 – 6.69 = 11.13 ft	505,892 – 189,922 = 315,970 gals		
Average flow rate from tank drop		315,970 gal/ (3 hr × 60 min/hr) = 1,755.4 gpm		
Average flow rate measured by flowmeter				1,694.8 gpm

NOTE: Production meter error from the drop test = [(drop test flow rate – meter flow rate) / drop test flow rate] × 100% = [(1,755.9 – 1,694.8)/1,755.9] × 100% = 3.48%, under-registration

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

of the production meter flow rates was determined to be 1,694.8 gpm. Because the test flow rate is higher than the rate produced by the production flowmeter, it is assumed that the production flowmeter is under-registering flow—in this case, by approximately 3.48 percent.

To achieve accurate results from a drop test, several controls must be in place. Because all incoming water into the clearwell must be halted, it is important that influent valves tightly seal water when they are closed. Water "passing" by the valves will compromise the test. All of the water in the clearwell must exit the storage facility via the effluent pipeline housing the production flowmeter being tested. This may require temporarily closing valves on any additional pipelines that also emanate from the clearwell or storage facility. If leaks are known to exist in the clearwell walls, then a portion of the water will leak out of the basin and not pass through the flowmeter. This will compromise the accuracy of the drop test by making it appear that the flowmeter is under-registering the flow. To confirm the existence of leakage from the clearwell, close both influent and effluent valves surrounding the clearwell for a brief period of time (assuming that it is feasible to temporarily resupply the service district from another source). If the clearwell level drops appreciably when all flow is halted, then leakage exists and a leakage rate can be measured over this brief period of time, and the leakage rate can be factored into the drop test results. Ultimately, leaks in the clearwell should be located and repaired.

Lastly, it is critical that the water level measurement is accurate because the calculation of the test volume and flow rate depends on knowing the level change. It is recommended that, prior to conducting a drop test, the level instrumentation be carefully checked, and the level sensor calibrated to ensure that it is providing an accurate measure of the clearwell water level. If necessary, physical measurements of the clearwell level could be taken by using a calibrated "dipstick" inserted into the clearwell and read every 15 minutes. Also, a temporary calibrated pressure gauge, with an attached data logger, can provide accurate data to obtain clearwell level measurements based on pressure head readings. The temporary pressure gauge should be installed inline (with a corporation stop or similar tapping device), upstream from the permanent flowmeter being tested. Physical measures of the clearwell geometry are also advised rather than reliance on construction drawing dimensions. Care must be given to minimize measurement error for both clearwell geometry and level.

For water utilities that have an acceptable configuration, drop tests can be conducted periodically to serve as a reliable accuracy check for production flowmeters. Unfortunately, drop tests are not possible for all water utilities because their systems are not easily isolated between a single storage facility and a single production flowmeter. In other cases, influent supply cannot be feasibly halted, or influent valving does not exist to halt the supply flow. However, the clearwell drop test is a valid means of production meter accuracy testing available to water utilities.

Notes on compound error rates. When testing flowmeter accuracy, it is important to consider the impact of compound error rates when using a portable calibrated flowmeter to compare to a permanently installed flowmeter. For instance, a properly factory-calibrated, portable ultrasonic flowmeter could have an accuracy specification of ±2 percent. When comparing flow data from this meter to that of another meter with similar accuracy specification, the overall error rate could rise to as high as 4 percent. For instance, if precisely 100 gallons of water were passed through two such flowmeters in series, the test ultrasonic meter could display a value between 98 and 102 gallons and still remain within its acceptable range of error based on factory calibration. The permanent flowmeter being tested could report a similar range.

When testing using two meters in series, the person conducting the test generally does not know the exact volume of water passing through the meters and uses the difference between the meters to establish accuracy. In these cases, the person conducting the test may

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

assume that the test meter (portable ultrasonic meter, Pitot rod, or similar device) is the baseline and that any difference between this meter and the permanent meter being tested represents the absolute error. If exactly 100 gallons were to pass through this installation, it is possible that one meter would display 98 gallons and the other would display 102 gallons, and the test would register a difference of 4 percent, when the actual difference is 2 percent.

Perhaps the ideal testing condition is to pass a known volume of water through the flowmeter and compare the known volume with the quantity registered by the flowmeter. Thus, the clearwell drop test and testing of meters in a laboratory using a test tank are two ways of reducing error due to test meter error compounding.

Test the Flowmeter In-Place: Compare Volume to Portable Meter Installation

This is perhaps the most common approach for testing large flowmeters that are not practical to transport to a meter testing facility. The approach relies on the use of a portable calibrated metering device that generates a flow measurement over a defined test period. The volume measurements of the subject flowmeter and the test meter are then compared, with any difference expressed as a percentage. The type of portable meter varies with the size and flow range of the production meter.

Portable meter test apparatus. This approach can usually be feasibly used to test turbine or propeller meters of sizes 3-in. to 8-in. in-place if the flow does not exceed the practical limits of the meter test apparatus. In this way, the production meter test is executed in the same manner as testing customer meters for commercial or industrial buildings. Many water utilities have an existing program for testing of these types of meters for customer accounts, and the same method can be applied for smaller production meters. A calibrated test meter is typically mounted inside a van or truck and water is passed through the subject flowmeter and the test meter in series and the volumes compared. (See Figures 4-16 to 4-19.) Meter testing services also exist to conduct this type of testing on a periodic basis. Details on this type of testing can be found in *Water Loss Control* (Thornton et al. 2008). Because this method is feasible only for relatively small meters, the approach may be applicable for only smaller water utilities that have relatively low production flows and smaller flowmeters, typically of size 8 in. or smaller. Note: To employ this method, test ports must exist on the flowmeter, and normal operational service on the subject water pipeline will be interrupted.

Inline flow comparison using portable test meters. This technique is typically used to determine the accuracy of large flowmeters. Many water utilities in high-density urban and suburban areas supply relatively high volumes of water through metered production pipelines at the effluent of water treatment plants, pumping stations, or reservoirs. The larger size of the flowmeters and the high volumes of flow passing through them make these installations impractical for testing using a portable test apparatus, where water flowed through the test apparatus must be deposed of as waste. Thus, these flowmeters must be tested in-place with water supplied through the meter and host pipeline within the routine expected range of flows.

Inline flow measurement uses a portable calibrated meter inserted into, or strapped onto, the host pipeline, typically downstream of the subject flowmeter to provide a volume measurement in series with the subject flowmeter. The test meter is usually attached to a data recording device so flows can be measured and logged for a period of, minutes, hours, or days. This has the advantage of allowing the flow rate and volumetric measurements to be gathered for an appropriate amount of time. Shorter duration tests (10-30 minutes), at single or multiple flow rates, can be conducted to establish the meter's accuracy within its normal operating range, but such brief periods may not fully capture the performance of the flowmeter. Longer tests can be conducted for trending purposes, to observe the normal

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

variation from high to low flows occurring in the production pipeline over a period of time. To measure daily operational trending patterns, however, it is recommended that inline flow measurements be conducted for a minimum of 24 hours for each flowmeter that is tested for accuracy. When necessary, these tests should be conducted for multiple days if the flow varies widely between weekday and weekend periods. If flows vary widely across the seasons, consideration should be given to conduct distinct inline measurement in each season. If production flows are relatively constant at a given water utility, shorter durations for the inline flow measurements may be considered, as previously stated. For the purposes of test integrity and repeatability, sound judgement should be employed when deciding on the testing duration.

One of the earliest portable flow measuring devices invented for commercial use is the insertion Pitot rod invented in 1896 by Edward Shaw Cole, who later founded a company that provided flow measurement and meter testing services for many years. The Pitot rod is based on the hydraulic principle discovered in 1730 by Henri Pitot who found that the differential pressure in two tubes inserted into the flow in upstream and downstream positions is proportional to the square of the velocity of the flow. Thus, by obtaining a differential pressure measurement, the velocity of flow can be determined. By measuring the cross-sectional area of the pipeline, the flow rate in the pipeline can be determined by using the Continuity Equation given here:

$$Q = V * A \qquad \text{(A-2)}$$

Where:

Q = flow rate (length^3/time)
V = velocity (length/time)
A = cross-sectional area of the pipeline (length^2)



(a) (b)

Figure A-9 Insertion Pitot rods in-place to measure flows for comparison of water flow in large flowmeters.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

The commercially designed Pitot rod is a portable metering device that can be inserted into an active pipeline via a standard tapped ferrule (corporation stop) connection in the pipeline (see Figure A-9). The accuracy of these devices is generally ±2 percent across a wide range of velocities and flows, and they are very useful for testing large flowmeter installations that have no other practical means for verifying flow.

Insertion flowmeters are also advantageous because they can be moved to various depths inside the pipeline and are able to measure velocity across the profile of the pipe. A separate insertion pipe caliper is used to obtain an exact measurement of the internal diameter of the pipe at the point of the measurement. The diameter can then be used to obtain an accurate measure of the pipeline cross-sectional area at the point of flow measurement. Because many pipelines suffer a buildup of corrosion products on the internal walls of the pipe, the actual internal diameter is often less than the internal diameter at the time of manufacture. It is important to obtain an exact measurement of the internal pipe diameter to quantify an accurate measure of the flow in the pipeline.

Once velocity measurements are taken across the pipe, a velocity profile can be plotted. The velocity profile is created by graphically plotting the velocity values measured across the diameter of the pipe. The shape of the velocity profile is dependent on several factors. Under ultra-low velocity conditions, in a very smooth-walled pipe, flow is uniform, and the velocity profile takes the shape of a semicircle (Figure A-10). At higher, measurable velocities, and for many water pipelines that have been in use for long periods of time, internal corrosion products built up on the pipe wall produce the commonly found bullet-shaped velocity profile shown in Figure A-10, with the greatest water velocity occurring at the center of the pipeline and lower velocities nearer to the pipe wall due to the friction effects of the pipe wall surface. The velocity profile can exhibit a nonuniform profile due to a pipeline feature (which may or may not be known to the water utility) that is disrupting the streamlined flow in the pipe. This can be due to devices such as a butterfly valve downstream of the measuring point (Figure A-12) or a partially closed gate valve downstream of the measuring point (Figure A-13). The velocity profile taken at measuring points that are near pipe bends may also produce a nonuniform, or skewed, shape such as that shown in Figure A-14. Flowmeters sited at locations of a skewed velocity profile may suffer compromised accuracy. Thus, knowing the velocity profile is essential to understanding the level of accuracy of the flowmeter.

The *pipe factor* is determined as the ratio of the average of the velocity values across the pipe to the value of the velocity at the centerline. The pipe factor is constant over all rates of

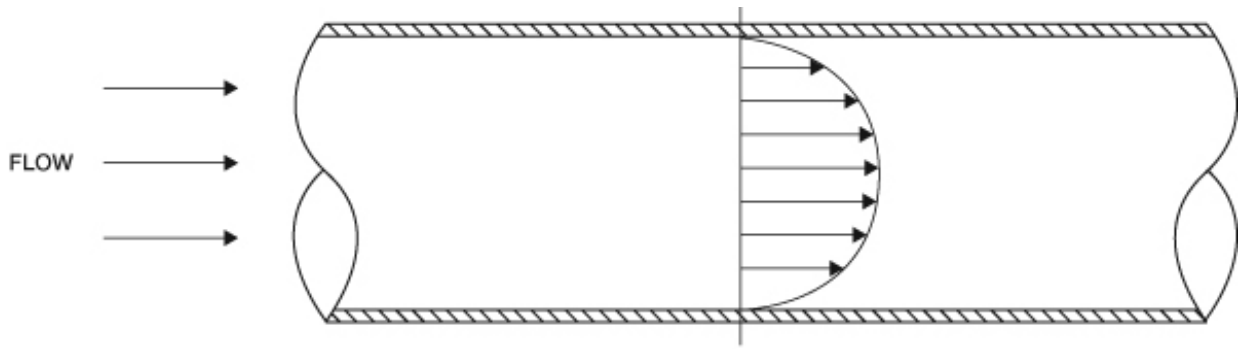


Figure A-10 Velocity profile in a smooth pipeline with unobstructed, uniform flow at ultra-low velocities

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

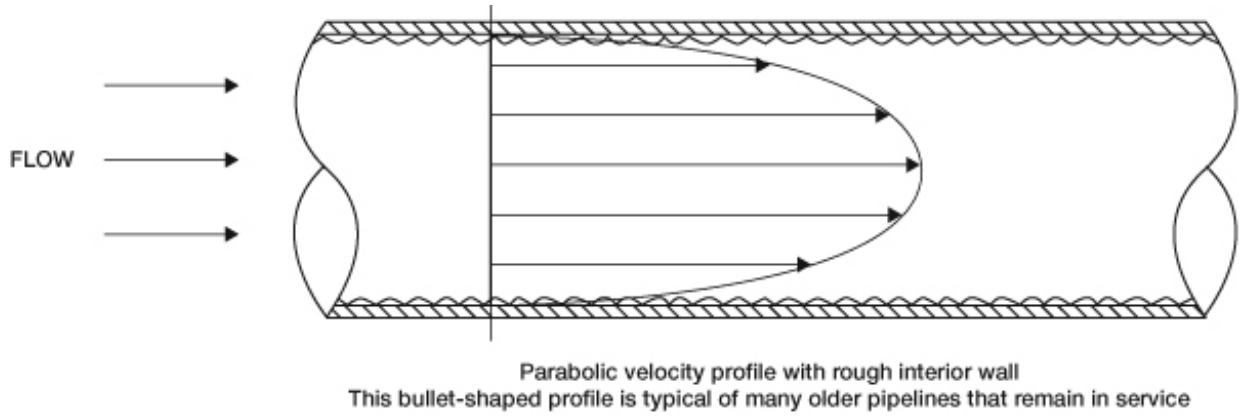


Figure A-11 Velocity profile in pipeline with moderate internal corrosion, typical of measurable turbulent flow

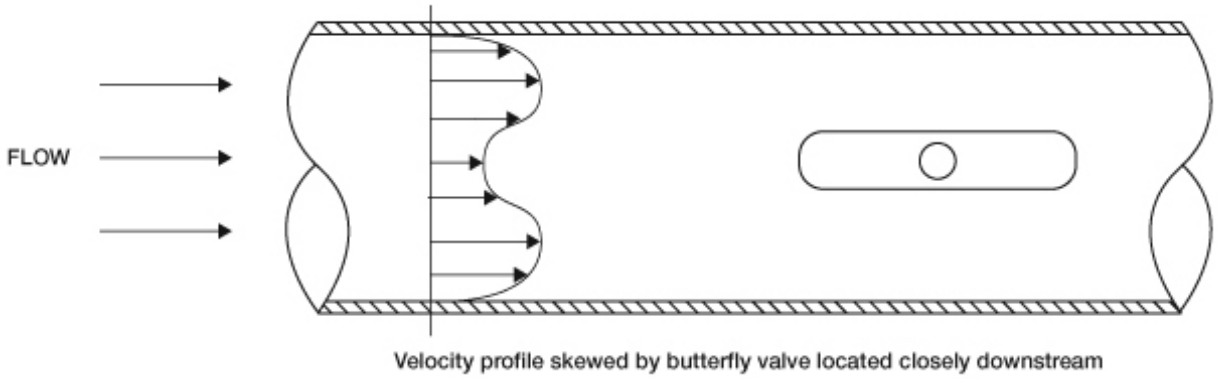


Figure A-12 Velocity profile upstream of a butterfly valve in a pipeline

flow for a uniform flow profile (Walski 1984). The pipe factor is used in the calculations to determine the flow measured during the designated test period. Once all this baseline data is gathered, the test period can be started by inserting the meter probes into the pipeline at the centerline velocity and recording the pressure difference occurring during the test period. At the end of the test period, the differential pressure values can be converted to velocity values and the flow rate calculated for time increments during the test period. The total flow recorded during the test period by the insertion meter can be compared to the volume measured by the subject flowmeter and the difference expressed as a percentage of inaccuracy of the flowmeter. Flow rates at the peak and minimum periods can also be compared.

Because Pitot-rod-type insertion meters are used at standard ferrule (corporation stop) connections on the water pipeline, some water utilities have constructed permanent meter test locations with a ferrule tapped into the pipeline, and a manhole or chamber constructed around the ferrule. In this way, the insertion site is protected and available for periodic use. Like proper location selection for a permanent flowmeter, care and consideration for

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

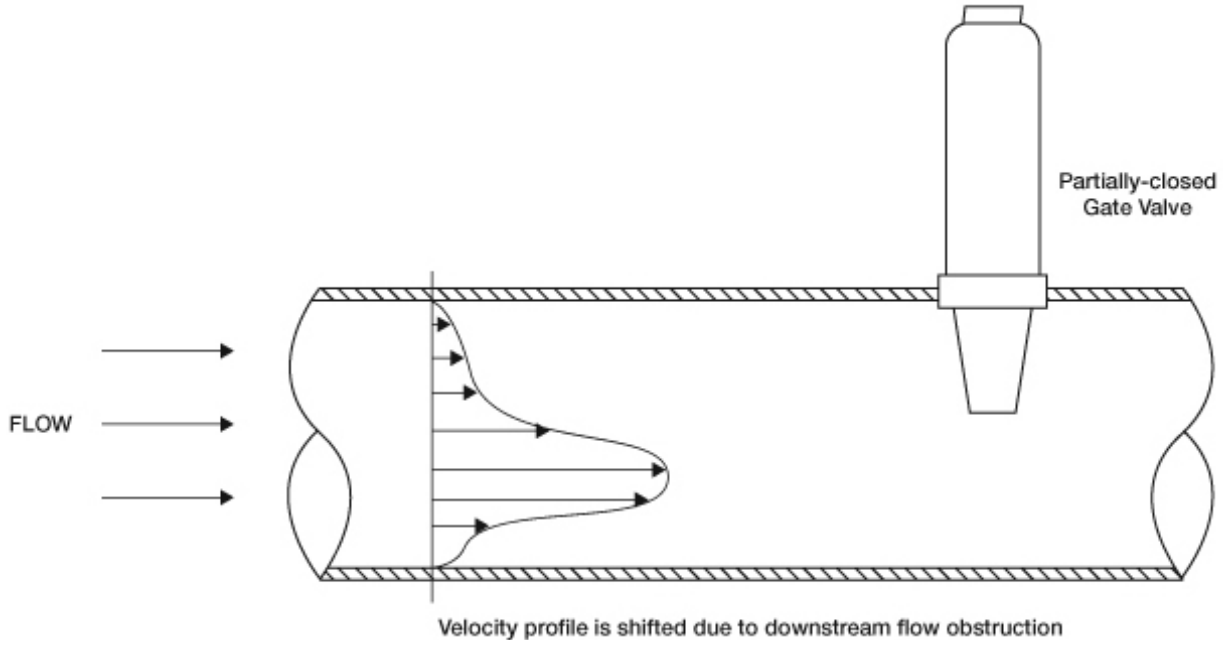


Figure A-13 Velocity profile upstream of a partially closed gate valve in a pipeline

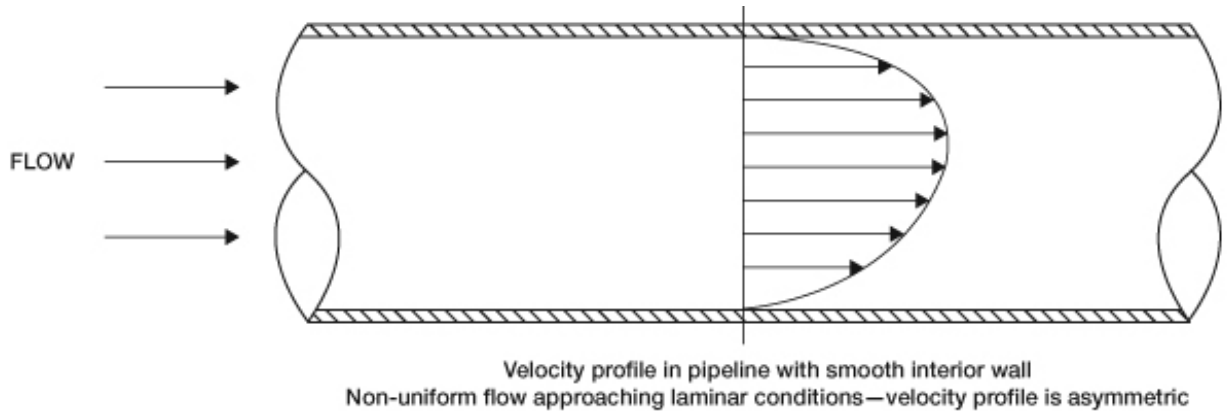


Figure A-14 Velocity profile skewed by close proximity to a bend in the pipeline at ultra-low velocities.

adequate upstream clearance should be given when selecting a location for the test site ferrule(s). Ideally, one test location is installed on each production pipeline downstream of the production meter. Space limitations may not allow a test location on individual production pipelines, but it may be possible to create a test location on a common pipeline, or header, downstream from a bank or manifold of metered pipelines. This is common for production pipelines at water treatment plants and pumping stations. Depending on the configuration of pipelines and meters, it may be necessary to halt flow in individual pipelines by closing line valves or halting individual pumps to force flow through a single

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

pipeline. In such cases, it is important to make certain that closed valves tightly seal water and do not allow any water passage across the valve. Every pipeline configuration has unique features, so the piping and flow patterns should be carefully evaluated in advance of testing to identify the most representative and effective test location and test duration to achieve a meaningful assessment of production meter accuracy.

When conducting inline flow accuracy testing, it is important to assess not only the measurements produced by the flowmeter, but also the entire data trail from the flowmeter installation to all data systems, and final reporting. The signal produced by the flowmeter is typically routed through a secondary device such as a differential pressure cell or other electronic interface, and then to data systems (SCADA, other) which archives the data or transfers it to a separate data historian. Water utility staff typically rely on data system reports that present daily flows in an hour-by-hour (or other time increment) listing. The archived data is ultimately the finished data that is used as the basis to produce the Water Supplied volume in the annual water audit. Water utilities should establish a procedure that confirms the flow quantities created by the flowmeter, as well as the quantity value at the secondary device and the final quantity archived in the data system or historian. It is not unusual to find differences in the value of the same parameter at each of these points in the data trail. A good metering management procedure is given in Table A-3.

While insertion meters are a practical means of obtaining flow measurement in series with an existing flowmeter, "strap-on" ultrasonic meters also offer portability and ease of use in generating flow measurements. The most common type of ultrasonic meter used for drinking water applications is the transit time meter, which uses pulse transmission to detect the velocity of flow. Transit time meters use a pair of transceivers to send and receive an ultrasonic pulse in the direction of the flow of water, followed by a return pulse against the direction of the flow. In a flowing liquid, the speed of the pulse directed downstream is increased by the speed of the stream. When directed upstream, the speed of the pulse is slowed by the flow. The time difference between the two pulse transmissions through the stream is a function of fluid velocity, which can be calculated (AWWA 2018). Each pulse represents a unit of volume. Over the testing period, the pulses will average out, but care should be taken that test time is sufficient to mitigate the effect of "partial" pulses at the beginning and ending of the test. Transit time meters require 7–10 pipe diameters of straight pipe upstream and 5 pipe diameters downstream for proper performance. It is imperative to locate this device where these straight upstream/downstream piping requirements can be met. Ultrasonic meters can be designed for permanently installed use or for portable use as a strap-on device. Users should be cautious about the location of the pipeline to install a strap-on ultrasonic meter since scale or corrosion product buildup inside the pipeline may compromise the accuracy of the meter. The user should have knowledge that the strap-on location is relatively unaffected by internal corrosion or scale buildup. This type of meter is also sensitive to noise and vibration, so care must be taken to locate the meter in a stable, quiet location.

Use a Meter Testing Facility

This approach may be the most precise means to determine the accuracy of a meter at various flow ranges. Several private companies provide reliable meter testing services. These companies have precise test equipment and can conduct meter testing under carefully controlled conditions. Many water utilities also maintain meter test facilities. These facilities, however, are often limited to the smaller-sized meters (less than 12-in.) used for customer accounts and may not accommodate larger-sized production flowmeters. Although this approach can provide highly accurate test results, it has the disadvantage of requiring the effort to carefully remove the meter from service, transport it to the meter testing facility, conduct testing, and return the meter to its location within the

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-3 Production flowmeter system procedure for inline meter accuracy verification

1. Inspect the metering installation. Confirm that the location and piping configuration of the flowmeter and related instrumentation matches as-built drawings. Make note of any differences found at the site.
2. Record information about the flowmeter and installation (model, type, serial number, date installed). It is important to keep detailed records of all equipment, and maintenance and testing activities.
3. Confirm the meter range. This is the maximum output value of the meter and the minimum output value. Are these values properly configured in the supervisory control and data acquisition (SCADA) system?
4. Inspect the differential pressure (DP) transmitters, or cells, if these exist. These devices should be mounted at the same elevation as the meter and near the meter. Check the connections of water pressure sensing lines from the flowmeter to the transmitter. Connections should be secure and not leaking any water. The sensing lines should flow freely when flushing, as blockages will affect the flowmeter performance.
5. Calibrate the DP transmitters and related instrumentation to the SCADA system. Ensure that a signal can be read at the transmitter and the value reliably reported in the SCADA system.
6. Assess the piping configuration. Does a flowmeter exist on each pipeline that extends from the water source (often downstream from a pump or stored water clearwell)? Identify a location to install a ferrule (corporation stop) on the pipeline (for insertion meter use) or a location to install a strap-on ultrasonic meter. If a location downstream from the flowmeter is not available, an alternative location further downstream on a "header" pipeline may be the next best location. The ferrule (tap) location needs to be suitable, ideally with at least 10 pipe diameters of unobstructed pipe upstream and 5 pipe diameters downstream. Avoid locations where the velocity profile is unstable and not fully developed. If a location appears to be unfavorable, look for an alternative location for the tap, or an alternative testing method.
7. Be mindful to monitor any "controls" that are employed to conduct the testing. For instance, it may be necessary to close valves on connecting piping to isolate flow into a single pipeline or header pipe during a flowmeter accuracy verification test. It is important that such valves close fully and do not seep or "pass" flow across the valve. This stray flow might not be measured and would therefore corrupt the integrity of the accuracy test process.
8. By using a Pitot-rod-type insertion meter as the test meter, the internal pipe diameter can be calipered, and velocity measurements gathered across the diameter of the pipe to construct a velocity profile. Confirm that the velocity profile appears like a "bullet" shape. If an unusual shape is obtained for the velocity profile, investigate to confirm whether an obstruction such as a partially closed valve exists downstream of the flowmeter. The function of flowmeters in congested piping locations with hydraulic disturbances may be improved by installing straightening vanes upstream of the meter to smooth the velocity profile of the flow.
9. Conduct the inline flow measurement using the test meter (Pitot rod or ultrasonic), over an appropriate time interval. When possible, the goal should be to capture the full range of flows normally encountered in the pipeline. The longer the duration of the test, the better the test results will be.
10. When the test meter is installed, compare its readings with the signal at the DP cell and the SCADA system to ensure that the portable meter is properly installed and producing a reasonable reading.
11. At the end of the measurement period, collect the data logged by the test meter and compare it to the SCADA system readings (and DP cell reading logged during the process). Note the variance of these values over the test period. Aggregate the data over the test period and obtain the average variance, measured in percentage over-registration or under-registration.
12. Define actionable levels for further action by the utility. For example, findings of ±3 percent (97–103 percent) may warrant no action. Findings of ±6 percent may warrant further inspection of all equipment, recalibration of DP transmitters, and possibly a retest. Findings of more than ±10 percent may warrant the above rechecks, retesting, and consideration of flowmeter replacement, depending on the age of the flowmeter and other factors.

water supply system. Additionally, while testing is likely to be highly accurate, the meter will be tested under laboratory conditions, rather than field conditions. *If factors exist in the field that adversely impact the accuracy of the meter, then this condition will not be present when testing under laboratory conditions at the meter testing facility.* If unusual in-situ conditions are known to exist at the flowmeter installation, this method may not be the most appropriate approach, and one of the other methods described below should be considered instead.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Version 6.0 of the AWWA Free Water Audit Software defines in situ accuracy testing of flowmeters as the desired process that qualifies for a high data grade and testing meters at a test facility results in a low data grading in the Audit Software.

Use of a meter testing facility may be feasible only for production meters up to size 20 in., so this approach will not likely be possible for many medium- and large-sized water utilities with larger production meters. Where feasible, this method of meter testing may be accomplished by rotating a small complement of meters out-of-service periodically for testing on a regular basis.

Compare Volume Measurements from the Primary Meter to Other Permanently Installed Flowmeters Existing in Series

Like the use of inline flow measurement that obtains flow values on the host pipeline in series with the subject flowmeter, this method relies on meters in series. However, this approach is only available to water utilities that have *permanently installed* meters in series. This method is not applicable to water utilities that have only a single flowmeter for the water that they supply to the distribution system. With two or more production flowmeters installed in series, the water utility can compare the values on an ongoing basis in a structured format known as the *mass balance technique*. It should be emphasized that this approach is *not* strictly a meter accuracy verification tool, but instead is an indicator of the level of control in the flow management process. It serves as a good surveillance tool to help detect when data from one or more flowmeters begins to "stray," thereby indicating the beginnings of meter accuracy degradation or other metering system upset. While use of the mass balance technique can assist utility operations, it is nonetheless recommended that the water utility attempt to verify flowmeter accuracy independently via one of the above methods and employ the mass balance technique as an additional monitoring check. Still, the mass balance technique alone offers distinct advantages of production flow validation if it is available to a water utility. It is a very useful approach if accuracy verification of individual meters is difficult. Granted, it is more expensive to install additional permanent meters in series than to use portable meters. Therefore, the water utility manager must decide whether the cost to implement such an approach is worthwhile. In some cases, production flowmeters have been installed in series because source water pumping is located a considerable distance from water treatment works and metering was desired at each step of the process. If metering exists in series—even if the meters are located several miles or kilometers apart—the engineer or operator can employ the mass balance technique.

Example of the mass balance technique. In addition to verifying most of its production flowmeters individually, the Philadelphia Water Department (PWD) employs the mass balance technique to serve as a monitoring and alert system and as a basis for flow adjustments from data produced by certain flowmeters that have a known level of inaccuracy. PWD supplies water from three water treatment plants across a water distribution system that includes more than 3,000 miles of piping. PWD's largest water treatment plant draws water from the Delaware River, while two smaller plants take water from the Schuylkill River. A schematic of the supply configuration for PWD's Queen Lane Water Treatment Plant, which draws water from the Schuylkill River, is shown in Figure A-15. The Queen Lane system includes three banks of flowmeters in series: (A) at the raw water pumping station at the Schuylkill River, (B) between the raw water sedimentation basin and the treatment building of the plant, and (C) at the plant effluent, where the finished drinking water is sent to the water distribution system. Note that the A meters at the Schuylkill River are located approximately one mile from the B and C meters, which are on the grounds of the Queen Lane plant.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

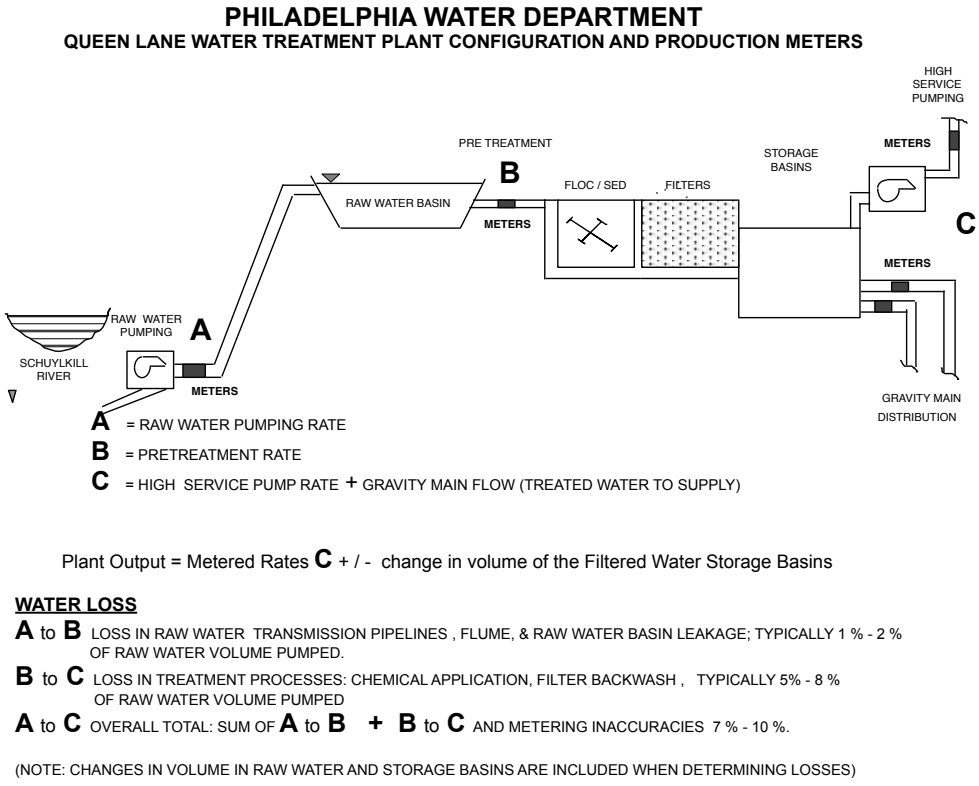


Figure A-15 Schematic of the water supply configuration of the Queen Lane Water Treatment Plant in Philadelphia, Pa.

PWD employs the mass balance technique in a structured manner by compiling production flow data from its SCADA system, which generates a routine mass balance report that is reviewed each business day. A copy of the report for the month of June 2012 is shown in Table A-4. As shown in the report, a daily volume of water measured by the A, B, and C banks of production flowmeters are listed (in million gallons, or MG). The differences in the values from the A to B meters, B to C meters, and A to C meters are calculated and displayed, both by difference in volume and by percentage. The average volumes for the month were reported as 62.03 mgd at the A meters (raw water), 60.97 mgd at the B meters (pretreatment), and 56.42 at the C meters (plant effluent). It is important to recognize that differences in these values are normal and indicative of the process features of the water treatment system. Since the A meters register water "from the source" and are located a considerable distance from the B meters, there is an expected "loss" of water attributed to leakage in several large raw water transmission mains, seepage into a large raw water sedimentation basin, and an inherent level of inaccuracy of the two banks of meters. This difference is typically about 1 mgd or approximately 1.5–2.0 percent of the raw water volume.

The trend of higher volumes from the upstream flowmeters to lower volumes for downstream flowmeters is normal for meters in series, particularly when they are located some distance apart. This is common since pipeline leakage loss or treatment process loss will occur on the piping between the flowmeters. Water volumes passing downstream meters cannot be greater than their upstream counterparts since water cannot be "created" in transit through the production pipeline. If volumes registered by downstream flowmeters

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-4 Mass balance report for the Queen Lane Water Treatment Plant in Philadelphia, Pa., showing consistent production flow metering data reflecting stable operations and good meter accuracy.

PHILADELPHIA WATER DEPARTMENT
QUEEN LANE WATER TREATMENT PLANT—MASS BALANCE REPORT
Volumes shown in million gallons, MG

	A	B	C	A-B		B-C		A-C	
Date	Raw Water Pumping Volume ± Storage Elevation Change	Plant Pretreatment Volume	Plant Effluent Volume	MG	%	MG	%	MG	%
06/01/12	56.82	55.90	51.55	0.92	1.62	4.35	7.78	5.27	9.27
06/02/12	57.69	56.60	52.17	1.09	1.89	4.43	7.83	5.52	9.57
06/03/12	58.16	57.10	52.75	1.06	1.82	4.35	7.62	5.41	9.30
06/04/12	56.77	55.90	51.56	0.87	1.53	4.34	7.76	5.21	9.18
06/05/12	58.50	57.50	53.02	1.00	1.71	4.48	7.79	5.48	9.37
06/06/12	57.82	56.80	52.33	1.02	1.76	4.47	7.87	5.49	9.49
06/07/12	57.03	56.00	52.03	1.03	1.81	3.97	7.09	5.00	8.77
06/08/12	57.19	56.30	51.99	0.89	1.56	4.31	7.66	5.20	9.09
06/09/12	57.92	56.90	52.81	1.02	1.76	4.09	7.19	5.11	8.82
06/10/12	58.01	57.00	52.57	1.01	1.74	4.43	7.77	5.44	9.38
06/11/12	58.39	57.40	53.18	0.99	1.70	4.22	7.35	5.21	8.92
06/12/12	55.89	55.00	50.72	0.89	1.59	4.28	7.78	5.17	9.25
06/13/12	57.46	56.50	52.17	0.96	1.67	4.33	7.66	5.29	9.21
06/14/12	58.11	57.10	52.66	1.01	1.74	4.44	7.78	5.45	9.38
06/15/12	57.96	56.80	52.96	1.16	2.00	3.84	6.76	5.00	8.63
06/16/12	58.03	56.90	52.89	1.13	1.95	4.01	7.05	5.14	8.86
06/17/12	58.10	57.10	52.85	1.00	1.72	4.25	7.44	5.25	9.04
06/18/12	57.90	56.90	52.67	1.00	1.73	4.23	7.43	5.23	9.03
06/19/12	58.71	57.70	53.48	1.01	1.72	4.22	7.31	5.23	8.91
06/20/12	70.99	69.80	64.71	1.19	1.68	5.09	7.29	6.28	8.85
06/21/12	77.24	76.00	70.39	1.24	1.61	5.61	7.38	6.85	8.87
06/22/12	76.30	75.00	69.59	1.30	1.70	5.41	7.21	6.71	8.79
06/23/12	68.68	67.50	62.52	1.18	1.72	4.98	7.38	6.16	8.97
06/24/12	68.94	67.80	62.83	1.14	1.65	4.97	7.33	6.11	8.86
06/25/12	65.07	64.00	59.30	1.07	1.64	4.70	7.34	5.77	8.87
06/26/12	63.07	62.00	57.38	1.07	1.70	4.62	7.45	5.69	9.02
06/27/12	63.80	62.70	58.29	1.10	1.72	4.41	7.03	5.51	8.64
06/28/12	66.82	65.70	60.54	1.12	1.68	5.16	7.85	6.28	9.40
06/29/12	70.85	69.70	64.26	1.15	1.62	5.44	7.80	6.59	9.30
06/30/12	72.75	71.50	66.40	1.25	1.72	5.10	7.13	6.35	8.73
Total	1,860.97	1,829.10	1,692.57	31.87		136.53		168.40	
Average	62.03	60.97	56.42	1.06	1.72	4.55	7.47	5.61	9.06
Maximum	77.24	76.00	70.39	1.30	2.00	5.61	7.87	6.85	9.57
Minimum	55.89	55.00	50.72	0.87	1.53	3.84	6.76	5.00	8.63

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

are greater than upstream flowmeters, then it is a given that downstream and/or upstream flowmeters have appreciable inaccuracies.

The difference between the B and C meters represents water volumes drawn from the process to assist in water treatment operations. Some of the water used for filter backwashing and other plant processes is routinely flushed to waste and appears as a "loss" or difference in the B and C volumes. Under normal operating conditions, the difference between the B and C columns is approximately 4.55 mgd, or approximately 7.5 percent at the Queen Lane plant. On a day-to-day basis, the relative differences of the volumes in the A, B, and C columns are generally constant or experience only a small degree of variation. If a more pronounced variation occurs—either suddenly or incrementally over a period of days or weeks—PWD is effectively alerted to either an upset of the treatment plant process, an increased pipeline leakage loss, or a loss of accuracy in the metering of water volumes in the A, B, or C columns. Note that the differences displayed here are valid for Philadelphia's Queen Lane Water Treatment Plant, but the flow patterns and metering locations at any given water supply system are unique. Therefore, the volume difference data is also unique to that system's configuration and operation.

Table A-5 shows hypothetical data for July 2012 for PWD's Queen Lane plant. Like the data displayed in Table A-4, relatively consistent differences in the A, B, and C volumes are listed for the period July 1–July 24, 2012. However, as highlighted in bold on the report,

Table A-5 Mass balance report for the Queen Lane Water Treatment Plant in Philadelphia, Pa., showing unusual data variation suggesting unstable operations or compromised meter accuracy for the period July 25–31, 2012

PHILADELPHIA WATER DEPARTMENT
QUEEN LANE WATER TREATMENT PLANT—MASS BALANCE REPORT
Volumes shown in million gallons, MG

	A	B	C	A-B		B-C		A-C	
Date	Raw Water Pumping Volume ± Storage Elevation Change	Plant Pretreatment Volume	Plant Effluent Volume	MG	%	MG	%	MG	%
07/01/12	72.05	70.95	65.38	1.10	1.53	5.57	7.85	6.67	9.26
07/02/12	72.28	71.17	65.85	1.11	1.54	5.32	7.48	6.43	8.90
07/03/12	72.35	71.23	66.05	1.12	1.55	5.18	7.27	6.30	8.71
07/04/12	69.86	68.48	63.16	1.38	1.98	5.32	7.77	6.70	9.59
07/05/12	71.85	70.64	65.50	1.21	1.68	5.14	7.28	6.35	8.84
07/06/12	72.05	70.95	65.65	1.10	1.53	5.30	7.47	6.40	8.88
07/07/12	74.80	73.45	68.02	1.35	1.80	5.43	7.39	6.78	9.06
07/08/12	75.28	74.14	69.07	1.14	1.51	5.07	6.84	6.21	8.25
07/09/12	76.44	75.25	70.12	1.19	1.56	5.13	6.82	6.32	8.27
07/10/12	77.02	75.69	70.22	1.33	1.73	5.47	7.23	6.80	8.83
07/11/12	76.24	75.22	69.77	1.02	1.34	5.45	7.25	6.47	8.49
07/12/12	73.82	72.43	67.29	1.39	1.88	5.14	7.10	6.53	8.85
07/13/12	71.62	70.23	65.37	1.39	1.94	4.86	6.92	6.25	8.73
07/14/12	71.28	70.06	65.02	1.22	1.71	5.04	7.19	6.26	8.78
07/15/12	70.33	69.09	64.23	1.24	1.76	4.86	7.03	6.10	8.67
07/16/12	70.11	68.85	63.87	1.26	1.80	4.98	7.23	6.24	8.90
07/17/12	70.08	68.91	63.82	1.17	1.67	5.09	7.39	6.26	8.93
07/18/12	69.88	68.47	63.34	1.41	2.02	5.13	7.49	6.54	9.36

(*continued*)

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-5 Mass balance report for the Queen Lane Water Treatment Plant in Philadelphia, Pa., showing unusual data variation suggesting unstable operations or compromised meter accuracy for the period July 25–31, 2012 (*Continued*)

PHILADELPHIA WATER DEPARTMENT
QUEEN LANE WATER TREATMENT PLANT—MASS BALANCE REPORT
Volumes shown in million gallons, MG

	A	B	C	A-B		B-C		A-C	
Date	Raw Water Pumping Volume ± Storage Elevation Change	Plant Pretreatment Volume	Plant Effluent Volume	MG	%	MG	%	MG	%
07/19/12	69.43	68.22	62.98	1.21	1.74	5.24	7.68	6.45	9.29
07/20/12	68.95	67.77	62.45	1.18	1.71	5.32	7.85	6.50	9.43
07/21/12	68.04	66.92	61.90	1.12	1.65	5.02	7.50	6.14	9.02
07/22/12	67.96	66.63	61.67	1.33	1.96	4.96	7.44	6.29	9.26
07/23/12	69.23	68.11	63.23	1.12	1.62	4.88	7.16	6.00	8.67
07/24/12	69.98	68.44	63.56	1.54	2.20	4.88	7.13	6.42	9.17
07/25/12	70.88	69.56	64.22	1.32	1.86	5.34	7.68	6.66	9.40
07/26/12	72.05	70.88	65.23	1.17	1.62	5.65	7.97	6.82	9.47
07/27/12	73.17	71.77	65.94	1.40	1.91	5.83	8.12	7.23	9.88
07/28/12	74.56	73.23	67.04	1.33	1.78	6.19	8.45	7.52	10.09
07/29/12	76.77	75.22	68.67	1.55	2.02	6.55	8.71	8.10	10.55
07/30/12	75.88	74.48	68.05	1.40	1.85	6.43	8.63	7.83	10.32
07/31/12	74.43	73.13	66.11	1.30	1.75	7.02	9.60	8.32	11.18
Total	2,238.67	2,199.57	2,032.78	39.10		166.79		205.89	
Average	72.22	70.95	65.57	1.26	1.75	5.38	7.58	6.64	9.19
Maximum	77.02	75.69	70.22	1.55	2.20	7.02	9.60	8.32	11.18
Minimum	67.96	66.63	61.67	1.02	1.34	4.86	6.82	6.00	8.25

a small but noticeable increase in the difference between the B and C values is apparent for the period July 25–July 31. This difference is evidenced by an increase in the percentage difference from the A to C volumes. Typically, the difference in the A and C volumes, by percentage, is approximately 9 percent. However, the percentage difference for the period July 25–31 averages 10.3 percent and shows a steady increase across all of these days, reaching more than 11 percent on July 31. This trend suggests that the C volume—the treated water effluent—appears to be under-registering a portion of flow. PWD operates six effluent water production flowmeters in the C bank. The data from the six flowmeters are added to give the C total. It is likely that one of these flowmeters encountered a problem around July 24 or July 25 and ultimately is causing the C total to be lower than expected. Note that the problem was identified by observing the unusual pattern in the percentage difference, not the volume difference. While the suspected problem is likely an under-registering production flowmeter, the total flow for the C flowmeters gradually increased during the late part of July since all water production was increasing, likely due to increasing water demand during hot July weather. If one looked only at the C volume alone, it would not be evident that the C volume was lower than might be expected. Thus, although monitoring water volumes is important, looking at volumes alone may not be sufficient to detect anomalies. By monitoring the percentage difference of the mass balance technique in addition to the volumes, the water utility can identify subtle changes in production meter data.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

PWD's daily review of mass balance data from across its system provides the capability to flag the data anomaly occurring in the C volume at the Queen Lane plant. More detailed data from the six individual C flowmeters can then be reviewed to see if one- or two-meters' data were producing inordinately low readings. Often, in such cases, this is the finding. Technicians can then be dispatched to inspect the flow metering installation to correct any problems that they find in the field. Once corrected, it is expected that the difference in the A, B, and C values will once again conform to the normal variation pattern.

The mass balance technique can be used if two or more permanently installed and well-functioning production meters exist in series. It can be a highly effective means to detect often subtle changes in metered data that reflect an emerging problem with a flowmeter or the subsequent flow data trail. Since relatively few water utilities have multiple meters installed in series, this technique is not available to all water utility managers. However, for those systems that have meters in series—or those that choose to install them—the mass balance technique offers a reliable capability for improved production meter management.

BALANCING FLOWS ACROSS WATER SUPPLY SYSTEM TO OBTAIN THE FINAL WATER SUPPLIED VOLUME

Production flows include the water supplied from treated water sources such as water treatment plants, well fields, and import/export flows. These flows are considered inputs to the water distribution system (except for exports, which leave the system). To determine how much water enters the distribution system on a given day (the Water Supplied volume), the net volume of water going into and out of storage tanks, as well as across multiple zones or districts, must be accurately derived and used to "balance" flows across the entire water supply system.

If a system is small and simply configured with one source of supply and no storage tanks, then the Water Supplied volume is derived merely from the data registered by the flowmeter on the single source of supply. However, if a system is supplied by multiple sources, includes one or more water storage tanks, and/or is segmented into pressure zones or district metered areas (DMAs) that provide some of the metered data, then the net impact of the shifting of flows must be "balanced" across the system to determine how much water entered the water distribution system. Many water utilities operate supply and distribution systems that include multiple pressure zones, DMAs, and water storage tanks or reservoirs while importing and exporting flows. It is not uncommon for even small water utilities to include dozens of inflows and outflows that require accurate tracking to determine the Water Supplied volume for a given day. Because of this complexity, it is good practice for utility personnel to monitor and balance supply flows on each business day. Reviewing such data less frequently—on a monthly or annual basis—results in a strong likelihood that data disruptions will be missed, inducing error into the annual Water Supplied volume. It is also important to conduct regular meter accuracy tests on the flowmeters that measure import and export supplies and flow in/out of water storage tanks and across pressure districts and DMAs. Depending on the system configuration and the location of flowmeters, flow measurement error from any of these flowmeters can upset the flow balancing process across the entire system. Similarly, instrumentation that measures the water level in storage tanks should be regularly calibrated to ensure accuracy in the tracking of water storage levels.

Chapter 2 provides an example of flow balancing from the four water storage tanks for CWC (see Table 2-5). In this example, the water volumes in the four tanks were observed at the beginning of the first day of the audit year and the end of the last day of the audit year (start and end volumes). For utility water storage facilities that "float" on the water

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

distribution system pressure, water levels should be fluctuated on a daily basis to prevent stagnation of the water. Over the course of long periods of time, their net elevation change is little if they are operated in a consistent manner. As an example, many utilities strive to achieve set water storage levels at midnight of each day, and perhaps also at a set morning hour such as 4:00 a.m., prior to the onset of the heavy morning water demand. If operators achieve the same midnight tank level day to day, then there is no net impact to the supply of water to the distribution system. The amount of water that flows into the tank on a given day is offset by the same amount of water that flows out of the tank during the day. Thus, if consistent, repeatable operations are in place throughout the year, the approach used for CWC in chapter 2 is adequate. However, tank operations are not always routine and repeatable for each of the utility's tanks for each day of the year. Thus, it is advisable to use a more detailed tracking, as discussed below.

For most water utilities, variations and/or disruptions in tank operating conditions are a periodic occurrence throughout the year. For example, many storage tanks are taken out of service for inspection and cleaning for several days each year, and the volume of water flushed and filled in the operation should be distinctly tabulated. Occasionally—usually due to the failure of a level sensor or operator error—tanks may be overfilled with water wasted to an overflow pipe until the problem is identified and corrected. Tank level data transmitted to the SCADA system may occasionally be disrupted because of malfunction of the water level sensor or communication link to the SCADA system. In these instances, the storage facility water level data will not be accurate and should be corrected to obtain the true volume of water flowing into or out of the tank on the day(s) of the unusual operations. Therefore, it is good practice to monitor storage tank data every business day. In this way, the circumstances of individual disruptions can be identified and corrections to the data can be made close to the time of the event, with corrected data included in weekly and monthly reports. If data are reviewed on an infrequent basis—or not at all—then inaccuracies caused by storage level data disruption will be missed and the annual Water Supplied volume will be in error.

In addition to water flows into and out of water storage facilities, flows are often routed into and out of pressure zones and DMAs. These flows should also be tracked every business day and problems corrected to achieve the greatest degree of accuracy in the Water Supplied volume. Distribution systems that are highly sectorized with many zones and DMAs often rely on flowmeters tracking flows into and out of these discrete areas. It is common in many utilities to sum the daily flow from all pressure zones and DMAs and compare ("balance") them with the volume total from production flowmeters, along with an adjustment for changes in water storage tank volumes. Since all flowmeters are subject to some degree of inaccuracy, and flowmeter data is subject to occasional disruption, the data from the meters into these sectors should be reviewed on a business-day basis and corrected when such problems are identified. In a general sense, the sum of the flows tabulated from all zones and DMAs should equal the total of the flow supplied to the distribution system from the finished water production flowmeters, including the net change in water storage tank levels. If the total of the individual zones does not equal the total of flow into supply, then metering or data issues likely exist in sector meters and/or production flowmeters, and these should be investigated and corrected.

Table A6 shows an example water supply balancing report for CWC, and Table A-7 shows how data for CWC was disrupted on Sept. 3, 2020. For illustrative purposes, data for a 10-day period in the month of September 2020 is listed, although such a report is usually configured to report data from the entire month. The report lists the flows from the two supply sources, volume into or out of the system's four water storage tanks, and the water imported from a neighboring water utility via the City Intertie interconnection piping. As seen by reading from left to right for each day, the water from the two primary sources is added, the storage tanks volume changes are tabulated (flow out of storage is

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-6 Water supply balancing report for County Water Company

County Water Company—Water Supply Report Quantities in million gallons, MG											
Date	Source 1 Mountain Creek	Source 2 Well Field	Volume From Own Sources	Storage 1 Apple Hill Vol Chg*	Storage 2 Cedar Ridge Vol Chg	Storage 3 Monument Road Vol Chg	Storage 4 Davis Vol Chg	Volume to Supply ± Storage	Source 3 City Intertie Imported Supply	Data Correction	Water Supplied
9/1/2020	12.00	1.10	13.10	+0.04	–0.08	+0.28	-0.02	12.88	1.10	0.00	13.98
9/2/2020	12.18	1.09	13.27	–0.03	–0.03	–0.04	-0.07	13.44	1.04	0.00	14.48
9/3/2020	11.78	1.02	12.80	+0.02	+0.06	+0.05	+0.06	12.61	0.89	–0.30	13.80
9/4/2020	11.68	0.77	12.45	+0.02	+0.11	+0.06	+0.03	12.23	0.86	0.00	13.09
9/5/2020	11.21	0.79	12.00	–0.03	–0.05	–0.23	-0.02	12.33	0.83	0.00	13.16
9/6/2020	11.66	0.81	12.47	+0.04	+0.08	–0.03	+0.13	12.25	0.88	0.00	13.13
9/7/2020	11.58	0.84	12.42	+0.02	+0.04	+0.06	0.00	12.30	0.84	0.00	13.14
9/8/2020	11.62	0.75	12.37	–0.04	–0.06	–0.03	–0.10	12.60	0.94	0.00	13.54
9/9/2020	11.48	0.78	12.26	–0.02	+0.03	+0.05	+0.04	12.16	0.91	0.00	13.07
9/10/2020	11.71	0.88	12.59	+0.03	–0.09	–0.05	–0.03	12.73	0.87	0.00	13.60
Total	116.90	8.83	125.73	+0.05	+0.01	+0.12	+0.02	125.53	9.16	–0.30	134.99
Average	11.69	0.88	12.57	+0.01	0.0	+0.01	0.00	12.55	0.92	–0.03	13.50
Maximum	12.18	1.10	13.27	+0.04	+0.11	+0.28	+0.13	13.44	1.10	–0.30	14.54
Minimum	11.21	0.77	12.00	–0.04	–0.09	–0.23	–0.10	12.16	0.83	0.00	13.07

NOTE: An increase in tank storage over 24 hours means volume is subtracted from supply; decrease in tank storage results in volume added to supply. The data correction shown on 9/3/2020 was due to an overflow event at Storage 3 (Monument Road), which caused water to be wasted to the tank overflow for 5 hours (see Table A-7).

*Vol Chg = Volume Change

Copyright © 2024 by American Water Works Association. All Rights Reserved

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-7 Water storage tank report for County Water Company showing corrections for Monument Road storage tank overflow event on September 3, 2020

County Water Company—Water Storage Tank Report Hourly Water Storage Tank Levels and Volumes for: Sept. 3, 2020										
Time (hours)	Storage Tank 1 Apple Hill 50,000-gal capacity 12-ft depth		Storage Tank 2 Cedar Ridge 300,000-gal capacity 15-ft depth		Storage Tank 3 Monument Road 1,000,000-gal capacity 20-ft depth		Storage Tank 4 Davis 250,000-gal capacity 18-ft depth		Corrected Data (gal)	Comments
0:00	2.13	8,884	6.56	131,228	18.82	941,000	9.21	127,942		
1:00	2.33	9,713	6.73	134,623	18.89	944,382	9.39	130,345		
2:00	2.62	10,927	6.86	137,238	18.95	947,602	9.62	133,561		
3:00	2.96	12,321	7.01	140,178	19.00	950,201	9.81	136,289		
4:00	3.40	14,167	7.19	143,876	19.06	952,945	10.09	140,087		
5:00	3.70	15,423	7.36	147,254	19.12	956,231	10.39	144,321		
6:00	3.90	16,231	7.38	147,567	19.24	961,764	10.41	144,567		
7:00	3.75	15,641	7.34	146,856	19.37	968,295	10.33	143,478		
8:00	3.58	14,923	7.31	146,112	19.47	973,719	10.44	144,987		
9:00	3.54	14,754	7.34	146,897	19.57	978,345	10.54	146,327		
10:00	3.56	14,827	7.41	148,223	19.60	980,112	10.78	149,725		
11:00	3.65	15,221	7.52	150,342	19.60	980,112	11.07	153,765		Tank 3 level disrupted
12:00	3.64	15,153	7.66	153,123	19.60	980,112	11.44	158,927		
13:00	3.76	15,674	7.81	156,234	19.60	980,112	11.77	163,495		
14:00	3.90	16,239	7.96	159,278	19.60	980,112	12.02	166,873		
15:00	4.18	17,428	8.11	162,187	19.60	980,112	12.22	169,652	1,000,000	Tank 3 overflowing
16:00	4.49	18,693	8.24	164,734	19.60	980,112	12.45	172,845	1,055,285	Tank 3 level
17:00	4.62	19,231	8.38	167,549	19.60	980,112	12.67	175,927	1,110,570	Tank 3 level

(continued)

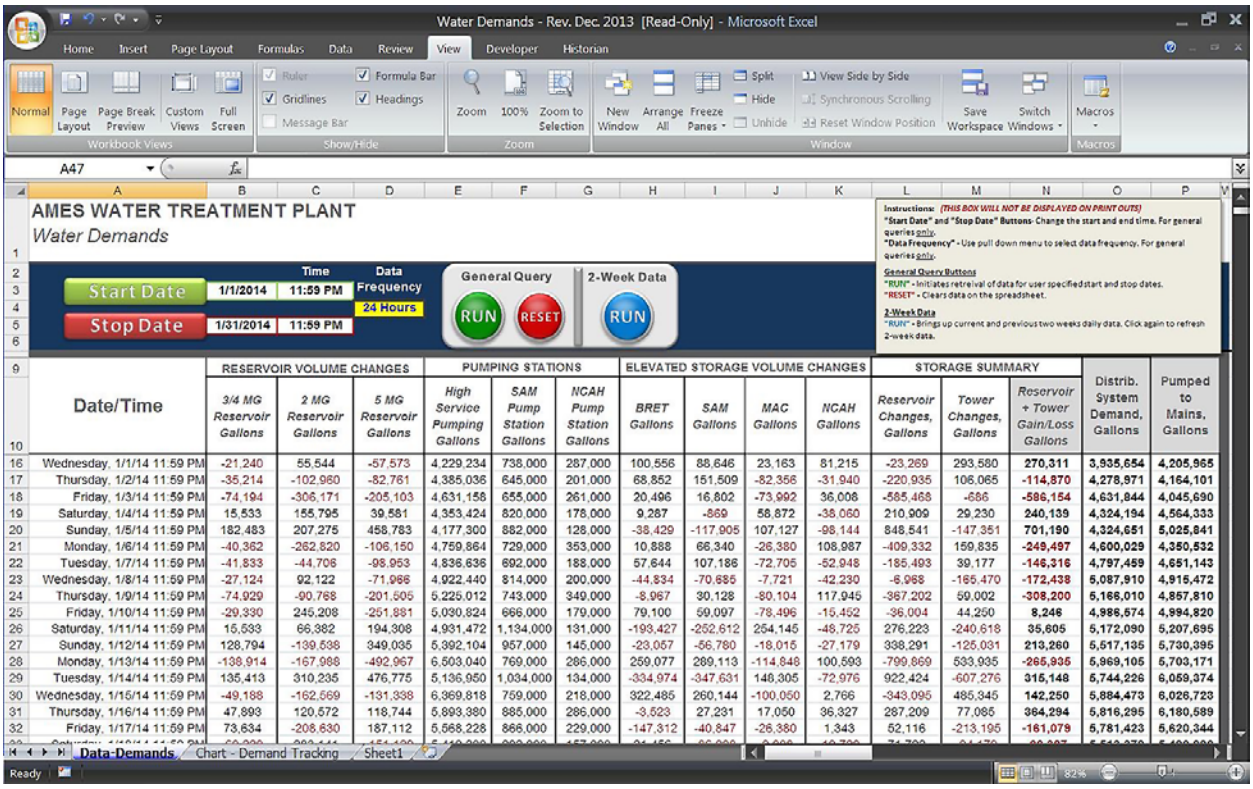
Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-7 Water storage tank report for County Water Company showing corrections for Monument Road storage tank overflow event on September 3, 2020 (*Continued*)

County Water Company—Water Storage Tank Report Hourly Water Storage Tank Levels and Volumes for: Sept. 3, 2020										
Time (hours)	Storage Tank 1 Apple Hill 50,000-gal capacity 12-ft depth		Storage Tank 2 Cedar Ridge 300,000-gal capacity 15-ft depth		Storage Tank 3 Monument Road 1,000,000-gal capacity 20-ft depth		Storage Tank 4 Davis 250,000-gal capacity 18-ft depth		Corrected Data (gal)	Comments
18:00	4.43	18,457	8.52	170,381	19.60	980,112	12.78	177,478	1,165,855	Tank 3 level
19:00	4.62	19,234	8.65	172,987	19.60	980,112	12.91	179,234	1,228,140	Tank 3 level
20:00	4.96	20,672	8.78	175,623	20.00	1,000,000	13.07	181,567	1,296,425	Tank 3 overflow halted
21:00	5.37	22,389	8.95	179,034	19.97	998,256	13.25	183,954		
22:00	5.90	24,567	9.21	184,112	19.89	994,538	13.37	185,678		
23:00	6.36	26,489	9.39	187,896	19.79	989,745	13.45	186,822		
0:00	6.91	28,792	9.58	191,548	19.75	987,634	13.57	188,453		
Tank elevation difference	4.78		3.02		0.93		4.36			
Tank volume difference		19,908		60,320		46,634		60,511	296,425	Tank 3 overflow volume (real loss)

Copyright © 2024 by American Water Works Association. All Rights Reserved

Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66



Date/Time	RESERVOIR VOLUME CHANGES 3/4 MG Reservoir Gallons	2 MG Reservoir Gallons	5 MG Reservoir Gallons	PUMPING STATIONS High Service Pumping Gallons	SAM Pump Station Gallons	NCAH Pump Station Gallons	ELEVATED STORAGE VOLUME CHANGES BRET Gallons	SAM Gallons	MAC Gallons	NCAH Gallons	STORAGE SUMMARY Reservoir Changes, Gallons	Tower Changes, Gallons	Reservoir + Tower Gain/Loss Gallons	Distrib. System Demand, Gallons	Pumped to Mains, Gallons
Wednesday, 1/1/14 11:59 PM	-21,240	55,544	-57,573	4,229,234	738,000	287,000	100,556	88,646	23,163	81,215	-23,269	293,580	270,311	3,935,654	4,205,965
Thursday, 1/2/14 11:59 PM	-35,214	-102,960	-82,761	4,385,036	645,000	201,000	68,852	151,509	-82,356	-31,940	-220,935	106,065	-114,870	4,278,971	4,164,101
Friday, 1/3/14 11:59 PM	-74,194	-306,171	-205,103	4,631,158	655,000	261,000	20,496	16,802	-73,992	36,008	-585,468	-686	-586,154	4,631,844	4,045,690
Saturday, 1/4/14 11:59 PM	15,533	155,795	39,581	4,353,424	820,000	178,000	9,287	-869	58,872	-38,060	210,909	29,230	240,139	4,324,194	4,564,333
Sunday, 1/5/14 11:59 PM	182,483	207,275	458,783	4,177,300	882,000	128,000	-38,429	-117,905	107,127	-98,144	848,541	-147,351	701,190	4,324,651	5,025,841
Monday, 1/6/14 11:59 PM	-40,362	-262,820	-106,150	4,759,864	729,000	353,000	10,888	66,340	-26,380	108,987	-409,332	159,835	-249,497	4,600,029	4,350,532
Tuesday, 1/7/14 11:59 PM	-41,833	-44,706	-98,953	4,836,636	692,000	188,000	57,644	107,186	-72,705	-52,948	-185,493	39,177	-146,316	4,797,459	4,651,143
Wednesday, 1/8/14 11:59 PM	-27,124	92,122	-71,986	4,922,440	814,000	200,000	-44,834	-70,685	-7,721	-42,230	-6,968	-165,470	-172,438	5,087,910	4,915,472
Thursday, 1/9/14 11:59 PM	-74,929	-90,768	-201,505	5,225,012	743,000	349,000	-8,967	30,128	-80,104	117,945	-367,202	59,002	-308,200	5,166,010	4,857,810
Friday, 1/10/14 11:59 PM	-29,330	245,208	-251,881	5,030,824	666,000	179,000	79,100	59,097	-78,496	-15,452	-36,004	44,250	8,246	4,986,574	4,994,820
Saturday, 1/11/14 11:59 PM	15,533	66,382	194,308	4,931,472	1,134,000	131,000	-193,427	-252,612	254,145	-48,725	276,223	-240,618	35,605	5,172,090	5,207,695
Sunday, 1/12/14 11:59 PM	128,794	-139,536	349,035	5,392,104	957,000	145,000	-23,057	-56,780	-18,015	-27,179	338,291	-125,031	213,260	5,517,135	5,730,395
Monday, 1/13/14 11:59 PM	-138,914	-167,988	-492,967	6,503,040	769,000	286,000	259,077	289,113	-114,848	100,593	-799,869	533,935	-265,935	5,969,105	5,703,171
Tuesday, 1/14/14 11:59 PM	135,413	310,235	476,775	5,136,950	1,034,000	134,000	-334,974	-347,631	146,305	-72,976	922,424	-607,276	315,148	5,744,226	6,059,374
Wednesday, 1/15/14 11:59 PM	-49,188	-162,569	-131,338	6,369,818	759,000	218,000	322,485	260,144	-100,050	2,766	-343,095	485,345	142,250	5,884,473	6,026,723
Thursday, 1/16/14 11:59 PM	47,893	120,572	118,744	5,893,380	885,000	286,000	-3,523	27,231	17,050	36,327	287,209	77,085	364,294	5,816,295	6,180,589
Friday, 1/17/14 11:59 PM	73,634	-208,630	187,112	5,568,228	866,000	229,000	-147,312	-40,847	-26,380	1,343	52,116	-213,195	-161,079	5,781,423	5,620,344

Courtesy of the City of Ames, Iowa

Figure A-17 Ames, Iowa, SCADA system water supply tracking report

added to the volume of water supplied; flow into storage is subtracted from the volume of water supplied), and the imported water from City Intertie is added to obtain the aggregate volume of water supplied for the CWC system each day.

The example from Table A-7 illustrates one of any number of minor upsets that can occur in a water distribution system over the course of a year. Utilities should employ a process of routine water supply data review and correction to ensure that the annual Water Supplied volume is as accurate as possible. While Tables A-6 and A-7 are representative of the fictitious County Water Company, Figure A-17 is a screenshot of an actual SCADA system report for the City of Ames, Iowa. As previously described, the key elements of balanced supplies are included in Figure A-17, with pumped flows and changes in reservoirs and storage tanks shown in tabular form with data listed on a daily basis. The general configuration of the report shown in Figure A-17 serves as a good example for water utilities to follow in balancing their supply flows. Such reports, however, must include the system components that are unique to the water utility and should be configured to reliably balance the flows across the water distribution system.

CWC personnel can reliably determine the Water Supplied volume to the system on a daily basis by using the report shown in Table A-6. Anomalies in the data are detected quickly and corrected on a regular basis. Such a problem was encountered on Sept. 3, 2020, when Storage Tank 3 overflowed for 5 hours. Table A-7 is a daily report of the water levels of all four of CWC's water storage tanks, with hourly readings recorded. At 10:00 hours, the level reading for Storage Tank 3 became locked at 19.60 ft due to a disruption in the reading coming from the water level sensor. Unfortunately, the operator on duty did not initially recognize this situation and continued to fill water into Storage Tank 3.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

After 5 hours, the tank—which was already at a high level—reached capacity and began to overflow. This condition lasted for several hours until the event was detected and fill operations halted flow into Storage Tank 3 at 20:00 hours.

Because of the failure of the level sensor and operator error, the recorded tank level for the day was inaccurate, and a certain volume of water was overflowed to waste. Water lost to an overflow is a real loss. Since the event was detected within several hours, the water utility manager reviewed the data and made a correction to account for the volume of water lost in the overflow. This was done by estimating the flow rate into Storage Tank 3 after 10:00 hours. It was assumed that the average flow rate in the tank from 7:00–10:00 hours was maintained throughout the day. Thus, the tank was estimated to be full by 15:00 hours (3:00 p.m.) with an equivalent volume of water continuing to flow until 20:00 hours. The estimated volume of water lost to the overflow was 296,425 gallons, or approximately 0.3 mil gal. This volume was added to the volume of water supplied for Sept. 3, 2020, and recorded as part of the real losses for the audit year.

Because the event was contained to several hours and data on the tank level and overflow volume was carefully tabulated, a correction was added to the volume of water supplied shortly after the incident on Sept. 3, 2020. By providing close monitoring of water supply data, timely corrections to data anomalies can be implemented and a high degree of accuracy of production data can be maintained throughout the audit year. If the water utility merely used the tank levels from the first day of the audit year and last day of the audit year (as shown in Table 2-5), the adjustment needed from the above event would be omitted, causing a degree of error in the annual Water Supplied volume. Although the data anomaly from this single event is small, water utilities may have such upsets occurring periodically at several water storage tanks throughout the year, and the data difference can become appreciable.

VALIDATING SCADA SYSTEM OUTPUT DATA (CORRECTIVE DATA ADJUSTMENTS)

Many water utilities employ SCADA systems—or similar data collection and archival systems—to collect and store data from production flowmeters and other distribution system instrumentation on a continuous, real-time basis. SCADA systems allow operators to closely monitor the performance of the water distribution system and provide the ability to detect system variations and upsets such as fluctuations or loss of pressure, pump failure, power outages, or other common disruptions. SCADA systems also serve as repositories of large amounts of water distribution data that are continuously collected and archived in a historian or other data repository. Sensors are typically installed at key locations in the water distribution system to collect data on water pressure, flowrates, water storage tank levels, and elevations of rivers, lakes, streams, or wells that provide source water. They also monitor system status such as the pump status (on/off), position of large flow-regulating valves, electric power supplies, and many other system parameters. The sensors and related instrumentation are vulnerable to mechanical failure, vandalism, harsh weather, and other factors that can disrupt their performance.

By leveraging the extensive capabilities that are available from most SCADA systems to send alerts and display data in output reports, system operators can learn a great deal about the function of the water distribution system. Ideally, to assess supply operations across the system, output reports should be created to display balanced water flows throughout the entire water distribution system, including production pipeline flows, import/export flows, flow into and out of individual water pressure zones, DMAs, and water storage tanks. A variety of utility reports, such as the mass balance report (see Table A-3), should also be configured and generated on a routine basis.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Depending on the nature of an upset or malfunction that can befall any device, the data generated by the equipment can become skewed or completely corrupted. Hence, even though the hydraulics of the system (flow and pressure) may be functioning normally on a given day, the data delivered to the SCADA system can be in error if a sensor fails or the data collection device malfunctions. One type of disruption is the "data gap" that occurs when a field device malfunctions and temporarily produces no data. Such disruptions happen periodically in all water utilities. Usually, these events are quickly detected by operators, and technicians are efficient in restoring the function of the device and resuming the data stream. But for periods of several hours or perhaps several days, there may be a lack of data reported to the SCADA system. Unfortunately, if the system operator or engineer does not routinely scrutinize and correct SCADA output reports, the reporting will misstate the Water Supplied volume to a particular part of the system. To maintain accurate reporting of water supply data, the water utility should

- give supply data accuracy a high priority in its operations—the data displayed in output reports should accurately reflect the hydraulic conditions in the water supply systems;
- establish written procedures for routine, systematic review of all data, with clear guidelines for making corrective adjustments to all data that are found to be in error;
- configure SCADA system data archival and reporting functions to allow corrective adjustments to the data to be incorporated as needed; and
- fully document the nature of data gaps that occur, as well as the adjustments and description of corrective actions taken to restore the integrity of the reported data.

If the water utility implements a structured process as described above, it will better ensure that the data displayed in the final output reports accurately reflect the water supply conditions occurring in the utility for the reported time period.

Data Gaps in Pumped Supply Reporting and Corrective Adjustments

Table A-8 shows a SCADA system output report that logs pumped flow into a water pressure district on a given day. The data presented in the two left columns (time, flow rate) is typical of a water pressure zone report. However, as part of the example, this table has been expanded to allow corrective adjustments to be displayed by adding two columns that show how data adjustments are used to correct for a data gap that occurred.

In this scenario, pumped flow supplies one of the water pressure zones in a large water utility. The supply typically averages approximately 8.9 mgd but varies from approximately 7.9 mgd during the minimum nighttime hours to more than 9.3 mgd during daytime peak hours. As shown in Column A, the pumped flow reading into the zone became zero at 4:00 a.m. SCADA systems will typically send an alert to the system operator when such a condition occurs. The operator must check other SCADA data and indicators to gain insight to the cause of the zero reading, first by checking other hydraulic indicators for the supply to the zone. In this case, the operator observed the SCADA system indication that the pump remains in the "ON" status, and the water pressure sensors at the pump discharge and key points in the zone are within a normal range. Thus, most indicators show that normal hydraulic operating conditions exist; therefore, the system is most likely still pumping water and all demands are being met. However, the reading from the flowmeter has been disrupted and displays zero. The most likely reason for the zero reading is a failure of the flowmeter or disruption in the communication of the data to the SCADA system. This might occur due to an external problem such as a lightning strike or severe weather impacts, or merely a malfunction of the flowmeter or communication equipment.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Table A-8 Example of a data gap and adjustments for pumped water flow to a typical pressure zone operated by a large water utility.

8/15/2020 (hours)	Column A Pumping Rate (mgd) (raw data transmitted to the SCADA system, showing a data gap)	Column B Pumping Rate (mgd) (actual flow through the flowmeter)	Column C Pumping Rate (mgd) (with corrective adjustments incorporated)	
0:00	8.69	8.69	8.69	
1:00	8.65	8.65	8.65	
2:00	8.32	8.32	8.32	
3:00	8.11	8.11	8.11	
4:00	0	7.94	8	
5:00	0	8.02	8	
6:00	0	8.44	8	**Approximate**
7:00	0	**Data Gap** 8.98	9	**values entered**
8:00	0	9.34	9.3	
9:00	0	9.25	9.3	
10:00	0	9.17	9.3	
11:00	9.12	9.12	9.12	
12:00	9.27	9.27	9.27	
13:00	9.22	9.22	9.22	
14:00	9.08	9.08	9.08	
15:00	8.99	8.99	8.99	
16:00	9.14	9.14	9.14	
17:00	9.18	9.18	9.18	
18:00	9.25	9.25	9.25	
19:00	9.22	9.22	9.22	
20:00	8.82	8.82	8.82	
21:00	8.78	8.78	8.78	
22:00	8.75	8.75	8.75	
23:00	8.71	8.71	8.71	
0:00	8.68	8.68	8.68	
Total of readings	159.98	221.12	220.88	
Average pumped volume	6.40	8.84	8.83	
Difference		2.44	–0.01	

If the operator had found that other hydraulic indicators were also disrupted—such as the pump in the "OFF" status and/or reduced pressure throughout the zone, then the problem would have most likely been a pump failure and lack of water supply to the zone, not a data outage from a malfunctioning instrument. In the above scenario, however, since the operator confirmed that all other hydraulic indicators are normal, water supply is being maintained and customers are reliably receiving water service. Hydraulically, the system is fine, but no flow data is being transmitted to the SCADA system.

As shown in Table A-8, the flow reading went to zero at 4:00 a.m. on August 15. Most water utilities do not have technicians working at that hour of the morning. Hence, the

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

reading remained zero until the start of the workday around 7:00 a.m. or 8:00 a.m. Technicians were provided a work order to investigate the disruption of the flowmeter reading. Over the next several hours, the technicians traveled to the flowmeter site, investigated the failure, and corrected the problem. The correct flow reading was restored to SCADA operations by 11:00 a.m.

Column A in Table A-8 shows the *registered* flow readings recorded in the SCADA system. Column B shows the *actual* flow passing through the flowmeter just downstream of the pump. (Note: These actual values are shown here for illustrative purposes. However, such readings might be available from a local chart recorder or flow totalizer at the pumping station.) Unfortunately, because of the 7-hour equipment outage, the pumped total for the day listed at the bottom of Column A greatly understates the actual flow into the zone, registering only 6.40 mil gal instead of the actual pumped volume of 8.84 mil gal listed in Column B. Thus, the outage understates the flow by 2.44 mil gal. If left uncorrected, the daily flow for the pressure zone will be averaged into the yearly flow balance and will create an understatement of the Water Supplied volume. Fortunately, a straightforward approach exists to correct the data in Column A to create final, archived data that is close to the actual flow supplied to the zone. Corrected hourly values of the flow are included in Column C of Table A-8. These values are approximate flow values that are taken as typical flow rates occurring at the respective hours of 4:00–11:00 a.m. on a typical day. Most system operators are familiar with the daily variations in the water supply data. By looking at pumped flow on several previous days, the operator can enter approximate values for the hours of 4:00–11:00 a.m. The approximate flow rates entered into Column C result in an adjusted 24-hour total of 8.83 mil gal, a difference of only 0.01 mil gal from the actual flow listed in Column B, and much more accurate than a difference of 2.44 mil gal if the data were left uncorrected.

The above example shows how data gaps can create inaccuracies in supply data. Because a data gap, or skewed data (as discussed previously in the discussion of the mass balance technique) can occur at any time, water utility personnel must be vigilant in monitoring supply data on a business-day basis and making corrective adjustments for data that are known to be inaccurate. If a water utility does not routinely review and correct supply data in the SCADA system archive, it will inevitably incur errors in the Water Supplied volume for the year. It will be very time-consuming and inefficient for utility personnel to attempt to review the entire year's records at the end of the year; thus, it is best to review and correct data in an ongoing manner throughout the course of the year.

EVALUATING PRODUCTION FLOW DATA FOR COUNTY WATER COMPANY

Chapter 2 instructs auditors on the step-by-step means to compile the annual water audit of a water utility. The example of the fictitious water utility—County Water Company—is used throughout chapter 2 and other sections of this manual, including this Appendix. The Audit Software Reporting Worksheet, Figure 2-5, is used as the means to enter the data that populates the water audit. Chapter 2 gives instruction to water utilities on gathering and in-putting data and on using the Interactive Data Grading worksheet, Figure 2-6, to grade the data inputs. Standardized performance indicators are calculated and displayed as shown in the dashboard worksheet in Figure 2-7. The Data Validity Score (DVS) for CWC is 67 as displayed on the dashboard in Figure 2-7.

While the basic instructions provided in chapter 2 allow for a reasonable quantification of water production values, the *validity* of the data will be much higher if the water auditor employs the more precise methods described in this Appendix. For example, the audit takes into account data adjustments from changes in distribution system water storage volumes over the audit period, as shown in Table 2-5. This cursory approach only looks at water storage tank levels at the beginning and end of the water audit year. This approach

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

fails to consider any data gaps, tank overflows, draining/refilling operations, and other changes in the normal operation of storage tanks during the year. As described in this Appendix, water utilities can obtain a more accurate adjustment quantity if they monitor water storage tank data on a daily basis and take into account storage data variations that invariably occur throughout the year. If data are reviewed only once each year, the audit will miss these data anomalies and, ultimately, the Water Supplied volume will be a less accurate and a less valid quantity.

SUMMARY

A variety of means for testing the accuracy of production flowmeters is described in this Appendix. Similarly, guidance is given on calibration of secondary instrumentation, balancing flows across the entire water distribution system, and accounting for data upsets. In the example of CWC, the auditor can employ the more extensive approaches to data quantification described in this Appendix. In doing so, all the values entered into the Audit Software will be more accurate and representative of utility operations.

The reliability of the water audit improves in evolutionary fashion if the water utility takes a continuous improvement approach. Monitoring and correcting data on a daily basis certainly involves more effort than more cursory reviews on a quarterly or annual basis. Flowmeter accuracy testing requires effort in time and money. But, if the water utility can gradually implement best practices for production flow data management—even if this occurs slowly over a period of years—the resulting data will be more accurate, trustworthy, and representative of utility operations. The water audit will be a more reliable accountability tool, and it will allow for astute planning of loss control activities and tracking of improvements.

The data generated by production meters and conveyed through the subsequent data trail produce the most important volumes that are entered into the annual water audit: the *Volume From Own Sources, Water Imported volume and Water Exported volume*. Thus, it is essential that water utility managers give priority to the management of primary production flowmeters, secondary instrumentation, data communications and water storage facility data, and accurate reporting that properly balances flows across the water distribution system. Failure to address any of these functions can result in some degree of error in the annual Water Supplied volume quantified in the water audit. Since these volumes are the largest in the water audit, even small degrees of error can represent significant water volumes, and every effort should be made to ensure that the data are as accurate as possible. The guidance provided in this Appendix gives water utility personnel the tools and approaches that will allow them to manage production metering equipment and data to produce highly accurate data for the annual water audit.

REFERENCES

AWWA (American Water Works Association). 2018. Manual M33. *Flowmeters in Water Supply*. Denver, Colo.: AWWA.

AWWA (American Water Works Association). 2012. Manual M6. *Water Meters—Selection, Installation, Testing and Maintenance*. Denver, Colo.: AWWA.

Department of Ecology, State of Washington. 2015. *Liquid Flowmeters—A Guide for Selecting a Flowmeter for Pressurized Systems*. www.ecy.wa.gov/programs/wr/measuring/images/pdf/gsfps.pdf (accessed Jan. 29, 2015).

Thornton, J., R. Sturm, and G. Kunkel. 2008. *Water Loss Control*. New York: McGraw-Hill.

Walski, T. 1984. *Analysis of Water Distribution Systems*. New York: Van Nostrand Reinhold.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

This page intentionally blank.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Appendix B

Assessing Water Resources Management

One of the key benefits of water loss control is the efficient utilization of precious and often limited water resources. Improving efficiency in water use and reducing water loss can help to extend limited water supplies, maintain reliable deliveries to water users, defer the need for costly water supply projects, and help utilities attain a better financial footing to meet future water supply goals. The primary focus of this manual is generally to assess only a portion of the water supply process— the treated water in utility distribution systems that is conveyed to customers. However, utilities can also use water auditing techniques to examine the entire water production system, from the withdrawal of water from its original source, through the treatment process, and continuing to the customer's end use. Losses can be assessed on both untreated source (raw) water upstream of the treatment plant, losses at the treatment plant itself, and losses within the distribution and customer metering systems. This can help a utility to develop a better understanding of water losses throughout the whole production system, and better manage limited water supplies to meet conservation and water resource management goals.

AWWA provides guidance on the management of water resources for regions and communities in its guidance manual *Water Resources Planning* (M50; AWWA 2017). AWWA Manual M50 provides insight to the value of water resources, resource development, water reuse, and recycling. Note that the AWWA water audit methodology can also be applied to recycled (reclaimed) water distribution systems but these system audits are not detailed in this manual.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

AUDITING THE UNTREATED (RAW) WATER SUPPLY VOLUME

Most water utilities obtain untreated (raw) water volumes from a natural water source or receive water from a regional water supply system such as a large reservoir. Water utilities with an untreated water supply typically provide a water treatment process to produce water of acceptable drinking water quality, and then deliver that water through the water distribution system to consumers. The standard AWWA water balance shown in Figure B-1 does not consider water losses in the raw water transmission system or the treatment process. For many water systems, the treatment works are located close to the raw water source, and losses in raw water conveyance are minimal. However, some utilities utilize large water supply networks such as aqueducts or transmission pipelines conveying untreated water for long distances—up to hundreds of miles—with losses of untreated water occurring in these systems. Some of these systems are operated by separate water authorities – or wholesale suppliers – and a separate water audit may be conducted specifically on these systems (see discussion in Chapter 2). For such sizeable systems, even relatively low rates of loss can translate to a significant volume of missing water. Additionally, with the advent of affordable membrane treatment systems, many utilities are now employing reverse osmosis and similar treatment processes that experience significant losses in the treatment process. Thus, many utilities and wholesale suppliers can benefit from looking more closely at raw water supply and treatment losses in the water audit process.

Depending on the specific system configuration, the water auditor has the option to either (1) compile a separate audit on the raw water process or its components (e.g., raw water and treatment plant) separately from the treated water system, or (2) compile a single audit that takes into account the losses in the raw water network and the treatment process as well as the treated water network. In the second case, a modified water balance

<table>
<tr><td rowspan="12">Volume from Own Sources (corrected for known errors)

Water Imported (corrected for known errors)</td><td rowspan="12">System Input Volume</td><td>Water Exported (corrected for known errors)</td><td colspan="3">Billed Water Exported</td><td>Revenue Water</td></tr>
<tr><td rowspan="5">Water Supplied</td><td rowspan="4">Authorized Consumption</td><td rowspan="2">Billed Authorized Consumption</td><td>Billed Metered Consumption</td><td rowspan="2">Revenue Water</td></tr>
<tr><td>Billed Unmetered Consumption</td></tr>
<tr><td rowspan="2">Unbilled Authorized Consumption</td><td>Unbilled Metered Consumption</td><td rowspan="4">NRW</td></tr>
<tr><td>Unbilled Unmetered Consumption</td></tr>
<tr><td rowspan="7">Water Losses</td><td rowspan="2">Apparent Losses</td><td>Systematic Data Handling Errors</td></tr>
<tr><td>Customer Metering Inaccuracies</td></tr>
<tr><td rowspan="5">Water Supplied</td><td rowspan="3">Real Losses</td><td>Unauthorized Consumption</td><td rowspan="4">NRW</td></tr>
<tr><td>Leakage on Transmission and Distribution Mains</td></tr>
<tr><td>Leakage and Overflows at Utility's Storage Tanks</td></tr>
<tr><td></td><td>Leakage on Service Connections up to the Point of Customer Metering</td></tr>
</table>

Figure B-1 Standard AWWA water balance

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

<table>
<tr><th></th><th colspan="5">Raw Water Losses and Treatment Losses</th><th>Raw Water Losses</th></tr>
<tr><td rowspan="10">Volume from Own Sources (corrected for known errors)

Water Imported (corrected for known errors)</td><td rowspan="3">System Input Volume</td><td>Water Exported (corrected for known errors)</td><td colspan="3">Billed Water Exported</td><td>Revenue Water</td></tr>
<tr><td rowspan="4">Water Supplied</td><td rowspan="3">Authorized Consumption</td><td>Billed Authorized Consumption</td><td>Billed Metered Consumption</td><td rowspan="2">Revenue Water</td></tr>
<tr><td>Billed Unmetered Consumption</td></tr>
<tr><td rowspan="2">Unbilled Authorized Consumption</td><td>Unbilled Metered Consumption</td><td rowspan="4">NRW</td></tr>
<tr><td>Unbilled Unmetered Consumption</td></tr>
<tr><td rowspan="7">System Input Volume</td><td rowspan="3">Water Losses</td><td rowspan="2">Apparent Losses</td><td>Systematic Data Handling Errors</td></tr>
<tr><td>Customer Metering Inaccuracies</td></tr>
<tr><td rowspan="3">Water Supplied</td><td>Unauthorized Consumption</td><td rowspan="4">NRW</td></tr>
<tr><td rowspan="3">Real Losses</td><td>Leakage on Transmission and/or Distribution Mains</td></tr>
<tr><td>Leakage and Overflows at Utility's Storage Tanks</td></tr>
<tr><td></td><td>Leakage on Service Connections up to the Point of Customer Metering</td></tr>
</table>

Figure B-2 Modified AWWA water balance showing raw water withdrawal, utilization, and losses

can be used, as shown in Figure B-2. In either approach, supply and production metering accuracy, real (leakage) losses, evaporation from open conveyance systems (canals, aqueducts), and treatment process losses should be considered in the audit. The auditor should be mindful to calculate a separate cost to value the raw water losses, because raw water has less value than treated water. The water balance in Figure B-2 closely matches that used by several state and regional water oversight agencies in the United States.

Assessing raw water production adds steps to the water audit process, and most utilities can compile a useful water audit based solely on finished water delivered to the distribution system. In some cases, however, taking the raw water and treatment system into account in a water audit can provide a more detailed and meaningful water audit and improve a utility's understanding of its water balance. Utilities with the following conditions should consider assessing raw water and treatment losses as part of their water audit process:

- Any utility with a significant discrepancy between raw water production or imported water volumes and the volume of finished water delivered to the distribution system;
- Utilities with high-loss treatment systems, such as reverse osmosis;
- Utilities with substantial finished water storage prior to the measured point of entry in the distribution system;

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

- Utilities or wholesale water suppliers with extensive raw water transmission networks;
- Utilities with multiple types of water sources or treatment systems for one distribution system;
- Utilities with limited raw water allocations, limited water supplies, or projected water supply shortfalls in the utility's planning horizon;
- Utilities with expensive raw water costs or utilities purchasing raw water from a regional supply system; or
- Utilities that make annual or periodic condition assessments of raw water conveyance facilities and/or treatment works, where water auditing may be integrated with asset management plans.

A raw water audit may not require the use of a standard water audit tool but can simply be a water balance on the raw water production and treatment system. Regardless of the tools used to audit the raw water system, the goals of a raw water audit are: 1) provide an accurate finished water input volume for the distribution system water audit, and 2) quantify raw water losses including leakage from pipelines, leakage and evaporation from open channel conveyance and storage systems (canals, aqueducts, reservoirs, etc.), metering inaccuracies, and treatment losses in the water treatment plant.

In many facilities, a portion of the annual raw water withdrawal volume does not enter the distribution system as finished, treated water. These are the *raw water system losses* and are comprised of *raw water losses* (detailed above) and *treatment losses*. Equations B-1 and B-2 give the mathematical relationship between the three components.

Treatment losses consist of water that is used or purged in the treatment process for backwashing filters, flushing, membrane concentrate, or other process uses. The simplest way to determine annual treatment loss is to subtract the annual finished water effluent volume (metered value) from the annual volume of treatment plant influent, if measured, or otherwise subtract the raw water conveyance loss from the raw water withdrawals (metered value). Where possible, these values should be corrected for metering error. In larger facilities, or in membrane-based treatment facilities, it can also be beneficial to meter and record volumes lost or consumed by individual treatment processes.

$$\text{raw water loss}\ (\text{mil gal}) = \text{raw water withdrawals} - \text{treatment plant influent} \qquad \text{(B-1)}$$

$$\text{treatment Loss}\ (\text{mil gal}) = \text{treatment plant influent} - \text{treatment plant effluent} \qquad \text{(B-2)}$$

When auditing is performed on the raw water supply as a separate and distinct water audit, all sources of water consumption and loss should be included. Potential sources of water consumption and losses to quantify in a raw water audit include:

- Volumes of raw water supplied to other users (utilities, agricultural users, etc.)
- Raw water real losses (pipeline leakage, leakage and evaporation from open conveyance and storage systems (canals, aqueducts, reservoirs), tank leakage, water lost back to the source via faulty check valves, etc.)
- Apparent losses (meter inaccuracies for wholesale raw water customers)
- Production meter inaccuracies on individual wells or surface water intake pumps

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

WATER LOSSES AND MANAGING WATER RESOURCES

In many cases, utilities are limited in the amount of water they may withdraw via regional permitting or water rights systems. This is often expressed as a water supply allocation. In these instances, controlling raw water losses, treatment losses, and distribution system losses can help utilities manage withdrawals to stay within their allowable water supply allocation. The difference between annual raw water withdrawal and the allowable yearly water supply allocation is sometimes referred to as the *remaining allocation* or *headroom* for the water utility.

Water loss control is a powerful tool to help utilities stay within their water supply allocation and reduce or defer the need for costly alternative water supply projects. As utilities implement water loss control programs, it can be useful to measure each of the sources of loss and consumption in the utility in relation to the water supply allocation. Once the raw water losses, treatment losses, and distribution system losses are quantified, the utility can compare these volumes to the total available water supply or water allocation and make a plan to reduce water losses or consumption over time. The goal of this plan should be to set loss reduction targets to ensure the utility stays within its available water allocation over the planning horizon. Another useful analytic technique would be tracking and graphing revenue water, Non-Revenue Water, and raw water losses in relation to the total allocation. This can allow the utility to track progress and ensure that water supply targets are being achieved.

ECONOMICS OF WATER LOSSES AND WATER RESOURCE MANAGEMENT

Utility economics play an important role in both water loss control and water resource management. When utilities are using water loss control as a strategy to manage limited water supplies, they must accurately value the cost of water at various stages of production. When evaluating the cost of real and apparent losses in the raw water system, costs will differ from those in a standard distribution system water audit. For raw water real losses, losses can be valued at the cost to transport the raw water to the water treatment plant (pumping from source, transmission costs) as well as any import or wholesale fees. As water flows through the treatment process and its quality improves, its value also increases. While water does not reach its full value until it leaves the treatment process, losses within a treatment plant can be generally valued at the total variable production cost (including both pumping and treatment costs) similarly to real losses in the distribution system in a typical water audit. Raw water apparent losses that occur from under-registering wholesale raw water meters should be valued as the billed cost to wholesale raw water customers (usually other utilities), representing a loss of revenue to the supplier of the raw water. Some raw water meters may over-register flow, possibly making conveyance losses appear higher than reality, or over-charging of the customer purchasing the raw water. Obviously, each utility will need to make smart decisions on how to value losses at each stage of the raw water and treatment system, and, in some cases a detailed assessment of plant operations can assign more detailed costs to water losses at specific points in the raw water system and treatment trains. The cost of distribution system real and apparent losses is covered in detail in Chapter 2.

In addition to the typical impact to annual utility finances, water loss control for water resource management has some additional factors to consider when valuing water. In some cases, utilities that purchase water from a wholesale authority or supplier may have increasing imported water fees based on allocation, total withdrawals, or peak demand. In this case, the cost of increased import fees or higher wholesale prices can be included in valuing all types of real losses. Additionally, if a utility is nearing capacity at its existing water supply or treatment facility, real loss control can be used to defer the need for

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

costly alternative water supplies or new production facilities. In this case, real water losses should be valued at the cost of the new, more expensive water supply when making economic decisions based upon loss control efforts and future water supplies. In all cases, the annual volume and costs of each type of loss should be evaluated within the scope of the water loss control program to meet both water supply and economic goals.

SUMMARY

As populations increase regionally, especially in areas experiencing water scarcity, it is critical that water resource planners and water utility managers address the sustainability of precious and limited water resources. A robust, thorough, and holistic water audit can include the investigation of use and loss components throughout the entire water production system to protect and control these resources for the future. Currently, many drinking water utilities must balance competing demands regarding growing populations, economic development, water rights, water affordability, climate change impacts, resource withdrawal/recharge imbalances, irrigation demands, and environmental protections; as well as political, social, and financial pressures. Having reliable data from a comprehensive water resource audit allows managers to make informed decisions on these complex issues.

REFERENCE

AWWA (American Water Works Association). 2017. Manual M50. *Water Resources Planning*. Denver, Colo.: AWWA.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Glossary

active leakage control: A proactive program implemented to identify and control unreported leakage in water distribution systems. Active leakage control includes determination of the nature and magnitude of unreported leakage and appropriate corrective actions.

apparent losses: Nonphysical (paper) losses that occur when water is successfully delivered to a water user but is not measured or recorded accurately, due to systematic data handling errors in meter reading and billing processes, customer metering inaccuracies, or unauthorized consumption. Apparent losses result in uncaptured revenue for the water utility and distortion of customer consumption data.

authorized consumption: The volume of water taken by registered customers, the water supplier, and others who are implicitly or explicitly authorized to do so by the water supplier for residential, commercial, industrial, institutional, agricultural, municipal, or other approved purposes. Authorized consumption may be billed or unbilled, metered, or unmetered.

automatic meter reading (AMR) and advanced metering infrastructure (AMI): These technologies provide electronic reading of customers' water meters and transfer of the data to a central location. AMR uses radio communications to obtain water meter readings and status data from endpoint devices connected to water meters. AMR typically employs walk-by or drive-by data collection, usually conducted every 30, 60, or 90 days. AMI is an integrated system of water meters, communication networks, and data management systems that enable two-way communication between meter and endpoint devices and the utility's central location. In the AMI environment, a permanent or fixed communication network is used, and meter readings are transmitted on a schedule or on demand, as frequently as hourly. The high granularity of AMI data may be used to create customer consumption profiles and to detect unusual changes in customer flow patterns. AMI also provides additional capabilities such as customer leak alerts, empty-pipe alerts, and supports remote disconnect (shut-off) valves.

average operating pressure (AOP): An estimated value of the composite water distribution system pressure that is used as an input in the water audit and is a component of the Unavoidable Annual Real Losses (UARL) calculation. AOP may be estimated using data from a calibrated distribution system hydraulic model or from static pressure measurements in the distribution system, or as a weighted composite of the average pressure of individual system pressure zones.

average pressure zone point (AZP): A determined reference point most representative of the average pressure across a District Metered Area or similar zone. It is used as a pressure monitoring point.

awareness, location, and repair times (ALR): The distinct time periods associated with response to a water distribution system failure (break or leak) broken into separate components, each of which should be measured and reported to conduct a leakage component analysis.

background leakage: The volumes from small leaks and weeps at pipe joints, valves, or other water system appurtenances at flow rates too low to be detected by traditional acoustic

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

leak detection methods of an active leakage control program. Background leakage is sensitive to pressure levels. Higher levels of background leakage are often found in water distribution systems with infrastructure in poor condition.

billed authorized consumption: The volume of authorized delivered water that is billed by the water utility. This includes both metered and unmetered consumption.

billed metered authorized consumption (BMAC): The volume of delivered water that is metered and billed to customers, including all groups of customers such as residential, commercial, industrial, or institutional, or to temporary or seasonal customers. The BMAC does not include water supplied to neighboring water utilities (water exported) that is metered and billed.

billed unmetered authorized consumption (BUAC): The volume of billed authorized consumption that is calculated based on estimates or norms from water usage sites that are unmetered. This does not include water supplied to neighboring water utilities (water exported) that is unmetered and billed.

bottom-up water audit approach: This water audit approach requires collection or generation of field-level data on individual loss components to quantify loss volumes and cost impacts as inputs to the water audit. A bottom-up approach is more resource-intensive and expected to provide a more-accurate and reliable water audit than the top-down water audit approach.

break: A term often used to identify a significant rupture in a pressurized pipeline that typically results in visible, disruptive, above-ground water requiring a prompt response from the water utility.

calibration: The act of performing a process used to correlate the output readings of a measuring device against a known quantitative standard and adjust the device settings as needed to ensure its accuracy. Production flowmeters should be regularly flow-tested for accuracy (*verification*) and their related instruments, or secondary devices, should be *calibrated* regularly. Calibration ensures that the secondary device functions properly but does not ensure flowmeter accuracy.

comparative meter flow test: An accuracy test of a flowmeter performed by comparing the meter-generated volume to the volume generated by a second calibrated metering device (often a temporary portable meter) installed in series with the first flow meter, over a designated period.

critical point (CP): The location in a zone or pressure-managed area of the water distribution system with the lowest pressure, caused by topography and/or hydraulic losses through the distribution system. Adequate pressure must be maintained at this point when introducing pressure management.

current annual real loss (CARL): A calculated volume of water lost from the system from reported leaks, unreported leaks, background losses, and storage tank overflows over a year.

customer metering inaccuracies (CMI): Apparent losses caused by the collective under- registration, over-registration, or malfunction of customer water meters. Specifically, inaccurate measurement of customer consumption can occur from meter wear, improper size or type of meter for the customer usage, improper installation, damage from aggressive water quality, malfunction, and other causes.

customer retail unit charge (CRUC): The weighted, average commodity that customers pay for water service in cost per unit volume of water consumed.

Data Validity Score (DVS): A measure of the overall trustworthiness of data entered in the AWWA Free Water Audit Software. The DVS is a composite value calculated from Data Grading values of data integrity assigned by the Audit Software for each input component. Since the use of the AWWA Free Water Audit Software is a desktop activity, the Data Gradings and the DVS are a good high-level representation of water audit data quality, but they do not necessarily confirm the degree of accuracy of the underlying data sources

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

maintained by the water utility. A higher DVS represents higher data trustworthiness.

demand: The amount of water needed to meet consumption, while sustaining adequate flow and pressure levels in a water distribution system. Components of demand include authorized consumption by customers or others permitted by the water utility to use water, real (leakage) losses, fire flow demands, and unauthorized consumption.

demand management: Strategic practices to control or reduce demand to support long-term sustainability of water resources. Demand management measures include water use conservation practices, minimization of water waste and leakage, conservation-oriented pricing, water recycling, and public education.

district metered area (DMA): A hydraulically discrete part of a water distribution system, with water supplied by one or more supply mains that are metered and closely monitored on a permanent basis. DMAs are used to analyze flows during minimum consumption periods (night flow analysis) to distinguish legitimate consumption from leakage and to provide an alert to newly emerging leaks and a more-reliable quantification of leakage volumes.

economic level of apparent losses (ELAL): The volume of apparent losses at which the sum of the cost of apparent loss reduction actions and the value of lost revenue caused by apparent losses is at a minimum. Reducing apparent losses below the ELAL is not cost-effective, because the cost of the loss abatement activities exceeds the value of water saved. ELAL may be used for apparent loss reduction target-setting.

economic level of leakage (ELL): The volume of real losses (leakage) at which the sum of the cost of real loss reduction activities and the value of the real losses is at a minimum. Reducing leakage levels below the ELL is not cost-effective, because the cost of the leak abatement activities exceeds the value of water saved. ELL may be used for leakage reduction target-setting.

failure: Referencing the distribution system, a general term used to encompass the terms *breaks* and *leaks collectively*.

flow test/fire flow test: A test conducted to determine the flow rate available from a specific location in the distribution system, typically performed by measuring flow from an open fire hydrant and logging the pressure reduction in the immediate area. This test may be used to quantify the system's capacity to deliver water for fire-fighting purposes.

infrastructure condition factor (ICF): The ratio between the actual volume of background leakage in a zone or district metered area and the calculated unavoidable background leakage volume of a well-maintained system.

infrastructure leakage index (ILI): The ratio of current annual real losses (CARL) to unavoidable annual real losses (UARL), used as a benchmarking indicator. Since the average operating pressure is a component of the UARL, the ILI is most reliable once any desired changes in system pressure have been implemented.

leak noise monitor: A leak detection device that measures sound characteristics of leak noise. Leak noise monitors are deployed in the distribution system and alert utility operators to newly emerging leak noise, thus minimizing the time for the operator to become aware of new leaks.

leakage: The water escaping from the pressurized distribution system caused by defects, ruptures, or failures in piping, pipe joints, valves, hydrants, taps, and other system appurtenances. Leakage occurrences are classified as reported leaks, unreported leakage, and background leakage.

leakage component analysis: A means to analyze the leakage occurring in a distribution system and quantify its components of background leakage, unreported leakage, and reported leakage. This analysis typically assesses leakage events in their three component phases—the awareness period, the location period, and the repair period.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

leakage management: The activities undertaken by water utilities to economically minimize real losses and inhibit new leaks from forming. These activities include leak detection surveys, use of district metered areas and minimum-hour (night flow) analysis, pressure management, system rehabilitation, and effective repair policies.

minimum-hour flow: The amount of water flowing into a discrete zone or district metered area during a 60-minute period of lowest consumptive demand, measured to help estimate unreported and background leakage.

minimum night flow (MNF): The amount of water flowing into a discrete zone or district metered area, typically between the hours of 2:00 a.m. and 4:00 a.m., taken as the period of lowest consumptive demand in areas of mostly residential water consumption. In areas with high non-residential water consumption, the period of minimum flow may not occur during night hours.

night flow analysis: A technique used to quantify leakage in a distribution system by measuring flows into a discrete zone or district metered area during minimum demand hours, less any change in storage volumes, and accounting for legitimate night consumption by customers. In areas with high non-residential water consumption, the period of minimum flow may not occur during night hours.

non-revenue water (NRW): Those components of system input volume that are not billed and produce no revenue. NRW equals unbilled authorized consumption plus apparent losses and real losses.

passive leakage control: A leakage control approach in which no systematic attempt is made by a water utility to be aware of, locate, or repair unreported or hidden leaks, or to manage background leakage. Instead, the utility addresses only visible, reported leaks and breaks (failures).

pressure managed area (PMA): Part of a district metered area (DMA), or the whole DMA, that is subject to pressure management.

pressure management: Optimization of pressures in a water distribution system to minimize leakage losses and surge-related damage while maintaining adequate water service, including fire-fighting flows. Usually employed to mitigate excessive pressure, pressure management can be very effective in reducing background leakage and inhibiting the occurrence of visible breaks.

pressure-reducing valve (PRV): A flow control valve used to allow water to flow from a higher-pressure zone to a lower-pressure zone so as not to exceed a set maximum pressure in the lower-pressure zone. A PRV may be outfitted with a variety of settings, features, and controllers enabling it to provide a wide range of pressure management capabilities.

rate of rise of unreported leakage (RR): The rate at which unreported leakage increases with time under a policy of passive leakage control at a specified average system pressure because of pipe deterioration and environmental effects. The rate of rise is not necessarily linear, as it can change because of seasonal temperature change, water hammer, land movement, and other impacts. It is usually expressed in a volume per day in a year, a volume per service connection per day in a year, or a volume per mile of mains per day in a year.

real losses: The physical water losses (leakage) from the pressurized distribution system and water storage facilities up to the customer meter or transfer point. Real losses include leakage from water mains, appurtenances, and service connections and storage tank overflows.

reported leaks: Those leakage events reported to the water utility by its employees (outside of active leakage control activities), the general public, or other parties as a result of water showing on the ground surface or other visible places, or from consumer complaints, such as poor pressure or noise in plumbing systems.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

revenue water: The volume of authorized consumption that is billed and produces revenue, including billed metered consumption and billed unmetered consumption.

service connection: The pipe connecting the water main supplying water to the measurement (customer metering) point, or the customer curb stop, as applicable, and supplying water to a customer's premises.

sounding: Determination of leak noise generated from pressurized water piping systems, leading to precise pinpointing of reported and unreported leaks. Leaks from metal pipes provide a stronger noise while leaks on plastic piping generate softer sounds and are more difficult to detect.

step testing: A test performed by gradually closing a valve on the sole input supply main and measuring successive flow and pressure reductions in an isolated zone or District Metered Area of a water distribution system. Both pressure and flow should be monitored during step testing because leakage rates are affected by the change in pressure that occurs as sections of the test grid are closed. Step testing may be used to estimate leakage and the potential for pressure management in a zone or DMA.

system input volume (SIV): The volume of water that has been treated and delivered to the utility's water distribution system. It is equal to the water utility's *Volume From Own Sources (VOS)* plus its *Water Imported Volume (WI)* during the audit period, plus or minus the net change in water storage where applicable and significant.

systematic data handling errors (SDHE): Customer consumption and billing data errors that occur in the water utility's business processes. These are apparent water losses caused by procedural, hardware, or software errors existing in the meter reading, data transfer, accounting, end user reporting, or archival function of customer consumption management.

top-down water audit approach: A method of compiling an annual water balance using readily available data, existing records, and estimated values, regardless of how complete and reliable they are. Because some records may be lacking, incomplete, or of poor accuracy, the top-down water audit is less accurate than a water audit done using a bottom-up approach where information is generated through detailed analysis and testing.

total annual cost of operating the water system (TAOC): Costs that include those for operations, maintenance, and any annually incurred costs for long-term upkeep of the system, such as repayment of capital bonds for infrastructure rehabilitation or renewal. Typical costs include employee salaries and benefits, materials, equipment, insurance, fees, chemicals, power, fuel, depreciation costs of assets, and all other costs that exist to sustain the drinking water supply. This is an optional input to version 6.0 of the AWWA Free Water Audit Software.

tracer gas method: A non-acoustic leak detection method often used for pinpointing small leaks in new pipelines undergoing hydrostatic testing but also applicable to leak pinpointing on pipelines in active service. A gas is injected into a section of water main that is believed to have a leak. At the point of leakage, the gas returns to its gaseous form, permeates directly to the surface, and can be detected above the surface of the pipeline, thereby indicating the location of the leak. Caution is urged in using this method because a high degree of operator knowledge and experience is needed to safely employ a tracer gas method for leak detection.

unauthorized consumption (UC): Any water taken from the water distribution system without the authorization of the water utility. Unauthorized consumption is a component of apparent losses and includes unpermitted water withdrawn from fire hydrants, illegal connections, bypasses to customer meters, meter or meter reading equipment tampering, or similar actions.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

unavoidable annual real losses (UARL): A reference measure of real losses in water utilities that cannot be eliminated based on current best management practice and technology. UARL represents the lowest loss technically achievable in a water utility based on its key characteristics. The UARL calculation was empirically derived by the Water Loss Specialist Group of the International Water Association; is based on measured frequencies, flow rates, and durations of background losses, reported leaks, and unreported leaks in well-maintained water distribution systems; and includes the pressure-leakage relationship. The UARL calculation may be less accurate for small systems with fewer than 3,000 service connections, and systems with very low or very high average operating pressure.

unavoidable background leakage (UBL): That portion of the background leakage (tiny weeps and seeps at pipe fittings, valves, fire hydrant connections, and customer service connection piping joints that are acoustically undetectable) that exists below the threshold that current best pressure management technology can address.

unbilled authorized consumption: All consumption that is not billed, but still authorized by the water utility. This includes unbilled metered and unmetered consumption.

unbilled metered authorized consumption (UMAC): Metered consumption that is authorized by the water utility, but, for any reason, is deemed by utility policy to be unbilled. This might include metered water consumed by the utility itself in treatment or distribution operations, or metered water provided to a civic institution free of charge.

unbilled unmetered authorized consumption (UUAC): Any authorized consumption that is not billed and is unmeasured. This component typically includes water used in activities such as firefighting, flushing of water mains and sewers, street cleaning, and fire flow tests conducted by the water utility.

unreported leaks: Leaks, usually hidden, that are found only if a water utility has an active leakage control program.

variable production cost (VPC): The costs to produce and supply the next unit of water (e.g., dollars per million gallons). This cost is determined by calculating the summed unit costs for ground and surface water treatment, bulk water purchased (if applicable), and all power used for pumping from the source to the customer. It may also include other related unit costs (such as residuals disposal costs) that apply to the production of drinking water.

volume from own sources (VOS): The volume of water withdrawn from water resources (rivers, lakes, streams, aquifers, etc.) controlled by the water utility, to be treated and available for potable water distribution. Along with Water Imported, it is a component of System Input Volume. If metering exists only at the raw water source, an adjustment for water used in the treatment process should be included to account for water consumed in treatment operations.

volumetric flow test: A procedure used to verify the accuracy of a water meter by passing a known volume of water through the meter and comparing it to the volume derived from the flow measurements of the meter for the test period. Volumetric flow testing is a meter *verification* process that is distinct and separate from the *calibration* of secondary devices (such as differential pressure cells) that might be connected to the flowmeter.

water audit: A thorough examination of the validity and accuracy of water utility data, records, accounts, policies, and practices regarding the volume and cost of water moved from source to treatment to distribution and customer consumption, ultimately distinguishing volumes reaching customers and revenue generated from volumes and cost of loss. Water audits can be performed in top-down (desktop assessment of records) or bottom-up (detailed field measurements and investigations to confirm records) fashion.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

water balance: The summary of key water audit data that shows water management from source to customer, with the sum of quantities in all columns equal and thus balancing. The standard water balance is shown in the figure.

water consumption: Water that reaches the customer destination. Consumption does not include water that is lost to leakage in the distribution system but does include leakage and water waste that occurs downstream of the customer metering or transfer point. Consumption can be authorized or unauthorized and may be billed or unbilled.

water exported (WE): The volume of bulk water conveyed and sold or transferred by the water utility to a neighboring water system that exists outside of its service area. Typically, the water is metered at the custody transfer point of interconnection between the two utilities.

water imported (WI): The volume of bulk water purchased or acquired by a water utility from a neighboring water utility or regional water authority, treated and available for distribution. Along with Volume From Own Sources, it is a component of System Input Volume. Typically, the water is metered at the custody transfer point of interconnection between the two entities.

water loss: The sum of apparent and real losses, calculated as the difference between the Water Supplied volume and Authorized Consumption volume. Water loss is considered as a total volume for the whole system, but may be calculated for partial systems, such as individual pressure zones or District Metered Areas during the water audit period.

water supplied: The volume of treated and delivered water supplied to the water distribution system of the water utility. It is equal to the Volume From Own Sources, plus the volume of Water Imported, minus the volume of Water Exported. If the water audit is conducted for a wholesale water agency that sells untreated water, then this quantity reflects the measure of the raw water, typically metered at the source.

water withdrawal: The process of drawing, or abstracting, water from a water source such as a well, lake, stream, river, quarry, or other source in a given period of time.

WLCC: The Water Loss Control Committee of the American Water Works Association.

<table>
<tr><td rowspan="13">Volume From Own Sources (corrected for known errors)

Water Imported (corrected for known errors)</td><td rowspan="13">System Input Volume</td><td>Water Exported (corrected for known errors)</td><td colspan="3">Billed Water Exported</td><td>Revenue Water</td></tr>
<tr><td rowspan="12">Water Supplied</td><td rowspan="4">Authorized Consumption</td><td rowspan="2">Billed Authorized Consumption</td><td>Billed Metered Consumption</td><td rowspan="2">Revenue Water</td></tr>
<tr><td>Billed Unmetered Consumption</td></tr>
<tr><td rowspan="2">Unbilled Authorized Consumption</td><td>Unbilled Metered Consumption</td><td rowspan="6">Non-Revenue Water</td></tr>
<tr><td>Unbilled Unmetered Consumption</td></tr>
<tr><td rowspan="6">Water Losses</td><td rowspan="3">Apparent Losses</td><td>Customer Metering Inaccuracies</td></tr>
<tr><td>Unauthorized Consumption</td></tr>
<tr><td>Systematic Data Handling Errors</td></tr>
<tr><td rowspan="3">Real Losses</td><td>Leakage on Transmission and Distribution Mains</td></tr>
<tr><td>Leakage and Overflows at Utility's Storage Tanks</td></tr>
<tr><td>Leakage on Service Connections up to the Point of Customer Metering</td></tr>
</table>

Note: Not to scale. All data in volume for the period of reference, typically one year.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Reference

WRF (Water Research Foundation). 2014. *Real Loss Component Analysis: A Tool for Economic Water Loss Control.* Project 4372a. Denver, Colo.: WRF.

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

Index

Note: *f* indicates figure; *t* indicates table

A

acoustic detection techniques, 183
acoustic leak detection, principles, 220
acoustic leak detection equipment, 222
acoustic leak detection surveys, 208
acoustic leak detection surveys, limitations, 232
active and inactive service connections, number, 61
active leakage control, 219
actual consumption in 2019, 49
addressing apparent losses, 293
addressing real losses, 295
AMR/AMI data, 35
AMR and AMI systems, benefits, 45
AMR meter reading, 116*f*
AOP, summary, 171*t*
apparent loss control strategy, 152*f*
apparent losses, 89
apparent losses, problems created, 101
appropriate meter sizing, 97
assessing water resources management, 339
assessment of water loss, 13
audit and control apparent losses, 86
auditing process, 17
audit software, 18
Audit Software, 75
audit software worksheet, 30
automatic meter reading, 109
available water resources, 4
average system pressure, 66*t*
AWWA Free Water Audit Software, 77*f*
AWWA free water audit software, 18*f*
AWWA Free Water Audit Software, 5
AWWA free water audit software, acronym key for terms, 21*t*
AWWA performance indicators, range of values, 81*t*
AWWA water audit, 287*t*
AWWA water audit compiler, 24
AWWA water audit methodology, 5

B

backflow preventers, 128
balancing flows across water supply system, 327
bank of water meters, 99*f*
benchmarking, 22
benchmarking progress, 5
benefit-cost ratio approach, 212
big box retail store customer consumption meter data, 136*f*
billed consumption, 50
billed metered authorized consumption, 33
billed unmetered authorized consumption, 38
billing adjustments, 45
billing functions, 47
billing system operations, 46
bottom-up field measurements, 11
bypass meter isolation valves, 144*f*

C

CAM implementation, 209
climate change impacts, 7
CMI, 125
CMI, commercial/industrial water rates, 18
communicating with community stakeholders-public relations, 282
compare volume measurements from the primary meter, 322
comparison of water flow in large flowmeters, 316*f*
compile the water balance, 82
component analysis summary CWC, 200*f*
compound error rates, 314
compound meter disturbed the flow profile, 130*f*
considered authorized consumption, 42
consumption underregistration and underbilling, billing patterns indicative, 118
control of real (leakage) losses, four-pillar approach, 218*f*
conventional DMAs, 179

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

correlating acoustic monitoring, excavated break detected, 210*f*
cost-benefit analysis, 213
cost curve for a mechanical water meter replacement program, 150*f*
County Water Company, AWWA free water audit software, 19*f*
County Water Company, meter testing data from a random sample of 50 meters, 57*t*
County Water Company, total metered water consumption by category, 34*t*
County Water Company, water loss control planning guide worksheet completed, 23*f*
County Water Company, water meters, 56*t*
CRUC, Volume-weighted average calculation, 71*t*
current annual real losses volume, 60
customer account inventory, 10
customer account management, 95
customer accounts, Number, 34*t*
customer billing data-mining analysis case study, 120
customer billing management technologies, 33
customer billing process, 51
customer billing system, 50*t*, 51*t*, 109
customer billing systems, 9
customer consumption, 48*t*
customer consumption, 9
customer consumption data collection, 55
customer expectations, 7
customer meter, 35
customer meter inaccuracy, 96
customer meter population demographics, 127*t*
customer meter readings, 36
customer meters, 53
customer meter testing program, 294
customer population, 38
customer premise, customer meter located, 64*f*
customer retail unit charge, 69
customer service connection piping leak repair policy, 264
customer service line, 62
customer service line, average length, 63*f*
customer service line piping leakage, 169
CWC, changes in reservoir/tank storage, 27*t*
CWC, source water measuring devices, 25*t*
CWC, total water supply in million gallons, 25*t*
CWC auditor, 43, 52

D

dashboard for county water company, AWWA free water audit software, 20*f*
data, routine annual collection, 17
data analysis errors, 46
data analytics software, 134
data corruption, 27
data gap and adjustments for pumped water flow, example, 335*t*
data grade for UMAC, 39
data grading, 30
data-mining analysis chart for the City of Wylie, Tex., 122*f*
data transfer errors, 44
default options, 84
determine system attributes, 73
developing a nonrevenue management program, 291
differential pressure transmitters, bank, 306*f*
discovered leaks, 180
discrete pressure zone or DMA, system boundaries, 15*f*
distribution system flow metering, 174*f*
district metered areas, 173, 235
DMA boundaries, 176
DMA boundary, 178
DMA 5 in Philadelphia, 258*f*
DMA planning, 175
DMA/PMA configuration, 257
drinking water supply systems, metering locations, 16*t*

E

economic intervention frequency, 196
economic level of leakage, 201
economics of leak noise monitors, 226
economic unreported real losses for regular survey, 206*f*
equitable water rates, 4
establishing a cross-functional NRW management group or task force, 275
evaluating production flow data, 336
evaluations of customer consumption, 14
example leak noise signature, 167*f*
example tabulations of a drop test, 313*t*
excessive water pressure, 296

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

F

final leak detection survey, 260*f*
finishing nail, 145*f*
fire hydrants, 42
fire hydrant usage policy, 42
flow-based dynamic modulation, 251*f*
flow-based dynamic modulation mode, 250*f*
flowmeter in-place, 315
flowmeter installation, 308*f*
flow step tests to localize leakage, 237
four-pillar approach to the control of apparent losses, 149*f*

G

gallons, apparent water loss, 108*f*
geographic information systems, 17
granularity, 94
graph produced from Wylie, Tex., 135*f*
ground-penetrating radar, 234
growing populations, 7

H

high or continuous water consumption, proactive detection, 110
high-quality water, 4
hose connected to large meter test port, 132*f*
how to locate leaks, 182
hydrophone retrofitted to fire hydrant base, 186*f*

I

IDG questions, 85
impact of CAM on RR and leakage reduction, 211*f*
implementing water loss control programs in small systems, 286
importer water utility, 31
3-in compound meter, 146*f*
4-in horizontal turbine meters for an Illinois water utility, data logging results, 138*t*
initial corrections, 156
inline meter accuracy verification, production flowmeter system procedure, 321*t*
3-in meters in Wylie, Tex., data-mining analysis chart, 124*f*
insertion magnetic flowmeter, 304*f*
installation of a PRV chamber, 177*f*
installing flowmeters, 308
interactive data grading for volume, AWWA free water audit software, 20*f*

J

joints and cracks, leakage losses, 181*t*
jumper pipe, 108*f*
jurisdictions, 32

L

lag-time adjustments, 36
large customer meter accuracy testing program, 131
large meters, example volume percentages, 58*t*
large meters for County Water Company, meter test data, 58*t*
large meters for CWC, turbine meter test data, 133*t*
large meter test truck, interior, 132*f*
large water main break, 164*f*
large water meter error, calculation, 58*t*
LCA model results, summarize, 199
leakage and appropriate intervention tools, components, 168*f*
leakage component analysis, 11
leakage components in a water distribution system, 192
leakage detection techniques, 217
leakage management activities, 2
leakage management program, 188
leakage management results, 269
leakage unit rates from the WARD, 189*t*
leak correlation, 184*f*
leak detection by survey for CWC, economic intervention frequency, 207*f*
leak detection interventions, 202
leak detection survey, completion, 259*f*
leak detection survey daily log, 268
leak detection surveys, 211
leak detection surveys, economic intervention frequency, 203
leak noise data loggers, 225
leak noise monitors, 185*f*
leak noise monitors, 224
leak repair techniques, 262
life-line water rates, 100
located by acoustic leak detection, types of leaks

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

customer meter leaks, 221
customer service connections, 221
premise plumbing, 222
system appurtenances, 221
water mains, 221
long-term loss control, 22
Lp, data grade, 63

M

managing unbilled authorized consumption, 292
managing water resources, 343
manual acoustic leak detection surveys, 231
manual meter reading, 44
map of fire hydrants, 67*t*
mechanical positive displacement meter replacement, 151
meter accuracy testing protocols, 54
meter data management, 112
meter demographics, 126
metered consumption volumes, 35
metered customer population, 38
metering fire services, 140
meter lag correction, 37*f*
meter-reading process, 93
meter-reading success rate, 45
meter replacement, 139
meter test bench for testing small water meters, 129*f*
meter testing facility, 29
meter-testing laboratories, 30
minimum-hour flow analysis, 236
modified AWWA water balance, 341*f*
moisture, accumulation, 93*f*
monthly NRW report for CWC, 281*t*

N

negative consumption number, 49
noise interference, 223
non-acoustic leak detection, 233
nonpaying customers, 102
nonrevenue water, 1
nonrevenue water management, 3
nonrevenue water (NRW) management program, 273
nonrevenue water policy, 5
North American water utilities, 2
NRW management program, 283
NRW management program, 2
NRW management program, benefits, 274
NRW management program, launching, 278
NRW management program activity planning matrix, 279*t*
NRW management progress, 280

O

open bypass line valve, 143*f*
operational perspectives, 13
optimized pressure management, 255
organizational structure, 276
organizing an acoustic leak detection program, 229
orifice plate flowmeter installation, cross-sectional view, 305*f*
other adjustments, 27

P

performance indicators, 76
performance indicators, 13
performance metrics, 6
piping distribution system, 162
policy and procedure shortcomings, 96
poor flowmeter installation configurations, problems, 309
portion of revenue, 43
positive water consumption, 59
predictive techniques, 187
preliminary benefit-cost analysis, 244
pressure and flow variations in a DMA, 241*f*
pressure control devices, 248*f*
pressure management, 239
pressure management, 242
pressure management program, 2
pressure management program, 270
pressure management strategy, refinement, 259*f*
pressure management system using PRVs, installation, 253
pressure-reducing valve, 129*f*
pressure-reducing valves, 246
pressure relief systems, 245
pressure stabilization and reduction strategies, 240
pressure zones and DMAs, 328
pressure zones in the water service area, 247*f*
primary and secondary device accuracy and calibration, 307
procedure shortcomings for CWC, 52

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

production flowmeter management, 26
production flowmeters, 24
production meters, data generated, 337
proper meter installation, 53
public relations flyer cautioning communities, 142*f*
pumped supply reporting, data gaps, 334

Q

quantify authorized consumption, 32
quantify individual leak types, 198
quantifying leakage by components, 191
Queen Lane Water Treatment Plant, 325*t*
Queen Lane Water Treatment Plant, mass balance report, 324*t*

R

ratepayers, 3
raw water supply, 342
real (leakage) losses, control, 190*f*
recommendations for desired upstream, 310*t*
reliable distribution system failure data collection, importance, 193
reliable meter reading, 47
repair clamps, 263*f*
repairs, speed and quality, 261
reporting of water production volumes, 302
reservoir/tank storage changes, 28
residential meter customer accounts, 123*f*
residential meters, 54
residential meter service, 147*f*
residential water meter error, calculation, 57*t*
residential water-meter error, sample calculation, 57*t*
resource needs, 277
revenue-dependent field values, 10
revenue protection plan, 153
revenue protection program summary, 155
revenue recovery, 3
routine data-collection processes, 17
RR of leakage and changing leakage levels, 205*f*
rulemaking process, 6

S

Sahara technology, 228*f*
sample revenue protection plan, 154
SCADA system, 29
SCADA system water supply tracking report, 332*f*
SDHE, 43
SDHE, sources, 113*f*
secondary device calibration, 303
single-family residential water consumption, 121*t*
small leak on customer service piping, 165*f*
small systems, 285
small water utilities, accountability, 299
sniffer dogs, 235
source-specific VPCs, 72
source water meters, 26
standard water audit, 18
strict auditing procedures, 4
systematic data handling errors, 105
systematic data-handling errors, 91
systematic data-handling errors, 11
systematic data handling errors, sources, 114*f*
system boundaries, 84
system boundaries, 14
system schematic using fixed outlet control mode, 249*f*

T

tank leaks, 160*f*
tank storage, 27
target-setting, 22
tethered inline leak detection technology, use, 228*f*
tethered systems, 227
time-based modulation, 252
time on leakage losses, effect, 163
top-down water audit data, 14
total supply, 26
treated water clearwell, schematic, 312*f*
treated water distribution system, identifying system boundaries, 15*f*
typical pressure management and DMA supply installation, 254*f*

U

UARL calculation, component values, 75*t*
UARL calculation, standard unit values used, 75*t*
UC, 141
unauthorized consumption, 9, 107
unauthorized consumption, criminal prosecution, 148
unavoidable annual real losses, 74

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

unbilled authorized consumption, 33
unbilled consumption, 39
Unbilled Unmetered Authorized Consumption, 40
Unbilled Unmetered Authorized Consumption, individual estimates, 41*t*
uncontrolled apparent losses in a water utility over time, impact, 102*f*
underbilling customers, 119
unmetered customer properties, 64*f*
unnecessary water withdrawals, 161
untreated (RAW) water supply volume, 340
use a meter testing facility, 320
utility staffing changes, 111
utility with AMI meter reading, 117*f*
utility with manual meter reading, billing process chart, 115*f*
UUAC, volume, 40
UUAC default values, history, 41

V

validating production flowmeter data, 301
validating Scada system output data, 333
value-based performance indicators, 80
variable production cost, 70
various leakage types, example, 166*f*
velocity profile in a smooth pipeline, 317*f*
velocity profile in pipeline with moderate internal corrosion, 318*f*
velocity profile skewed by close proximity, 319*f*
volume-based performance indicators, 78
volume from own sources in million gallons for CWC, 30*t*
volume of logged consumption, 137
volume of UMAC, 39
VOSEA volume, 30
VOS error adjustment, 28
VOS quantity, 24

W

waste during large meter testing operation, water being discharged, 132*f*
water accountability, 6, 7
water accounting practices, 1
water audit, 106
water audit, 36, 42
water audit, system boundaries, 14*f*
water audit outputs, 85
water audit results, 297
water audits, 3
water audit using the audit software, 290
water balance, 12*f*
water balance calculation, 11
water balance for county water company-2020 calendar year, 12*f*
water balance terms and definitions, 13*t*
water conservation programs, 256
water consumption in Wylie, Tex, 121*t*
water distribution system, 1
water distribution system, calculating average pressure, 68
water distribution systems, 159, 172
water exported error adjustment, 32
Water Imported (WI) from outside sources, 31
water loss control, financial resources, 285
Water Loss Control Planning Guide, 22
water losses, economics, 343
water mains repair frequency, 243*f*
water meter installed upside down, 98*f*
water meter readings, 92
water meters, 46
water pressure and leakage, 170
water scarcity, 344
water source, 26
water sources, 6
water storage tank drop test, 311
water storage tank report, 330*t*
water supplied, annual volumes, 298*f*
water supplied downstream, 9
water supply balancing report, 329*t*
water supply configuration, schematic, 323*f*
water supply operations, 40, 284
water systems, 10
water treatment plants, 14
water utilities, 1
water utility, billing process, 90
water utility distribution system, determining average operating pressure, 65*t*
water utility operator, 113
wholesale water systems, 83
WI error adjustment, 31
WRF project 4372-b, 194*f*
WRF Project 4372-b, leak repair data collection guide, 195*f*
WWAWC's water distribution system, 238

Z

zonal water balance, 10

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

AWWA Manuals

M1, *Principles of Water Rates, Fees, and Charges*, #30001
M2, *Instrumentation and Control*, #30002
M3, *Safety Management for Water Utilities*, #30003
M4, *Water Fluoridation Principles and Practices*, #30004
M5, *Water Utility Management*, #30005
M6, *Water Meters—Selection, Installation, Testing, and Maintenance*, #30006
M7, *Problem Organisms in Water: Identification and Treatment*, #30007
M9, *Concrete Pressure Pipe*, #30009
M11, *Steel Pipe—A Guide for Design and Installation*, #30011
M12, *Simplified Procedures for Water Examination*, #30012
M14, *Backflow Prevention and Cross-Connection Control: Recommended Practices*, #30014
M17, *Fire Hydrants: Installation, Field Testing, and Maintenance*, #30017
M19, *Emergency Planning for Water and Wastewater Utilities*, #30019
M20, *Water Chlorination/Chloramination Practices and Principles*, #30020
M21, *Groundwater*, #30021
M22, *Sizing Water Service Lines and Meters*, #30022
M23, *PVC Pipe—Design and Installation*, #30023
M24, *Planning for the Distribution of Reclaimed Water*, #30024
M25, *Flexible-Membrane Covers and Linings for Potable-Water Reservoirs*, #30025
M27, *External Corrosion Control for Infrastructure Sustainability*, #30027
M28, *Rehabilitation of Water Mains*, #30028
M29, *Water Utility Capital Financing*, #30029
M30, *Precoat Filtration*, #30030
M31, *Distribution System Requirements for Fire Protection*, #30031
M32, *Computer Modeling of Water Distribution Systems*, #30032
M33, *Flowmeters in Water Supply*, #30033
M36, *Water Audits and Loss Control Programs*, #30036
M37, *Operational Control of Coagulation and Filtration Processes*, #30037
M38, *Electrodialysis and Electrodialysis Reversal*, #30038
M41, *Ductile-Iron Pipe and Fittings*, #30041
M42, *Steel Water-Storage Tanks*, #30042
M44, *Distribution Valves: Selection, Installation, Field Testing, and Maintenance*, #30044
M45, *Fiberglass Pipe Design*, #30045
M46, *Reverse Osmosis and Nanofiltration*, #30046
M47, *Capital Project Delivery*, #30047
M48, *Waterborne Pathogens*, #30048
M49, *Quarter-Turn Valves: Head Loss, Torque, and Cavitation Analysis*, #30049
M50, *Water Resources Planning*, #30050
M51, *Air Valves: Air-Release, Air/Vacuum and Combination*, #30051
M52, *Water Conservation Programs—A Planning Manual*, #30052
M53, *Microfiltration and Ultrafiltration Membranes for Drinking Water*, #30053
M54, *Developing Rates for Small Systems*, #30054
M55, *PE Pipe—Design and Installation*, #30055
M56, *Nitrification Prevention and Control in Drinking Water*, #30056
M57, *Algae: Source to Treatment*, #30057
M58, *Internal Corrosion Control in Water Distribution Systems*, #30058
M60, *Drought Preparedness and Response*, #30060
M61, *Desalination of Seawater*, #30061
M62, *Membrane Applications for Water Reuse*, #30062
M63, *Aquifer Storage and Recovery*, #30063
M65, *On-Site Generation of Hypochlorite*, #30065

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66

M66, *Cylinder and Vane Actuators and Controls—Design and Installation*, #30066

M68, *Water Quality in Distribution Systems*, #30068

M69, *Inland Desalination and Concentrate Management*, #30069

M77, *Condition Assessment of Water Mains*, #30077

Copyright © 2024 by American Water Works Association. All Rights Reserved
Accessed by account: Olsson, Inc. | User: Jon Cronin | Date: Wed Oct 07 04:57:18 UTC 2026 | IP address: 149.106.110.66